TogetherWeGo: complete source code

FBLA Mobile Application Development 2026-27, “Together We Go: Group Trip Planner”. Expo SDK 57 / React Native / TypeScript, with a small Node sync server. 241 files, 24,808 lines. Photos and fonts are listed in ATTRIBUTIONS.md and are not printed.

1. README.md
2. ATTRIBUTIONS.md
3. package.json
4. app.json
5. eas.json
6. tsconfig.json
7. eslint.config.js
8. jest.config.js
9. jest.setup.js
10. src/app/(tabs)/_layout.tsx
11. src/app/(tabs)/budget.tsx
12. src/app/(tabs)/chat.tsx
13. src/app/(tabs)/explore.tsx
14. src/app/(tabs)/itinerary.tsx
15. src/app/(tabs)/profile.tsx
16. src/app/_layout.tsx
17. src/app/about.tsx
18. src/app/airport.tsx
19. src/app/bucket-list.tsx
20. src/app/crew.tsx
21. src/app/currency.tsx
22. src/app/destination/[id].tsx
23. src/app/destinations.tsx
24. src/app/documents.tsx
25. src/app/emergency.tsx
26. src/app/flights.tsx
27. src/app/forgot-password.tsx
28. src/app/help.tsx
29. src/app/how-it-works.tsx
30. src/app/index.tsx
31. src/app/join.tsx
32. src/app/map.tsx
33. src/app/nearby.tsx
34. src/app/new-trip.tsx
35. src/app/notifications.tsx
36. src/app/onboarding.tsx
37. src/app/packing.tsx
38. src/app/panorama.tsx
39. src/app/personalize.tsx
40. src/app/photos.tsx
41. src/app/phrasebook.tsx
42. src/app/quiz.tsx
43. src/app/recap.tsx
44. src/app/settings.tsx
45. src/app/sign-in.tsx
46. src/app/sign-up.tsx
47. src/app/survey.tsx
48. src/app/trips.tsx

49. src/app/weather.tsx
50. src/components/CurrencyConverter.tsx
51. src/components/CurrencyPickerSheet.tsx
52. src/components/GuidedTour.tsx
53. src/components/Logo.tsx
54. src/components/QrCode.tsx
55. src/components/ReminderService.tsx
56. src/components/ShareSheet.tsx
57. src/components/SkylineHero.tsx
58. src/components/TabBar.tsx
59. src/components/auth/GoogleSheet.tsx
60. src/components/auth/legal.ts
61. src/components/budget/AddExpenseSheet.tsx
62. src/components/budget/BalancesCard.tsx
63. src/components/budget/DailyPlanner.tsx
64. src/components/budget/ExpenseDetailSheet.tsx
65. src/components/budget/ExpensesSection.tsx
66. src/components/budget/SettleSheet.tsx
67. src/components/budget/SpendSummary.tsx
68. src/components/budget/styles.ts
69. src/components/chat/ChatBubble.tsx
70. src/components/chat/ChatHeader.tsx
71. src/components/chat/Composer.tsx
72. src/components/chat/EmptyChat.tsx
73. src/components/crew/AddMemberSheet.tsx
74. src/components/crew/ChecklistSection.tsx
75. src/components/crew/DocsSection.tsx
76. src/components/crew/InviteCard.tsx
77. src/components/crew/InviteQrSheet.tsx
78. src/components/crew/MembersSection.tsx
79. src/components/crew/NewPollSheet.tsx
80. src/components/crew/PollsSection.tsx
81. src/components/crew/RoleSheet.tsx
82. src/components/documents/DocFileBody.tsx
83. src/components/documents/DocRow.tsx
84. src/components/explore.tsx
85. src/components/flights/AddFlightSheet.tsx
86. src/components/home/BudgetTile.tsx
87. src/components/home/DiscoverSection.tsx
88. src/components/home/GlanceTiles.tsx
89. src/components/home/HomeHero.tsx
90. src/components/home/NearbySection.tsx
91. src/components/home/PicksSection.tsx
92. src/components/home/QuickTiles.tsx
93. src/components/home/UpNextSection.tsx
94. src/components/home/__tests__/picks.test.ts
95. src/components/home/picks.ts
96. src/components/join/QrScanner.tsx
97. src/components/map/OfflineRadar.tsx
98. src/components/map/TripMap.tsx
99. src/components/map/TripMap.web.tsx
100. src/components/map/types.ts
101. src/components/panorama/Scene.tsx
102. src/components/panorama/shapes.ts
103. src/components/personalize/options.ts

104. src/components/personalize/palette.ts
105. src/components/personalize/parts.tsx
106. src/components/personalize/steps/StepDone.tsx
107. src/components/personalize/steps/StepHello.tsx
108. src/components/personalize/steps/StepHome.tsx
109. src/components/personalize/steps/StepInterests.tsx
110. src/components/personalize/steps/StepStyle.tsx
111. src/components/personalize/steps/StepVibe.tsx
112. src/components/personalize/steps/StepYou.tsx
113. src/components/personalize/styles.ts
114. src/components/photos/PhotoTile.tsx
115. src/components/photos/PhotoViewer.tsx
116. src/components/pickers.tsx
117. src/components/settings/AccessibilityCard.tsx
118. src/components/settings/AssistantCard.tsx
119. src/components/settings/DataCard.tsx
120. src/components/settings/DemoCard.tsx
121. src/components/settings/NotificationsCard.tsx
122. src/components/settings/ProfileCard.tsx
123. src/components/sync/SyncBadge.tsx
124. src/components/sync/SyncSettingsCard.tsx
125. src/components/sync/SyncTripCard.tsx
126. src/components/trip/ActivityCard.tsx
127. src/components/trip/AddActivitySheet.tsx
128. src/components/trip/MemberPickerSheet.tsx
129. src/components/trip/NoTrip.tsx
130. src/components/trip/TripMenuSheet.tsx
131. src/components/trip/meta.ts
132. src/components/ui/Button.tsx
133. src/components/ui/Donut.tsx
134. src/components/ui/Input.tsx
135. src/components/ui/Press.tsx
136. src/components/ui/Screen.tsx
137. src/components/ui/Segmented.tsx
138. src/components/ui/Sheet.tsx
139. src/components/ui/Slider.tsx
140. src/components/ui/T.tsx
141. src/components/ui/bits.tsx
142. src/components/ui/feedback.tsx
143. src/components/ui/motion.tsx
144. src/hooks/__tests__/useReduceMotion.test.ts
145. src/hooks/useReduceMotion.ts
146. src/hooks/useTrip.ts
147. src/i18n/index.ts
148. src/i18n/strings.ts
149. src/services/__tests__/assistant.test.ts
150. src/services/__tests__/pretendFriends.test.ts
151. src/services/assistant.ts
152. src/services/currency.ts
153. src/services/files.ts
154. src/services/location.ts
155. src/services/notifications.ts
156. src/services/pretendFriends.ts
157. src/services/print.ts
158. src/services/secrets.ts

159. src/services/share.ts
160. src/services/weather.ts
161. src/store/__tests__/auth.test.ts
162. src/store/__tests__/trips.test.ts
163. src/store/seed.ts
164. src/store/selectors.ts
165. src/store/slices/app.ts
166. src/store/slices/auth.ts
167. src/store/slices/budget.ts
168. src/store/slices/demo.ts
169. src/store/slices/files.ts
170. src/store/slices/groupTools.ts
171. src/store/slices/helpers.ts
172. src/store/slices/itinerary.ts
173. src/store/slices/social.ts
174. src/store/slices/trips.ts
175. src/store/slices/types.ts
176. src/store/store.ts
177. src/store/types.ts
178. src/store/useAppStore.ts
179. src/sync/__tests__/blobs.test.ts
180. src/sync/__tests__/crdt.test.ts
181. src/sync/__tests__/e2e.test.ts
182. src/sync/__tests__/engine.test.ts
183. src/sync/__tests__/invite.test.ts
184. src/sync/__tests__/secureRandom.test.ts
185. src/sync/__tests__/transport.test.ts
186. src/sync/actions.ts
187. src/sync/adapter.ts
188. src/sync/blobCache.ts
189. src/sync/blobs.ts
190. src/sync/crdt.ts
191. src/sync/engine.ts
192. src/sync/fileio.ts
193. src/sync/hlc.ts
194. src/sync/invite.ts
195. src/sync/notify.ts
196. src/sync/prefs.ts
197. src/sync/runtime.ts
198. src/sync/schema.ts
199. src/sync/secureRandom.ts
200. src/sync/status.ts
201. src/sync/storage.ts
202. src/sync/testing.ts
203. src/sync/timers.ts
204. src/sync/transport.ts
205. src/sync/useFile.ts
206. src/sync/view.ts
207. src/theme/index.ts
208. src/utils/__tests__/money.test.ts
209. src/utils/__tests__/password.test.ts
210. src/utils/__tests__/qr.test.ts
211. src/utils/__tests__/time.test.ts
212. src/utils/__tests__/validation.test.ts
213. src/utils/format.ts

214. src/utils/geo.ts
215. src/utils/money.ts
216. src/utils/password.ts
217. src/utils/qr.ts
218. src/utils/time.ts
219. src/utils/validation.ts
220. server/__tests__/server.test.js
221. server/index.js
222. server/store.js
223. e2e/README.md
224. e2e/lib.js
225. e2e/motion.js
226. e2e/nesting.js
227. e2e/overflow.js
228. e2e/run.js
229. e2e/serve.js
230. e2e/twophones.js
231. e2e/verify.js
232. e2e/verify2.js
233. e2e/verify3.js
234. e2e/verify4.js
235. scripts/source-pdf.js
236. docs/BUILDING.md
237. docs/DEMO_SCRIPT.md
238. docs/JUDGE_CHECKLIST.md
239. docs/PLANNING.md
240. docs/RATING_SHEET.md
241. docs/SYNC.md

README.md

```
# TogetherWeGo ✈️ — Group Trip Planner

**FBLA Mobile Application Development 2026–27 · Topic: “Together We Go: Group Trip Planner”**

TogetherWeGo helps a group of friends or family plan, budget and enjoy a shared trip — from the first idea, through the trip itself, to the final recap. It runs natively on **iPhone and Android** (Expo / React Native, TypeScript) and keeps working **offline**, which matters at a conference venue with unreliable wifi.

<p>
<img src="docs/screenshots/03-home-night.jpg" width="170" alt="Home with the night skyline hero"/>
<img src="docs/screenshots/04-home-day.jpg" width="170" alt="Home by day"/>
<img src="docs/screenshots/05-plan.jpg" width="170" alt="The Plan"/>
<img src="docs/screenshots/06-vote.jpg" width="170" alt="Vote tab with Lock in"/>
<img src="docs/screenshots/07-budget.jpg" width="170" alt="Budget with who owes who"/>
<img src="docs/screenshots/08-chat-scout-ai.jpg" width="170" alt="Chat: Crew and Scout AI"/>
<img src="docs/screenshots/09-more.jpg" width="170" alt="More menu"/>
<img src="docs/screenshots/19-360-day.jpg" width="170" alt="360° view of Paris by day"/>
<img src="docs/screenshots/20-360-night.jpg" width="170" alt="360° view of Paris at night with the lights on"/>
<img src="docs/screenshots/15-make-it-yours.jpg" width="170" alt="Make it yours: pick a colour theme"/>
<img src="docs/screenshots/16-midnight-theme.jpg" width="170" alt="Home in the Midnight dark theme"/>
<img src="docs/screenshots/22-sync-chat.jpg" width="170" alt="Group chat on a phone synced with a friend's phone"/>
<img src="docs/screenshots/23-invite-qr.jpg" width="170" alt="Invite QR code"/>
<img src="docs/screenshots/24-sync-photos.jpg" width="170" alt="Photo dump showing a friend's photo"/>
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## 1. Run it on your phone (5 minutes)

**You need:** a laptop with [Node.js 20+](https://nodejs.org) and **Expo Go** on your phone. This project uses **Expo SDK 57**, so it must be the SDK 57 build of Expo Go:

* **Android:** install it from [expo.dev/go](https://expo.dev/go) (pick SDK 57; the Play Store copy can lag behind a new SDK).
* **iPhone:** Apple's App Store copy of Expo Go stops at SDK 54, so build your own with `npx eas-cli@latest go` and install it through TestFlight. That needs the paid Apple Developer Program, and each teammate's Apple ID must be added as an *internal tester* (App Store Connect → Expo Go → TestFlight). Expo's walkthrough: [Set up your environment → iOS device with Expo Go](https://docs.expo.dev/get-started/set-up-your-environment/).
* **No Apple account?** Use an Android phone, or the web build on the iPhone ([docs/BUILDING.md](docs/BUILDING.md)).

```bash
```
```

```
cd together-we-go
npm install
npx expo start            # same wifi for phone + laptop
# or, on school / conference wifi that blocks devices from seeing each other:
npx expo start --tunnel
```

Scan the QR code (iPhone: Camera app · Android: Expo Go → *Scan QR code*).

On an iPhone, sign in first. Expo Go only opens a project from your laptop when the laptop and the phone use the *same* Expo accoun
t: run `npx expo login` on the laptop (check with `npx expo whoami`), then tap the account icon (top right) in Expo Go and sign in with
that account. Android phones do not need this.

A teammate cannot open the app?
One running `npx expo start` can serve several phones. Match what the teammate's phone says:

The phone shows	Fix
You need to be signed in to Expo Go and Expo CLI… or *You're signed in to Expo CLI as … and to Expo Go as …* (iPhone only)	Use the
same Expo account in both places: `npx expo whoami` on the laptop, account icon in Expo Go, then tap *Try again* (restart `npx expo star
t` if you changed the laptop's account). A teammate who wants their own account runs the app from their own laptop instead: `git clone`,
`npm install`, `npx expo login`, `npx expo start`.
Project is incompatible with this version of Expo Go or *requires a newer version of Expo Go*	Wrong Expo Go version: install the S
DK 57 build (see above).
It keeps spinning, or *Could not connect to the server* / *Network request failed*	Same wifi as the laptop (a phone hotspot is safes
t). Windows: allow Node.js on **private** networks in the firewall prompt. Turn VPNs off. iPhone: *Settings → Expo Go → Local Network* o
n. Not on the same wifi at all? Start with `npx expo start --tunnel`.
Scanning does nothing	iPhone: scan with the Camera app. In Expo Go choose *Enter URL manually* and type the `exp://…` address printe
d under the QR code.

Demo login (also shown on the sign-in screen — tap the box to fill it in):

Email	Password
`maya.chen@togetherwego.app`	`TravelTogether2026!`

The sample trip is generated **relative to today**, so “Amalfi Squad” (4 friends, 6 days in Positano, Italy) always starts in 3 days; a
second 10-day “Tokyo & Kyoto Explorer” trip, a Bali plan and a finished Barcelona trip show multi-trip management. *Settings → Demo time
line* rebuilds it as **upcoming**, **in progress (Day 2)** or **just finished** to show every stage.

Plan together on different phones
Friends can each use their own phone and see the same trip, live. Start the small sync server on your computer in a **second terminal**:

```bash
npm run server        # prints the address your phones will use, e.g. http://192.168.1.20:8787
```

Then on phone A: *More → Crew & roles → Share this trip across phones*. On phone B: *Join a trip* and type the invite code (or scan its
QR code). Chat, votes, the plan, expenses, photos, documents and receipts now update on both, and catch up after being offline. During d
evelopment the phones find the server by themselves; otherwise type its address in *Settings → Sync across phones*. Details, the merge r
ule and the limits are in **docs/SYNC.md**.

For the competition: run without Expo Go or venue wifi
Expo Go needs the laptop's dev server. For an installed copy, build an Android `.apk` with EAS (free account) or use Expo Go over a phon
e hotspot. Steps, the iPhone options and what each choice costs are in **docs/BUILDING.md**. Once installed, everyth
ing except live weather, live exchange rates, map tiles and the optional Claude AI works offline — and those fall back to cached or esti
mated data automatically.

Other scripts: `npm test` (unit tests), `npm run e2e` (browser tests, see [`e2e/README.md`](e2e/README.md)), `npm run web` (browser prev
iew), `npx tsc --noEmit` (type-check), `npm run lint`.

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2. What it does — mapped to the brief

Brief area	Feature in TogetherWeGo	Where
Collaboration	Trips with auto-generated invite codes (`AML-2026`), join by code, deep link or **QR code**	Home code pill · Mor
e → Crew & roles · *Join a trip*
	Live sync across phones: chat, votes, plan, expenses, checklist, photos, documents and receipts, with offline catch-up	More →
Crew & roles → *Share this trip across phones* · docs/SYNC.md
	Roles: Organizer, Treasurer, Navigator, Food Lead, Photographer… with descriptions	More → Crew & roles
	Shared checklist with assignees	More → Crew & roles
	Voting: 👍/👎 on activities (majority auto-confirms) + multi-option polls	Plan → Vote tab (progress bar + **Lock in**) · More → Cr
ew & roles (polls)
Communication	Group trip chat with typing indicators	Chat tab
	Trip Assistant (AI chatbot) — answers from live trip data: budget, who owes who, weather, food nearby, next activity, flights, t
ime zones, currency, phrases, emergencies. Works offline; optional Claude mode with your own key	Chat tab (✨ button or quick chips)
Scheduling	Day-by-day itinerary/calendar with timeline, status pills, assignees	Plan tab
	Members upload flight tickets (photo/PDF); everyone’s arrivals & departures in one list	Flights
	Time zones: times stored as exact instants; toggle trip time ↔ your phone’s time	More → Flights & arrivals
	Trip map with itinerary pins, places and airports + **offline radar map** (no tiles needed)	Trip map
	Live reminders: tap 🔔 on an activity → real phone notification before it starts (10/30/60/120 min) + live countdown on Home	A
ctivity card · Settings
Budgeting	Group budget cap, spent so far, per-person, % used	Budget tab
	Spending breakdown by category (Wanderlog-style bar)	Budget tab

	Currency converter (live rates, cached offline) + **home ↔ local currency view**	Budget tab · Currency
	Expenses in any currency, split between chosen members, **settle-up with fewest payments**	Budget tab
	Daily spending planner + “budget finds” within budget & radius	Budget tab
	Printable planner (print or save as PDF)	Plan → printer button · More → Printable Planner
Organization	Offline document vault (PDFs, photos, notes)	More → Offline Docs
	Weather (7-day forecast, best outdoor day, packing tips)	Weather
	Choose your own radius → popular places nearby	Home · More → Explore Nearby
	Photo dump with likes, captions, per-day filter, share to socials	Photo dump
	Activity filters (adventure, sightseeing, food…) **and** time filters (under 1 h, 1–2 h, 2–4 h, half day, full day)	Plan · Nearby
Other	Interactive travel quiz with timer, streak bonus and group leaderboard	Travel quiz
	6 languages: English, Español, Français, 日本語, हिन्दी, 中文	Settings → Language
	Bucket list with inspiration	Bucket list
	Tracks location → nearest airport, distance, drive time, directions	Nearest airport
	Recommendation survey → ranked destinations with photography	For you
	Offline map	Trip map → Offline
Trip management	Multiple trips, switch / create / edit / leave / delete, planning → active → completed	My trips

Extras to stand out
- **360° city views** — drag (or use the arrows / auto-rotate) to spin all the way around 10 popular cities: Positano, Tokyo, Paris, New York, London, Dubai, Sydney, Singapore, Rio and Rome. Three parallax skyline layers, famous landmarks (tap a landmark chip to turn to it and read a fact), a compass readout and a **day / night toggle**: at night the windows light up, the landmarks glow, cars stream along the street and stars and a moon come out. Open it from Home (360° chip), a destination page or More → 360° City Views.
- **Skyline hero that follows the clock** — the Home photo switches between a sunny day look (sun glare, drifting clouds) and a blue-hour night look (stars, moon) based on the local time in the trip’s city, with a sun/moon chip to preview the other half of the day.
- **Lock in voting** — Plan → Vote shows each proposal’s vote count, a progress bar towards a majority and a **Lock in** button (organizer only) that confirms it and turns its reminder on.
- **Scout AI tab** — the group chat stays human; the Trip Assistant has its own tab with quick questions (budget, food nearby, weather, who owes who…).
- **Today card** — once the trip starts, Home shows “Today · Day 2 of 6” with a tick-off list.
- **Who’s bringing what** — the packing list assigns every item to a person, with a “Split the rest fairly” button.
- **Receipt photos** on expenses, and a shareable **story card** for the trip recap.
- **First-run tour** and a **How it works** page (architecture, security, validation and a rubric map) inside the app.
- **Make it yours** — a 7-step animated personalization survey: 6 colour themes (Forest, Ocean, Sunset, Lavender, Sakura, **Midnight dark mode**), card shape (Rounded / Bubbly / Crisp), travel interests with photos, pace & budget style, which Explore sections to show, nickname + emoji avatar, language and home currency — with a live preview and a confetti finish; on the last page, tapping any answer jumps back to change it. New accounts get it right after sign-up; it's always in *Settings*, the 🎨 button on Explore and the menu.
- **“Picked for you”** — Home ranks destinations against your interests and budget style and shows a match %.
- **Motion everywhere** — animated home hero (a plane flies a dotted route past drifting clouds and a pulsing sun), staggered card entrances, a tab highlight that glides between tabs, Ken Burns zoom on onboarding photos, floating logo, pulsing live-reminder bell. One *Animations* switch turns it all off for people who prefer less motion.
- **Trip recap** — stats (days, km between stops, votes, photos), highlights, final balances, rating, shareable summary.
- **Emergency & safety** — local police/ambulance numbers (tap to call), share live location, hospital & embassy finder.
- **Phrasebook** with **text-to-speech** pronunciation (Japanese, French, Spanish, Italian, Korean, Thai, Portuguese).
- **Smart packing list** — suggestions from the destination and the forecast (e.g. “IC card (Suica)”, umbrella when rain is likely).
- **Demo timeline** switch to show the whole trip lifecycle live.
- **Social media integration** — direct share into WhatsApp, Instagram, X, Facebook, Messenger, Telegram, Messages, Email + the native share sheet; photos and the **recap story card share as pictures**; invites are a code, a link and a **QR code** that a friend can scan from inside the app (or with their camera), and the link carries the sync server so their phone connects by itself. See *Honest limitations* for what needs developer accounts.
- Accessibility: screen-reader labels on every control, adjustable text size, haptics toggle, high-contrast palette, and the phone's **Reduce Motion** setting is respected (no animations, no auto-rotating 360° view).

Judges: [`docs/RATING_SHEET.md`](docs/RATING_SHEET.md) lists every line of the FBLA rating sheet with where it is shown. The 7-minute script is in [`docs/DEMO_SCRIPT.md`](docs/DEMO_SCRIPT.md).

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3. How it’s built

```
src/
  app/            Screens (Expo Router — every file is a route)
    (tabs)/       Home · Plan · Budget · Chat · More
    *.tsx         Feature screens (flights, map, weather, photos, …)
  components/     Reusable UI kit (ui/), trip widgets (trip/), map (native + web)
  store/          Zustand store: data model, actions, selectors, demo seed
  services/       Side effects: weather, currency, notifications, assistant, files, print, share, location
  data/           Offline datasets: 32 destinations, curated places, 100+ airports, phrases, quiz…
  i18n/           Translations (6 languages) + useT() hook
  utils/          Validation, time zones, money maths, passwords, QR codes, formatting
  sync/           Phone-to-phone sync: merge rules, engine, files, invites (see docs/SYNC.md)
server/           The sync server (plain Node, no dependencies) and its tests
e2e/              Browser tests that drive the web build, including a two-phone sync run
docs/             Planning, rating-sheet map, demo script, sync, building, judge checklist
```

- **Architecture:** layered MVVM-style. Screens (views) read state through selector hooks (view-model) and call store actions; services wrap every external API/device capability so screens never talk to the network directly. One-way data flow.
- **State & storage:** Zustand + Immer, persisted to AsyncStorage → the whole app works offline and survives restarts.
- **Navigation:** Expo Router with `Stack.Protected` guards (signed-out users can only see onboarding/auth).
- **Platform-specific code:** `TripMap.tsx` (react-native-maps) vs `TripMap.web.tsx` (OpenStreetMap embed).
- **Theming:** `src/theme` holds 6 palettes and 3 corner styles. The saved choice is read synchronously before any screen's styles are created, so every `StyleSheet` picks up the user's palette; applying a new look restarts the UI once (`DevSettings.reload` on phones, a p

age reload on web).
- **Animations:** `src/components/ui/motion.tsx` (FadeIn, Float, Pulse, looping values) on React Native's `Animated` API with the native driver on iOS/Android.
- **React Compiler** enabled; the code passes the compiler's lint rules (`npm run lint` → 0 problems) and `tsc --noEmit` with `strict: true`.
- **Small files.** The store is split into slices (`src/store/slices`), and the big screens into components (`src/components/home`, `crew`, `settings`, `chat`, `photos`, …). No screen file is over 300 lines.

Testing
- `npm test` runs **unit tests** for the money maths (exact settle-up), validation, time zones, password hashing (against the official RFC test vector), the store, the assistant (including the Claude request), invites and QR codes, and the whole sync stack (merge convergence on random runs, the real server over real HTTP).
- `npm run e2e` runs **browser tests** against the web build: every main flow, a sideways-overflow audit of every screen at three sizes, a check that no button sits inside another button, Reduce Motion, and two phones syncing through the real server. See [`docs/PLANNING.md`](docs/PLANNING.md) §9 for what each covers.
- Not covered: the parts that need a physical phone (native file upload and download, the camera QR scanner, "share as picture", notifications). They are built into the iPhone bundle but were not run on a device here.

Data handling & security
- Passwords are **salted and stretched with PBKDF2-HMAC-SHA256** (`@noble/hashes`, rounds stored in the hash, old hashes upgraded at next sign-in) — never stored in plain text. Covered by unit tests, including the official RFC test vector.
- "Remember me" off → the session is not persisted.
- The optional Anthropic API key lives in the **iOS Keychain / Android Keystore** (expo-secure-store), never in the app database.
- Documents are copied into the app's private storage so they stay available offline.
- Network calls have timeouts and fall back to cached or bundled data.
- **Sync privacy:** your Scout AI chat, other people's email addresses and flight ticket files never leave your phone. A shared trip is kept on the sync server you run yourself, behind a 24-character random id that only the invite code reveals (code look-ups are rate limited). There are no accounts and the traffic is plain HTTP, so run the server only on a network you trust.

Input validation (syntactic + semantic)
Email format, password strength meter, matching passwords, names (letters only), money amounts (`25` / `25.50`, positive, sane max), trip dates (end after start, ≤ 60 days, not in the past), invite-code format `ABC-1234`, flight numbers (`NH 175`), IATA codes checked against the airport database, arrival-after-departure across time zones, seat format, overlapping-activity warning, duplicate poll options, file-size limits.

---

4. Libraries, sources & licences
See **ATTRIBUTIONS.md** for every library, API, font, icon set and photo source. Planning documents: **docs/PLANNING.md**. Presentation script: **docs/DEMO_SCRIPT.md**.

5. Honest limitations
- **Sync needs a server you run.** Phones share a trip through `npm run server` on a computer they can reach (same wifi, a hotspot, or a typed address). There is no hosted service, no accounts, and the traffic is plain HTTP on your own network. See [docs/SYNC.md](docs/SYNC.md#limits-the-honest-list). Without a server the app still works fully on one phone (invite codes then join the demo directory, try `SEO-5521`).
- **Last writer wins per field.** Two people changing the *same* field at the same moment keep the later change. Chat messages, votes, expenses and photos are separate items, so they never clash.
- **Social media:** sharing uses the phone's share sheet and the apps' own link formats. Signing in with, or posting to, Instagram/Facebook/X *from inside the app* needs developer accounts and app review (OAuth credentials) that a student project cannot hold, so it is not simulated.
- **Not run on a physical phone:** on-device file upload/download, the camera QR scanner and "share as picture" (see Testing).
- The 360° city views are illustrated scenes drawn in code (SVG), not photographs, so they work offline and need no image licences. The Home hero's day/night effect is a bundled photo with code-drawn light, stars and clouds.
- "Continue with Google" is a simulated account chooser (it says so on screen); a store build would use Google OAuth (expo-auth-session).
- Claude AI mode requires the user's own Anthropic API key; without it the built-in offline assistant answers.
- Standalone Android builds need a Google Maps API key for the native map (Expo Go already includes one); the offline radar map needs nothing.
- Passwords live in accounts on the phone, hashed with PBKDF2 (10,000 rounds: a count chosen so sign-in stays quick in a phone's JavaScript engine, where a server would use 600,000 or Argon2id).

ATTRIBUTIONS.md

Attributions, sources & licences

TogetherWeGo was written for the FBLA Mobile Application Development event (2026–27). This file documents every third-party library, data source, font, icon set and image used, as required by the rating sheet's *Documentation and copyright compliance* item.

Starter template
The project was scaffolded with `npx create-expo-app` (default template, MIT licence — see `LICENSE`, © 650 Industries / Expo). The template's example screens were removed; all screens, components, data and services in `src/` are original work for this project.

Libraries (all open source)
Package	Version	Licence
`expo` (SDK 57) and Expo modules: `expo-router`, `expo-notifications`, `expo-location`, `expo-image`, `expo-image-picker`, `expo-document-picker`, `expo-file-system`, `expo-print`, `expo-sharing`, `expo-speech`, `expo-crypto`, `expo-secure-store`, `expo-camera` (scanning invite QR codes), `expo-build-properties`, `expo-clipboard`, `expo-haptics`, `expo-linear-gradient`, `expo-linking`, `expo-localization`, `expo-asset`, `expo-font`, `expo-splash-screen`, `expo-status-bar`, `expo-system-ui`, `expo-constants`	57.x	MIT
`react`, `react-dom`	19.2.3	MIT
`react-native`	0.86.3	MIT
`react-native-web`	0.21.3	MIT
`react-native-maps`	1.27.2	MIT
`react-native-svg` (spending donut chart)	15.x	MIT
`react-native-safe-area-context`, `react-native-screens`, `react-native-gesture-handler`, `react-native-reanimated`, `react-native-wor

klets` | – | MIT |
`@react-native-async-storage/async-storage`	2.2.0	MIT
`zustand`	5.0.15	MIT
`immer`	11.1.21	MIT
`@noble/hashes` (PBKDF2-SHA256 password hashing; audited, no dependencies)	2.4.0	MIT
`qrcode-generator` (draws the invite QR code; Kazuhiko Arase)	1.5.x	MIT
`react-native-view-shot` (turns the recap story card into a picture to share)	5.1.0	MIT
`jest`, `jest-expo`, `@types/jest` (unit tests, development only)	29.x / 57.x	MIT
`jsqr` (reads a QR code back, so the tests prove the codes can be scanned; development only)	1.4.0	Apache-2.0
`playwright-core` (browser tests, development only)	1.x	Apache-2.0

“QR Code” is a registered trademark of DENSO WAVE INCORPORATED. The sync server in `server/` uses only Node.js built-in modules.
| `@expo/vector-icons` (Ionicons, Material Community Icons) | 15.1.1 | MIT |
| `@expo-google-fonts/manrope` (Manrope by Mikhail Sharanda), `@expo-google-fonts/anton` (Anton by Vernon Adams) | 0.4.x | MIT (package) / SIL Open Font License 1.1 (fonts) |

APIs & data
Data	Source	Terms
Weather forecasts	[Open-Meteo](https://open-meteo.com)	Free, no key, CC BY 4.0 attribution
Exchange rates	[ExchangeRate-API open endpoint](https://www.exchangerate-api.com/docs/free) (`open.er-api.com`)	Free with attribution; cached for 6 h
Maps (phone)	Apple Maps (iOS) / Google Maps (Android) via react-native-maps	Platform map terms
Maps (web preview)	[OpenStreetMap](https://www.openstreetmap.org/copyright) embed	© OpenStreetMap contributors, ODbL
Airport coordinates & codes	Public IATA codes and coordinates (compiled by hand, 100+ airports)	Factual data
Emergency numbers	Official tourism / government pages, e.g. [JNTO emergency numbers](https://www.jnto.go.jp/safety-tips/eng/emergency-numbers.html)	Factual data
Destination & place facts	Summarised in our own words from public tourism information	Original text
AI chatbot (optional)	[Anthropic Claude API](https://docs.claude.com) — only when a user adds their own API key	Anthropic usage policies

The built-in offline Trip Assistant is original rule-based code (`src/services/assistant.ts`) that reads the trip’s own data.

Photography
All photos are from [Unsplash](https://unsplash.com) and used under the [Unsplash License](https://unsplash.com/license) (free for commercial and non-commercial use, no permission needed). They are bundled in `assets/images/photos/` (resized/compressed) so the app looks complete offline. Source URLs:

File	Source
`amalfi.jpg`	https://images.unsplash.com/photo-1612698093158-e07ac200d44e
`tokyo.jpg`	https://images.unsplash.com/photo-1536098561742-ca998e48cbcc
`kyoto.jpg`	https://images.unsplash.com/photo-1493976040374-85c8e12f0c0e
`paris.jpg`	https://images.unsplash.com/photo-1502602898657-3e91760cbb34
`bali.jpg`	https://images.unsplash.com/photo-1537996194471-e657df975ab4
`bangkok.jpg`	https://images.unsplash.com/photo-1563492065599-3520f775eeed
`singapore.jpg`	https://images.unsplash.com/photo-1525625293386-3f8f99389edd
`cape-town.jpg`	https://images.unsplash.com/photo-1580060839134-75a5edca2e99
`new-york.jpg`	https://images.unsplash.com/photo-1534430480872-3498386e7856
`santorini.jpg`	https://images.unsplash.com/photo-1613395877344-13d4a8e0d49e
`rome.jpg`	https://images.unsplash.com/photo-1552832230-c0197dd311b5
`london.jpg`	https://images.unsplash.com/photo-1513635269975-59663e0ac1ad
`iceland.jpg`	https://images.unsplash.com/photo-1504893524553-b855bce32c67
`swiss-alps.jpg`	https://images.unsplash.com/photo-1530122037265-a5f1f91d3b99
`maldives.jpg`	https://images.unsplash.com/photo-1514282401047-d79a71a590e8
`machu-picchu.jpg`	https://images.unsplash.com/photo-1526392060635-9d6019884377
`ha-long-bay.jpg`	https://images.unsplash.com/photo-1528127269322-539801943592
`dubai.jpg`	https://images.unsplash.com/photo-1512453979798-5ea266f8880c
`barcelona.jpg`	https://images.unsplash.com/photo-1583422409516-2895a77efded
`sydney.jpg`	https://images.unsplash.com/photo-1506973035872-a4ec16b8e8d9
`seoul.jpg`	https://images.unsplash.com/photo-1517154421773-0529f29ea451
`istanbul.jpg`	https://images.unsplash.com/photo-1541432901042-2d8bd64b4a9b
`amsterdam.jpg`	https://images.unsplash.com/photo-1534351590666-13e3e96b5017
`prague.jpg`	https://images.unsplash.com/photo-1519677100203-a0e668c92439
`lisbon.jpg`	https://images.unsplash.com/photo-1585208798174-6cedd86e019a
`banff.jpg`	https://images.unsplash.com/photo-1561134643-668f9057cce4
`marrakech.jpg`	https://images.unsplash.com/photo-1597212618440-806262de4f6b
`rio.jpg`	https://images.unsplash.com/photo-1483729558449-99ef09a8c325
`mexico-city.jpg`	https://images.unsplash.com/photo-1512813195386-6cf811ad3542
`grand-canyon.jpg`	https://images.unsplash.com/photo-1615551043360-33de8b5f410c
`cinque-terre.jpg`	https://images.unsplash.com/photo-1499678329028-101435549a4e
`venice.jpg`	https://images.unsplash.com/photo-1514890547357-a9ee288728e0
`mount-fuji.jpg`	https://images.unsplash.com/photo-1528164344705-47542687000d
`onboard-1.jpg`	https://images.unsplash.com/photo-1501555088652-021faa106b9b
`onboard-2.jpg`	https://images.unsplash.com/photo-1469854523086-cc02fe5d8800
`onboard-3.jpg`	https://images.unsplash.com/photo-1507525428034-b723cf961d3e
`lagoon.jpg`	https://images.unsplash.com/photo-1506929562872-bb421503ef21
`shibuya.jpg`	https://images.unsplash.com/photo-1542051841857-5f90071e7989
`akihabara.jpg`	https://images.unsplash.com/photo-1540959733332-eab4deabeeaf
`fushimi-inari.jpg`	https://images.unsplash.com/photo-1478436127897-769e1b3f0f36
`omoide-yokocho.jpg`	https://images.unsplash.com/photo-1554797589-7241bb691973
`lantern-alley.jpg`	https://images.unsplash.com/photo-1528360983277-13d401cdc186
`fuji-blossom.jpg`	https://images.unsplash.com/photo-1490806843957-31f4c9a91c65
`tokyo-skyline.jpg`	https://images.unsplash.com/photo-1513407030348-c983a97b98d8
`kyoto-skyline.jpg`	https://images.unsplash.com/photo-1545569341-9eb8b30979d9
`sushi.jpg`	https://images.unsplash.com/photo-1579871494447-9811cf80d66c

`ramen.jpg`	https://images.unsplash.com/photo-1569718212165-3a8278d5f624
`planning.jpg`	https://images.unsplash.com/photo-1488646953014-85cb44e25828
`avatar-maya.jpg`	https://images.unsplash.com/photo-1494790108377-be9c29b29330
`avatar-diego.jpg`	https://images.unsplash.com/photo-1507003211169-0a1dd7228f2d
`avatar-priya.jpg`	https://images.unsplash.com/photo-1534528741775-53994a69daeb
`avatar-sam.jpg`	https://images.unsplash.com/photo-1500648767791-00dcc994a43e
`avatar-ana.jpg`	https://images.unsplash.com/photo-1438761681033-6461ffad8d80
`avatar-noah.jpg`	https://images.unsplash.com/photo-1506794778202-cad84cf45f1d

Portrait photos are used only as avatars for the fictional demo group (Maya, Diego, Priya, Sam, Ana, Noah). All demo people, trips, chats and bookings are fictional sample data.

App icon & logo
Original design for this project: forest-green tile, the Material Community Icons “airplane-takeoff” glyph (Apache 2.0 / MIT via `@expo/vector-icons`) and three dots representing the travel group. Rendered to PNG for the app icon, Android adaptive icon and splash screen.

Design reference
The visual style (cream background, forest-green accents, pill navigation, card layouts) follows the reference prototype video provided by the team.

Developer tooling
- **Ponytail** skills (`.claude/skills/ponytail*`, from [DietrichGebert/ponytail](https://github.com/DietrichGebert/ponytail) @ 9cc65d0, MIT licence) — coding-style instructions for the Claude Code assistant used while building the app. Not part of the app itself.
- The 360° city scenes (`src/components/panorama`, `src/data/panoramas.ts`) are original code-drawn illustrations of public landmarks. Landmark facts were written from public sources (e.g. official tower and building heights).

package.json

```
{
  "name": "together-we-go",
  "main": "expo-router/entry",
  "version": "1.0.0",
  "dependencies": {
    "@expo-google-fonts/anton": "^0.4.2",
    "@expo-google-fonts/manrope": "^0.4.2",
    "@expo/vector-icons": "^15.0.2",
    "@noble/hashes": "^2.4.0",
    "@react-native-async-storage/async-storage": "2.2.0",
    "expo": "~57.0.30",
    "expo-asset": "~57.0.18",
    "expo-build-properties": "~57.0.22",
    "expo-camera": "~57.0.6",
    "expo-clipboard": "~57.0.2",
    "expo-constants": "~57.0.20",
    "expo-crypto": "~57.0.4",
    "expo-document-picker": "~57.0.3",
    "expo-file-system": "~57.0.7",
    "expo-font": "~57.0.4",
    "expo-haptics": "~57.0.3",
    "expo-image": "~57.0.5",
    "expo-image-picker": "~57.0.20",
    "expo-linear-gradient": "~57.0.2",
    "expo-linking": "~57.0.13",
    "expo-localization": "~57.0.2",
    "expo-location": "~57.0.21",
    "expo-notifications": "~57.0.23",
    "expo-print": "~57.0.2",
    "expo-router": "~57.0.28",
    "expo-secure-store": "~57.0.4",
    "expo-sharing": "~57.0.23",
    "expo-speech": "~57.0.3",
    "expo-splash-screen": "~57.0.10",
    "expo-status-bar": "~57.0.1",
    "expo-system-ui": "~57.0.4",
    "immer": "^11.1.21",
    "qrcode-generator": "^1.5.2",
    "react": "19.2.3",
    "react-dom": "19.2.3",
    "react-native": "0.86.3",
    "react-native-gesture-handler": "~2.32.0",
    "react-native-maps": "1.27.2",
    "react-native-reanimated": "4.5.1",
    "react-native-safe-area-context": "~5.7.0",
    "react-native-screens": "~4.26.0",
    "react-native-svg": "15.15.4",
    "react-native-view-shot": "5.1.0",
    "react-native-web": "~0.21.0",
    "react-native-worklets": "0.10.1",
    "zustand": "^5.0.15"
  },
  "devDependencies": {
    "@types/jest": "29.5.14",
    "@types/react": "~19.2.2",
    "eslint": "^9.0.0",
    "eslint-config-expo": "~57.0.2",
    "jest": "~29.7.0",
```

```
    "jest-expo": "~57.0.6",
    "jsqr": "^1.4.0",
    "playwright-core": "^1.56.1",
    "typescript": "~6.0.3"
  },
  "scripts": {
    "start": "expo start",
    "android": "expo start --android",
    "ios": "expo start --ios",
    "web": "expo start --web",
    "server": "node server/index.js",
    "lint": "expo lint",
    "test": "jest",
    "e2e": "node e2e/run.js",
    "source-pdf": "node scripts/source-pdf.js"
  },
  "private": true
}
```

app.json

```
{
  "expo": {
    "name": "TogetherWeGo",
    "slug": "together-we-go",
    "description": "Plan, budget and enjoy group trips together.",
    "version": "1.0.0",
    "orientation": "portrait",
    "icon": "./assets/images/icon.png",
    "scheme": "togetherwego",
    "userInterfaceStyle": "light",
    "backgroundColor": "#F1F2F4",
    "ios": {
      "bundleIdentifier": "com.togetherwego.app",
      "supportsTablet": true,
      "infoPlist": {
        "NSAppTransportSecurity": {
          "NSAllowsLocalNetworking": true
        },
        "NSLocalNetworkUsageDescription": "TogetherWeGo connects to your group's sync server on your wifi so friends see the same trip."
      }
    },
    "android": {
      "package": "com.togetherwego.app",
      "adaptiveIcon": {
        "backgroundColor": "#1F4D25",
        "foregroundImage": "./assets/images/android-icon-foreground.png",
        "backgroundImage": "./assets/images/android-icon-background.png",
        "monochromeImage": "./assets/images/android-icon-monochrome.png"
      },
      "predictiveBackGestureEnabled": false
    },
    "web": {
      "output": "single",
      "favicon": "./assets/images/favicon.png",
      "name": "TogetherWeGo",
      "themeColor": "#0B0B0D",
      "backgroundColor": "#F1F2F4"
    },
    "plugins": [
      "expo-router",
      [
        "expo-splash-screen",
        {
          "backgroundColor": "#F1F2F4",
          "image": "./assets/images/splash-icon.png",
          "imageWidth": 160
        }
      ],
      [
        "expo-location",
        {
          "locationWhenInUsePermission": "TogetherWeGo uses your location to show places nearby, the closest airport and to share your l
ocation in an emergency."
        }
      ],
      [
        "expo-image-picker",
        {
          "photosPermission": "TogetherWeGo lets you add trip photos and flight tickets from your library.",
          "cameraPermission": "TogetherWeGo uses the camera to take photos for the group photo dump and to scan the invite QR code from
a friend."
        }
      ],
      [
        "expo-notifications",
```

```
        {
          "color": "#0B0B0D"
        }
      ],
      [
        "expo-camera",
        {
          "cameraPermission": "TogetherWeGo uses the camera to take photos for the group photo dump and to scan the invite QR code from
a friend.",
          "recordAudioAndroid": false
        }
      ],
      "expo-sharing",
      "expo-localization",
      "expo-secure-store",
      "expo-asset",
      "expo-font",
      [
        "expo-build-properties",
        {
          "android": {
            "usesCleartextTraffic": true
          }
        }
      ]
    ],
    "experiments": {
      "typedRoutes": true,
      "reactCompiler": true
    }
  }
}
```

eas.json

```
{
  "cli": {
    "version": ">= 16.0.0",
    "appVersionSource": "remote"
  },
  "build": {
    "preview": {
      "distribution": "internal",
      "android": {
        "buildType": "apk"
      }
    },
    "production": {
      "autoIncrement": true
    }
  },
  "submit": {
    "production": {}
  }
}
```

tsconfig.json

```
{
  "extends": "expo/tsconfig.base",
  "compilerOptions": {
    "strict": true,
    "paths": {
      "@/*": [
        "./src/*"
      ],
      "@/assets/*": [
        "./assets/*"
      ]
    },
    "types": [
      "jest",
      "node"
    ]
  },
  "include": [
    "**/*.ts",
    "**/*.tsx",
    ".expo/types/**/*.ts",
    "expo-env.d.ts"
  ],
  "exclude": [
    "node_modules",
    "dist",
    "server"
```

```
  ]
}
```

eslint.config.js

```
// https://docs.expo.dev/guides/using-eslint/
const { defineConfig } = require('eslint/config');
const expoConfig = require("eslint-config-expo/flat");

const node = ['require', 'module', 'exports', 'process', 'Buffer', '__dirname', 'console', 'setTimeout', 'clearTimeout', 'URL', 'fetch',
'document', 'window', 'localStorage', 'Image', 'btoa', 'XMLSerializer', 'getComputedStyle'];
const jest = ['describe', 'it', 'expect', 'beforeEach', 'afterEach', 'beforeAll', 'afterAll', 'jest'];

module.exports = defineConfig([
  expoConfig,
  {
    ignores: ["dist/*", "e2e/.dist/**", "e2e/shots/**", "server/data/**"],
  },
  {
    // the sync server, the browser tests and the helper scripts are plain Node (CommonJS), not part of the app bundle
    files: ["server/**/*.js", "e2e/**/*.js", "scripts/**/*.js", "jest.config.js", "jest.setup.js"],
    languageOptions: {
      sourceType: 'commonjs',
      globals: Object.fromEntries([...node, ...jest].map((name) => [name, 'readonly'])),
    },
  },
]);
```

jest.config.js

```
// Unit tests run with jest-expo so React Native modules are mocked the same way Expo does it.
const preset = require('jest-expo/jest-preset');

module.exports = {
  preset: 'jest-expo',
  setupFiles: ['<rootDir>/jest.setup.js'],
  moduleNameMapper: { '^@/(.*)$': '<rootDir>/src/$1' },
  // immer and zustand ship ES modules, so they must go through Babel like the Expo packages do.
  transformIgnorePatterns: [
    preset.transformIgnorePatterns[0].replace('standard-navigation))', 'standard-navigation|immer|zustand|@noble))'),
    ...preset.transformIgnorePatterns.slice(1),
  ],
  testPathIgnorePatterns: ['/node_modules/', '/dist/'],
};
```

jest.setup.js

```
// AsyncStorage has no native module under Jest, so use the in-memory mock that ships with it.
jest.mock('@react-native-async-storage/async-storage', () => require('@react-native-async-storage/async-storage/jest/async-storage-moc
k'));

// expo-crypto needs its native module; stand in with Node's crypto so salts are real random values.
jest.mock('expo-crypto', () => ({
  randomUUID: () => require('crypto').randomUUID(),
  getRandomBytes: (n) => new Uint8Array(require('crypto').randomBytes(n)),
}));
```

src/app/(tabs)/_layout.tsx

```
import { Tabs } from 'expo-router/js-tabs';

import { TabBar } from '@/components/TabBar';
import { colors } from '@/theme';

export default function TabsLayout() {
  return (
    <Tabs
      initialRouteName="explore"
      tabBar={(props) => <TabBar {...props} />}
      screenOptions={{ headerShown: false, sceneStyle: { backgroundColor: colors.bg } }}>
      <Tabs.Screen name="explore" />
      <Tabs.Screen name="itinerary" />
      <Tabs.Screen name="budget" />
      <Tabs.Screen name="chat" />
      <Tabs.Screen name="profile" />
    </Tabs>
  );
}
```

src/app/(tabs)/budget.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { useState } from 'react';
import { View } from 'react-native';
```

```
import { AddExpenseSheet } from '@/components/budget/AddExpenseSheet';
import { BalancesCard } from '@/components/budget/BalancesCard';
import { DailyPlanner } from '@/components/budget/DailyPlanner';
import { ExpenseDetailSheet } from '@/components/budget/ExpenseDetailSheet';
import { ExpensesSection } from '@/components/budget/ExpensesSection';
import { SettleSheet } from '@/components/budget/SettleSheet';
import { SpendSummary } from '@/components/budget/SpendSummary';
import { styles } from '@/components/budget/styles';
import { CurrencyConverter } from '@/components/CurrencyConverter';
import { NoTrip } from '@/components/trip/NoTrip';
import { Card } from '@/components/ui/bits';
import { Press } from '@/components/ui/Press';
import { Screen } from '@/components/ui/Screen';
import { Segmented } from '@/components/ui/Segmented';
import { T } from '@/components/ui/T';
import { destinationById } from '@/data/destinations';
import { useRatesLoader } from '@/hooks/useTrip';
import { useT } from '@/i18n';
import { convert, useRates } from '@/services/currency';
import type { Expense } from '@/store/types';
import { totalSpent, useActiveTrip, useAppStore } from '@/store/useAppStore';
import { colors } from '@/theme';
import { money } from '@/utils/format';
import { daysBetween, todayInTz } from '@/utils/time';

export default function Budget() {
  const trip = useActiveTrip();
  if (!trip) return <NoTrip />;
  return <BudgetInner key={trip.id} />;
}

/** The Budget tab. It owns the few things the parts share (home/local view, which sheet is open) and hands the rest to components/budge
t. */
function BudgetInner() {
  const t = useT();
  useRatesLoader();
  const trip = useActiveTrip()!;
  const rates = useRates();
  const home = useAppStore((s) => s.settings.homeCurrency);
  const userId = useAppStore((s) => s.session?.userId ?? '');
  const dest = destinationById(trip.destinationId);
  const local = dest?.currency ?? 'EUR';

  const [view, setView] = useState<'home' | 'local'>('home');
  const [conv, setConv] = useState<[string, string]>([home, local === home ? 'EUR' : local]);
  const [detail, setDetail] = useState<Expense | null>(null);
  const [settleOpen, setSettleOpen] = useState(false);
  const [addOpen, setAddOpen] = useState(false);
  const [today] = useState(() => todayInTz(dest?.tz ?? 'UTC'));

  const viewCode = view === 'home' ? home : local;
  const fmt = (usd: number) => money(convert(usd, 'USD', viewCode, rates), viewCode);

  const spent = totalSpent(trip);
  const catTotals: Record<string, number> = {};
  trip.expenses.filter((e) => !e.settlement).forEach((e) => (catTotals[e.category] = (catTotals[e.category] ?? 0) + e.amount));
  const byCat = Object.entries(catTotals).sort((a, b) => b[1] - a[1]);
  const tripLen = daysBetween(trip.startDate, trip.endDate) + 1;
  const daysLeft = Math.max(1, Math.min(tripLen, daysBetween(today, trip.endDate) + 1));

  return (
    <Screen bottomInset={30}>
      <View style={styles.head}>
        <View style={{ flex: 1 }}>
          <T variant="kicker" color={colors.textSecondary} numberOfLines={1}>
            {trip.name}
          </T>
          <T variant="display" accessibilityRole="header">
            {t('budget.title')}
          </T>
        </View>
        <Press onPress={() => setAddOpen(true)} style={styles.plus} accessibilityLabel="Add expense" testID="add-expense-open">
          <Ionicons name="add" size={26} color={colors.onPrimary} />
        </Press>
      </View>

      <Segmented
        options={[
          { key: 'home', label: `Home · ${home}` },
          { key: 'local', label: `Local · ${local}` },
        ]}
        value={view}
        onChange={setView}
      />

      <SpendSummary byCat={byCat} spent={spent} budget={trip.budget} remaining={trip.budget - spent} per={spent / Math.max(1, trip.membe
```

```
rs.length)} daysLeft={daysLeft} fmt={fmt} />
      <BalancesCard trip={trip} userId={userId} fmt={fmt} onSettle={() => setSettleOpen(true)} />

      <T variant="headline" accessibilityRole="header" style={styles.section}>
        Converter
      </T>
      <Card style={styles.card}>
        <CurrencyConverter from={conv[0]} to={conv[1]} onChange={(a, b) => setConv([a, b])} />
      </Card>

      <DailyPlanner trip={trip} />
      <ExpensesSection trip={trip} fmt={fmt} onOpen={setDetail} />

      <AddExpenseSheet visible={addOpen} onClose={() => setAddOpen(false)} trip={trip} userId={userId} />
      <ExpenseDetailSheet key={detail?.id ?? 'none'} detail={detail} onClose={() => setDetail(null)} trip={trip} fmt={fmt} />
      <SettleSheet visible={settleOpen} onClose={() => setSettleOpen(false)} trip={trip} />
    </Screen>
  );
}
```

src/app/(tabs)/chat.tsx

```
import { useFocusEffect } from 'expo-router';
import { useCallback, useEffect, useRef, useState } from 'react';
import { FlatList, KeyboardAvoidingView, Platform, ScrollView, StyleSheet, View } from 'react-native';

import { ChatBubble } from '@/components/chat/ChatBubble';
import { ChatHeader, type ChatTab } from '@/components/chat/ChatHeader';
import { Composer } from '@/components/chat/Composer';
import { EmptyChat } from '@/components/chat/EmptyChat';
import { NoTrip } from '@/components/trip/NoTrip';
import { Chip } from '@/components/ui/bits';
import { toast } from '@/components/ui/feedback';
import { tap } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import { useT } from '@/i18n';
import { answerLocally, askClaude, type AssistantContext } from '@/services/assistant';
import { friendReply, planFriendReply } from '@/services/pretendFriends';
import { getApiKey } from '@/services/secrets';
import { useSyncStatus } from '@/sync/status';
import { useActiveTrip, useAppStore } from '@/store/useAppStore';
import type { ChatMessage, Trip } from '@/store/types';
import { MAX_WIDTH, colors } from '@/theme';

const QUICK = ['chat.q.budget', 'chat.q.food', 'chat.q.weather', 'chat.q.next'];

const inAiThread = (m: ChatMessage) => m.channel === 'ai' || m.ai === true;

export default function Chat() {
  const trip = useActiveTrip();
  if (!trip) return <NoTrip />;
  return <ChatInner key={trip.id} />;
}

function ChatInner() {
  const t = useT();
  const trip = useActiveTrip()!;
  const userId = useAppStore((s) => s.session?.userId ?? '');
  const send = useAppStore((s) => s.sendMessage);
  const aiMode = useAppStore((s) => s.settings.aiMode);
  const markKindRead = useAppStore((s) => s.markKindRead);
  const synced = useSyncStatus((s) => s.shared.includes(trip.id)); // real friends are on the other end: no pretend replies
  const [text, setText] = useState('');
  const [tab, setTab] = useState<ChatTab>('crew');
  const [typing, setTyping] = useState<string | null>(null);
  const list = useRef<FlatList<ChatMessage>>(null);
  const timers = useRef<ReturnType<typeof setTimeout>[]>([]);

  useFocusEffect(
    useCallback(() => {
      markKindRead('chat');
    }, [markKindRead]),
  );
  useEffect(() => () => timers.current.forEach(clearTimeout), []);

  const later = (fn: () => void, ms: number) => timers.current.push(setTimeout(fn, ms));

  const askAssistant = async (question: string, history: ChatMessage[]) => {
    setTyping('assistant');
    const s = useAppStore.getState();
    const live = s.trips.find((x) => x.id === trip.id) as Trip;
    const ctx: AssistantContext = { trip: live, userId, weather: s.weather[live.destinationId], rates: s.rates, homeCurrency: s.setting
s.homeCurrency };
    let reply = '';
    let source: 'local' | 'claude' = 'local';
    if (aiMode === 'claude') {
```

```
      const key = await getApiKey();
      if (key) {
        try {
          reply = await askClaude(key, question, history, ctx);
          source = 'claude';
        } catch {
          toast('Claude is unavailable — answered offline instead', { tone: 'warn', icon: 'cloud-offline-outline' });
        }
      }
    }
    if (!reply) {
      await new Promise((r) => setTimeout(r, 650));
      reply = answerLocally(question, ctx);
    }
    setTyping(null);
    send(trip.id, { authorId: 'assistant', ai: true, aiSource: source, channel: 'ai', text: reply });
    tap();
  };

  const submit = (raw?: string) => {
    const msg = (raw ?? text).trim();
    if (!msg) return;
    if (msg.length > 1000) return toast('Messages can be up to 1000 characters', { tone: 'warn' });
    setText('');
    if (tab === 'ai') {
      send(trip.id, { authorId: userId, text: msg, channel: 'ai' });
      askAssistant(msg, trip.messages.filter(inAiThread));
      return;
    }
    send(trip.id, { authorId: userId, text: msg });
    const plan = synced ? null : planFriendReply(trip.members.filter((m) => m.id !== userId && !m.invited));
    if (plan) {
      later(() => setTyping(plan.friend.id), 900);
      later(() => {
        setTyping(null);
        send(trip.id, { authorId: plan.friend.id, text: friendReply(msg) });
      }, plan.delay);
    }
  };

  const isAi = tab === 'ai';
  const data = trip.messages.filter((m) => inAiThread(m) === isAi);
  const typingFor = typing === 'assistant' ? (isAi ? 'Scout AI' : null) : isAi ? null : typing ? trip.members.find((m) => m.id === typin
g)?.name.split(' ')[0] : null;

  return (
    <View style={styles.root}>
      <KeyboardAvoidingView behavior={Platform.OS === 'ios' ? 'padding' : undefined} style={styles.column}>
        <ChatHeader trip={trip} tab={tab} onTab={setTab} synced={synced} aiMode={aiMode} />

        <FlatList
          key={tab}
          ref={list}
          data={data}
          keyExtractor={(m) => m.id}
          renderItem={({ item, index }) => <ChatBubble item={item} prev={data[index - 1]} trip={trip} userId={userId} />}
          contentContainerStyle={{ paddingHorizontal: 16, paddingVertical: 12, flexGrow: 1 }}
          onContentSizeChange={() => list.current?.scrollToEnd({ animated: true })}
          onLayout={() => list.current?.scrollToEnd({ animated: false })}
          keyboardShouldPersistTaps="handled"
          ListEmptyComponent={<EmptyChat ai={isAi} />}
          ListFooterComponent={
            typingFor ? (
              <View style={styles.typing}>
                <T variant="caption" color={colors.textMuted}>
                  {t('chat.typing', { name: typingFor })}
                </T>
              </View>
            ) : null
          }
        />

        {isAi ? (
          <ScrollView horizontal showsHorizontalScrollIndicator={false} keyboardShouldPersistTaps="handled" contentContainerStyle={{ ga
p: 6, paddingHorizontal: 16, paddingBottom: 8 }} style={{ flexGrow: 0 }}>
            {QUICK.map((k) => (
              <Chip key={k} small label={t(k)} onPress={() => submit(t(k))} />
            ))}
            <Chip small label="Who owes who?" onPress={() => submit('Who owes who?')} />
            <Chip small label="Convert 50 USD" onPress={() => submit('Convert 50 USD')} />
          </ScrollView>
        ) : null}

        <Composer value={text} onChange={setText} onSend={() => submit()} placeholder={isAi ? 'Ask Scout anything…' : 'Message the cre
w…'} />
      </KeyboardAvoidingView>
```

```
    </View>
  );
}

const styles = StyleSheet.create({
  root: { flex: 1, backgroundColor: colors.bg, alignItems: 'center' },
  column: { flex: 1, width: '100%', maxWidth: MAX_WIDTH },
  typing: { paddingHorizontal: 44, paddingTop: 10 },
});
```

src/app/(tabs)/explore.tsx

```
import { Ionicons } from '@expo/vector-icons';
import * as Clipboard from 'expo-clipboard';
import { router } from 'expo-router';
import { useState } from 'react';
import { ScrollView, StyleSheet, View, useWindowDimensions } from 'react-native';

import { GuidedTour } from '@/components/GuidedTour';
import { BudgetTile } from '@/components/home/BudgetTile';
import { DiscoverSection } from '@/components/home/DiscoverSection';
import { GlanceTiles } from '@/components/home/GlanceTiles';
import { HomeHero } from '@/components/home/HomeHero';
import { NearbySection } from '@/components/home/NearbySection';
import { PicksSection } from '@/components/home/PicksSection';
import { QuickTiles } from '@/components/home/QuickTiles';
import { UpNextSection } from '@/components/home/UpNextSection';
import { TripMenuSheet } from '@/components/trip/TripMenuSheet';
import { Card } from '@/components/ui/bits';
import { Button } from '@/components/ui/Button';
import { toast } from '@/components/ui/feedback';
import { Press, tap } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import { destinationById } from '@/data/destinations';
import { useNow, useWeather } from '@/hooks/useTrip';
import { useT } from '@/i18n';
import { nextUpcoming } from '@/services/notifications';
import { totalSpent, tripStatus, useActiveTrip, useAppStore, useCurrentUser } from '@/store/useAppStore';
import { MAX_WIDTH, colors } from '@/theme';
import { daysBetween, deviceTimeZone, greetingFor, todayInTz } from '@/utils/time';

/** Home: a hero picture, then the things worth seeing at a glance. Each block is its own component in components/home. */
export default function Home() {
  const t = useT();
  const { width } = useWindowDimensions();
  const user = useCurrentUser();
  const trip = useActiveTrip();
  const unread = useAppStore((s) => s.notifications.filter((n) => !n.read).length);
  const personal = useAppStore((s) => s.personal);
  const setToured = useAppStore((s) => s.setToured);
  const now = useNow(30000);
  const [menu, setMenu] = useState(false);

  const dest = destinationById(trip?.destinationId ?? 'positano');
  const tz = dest?.tz ?? 'UTC';
  const weather = useWeather(dest?.id);
  const status = trip ? tripStatus(trip) : null;
  const show = personal.sections;
  const colW = Math.min(width, MAX_WIDTH) - 40;
  const today = todayInTz(tz);
  const tripLen = trip ? daysBetween(trip.startDate, trip.endDate) + 1 : 0;
  const daysToGo = trip ? Math.max(0, daysBetween(today, trip.startDate)) : 0;
  const dayNo = trip ? Math.min(tripLen, Math.max(1, daysBetween(trip.startDate, today) + 1)) : 0;
  const votingCount = trip ? trip.activities.filter((a) => a.status === 'voting').length : 0;
  const firstName = personal.nickname || (user?.name.split(' ')[0] ?? 'traveller');
  const todays = trip && status === 'active' ? trip.activities.filter((a) => a.date === today && a.status !== 'voting').sort((a, b) =>
a.time.localeCompare(b.time)) : [];

  const copyCode = async () => {
    if (!trip) return;
    await Clipboard.setStringAsync(trip.inviteCode).catch(() => {});
    tap('success');
    toast(`Invite code ${trip.inviteCode} copied`, { icon: 'copy-outline' });
  };

  return (
    <View style={styles.root}>
      <ScrollView style={styles.scroll} showsVerticalScrollIndicator={false} contentContainerStyle={{ paddingBottom: 40 }}>
        <View style={styles.column}>
          <HomeHero trip={trip} dest={dest} tz={tz} status={status} daysToGo={daysToGo} unread={unread} onMenu={() => setMenu(true)} onC
opyCode={copyCode} />

          <View style={styles.sheet}>
            <T variant="small" color={colors.textSecondary} style={{ marginBottom: 14 }}>
              {t(greetingFor(new Date(now).getHours()))}, {firstName} {personal.emoji}
            </T>
```

```
            {!trip ? (
              <Card style={{ gap: 12, marginBottom: 14 }}>
                <T variant="headline">Plan your first group trip</T>
                <T variant="small" color={colors.textSecondary}>
                  Create a trip, invite friends with a code and start planning together.
                </T>
                <View style={{ flexDirection: 'row', gap: 8 }}>
                  <Button label="Create trip" size="md" icon="add" onPress={() => router.push('/new-trip')} style={{ flex: 1 }} />
                  <Button label="Join" size="md" variant="secondary" icon="people-outline" onPress={() => router.push('/join')} style={{
flex: 1 }} />
                </View>
              </Card>
            ) : null}

            {show.glance ? <GlanceTiles dest={dest} weather={weather} now={now} tz={tz} homeTz={deviceTimeZone()} /> : null}
            {trip && show.budget ? <BudgetTile trip={trip} spent={totalSpent(trip)} /> : null}
            {trip && show.upnext ? <UpNextSection trip={trip} status={status} todays={todays} dayNo={dayNo} tripLen={tripLen} next={next
Upcoming(trip)} now={now} /> : null}
            {trip && show.quick ? <QuickTiles votingCount={votingCount} /> : null}
            {show.forYou ? <PicksSection personal={personal} /> : null}
            {trip && dest && show.nearby ? <NearbySection trip={trip} dest={dest} /> : null}
            {show.discover ? <DiscoverSection dest={dest} colW={colW} /> : null}

            <Press onPress={() => router.push('/personalize')} style={styles.customize} accessibilityLabel="Customize this page">
              <Ionicons name="color-palette-outline" size={16} color={colors.text} />
              <T variant="small" weight="bold">
                Customize this page
              </T>
            </Press>
          </View>
        </View>
      </ScrollView>

      {trip ? <TripMenuSheet visible={menu} onClose={() => setMenu(false)} trip={trip} /> : null}
      <GuidedTour visible={!personal.toured && !!trip} onClose={() => setToured(true)} />
    </View>
  );
}

const styles = StyleSheet.create({
  root: { flex: 1, backgroundColor: colors.bg, alignItems: 'center' },
  // width 100% so the page is exactly as wide as the phone (the root centres its children, which would otherwise let a wide card row st
retch the page sideways)
  scroll: { width: '100%' },
  column: { width: '100%', maxWidth: MAX_WIDTH },
  sheet: { marginTop: -28, borderTopLeftRadius: 28, borderTopRightRadius: 28, backgroundColor: colors.bg, paddingHorizontal: 20, padding
Top: 22 },
  customize: { flexDirection: 'row', alignItems: 'center', justifyContent: 'center', gap: 6, marginTop: 26, paddingVertical: 13, borderR
adius: 999, borderWidth: 1, borderStyle: 'dashed', borderColor: colors.borderStrong },
});
```

src/app/(tabs)/itinerary.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { useLocalSearchParams } from 'expo-router';
import { useState } from 'react';
import { ScrollView, StyleSheet, View } from 'react-native';

import { ActivityCard } from '@/components/trip/ActivityCard';
import { AddActivitySheet } from '@/components/trip/AddActivitySheet';
import { MemberPickerSheet } from '@/components/trip/MemberPickerSheet';
import { activityFilters, typeEmoji } from '@/components/trip/meta';
import { NoTrip } from '@/components/trip/NoTrip';
import { Chip, EmptyState, IconButton } from '@/components/ui/bits';
import { toast } from '@/components/ui/feedback';
import { Press } from '@/components/ui/Press';
import { Screen } from '@/components/ui/Screen';
import { Segmented } from '@/components/ui/Segmented';
import { T } from '@/components/ui/T';
import { destinationById } from '@/data/destinations';
import { useT } from '@/i18n';
import { printPlanner } from '@/services/print';
import { useActiveTrip, useAppStore } from '@/store/useAppStore';
import type { Activity, ActivityType } from '@/store/types';
import { colors } from '@/theme';
import { LENGTH_FILTERS, matchesLength, type LengthFilter } from '@/utils/format';
import { dayLabel, todayInTz, tripDays, tzAbbrev, utcOffsetLabel } from '@/utils/time';

export default function Plan() {
  const trip = useActiveTrip();
  if (!trip) return <NoTrip />;
  return <PlanInner key={trip.id} />;
}

function PlanInner() {
```

```
  const t = useT();
  const trip = useActiveTrip()!;
  const params = useLocalSearchParams<{ tab?: string }>();
  const toggleAssign = useAppStore((s) => s.toggleAssign);
  const dest = destinationById(trip.destinationId);
  const tz = dest?.tz ?? 'UTC';
  const days = tripDays(trip.startDate, trip.endDate);
  const today = todayInTz(tz);

  const [tab, setTab] = useState<'itinerary' | 'vote'>(params.tab === 'vote' ? 'vote' : 'itinerary');
  const [seenTab, setSeenTab] = useState(params.tab);
  // Home's "Vote" tile navigates here with ?tab=vote while this screen is already mounted.
  if (params.tab !== seenTab) {
    setSeenTab(params.tab);
    if (params.tab) setTab(params.tab === 'vote' ? 'vote' : 'itinerary');
  }
  const [day, setDay] = useState<'all' | string>('all');
  const [cat, setCat] = useState<'all' | ActivityType>('all');
  const [len, setLen] = useState<LengthFilter>('any');
  const [addOpen, setAddOpen] = useState(false);
  const [editing, setEditing] = useState<Activity | null>(null);
  const [assigning, setAssigning] = useState<Activity | null>(null);

  const selectedDay = day === 'all' || days.includes(day) ? day : 'all';
  const matches = (a: Activity) => (selectedDay === 'all' || a.date === selectedDay) && (cat === 'all' || a.type === cat) && matchesLeng
th(a.durationMin, len);
  const byTime = (a: Activity, b: Activity) => (a.date + a.time).localeCompare(b.date + b.time);
  const voting = trip.activities.filter((a) => a.status === 'voting');
  const shown = trip.activities.filter((a) => (tab === 'vote' ? a.status === 'voting' : a.status !== 'voting') && matches(a)).sort(byTim
e);
  const assigningLive = assigning ? trip.activities.find((a) => a.id === assigning.id) ?? null : null;

  const print = async () => {
    try {
      await printPlanner(trip);
    } catch {
      toast('Allow pop-ups to print the planner', { tone: 'warn' });
    }
  };

  return (
    <Screen
      bottomInset={30}
      header={
        <View style={styles.header}>
          <View style={{ flex: 1 }}>
            <T variant="kicker" color={colors.textSecondary} numberOfLines={1}>
              {dest ? `${dest.city}, ${dest.country}` : trip.name}
            </T>
            <T variant="display" accessibilityRole="header">
              The plan
            </T>
          </View>
          <IconButton icon="print-outline" label="Print planner" onPress={print} size={44} />
          <IconButton
            icon="add"
            label="Add activity"
            onPress={() => {
              setEditing(null);
              setAddOpen(true);
            }}
            size={44}
            bg={colors.primary}
            color={colors.onPrimary}
            bordered={false}
          />
        </View>
      }>
      <Segmented
        value={tab}
        onChange={setTab}
        options={[
          { key: 'itinerary', label: 'Itinerary' },
          { key: 'vote', label: `Vote (${voting.length})` },
        ]}
      />

      <ScrollView horizontal showsHorizontalScrollIndicator={false} contentContainerStyle={styles.chips} style={styles.chipScroll}>
        <Chip label="All days" active={selectedDay === 'all'} onPress={() => setDay('all')} />
        {days.map((d, i) => (
          <Chip key={d} label={`Day ${i + 1} · ${dayLabel(d)}`} active={selectedDay === d} onPress={() => setDay(d)} />
        ))}
      </ScrollView>
      <ScrollView horizontal showsHorizontalScrollIndicator={false} contentContainerStyle={styles.chips} style={styles.chipScroll}>
        {activityFilters.map((f) => (
          <Chip key={f.key} small label={f.key === 'all' ? t(f.labelKey) : `${typeEmoji[f.key]} ${t(f.labelKey)}`} active={cat === f.ke
```

```
y} onPress={() => setCat(f.key)} />
        ))}
      </ScrollView>
      <ScrollView horizontal showsHorizontalScrollIndicator={false} contentContainerStyle={styles.chips} style={styles.chipScroll}>
        {LENGTH_FILTERS.map((f) => (
          <Chip key={f.key} small icon={f.key === 'any' ? undefined : 'timer-outline'} label={t(f.labelKey)} active={len === f.key} onPr
ess={() => setLen(f.key)} />
        ))}
      </ScrollView>

      <View style={styles.tzRow}>
        <Ionicons name="globe-outline" size={13} color={colors.textMuted} />
        <T variant="caption" color={colors.textMuted}>
          Times are in {dest?.city ?? 'local'} time ({tzAbbrev(tz)}, {utcOffsetLabel(tz)}){days.includes(today) ? ' · today is on the tr
ip' : ''}
        </T>
      </View>

      <View style={{ marginTop: 10 }}>
        {shown.length ? (
          shown.map((a) => (
            <ActivityCard
              key={a.id}
              trip={trip}
              activity={a}
              mode={tab === 'vote' ? 'vote' : 'plan'}
              onEdit={(x) => {
                setEditing(x);
                setAddOpen(true);
              }}
              onAssign={setAssigning}
            />
          ))
        ) : (
          <EmptyState
            icon={tab === 'vote' ? 'thumbs-up-outline' : 'calendar-outline'}
            title={tab === 'vote' ? 'Nothing to vote on' : (day !== 'all' || cat !== 'all' || len !== 'any') && trip.activities.length ?
t('itin.noMatch') : t('itin.noActivities')}
            body={tab === 'vote' ? 'Add an activity and mark it “voting” — the crew decides together.' : (day !== 'all' || cat !== 'all'
|| len !== 'any') && trip.activities.length ? 'Try a different day, category or length.' : t('itin.noActivitiesSub')}
          />
        )}
      </View>

      <Press
        onPress={() => {
          setEditing(null);
          setAddOpen(true);
        }}
        style={styles.addRow}
        accessibilityLabel={t('itin.addActivity')}
        testID="add-activity">
        <Ionicons name="add-circle" size={20} color={colors.primary} />
        <T variant="small" weight="bold" color={colors.primary}>
          {t('itin.addActivity')}
        </T>
      </Press>

      <AddActivitySheet visible={addOpen} onClose={() => setAddOpen(false)} trip={trip} date={selectedDay === 'all' ? days[0] : selected
Day} editing={editing} />
      <MemberPickerSheet visible={!!assigning} onClose={() => setAssigning(null)} members={trip.members} selected={assigningLive?.assign
ed ?? []} onToggle={(id) => assigning && toggleAssign(trip.id, assigning.id, id)} />
    </Screen>
  );
}

const styles = StyleSheet.create({
  header: { flexDirection: 'row', alignItems: 'flex-end', gap: 10, paddingHorizontal: 20, paddingTop: 8, paddingBottom: 14 },
  chipScroll: { marginHorizontal: -20, marginTop: 12, flexGrow: 0 },
  chips: { gap: 8, paddingHorizontal: 20 },
  tzRow: { flexDirection: 'row', alignItems: 'center', gap: 6, marginTop: 14 },
  addRow: { flexDirection: 'row', alignItems: 'center', justifyContent: 'center', gap: 6, paddingVertical: 12, marginTop: 4 },
});
```

src/app/(tabs)/profile.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { router } from 'expo-router';
import { StyleSheet, View } from 'react-native';

import { Avatar, Card, type IconName } from '@/components/ui/bits';
import { toast } from '@/components/ui/feedback';
import { Press } from '@/components/ui/Press';
import { Screen } from '@/components/ui/Screen';
import { T } from '@/components/ui/T';
import { useT } from '@/i18n';
```

```
import { printPlanner } from '@/services/print';
import { useActiveTrip, useCurrentUser } from '@/store/useAppStore';
import { colors, radius } from '@/theme';

type MenuRow = { icon: IconName; title: string; sub?: string; href?: string; onPress?: () => void };

export default function More() {
  const t = useT();
  const trip = useActiveTrip();
  const user = useCurrentUser();

  const printTrip = async () => {
    if (!trip) return toast('Create or join a trip to print its planner', { tone: 'info' });
    try {
      await printPlanner(trip);
    } catch {
      toast('Allow pop-ups to print the planner', { tone: 'warn' });
    }
  };

  const groups: { title: string; rows: MenuRow[] }[] = [
    {
      title: 'Trip',
      rows: [
        { icon: 'people-outline', title: 'Crew & roles', sub: 'Invite code, roles, votes & checklist', href: '/crew' },
        { icon: 'airplane-outline', title: 'Flights & arrivals', sub: 'Everyone’s arrival & departure times', href: '/flights' },
        { icon: 'swap-horizontal-outline', title: 'My trips', sub: 'Switch, create or join a trip', href: '/trips' },
        { icon: 'trophy-outline', title: 'Trip recap', sub: 'Stats, highlights & share card', href: '/recap' },
      ],
    },
    {
      title: 'Get around',
      rows: [
        { icon: 'map-outline', title: 'Trip Map', sub: 'Pins + nearest airport', href: '/map' },
        { icon: 'globe-outline', title: '360° City Views', sub: 'Day & night skylines with the lights on', href: '/panorama' },
        { icon: 'compass-outline', title: 'Explore Nearby', sub: 'Food & sights by radius', href: '/nearby' },
        { icon: 'partly-sunny-outline', title: 'Weather', sub: '7-day forecast', href: '/weather' },
        { icon: 'cash-outline', title: 'Currency', sub: 'Live converter', href: '/currency' },
        { icon: 'locate-outline', title: 'Nearest airport', sub: 'Based on your location', href: '/airport' },
      ],
    },
    {
      title: 'Keep it safe',
      rows: [
        { icon: 'folder-open-outline', title: 'Offline Docs', sub: 'Passports, bookings — no wifi needed', href: '/documents' },
        { icon: 'camera-outline', title: 'Photo Dump', sub: 'Everyone’s shots in one place', href: '/photos' },
        { icon: 'print-outline', title: 'Printable Planner', sub: 'Paper itinerary', onPress: printTrip },
        { icon: 'medkit-outline', title: 'Emergency', sub: 'Local numbers & safety', href: '/emergency' },
        { icon: 'briefcase-outline', title: 'Packing list', sub: 'Smart suggestions', href: '/packing' },
      ],
    },
    {
      title: 'Fun',
      rows: [
        { icon: 'star-outline', title: 'Bucket List', sub: 'Must-dos for the trip', href: '/bucket-list' },
        { icon: 'game-controller-outline', title: 'Trip Quiz', sub: 'Test the crew', href: '/quiz' },
        { icon: 'sparkles-outline', title: 'Discover', sub: 'Survey → your dream destination', href: '/survey' },
        { icon: 'language-outline', title: 'Phrasebook', sub: 'Speak the basics', href: '/phrasebook' },
        { icon: 'color-palette-outline', title: 'Make it yours', sub: 'Themes, interests & home screen', href: '/personalize' },
        { icon: 'settings-outline', title: 'Language & Settings', sub: '6 languages', href: '/settings' },
      ],
    },
    {
      title: 'Help',
      rows: [
        { icon: 'bulb-outline', title: 'How it works', sub: 'A tour of the app', href: '/how-it-works' },
        { icon: 'help-circle-outline', title: 'Help & FAQ', href: '/help' },
        { icon: 'information-circle-outline', title: 'About', href: '/about' },
      ],
    },
  ];

  return (
    <Screen bottomInset={30}>
      <View style={styles.head}>
        {trip ? (
          <T variant="kicker" color={colors.textSecondary} numberOfLines={1}>
            {trip.name}
          </T>
        ) : null}
        <T variant="display" accessibilityRole="header">
          {t('nav.profile')}
        </T>
      </View>

      <Card padded={false} style={styles.card}>
```

```
        <Press onPress={() => router.push('/settings')} style={styles.profile} scaleTo={0.99} accessibilityLabel="Open your profile and
settings">
          <Avatar name={user?.name ?? 'You'} src={user?.avatar} size={52} />
          <View style={{ flex: 1 }}>
            <T variant="title" weight="bold" numberOfLines={1}>
              {user?.name ?? 'You'}
            </T>
            <T variant="caption" color={colors.textSecondary} numberOfLines={1}>
              {[user?.email, user?.homeCity].filter(Boolean).join(' · ')}
            </T>
          </View>
          <Ionicons name="chevron-forward" size={16} color={colors.textMuted} />
        </Press>
      </Card>

      {groups.map((g) => (
        <View key={g.title}>
          <T variant="kicker" color={colors.textSecondary} style={styles.kicker}>
            {g.title}
          </T>
          <Card padded={false} style={styles.card}>
            {g.rows.map((r, i) => (
              <Press
                key={r.title}
                onPress={() => (r.onPress ? r.onPress() : router.push(r.href as never))}
                style={[styles.row, i > 0 && styles.line]}
                scaleTo={0.99}
                accessibilityLabel={r.sub ? `${r.title}. ${r.sub}` : r.title}>
                <View style={styles.icon}>
                  <Ionicons name={r.icon} size={19} color={colors.text} />
                </View>
                <View style={{ flex: 1 }}>
                  <T variant="title" weight="bold">
                    {r.title}
                  </T>
                  {r.sub ? (
                    <T variant="caption" color={colors.textSecondary} numberOfLines={1}>
                      {r.sub}
                    </T>
                  ) : null}
                </View>
                <Ionicons name="chevron-forward" size={16} color={colors.textMuted} />
              </Press>
            ))}
          </Card>
        </View>
      ))}
    </Screen>
  );
}

const styles = StyleSheet.create({
  head: { marginTop: 12, marginBottom: 14 },
  card: { borderRadius: radius.xl },
  profile: { flexDirection: 'row', alignItems: 'center', gap: 14, paddingHorizontal: 16, paddingVertical: 14 },
  kicker: { marginTop: 24, marginBottom: 8, marginLeft: 6 },
  row: { flexDirection: 'row', alignItems: 'center', gap: 14, paddingHorizontal: 16, paddingVertical: 13 },
  line: { borderTopWidth: 1, borderTopColor: colors.border },
  icon: { width: 40, height: 40, borderRadius: 20, backgroundColor: colors.bgAlt, alignItems: 'center', justifyContent: 'center' },
});
```

src/app/_layout.tsx

```
import { Anton_400Regular } from '@expo-google-fonts/anton';
import { Manrope_400Regular, Manrope_500Medium, Manrope_600SemiBold, Manrope_700Bold, Manrope_800ExtraBold, useFonts } from '@expo-googl
e-fonts/manrope';
import { Stack } from 'expo-router';
import * as SplashScreen from 'expo-splash-screen';
import { StatusBar } from 'expo-status-bar';
import { useEffect } from 'react';
import { SafeAreaProvider } from 'react-native-safe-area-context';

import { ReminderService } from '@/components/ReminderService';
import { DialogHost, ToastHost, toast } from '@/components/ui/feedback';
import { useWatchOsReduceMotion } from '@/hooks/useReduceMotion';
import { initNotifications } from '@/services/notifications';
import { startSync } from '@/sync/runtime';
import { useAppStore } from '@/store/useAppStore';
import { colors, isDark } from '@/theme';

SplashScreen.preventAutoHideAsync().catch(() => {});

export default function RootLayout() {
  const [fontsLoaded, fontError] = useFonts({ Anton_400Regular, Manrope_400Regular, Manrope_500Medium, Manrope_600SemiBold, Manrope_700B
old, Manrope_800ExtraBold });
  const hydrated = useAppStore((s) => s.hydrated);
```

```
  const seeded = useAppStore((s) => s.seeded);
  const signedIn = useAppStore((s) => !!s.session);
  const ensureSeed = useAppStore((s) => s.ensureSeed);

  useWatchOsReduceMotion();
  useEffect(() => {
    initNotifications();
  }, []);

  useEffect(() => {
    if (hydrated) ensureSeed();
  }, [hydrated, ensureSeed]);

  const ready = (fontsLoaded || !!fontError) && hydrated && seeded;
  useEffect(() => {
    if (!ready) return;
    SplashScreen.hideAsync().catch(() => {});
    void startSync(); // phone-to-phone sync for trips that were shared (does nothing until you share or join one)
    // Back from the reload that applies a new theme: say so, then forget the one-off session.
    const { restyled, setRestyled } = useAppStore.getState();
    if (restyled) {
      setRestyled(false);
      setTimeout(() => toast('Your new look is live ✨', { icon: 'sparkles' }), 600);
    }
  }, [ready]);

  if (!ready) return null;

  return (
    <SafeAreaProvider>
      <StatusBar style={isDark ? 'light' : 'dark'} />
      <Stack screenOptions={{ headerShown: false, contentStyle: { backgroundColor: colors.bg }, animation: 'slide_from_right' }}>
        <Stack.Screen name="index" />
        <Stack.Protected guard={!signedIn}>
          <Stack.Screen name="onboarding" options={{ animation: 'fade' }} />
          <Stack.Screen name="sign-in" options={{ animation: 'fade' }} />
          <Stack.Screen name="sign-up" />
          <Stack.Screen name="forgot-password" />
        </Stack.Protected>
        <Stack.Protected guard={signedIn}>
          <Stack.Screen name="(tabs)" options={{ animation: 'fade' }} />
          <Stack.Screen name="destination/[id]" />
          <Stack.Screen name="destinations" />
          <Stack.Screen name="nearby" />
          <Stack.Screen name="trips" />
          <Stack.Screen name="new-trip" options={{ presentation: 'modal', animation: 'slide_from_bottom' }} />
          <Stack.Screen name="join" options={{ presentation: 'modal', animation: 'slide_from_bottom' }} />
          <Stack.Screen name="flights" />
          <Stack.Screen name="crew" />
          <Stack.Screen name="how-it-works" />
          <Stack.Screen name="panorama" options={{ animation: 'fade' }} />
          <Stack.Screen name="map" />
          <Stack.Screen name="weather" />
          <Stack.Screen name="photos" />
          <Stack.Screen name="bucket-list" />
          <Stack.Screen name="quiz" />
          <Stack.Screen name="survey" />
          <Stack.Screen name="documents" />
          <Stack.Screen name="packing" />
          <Stack.Screen name="currency" />
          <Stack.Screen name="notifications" />
          <Stack.Screen name="settings" />
          <Stack.Screen name="emergency" />
          <Stack.Screen name="phrasebook" />
          <Stack.Screen name="recap" />
          <Stack.Screen name="airport" />
          <Stack.Screen name="help" />
          <Stack.Screen name="about" />
          <Stack.Screen name="personalize" options={{ presentation: 'modal', animation: 'slide_from_bottom' }} />
        </Stack.Protected>
      </Stack>
      {signedIn ? <ReminderService /> : null}
      <ToastHost />
      <DialogHost />
    </SafeAreaProvider>
  );
}
```

src/app/about.tsx

```
import * as Linking from 'expo-linking';
import { StyleSheet, View } from 'react-native';

import { LogoMark } from '@/components/Logo';
import { Card } from '@/components/ui/bits';
import { Header, Screen } from '@/components/ui/Screen';
```

```
import { T } from '@/components/ui/T';
import { colors } from '@/theme';

const SOURCES: [string, string, string][] = [
  ['Weather', 'Open-Meteo forecast API (CC BY 4.0)', 'https://open-meteo.com'],
  ['Exchange rates', 'ExchangeRate-API open access endpoint', 'https://www.exchangerate-api.com/docs/free'],
  ['Maps', 'Apple Maps / Google Maps via react-native-maps; OpenStreetMap on web (ODbL)', 'https://www.openstreetmap.org/copyright'],
  ['Photography', 'Unsplash — free to use under the Unsplash License (see ATTRIBUTIONS.md)', 'https://unsplash.com/license'],
  ['Fonts', 'Manrope (Mikhail Sharanda) and Anton (Vernon Adams) — SIL Open Font License', 'https://fonts.google.com'],
  ['Icons', 'Ionicons & Material Community Icons (MIT) via @expo/vector-icons', 'https://icons.expo.fyi'],
  ['Emergency numbers', 'Official tourism / government sources per country (e.g. JNTO for Japan)', 'https://www.jnto.go.jp/safety-tips/eng/emergency-numbers.html'],
  ['AI (optional)', 'Anthropic Claude API — only when the user adds their own key', 'https://docs.claude.com'],
];

const STACK = [
  'Expo SDK 57 + React Native 0.86 (iOS & Android, TypeScript)',
  'Expo Router — file-based navigation with protected routes',
  'Zustand + Immer — state; AsyncStorage — offline persistence',
  '@noble/hashes (PBKDF2 password hashing), expo-crypto (random salts), expo-secure-store (API key)',
  'expo-notifications, expo-location, expo-image-picker, expo-document-picker',
  'expo-print, expo-sharing, expo-speech, expo-file-system, expo-camera, react-native-maps',
  'qrcode-generator (invite QR codes, MIT), react-native-view-shot (share the recap as a picture)',
  'A small Node sync server in server/ (no dependencies) keeps friends' phones in step; Jest + Playwright for tests',
];

export default function About() {
  return (
    <Screen header={<Header title="About" subtitle="Credits, sources & how it's built" />}>
      <View style={styles.brand}>
        <LogoMark size={64} />
        <T variant="h2">TogetherWeGo</T>
        <T variant="small" color={colors.textSecondary} center>
          Together We Go: Group Trip Planner · FBLA Mobile Application Development 2026–27 · v1.0
        </T>
      </View>
      <Card style={{ gap: 10 }}>
        <T variant="title">Architecture</T>
        <T variant="small" color={colors.textSecondary}>
          Layered MVVM-style design: screens (views) in src/app, reusable components in src/components, a single store (model + view-model selectors) in src/store, and side-effect services (weather, currency, notifications, AI, files) in src/services. Data flows one way: screens call store actions → state updates → UI re-renders.
        </T>
        {STACK.map((s) => (
          <T key={s} variant="caption">
            • {s}
          </T>
        ))}
      </Card>
      <Card style={{ gap: 12, marginTop: 14 }}>
        <T variant="title">Data sources & licences</T>
        {SOURCES.map(([k, v, url]) => (
          <View key={k}>
            <T variant="small" weight="semibold">
              {k}
            </T>
            <T variant="caption" color={colors.textSecondary}>
              {v}
            </T>
            <T variant="caption" color={colors.primary} onPress={() => Linking.openURL(url)} accessibilityRole="link">
              {url}
            </T>
          </View>
        ))}
      </Card>
      <T variant="caption" color={colors.textMuted} center style={{ marginTop: 14 }}>
        Demo people, trips and chats are fictional sample data. Destination facts are summarised from public tourism information.
      </T>
    </Screen>
  );
}

const styles = StyleSheet.create({
  brand: { alignItems: 'center', gap: 8, marginBottom: 18 },
});
```

src/app/airport.tsx

```
import { Ionicons } from '@expo/vector-icons';
import * as Linking from 'expo-linking';
import { useState } from 'react';
import { ActivityIndicator, StyleSheet, View } from 'react-native';

import { Button } from '@/components/ui/Button';
import { Card, Chip, Pill } from '@/components/ui/bits';
import { Header, Screen } from '@/components/ui/Screen';
```

```
import { T } from '@/components/ui/T';
import { destinationById } from '@/data/destinations';
import { getLocation, type LocationResult } from '@/services/location';
import { useNow } from '@/hooks/useTrip';
import { useActiveTrip } from '@/store/useAppStore';
import { colors, radius } from '@/theme';
import { bearingDeg, compassPoint, driveMinutes, kmLabel, nearestAirports } from '@/utils/geo';
import { partsInTz, time12, tzAbbrev } from '@/utils/time';

/** Tracks your location and finds the closest airports (works offline — the airport list is bundled). */
export default function AirportScreen() {
  const trip = useActiveTrip();
  const dest = destinationById(trip?.destinationId ?? 'tokyo')!;
  const [from, setFrom] = useState<'trip' | 'me'>('trip');
  const [loc, setLoc] = useState<LocationResult | null>(null);
  const [loading, setLoading] = useState(false);
  const now = useNow(60000);

  const useMine = async () => {
    setFrom('me');
    if (loc) return;
    setLoading(true);
    setLoc(await getLocation(dest));
    setLoading(false);
  };

  const origin = from === 'me' && loc?.source === 'device' ? loc : { lat: dest.lat, lng: dest.lng };
  const list = nearestAirports(origin, 6);

  return (
    <Screen header={<Header title="Nearest airport" subtitle={from === 'me' ? 'From your current location' : `From central ${dest.city}
`} />}>
      <View style={{ flexDirection: 'row', gap: 8, marginBottom: 14 }}>
        <Chip icon="business-outline" label={`${dest.city} centre`} active={from === 'trip'} onPress={() => setFrom('trip')} />
        <Chip icon="locate-outline" label="My location" active={from === 'me'} onPress={useMine} />
      </View>
      {loading ? (
        <View style={{ padding: 30, alignItems: 'center', gap: 8 }}>
          <ActivityIndicator color={colors.primary} />
          <T variant="small" color={colors.textSecondary}>
            Finding you…
          </T>
        </View>
      ) : null}
      {from === 'me' && loc?.source === 'fallback' ? (
        <Card style={{ marginBottom: 12, backgroundColor: colors.orangeSoft, borderColor: colors.orangeSoft }}>
          <T variant="small">Location permission is off, so we’re using central {dest.city}. Enable location in your phone settings to u
se GPS.</T>
        </Card>
      ) : null}
      <View style={{ gap: 10 }}>
        {list.map((a, i) => {
          const local = time12(partsInTz(now, a.tz).hhmm);
          return (
            <Card key={a.iata} style={[{ gap: 8 }, i === 0 && { borderColor: colors.primary, borderWidth: 2 }]}>
              <View style={styles.row}>
                <View style={[styles.code, i === 0 && { backgroundColor: colors.primary }]}>
                  <T variant="title" weight="bold" color={i === 0 ? colors.white : colors.primary}>
                    {a.iata}
                  </T>
                </View>
                <View style={{ flex: 1 }}>
                  <T variant="small" weight="semibold">
                    {a.name}
                  </T>
                  <T variant="caption" color={colors.textSecondary}>
                    {a.city}, {a.country}
                  </T>
                </View>
                {i === 0 ? <Pill label="Closest" /> : null}
              </View>
              <View style={[styles.row, { gap: 14 }]}>
                <View style={styles.stat}>
                  <Ionicons name="navigate-outline" size={13} color={colors.textSecondary} />
                  <T variant="caption">
                    {kmLabel(a.km)} {compassPoint(bearingDeg(origin, a))}
                  </T>
                </View>
                <View style={styles.stat}>
                  <Ionicons name="car-outline" size={13} color={colors.textSecondary} />
                  <T variant="caption">~{driveMinutes(a.km)} min</T>
                </View>
                <View style={styles.stat}>
                  <Ionicons name="time-outline" size={13} color={colors.textSecondary} />
                  <T variant="caption">
                    {local} {tzAbbrev(a.tz)}
```

```
                </T>
              </View>
            </View>
            <Button
              label="Directions"
              icon="navigate"
              size="sm"
              variant={i === 0 ? 'primary' : 'soft'}
              onPress={() => Linking.openURL(`https://www.google.com/maps/dir/?api=1&destination=${a.lat},${a.lng}`)}
            />
          </Card>
        );
      })}
    </View>
  </Screen>
 );
}

const styles = StyleSheet.create({
  row: { flexDirection: 'row', alignItems: 'center', gap: 12 },
  code: { width: 56, height: 44, borderRadius: radius.sm, backgroundColor: colors.primarySoft, alignItems: 'center', justifyContent: 'ce
nter' },
  stat: { flexDirection: 'row', alignItems: 'center', gap: 4 },
});
```

src/app/bucket-list.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { Image } from 'expo-image';
import { router } from 'expo-router';
import { useState } from 'react';
import { ScrollView, StyleSheet, TextInput, View } from 'react-native';

import { Card, EmptyState, ProgressBar } from '@/components/ui/bits';
import { confirmAction, toast } from '@/components/ui/feedback';
import { Press, tap } from '@/components/ui/Press';
import { Header, Screen } from '@/components/ui/Screen';
import { T } from '@/components/ui/T';
import { destinationById, destinations } from '@/data/destinations';
import { imageSource } from '@/data/images';
import { useAppStore } from '@/store/useAppStore';
import { colors, fonts, radius } from '@/theme';

export default function BucketList() {
  const bucket = useAppStore((s) => s.bucket);
  const favorites = useAppStore((s) => s.favorites);
  const add = useAppStore((s) => s.addBucket);
  const toggle = useAppStore((s) => s.toggleBucket);
  const del = useAppStore((s) => s.deleteBucket);
  const [text, setText] = useState('');
  const [err, setErr] = useState<string | null>(null);
  const done = bucket.filter((b) => b.done).length;
  const ideas = destinations.filter((d) => !bucket.some((b) => b.destinationId === d.id)).sort((a, b) => Number(favorites.includes(b.i
d)) - Number(favorites.includes(a.id)) || b.rating - a.rating).slice(0, 8);

  const submit = () => {
    const t = text.trim();
    if (t.length < 3) return setErr('Write a few words about your dream');
    if (t.length > 80) return setErr('Keep it under 80 characters');
    add(t);
    setText('');
    setErr(null);
    tap('success');
    toast('Added to your bucket list ✨');
  };

  return (
    <Screen header={<Header title="Bucket list" subtitle={`${done} of ${bucket.length} dreams ticked off`} />}>
      <Card style={{ gap: 8 }}>
        <ProgressBar value={bucket.length ? done / bucket.length : 0} height={8} />
        <View style={{ flexDirection: 'row', gap: 8, marginTop: 6 }}>
          <TextInput value={text} onChangeText={(v) => { setText(v); setErr(null); }} placeholder="Someday I want to…" placeholderTextCo
lor={colors.textMuted} style={[styles.input, err && { borderColor: colors.red }]} onSubmitEditing={submit} returnKeyType="done" accessib
ilityLabel="New bucket list item" maxLength={90} />
          <Press onPress={submit} style={styles.addBtn} accessibilityLabel="Add to bucket list">
            <Ionicons name="add" size={22} color={colors.white} />
          </Press>
        </View>
        {err ? (
          <T variant="caption" color={colors.red}>
            {err}
          </T>
        ) : null}
      </Card>

      <T variant="h3" style={{ marginTop: 18, marginBottom: 10 }}>
```

```
        Inspiration
      </T>
      <ScrollView horizontal showsHorizontalScrollIndicator={false} contentContainerStyle={{ gap: 10, paddingHorizontal: 20 }} style={{
marginHorizontal: -20 }}>
        {ideas.map((d) => (
          <Press
            key={d.id}
            onPress={() => {
              add(`Visit ${d.city}, ${d.country}`, d.id);
              toast(`${d.city} added ✨`);
            }}
            style={styles.idea}
            accessibilityLabel={`Add ${d.city} to bucket list`}>
            <Image source={imageSource(d.image)} style={StyleSheet.absoluteFill} contentFit="cover" />
            <View style={styles.ideaShade} />
            <Ionicons name="add-circle" size={22} color={colors.white} style={{ position: 'absolute', top: 8, right: 8 }} />
            <T variant="small" weight="bold" color={colors.white} style={{ position: 'absolute', left: 10, bottom: 10 }}>
              {d.city}
            </T>
          </Press>
        ))}
      </ScrollView>

      <T variant="h3" style={{ marginTop: 18, marginBottom: 10 }}>
        My list
      </T>
      {bucket.length ? (
        <View style={{ gap: 8 }}>
          {bucket.map((b) => {
            const d = b.destinationId ? destinationById(b.destinationId) : undefined;
            return (
              <View key={b.id} style={[styles.item, b.done && { opacity: 0.7 }]}>
                <Press onPress={() => toggle(b.id)} accessibilityRole="checkbox" accessibilityState={{ checked: b.done }} accessibilityL
abel={b.text} hitSlop={6}>
                  <Ionicons name={b.done ? 'checkmark-circle' : 'ellipse-outline'} size={24} color={b.done ? colors.primary : colors.bor
derStrong} />
                </Press>
                {d ? <Image source={imageSource(d.image)} style={styles.thumb} contentFit="cover" /> : null}
                <Press style={{ flex: 1 }} onPress={() => (d ? router.push({ pathname: '/destination/[id]', params: { id: d.id } }) : to
ggle(b.id))}>
                  <T variant="small" weight="semibold" style={b.done && { textDecorationLine: 'line-through' }}>
                    {b.text}
                  </T>
                  {d ? (
                    <T variant="caption" color={colors.textSecondary}>
                      {d.country} · best {d.bestMonths.split(',')[0]}
                    </T>
                  ) : null}
                </Press>
                <Press onPress={() => confirmAction('Remove from bucket list?', b.text, () => del(b.id), { confirmLabel: 'Remove', destr
uctive: true })} hitSlop={8} accessibilityLabel="Remove">
                  <Ionicons name="close" size={18} color={colors.textMuted} />
                </Press>
              </View>
            );
          })}
        </View>
      ) : (
        <EmptyState icon="flag-outline" title="Dream big" body="Add places and experiences you want to do with your crew one day." />
      )}
    </Screen>
  );
}

const styles = StyleSheet.create({
  input: { flex: 1, height: 46, borderRadius: radius.md, borderWidth: 1, borderColor: colors.border, backgroundColor: colors.cardMuted,
paddingHorizontal: 14, fontFamily: fonts.regular, fontSize: 15, color: colors.text },
  addBtn: { width: 46, height: 46, borderRadius: 23, backgroundColor: colors.primary, alignItems: 'center', justifyContent: 'center' },
  idea: { width: 120, height: 150, borderRadius: radius.lg, overflow: 'hidden' },
  ideaShade: { ...StyleSheet.absoluteFill, backgroundColor: 'rgba(0,0,0,0.25)' },
  item: { flexDirection: 'row', alignItems: 'center', gap: 10, padding: 12, borderRadius: radius.md, backgroundColor: colors.card, borde
rWidth: 1, borderColor: colors.border },
  thumb: { width: 44, height: 44, borderRadius: 10 },
});
```

src/app/crew.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { router } from 'expo-router';
import { useState } from 'react';
import { StyleSheet, View } from 'react-native';

import { ShareSheet } from '@/components/ShareSheet';
import { AddMemberSheet } from '@/components/crew/AddMemberSheet';
import { ChecklistSection } from '@/components/crew/ChecklistSection';
import { DocsSection } from '@/components/crew/DocsSection';
```

```
import { InviteCard } from '@/components/crew/InviteCard';
import { MembersSection } from '@/components/crew/MembersSection';
import { NewPollSheet } from '@/components/crew/NewPollSheet';
import { PollsSection } from '@/components/crew/PollsSection';
import { RoleSheet } from '@/components/crew/RoleSheet';
import { SyncTripCard } from '@/components/sync/SyncTripCard';
import { MemberPickerSheet } from '@/components/trip/MemberPickerSheet';
import { NoTrip } from '@/components/trip/NoTrip';
import { Card, Pill } from '@/components/ui/bits';
import { toast, confirmAction } from '@/components/ui/feedback';
import { Press } from '@/components/ui/Press';
import { Header, Screen } from '@/components/ui/Screen';
import { T } from '@/components/ui/T';
import { destinationById } from '@/data/destinations';
import { inviteMessage } from '@/services/share';
import { tripStatus, useActiveTrip, useAppStore, useMyTrips } from '@/store/useAppStore';
import type { Member } from '@/store/types';
import { colors } from '@/theme';
import { rangeLabel } from '@/utils/time';

export default function Crew() {
  const trip = useActiveTrip();
  if (!trip) return <NoTrip />;
  return <Hub key={trip.id} />;
}

/** Crew & roles: who is on the trip, the invite (code, QR, sync), polls, the shared checklist and documents. */
function Hub() {
  const trip = useActiveTrip()!;
  const myTrips = useMyTrips();
  const userId = useAppStore((s) => s.session?.userId ?? '');
  // Actions are stable references, so read them without subscribing to the whole store.
  const s = useAppStore.getState();
  const owner = trip.ownerId === userId;
  const dest = destinationById(trip.destinationId);

  const [shareOpen, setShareOpen] = useState(false);
  const [roleFor, setRoleFor] = useState<Member | null>(null);
  const [addOpen, setAddOpen] = useState(false);
  const [pollOpen, setPollOpen] = useState(false);
  const [assignTaskId, setAssignTaskId] = useState<string | null>(null);

  const status = tripStatus(trip);
  const assignee = trip.tasks.find((k) => k.id === assignTaskId)?.assignee;

  return (
    <Screen header={<Header title="Crew & roles" subtitle="Invite code, roles, votes and checklist" backLabel="More" />} bottomInset={3
0}>
      {/* Trip switcher */}
      <Card padded={false}>
        <Press onPress={() => router.push('/trips')} style={styles.tripRow} accessibilityLabel="Switch trip">
          <View style={styles.tripIcon}>
            <Ionicons name="airplane" size={18} color={colors.white} />
          </View>
          <View style={{ flex: 1 }}>
            <T variant="title" numberOfLines={1}>
              {trip.name}
            </T>
            <T variant="caption" color={colors.textSecondary} numberOfLines={1}>
              {dest?.city} · {rangeLabel(trip.startDate, trip.endDate)}
            </T>
          </View>
          <Pill label={status === 'planning' ? 'Planning' : status === 'active' ? 'In progress' : 'Completed'} tone={status === 'active'
? 'green' : status === 'planning' ? 'blue' : 'gray'} />
          <T variant="caption" weight="semibold" color={colors.textSecondary}>
            {myTrips.length > 1 ? `+${myTrips.length - 1}` : ''}
          </T>
          <Ionicons name="chevron-forward" size={16} color={colors.textMuted} />
        </Press>
      </Card>

      <InviteCard trip={trip} owner={owner} onShare={() => setShareOpen(true)} />
      <SyncTripCard trip={trip} />
      <MembersSection trip={trip} userId={userId} onPick={setRoleFor} onAdd={() => setAddOpen(true)} />
      <PollsSection trip={trip} userId={userId} owner={owner} onNew={() => setPollOpen(true)} />
      <ChecklistSection trip={trip} onAssign={setAssignTaskId} />
      <DocsSection trip={trip} />

      <ShareSheet visible={shareOpen} onClose={() => setShareOpen(false)} title="Invite your group" message={inviteMessage(trip.name, tr
ip.inviteCode)} preview={inviteMessage(trip.name, trip.inviteCode)} />

      <RoleSheet
        member={roleFor}
        canEdit={owner || roleFor?.id === userId}
        canRemove={owner && roleFor?.id !== trip.ownerId}
        onClose={() => setRoleFor(null)}
```

```
        onPick={(role) => {
          if (roleFor) {
            s.setRole(trip.id, roleFor.id, role);
            toast(`${roleFor.name.split(' ')[0]} is now ${role}`);
          }
          setRoleFor(null);
        }}
        onRemove={() => {
          const m = roleFor!;
          setRoleFor(null);
          confirmAction(`Remove ${m.name}?`, 'They'll lose access to this trip.', () => {
            s.removeMember(trip.id, m.id);
            toast(`${m.name.split(' ')[0]} removed`);
          }, { destructive: true, confirmLabel: 'Remove' });
        }}
      />
      <AddMemberSheet
        visible={addOpen}
        onClose={() => setAddOpen(false)}
        onAdd={(n, e) => s.addMember(trip.id, n, e)}
        onShare={() => {
          setAddOpen(false);
          setTimeout(() => setShareOpen(true), 250);
        }}
      />
      <NewPollSheet visible={pollOpen} onClose={() => setPollOpen(false)} onCreate={(q, o) => s.createPoll(trip.id, q, o)} />
      <MemberPickerSheet
        visible={!!assignTaskId}
        single
        title="Assign task"
        onClose={() => setAssignTaskId(null)}
        members={trip.members}
        selected={assignee ? [assignee] : []}
        onToggle={(id) => assignTaskId && s.assignTask(trip.id, assignTaskId, id)}
      />
    </Screen>
  );
}

const styles = StyleSheet.create({
  tripRow: { flexDirection: 'row', alignItems: 'center', gap: 10, padding: 14 },
  tripIcon: { width: 40, height: 40, borderRadius: 20, backgroundColor: colors.primary, alignItems: 'center', justifyContent: 'center'
},
});
```

src/app/currency.tsx

```
import { useState } from 'react';
import { StyleSheet, View } from 'react-native';

import { CurrencyConverter } from '@/components/CurrencyConverter';
import { Card, Pill } from '@/components/ui/bits';
import { toast } from '@/components/ui/feedback';
import { Header, Screen } from '@/components/ui/Screen';
import { T } from '@/components/ui/T';
import { destinationById } from '@/data/destinations';
import { useRatesLoader } from '@/hooks/useTrip';
import { convert, fetchRates, useRates } from '@/services/currency';
import { useActiveTrip, useAppStore } from '@/store/useAppStore';
import { colors } from '@/theme';
import { money, timeAgo } from '@/utils/format';

const QUICK = [1, 5, 10, 20, 50, 100, 500, 1000];

export default function Currency() {
  useRatesLoader();
  const trip = useActiveTrip();
  const home = useAppStore((s) => s.settings.homeCurrency);
  const rates = useRates();
  const local = destinationById(trip?.destinationId ?? 'tokyo')?.currency ?? 'JPY';
  const [pair, setPair] = useState<[string, string]>([home, local === home ? 'EUR' : local]);
  const [refreshing, setRefreshing] = useState(false);

  return (
    <Screen
      header={<Header title="Currency converter" subtitle={rates.live ? `Live rates · updated ${timeAgo(rates.fetchedAt)}` : 'Offline re
ference rates'} />}
      onRefresh={async () => {
        setRefreshing(true);
        const r = await fetchRates(true);
        setRefreshing(false);
        toast(r.live ? 'Rates updated' : 'Offline — using saved rates', { tone: r.live ? 'ok' : 'warn' });
      }}
      refreshing={refreshing}>
      <Card>
        <CurrencyConverter from={pair[0]} to={pair[1]} onChange={(a, b) => setPair([a, b])} />
```

```
      </Card>

      <View style={styles.head}>
        <T variant="h3">Quick reference</T>
        <Pill label={`${pair[0]} → ${pair[1]}`} />
      </View>
      <Card padded={false}>
        {QUICK.map((n, i) => (
          <View key={n} style={[styles.row, i % 2 === 1 && { backgroundColor: colors.cardMuted }]}>
            <T variant="small" weight="semibold">
              {money(n, pair[0])}
            </T>
            <T variant="small">{money(convert(n, pair[0], pair[1], rates), pair[1], { decimals: true })}</T>
          </View>
        ))}
      </Card>
      <T variant="caption" color={colors.textMuted} style={{ marginTop: 12 }}>
        Rates from open.er-api.com, cached on your phone so the converter keeps working offline. Your home currency ({home}) can be chan
ged in Settings.
      </T>
    </Screen>
  );
}

const styles = StyleSheet.create({
  head: { flexDirection: 'row', justifyContent: 'space-between', alignItems: 'center', marginTop: 20, marginBottom: 10 },
  row: { flexDirection: 'row', justifyContent: 'space-between', paddingHorizontal: 16, paddingVertical: 11 },
});
```

src/app/destination/[id].tsx

```
import { Ionicons } from '@expo/vector-icons';
import { Image } from 'expo-image';
import { LinearGradient } from 'expo-linear-gradient';
import { router, useLocalSearchParams } from 'expo-router';
import { useState } from 'react';
import { ScrollView, StyleSheet, View } from 'react-native';
import { useSafeAreaInsets } from 'react-native-safe-area-context';

import { AddToTripSheet, FavStar } from '@/components/explore';
import { placeCategoryIcon } from '@/components/trip/meta';
import { Button } from '@/components/ui/Button';
import { panoById } from '@/data/panoramas';
import { Card, EmptyState, IconButton, Pill, type IconName } from '@/components/ui/bits';
import { Press } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import { destinationById } from '@/data/destinations';
import { imageSource } from '@/data/images';
import { placesFor } from '@/data/places';
import { currencyByCode } from '@/data/reference';
import { useNow, useWeather } from '@/hooks/useTrip';
import { shareText } from '@/services/share';
import { weatherInfo } from '@/services/weather';
import { MAX_WIDTH, colors, radius } from '@/theme';
import { durationLabel, money, priceTier } from '@/utils/format';
import { distanceKm, kmLabel, nearestAirports } from '@/utils/geo';
import { hoursAhead, deviceTimeZone, partsInTz, time12, utcOffsetLabel } from '@/utils/time';

export default function DestinationScreen() {
  const { id } = useLocalSearchParams<{ id: string }>();
  const d = destinationById(String(id));
  const insets = useSafeAreaInsets();
  const weather = useWeather(d?.id);
  const [adding, setAdding] = useState<string | null>(null);
  const now = useNow(60000);
  if (!d) return <EmptyState icon="alert-circle-outline" title="Destination not found" />;

  const places = placesFor(d.id)
    .map((p) => ({ p, km: distanceKm(d, p) }))
    .sort((a, b) => b.p.rating - a.p.rating)
    .slice(0, 6);
  const air = nearestAirports(d, 1)[0];
  const diff = hoursAhead(d.tz, deviceTimeZone());
  const cur = currencyByCode(d.currency);

  const facts: [IconName, string, string][] = [
    ['calendar-outline', 'Best time', d.bestMonths],
    ['wallet-outline', 'Daily cost', `~${money(d.dailyCost)} pp`],
    ['cash-outline', 'Currency', `${d.currency}${cur ? ` (${cur.symbol})` : ''}`],
    ['language-outline', 'Language', d.language],
    ['time-outline', 'Local time', `${time12(partsInTz(now, d.tz).hhmm)} · ${utcOffsetLabel(d.tz)}`],
    ['airplane-outline', 'Airport', `${air.iata} · ${kmLabel(air.km)}`],
  ];

  return (
    <View style={styles.root}>
```

```
      <ScrollView style={{ width: '100%' }} contentContainerStyle={{ paddingBottom: insets.bottom + 100 }} showsVerticalScrollIndicator={false}>
        <View style={styles.hero}>
          <Image source={imageSource(d.image)} style={StyleSheet.absoluteFill} contentFit="cover" transition={250} />
          <LinearGradient colors={['rgba(0,0,0,0.35)', 'rgba(0,0,0,0)', 'rgba(0,0,0,0.55)']} locations={[0, 0.4, 1]} style={StyleSheet.absoluteFill} />
          <View style={[styles.heroTop, { top: insets.top + 8 }]}>
            <IconButton icon="chevron-back" label="Back" onPress={() => (router.canGoBack() ? router.back() : router.replace('/(tabs)/explore'))} bg="rgba(255,255,255,0.9)" bordered={false} />
            <View style={{ flexDirection: 'row', gap: 8 }}>
              <IconButton icon="share-social-outline" label={`Share ${d.city}`} onPress={() => shareText(`${d.city}, ${d.country} 🌍 ${d.blurb} — found on TogetherWeGo`)} bg="rgba(255,255,255,0.9)" bordered={false} />
              <FavStar id={d.id} size={40} />
            </View>
          </View>
          <View style={styles.heroText}>
            <T variant="display" color={colors.white}>
              {d.city}
            </T>
            <View style={styles.row}>
              <Ionicons name="location-sharp" size={13} color="rgba(255,255,255,0.9)" />
              <T variant="small" color="rgba(255,255,255,0.9)">
                {d.country}
              </T>
              <Ionicons name="star" size={13} color={colors.star} style={{ marginLeft: 8 }} />
              <T variant="small" weight="bold" color={colors.white}>
                {d.rating} <T variant="caption" color="rgba(255,255,255,0.8)">({d.reviews.toLocaleString()} reviews)</T>
              </T>
            </View>
          </View>
        </View>

        <View style={styles.body}>
          <View style={styles.tags}>
            {d.tags.map((t) => (
              <Pill key={t} label={t === 'best' ? 'Top pick' : t[0].toUpperCase() + t.slice(1)} tone={t === 'best' ? 'yellow' : 'green'} />
            ))}
            <Pill label={'$'.repeat(d.priceLevel)} tone="gray" />
          </View>
          <T variant="body" color={colors.textSecondary}>
            {d.blurb}
          </T>

          {weather ? (
            <Press onPress={() => router.push({ pathname: '/weather', params: { id: d.id } })} style={styles.weather}>
              <Ionicons name={weatherInfo(weather.current.code).icon} size={26} color="#E8A23A" />
              <View style={{ flex: 1 }}>
                <T variant="title">
                  {weather.current.temp}°C · {weatherInfo(weather.current.code).label}
                </T>
                <T variant="caption" color={colors.textSecondary}>
                  Next 3 days: {weather.daily.slice(1, 4).map((x) => `${x.max}°`).join(' · ')} {weather.live ? '' : '(estimate)'}
                </T>
              </View>
              <Ionicons name="chevron-forward" size={16} color={colors.textMuted} />
            </Press>
          ) : null}

          <View style={styles.facts}>
            {facts.map(([icon, label, value]) => (
              <View key={label} style={styles.fact}>
                <Ionicons name={icon} size={16} color={colors.primary} />
                <T variant="micro" color={colors.textMuted}>
                  {label}
                </T>
                <T variant="caption" weight="semibold" numberOfLines={2}>
                  {value}
                </T>
              </View>
            ))}
          </View>
          <T variant="caption" color={colors.textMuted}>
            {diff === 0 ? 'Same time zone as you.' : `${Math.abs(diff)}h ${diff > 0 ? 'ahead of' : 'behind'} your time zone.`}
          </T>

          <T variant="h3" style={{ marginTop: 6 }}>
            Top things to do
          </T>
          {places.map(({ p, km }) => (
            <Card key={p.id} style={styles.placeRow} onPress={() => setAdding(d.id)}>
              <View style={styles.placeIcon}>
                <Ionicons name={placeCategoryIcon[p.category]} size={17} color={colors.primary} />
              </View>
              <View style={{ flex: 1 }}>
                <T variant="small" weight="semibold">
```

```
                {p.name}
              </T>
              <T variant="caption" color={colors.textSecondary} numberOfLines={2}>
                {p.blurb}
              </T>
              <T variant="caption" color={colors.textMuted} style={{ marginTop: 2 }}>
                ★ {p.rating} · {durationLabel(p.durationMin)} · {priceTier(p.cost)} · {kmLabel(km)} from centre
              </T>
            </View>
          </Card>
        ))}
      </View>
    </ScrollView>
    <View style={[styles.footer, { paddingBottom: insets.bottom + 12 }]}>
      {panoById(d.id) ? <Button label="360° view" icon="globe-outline" variant="secondary" full={false} onPress={() => router.push({ p
athname: '/panorama', params: { city: d.id } })} /> : null}
      <Button label="Add to Trip" icon="add-circle-outline" onPress={() => setAdding(d.id)} style={{ flex: 1 }} />
    </View>
    <AddToTripSheet destinationId={adding} onClose={() => setAdding(null)} />
  </View>
 );
}

const styles = StyleSheet.create({
 root: { flex: 1, backgroundColor: colors.bg },
 hero: { height: 360, width: '100%', maxWidth: MAX_WIDTH, alignSelf: 'center' },
 heroTop: { position: 'absolute', left: 16, right: 16, flexDirection: 'row', justifyContent: 'space-between' },
 heroText: { position: 'absolute', left: 20, right: 20, bottom: 36, gap: 4 },
 row: { flexDirection: 'row', alignItems: 'center', gap: 4 },
 body: { marginTop: -22, backgroundColor: colors.bg, borderTopLeftRadius: 26, borderTopRightRadius: 26, padding: 20, gap: 14, width: '1
00%', maxWidth: MAX_WIDTH, alignSelf: 'center' },
 tags: { flexDirection: 'row', flexWrap: 'wrap', gap: 6 },
 weather: { flexDirection: 'row', alignItems: 'center', gap: 12, backgroundColor: colors.yellowSoft, padding: 14, borderRadius: radius.
lg },
 facts: { flexDirection: 'row', flexWrap: 'wrap', gap: 8 },
 fact: { width: '31.5%', backgroundColor: colors.card, borderRadius: radius.md, padding: 10, gap: 3, borderWidth: 1, borderColor: color
s.border },
 placeRow: { flexDirection: 'row', gap: 12, padding: 14 },
 placeIcon: { width: 36, height: 36, borderRadius: 10, backgroundColor: colors.primarySofter, alignItems: 'center', justifyContent: 'ce
nter' },
 footer: { position: 'absolute', left: 0, right: 0, bottom: 0, paddingHorizontal: 20, paddingTop: 12, backgroundColor: 'rgba(244,244,23
8,0.96)', borderTopWidth: 1, borderTopColor: colors.border, flexDirection: 'row', alignItems: 'center', alignSelf: 'center', maxWidth: M
AX_WIDTH, width: '100%' },
});
```

src/app/destinations.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { useLocalSearchParams } from 'expo-router';
import { useMemo, useState } from 'react';
import { ScrollView, StyleSheet, TextInput, View, useWindowDimensions } from 'react-native';

import { PlaceCard } from '@/components/explore';
import { Chip, EmptyState } from '@/components/ui/bits';
import { Header, Screen } from '@/components/ui/Screen';
import { T } from '@/components/ui/T';
import { destinations, exploreCategories, type DestinationTag } from '@/data/destinations';
import { useT } from '@/i18n';
import { useAppStore } from '@/store/useAppStore';
import { MAX_WIDTH, colors, fonts, radius } from '@/theme';

type Sort = 'rating' | 'cheap' | 'name';

export default function Destinations() {
 const t = useT();
 const params = useLocalSearchParams<{ q?: string; cat?: string }>();
 const { width } = useWindowDimensions();
 const favorites = useAppStore((s) => s.favorites);
 const [q, setQ] = useState(params.q ?? '');
 const [cat, setCat] = useState<'all' | DestinationTag | 'fav'>((params.cat as DestinationTag) ?? 'all');
 const [sort, setSort] = useState<Sort>('rating');
 const colW = Math.min(width, MAX_WIDTH) - 40;

 const list = useMemo(() => {
   const s = q.trim().toLowerCase();
   return destinations
     .filter((d) => (cat === 'all' ? true : cat === 'fav' ? favorites.includes(d.id) : d.tags.includes(cat)))
     .filter((d) => !s || `${d.city} ${d.country} ${d.highlights.join(' ')}`.toLowerCase().includes(s))
     .sort((a, b) => (sort === 'rating' ? b.rating - a.rating : sort === 'cheap' ? a.dailyCost - b.dailyCost : a.city.localeCompare(b.c
ity)));
 }, [q, cat, sort, favorites]);

 return (
   <Screen header={<Header title="All destinations" subtitle={`${list.length} places to explore`} />}>
     <View style={styles.search}>
       <Ionicons name="search" size={18} color={colors.textMuted} />
```

```
        <TextInput value={q} onChangeText={setQ} placeholder={t('explore.search')} placeholderTextColor={colors.textMuted} style={style
s.input} accessibilityLabel="Search destinations" />
      </View>
      <ScrollView horizontal showsHorizontalScrollIndicator={false} contentContainerStyle={{ gap: 8, paddingHorizontal: 20 }} style={{ m
arginHorizontal: -20, marginTop: 12 }}>
        <Chip label="★ Favourites" active={cat === 'fav'} onPress={() => setCat('fav')} />
        {exploreCategories.map((c) => (
          <Chip key={c.key} label={t(c.labelKey)} image={c.image} active={cat === c.key} onPress={() => setCat(c.key)} />
        ))}
      </ScrollView>
      <View style={styles.sortRow}>
        <T variant="caption" color={colors.textSecondary}>
          Sort by
        </T>
        {(
          [
            ['rating', 'Top rated'],
            ['cheap', 'Budget-friendly'],
            ['name', 'A–Z'],
          ] as [Sort, string][]
        ).map(([k, l]) => (
          <Chip key={k} small label={l} active={sort === k} onPress={() => setSort(k)} />
        ))}
      </View>
      {list.length ? (
        <View style={styles.grid}>
          {list.map((d) => (
            <View key={d.id} style={{ width: (colW - 12) / 2, gap: 4 }}>
              <PlaceCard d={d} height={170} />
              <T variant="caption" color={colors.textSecondary}>
                ~${d.dailyCost}/day · {d.bestMonths.split(',')[0]}
              </T>
            </View>
          ))}
        </View>
      ) : (
        <EmptyState icon="search-outline" title="No matches" body={cat === 'fav' ? 'Tap the ☆ on any place to save it here.' : 'Try anot
her search or category.'} />
      )}
    </Screen>
  );
}

const styles = StyleSheet.create({
  search: { flexDirection: 'row', alignItems: 'center', gap: 10, backgroundColor: colors.card, borderRadius: radius.md, borderWidth: 1,
borderColor: colors.border, paddingHorizontal: 14, height: 48 },
  input: { flex: 1, fontFamily: fonts.regular, fontSize: 15, color: colors.text, height: '100%', ...({ outlineStyle: 'none' } as object)
},
  sortRow: { flexDirection: 'row', alignItems: 'center', gap: 6, marginVertical: 14 },
  grid: { flexDirection: 'row', flexWrap: 'wrap', gap: 12 },
});
```

src/app/documents.tsx

```
import { Ionicons } from '@expo/vector-icons';
import * as Linking from 'expo-linking';
import * as Sharing from 'expo-sharing';
import { useLocalSearchParams } from 'expo-router';
import { useState } from 'react';
import { Platform, StyleSheet, View } from 'react-native';

import { DocFileBody } from '@/components/documents/DocFileBody';
import { DocRow } from '@/components/documents/DocRow';
import { NoTrip } from '@/components/trip/NoTrip';
import { Button } from '@/components/ui/Button';
import { Card, EmptyState } from '@/components/ui/bits';
import { confirmAction, toast } from '@/components/ui/feedback';
import { Input } from '@/components/ui/Input';
import { tap } from '@/components/ui/Press';
import { Header, Screen } from '@/components/ui/Screen';
import { Sheet } from '@/components/ui/Sheet';
import { T } from '@/components/ui/T';
import { FileTooLargeError, deleteLocalFile, pickDocument, pickImage, sizeLabel } from '@/services/files';
import { useActiveTrip, useAppStore } from '@/store/useAppStore';
import type { TripDoc } from '@/store/types';
import { colors, radius } from '@/theme';
import { timeAgo } from '@/utils/format';
import { firstError, maxLen, required } from '@/utils/validation';

/** Offline document vault: tickets, bookings, passport copies and notes. */
export default function Documents() {
  const params = useLocalSearchParams<{ open?: string }>();
  const trip = useActiveTrip();
  const userId = useAppStore((s) => s.session?.userId ?? '');
  const addDoc = useAppStore((s) => s.addDoc);
  const deleteDoc = useAppStore((s) => s.deleteDoc);
```

```
  const [open, setOpen] = useState<TripDoc | null>(() => (params.open ? trip?.docs.find((d) => d.id === params.open) ?? null : null));
  const [noteOpen, setNoteOpen] = useState(false);
  const [name, setName] = useState('');
  const [note, setNote] = useState('');
  const [errors, setErrors] = useState<Record<string, string | null>>({});

  if (!trip) return <NoTrip />;

  const upload = async (kind: 'file' | 'photo') => {
    try {
      const f = kind === 'photo' ? await pickImage() : await pickDocument();
      if (!f) return;
      addDoc(trip.id, { name: f.name.replace(/\.[^.]+$/, ''), kind: f.kind, uri: f.uri, size: f.size, offline: true });
      tap('success');
      toast('Saved to the vault — available offline');
    } catch (e) {
      toast(e instanceof FileTooLargeError ? e.message : 'Couldn’t save that file', { tone: 'warn' });
    }
  };

  const openFile = async (uri: string, title: string) => {
    try {
      if (Platform.OS !== 'web' && (await Sharing.isAvailableAsync())) await Sharing.shareAsync(uri, { dialogTitle: title });
      else await Linking.openURL(uri);
    } catch {
      toast('No app available to open this file', { tone: 'warn' });
    }
  };

  const used = trip.docs.reduce((n, d) => n + (d.size ?? (d.note?.length ?? 0)), 0);

  return (
    <Screen header={<Header title="Documents" subtitle={`${trip.docs.length} items · all available offline`} />}>
      <Card style={styles.banner}>
        <Ionicons name="cloud-done-outline" size={22} color={colors.primary} />
        <View style={{ flex: 1 }}>
          <T variant="small" weight="semibold">
            Works without wifi or data
          </T>
          <T variant="caption" color={colors.textSecondary}>
            Files are copied into TogetherWeGo’s private storage on this phone · {sizeLabel(used) || '0 KB'} used
          </T>
        </View>
      </Card>
      <View style={{ flexDirection: 'row', gap: 8, marginTop: 12 }}>
        <Button label="File" icon="document-attach-outline" size="md" onPress={() => upload('file')} style={{ flex: 1 }} />
        <Button label="Photo" icon="image-outline" size="md" variant="secondary" onPress={() => upload('photo')} style={{ flex: 1 }} />
        <Button label="Note" icon="create-outline" size="md" variant="secondary" onPress={() => setNoteOpen(true)} style={{ flex: 1 }} /
>
      </View>

      <View style={{ gap: 8, marginTop: 16 }}>
        {trip.docs.length ? (
          trip.docs.map((d) => <DocRow key={d.id} doc={d} trip={trip} onPress={() => setOpen(d)} />)
        ) : (
          <EmptyState icon="folder-open-outline" title="No documents yet" body="Add tickets, bookings and passport copies so they’re alw
ays on hand." />
        )}
      </View>

      <Sheet visible={!!open} onClose={() => setOpen(null)} title={open?.name} subtitle={open ? `${open.kind === 'note' ? 'Note' : open.
kind.toUpperCase()} · added ${timeAgo(open.addedAt)}` : undefined}>
        {open?.kind === 'note' ? (
          <View style={styles.note}>
            <T variant="bodySm" selectable>
              {open.note}
            </T>
          </View>
        ) : null}
        {open ? <DocFileBody doc={open} adderName={trip.members.find((m) => m.id === open.addedBy)?.name.split(' ')[0] ?? 'a friend'} on
Open={(uri) => openFile(uri, open.name)} /> : null}
        {open && (open.addedBy === userId || trip.ownerId === userId) ? (
          <Button
            label="Delete"
            variant="danger"
            icon="trash-outline"
            onPress={() => {
              const d = open;
              setOpen(null);
              confirmAction('Delete document?', `“${d.name}” will be removed from the vault.`, () => {
                deleteLocalFile(d.uri);
                deleteDoc(trip.id, d.id);
                toast('Document deleted');
              }, { destructive: true, confirmLabel: 'Delete' });
            }}
          />
```

```
        ) : null}
      </Sheet>

      <Sheet
        visible={noteOpen}
        onClose={() => setNoteOpen(false)}
        title="New note"
        subtitle="Addresses, booking refs, allergy cards…"
        footer={
          <Button
            label="Save note"
            icon="checkmark"
            onPress={() => {
              const e = { name: firstError(name, required('a title'), maxLen(40, 'Title')), note: firstError(note, required('some tex
t'), maxLen(2000, 'Note')) };
              setErrors(e);
              if (e.name || e.note) return tap('warning');
              addDoc(trip.id, { name: name.trim(), kind: 'note', note: note.trim(), offline: true });
              setName('');
              setNote('');
              setNoteOpen(false);
              toast('Note saved offline');
            }}
          />
        }>
        <Input label="Title" value={name} onChangeText={setName} error={errors.name} placeholder="e.g. Hotel address in Japanese" />
        <Input label="Note" value={note} onChangeText={setNote} error={errors.note} multiline placeholder="Write anything you'll need of
fline…" style={{ minHeight: 140 }} />
      </Sheet>
    </Screen>
  );
}

const styles = StyleSheet.create({
  banner: { flexDirection: 'row', alignItems: 'center', gap: 12, backgroundColor: colors.primarySofter, borderColor: colors.primaryLine
},
  note: { backgroundColor: colors.cardMuted, padding: 14, borderRadius: radius.md, borderWidth: 1, borderColor: colors.border },
});
```

src/app/emergency.tsx

```
import { Ionicons } from '@expo/vector-icons';
import * as Linking from 'expo-linking';
import { router } from 'expo-router';
import { useState } from 'react';
import { StyleSheet, View } from 'react-native';

import { ShareSheet } from '@/components/ShareSheet';
import { Button } from '@/components/ui/Button';
import { Card, ListRow } from '@/components/ui/bits';
import { toast } from '@/components/ui/feedback';
import { Press } from '@/components/ui/Press';
import { Header, Screen } from '@/components/ui/Screen';
import { T } from '@/components/ui/T';
import { destinationById } from '@/data/destinations';
import { emergencyByCountry } from '@/data/reference';
import { getLocation } from '@/services/location';
import { useActiveTrip } from '@/store/useAppStore';
import { colors, radius } from '@/theme';

/** Safety screen — everything here works offline except the map searches. */
export default function Emergency() {
  const trip = useActiveTrip();
  const dest = destinationById(trip?.destinationId ?? 'tokyo')!;
  const info = emergencyByCountry[dest.countryCode] ?? emergencyByCountry.DEFAULT;
  const [shareMsg, setShareMsg] = useState<string | null>(null);

  const call = (n: string) => Linking.openURL(`tel:${n}`).catch(() => toast(`Dial ${n} from your phone`, { tone: 'warn' }));

  const shareLocation = async () => {
    const loc = await getLocation(dest);
    if (loc.source !== 'device') toast('GPS unavailable — sharing the trip centre instead', { tone: 'warn' });
    setShareMsg(`🚨 I need help. My location: https://maps.google.com/?q=${loc.lat.toFixed(5)},${loc.lng.toFixed(5)} — sent from Togethe
rWeGo (${trip?.name ?? 'trip'})`);
  };

  return (
    <Screen header={<Header title="Emergency & safety" subtitle={`${dest.country} · works offline`} />}>
      <View style={styles.grid}>
        {[
          ['Police', info.police, 'shield-outline', '#2B5FB0'],
          ['Ambulance', info.ambulance, 'medkit-outline', colors.red],
          ['Fire', info.fire, 'flame-outline', colors.orange],
        ].map(([label, num, icon, color]) => (
          <Press key={label} onPress={() => call(num)} style={[styles.callBox, { borderColor: `${color}55` }]} accessibilityLabel={`Call
${label}, ${num}`}>
```

```
            <Ionicons name={icon as 'shield-outline'} size={24} color={color} />
            <T variant="h1" color={color}>
              {num}
            </T>
            <T variant="caption" weight="semibold" color={colors.textSecondary}>
              {label}
            </T>
          </Press>
        ))}
      </View>

      <Button label="Share my live location with the group" icon="location" variant="danger" onPress={shareLocation} style={{ marginTop:
14 }} />

      <Card style={{ marginTop: 14, gap: 8 }}>
        <T variant="title">Good to know in {dest.country}</T>
        {info.tips.map((tip) => (
          <View key={tip} style={{ flexDirection: 'row', gap: 8 }}>
            <Ionicons name="information-circle-outline" size={16} color={colors.primary} style={{ marginTop: 2 }} />
            <T variant="small" style={{ flex: 1 }}>
              {tip}
            </T>
          </View>
        ))}
      </Card>

      <Card style={{ marginTop: 14 }}>
        <ListRow icon="medical-outline" label="Find the nearest hospital" sub="Opens maps around you" tint={colors.red} onPress={() => L
inking.openURL('https://www.google.com/maps/search/hospital+near+me')} />
        <ListRow icon="flag-outline" label="Find my embassy" sub={`Embassies and consulates in ${dest.city}`} onPress={() => Linking.ope
nURL(`https://www.google.com/maps/search/embassy+${encodeURIComponent(dest.city)}`)} />
        <ListRow icon="document-lock-outline" label="Passport copies & insurance" sub="Your offline document vault" onPress={() => route
r.push('/documents')} />
        <ListRow icon="language-outline" label="Emergency phrases" sub={`“Help!” and more in ${dest.language}`} onPress={() => router.pu
sh('/phrasebook')} />
        <ListRow icon="airplane-outline" label="Nearest airport" onPress={() => router.push('/airport')} />
      </Card>

      <ShareSheet visible={!!shareMsg} onClose={() => setShareMsg(null)} title="Send your location" message={shareMsg ?? ''} preview={sh
areMsg ?? ''} />
    </Screen>
  );
}

const styles = StyleSheet.create({
  grid: { flexDirection: 'row', gap: 10 },
  callBox: { flex: 1, alignItems: 'center', gap: 2, paddingVertical: 16, borderRadius: radius.lg, backgroundColor: colors.card, borderWi
dth: 1.5 },
});
```

src/app/flights.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { Image } from 'expo-image';
import * as Linking from 'expo-linking';
import { useState } from 'react';
import { StyleSheet, View } from 'react-native';

import { AddFlightSheet } from '@/components/flights/AddFlightSheet';
import { NoTrip } from '@/components/trip/NoTrip';
import { Button } from '@/components/ui/Button';
import { Avatar, Card } from '@/components/ui/bits';
import { confirmAction, toast } from '@/components/ui/feedback';
import { Press } from '@/components/ui/Press';
import { Header, Screen } from '@/components/ui/Screen';
import { T } from '@/components/ui/T';
import { airportByIata } from '@/data/airports';
import { destinationById } from '@/data/destinations';
import { useActiveTrip, useAppStore } from '@/store/useAppStore';
import type { Flight, Member } from '@/store/types';
import { colors, radius } from '@/theme';
import { durationLabel } from '@/utils/format';
import { dayLabel, deviceTimeZone, formatInTz, partsInTz, time12, tzAbbrev, utcOffsetLabel } from '@/utils/time';

const AVATAR_COLORS = ['#F2553D', '#1E88E5', '#2FBF5B', '#E5A100', '#8E5CD6', '#D94F86'];

type Person = { id: string; name: string; member?: Member; flights: Flight[] };

export default function Flights() {
  const trip = useActiveTrip();
  const userId = useAppStore((s) => s.session?.userId ?? '');
  const showTripTime = useAppStore((s) => s.settings.showTripTime);
  const updateSettings = useAppStore((s) => s.updateSettings);
  const deleteFlight = useAppStore((s) => s.deleteFlight);
  const [adding, setAdding] = useState(false);
  if (!trip) return <NoTrip />;
```

```
  const dest = destinationById(trip.destinationId)!;
  const tz = showTripTime ? dest.tz : deviceTimeZone();
  const sorted = trip.flights.slice().sort((a, b) => (a.direction === 'arrival' ? a.arriveUtc : a.departUtc) - (b.direction === 'arriva
l' ? b.arriveUtc : b.departUtc));
  const arrivals = sorted.filter((f) => f.direction === 'arrival');
  const firstLand = arrivals[0];
  const lastLand = arrivals[arrivals.length - 1];

  const everyone: Person[] = trip.members.map((m) => ({ id: m.id, name: m.id === userId ? 'You' : m.name, member: m, flights: sorted.fil
ter((f) => f.memberId === m.id) }));
  const former = sorted.filter((f) => !trip.members.some((m) => m.id === f.memberId));
  if (former.length) everyone.push({ id: 'former', name: 'Former member', flights: former });
  const people = [...everyone.filter((p) => p.flights.length), ...everyone.filter((p) => !p.flights.length)];

  const when = (utc: number) => {
    const p = partsInTz(utc, tz);
    return `${time12(p.hhmm)} local · ${dayLabel(p.ymd)}`;
  };

  const detail = (f: Flight) => {
    const from = airportByIata(f.from);
    const to = airportByIata(f.to);
    const canDelete = f.memberId === userId || trip.ownerId === userId;
    const extra = [f.seat ? `Seat ${f.seat}` : null, f.confirmation ? `Conf. ${f.confirmation}` : null, from && to ? `Airport time ${tim
e12(partsInTz(f.departUtc, from.tz).hhmm)} → ${time12(partsInTz(f.arriveUtc, to.tz).hhmm)}` : null]
      .filter(Boolean)
      .join(' · ');
    return (
      <View key={f.id} style={styles.detail}>
        <Ionicons name={f.direction === 'arrival' ? 'airplane' : 'airplane-outline'} size={14} color={colors.textMuted} />
        <View style={{ flex: 1 }}>
          <T variant="caption" weight="semibold">
            {f.direction === 'arrival' ? 'Arrival' : 'Departure'} · {f.flightNo} · {f.from} → {f.to}
            <T variant="caption" color={colors.textSecondary}>
              {` · ${durationLabel(Math.round((f.arriveUtc - f.departUtc) / 60000))}`}
            </T>
          </T>
          {extra ? (
            <T variant="caption" color={colors.textMuted}>
              {extra}
            </T>
          ) : null}
        </View>
        {f.ticketUri ? (
          <Press onPress={() => Linking.openURL(f.ticketUri!).catch(() => toast('Ticket saved offline in Documents'))} style={styles.tic
ket} accessibilityLabel={`Open ticket ${f.ticketName ?? ''}`}>
            {/\.(png|jpe?g|webp|heic)$/i.test(f.ticketName ?? '') || f.ticketUri.startsWith('data:image') ? (
              <Image source={{ uri: f.ticketUri }} style={{ width: 22, height: 22, borderRadius: 6 }} />
            ) : (
              <Ionicons name="document-attach-outline" size={15} color={colors.text} />
            )}
            <T variant="caption" weight="semibold" numberOfLines={1} style={{ maxWidth: 80 }}>
              {f.ticketName ?? 'Ticket'}
            </T>
          </Press>
        ) : null}
        {canDelete ? (
          <Press
            onPress={() => confirmAction('Remove flight?', `${f.flightNo} will be removed from the trip.`, () => deleteFlight(trip.id,
f.id), { destructive: true, confirmLabel: 'Remove' })}
            hitSlop={8}
            accessibilityLabel={`Remove flight ${f.flightNo}`}>
            <Ionicons name="trash-outline" size={17} color={colors.textMuted} />
          </Press>
        ) : null}
      </View>
    );
  };

  const personCard = (p: Person) => {
    const arr = p.flights.filter((f) => f.direction === 'arrival');
    const dep = p.flights.filter((f) => f.direction === 'departure');
    const labels = [...new Set([arr[0], dep[0]].filter((f): f is Flight => !!f).map((f) => `${f.airline} ${f.flightNo}`))];
    const bg = AVATAR_COLORS[(p.name.charCodeAt(0) + p.name.length) % AVATAR_COLORS.length];
    return (
      <Card key={p.id} style={styles.person}>
        <View style={styles.row}>
          {p.member?.avatar ? (
            <Avatar name={p.name} src={p.member.avatar} size={34} />
          ) : (
            <View style={[styles.dot, { backgroundColor: bg }]} accessibilityLabel={p.name}>
              <T variant="small" weight="bold" color={colors.white}>
                {p.name.trim().charAt(0).toUpperCase()}
              </T>
            </View>
          )}
```

```
          <T variant="title" weight="bold" numberOfLines={1} style={{ flex: 1 }}>
            {p.name}
          </T>
          <View style={{ alignItems: 'flex-end', gap: 4 }}>
            {labels.map((l) => (
              <View key={l} style={styles.flightPill}>
                <T variant="micro" weight="bold" numberOfLines={1}>
                  {l}
                </T>
              </View>
            ))}
          </View>
        </View>
        {p.flights.length ? (
          <>
            {Array.from({ length: Math.max(arr.length, dep.length) }, (_, i) => (
              <View key={i} style={styles.leg}>
                <View style={{ flex: 1 }}>
                  <T variant="kicker" color={colors.textMuted} style={styles.legLabel}>
                    Arrives
                  </T>
                  <T variant="small" weight="bold" color={arr[i] ? colors.text : colors.textMuted}>
                    {arr[i] ? when(arr[i].arriveUtc) : '—'}
                  </T>
                </View>
                <Ionicons name="airplane" size={16} color={colors.textMuted} />
                <View style={{ flex: 1, alignItems: 'flex-end' }}>
                  <T variant="kicker" color={colors.textMuted} style={styles.legLabel}>
                    Departs
                  </T>
                  <T variant="small" weight="bold" color={dep[i] ? colors.text : colors.textMuted}>
                    {dep[i] ? when(dep[i].departUtc) : '—'}
                  </T>
                </View>
              </View>
            ))}
            <View style={{ gap: 2 }}>{[...arr, ...dep].map(detail)}</View>
          </>
        ) : (
          <T variant="small" color={colors.textMuted}>
            No flight added yet
          </T>
        )}
      </Card>
    );
  };

  return (
    <Screen
      header={
        <Header
          title="Flights"
          subtitle={`Times shown in ${tz}`}
          right={<Button label="Mine" icon="cloud-upload-outline" size="sm" full={false} onPress={() => setAdding(true)} accessibilityHi
nt="Add your flight or upload a ticket" testID="add-flight" />}
        />
      }>
      <Press onPress={() => updateSettings({ showTripTime: !showTripTime })} style={styles.tz} accessibilityLabel={`Showing ${showTripTi
me ? `${dest.city} time` : 'your time'}. Tap to switch time zone`}>
        <Ionicons name="globe-outline" size={15} color={colors.blue} />
        <T variant="caption" weight="semibold" color={colors.blue}>
          {showTripTime ? `${dest.city} time` : 'Your time'} · {tzAbbrev(tz)} ({utcOffsetLabel(tz)}) · tap to switch
        </T>
        <Ionicons name="swap-horizontal" size={15} color={colors.blue} />
      </Press>

      <View style={{ gap: 12 }}>{people.map(personCard)}</View>

      {firstLand && lastLand ? (
        <Card style={styles.summary}>
          <Ionicons name="people-outline" size={20} color={colors.text} />
          <T variant="caption" style={{ flex: 1 }}>
            First arrival {formatInTz(firstLand.arriveUtc, tz)}, last {formatInTz(lastLand.arriveUtc, tz)} — the group is complete{' '}
            {durationLabel(Math.round((lastLand.arriveUtc - firstLand.arriveUtc) / 60000))} after the first landing.
          </T>
        </Card>
      ) : null}

      <AddFlightSheet visible={adding} onClose={() => setAdding(false)} />
    </Screen>
  );
}

const styles = StyleSheet.create({
  row: { flexDirection: 'row', alignItems: 'center', gap: 10 },
  tz: { flexDirection: 'row', alignItems: 'center', alignSelf: 'flex-start', gap: 6, marginBottom: 14, paddingVertical: 4 },
```

```
  person: { gap: 12, borderRadius: radius.xl, paddingHorizontal: 16, paddingVertical: 14 },
  dot: { width: 34, height: 34, borderRadius: 17, alignItems: 'center', justifyContent: 'center' },
  flightPill: { backgroundColor: colors.bgAlt, borderRadius: radius.pill, paddingHorizontal: 10, paddingVertical: 4, maxWidth: 170 },
  leg: { flexDirection: 'row', alignItems: 'center', gap: 10 },
  legLabel: { fontSize: 9.5, lineHeight: 12, letterSpacing: 1 },
  detail: { flexDirection: 'row', alignItems: 'center', gap: 8, paddingTop: 10, marginTop: 2, borderTopWidth: 1, borderTopColor: colors.
border },
  ticket: { flexDirection: 'row', alignItems: 'center', gap: 6, paddingHorizontal: 8, paddingVertical: 5, borderRadius: radius.pill, bac
kgroundColor: colors.bgAlt },
  summary: { marginTop: 14, flexDirection: 'row', gap: 10, alignItems: 'center', borderRadius: radius.xl },
});
```

src/app/forgot-password.tsx

```
import { router } from 'expo-router';
import { useState } from 'react';
import { View } from 'react-native';

import { Button } from '@/components/ui/Button';
import { Card } from '@/components/ui/bits';
import { showDialog, toast } from '@/components/ui/feedback';
import { Input } from '@/components/ui/Input';
import { tap } from '@/components/ui/Press';
import { Header, Screen } from '@/components/ui/Screen';
import { T } from '@/components/ui/T';
import { useAppStore } from '@/store/useAppStore';
import { colors } from '@/theme';
import { validateEmail, validatePassword } from '@/utils/validation';

/** Two-step reset: request a 6-digit code, then set a new password. Works fully offline. */
export default function ForgotPassword() {
  const accountExists = useAppStore((s) => s.accountExists);
  const resetPassword = useAppStore((s) => s.resetPassword);
  const [step, setStep] = useState<1 | 2>(1);
  const [email, setEmail] = useState('');
  const [code, setCode] = useState('');
  const [sent, setSent] = useState('');
  const [pw, setPw] = useState('');
  const [errors, setErrors] = useState<Record<string, string | null>>({});
  const [loading, setLoading] = useState(false);

  const sendCode = () => {
    const e = validateEmail(email);
    if (e) return setErrors({ email: e });
    if (!accountExists(email)) return setErrors({ email: 'We couldn’t find an account with that email.' });
    const c = String(Math.floor(100000 + Math.random() * 900000));
    setSent(c);
    setErrors({});
    setStep(2);
    showDialog('Check your inbox', `We sent a 6-digit code to ${email.trim()}.\n\nDemo build: your code is ${c}`, [{ label: 'OK' }], 'ma
il-unread-outline');
  };

  const reset = async () => {
    const e = {
      code: !/^\d{6}$/.test(code) ? 'Enter the 6-digit code' : code !== sent ? 'That code doesn’t match' : null,
      pw: validatePassword(pw),
    };
    setErrors(e);
    if (e.code || e.pw) return tap('warning');
    setLoading(true);
    const r = await resetPassword(email, pw);
    setLoading(false);
    if (!r.ok) return setErrors({ code: r.error });
    tap('success');
    toast('Password updated — sign in with your new password');
    router.back();
  };

  return (
    <Screen header={<Header title="Reset password" subtitle={step === 1 ? 'Step 1 of 2 · verify email' : 'Step 2 of 2 · new password'} /
>}>
      <Card>
        <T variant="bodySm" color={colors.textSecondary} style={{ marginBottom: 14 }}>
          {step === 1 ? 'Enter the email you signed up with and we’ll send you a reset code.' : `Enter the code sent to ${email.trim()}
and choose a new password.`}
        </T>
        {step === 1 ? (
          <View style={{ gap: 14 }}>
            <Input label="Email address" icon="mail-outline" value={email} onChangeText={setEmail} keyboardType="email-address" autoCapi
talize="none" error={errors.email} />
            <Button label="Send reset code" onPress={sendCode} icon="paper-plane" />
          </View>
        ) : (
          <View style={{ gap: 14 }}>
            <Input label="6-digit code" icon="key-outline" value={code} onChangeText={(v) => setCode(v.replace(/\D/g, '').slice(0, 6))}
```

```
keyboardType="number-pad" error={errors.code} />
            <Input label="New password" icon="lock-closed-outline" value={pw} onChangeText={setPw} secureToggle error={errors.pw} hint
="8+ characters with a letter and a number" />
            <Button label="Update password" onPress={reset} loading={loading} />
            <Button label="Resend code" variant="ghost" size="md" onPress={sendCode} />
          </View>
        )}
      </Card>
    </Screen>
  );
}
```

src/app/help.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { router, type Href } from 'expo-router';
import { useState } from 'react';
import { StyleSheet, View } from 'react-native';

import { Card, type IconName } from '@/components/ui/bits';
import { Press } from '@/components/ui/Press';
import { Header, Screen } from '@/components/ui/Screen';
import { T } from '@/components/ui/T';
import { colors, radius, shadow } from '@/theme';

const JOURNEY: { icon: IconName; title: string; body: string; href: Href }[] = [
  { icon: 'add-circle-outline', title: '1. Create or join a trip', body: 'Pick a destination, dates and a group budget — or enter a frie
nd's invite code.', href: '/trips' },
  { icon: 'share-social-outline', title: '2. Invite your group', body: 'More → Crew & roles → Share sends it straight to WhatsApp, Insta
gram, Messages and more.', href: '/crew' },
  { icon: 'people-outline', title: '3. Give everyone a role', body: 'Treasurer, Navigator, Food Lead… tap a member to assign who owns wh
at.', href: '/crew' },
  { icon: 'map-outline', title: '4. Build the plan', body: 'In the Plan tab, add activities per day, filter by type or length, and vote
with "I'm in". Majority confirms.', href: '/(tabs)/itinerary' },
  { icon: 'airplane-outline', title: '5. Add flights & tickets', body: 'Everyone's arrivals show in one list, converted to the trip's ti
me zone automatically.', href: '/flights' },
  { icon: 'wallet-outline', title: '6. Track the budget', body: 'In the Budget tab, log expenses in any currency, see the breakdown and
settle up with the fewest payments.', href: '/(tabs)/budget' },
  { icon: 'chatbubbles-outline', title: '7. Chat & ask the assistant', body: 'In the Chat tab, message the group, or tap ✨ to ask the T
rip Assistant about budget, weather, food or what's next.', href: '/(tabs)/chat' },
  { icon: 'notifications-outline', title: '8. Get live reminders', body: 'Tap the bell on any activity — everyone gets an alert before i
t starts.', href: '/settings' },
  { icon: 'trophy-outline', title: '9. Wrap it up', body: 'Mark the trip complete for a shareable recap, final balances and the photo du
mp.', href: '/recap' },
];

const FAQ = [
  ['Does it work without internet?', 'Yes. Trips, chats, documents, the phrasebook, emergency numbers, the offline map and the Trip Assi
stant all work offline. Weather and currency rates refresh when you're back online and are cached meanwhile.'],
  ['Where is my data stored?', 'On your phone, in TogetherWeGo's private storage. Passwords are salted and hashed; an optional AI key go
es in the secure keychain.'],
  ['How do time zones work?', 'Itinerary times are in the destination's local time. Flights are stored as exact moments, so they display
correctly in trip time or your own time — toggle it in Flights.'],
  ['How does voting work?', 'Activities marked "Voting" become "Confirmed" once more than half the group taps "I'm in". Polls in More le
t the group pick between options.'],
  ['Can I change the language?', 'Settings → Language: English, Español, Français, 日本語, हिन्दी and 中文.'],
  ['How do I try the demo?', 'Sign in with the demo account on the sign-in screen. Settings → Demo timeline switches the sample trip bet
ween upcoming, in progress and completed.'],
];

export default function Help() {
  const [open, setOpen] = useState<number | null>(0);
  return (
    <Screen header={<Header title="How to use TogetherWeGo" subtitle="Your group trip, from first idea to final recap" />}>
      <Card onPress={() => router.push('/how-it-works')} style={styles.how} accessibilityLabel="How it works">
        <View style={styles.howIcon}>
          <Ionicons name="construct-outline" size={20} color={colors.onAccent} />
        </View>
        <View style={{ flex: 1 }}>
          <T variant="title" weight="bold">
            How it works
          </T>
          <T variant="caption" color={colors.textSecondary}>
            How the app is built, how your data is kept safe, and where to find every feature
          </T>
        </View>
        <Ionicons name="chevron-forward" size={16} color={colors.textMuted} />
      </Card>
      <T variant="headline" style={{ marginTop: 22, marginBottom: 10 }} accessibilityRole="header">
        The journey
      </T>
      <View style={{ gap: 8 }}>
        {JOURNEY.map((s) => (
          <Card key={s.title} onPress={() => router.push(s.href)} style={styles.step} accessibilityLabel={s.title}>
            <View style={styles.icon}>
              <Ionicons name={s.icon} size={18} color={colors.onAccent} />
```

```
            </View>
            <View style={{ flex: 1 }}>
              <T variant="small" weight="semibold">
                {s.title}
              </T>
              <T variant="caption" color={colors.textSecondary}>
                {s.body}
              </T>
            </View>
            <Ionicons name="chevron-forward" size={16} color={colors.textMuted} />
          </Card>
        ))}
      </View>
      <T variant="headline" style={{ marginTop: 22, marginBottom: 10 }} accessibilityRole="header">
        FAQ
      </T>
      <View style={{ gap: 8 }}>
        {FAQ.map(([q, a], i) => (
          <Press key={q} onPress={() => setOpen(open === i ? null : i)} style={styles.faq} accessibilityState={{ expanded: open === i }}
>
            <View style={{ flexDirection: 'row', alignItems: 'center', gap: 8 }}>
              <T variant="small" weight="semibold" style={{ flex: 1 }}>
                {q}
              </T>
              <Ionicons name={open === i ? 'chevron-up' : 'chevron-down'} size={16} color={colors.textMuted} />
            </View>
            {open === i ? (
              <T variant="small" color={colors.textSecondary} style={{ marginTop: 6 }}>
                {a}
              </T>
            ) : null}
          </Press>
        ))}
      </View>
    </Screen>
  );
}

const styles = StyleSheet.create({
  how: { flexDirection: 'row', alignItems: 'center', gap: 12, padding: 14, borderRadius: radius.xl },
  howIcon: { width: 42, height: 42, borderRadius: 21, backgroundColor: colors.accent, alignItems: 'center', justifyContent: 'center' },
  step: { flexDirection: 'row', alignItems: 'center', gap: 12, padding: 14, borderRadius: radius.xl },
  icon: { width: 38, height: 38, borderRadius: 19, backgroundColor: colors.accent, alignItems: 'center', justifyContent: 'center' },
  faq: { padding: 14, borderRadius: radius.xl, backgroundColor: colors.card, borderWidth: 1, borderColor: colors.border, ...shadow },
});
```

src/app/how-it-works.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { router, type Href } from 'expo-router';
import type { ReactNode } from 'react';
import { StyleSheet, View } from 'react-native';

import { Button } from '@/components/ui/Button';
import { Card, Divider, type IconName } from '@/components/ui/bits';
import { Press } from '@/components/ui/Press';
import { Header, Screen } from '@/components/ui/Screen';
import { T } from '@/components/ui/T';
import { useAppStore } from '@/store/useAppStore';
import { colors, radius } from '@/theme';

const LAYERS: { icon: IconName; title: string; where: string; body: string }[] = [
  { icon: 'phone-portrait-outline', title: 'Screens', where: 'src/app · src/components', body: 'Show the data and turn taps into action
s.' },
  { icon: 'funnel-outline', title: 'Selector hooks', where: 'useActiveTrip · balances · settlePlan', body: 'Work out exactly what each s
creen needs.' },
  { icon: 'server-outline', title: 'Zustand + Immer store', where: 'src/store', body: 'One source of truth. Saved to AsyncStorage on eve
ry change.' },
  { icon: 'sync-outline', title: 'Sync engine', where: 'src/sync · server', body: 'Keeps a shared trip identical on every phone, even af
ter being offline.' },
  { icon: 'cloud-outline', title: 'Services', where: 'src/services', body: 'Weather, currency, notifications, AI, files and sharing.' },
];

const SYNC: [IconName, string, string][] = [
  ['create-outline', 'Every change is a tiny note', 'Adding an expense or sending a message becomes a small stamped change, not a whole
copy of the trip.'],
  ['git-merge-outline', 'Newest change wins, per field', 'Two people can edit different things at once and both survive. Phones that swa
p their changes end up identical, in any order.'],
  ['cloud-offline-outline', 'Works with no signal', 'Changes wait on the phone and are sent when you are back online. Nothing is lost if
the app closes.'],
  ['images-outline', 'Photos and files travel too', 'Photos, documents and receipts are sent to the server separately, then saved on eve
ry phone so they open offline.'],
  ['server-outline', 'Your own server, no accounts', 'A small program on your computer (npm run server) passes changes between phones. A
trip is protected by a long random id that only the invite code reveals.'],
];
```

```
const SECURITY: [IconName, string, string][] = [
  ['key-outline', 'Passwords are never stored', 'Each one is salted and stretched with PBKDF2-SHA256 before it is saved, so even the sto
red copy can’t be read back.'],
  ['lock-closed-outline', 'API key in the keychain', 'The optional AI key lives in the iOS Keychain or Android Keystore (expo-secure-sto
re), not in the app database.'],
  ['cloud-offline-outline', 'Offline-first', 'Trips, chats, documents and maps are saved on the phone. Weather and exchange rates are ca
ched for when you lose signal.'],
  ['shield-checkmark-outline', 'Your data stays with you', 'Trips and chats are stored on the device. A trip you share is also kept on t
he sync server you chose. Only weather, exchange-rate and optional AI requests reach the wider internet.'],
  ['eye-off-outline', 'Private stays private', 'Your Scout AI chat, other people’s email addresses and your flight tickets are never sen
t to the sync server.'],
];

const VALIDATION: [string, string, string][] = [
  ['Email', 'Must look like name@mail.com', 'That email doesn’t look right'],
  ['Password', '8+ characters with a letter and a number', 'Use at least 8 characters'],
  ['Expense amount', 'Numbers only, above 0 and at most 1,000,000', 'Use numbers only, e.g. 25 or 25.50'],
  ['Trip dates', 'End on or after the start, up to 60 days', 'End date must be on or after the start date'],
  ['Invite code', 'Three letters, a dash, four digits', 'Codes look like ABC-1234'],
  ['Flight number', 'Airline code plus 1 to 4 digits', 'Flight numbers look like NH 175'],
  ['Packing item', '50 characters or fewer, no duplicates', 'That’s already on the list'],
];

const RUBRIC: { icon: IconName; area: string; where: string; href: Href }[] = [
  { icon: 'people-outline', area: 'Collaboration', where: 'More: members, roles, polls and checklist, plus live sync across phones. Pla
n: voting. Packing: who’s bringing what.', href: '/crew' },
  { icon: 'chatbubbles-outline', area: 'Communication', where: 'Chat tab: group chat and the Scout AI assistant.', href: '/(tabs)/chat'
},
  { icon: 'calendar-outline', area: 'Scheduling', where: 'Plan tab: day-by-day itinerary, reminders, flights and time zones.', href: '/
(tabs)/itinerary' },
  { icon: 'wallet-outline', area: 'Budgeting', where: 'Budget tab: expenses, split, settle up and currency converter.', href: '/(tabs)/b
udget' },
  { icon: 'folder-open-outline', area: 'Organization', where: 'More: offline docs, packing list, weather, photo dump and printable plann
er.', href: '/documents' },
  { icon: 'sparkles-outline', area: 'Other features', where: 'Themes and languages, trip quiz, bucket list, phrasebook and the recap sto
ry card.', href: '/personalize' },
  { icon: 'airplane-outline', area: 'Trip management', where: 'Home: switch trips. Create or join with an invite code. Recap when the tr
ip ends.', href: '/trips' },
];

const LIBRARIES = [
  'Expo SDK 57',
  'React Native 0.86',
  'TypeScript',
  'Expo Router',
  'Zustand',
  'Immer',
  'AsyncStorage',
  'expo-crypto',
  'expo-secure-store',
  'expo-notifications',
  'expo-location',
  'expo-image-picker',
  'expo-document-picker',
  'expo-file-system',
  'expo-print',
  'expo-sharing',
  'expo-camera',
  'react-native-view-shot',
  'qrcode-generator',
  '@noble/hashes',
  'expo-speech',
  'expo-haptics',
  'react-native-maps',
  'react-native-svg',
  'Reanimated',
  'Anthropic Messages API via fetch (optional)',
  'Jest',
];

function Section({ icon, title, children }: { icon: IconName; title: string; children: ReactNode }) {
  return (
    <Card style={styles.card}>
      <View style={styles.sectionHead}>
        <View style={styles.sectionIcon}>
          <Ionicons name={icon} size={17} color={colors.onAccent} />
        </View>
        <T variant="headline" accessibilityRole="header" style={{ flex: 1 }}>
          {title}
        </T>
      </View>
      {children}
    </Card>
  );
}
```

```
export default function HowItWorks() {
  const setToured = useAppStore((s) => s.setToured);
  return (
    <Screen header={<Header title="How it works" subtitle="How TogetherWeGo is built and where to find everything" backLabel="More" />}>
      <View style={{ gap: 14 }}>
        <Section icon="layers-outline" title="Architecture">
          <T variant="small" color={colors.textSecondary}>
            A layered, MVVM-style design. Data flows one way: a screen calls a store action, the state updates, and the screen redraws.
          </T>
          <View>
            {LAYERS.map((l, i) => (
              <View key={l.title}>
                <View style={styles.layer}>
                  <View style={styles.layerIcon}>
                    <Ionicons name={l.icon} size={17} color={colors.primary} />
                  </View>
                  <View style={{ flex: 1 }}>
                    <T variant="small" weight="bold">
                      {l.title}
                    </T>
                    <T variant="caption" weight="semibold" color={colors.textMuted}>
                      {l.where}
                    </T>
                    <T variant="caption" color={colors.textSecondary}>
                      {l.body}
                    </T>
                  </View>
                </View>
                {i < LAYERS.length - 1 ? <Ionicons name="arrow-down" size={14} color={colors.borderStrong} style={{ alignSelf: 'center',
marginVertical: 3 }} /> : null}
              </View>
            ))}
          </View>
        </Section>

        <Section icon="sync-outline" title="Sync across phones">
          <T variant="small" color={colors.textSecondary}>
            Friends on different phones see the same plan, chat, votes and budget, live.
          </T>
          {SYNC.map(([icon, title, body]) => (
            <View key={title} style={styles.row}>
              <Ionicons name={icon} size={20} color={colors.primary} style={{ marginTop: 1 }} />
              <View style={{ flex: 1 }}>
                <T variant="small" weight="bold">
                  {title}
                </T>
                <T variant="caption" color={colors.textSecondary}>
                  {body}
                </T>
              </View>
            </View>
          ))}
        </Section>

        <Section icon="shield-checkmark-outline" title="Data & security">
          {SECURITY.map(([icon, title, body]) => (
            <View key={title} style={styles.row}>
              <Ionicons name={icon} size={20} color={colors.primary} style={{ marginTop: 1 }} />
              <View style={{ flex: 1 }}>
                <T variant="small" weight="bold">
                  {title}
                </T>
                <T variant="caption" color={colors.textSecondary}>
                  {body}
                </T>
              </View>
            </View>
          ))}
        </Section>

        <Section icon="checkmark-done-outline" title="Input validation">
          <T variant="small" color={colors.textSecondary}>
            Every form checks the format and whether the value makes sense, then shows a plain message next to the field.
          </T>
          {VALIDATION.map(([field, rule, message], i) => (
            <View key={field}>
              {i ? <Divider style={{ marginBottom: 10 }} /> : null}
              <T variant="small" weight="bold">
                {field}
              </T>
              <T variant="caption" color={colors.textSecondary}>
                {rule}
              </T>
              <View style={styles.msg}>
                <Ionicons name="alert-circle" size={13} color={colors.red} />
                <T variant="caption" color={colors.red} style={{ flex: 1 }}>
```

```
                {message}
              </T>
            </View>
          </View>
        ))}
      </Section>

      <Section icon="ribbon-outline" title="Rubric map">
        <T variant="small" color={colors.textSecondary}>
          Each competition area and where to find it. Tap a row to open it.
        </T>
        {RUBRIC.map((r, i) => (
          <View key={r.area}>
            {i ? <Divider style={{ marginBottom: 10 }} /> : null}
            <Press onPress={() => router.push(r.href)} style={styles.row} accessibilityLabel={`${r.area}. ${r.where}`}>
              <Ionicons name={r.icon} size={20} color={colors.primary} style={{ marginTop: 1 }} />
              <View style={{ flex: 1 }}>
                <T variant="small" weight="bold">
                  {r.area}
                </T>
                <T variant="caption" color={colors.textSecondary}>
                  {r.where}
                </T>
              </View>
              <Ionicons name="chevron-forward" size={16} color={colors.textMuted} style={{ marginTop: 2 }} />
            </Press>
          </View>
        ))}
      </Section>

      <Section icon="cube-outline" title="Libraries used">
        <View style={styles.tags}>
          {LIBRARIES.map((l) => (
            <View key={l} style={styles.tag}>
              <T variant="caption" weight="semibold">
                {l}
              </T>
            </View>
          ))}
        </View>
      </Section>
    </View>

    <Button
      label="Replay the app tour"
      icon="play-circle-outline"
      style={{ marginTop: 18 }}
      onPress={() => {
        setToured(false);
        router.navigate('/(tabs)/explore');
      }}
    />
  </Screen>
 );
}

const styles = StyleSheet.create({
  card: { borderRadius: radius.xl, gap: 12 },
  sectionHead: { flexDirection: 'row', alignItems: 'center', gap: 10 },
  sectionIcon: { width: 32, height: 32, borderRadius: 16, backgroundColor: colors.accent, alignItems: 'center', justifyContent: 'center'
},
  layer: { flexDirection: 'row', alignItems: 'center', gap: 12, padding: 12, borderRadius: radius.md, backgroundColor: colors.cardMuted,
borderWidth: 1, borderColor: colors.border },
  layerIcon: { width: 36, height: 36, borderRadius: 18, backgroundColor: colors.card, alignItems: 'center', justifyContent: 'center', bo
rderWidth: 1, borderColor: colors.border },
  row: { flexDirection: 'row', alignItems: 'flex-start', gap: 12 },
  msg: { flexDirection: 'row', alignItems: 'center', gap: 5, marginTop: 5, paddingHorizontal: 10, paddingVertical: 5, borderRadius: radi
us.sm, backgroundColor: colors.redSoft, alignSelf: 'flex-start' },
  tags: { flexDirection: 'row', flexWrap: 'wrap', gap: 6 },
  tag: { paddingHorizontal: 11, paddingVertical: 6, borderRadius: radius.pill, backgroundColor: colors.bgAlt },
});
```

src/app/index.tsx

```
import { Redirect } from 'expo-router';

import { useAppStore } from '@/store/useAppStore';

/** Entry point: routes to onboarding, sign-in or the app depending on state. */
export default function Index() {
  const signedIn = useAppStore((s) => !!s.session);
  const hasOnboarded = useAppStore((s) => s.hasOnboarded);
  if (signedIn) return <Redirect href="/(tabs)/explore" />;
  return <Redirect href={hasOnboarded ? '/sign-in' : '/onboarding'} />;
}
```

src/app/join.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { router, useLocalSearchParams } from 'expo-router';
import { useState } from 'react';
import { Platform, StyleSheet, TextInput, View } from 'react-native';

import { QrScanner } from '@/components/join/QrScanner';
import { Button } from '@/components/ui/Button';
import { Card } from '@/components/ui/bits';
import { confirmAction, toast } from '@/components/ui/feedback';
import { Press, tap } from '@/components/ui/Press';
import { Header, Screen } from '@/components/ui/Screen';
import { T } from '@/components/ui/T';
import { joinTripAnywhere } from '@/sync/actions';
import { parseInviteUrl } from '@/sync/invite';
import { effectiveServerUrl, useSyncPrefs } from '@/sync/prefs';
import { normalizeServerUrl } from '@/sync/transport';
import { colors, fonts, radius } from '@/theme';
import { normalizeInvite, validateInvite } from '@/utils/validation';

/** Join a trip with an invite code (also opened by togetherwego://join?code=XXX-1234 links and by scanning the invite QR code). */
export default function Join() {
  const params = useLocalSearchParams<{ code?: string; server?: string }>();
  const [code, setCode] = useState(params.code ? normalizeInvite(String(params.code)) : '');
  // The sync server a friend's invite names. It is only used after this person agrees to it.
  const [inviteServer, setInviteServer] = useState<string | null>(params.server ? normalizeServerUrl(String(params.server)) : null);
  const [scanning, setScanning] = useState(false);
  const [error, setError] = useState<string | null>(null);
  const [busy, setBusy] = useState(false);
  useSyncPrefs((s) => s.serverUrl); // re-check when the address changes
  const newServer = inviteServer && inviteServer !== effectiveServerUrl() ? inviteServer : null;

  const join = async () => {
    setBusy(true);
    const r = await joinTripAnywhere(code);
    setBusy(false);
    if (!r.ok) {
      setError(r.error);
      return tap('warning');
    }
    tap('success');
    toast(r.synced ? 'You’re in! 🎉 This trip now syncs live with your friends' : r.offlineCopy ? 'Joined the copy on this phone. The sy
nc server was not reachable.' : 'You’re in! 🎉 Say hi in the trip chat');
    router.dismissAll?.();
    router.replace('/(tabs)/chat');
  };

  const submit = () => {
    if (busy) return;
    const e = validateInvite(code);
    if (e) {
      setError(e);
      return tap('warning');
    }
    if (!newServer) return void join();
    confirmAction(
      'Use your friend’s sync server?',
      `This invite shares the trip through ${newServer.replace(/^https?:\/\//, '')}. Its chat, plans and photos are kept on that compute
r, so only go on if you know who runs it.`,
      () => {
        useSyncPrefs.getState().setServerUrl(newServer);
        void join();
      },
      { confirmLabel: 'Use it and join' },
    );
  };

  const scanned = (text: string) => {
    setScanning(false);
    const found = parseInviteUrl(text);
    if (!found.code) return toast('That QR code is not a TogetherWeGo invite', { tone: 'warn', icon: 'qr-code-outline' });
    setCode(found.code);
    setError(null);
    if (found.server) setInviteServer(found.server);
    tap('success');
  };

  return (
    <Screen header={<Header title="Join a trip" subtitle="Enter the code your friend shared" />}>
      <View style={styles.hero}>
        <View style={styles.heroIcon}>
          <Ionicons name="people" size={30} color={colors.primary} />
        </View>
        <T variant="h2" center>
          Got an invite code?
```

```
        </T>
        <T variant="small" color={colors.textSecondary} center>
          Codes look like <T variant="small" weight="bold">AML-2026</T>. Ask the organizer to share it from More → Crew & roles.
        </T>
      </View>
      <TextInput
        value={code}
        onChangeText={(v) => {
          setCode(normalizeInvite(v));
          setError(null);
        }}
        placeholder="ABC-1234"
        placeholderTextColor={colors.borderStrong}
        autoCapitalize="characters"
        autoCorrect={false}
        maxLength={8}
        style={[styles.code, error && { borderColor: colors.red }]}
        accessibilityLabel="Invite code"
        onSubmitEditing={submit}
        returnKeyType="go"
        testID="invite-code"
      />
      {error ? (
        <T variant="small" color={colors.red} center style={{ marginTop: 8 }} accessibilityLiveRegion="assertive">
          {error}
        </T>
      ) : null}
      {newServer ? (
        <T variant="caption" color={colors.textSecondary} center style={{ marginTop: 8 }}>
          This invite uses a sync server at {newServer.replace(/^https?:\/\//, '')}. You will be asked before it is used.
        </T>
      ) : null}
      <Button label="Join trip" icon="enter-outline" onPress={submit} loading={busy} style={{ marginTop: 18 }} testID="join-submit" />
      {Platform.OS !== 'web' ? <Button label="Scan a QR code" icon="qr-code-outline" variant="secondary" onPress={() => setScanning(tru
e)} style={{ marginTop: 10 }} testID="join-scan" /> : null}
      {Platform.OS !== 'web' ? <QrScanner visible={scanning} onClose={() => setScanning(false)} onScan={scanned} /> : null}
      <Card style={{ marginTop: 18, gap: 8 }}>
        <T variant="small" weight="semibold">
          Try it with a demo trip
        </T>
        <T variant="caption" color={colors.textSecondary}>
          “Seoul Food Crawl” is waiting for new members. Tap to fill its code:
        </T>
        <Press onPress={() => setCode('SEO-5521')} style={styles.demo} accessibilityLabel="Use demo code SEO-5521">
          <T variant="title" weight="bold" color={colors.primary} style={{ letterSpacing: 2 }}>
            SEO-5521
          </T>
          <Ionicons name="arrow-forward-circle" size={22} color={colors.primary} />
        </Press>
      </Card>
    </Screen>
  );
}

const styles = StyleSheet.create({
  hero: { alignItems: 'center', gap: 8, marginVertical: 18 },
  heroIcon: { width: 70, height: 70, borderRadius: 35, backgroundColor: colors.primarySoft, alignItems: 'center', justifyContent: 'cente
r', marginBottom: 6 },
  code: { height: 72, borderRadius: radius.lg, borderWidth: 2, borderColor: colors.primaryLine, backgroundColor: colors.card, textAlign:
'center', fontFamily: fonts.bold, fontSize: 30, letterSpacing: 6, color: colors.text },
  demo: { flexDirection: 'row', alignItems: 'center', justifyContent: 'space-between', padding: 12, borderRadius: radius.md, backgroundC
olor: colors.primarySofter },
});
```

src/app/map.tsx

```
import { Ionicons } from '@expo/vector-icons';
import * as Linking from 'expo-linking';
import { router } from 'expo-router';
import { useMemo, useState } from 'react';
import { Platform, ScrollView, StyleSheet, View } from 'react-native';
import { useSafeAreaInsets } from 'react-native-safe-area-context';

import { OfflineRadar } from '@/components/map/OfflineRadar';
import { TripMap } from '@/components/map/TripMap';
import type { MapPoint } from '@/components/map/types';
import { typeMeta } from '@/components/trip/meta';
import { NoTrip } from '@/components/trip/NoTrip';
import { Chip, IconButton } from '@/components/ui/bits';
import { toast } from '@/components/ui/feedback';
import { Press } from '@/components/ui/Press';
import { Slider } from '@/components/ui/Slider';
import { T } from '@/components/ui/T';
import { destinationById } from '@/data/destinations';
import { placesFor } from '@/data/places';
import { getLocation } from '@/services/location';
```

```
import { useActiveTrip } from '@/store/useAppStore';
import { MAX_WIDTH, colors, radius, shadowStrong } from '@/theme';
import { durationLabel } from '@/utils/format';
import { distanceKm, kmLabel, nearestAirports } from '@/utils/geo';
import { dayLabel, time12, tripDays } from '@/utils/time';

export default function MapScreen() {
  const trip = useActiveTrip();
  const insets = useSafeAreaInsets();
  const [mode, setMode] = useState<'live' | 'offline'>('live');
  const [day, setDay] = useState<string>('all');
  const [layers, setLayers] = useState({ plan: true, places: true, airports: true });
  const [radiusKm, setRadiusKm] = useState(10);
  const [selected, setSelected] = useState<string | null>(null);
  const [me, setMe] = useState<{ lat: number; lng: number } | null>(null);
  const dest = destinationById(trip?.destinationId ?? 'tokyo')!;
  const [center, setCenter] = useState({ lat: dest.lat, lng: dest.lng });

  const points = useMemo<MapPoint[]>(() => {
    if (!trip) return [];
    const out: MapPoint[] = [];
    if (layers.plan)
      trip.activities
        .filter((a) => a.lat && a.lng && (day === 'all' || a.date === day))
        .forEach((a) => out.push({ id: a.id, lat: a.lat!, lng: a.lng!, title: a.title, subtitle: `${dayLabel(a.date)} · ${time12(a.tim
e)} · ${durationLabel(a.durationMin)}`, color: typeMeta(a.type).color, kind: 'activity' }));
    if (layers.places) {
      const planned = new Set(trip.activities.map((a) => a.placeId));
      [trip.destinationId, ...trip.extraDestinationIds]
        .flatMap((id) => placesFor(id))
        .filter((p) => !planned.has(p.id))
        .forEach((p) => out.push({ id: p.id, lat: p.lat, lng: p.lng, title: p.name, subtitle: `${p.category} · ★ ${p.rating}`, color: '#
9AA390', kind: 'place' }));
    }
    if (layers.airports) nearestAirports(center, 3).forEach((a) => out.push({ id: `air-${a.iata}`, lat: a.lat, lng: a.lng, title: `${a.i
ata} · ${a.name}`, subtitle: `${kmLabel(a.km)} away`, color: colors.blue, kind: 'airport' }));
    if (me) out.push({ id: 'me', lat: me.lat, lng: me.lng, title: 'You are here', subtitle: '', color: colors.red, kind: 'me' });
    return out;
  }, [trip, layers, day, center, me]);

  if (!trip) return <NoTrip />;
  const sel = points.find((p) => p.id === selected);
  const days = tripDays(trip.startDate, trip.endDate);

  const locate = async () => {
    const loc = await getLocation(dest);
    if (loc.source !== 'device') return toast('Location unavailable — showing the trip centre', { tone: 'warn' });
    setMe(loc);
    if (distanceKm(loc, dest) < 80) setCenter(loc);
    else toast(`You're ${kmLabel(distanceKm(loc, dest))} from ${dest.city}`, { icon: 'navigate-outline' });
  };

  return (
    <View style={styles.root}>
      <View style={styles.mapWrap}>
        {mode === 'live' ? (
          <TripMap center={center} points={points} radiusKm={radiusKm} selectedId={selected} onSelect={setSelected} showsUser={!!me} />
        ) : (
          <OfflineRadar center={center} points={points} radiusKm={radiusKm} selectedId={selected} onSelect={setSelected} />
        )}
      </View>

      <View style={[styles.top, { paddingTop: insets.top + 8 }]}>
        <View style={styles.topRow}>
          <IconButton icon="chevron-back" label="Back" onPress={() => (router.canGoBack() ? router.back() : router.replace('/(tabs)/expl
ore'))} bordered={false} />
          <View style={styles.seg}>
            {(['live', 'offline'] as const).map((m) => (
              <Press key={m} onPress={() => setMode(m)} style={[styles.segBtn, mode === m && styles.segOn]} accessibilityState={{ select
ed: mode === m }}>
                <Ionicons name={m === 'live' ? 'map' : 'cloud-offline-outline'} size={13} color={mode === m ? colors.white : colors.text
Secondary} />
                <T variant="caption" weight="semibold" color={mode === m ? colors.white : colors.textSecondary}>
                  {m === 'live' ? 'Map' : 'Offline'}
                </T>
              </Press>
            ))}
          </View>
          <IconButton icon="locate-outline" label="Show my location" onPress={locate} bordered={false} />
        </View>
        <ScrollView horizontal showsHorizontalScrollIndicator={false} style={styles.chips} contentContainerStyle={{ gap: 6, paddingHoriz
ontal: 16 }}>
          <Chip small label="All days" active={day === 'all'} onPress={() => setDay('all')} />
          {days.map((d, i) => (
            <Chip key={d} small label={`Day ${i + 1}`} active={day === d} onPress={() => setDay(d)} />
          ))}
```

```
        </ScrollView>
        <ScrollView horizontal showsHorizontalScrollIndicator={false} style={styles.chips} contentContainerStyle={{ gap: 6, paddingHoriz
ontal: 16 }}>
          <Chip small icon="calendar-outline" label="Itinerary" active={layers.plan} onPress={() => setLayers((l) => ({ ...l, plan: !l.p
lan }))} />
          <Chip small icon="compass-outline" label="Places" active={layers.places} onPress={() => setLayers((l) => ({ ...l, places: !l.p
laces }))} />
          <Chip small icon="airplane-outline" label="Airports" active={layers.airports} onPress={() => setLayers((l) => ({ ...l, airport
s: !l.airports }))} />
        </ScrollView>
      </View>

      <View style={[styles.bottom, { paddingBottom: insets.bottom + 12 }]}>
        {sel ? (
          <View style={styles.selCard}>
            <View style={[styles.selDot, { backgroundColor: sel.color }]} />
            <View style={{ flex: 1 }}>
              <T variant="small" weight="semibold" numberOfLines={1}>
                {sel.title}
              </T>
              <T variant="caption" color={colors.textSecondary} numberOfLines={1}>
                {sel.subtitle} · {kmLabel(distanceKm(center, sel))} from centre
              </T>
            </View>
            <Press onPress={() => Linking.openURL(`https://www.google.com/maps/dir/?api=1&destination=${sel.lat},${sel.lng}`)} style={st
yles.go} accessibilityLabel="Get directions">
              <Ionicons name="navigate" size={16} color={colors.white} />
            </Press>
          </View>
        ) : (
          <T variant="caption" color={colors.textSecondary}>
            {points.length} pins · tap one for details{mode === 'offline' ? ' · works with no internet' : Platform.OS === 'web' ? ' · we
b preview uses OpenStreetMap' : ''}
          </T>
        )}
        <View style={styles.radiusRow}>
          <T variant="caption" weight="semibold">
            Radius {radiusKm} km
          </T>
          <View style={{ flex: 1 }}>
            <Slider value={radiusKm} min={2} max={40} onChange={setRadiusKm} label="Map radius" formatValue={(v) => `${v} km`} />
          </View>
        </View>
        {mode === 'live' && Platform.OS === 'web' && points.length ? (
          <ScrollView horizontal showsHorizontalScrollIndicator={false} contentContainerStyle={{ gap: 6 }}>
            {points.filter((p) => p.kind !== 'place').slice(0, 12).map((p) => (
              <Chip key={p.id} small label={p.title.slice(0, 22)} active={p.id === selected} onPress={() => setSelected(p.id)} />
            ))}
          </ScrollView>
        ) : null}
      </View>
    </View>
  );
}

const styles = StyleSheet.create({
  root: { flex: 1, backgroundColor: colors.bg },
  mapWrap: { ...StyleSheet.absoluteFill },
  top: { position: 'absolute', left: 0, right: 0, top: 0, gap: 8, alignItems: 'center' },
  // a centred column would size a row of chips to its content (wider than the screen); keep it as wide as the screen
  chips: { width: '100%', maxWidth: MAX_WIDTH, flexGrow: 0 },
  topRow: { flexDirection: 'row', alignItems: 'center', justifyContent: 'space-between', width: '100%', maxWidth: MAX_WIDTH, paddingHori
zontal: 16 },
  seg: { flexDirection: 'row', backgroundColor: colors.card, borderRadius: 999, padding: 3, ...shadowStrong },
  segBtn: { flexDirection: 'row', alignItems: 'center', gap: 4, paddingHorizontal: 12, paddingVertical: 7, borderRadius: 999 },
  segOn: { backgroundColor: colors.primary },
  bottom: { position: 'absolute', left: 12, right: 12, bottom: 0, backgroundColor: colors.card, borderTopLeftRadius: radius.xl, borderTo
pRightRadius: radius.xl, padding: 16, gap: 8, alignSelf: 'center', maxWidth: MAX_WIDTH, ...shadowStrong },
  selCard: { flexDirection: 'row', alignItems: 'center', gap: 10 },
  selDot: { width: 12, height: 12, borderRadius: 6 },
  go: { width: 38, height: 38, borderRadius: 19, backgroundColor: colors.primary, alignItems: 'center', justifyContent: 'center' },
  radiusRow: { flexDirection: 'row', alignItems: 'center', gap: 12 },
});
```

src/app/nearby.tsx

```
import { Ionicons } from '@expo/vector-icons';
import * as Linking from 'expo-linking';
import { useLocalSearchParams } from 'expo-router';
import { useMemo, useState } from 'react';
import { ScrollView, StyleSheet, View } from 'react-native';

import { AddActivitySheet, type ActivityDraft } from '@/components/trip/AddActivitySheet';
import { placeCategoryIcon, placeToActivityType } from '@/components/trip/meta';
import { Button } from '@/components/ui/Button';
import { Card, Chip, EmptyState, Pill } from '@/components/ui/bits';
```

```
import { toast } from '@/components/ui/feedback';
import { Press } from '@/components/ui/Press';
import { Header, Screen } from '@/components/ui/Screen';
import { Slider } from '@/components/ui/Slider';
import { T } from '@/components/ui/T';
import { destinationById } from '@/data/destinations';
import { placesFor, type PlaceCategory } from '@/data/places';
import { useT } from '@/i18n';
import { getLocation } from '@/services/location';
import { useActiveTrip } from '@/store/useAppStore';
import { colors, radius } from '@/theme';
import { LENGTH_FILTERS, durationLabel, lengthBucketLabel, matchesLength, money, priceTier, type LengthFilter } from '@/utils/format';
import { bearingDeg, compassPoint, distanceKm, kmLabel, type LatLng } from '@/utils/geo';
import { todayInTz, tripDays } from '@/utils/time';

const CATS: ('all' | PlaceCategory)[] = ['all', 'food', 'sightseeing', 'adventure', 'culture', 'nightlife', 'shopping', 'relax'];
type Sort = 'distance' | 'rating' | 'price';

export default function Nearby() {
  const t = useT();
  const params = useLocalSearchParams<{ focus?: string; maxCost?: string }>();
  const trip = useActiveTrip();
  const dest = destinationById(trip?.destinationId ?? 'tokyo')!;
  const [center, setCenter] = useState<LatLng & { label: string }>({ lat: dest.lat, lng: dest.lng, label: `Central ${dest.city}` });
  const [radiusKm, setRadiusKm] = useState(params.focus ? 25 : 6);
  const [cat, setCat] = useState<'all' | PlaceCategory>('all');
  const [len, setLen] = useState<LengthFilter>('any');
  const [maxCost, setMaxCost] = useState(params.maxCost ? Math.min(100, Number(params.maxCost)) : 100);
  const [sort, setSort] = useState<Sort>(params.focus ? 'distance' : 'rating');
  const [locating, setLocating] = useState(false);
  const [draft, setDraft] = useState<ActivityDraft | null>(null);

  const all = useMemo(() => (trip ? [trip.destinationId, ...trip.extraDestinationIds] : ['tokyo']).flatMap((id) => placesFor(id)), [tri
p]);
  const results = useMemo(
    () =>
      all
        .map((p) => ({ p, km: distanceKm(center, p) }))
        .filter(({ p, km }) => km <= radiusKm && (cat === 'all' || p.category === cat) && matchesLength(p.durationMin, len) && (maxCost
>= 100 || p.cost <= maxCost))
        .sort((a, b) => (sort === 'distance' ? a.km - b.km : sort === 'rating' ? b.p.rating - a.p.rating : a.p.cost - b.p.cost)),
    [all, center, radiusKm, cat, len, maxCost, sort],
  );
  const planned = new Set(trip?.activities.map((a) => a.placeId).filter(Boolean));

  const useMyLocation = async () => {
    setLocating(true);
    const loc = await getLocation(dest);
    setLocating(false);
    if (loc.source === 'device') {
      const far = distanceKm(loc, dest) > 60;
      setCenter({ lat: loc.lat, lng: loc.lng, label: 'My location' });
      if (far) {
        setRadiusKm(25);
        toast(`You're ${kmLabel(distanceKm(loc, dest))} from ${dest.city} — results will appear once you're there`, { tone: 'info', ico
n: 'navigate-outline' });
      } else toast('Using your current location');
    } else toast('Location unavailable — using the trip centre', { tone: 'warn' });
  };

  const today = trip ? todayInTz(dest.tz) : '';
  const defaultDay = trip ? (tripDays(trip.startDate, trip.endDate).includes(today) ? today : trip.startDate) : '';

  return (
    <Screen header={<Header title="Nearby" subtitle={`${results.length} places · ${center.label}`} />}>
      <Card style={{ gap: 10 }}>
        <View style={styles.between}>
          <T variant="title">Search radius</T>
          <T variant="title" color={colors.primary}>
            {radiusKm} km
          </T>
        </View>
        <Slider value={radiusKm} min={1} max={25} onChange={setRadiusKm} label="Search radius" formatValue={(v) => `${v} km`} />
        <View style={{ flexDirection: 'row', gap: 8 }}>
          <Chip small icon="business-outline" label={`${dest.city} centre`} active={center.label !== 'My location'} onPress={() => setCe
nter({ lat: dest.lat, lng: dest.lng, label: `Central ${dest.city}` })} />
          <Chip small icon="locate-outline" label={locating ? 'Locating…' : 'My location'} active={center.label === 'My location'} onPre
ss={useMyLocation} />
        </View>
        <View style={styles.between}>
          <T variant="small" weight="semibold">
            Max price per person
          </T>
          <T variant="small" weight="semibold" color={colors.primary}>
            {maxCost >= 100 ? 'Any' : maxCost === 0 ? 'Free only' : money(maxCost)}
          </T>
```

```
        </View>
        <Slider value={maxCost} min={0} max={100} step={5} onChange={setMaxCost} label="Maximum price per person" formatValue={(v) => (v
>= 100 ? 'Any' : `$${v}`)} />
      </Card>

      <ScrollView horizontal showsHorizontalScrollIndicator={false} contentContainerStyle={{ gap: 6, paddingHorizontal: 20 }} style={{ m
arginHorizontal: -20, marginTop: 14 }}>
        {CATS.map((c) => (
          <Chip key={c} small icon={c === 'all' ? undefined : placeCategoryIcon[c]} label={t(`cat.${c === 'all' ? 'all' : c}`)} active=
{cat === c} onPress={() => setCat(c)} />
        ))}
      </ScrollView>
      <ScrollView horizontal showsHorizontalScrollIndicator={false} contentContainerStyle={{ gap: 6, paddingHorizontal: 20 }} style={{ m
arginHorizontal: -20, marginTop: 8 }}>
        {LENGTH_FILTERS.map((f) => (
          <Chip key={f.key} small tone="soft" label={t(f.labelKey)} active={len === f.key} onPress={() => setLen(f.key)} />
        ))}
      </ScrollView>
      <View style={styles.sortRow}>
        <T variant="caption" color={colors.textSecondary}>
          Sort
        </T>
        {(['distance', 'rating', 'price'] as Sort[]).map((s) => (
          <Chip key={s} small label={s === 'distance' ? 'Closest' : s === 'rating' ? 'Top rated' : 'Cheapest'} active={sort === s} onPre
ss={() => setSort(s)} />
        ))}
      </View>

      {results.length ? (
        <View style={{ gap: 10 }}>
          {results.map(({ p, km }) => {
            const focus = params.focus === p.id;
            return (
              <Card key={p.id} style={[{ gap: 8 }, focus && { borderColor: colors.primary, borderWidth: 2 }]}>
                <View style={{ flexDirection: 'row', gap: 12 }}>
                  <View style={styles.icon}>
                    <Ionicons name={placeCategoryIcon[p.category]} size={18} color={colors.primary} />
                  </View>
                  <View style={{ flex: 1 }}>
                    <T variant="bodySm" weight="semibold">
                      {p.name}
                    </T>
                    <T variant="caption" color={colors.textSecondary}>
                      {p.area} · ★ {p.rating}
                    </T>
                  </View>
                  <View style={{ alignItems: 'flex-end' }}>
                    <T variant="small" weight="bold">
                      {kmLabel(km)}
                    </T>
                    <T variant="micro" color={colors.textMuted}>
                      {compassPoint(bearingDeg(center, p))}
                    </T>
                  </View>
                </View>
                <T variant="caption" color={colors.textSecondary}>
                  {p.blurb}
                </T>
                <View style={{ flexDirection: 'row', gap: 6, flexWrap: 'wrap' }}>
                  <Pill label={t(`cat.${p.category}`)} />
                  <Pill label={`${lengthBucketLabel(p.durationMin)} · ${durationLabel(p.durationMin)}`} tone="blue" icon="time-outline"
/>
                  <Pill label={p.cost ? `${priceTier(p.cost)} · ~${money(p.cost)}` : 'Free'} tone="yellow" />
                  {planned.has(p.id) ? <Pill label="On itinerary" tone="purple" icon="checkmark" /> : null}
                </View>
                <View style={{ flexDirection: 'row', gap: 8 }}>
                  {trip ? (
                    <Button
                      label="Add to plan"
                      size="sm"
                      icon="add"
                      style={{ flex: 1 }}
                      onPress={() =>
                        setDraft({ title: p.name, location: `${p.area}, ${destinationById(p.destinationId)?.city ?? ''}`, type: placeToA
ctivityType[p.category], durationMin: p.durationMin, costPerPerson: p.cost, lat: p.lat, lng: p.lng, placeId: p.id, notes: p.blurb })
                      }
                    />
                  ) : null}
                  <Press
                    onPress={() => Linking.openURL(`https://www.google.com/maps/search/?api=1&query=${p.lat},${p.lng}`)}
                    style={styles.mapBtn}
                    accessibilityLabel={`Directions to ${p.name}`}>
                    <Ionicons name="navigate-outline" size={16} color={colors.primary} />
                    <T variant="small" weight="semibold" color={colors.primary}>
                      Directions
                    </T>
```

```
                </Press>
              </View>
            </Card>
          );
        })}
      </View>
    ) : (
      <EmptyState icon="compass-outline" title="Nothing matches yet" body="Widen the radius, raise the max price or pick another categ
ory." />
    )}

    {trip ? <AddActivitySheet visible={!!draft} onClose={() => setDraft(null)} trip={trip} date={defaultDay} draft={draft} /> : null}
  </Screen>
 );
}

const styles = StyleSheet.create({
 between: { flexDirection: 'row', justifyContent: 'space-between', alignItems: 'center' },
 sortRow: { flexDirection: 'row', alignItems: 'center', gap: 6, marginVertical: 14 },
 icon: { width: 40, height: 40, borderRadius: 12, backgroundColor: colors.primarySofter, alignItems: 'center', justifyContent: 'center'
},
 mapBtn: { flexDirection: 'row', alignItems: 'center', gap: 6, paddingHorizontal: 14, height: 36, borderRadius: radius.md, borderWidth:
1, borderColor: colors.primaryLine },
});
```

src/app/new-trip.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { Image } from 'expo-image';
import { router, useLocalSearchParams } from 'expo-router';
import { useMemo, useState } from 'react';
import { ScrollView, StyleSheet, View } from 'react-native';

import { DatePickerSheet } from '@/components/pickers';
import { Button } from '@/components/ui/Button';
import { Card } from '@/components/ui/bits';
import { toast } from '@/components/ui/feedback';
import { Input } from '@/components/ui/Input';
import { Press, tap } from '@/components/ui/Press';
import { Header, Screen } from '@/components/ui/Screen';
import { T } from '@/components/ui/T';
import { destinationById, destinations } from '@/data/destinations';
import { imageSource } from '@/data/images';
import { useAppStore } from '@/store/useAppStore';
import { colors, radius } from '@/theme';
import { money } from '@/utils/format';
import { addDays, dayLabel, daysBetween, todayInTz } from '@/utils/time';
import { firstError, maxLen, parseAmount, required, validateAmount, validateTripDates } from '@/utils/validation';

/** Create a trip (or edit one with ?edit=<tripId>). */
export default function NewTrip() {
 const params = useLocalSearchParams<{ edit?: string; destination?: string }>();
 const editing = useAppStore((s) => (params.edit ? s.trips.find((t) => t.id === params.edit) : undefined));
 const createTrip = useAppStore((s) => s.createTrip);
 const updateTrip = useAppStore((s) => s.updateTrip);
 const today = todayInTz('UTC');

 const initialDest = editing?.destinationId ?? params.destination ?? 'tokyo';
 const [destId, setDestId] = useState(initialDest);
 const [name, setName] = useState(editing?.name ?? '');
 const [start, setStart] = useState(editing?.startDate ?? addDays(today, 30));
 const [end, setEnd] = useState(editing?.endDate ?? addDays(today, 36));
 const [budget, setBudget] = useState(editing ? String(editing.budget) : '');
 const [q, setQ] = useState('');
 const [picker, setPicker] = useState<'start' | 'end' | null>(null);
 const [errors, setErrors] = useState<Record<string, string | null | undefined>>({});

 const dest = destinationById(destId)!;
 const days = Math.max(1, daysBetween(start, end) + 1);
 const suggested = Math.round(dest.dailyCost * days * 4 + 900 * 4);
 const filtered = useMemo(() => destinations.filter((d) => !q || `${d.city} ${d.country}`.toLowerCase().includes(q.toLowerCase())),
[q]);

 const save = () => {
   const dateErr = validateTripDates(start, end);
   const e = {
     name: firstError(name, required('a trip name'), maxLen(40, 'Trip name')),
     budget: validateAmount(budget, { min: 50, max: 1_000_000, label: 'a group budget' }),
     start: dateErr.start ?? (!editing && daysBetween(today, start) < -1 ? 'Start date can’t be in the past' : null),
     end: dateErr.end,
   };
   setErrors(e);
   if (Object.values(e).some(Boolean)) return tap('warning');
   const b = parseAmount(budget)!;
   if (editing) {
     updateTrip(editing.id, { name: name.trim(), startDate: start, endDate: end, budget: b, destinationId: destId, cover: destId !== ed
```

```
iting.destinationId ? dest.image : editing.cover });
      toast('Trip updated');
      router.back();
    } else {
      createTrip({ name, destinationId: destId, startDate: start, endDate: end, budget: b });
      tap('success');
      toast('Trip created 🎉 — share the invite code with your group');
      router.dismissAll?.();
      router.replace('/crew');
    }
  };

  return (
    <Screen header={<Header title={editing ? 'Edit trip' : 'New trip'} subtitle={editing ? editing.name : 'Where is the group headed?'}
/>}>
      <View style={{ gap: 16 }}>
        <View style={{ gap: 8 }}>
          <T variant="small" weight="semibold">
            Destination
          </T>
          <Input placeholder="Search destinations…" icon="search" value={q} onChangeText={setQ} />
          <ScrollView horizontal showsHorizontalScrollIndicator={false} contentContainerStyle={{ gap: 10, paddingHorizontal: 20 }} style
={{ marginHorizontal: -20 }}>
            {filtered.map((d) => {
              const on = d.id === destId;
              return (
                <Press key={d.id} onPress={() => setDestId(d.id)} style={[styles.dest, on && styles.destOn]} accessibilityState={{ selec
ted: on }} accessibilityLabel={`${d.city}, ${d.country}`}>
                  <Image source={imageSource(d.image)} style={styles.destImg} contentFit="cover" />
                  {on ? (
                    <View style={styles.check}>
                      <Ionicons name="checkmark" size={14} color={colors.white} />
                    </View>
                  ) : null}
                  <T variant="caption" weight="semibold" numberOfLines={1} style={{ marginTop: 6 }}>
                    {d.city}
                  </T>
                  <T variant="micro" color={colors.textMuted} numberOfLines={1}>
                    {d.country}
                  </T>
                </Press>
              );
            })}
          </ScrollView>
        </View>

        <Input label="Trip name" placeholder={`e.g. ${dest.city} with the crew`} value={name} onChangeText={setName} error={errors.name}
maxLength={45} icon="create-outline" />

        <View style={{ flexDirection: 'row', gap: 10 }}>
          <Input label="Start date" icon="calendar-outline" value={dayLabel(start) + `, ${start.slice(0, 4)}`} onPressField={() => setPi
cker('start')} error={errors.start} containerStyle={{ flex: 1 }} />
          <Input label="End date" icon="calendar-outline" value={dayLabel(end) + `, ${end.slice(0, 4)}`} onPressField={() => setPicker
('end')} error={errors.end} containerStyle={{ flex: 1 }} />
        </View>
        <T variant="caption" color={colors.textSecondary}>
          {days} day{days === 1 ? '' : 's'} · local time in {dest.city} ({dest.tz.split('/').pop()?.replace('_', ' ')})
        </T>

        <Input label="Group budget (USD)" placeholder={String(suggested)} value={budget} onChangeText={setBudget} keyboardType="decimal-
pad" error={errors.budget} icon="wallet-outline" hint={`Tip: ~${money(dest.dailyCost)}/person/day in ${dest.city}. For 4 people incl. fl
ights ≈ ${money(suggested)}.`} />
        {!budget ? <Button label={`Use suggested ${money(suggested)}`} variant="soft" size="sm" onPress={() => setBudget(String(suggeste
d))} /> : null}

        {!editing ? (
          <Card style={{ flexDirection: 'row', gap: 10, alignItems: 'center' }}>
            <Ionicons name="key-outline" size={18} color={colors.primary} />
            <T variant="caption" color={colors.textSecondary} style={{ flex: 1 }}>
              We'll generate an invite code like <T variant="caption" weight="bold">{dest.city.slice(0, 3).toUpperCase()}-1234</T> so fr
iends can join instantly.
            </T>
          </Card>
        ) : null}

        <Button label={editing ? 'Save changes' : 'Create trip'} icon={editing ? 'checkmark' : 'airplane'} onPress={save} />
      </View>

      <DatePickerSheet visible={picker === 'start'} value={start} title="Start date" onClose={() => setPicker(null)} onChange={(v) => {
        setStart(v);
        if (daysBetween(v, end) < 0) setEnd(addDays(v, 5));
      }} />
      <DatePickerSheet visible={picker === 'end'} value={end} min={start} max={addDays(start, 60)} title="End date" onClose={() => setPi
cker(null)} onChange={setEnd} />
    </Screen>
  );
```

```
}

const styles = StyleSheet.create({
  dest: { width: 104, padding: 6, borderRadius: radius.md, borderWidth: 2, borderColor: 'transparent' },
  destOn: { borderColor: colors.primary, backgroundColor: colors.primarySofter },
  destImg: { width: '100%', height: 76, borderRadius: radius.sm },
  check: { position: 'absolute', top: 10, right: 10, width: 22, height: 22, borderRadius: 11, backgroundColor: colors.primary, alignItem
s: 'center', justifyContent: 'center' },
});
```

src/app/notifications.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { router, type Href } from 'expo-router';
import { useEffect } from 'react';
import { StyleSheet, View } from 'react-native';

import { Button } from '@/components/ui/Button';
import { EmptyState, type IconName } from '@/components/ui/bits';
import { Press } from '@/components/ui/Press';
import { Header, Screen } from '@/components/ui/Screen';
import { T } from '@/components/ui/T';
import { sendTestReminder } from '@/services/notifications';
import { useAppStore } from '@/store/useAppStore';
import type { AppNotification } from '@/store/types';
import { colors, radius } from '@/theme';
import { timeAgo } from '@/utils/format';
import { toast } from '@/components/ui/feedback';

const ICONS: Record<AppNotification['kind'], { icon: IconName; color: string }> = {
  reminder: { icon: 'alarm-outline', color: colors.orange },
  chat: { icon: 'chatbubble-ellipses-outline', color: colors.blue },
  vote: { icon: 'stats-chart-outline', color: colors.purple },
  expense: { icon: 'wallet-outline', color: colors.primary },
  system: { icon: 'information-circle-outline', color: colors.textSecondary },
  member: { icon: 'person-add-outline', color: colors.teal },
  photo: { icon: 'images-outline', color: colors.red },
};

export default function Notifications() {
  const list = useAppStore((s) => s.notifications);
  const markAllRead = useAppStore((s) => s.markAllRead);
  const clear = useAppStore((s) => s.clearNotifications);
  const setActive = useAppStore((s) => s.setActiveTrip);

  // Mark everything read when leaving the screen.
  useEffect(() => () => markAllRead(), [markAllRead]);

  return (
    <Screen
      header={
        <Header
          title="Notifications"
          subtitle={`${list.filter((n) => !n.read).length} unread`}
          right={
            list.length ? (
              <Press onPress={clear} hitSlop={8} accessibilityLabel="Clear all notifications">
                <T variant="small" weight="semibold" color={colors.textSecondary}>
                  Clear
                </T>
              </Press>
            ) : null
          }
        />
      }>
      <Button
        label="Send a test reminder (5 sec)"
        icon="alarm-outline"
        variant="soft"
        onPress={() => {
          sendTestReminder(5);
          toast('Test reminder scheduled — lock your phone to see it arrive', { icon: 'alarm-outline' });
        }}
      />
      <View style={{ gap: 8, marginTop: 14 }}>
        {list.length ? (
          list.map((n) => {
            const meta = ICONS[n.kind];
            return (
              <Press
                key={n.id}
                onPress={() => {
                  if (n.tripId) setActive(n.tripId);
                  if (n.route) router.push((n.route === '/(tabs)/profile' ? '/crew' : n.route) as Href);
                }}
                style={[styles.item, !n.read && styles.unread]}
```

```
                accessibilityLabel={`${n.title}. ${n.body}`}>
                <View style={[styles.icon, { backgroundColor: `${meta.color}1A` }]}>
                  <Ionicons name={meta.icon} size={18} color={meta.color} />
                </View>
                <View style={{ flex: 1 }}>
                  <T variant="small" weight="semibold">
                    {n.title}
                  </T>
                  <T variant="caption" color={colors.textSecondary}>
                    {n.body}
                  </T>
                  <T variant="micro" color={colors.textMuted} style={{ marginTop: 3 }}>
                    {timeAgo(n.ts)}
                  </T>
                </View>
                {!n.read ? <View style={styles.dot} /> : null}
              </Press>
            );
          })
        ) : (
          <EmptyState icon="notifications-off-outline" title="You're all caught up" body="Reminders, votes and chat mentions will show u
p here." />
        )}
      </View>
    </Screen>
  );
}

const styles = StyleSheet.create({
  item: { flexDirection: 'row', gap: 12, padding: 14, borderRadius: radius.md, backgroundColor: colors.card, borderWidth: 1, borderColo
r: colors.border },
  unread: { backgroundColor: colors.primarySofter, borderColor: colors.primaryLine },
  icon: { width: 38, height: 38, borderRadius: 19, alignItems: 'center', justifyContent: 'center' },
  dot: { width: 9, height: 9, borderRadius: 5, backgroundColor: colors.primary, marginTop: 4 },
});
```

src/app/onboarding.tsx

```
import { Image } from 'expo-image';
import { LinearGradient } from 'expo-linear-gradient';
import { router } from 'expo-router';
import { useEffect, useRef, useState } from 'react';
import { Animated, FlatList, StyleSheet, View, useWindowDimensions, type NativeScrollEvent, type NativeSyntheticEvent } from 'react-nati
ve';
import { useSafeAreaInsets } from 'react-native-safe-area-context';

import { LogoMark } from '@/components/Logo';
import { Button } from '@/components/ui/Button';
import { FadeIn, Float, nativeDriver, useMotion } from '@/components/ui/motion';
import { Press } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import { photos } from '@/data/images';
import { useT } from '@/i18n';
import { useAppStore } from '@/store/useAppStore';
import { MAX_WIDTH, colors } from '@/theme';

const slides = [
  { key: '1', image: photos['onboard-1'], title: 'Your Journey\nBegins Here', body: 'Welcome to a seamless way to explore the world with
the people you love.' },
  { key: '2', image: photos['onboard-2'], title: 'Plan Together,\nTravel Better', body: 'Coordinate itineraries, budgets, and memories —
all in one place.' },
  { key: '3', image: photos['onboard-3'], title: 'Every Trip,\nUnforgettable', body: 'From bucket lists to group chats — TogetherWeGo ha
s you covered.' },
];

export default function Onboarding() {
  const t = useT();
  const { width: winW, height } = useWindowDimensions();
  const width = Math.min(winW, MAX_WIDTH);
  const insets = useSafeAreaInsets();
  const [index, setIndex] = useState(0);
  const list = useRef<FlatList>(null);
  const complete = useAppStore((s) => s.completeOnboarding);
  const motion = useMotion();
  const [zoom] = useState(() => new Animated.Value(0));

  // Slow "Ken Burns" zoom on the current photo, restarted on every slide.
  useEffect(() => {
    zoom.setValue(0);
    if (!motion) return;
    const anim = Animated.timing(zoom, { toValue: 1, duration: 9000, useNativeDriver: nativeDriver });
    anim.start();
    return () => anim.stop();
  }, [index, motion, zoom]);
  const scale = zoom.interpolate({ inputRange: [0, 1], outputRange: [1, 1.14] });
```

```
  const finish = () => {
    complete();
    router.replace('/sign-in');
  };
  const next = () => {
    if (index >= slides.length - 1) return finish();
    list.current?.scrollToIndex({ index: index + 1, animated: true });
    setIndex(index + 1);
  };
  const onScroll = (e: NativeSyntheticEvent<NativeScrollEvent>) => {
    const i = Math.round(e.nativeEvent.contentOffset.x / width);
    if (i !== index) setIndex(i);
  };

  return (
    <View style={styles.root}>
      <View style={{ width, height, alignSelf: 'center' }}>
        <FlatList
          ref={list}
          data={slides}
          horizontal
          pagingEnabled
          bounces={false}
          showsHorizontalScrollIndicator={false}
          onMomentumScrollEnd={onScroll}
          onScroll={onScroll}
          scrollEventThrottle={32}
          getItemLayout={(_, i) => ({ length: width, offset: width * i, index: i })}
          keyExtractor={(s) => s.key}
          renderItem={({ item, index: i }) => (
            <View style={{ width, height, overflow: 'hidden' }}>
              <Animated.View style={[StyleSheet.absoluteFill, i === index ? { transform: [{ scale }] } : null]}>
                <Image source={item.image} style={StyleSheet.absoluteFill} contentFit="cover" transition={300} accessibilityIgnoresInver
tColors />
              </Animated.View>
              <LinearGradient colors={['rgba(0,0,0,0.05)', 'rgba(0,0,0,0)', 'rgba(8,16,6,0.78)']} locations={[0, 0.45, 1]} style={StyleS
heet.absoluteFill} />
            </View>
          )}
        />
        <View style={[styles.top, { top: insets.top + 8 }]}>
          <FadeIn from="down">
            <View style={styles.brand}>
              <Float amount={5} duration={2000}>
                <LogoMark size={26} />
              </Float>
              <T variant="small" weight="bold" color={colors.white}>
                TogetherWeGo
              </T>
            </View>
          </FadeIn>
          {index < slides.length - 1 ? (
            <Press onPress={finish} accessibilityLabel={t('common.skip')} hitSlop={10} style={styles.skip}>
              <T variant="small" weight="semibold" color={colors.white}>
                {t('common.skip')}
              </T>
            </Press>
          ) : null}
        </View>
        <View style={[styles.bottom, { paddingBottom: insets.bottom + 24 }]}>
          <FadeIn key={`k${index}`} delay={80} distance={26}>
            <View style={styles.kicker}>
              <T variant="micro" weight="bold" color="#1A1C17">
                {['✈️  PLAN', '🤝  TOGETHER', '📸  REMEMBER'][index]}
              </T>
            </View>
            <T variant="display" color={colors.white} style={{ fontSize: 34, lineHeight: 40 }} accessibilityRole="header">
              {slides[index].title}
            </T>
          </FadeIn>
          <FadeIn key={`b${index}`} delay={220} distance={20}>
            <T variant="body" color="rgba(255,255,255,0.85)" style={{ marginTop: 10, maxWidth: 320 }}>
              {slides[index].body}
            </T>
          </FadeIn>
          <View style={styles.dots} accessibilityLabel={`Page ${index + 1} of ${slides.length}`}>
            {slides.map((s, i) => (
              <View key={s.key} style={[styles.dot, i === index && styles.dotActive]} />
            ))}
          </View>
          <Button label={index === slides.length - 1 ? t('common.getStarted') : t('common.next')} onPress={next} iconRight="arrow-forwar
d" style={{ borderRadius: 999 }} />
        </View>
      </View>
    </View>
  );
```

```
}

const styles = StyleSheet.create({
  root: { flex: 1, backgroundColor: '#0F1A0D' },
  top: { position: 'absolute', left: 20, right: 20, flexDirection: 'row', justifyContent: 'space-between', alignItems: 'center' },
  brand: { flexDirection: 'row', alignItems: 'center', gap: 8, backgroundColor: 'rgba(255,255,255,0.18)', paddingLeft: 5, paddingRight:
12, paddingVertical: 5, borderRadius: 999 },
  kicker: { alignSelf: 'flex-start', backgroundColor: 'rgba(255,255,255,0.92)', paddingHorizontal: 10, paddingVertical: 5, borderRadius:
999, marginBottom: 12 },
  skip: { backgroundColor: 'rgba(0,0,0,0.25)', paddingHorizontal: 14, paddingVertical: 7, borderRadius: 999 },
  bottom: { position: 'absolute', left: 24, right: 24, bottom: 0 },
  dots: { flexDirection: 'row', gap: 6, marginTop: 22, marginBottom: 22 },
  dot: { width: 7, height: 7, borderRadius: 4, backgroundColor: 'rgba(255,255,255,0.45)' },
  dotActive: { width: 24, backgroundColor: colors.accent },
});
```

src/app/packing.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { useState } from 'react';
import { ScrollView, StyleSheet, TextInput, View } from 'react-native';

import { MemberPickerSheet } from '@/components/trip/MemberPickerSheet';
import { NoTrip } from '@/components/trip/NoTrip';
import { Button } from '@/components/ui/Button';
import { Avatar, Card, Chip, ProgressBar } from '@/components/ui/bits';
import { toast } from '@/components/ui/feedback';
import { Press, tap } from '@/components/ui/Press';
import { Header, Screen } from '@/components/ui/Screen';
import { Segmented } from '@/components/ui/Segmented';
import { T } from '@/components/ui/T';
import { destinationById } from '@/data/destinations';
import { packingBase } from '@/data/reference';
import { useWeather } from '@/hooks/useTrip';
import { useActiveTrip, useAppStore } from '@/store/useAppStore';
import { colors, fonts, radius } from '@/theme';

type Filter = 'all' | 'mine' | 'open';

const COUNTRY_EXTRAS: Record<string, string[]> = {
  JP: ['IC transit card (Suica / ICOCA)', 'Type A plug adapter', 'Coin purse — lots of cash & coins', 'Small towel (many restrooms lack
dryers)', 'Slip-on shoes for temples & ryokan'],
  FR: ['Type E plug adapter', 'Reusable shopping bag'],
  IT: ['Type L/F plug adapter', 'Scarf to cover shoulders in churches'],
  TH: ['Mosquito repellent', 'Light clothes covering knees for temples'],
  ID: ['Reef-safe sunscreen', 'Mosquito repellent'],
};

export default function Packing() {
  const trip = useActiveTrip();
  const me = useAppStore((s) => s.session?.userId);
  const toggle = useAppStore((s) => s.togglePacked);
  const assign = useAppStore((s) => s.assignPacking);
  const addItems = useAppStore((s) => s.addPackingItems);
  const del = useAppStore((s) => s.deletePackingItem);
  const weather = useWeather(trip?.destinationId);
  const [text, setText] = useState('');
  const [cat, setCat] = useState('Extras');
  const [err, setErr] = useState<string | null>(null);
  const [filter, setFilter] = useState<Filter>('all');
  const [pickId, setPickId] = useState<string | null>(null);
  if (!trip) return <NoTrip />;

  const dest = destinationById(trip.destinationId);
  const total = trip.packing.length;
  const packed = trip.packing.filter((p) => p.packed).length;
  const bringer = (broughtBy?: string) => trip.members.find((m) => m.id === broughtBy);
  const needsSomeone = trip.packing.filter((p) => !p.packed && !bringer(p.broughtBy));
  const shown = trip.packing.filter((p) => (filter === 'mine' ? p.broughtBy === me : filter === 'open' ? !p.packed && !bringer(p.brought
By) : true));
  const groups = Array.from(new Set(shown.map((p) => p.category)));
  const have = new Set(trip.packing.map((p) => p.text.toLowerCase()));
  const picking = trip.packing.find((p) => p.id === pickId);
  const suggestions = [
    ...(COUNTRY_EXTRAS[dest?.countryCode ?? ''] ?? []),
    ...(weather?.daily.some((d) => d.rain >= 50) ? ['Compact umbrella'] : []),
    ...(weather && Math.max(...weather.daily.map((d) => d.max)) >= 27 ? ['Sunglasses & hat'] : []),
    ...(weather && Math.min(...weather.daily.map((d) => d.min)) <= 10 ? ['Warm jacket'] : []),
  ].filter((s) => !have.has(s.toLowerCase()));

  const add = () => {
    const t = text.trim();
    if (!t) return setErr('Type an item to add');
    if (t.length > 50) return setErr('Keep items under 50 characters');
    if (have.has(t.toLowerCase())) return setErr('That’s already on the list');
    addItems(trip.id, [{ text: t, category: cat }]);
```

```
  setText('');
  setErr(null);
  tap('success');
};

const splitFairly = () => {
  if (!needsSomeone.length || !trip.members.length) return;
  const load: Record<string, number> = {};
  trip.members.forEach((m) => (load[m.id] = trip.packing.filter((p) => p.broughtBy === m.id).length));
  needsSomeone.forEach((p) => {
    const next = trip.members.reduce((a, b) => (load[b.id] < load[a.id] ? b : a));
    assign(trip.id, p.id, next.id);
    load[next.id] += 1;
  });
  tap('success');
  toast(`Assigned ${needsSomeone.length} item${needsSomeone.length === 1 ? '' : 's'}`);
};

return (
  <Screen header={<Header title="Packing list" subtitle={`Shared with ${trip.members.length} travellers`} />}>
    <Card style={[styles.card, styles.stack]}>
      <View style={styles.between}>
        <T variant="headline">Shared checklist</T>
        <View style={styles.pct}>
          <T variant="small" weight="bold" color={colors.onAccent}>
            {total ? Math.round((packed / total) * 100) : 0}%
          </T>
        </View>
      </View>
      <T variant="bodySm" weight="semibold">
        {packed} of {total} packed · {needsSomeone.length} still need someone
      </T>
      <ProgressBar value={total ? packed / total : 0} height={10} />
      {total ? (
        <View style={styles.wrap}>
          {trip.members.map((m) => (
            <View key={m.id} style={styles.load}>
              <Avatar name={m.name} src={m.avatar} size={22} />
              <T variant="caption" weight="semibold">
                {m.name.split(' ')[0]} · {trip.packing.filter((p) => p.broughtBy === m.id).length}
              </T>
            </View>
          ))}
        </View>
      ) : null}
      <Button label="Split the rest fairly" icon="shuffle-outline" size="md" disabled={!needsSomeone.length} onPress={splitFairly} />
    </Card>

    <View style={{ marginTop: 14 }}>
      <Segmented
        value={filter}
        onChange={setFilter}
        options={[
          { key: 'all', label: 'All' },
          { key: 'mine', label: 'Mine' },
          { key: 'open', label: 'Unassigned' },
        ]}
      />
    </View>

    {suggestions.length ? (
      <Card style={[styles.card, styles.stack, { marginTop: 14, backgroundColor: colors.accentSoft, borderColor: colors.accent }]}>
        <View style={styles.between}>
          <T variant="small" weight="bold" color={colors.onAccent}>
            Smart suggestions for {dest?.city}
          </T>
          <Press
            onPress={() => {
              addItems(trip.id, suggestions.map((s) => ({ text: s, category: 'Suggested' })));
              toast(`Added ${suggestions.length} items`);
            }}
            hitSlop={8}
            style={styles.addAll}>
            <T variant="caption" weight="bold" color={colors.onPrimary}>
              Add all
            </T>
          </Press>
        </View>
        <View style={styles.wrap}>
          {suggestions.map((s) => (
            <Chip key={s} small icon="add" label={s} onPress={() => addItems(trip.id, [{ text: s, category: 'Suggested' }])} />
          ))}
        </View>
      </Card>
    ) : null}
```

```
      <Card style={[styles.card, { marginTop: 14, gap: 8 }]}>
        <View style={{ flexDirection: 'row', gap: 8 }}>
          <TextInput value={text} onChangeText={(v) => { setText(v); setErr(null); }} placeholder="Add an item…" placeholderTextColor={c
olors.textMuted} style={[styles.input, err && { borderColor: colors.red }]} onSubmitEditing={add} returnKeyType="done" accessibilityLabe
l="New packing item" />
          <Press onPress={add} style={styles.addBtn} accessibilityLabel="Add item">
            <Ionicons name="add" size={22} color={colors.onPrimary} />
          </Press>
        </View>
        {err ? (
          <T variant="caption" color={colors.red}>
            {err}
          </T>
        ) : null}
        <ScrollView horizontal showsHorizontalScrollIndicator={false} contentContainerStyle={{ gap: 6 }}>
          {[...packingBase.map((g) => g.category), 'Weather', 'Suggested'].map((c) => (
            <Chip key={c} small label={c} active={cat === c} onPress={() => setCat(c)} />
          ))}
        </ScrollView>
      </Card>

      {groups.map((g) => (
        <View key={g} style={{ marginTop: 20 }}>
          <T variant="headline" style={{ marginBottom: 10 }} accessibilityRole="header">
            {g}
          </T>
          <Card padded={false} style={[styles.card, { paddingHorizontal: 14 }]}>
            {shown
              .filter((p) => p.category === g)
              .map((p, i, arr) => {
                const who = bringer(p.broughtBy);
                return (
                  <View key={p.id} style={[styles.item, i === arr.length - 1 && { borderBottomWidth: 0 }]}>
                    <Press onPress={() => toggle(trip.id, p.id)} style={styles.itemMain} accessibilityRole="checkbox" accessibilityState
={{ checked: p.packed }} accessibilityLabel={p.text}>
                      <Ionicons name={p.packed ? 'checkbox' : 'square-outline'} size={22} color={p.packed ? colors.primary : colors.bord
erStrong} />
                      <T variant="small" color={p.packed ? colors.textMuted : colors.text} style={[{ flex: 1 }, p.packed && { textDecora
tionLine: 'line-through' }]}>
                        {p.text}
                      </T>
                    </Press>
                    <Press onPress={() => setPickId(p.id)} hitSlop={6} style={[styles.who, !who && styles.whoEmpty]} accessibilityLabel=
{who ? `${p.text} is brought by ${who.name}. Change` : `Assign ${p.text}`}>
                      {who ? (
                        <>
                          <Avatar name={who.name} src={who.avatar} size={22} />
                          <T variant="caption" weight="semibold" numberOfLines={1} style={{ maxWidth: 56 }}>
                            {who.id === me ? 'Me' : who.name.split(' ')[0]}
                          </T>
                        </>
                      ) : (
                        <T variant="caption" weight="semibold" color={colors.textSecondary}>
                          Anyone
                        </T>
                      )}
                    </Press>
                    <Press onPress={() => del(trip.id, p.id)} hitSlop={8} accessibilityLabel={`Remove ${p.text}`}>
                      <Ionicons name="close" size={16} color={colors.textMuted} />
                    </Press>
                  </View>
                );
              })}
          </Card>
        </View>
      ))}
      {total && !shown.length ? (
        <Card style={[styles.card, { marginTop: 20, alignItems: 'center' }]}>
          <T variant="small" color={colors.textSecondary} center>
            {filter === 'mine' ? 'Nothing is assigned to you yet. Tap “Anyone” on an item to pick it up.' : 'Every item has someone brin
ging it. Nice teamwork!'}
          </T>
        </Card>
      ) : null}
      {!total ? (
        <Button
          label="Start with the essentials"
          icon="sparkles-outline"
          variant="secondary"
          style={{ marginTop: 16 }}
          onPress={() => addItems(trip.id, packingBase.flatMap((g) => g.items.map((text) => ({ text, category: g.category }))))}
        />
      ) : null}

      <MemberPickerSheet
        visible={!!picking}
```

```
        onClose={() => setPickId(null)}
        members={trip.members}
        selected={picking?.broughtBy ? [picking.broughtBy] : []}
        onToggle={(id) => picking && assign(trip.id, picking.id, picking.broughtBy === id ? undefined : id)}
        title={picking ? `Who’s bringing ${picking.text}?` : 'Who’s bringing it?'}
        single
      />
    </Screen>
  );
}

const styles = StyleSheet.create({
  card: { borderRadius: radius.xl },
  stack: { gap: 12 },
  between: { flexDirection: 'row', justifyContent: 'space-between', alignItems: 'center', gap: 8 },
  wrap: { flexDirection: 'row', flexWrap: 'wrap', gap: 6 },
  pct: { backgroundColor: colors.accent, borderRadius: radius.pill, paddingHorizontal: 12, paddingVertical: 4 },
  load: { flexDirection: 'row', alignItems: 'center', gap: 6, paddingRight: 10, paddingLeft: 3, height: 28, borderRadius: radius.pill, b
ackgroundColor: colors.bgAlt },
  addAll: { backgroundColor: colors.primary, borderRadius: radius.pill, paddingHorizontal: 12, height: 26, alignItems: 'center', justify
Content: 'center' },
  input: { flex: 1, height: 46, borderRadius: radius.pill, borderWidth: 1, borderColor: colors.border, backgroundColor: colors.cardMute
d, paddingHorizontal: 18, fontFamily: fonts.regular, fontSize: 15, color: colors.text },
  addBtn: { width: 46, height: 46, borderRadius: 23, backgroundColor: colors.primary, alignItems: 'center', justifyContent: 'center' },
  item: { flexDirection: 'row', alignItems: 'center', gap: 10, borderBottomWidth: 1, borderBottomColor: colors.border },
  itemMain: { flex: 1, flexDirection: 'row', alignItems: 'center', gap: 10, paddingVertical: 12 },
  who: { flexDirection: 'row', alignItems: 'center', gap: 5, height: 30, paddingLeft: 4, paddingRight: 10, borderRadius: radius.pill, ba
ckgroundColor: colors.bgAlt },
  whoEmpty: { paddingLeft: 10, backgroundColor: 'transparent', borderWidth: 1, borderStyle: 'dashed', borderColor: colors.borderStrong
},
});
```

src/app/panorama.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { LinearGradient } from 'expo-linear-gradient';
import { router, useFocusEffect, useLocalSearchParams } from 'expo-router';
import { useCallback, useEffect, useRef, useState } from 'react';
import { Animated, Easing, ScrollView, StyleSheet, View, useWindowDimensions } from 'react-native';
import { StatusBar } from 'expo-status-bar';
import { useSafeAreaInsets } from 'react-native-safe-area-context';

import { headingOf, K, layout, PanoramaScene, posForHeading } from '@/components/panorama/Scene';
import { timeOfDayAt } from '@/components/SkylineHero';
import { toast } from '@/components/ui/feedback';
import { useMotion } from '@/components/ui/motion';
import { Press, tap } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import { useNow } from '@/hooks/useTrip';
import { PANORAMAS, panoById } from '@/data/panoramas';
import { MAX_WIDTH, colors } from '@/theme';

const WIND = ['N', 'NNE', 'NE', 'ENE', 'E', 'ESE', 'SE', 'SSE', 'S', 'SSW', 'SW', 'WSW', 'W', 'WNW', 'NW', 'NNW'];
const wind = (deg: number) => WIND[Math.round(deg / 22.5) % 16];
/** Smallest signed angle from a to b, in degrees. */
const angleDiff = (a: number, b: number) => ((((b - a) % 360) + 540) % 360) - 180;

export default function PanoramaScreen() {
  const { city, focus } = useLocalSearchParams<{ city?: string; focus?: string }>();
  const { width, height } = useWindowDimensions();
  const insets = useSafeAreaInsets();
  const S = Math.min(width, MAX_WIDTH);
  const H = height;
  const now = useNow(60000);

  const [cityId, setCityId] = useState(panoById(city)?.id ?? PANORAMAS[0].id);
  const pano = panoById(cityId) ?? PANORAMAS[0];
  const startLm = pano.landmarks.find((l) => l.id === focus) ?? pano.landmarks[0];
  const [forced, setForced] = useState<'day' | 'night' | null>(null);
  const [auto, setAuto] = useState(true);
  const motion = useMotion(); // off when "Animations" is off or the phone asks apps to reduce motion: no auto-rotation, no coasting, no
glides
  const autoOn = auto && motion;
  const [heading, setHeading] = useState(startLm.heading);
  const mode = forced ?? timeOfDayAt(pano.tz, now);

  const [pos] = useState(() => new Animated.Value(posForHeading(startLm.heading, S)));
  const [night] = useState(() => new Animated.Value(mode === 'night' ? 1 : 0));
  const posRef = useRef(posForHeading(startLm.heading, S));
  const velRef = useRef(0);
  const dragRef = useRef(false);
  const busyRef = useRef(false);
  const idleRef = useRef(1000);
  const autoRef = useRef(true);
  const motionRef = useRef(true);
  const gestureRef = useRef({ x: 0, pos: 0, last: 0 });
```

```
  // Keep the mirrors of state that the animation loop and gesture handlers read.
  useEffect(() => {
    motionRef.current = motion;
    autoRef.current = autoOn;
  }, [autoOn, motion]);
  useEffect(() => {
    const id = pos.addListener(({ value }) => {
      posRef.current = value;
    });
    return () => pos.removeListener(id);
  }, [pos]);

  // Day ↔ night cross-fade.
  useEffect(() => {
    const anim = Animated.timing(night, { toValue: mode === 'night' ? 1 : 0, duration: motion ? 1200 : 0, easing: Easing.inOut(Easing.cu
bic), useNativeDriver: false });
    anim.start();
    return () => anim.stop();
  }, [mode, night, motion]);

  // One ~30 fps loop: coasting after a flick, gentle auto-rotation when idle, and the compass readout.
  // It only runs while this screen is the one being looked at, so it costs nothing behind another screen.
  useFocusEffect(
    useCallback(() => {
      let tick = 0;
      const id = setInterval(() => {
        tick++;
        if (!dragRef.current && !busyRef.current) {
          velRef.current = motionRef.current ? velRef.current * 0.94 : 0;
          idleRef.current++;
          const idle = idleRef.current > 105;
          const step = velRef.current + (autoRef.current && idle && Math.abs(velRef.current) < 0.2 ? 0.45 : 0);
          if (Math.abs(step) > 0.01) pos.setValue(posRef.current + step);
        }
        if (tick % 4 === 0) setHeading(Math.round(headingOf(posRef.current, S)) % 360);
      }, 33);
      return () => clearInterval(id);
    }, [pos, S]),
  );

  const glide = (delta: number) => {
    idleRef.current = 0;
    velRef.current = 0;
    if (!motion) {
      pos.setValue(posRef.current + delta); // no sweeping animation when motion is off
      return;
    }
    busyRef.current = true;
    Animated.timing(pos, { toValue: posRef.current + delta, duration: 800, easing: Easing.out(Easing.cubic), useNativeDriver: false }).s
tart(() => {
      busyRef.current = false;
    });
  };
  const nudge = (dir: 1 | -1) => {
    tap();
    glide((dir * S * 0.45) / K.mid);
  };
  const lookAt = (deg: number) => {
    tap();
    const period = layout(S).far;
    const mid = layout(S).mid;
    const want = (angleDiff(headingOf(posRef.current, S), deg) / 360) * mid;
    glide((want / K.mid) % period);
  };
  const pick = (id: string) => {
    tap();
    const p = panoById(id);
    if (!p) return;
    setCityId(p.id);
    setForced(null);
    const h = p.landmarks[0].heading;
    setHeading(h);
    const target = posForHeading(h, S);
    posRef.current = target;
    velRef.current = 0;
    pos.setValue(target);
  };

  const near = pano.landmarks.find((l) => Math.abs(angleDiff(heading, l.heading)) < 13);
  const top = insets.top + 10;

  return (
    <View style={styles.root}>
      <StatusBar style="light" />
      <View
```

```
        style={{ width: S, height: H, overflow: 'hidden', backgroundColor: '#05070F' }}
        onStartShouldSetResponder={() => true}
        onMoveShouldSetResponder={() => true}
        onResponderGrant={(e) => {
          dragRef.current = true;
          busyRef.current = false;
          idleRef.current = 0;
          velRef.current = 0;
          gestureRef.current = { x: e.nativeEvent.pageX, pos: posRef.current, last: e.nativeEvent.pageX };
        }}
        onResponderMove={(e) => {
          const g = gestureRef.current;
          const x = e.nativeEvent.pageX;
          idleRef.current = 0;
          velRef.current = -(x - g.last) / K.mid;
          g.last = x;
          pos.setValue(g.pos - (x - g.x) / K.mid);
        }}
        onResponderRelease={() => {
          dragRef.current = false;
        }}
        onResponderTerminate={() => {
          dragRef.current = false;
        }}>
        <PanoramaScene key={pano.id} pano={pano} night={night} pos={pos} S={S} H={H} />
        <LinearGradient pointerEvents="none" colors={['rgba(0,0,0,0.55)', 'rgba(0,0,0,0)']} style={[styles.fade, { top: 0, height: 200
}]} />
        <LinearGradient pointerEvents="none" colors={['rgba(0,0,0,0)', 'rgba(0,0,0,0.7)']} style={[styles.fade, { bottom: 0, height: 330
}]} />

        {/* Top bar */}
        <View style={[styles.top, { top }]}>
          <Press onPress={() => (router.canGoBack() ? router.back() : router.replace('/(tabs)/explore'))} style={styles.round} accessibi
lityLabel="Go back">
            <Ionicons name="chevron-back" size={20} color={colors.white} />
          </Press>
          <View style={{ flex: 1 }} />
          <Press
            onPress={() => (motion ? setAuto((a) => !a) : toast('Animations are off, so the view only moves when you drag it', { icon:
'accessibility-outline' }))}
            style={styles.chip}
            accessibilityRole="switch"
            accessibilityState={{ checked: autoOn }}
            accessibilityLabel="Auto rotate">
            <Ionicons name={autoOn ? 'sync' : 'pause'} size={14} color={colors.white} />
            <T variant="micro" weight="bold" color={colors.white}>
              {autoOn ? 'Auto' : 'Paused'}
            </T>
          </Press>
          <Press
            onPress={() => {
              tap();
              setForced(mode === 'night' ? 'day' : 'night');
            }}
            onLongPress={() => setForced(null)}
            style={[styles.chip, { backgroundColor: mode === 'night' ? '#1B2250' : 'rgba(255,255,255,0.28)' }]}
            accessibilityLabel={mode === 'night' ? 'Night view. Switch to day' : 'Day view. Switch to night'}
            accessibilityHint="Long press to follow local time">
            <Ionicons name={mode === 'night' ? 'moon' : 'sunny'} size={15} color={mode === 'night' ? '#FFE9A8' : '#FFFFFF'} />
            <T variant="micro" weight="bold" color={colors.white}>
              {mode === 'night' ? 'Night · lights on' : 'Day'}
              {forced ? '' : ' · live'}
            </T>
          </Press>
        </View>

        {/* Title + compass */}
        <View style={[styles.titleBlock, { top: top + 54 }]} pointerEvents="none">
          <T variant="kicker" color="rgba(255,255,255,0.85)">
            {pano.emoji} {pano.country}
          </T>
          <T variant="display" color={colors.white} style={styles.titleText} accessibilityRole="header">
            {pano.name}
          </T>
          <View style={styles.compass}>
            <Ionicons name="compass-outline" size={13} color={colors.white} />
            <T variant="micro" weight="bold" color={colors.white}>
              360° · {wind(heading)} {heading}°
            </T>
          </View>
        </View>

        {/* Arrows */}
        <Press onPress={() => nudge(-1)} style={[styles.arrow, { left: 10 }]} accessibilityLabel="Look left">
          <Ionicons name="chevron-back" size={22} color={colors.white} />
        </Press>
```

```
        <Press onPress={() => nudge(1)} style={[styles.arrow, { right: 10 }]} accessibilityLabel="Look right">
          <Ionicons name="chevron-forward" size={22} color={colors.white} />
        </Press>

        {/* Bottom: callout, landmark jumps, city picker */}
        <View style={[styles.bottom, { paddingBottom: insets.bottom + 12 }]}>
          <View style={styles.callout} accessibilityLiveRegion="polite">
            {near ? (
              <>
                <T variant="title" weight="extrabold" color={colors.white}>
                  {near.name}
                </T>
                <T variant="small" color="rgba(255,255,255,0.9)">
                  {near.fact}
                </T>
              </>
            ) : (
              <View style={{ flexDirection: 'row', alignItems: 'center', gap: 8 }}>
                <Ionicons name="hand-left-outline" size={18} color={colors.white} />
                <T variant="small" weight="semibold" color={colors.white} style={{ flex: 1 }}>
                  Drag to look around — it wraps all the way back to where you started.
                </T>
              </View>
            )}
          </View>
          <ScrollView horizontal showsHorizontalScrollIndicator={false} contentContainerStyle={styles.row} style={{ flexGrow: 0 }}>
            {pano.landmarks.map((l) => (
              <Press key={l.id} onPress={() => lookAt(l.heading)} style={[styles.lm, near?.id === l.id && { backgroundColor: colors.acce
nt }]} accessibilityLabel={`Look at ${l.name}`}>
                <Ionicons name="locate" size={13} color={near?.id === l.id ? colors.onAccent : colors.white} />
                <T variant="micro" weight="bold" color={near?.id === l.id ? colors.onAccent : colors.white}>
                  {l.name}
                </T>
              </Press>
            ))}
          </ScrollView>
          <ScrollView horizontal showsHorizontalScrollIndicator={false} contentContainerStyle={styles.row} style={{ flexGrow: 0 }}>
            {PANORAMAS.map((p) => {
              const on = p.id === pano.id;
              return (
                <Press
                  key={p.id}
                  onPress={() => pick(p.id)}
                  style={[styles.city, on && { backgroundColor: colors.white }]}
                  accessibilityRole="button"
                  accessibilityState={{ selected: on }}
                  accessibilityLabel={`${p.name} 360 degree view`}>
                  <T style={{ fontSize: 16, lineHeight: 20 }}>{p.emoji}</T>
                  <T variant="small" weight="bold" color={on ? '#111111' : colors.white}>
                    {p.name}
                  </T>
                </Press>
              );
            })}
          </ScrollView>
        </View>
      </View>
    </View>
  );
}

const glass = 'rgba(20,22,32,0.5)';
const styles = StyleSheet.create({
  root: { flex: 1, backgroundColor: '#000', alignItems: 'center', overflow: 'hidden' },
  fade: { position: 'absolute', left: 0, right: 0 },
  top: { position: 'absolute', left: 14, right: 14, flexDirection: 'row', alignItems: 'center', gap: 8 },
  round: { width: 42, height: 42, borderRadius: 21, backgroundColor: glass, alignItems: 'center', justifyContent: 'center' },
  chip: { flexDirection: 'row', alignItems: 'center', gap: 6, height: 38, paddingHorizontal: 12, borderRadius: 999, backgroundColor: gla
ss },
  titleBlock: { position: 'absolute', left: 20, right: 20, gap: 2 },
  titleText: { fontSize: 52, lineHeight: 56, textShadowColor: 'rgba(0,0,0,0.5)', textShadowRadius: 10 },
  compass: { flexDirection: 'row', alignItems: 'center', gap: 5, alignSelf: 'flex-start', marginTop: 6, paddingHorizontal: 10, paddingVe
rtical: 5, borderRadius: 999, backgroundColor: glass },
  arrow: { position: 'absolute', top: '46%', width: 40, height: 40, borderRadius: 20, backgroundColor: glass, alignItems: 'center', just
ifyContent: 'center' },
  bottom: { position: 'absolute', left: 0, right: 0, bottom: 0, gap: 10 },
  callout: { marginHorizontal: 16, padding: 14, borderRadius: 20, backgroundColor: glass, gap: 4, minHeight: 58, justifyContent: 'cente
r' },
  row: { gap: 8, paddingHorizontal: 16 },
  lm: { flexDirection: 'row', alignItems: 'center', gap: 5, height: 34, paddingHorizontal: 12, borderRadius: 999, backgroundColor: glass
},
  city: { flexDirection: 'row', alignItems: 'center', gap: 6, height: 42, paddingHorizontal: 14, borderRadius: 999, backgroundColor: gla
ss },
});
```

src/app/personalize.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { LinearGradient } from 'expo-linear-gradient';
import { router } from 'expo-router';
import { useEffect, useState } from 'react';
import { Animated, ScrollView, StyleSheet, View } from 'react-native';
import { useSafeAreaInsets } from 'react-native-safe-area-context';

import { toast } from '@/components/ui/feedback';
import { FadeIn } from '@/components/ui/motion';
import { Press } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import { STEPS } from '@/components/personalize/options';
import { paletteOf } from '@/components/personalize/palette';
import { StepDone } from '@/components/personalize/steps/StepDone';
import { StepHello } from '@/components/personalize/steps/StepHello';
import { StepHome } from '@/components/personalize/steps/StepHome';
import { StepInterests } from '@/components/personalize/steps/StepInterests';
import { StepStyle } from '@/components/personalize/steps/StepStyle';
import { StepVibe } from '@/components/personalize/steps/StepVibe';
import { StepYou } from '@/components/personalize/steps/StepYou';
import { type LangCode } from '@/i18n';
import { useAppStore, useCurrentUser } from '@/store/useAppStore';
import type { Personal } from '@/store/types';
import { MAX_WIDTH, appearance, applyAppearance, shadow, type CornerStyle, type ThemeKey } from '@/theme';

/**
 * "Make it yours": the 7-step personalization survey. This screen owns the answers and the
 * navigation; each step is its own component in components/personalize/steps.
 */
export default function Personalize() {
  const insets = useSafeAreaInsets();
  const user = useCurrentUser();
  const saved = useAppStore((s) => s.personal);
  const settings = useAppStore((s) => s.settings);
  const updatePersonal = useAppStore((s) => s.updatePersonal);
  const updateSettings = useAppStore((s) => s.updateSettings);
  const setRestyled = useAppStore((s) => s.setRestyled);

  const [step, setStep] = useState(0);
  // True after a row on the summary was tapped: Continue and Back then return straight to the summary.
  const [fromSummary, setFromSummary] = useState(false);
  const [theme, setTheme] = useState<ThemeKey>(appearance.theme);
  const [corners, setCorners] = useState<CornerStyle>(appearance.corners);
  const [p, setP] = useState<Personal>(() => ({
    ...saved,
    sections: { ...saved.sections },
    interests: [...saved.interests],
  }));
  const [lang, setLang] = useState<LangCode>(settings.language);
  const [currency, setCurrency] = useState(settings.homeCurrency);
  const [applying, setApplying] = useState(false);
  const [progress] = useState(() => new Animated.Value(0));

  useEffect(() => {
    Animated.spring(progress, {
      toValue: step / (STEPS.length - 1),
      useNativeDriver: false,
      friction: 9,
    }).start();
  }, [progress, step]);

  const set = (patch: Partial<Personal>) => setP((cur) => ({ ...cur, ...patch }));
  const pal = paletteOf(theme);
  const firstName = user?.name.split(' ')[0] ?? 'traveller';
  const shownName = p.nickname.trim() || firstName;
  const last = step === STEPS.length - 1;

  const toSummary = () => {
    setFromSummary(false);
    setStep(STEPS.length - 1);
  };
  const edit = (to: number) => {
    setFromSummary(true);
    setStep(to);
  };

  const next = () => {
    if (step === 2 && p.interests.length === 0) {
      toast('Pick at least one interest so we can tailor Explore', {
        tone: 'warn',
        icon: 'alert-circle-outline',
      });
      return;
    }
```

```
    if (step === 5 && p.nickname.trim().length > 20) {
      toast('Keep your nickname under 20 characters', {
        tone: 'warn',
        icon: 'alert-circle-outline',
      });
      return;
    }
    if (fromSummary) return toSummary();
    setStep((s) => Math.min(STEPS.length - 1, s + 1));
  };
  const back = () => (step === 0 ? router.back() : fromSummary ? toSummary() : setStep((s) => s - 1));

  const apply = () => {
    setApplying(true);
    updatePersonal({ ...p, nickname: p.nickname.trim(), done: true });
    updateSettings({ language: lang, homeCurrency: currency });
    const restyle = theme !== appearance.theme || corners !== appearance.corners;
    if (!restyle) {
      toast('All set — the app is now yours ✨', { icon: 'sparkles' });
      router.replace('/(tabs)/explore');
      return;
    }
    setRestyled(true);
    // Give the store a moment to save before the UI restarts with the new look.
    setTimeout(() => {
      if (!applyAppearance({ theme, corners })) {
        setApplying(false);
        toast('Saved! Close and reopen the app to see your new look.', {
          icon: 'refresh',
        });
        router.replace('/(tabs)/explore');
      }
    }, 450);
  };

  return (
    <View style={[styles.root, { backgroundColor: pal.bg, paddingTop: insets.top }]}>
      <View style={styles.column}>
        {/* Top bar */}
        <View style={styles.top}>
          <Press onPress={back} accessibilityLabel={step === 0 ? 'Close' : 'Back'} style={[styles.round, { backgroundColor: pal.card, bo
rderColor: pal.border }]}>
            <Ionicons name={step === 0 ? 'close' : 'chevron-back'} size={20} color={pal.text} />
          </Press>
          <View style={{ flex: 1 }}>
            <View style={[styles.track, { backgroundColor: pal.border }]}>
              <Animated.View
                style={[
                  styles.fill,
                  {
                    backgroundColor: pal.primary,
                    width: progress.interpolate({
                      inputRange: [0, 1],
                      outputRange: ['4%', '100%'],
                    }),
                  },
                ]}
              />
            </View>
            <T variant="micro" color={pal.textSecondary} style={{ marginTop: 5 }}>
              Step {step + 1} of {STEPS.length} · {STEPS[step]}
            </T>
          </View>
          {!last ? (
            <Press onPress={toSummary} hitSlop={8} accessibilityLabel="Skip to the end">
              <T variant="small" weight="semibold" color={pal.textSecondary}>
                Skip
              </T>
            </Press>
          ) : (
            <View style={{ width: 30 }} />
          )}
        </View>

        <ScrollView style={{ width: '100%' }} contentContainerStyle={{ padding: 20, paddingBottom: 30 }} keyboardShouldPersistTaps="hand
led" showsVerticalScrollIndicator={false}>
          <FadeIn key={step} from="left" distance={28} duration={380}>
            {step === 0 ? <StepHello pal={pal} p={p} firstName={firstName} /> : null}
            {step === 1 ? <StepVibe pal={pal} p={p} theme={theme} setTheme={setTheme} corners={corners} shownName={shownName} /> : null}
            {step === 2 ? <StepInterests pal={pal} p={p} set={set} /> : null}
            {step === 3 ? <StepStyle pal={pal} p={p} set={set} /> : null}
            {step === 4 ? <StepHome pal={pal} p={p} set={set} corners={corners} setCorners={setCorners} /> : null}
            {step === 5 ? <StepYou pal={pal} p={p} set={set} firstName={firstName} lang={lang} setLang={setLang} currency={currency} set
Currency={setCurrency} /> : null}
            {step === 6 ? <StepDone pal={pal} p={p} theme={theme} corners={corners} lang={lang} currency={currency} shownName={shownNam
e} onEdit={edit} /> : null}
```

```
          </FadeIn>
        </ScrollView>

        <View
          style={[
            styles.footer,
            {
              paddingBottom: insets.bottom + 14,
              backgroundColor: pal.bg,
              borderTopColor: pal.border,
            },
          ]}>
          <Press
            onPress={last ? apply : next}
            disabled={applying}
            accessibilityRole="button"
            accessibilityLabel={last ? 'Apply my style' : fromSummary ? 'Back to summary' : step === 0 ? 'Let’s go' : 'Continue'}
            style={[styles.cta, { opacity: applying ? 0.6 : 1 }]}>
            <LinearGradient colors={[pal.primary, pal.heroB]} start={{ x: 0, y: 0 }} end={{ x: 1, y: 0 }} style={[StyleSheet.absoluteFil
l, { borderRadius: 999 }]} />
            <T variant="title" weight="bold" color="#FFFFFF">
              {applying ? 'Applying your look…' : last ? 'Apply my style ✨' : fromSummary ? 'Back to summary' : step === 0 ? 'Let’s go'
: 'Continue'}
            </T>
            {!last ? <Ionicons name="arrow-forward" size={18} color="#FFFFFF" /> : null}
          </Press>
        </View>
      </View>
    </View>
  );
}

const styles = StyleSheet.create({
  root: { flex: 1, alignItems: 'center' },
  column: { flex: 1, width: '100%', maxWidth: MAX_WIDTH },
  top: {
    flexDirection: 'row',
    alignItems: 'center',
    gap: 12,
    paddingHorizontal: 20,
    paddingTop: 8,
    paddingBottom: 4,
  },
  round: {
    width: 40,
    height: 40,
    borderRadius: 20,
    borderWidth: 1,
    alignItems: 'center',
    justifyContent: 'center',
  },
  track: { height: 8, borderRadius: 4, overflow: 'hidden' },
  fill: { height: 8, borderRadius: 4 },
  footer: { paddingHorizontal: 20, paddingTop: 12, borderTopWidth: 1 },
  cta: {
    height: 56,
    borderRadius: 999,
    flexDirection: 'row',
    alignItems: 'center',
    justifyContent: 'center',
    gap: 8,
    overflow: 'hidden',
    ...shadow,
  },
});
```

src/app/photos.tsx

```
import { Image } from 'expo-image';
import { useMemo, useState } from 'react';
import { Modal, Platform, ScrollView, StyleSheet, View, useWindowDimensions } from 'react-native';

import { PhotoTile } from '@/components/photos/PhotoTile';
import { PhotoViewer } from '@/components/photos/PhotoViewer';
import { ShareSheet } from '@/components/ShareSheet';
import { NoTrip } from '@/components/trip/NoTrip';
import { Button } from '@/components/ui/Button';
import { Chip, EmptyState } from '@/components/ui/bits';
import { toast } from '@/components/ui/feedback';
import { Input } from '@/components/ui/Input';
import { tap } from '@/components/ui/Press';
import { Header, Screen } from '@/components/ui/Screen';
import { Sheet } from '@/components/ui/Sheet';
import { destinationById } from '@/data/destinations';
import { FileTooLargeError, pickImage } from '@/services/files';
import { useActiveTrip, useAppStore } from '@/store/useAppStore';
```

```
import type { Photo } from '@/store/types';
import { MAX_WIDTH } from '@/theme';
import { todayInTz, tripDays } from '@/utils/time';
import { maxLen } from '@/utils/validation';

type Filter = 'all' | 'mine' | 'liked' | string;

export default function Photos() {
  const trip = useActiveTrip();
  const { width } = useWindowDimensions();
  const userId = useAppStore((s) => s.session?.userId ?? '');
  const addPhoto = useAppStore((s) => s.addPhoto);
  const like = useAppStore((s) => s.toggleLikePhoto);
  const remove = useAppStore((s) => s.deletePhoto);
  const [filter, setFilter] = useState<Filter>('all');
  const [viewing, setViewing] = useState<Photo | null>(null);
  const [pending, setPending] = useState<string | null>(null);
  const [caption, setCaption] = useState('');
  const [captionErr, setCaptionErr] = useState<string | null>(null);
  const [shareFor, setShareFor] = useState<Photo | null>(null);

  const list = useMemo(() => {
    if (!trip) return [];
    const p = trip.photos.slice();
    if (filter === 'mine') return p.filter((x) => x.by === userId);
    if (filter === 'liked') return p.sort((a, b) => b.likes.length - a.likes.length);
    if (filter !== 'all') return p.filter((x) => x.day === filter);
    return p;
  }, [trip, filter, userId]);

  if (!trip) return <NoTrip />;
  const colW = Math.min(width, MAX_WIDTH) - 40;
  const tile = (colW - 8) / 3;
  const live = viewing ? trip.photos.find((p) => p.id === viewing.id) ?? null : null;
  const days = tripDays(trip.startDate, trip.endDate);

  const add = async (camera: boolean) => {
    try {
      const f = await pickImage(camera);
      if (f) {
        setPending(f.uri);
        setCaption('');
      }
    } catch (e) {
      toast(e instanceof FileTooLargeError ? e.message : e instanceof Error ? e.message : 'Couldn’t open photos', { tone: 'warn' });
    }
  };

  const today = todayInTz(destinationById(trip.destinationId)?.tz ?? 'UTC');

  return (
    <Screen header={<Header title="Photo dump" subtitle={`${trip.photos.length} photos · ${trip.name}`} />}>
      <View style={{ flexDirection: 'row', gap: 8 }}>
        <Button label="Add photos" icon="images-outline" size="md" onPress={() => add(false)} style={{ flex: 1 }} />
        {Platform.OS !== 'web' ? <Button label="Camera" icon="camera-outline" size="md" variant="secondary" onPress={() => add(true)} st
yle={{ flex: 1 }} /> : null}
      </View>
      <ScrollView horizontal showsHorizontalScrollIndicator={false} contentContainerStyle={{ gap: 6, paddingHorizontal: 20 }} style={{ m
arginHorizontal: -20, marginVertical: 14 }}>
        <Chip small label="All" active={filter === 'all'} onPress={() => setFilter('all')} />
        <Chip small label="❤️ Most liked" active={filter === 'liked'} onPress={() => setFilter('liked')} />
        <Chip small label="Mine" active={filter === 'mine'} onPress={() => setFilter('mine')} />
        {days.map((d, i) => (
          <Chip key={d} small label={`Day ${i + 1}`} active={filter === d} onPress={() => setFilter(d)} />
        ))}
      </ScrollView>
      {list.length ? (
        <View style={styles.grid}>
          {list.map((p) => (
            <PhotoTile key={p.id} photo={p} size={tile} who={trip.members.find((x) => x.id === p.by)?.name.split(' ')[0] ?? 'a friend'}
onPress={() => setViewing(p)} />
          ))}
        </View>
      ) : (
        <EmptyState icon="images-outline" title="No photos here yet" body="Add your best shots — everyone in the trip can like and share
them." />
      )}

      <Sheet
        visible={!!pending}
        onClose={() => setPending(null)}
        title="Add a caption"
        footer={
          <Button
            label="Post to photo dump"
            icon="cloud-upload-outline"
```

```
            onPress={() => {
              const e = maxLen(80, 'Caption')(caption);
              setCaptionErr(e);
              if (e || !pending) return;
              addPhoto(trip.id, { uri: pending, caption: caption.trim(), day: days.includes(today) ? today : trip.startDate });
              setPending(null);
              tap('success');
              toast('Photo added 📸');
            }}
          />
        }>
        {pending ? <Image source={{ uri: pending }} style={{ width: '100%', height: 220, borderRadius: 16 }} contentFit="cover" /> : nul
l}
        <Input placeholder="Say something about it…" value={caption} onChangeText={setCaption} error={captionErr} maxLength={90} />
      </Sheet>

      <Modal visible={!!live} transparent animationType="fade" onRequestClose={() => setViewing(null)}>
        {live ? (
          <PhotoViewer
            photo={live}
            trip={trip}
            userId={userId}
            onClose={() => setViewing(null)}
            onShare={() => setShareFor(live)}
            onLike={() => like(trip.id, live.id)}
            onDelete={() => {
              remove(trip.id, live.id);
              setViewing(null);
            }}
          />
        ) : null}
        <ShareSheet
          visible={!!shareFor}
          onClose={() => setShareFor(null)}
          title="Share this memory"
          message={`${shareFor?.caption ?? ''} 📸 from our trip “${trip.name}” — planned with TogetherWeGo`}
        />
      </Modal>
    </Screen>
  );
}

const styles = StyleSheet.create({
  grid: { flexDirection: 'row', flexWrap: 'wrap', gap: 4 },
});
```

src/app/phrasebook.tsx

```
import { Ionicons } from '@expo/vector-icons';
import * as Speech from 'expo-speech';
import { useState } from 'react';
import { ScrollView, StyleSheet, TextInput, View } from 'react-native';

import { Card, Chip, EmptyState } from '@/components/ui/bits';
import { toast } from '@/components/ui/feedback';
import { Press } from '@/components/ui/Press';
import { Header, Screen } from '@/components/ui/Screen';
import { T } from '@/components/ui/T';
import { destinationById } from '@/data/destinations';
import { phrasebook } from '@/data/reference';
import { copy } from '@/services/share';
import { useActiveTrip } from '@/store/useAppStore';
import { colors, fonts, radius } from '@/theme';

/** Offline phrasebook with native text-to-speech pronunciation. */
export default function Phrasebook() {
  const trip = useActiveTrip();
  const dest = destinationById(trip?.destinationId ?? 'tokyo');
  const initial = dest && phrasebook[dest.language] ? dest.language : 'Japanese';
  const [lang, setLang] = useState(initial);
  const [q, setQ] = useState('');
  const [speaking, setSpeaking] = useState<string | null>(null);
  const book = phrasebook[lang];
  const list = book.phrases.filter((p) => !q || `${p.en} ${p.local} ${p.roman ?? ''}`.toLowerCase().includes(q.toLowerCase()));

  const say = (text: string) => {
    Speech.stop();
    setSpeaking(text);
    Speech.speak(text, {
      language: book.speechCode,
      rate: 0.85,
      onDone: () => setSpeaking(null),
      onStopped: () => setSpeaking(null),
      onError: () => {
        setSpeaking(null);
        toast(`Install the ${book.lang} voice in your phone settings to hear this`, { tone: 'warn' });
```

```
      },
    });
  };

  return (
    <Screen header={<Header title="Phrasebook" subtitle={`${book.phrases.length} ${book.lang} phrases · tap 🔊 to hear them`} />}>
      <ScrollView horizontal showsHorizontalScrollIndicator={false} contentContainerStyle={{ gap: 6, paddingHorizontal: 20 }} style={{ m
arginHorizontal: -20 }}>
        {Object.keys(phrasebook).map((l) => (
          <Chip key={l} label={l} active={l === lang} onPress={() => setLang(l)} />
        ))}
      </ScrollView>
      <View style={styles.search}>
        <Ionicons name="search" size={17} color={colors.textMuted} />
        <TextInput value={q} onChangeText={setQ} placeholder="Search phrases…" placeholderTextColor={colors.textMuted} style={styles.inp
ut} accessibilityLabel="Search phrases" />
      </View>
      {list.length ? (
        <View style={{ gap: 10 }}>
          {list.map((p) => (
            <Card key={p.en} style={styles.row}>
              <View style={{ flex: 1, gap: 2 }}>
                <T variant="caption" color={colors.textSecondary}>
                  {p.en}
                </T>
                <T variant="h3" selectable>
                  {p.local}
                </T>
                {p.roman ? (
                  <T variant="small" color={colors.primary} style={{ fontStyle: 'italic' }}>
                    {p.roman}
                  </T>
                ) : null}
              </View>
              <Press onPress={() => copy(p.local, 'Phrase copied')} style={styles.iconBtn} accessibilityLabel={`Copy ${p.en}`}>
                <Ionicons name="copy-outline" size={17} color={colors.primary} />
              </Press>
              <Press onPress={() => say(p.local)} style={[styles.iconBtn, speaking === p.local && { backgroundColor: colors.primary }]}
accessibilityLabel={`Hear ${p.en} in ${book.lang}`}>
                <Ionicons name="volume-high-outline" size={18} color={speaking === p.local ? colors.white : colors.primary} />
              </Press>
            </Card>
          ))}
        </View>
      ) : (
        <EmptyState icon="language-outline" title="No phrases match" />
      )}
    </Screen>
  );
}

const styles = StyleSheet.create({
  search: { flexDirection: 'row', alignItems: 'center', gap: 8, height: 46, marginVertical: 14, paddingHorizontal: 14, borderRadius: rad
ius.md, borderWidth: 1, borderColor: colors.border, backgroundColor: colors.card },
  input: { flex: 1, fontFamily: fonts.regular, fontSize: 15, color: colors.text, height: '100%', ...({ outlineStyle: 'none' } as object)
},
  row: { flexDirection: 'row', alignItems: 'center', gap: 10 },
  iconBtn: { width: 40, height: 40, borderRadius: 20, backgroundColor: colors.primarySofter, alignItems: 'center', justifyContent: 'cent
er' },
});
```

src/app/quiz.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { Image } from 'expo-image';
import { useEffect, useMemo, useRef, useState } from 'react';
import { StyleSheet, View } from 'react-native';

import { ShareSheet } from '@/components/ShareSheet';
import { Button } from '@/components/ui/Button';
import { Avatar, Card, ProgressBar } from '@/components/ui/bits';
import { Press, tap } from '@/components/ui/Press';
import { Header, Screen } from '@/components/ui/Screen';
import { T } from '@/components/ui/T';
import { photos } from '@/data/images';
import { quizQuestions, type QuizQuestion } from '@/data/reference';
import { useActiveTrip, useAppStore, useCurrentUser } from '@/store/useAppStore';
import { colors, radius } from '@/theme';

const ROUND = 8;
const SECONDS = 20;

function shuffle<T>(a: T[]): T[] {
  const x = a.slice();
  for (let i = x.length - 1; i > 0; i--) {
    const j = Math.floor(Math.random() * (i + 1));
```

```
    [x[i], x[j]] = [x[j], x[i]];
  }
  return x;
}

export default function Quiz() {
  const best = useAppStore((s) => s.quizBest);
  const setBest = useAppStore((s) => s.setQuizBest);
  const trip = useActiveTrip();
  const user = useCurrentUser();
  const [phase, setPhase] = useState<'intro' | 'play' | 'done'>('intro');
  const [qs, setQs] = useState<QuizQuestion[]>([]);
  const [i, setI] = useState(0);
  const [picked, setPicked] = useState<number | null>(null);
  const [score, setScore] = useState(0);
  const [streak, setStreak] = useState(0);
  const [left, setLeft] = useState(SECONDS);
  const [share, setShare] = useState(false);
  const timer = useRef<ReturnType<typeof setInterval> | null>(null);

  const start = () => {
    setQs(shuffle(quizQuestions).slice(0, ROUND));
    setI(0);
    setScore(0);
    setStreak(0);
    setPicked(null);
    setLeft(SECONDS);
    setPhase('play');
  };

  // Running out of time counts as answer -1 (derived, so no extra state sync needed).
  const answer = picked ?? (phase === 'play' && left <= 0 ? -1 : null);
  const ticking = phase === 'play' && answer === null;

  useEffect(() => {
    if (!ticking) return;
    timer.current = setInterval(() => setLeft((l) => Math.max(0, l - 1)), 1000);
    return () => {
      if (timer.current) clearInterval(timer.current);
    };
  }, [ticking, i]);

  const choose = (k: number) => {
    if (answer !== null) return;
    setPicked(k);
    const q = qs[i];
    if (k === q.answer) {
      const bonus = Math.max(0, Math.round(left / 4));
      setScore((s) => s + 10 + bonus + streak * 2);
      setStreak((s) => s + 1);
      tap('success');
    } else {
      setStreak(0);
      tap('warning');
    }
  };

  const next = () => {
    if (answer === -1) setStreak(0);
    if (i + 1 >= qs.length) {
      setPhase('done');
      setBest(score);
      return;
    }
    setI(i + 1);
    setPicked(null);
    setLeft(SECONDS);
  };

  // Friendly leaderboard: group members get stable pseudo-scores, you get your real best.
  const board = useMemo(() => {
    const others = (trip?.members ?? []).filter((m) => m.id !== user?.id).map((m) => ({ name: m.name, avatar: m.avatar, score: 60 + ((m.
name.charCodeAt(0) * 7 + m.name.length * 13) % 90) }));
    return [...others, { name: user?.name ?? 'You', avatar: user?.avatar, score: Math.max(best, phase === 'done' ? score : 0), me: true
}].sort((a, b) => b.score - a.score);
  }, [trip, user, best, score, phase]);

  if (phase === 'intro' || phase === 'done') {
    const finished = phase === 'done';
    return (
      <Screen header={<Header title="Travel quiz" subtitle="Test your globe-trotting knowledge" />}>
        <View style={styles.hero}>
          <Image source={photos['fuji-blossom']} style={StyleSheet.absoluteFill} contentFit="cover" />
          <View style={styles.heroShade} />
          <T variant="kicker" color="rgba(255,255,255,0.85)">
            {finished ? 'Round complete' : `${ROUND} questions · ${SECONDS}s each`}
```

```
          </T>
          <T variant="display" color={colors.white}>
            {finished ? `${score} pts` : 'Trip Trivia'}
          </T>
          <T variant="small" color="rgba(255,255,255,0.9)">
            {finished ? (score >= 100 ? 'Legendary navigator! 🧭' : score >= 60 ? 'Seasoned traveller ✈️' : 'Every expert was once a tou
rist 🌱') : 'Faster answers and streaks earn bonus points.'}
          </T>
        </View>
        <Button label={finished ? 'Play again' : 'Start quiz'} icon="play" onPress={start} style={{ marginTop: 16 }} />
        {finished ? <Button label="Challenge the group" icon="share-social-outline" variant="secondary" onPress={() => setShare(true)} s
tyle={{ marginTop: 10 }} /> : null}
        <Card style={{ marginTop: 16, gap: 10 }}>
          <View style={{ flexDirection: 'row', justifyContent: 'space-between' }}>
            <T variant="title">Group leaderboard</T>
            <T variant="caption" color={colors.textSecondary}>
              Your best: {best}
            </T>
          </View>
          {board.map((b, k) => (
            <View key={b.name} style={[styles.boardRow, 'me' in b && styles.me]}>
              <T variant="small" weight="bold" style={{ width: 22 }}>
                {k === 0 ? '🥇' : k === 1 ? '🥈' : k === 2 ? '🥉' : k + 1}
              </T>
              <Avatar name={b.name} src={b.avatar} size={28} />
              <T variant="small" style={{ flex: 1 }}>
                {b.name.split(' ')[0]}
                {'me' in b ? ' (you)' : ''}
              </T>
              <T variant="small" weight="semibold">
                {b.score}
              </T>
            </View>
          ))}
        </Card>
        <ShareSheet visible={share} onClose={() => setShare(false)} title="Share your score" message={`I scored ${score} points on the T
ogetherWeGo travel quiz 🧭 Think you can beat me?`} />
      </Screen>
    );
  }

  const q = qs[i];
  return (
    <Screen header={<Header title={`Question ${i + 1} of ${qs.length}`} subtitle={`${score} pts${streak > 1 ? ` · 🔥 ${streak} streak` :
''}`} />}>
      <ProgressBar value={(i + (answer !== null ? 1 : 0)) / qs.length} height={6} />
      <View style={styles.timerRow}>
        <Ionicons name="timer-outline" size={16} color={left <= 5 ? colors.red : colors.textSecondary} />
        <View style={{ flex: 1 }}>
          <ProgressBar value={left / SECONDS} color={left <= 5 ? colors.red : colors.orange} height={5} />
        </View>
        <T variant="small" weight="bold" color={left <= 5 ? colors.red : colors.text} style={{ width: 28, textAlign: 'right' }}>
          {Math.max(0, left)}s
        </T>
      </View>
      <T variant="h2" style={{ marginVertical: 16 }} accessibilityRole="header">
        {q.q}
      </T>
      <View style={{ gap: 10 }}>
        {q.options.map((o, k) => {
          const isRight = answer !== null && k === q.answer;
          const isWrong = answer === k && k !== q.answer;
          return (
            <Press
              key={o}
              onPress={() => choose(k)}
              disabled={answer !== null}
              style={[styles.option, isRight && styles.right, isWrong && styles.wrong]}
              accessibilityLabel={o}
              accessibilityState={{ selected: answer === k }}>
              <View style={[styles.letter, isRight && { backgroundColor: colors.primary }, isWrong && { backgroundColor: colors.red }]}>
                <T variant="small" weight="bold" color={isRight || isWrong ? colors.white : colors.primary}>
                  {'ABCD'[k]}
                </T>
              </View>
              <T variant="bodySm" weight="medium" style={{ flex: 1 }}>
                {o}
              </T>
              {isRight ? <Ionicons name="checkmark-circle" size={20} color={colors.primary} /> : isWrong ? <Ionicons name="close-circle"
size={20} color={colors.red} /> : null}
            </Press>
          );
        })}
      </View>
      {answer !== null ? (
        <Card style={{ marginTop: 16, gap: 10, backgroundColor: answer === q.answer ? colors.primarySofter : colors.orangeSoft }}>
```

```
          <T variant="small" weight="semibold">
            {answer === q.answer ? 'Correct! 🎉' : answer === -1 ? 'Time's up ⏰' : 'Not quite'}
          </T>
          <T variant="small" color={colors.textSecondary}>
            {q.fact}
          </T>
          <Button label={i + 1 >= qs.length ? 'See results' : 'Next question'} iconRight="arrow-forward" onPress={next} />
        </Card>
      ) : null}
    </Screen>
  );
}

const styles = StyleSheet.create({
  hero: { height: 220, borderRadius: radius.xl, overflow: 'hidden', padding: 20, justifyContent: 'flex-end', gap: 4 },
  heroShade: { ...StyleSheet.absoluteFill, backgroundColor: 'rgba(10,25,10,0.45)' },
  boardRow: { flexDirection: 'row', alignItems: 'center', gap: 10, padding: 8, borderRadius: radius.sm },
  me: { backgroundColor: colors.primarySofter },
  timerRow: { flexDirection: 'row', alignItems: 'center', gap: 8, marginTop: 14 },
  option: { flexDirection: 'row', alignItems: 'center', gap: 12, padding: 14, borderRadius: radius.md, backgroundColor: colors.card, bor
derWidth: 1.5, borderColor: colors.border },
  right: { borderColor: colors.primary, backgroundColor: colors.primarySofter },
  wrong: { borderColor: colors.red, backgroundColor: colors.redSoft },
  letter: { width: 30, height: 30, borderRadius: 15, backgroundColor: colors.primarySoft, alignItems: 'center', justifyContent: 'center'
},
});
```

src/app/recap.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { Image } from 'expo-image';
import { LinearGradient } from 'expo-linear-gradient';
import { router } from 'expo-router';
import { useRef, useState } from 'react';
import { Platform, StyleSheet, View } from 'react-native';

import { ShareSheet } from '@/components/ShareSheet';
import { NoTrip } from '@/components/trip/NoTrip';
import { Button } from '@/components/ui/Button';
import { Avatar, Card, Pill, type IconName } from '@/components/ui/bits';
import { toast } from '@/components/ui/feedback';
import { Press } from '@/components/ui/Press';
import { FadeIn } from '@/components/ui/motion';
import { Header, Screen } from '@/components/ui/Screen';
import { T, type TProps } from '@/components/ui/T';
import { destinationById } from '@/data/destinations';
import { imageSource } from '@/data/images';
import { shareView } from '@/services/share';
import { settlePlan, totalSpent, tripStatus, useActiveTrip } from '@/store/useAppStore';
import { colors, radius, shadow } from '@/theme';
import { money } from '@/utils/format';
import { distanceKm, kmLabel } from '@/utils/geo';
import { daysBetween, rangeLabel } from '@/utils/time';

/** End-of-trip summary: the "completion" stage of the planning → travel → memories journey. */
export default function Recap() {
  const trip = useActiveTrip();
  const [share, setShare] = useState(false);
  const [rating, setRating] = useState(0);
  const card = useRef<View>(null);
  if (!trip) return <NoTrip />;

  const dest = destinationById(trip.destinationId)!;
  const status = tripStatus(trip);
  const days = daysBetween(trip.startDate, trip.endDate) + 1;
  const spent = totalSpent(trip);
  const stops = trip.activities.filter((a) => a.lat && a.lng).sort((a, b) => (a.date + a.time).localeCompare(b.date + b.time));
  const km = stops.reduce((sum, a, i) => (i ? sum + distanceKm(stops[i - 1] as { lat: number; lng: number }, a as { lat: number; lng: nu
mber }) : 0), 0);
  const votes = trip.activities.reduce((n, a) => n + Object.keys(a.votes).length, 0) + trip.polls.reduce((n, p) => n + p.options.reduce
((m, o) => m + o.votes.length, 0), 0);
  const topPhoto = trip.photos.slice().sort((a, b) => b.likes.length - a.likes.length)[0];
  const topActivity = trip.activities.filter((a) => a.type !== 'transport' && a.type !== 'stay').sort((a, b) => Object.values(b.votes).f
ilter((v) => v === 1).length - Object.values(a.votes).filter((v) => v === 1).length)[0];
  const byCat: Record<string, number> = {};
  trip.expenses.filter((e) => !e.settlement).forEach((e) => (byCat[e.category] = (byCat[e.category] ?? 0) + e.amount));
  const topCat = Object.entries(byCat).sort((a, b) => b[1] - a[1])[0];
  const photographer = trip.members
    .map((m) => ({ m, n: trip.photos.filter((p) => p.by === m.id).length }))
    .sort((a, b) => b.n - a.n)[0];
  const owed = settlePlan(trip);

  const perPerson = spent / Math.max(1, trip.members.length);
  const stats: [IconName, string, string][] = [
    ['calendar-outline', `${days}`, 'days'],
    ['flag-outline', `${trip.activities.length}`, 'activities'],
```

```
    ['navigate-outline', kmLabel(km), 'between stops'],
    ['wallet-outline', money(spent), 'spent together'],
    ['images-outline', `${trip.photos.length}`, 'photos'],
    ['thumbs-up-outline', `${votes}`, 'votes cast'],
  ];
  const storyStats: [string, string][] = [
    [`${days}`, 'days'],
    [`${trip.activities.length}`, 'activities'],
    [`${trip.photos.length}`, 'photos'],
    [money(perPerson), 'spent each'],
  ];

  const summary = `Our trip “${trip.name}” 🌏 ${days} days in ${dest.city} · ${trip.activities.length} activities · ${trip.photos.lengt
h} photos · ${kmLabel(km)} explored · ${money(perPerson)} each. Planned together on TogetherWeGo ✈️`;

  return (
    <Screen header={<Header title="Trip recap" subtitle={status === 'completed' ? 'Your memories, wrapped up' : 'A preview — the full re
cap unlocks when the trip ends'} />}>
      <View style={{ alignSelf: 'center', marginBottom: 14 }}>
        <Pill label={status === 'completed' ? 'Trip complete 🎉' : status === 'active' ? 'In progress' : 'Coming up'} tone={status ===
'completed' ? 'yellow' : 'green'} />
      </View>

      <FadeIn from="scale" style={{ alignItems: 'center' }}>
        <View ref={card} collapsable={false} style={styles.story} accessible accessibilityLabel={`Story card. ${summary}`}>
          <Image source={imageSource(trip.cover)} style={StyleSheet.absoluteFill} contentFit="cover" />
          <LinearGradient colors={['rgba(0,0,0,0.35)', 'rgba(0,0,0,0.05)', 'rgba(0,0,0,0.92)']} locations={[0, 0.3, 0.82]} style={StyleS
heet.absoluteFill} />
          <View style={styles.storyTag}>
            <Fixed variant="micro" weight="bold" color={colors.onAccent} style={{ letterSpacing: 1.2 }}>
              TRIP RECAP
            </Fixed>
          </View>
          <View style={styles.storyBottom}>
            <Fixed variant="display" color={colors.white} numberOfLines={3} style={{ fontSize: 40, lineHeight: 42 }}>
              {trip.name}
            </Fixed>
            <Fixed variant="small" weight="semibold" color={colors.white} style={{ marginTop: 8 }}>
              {dest.city}, {dest.country}
            </Fixed>
            <Fixed variant="caption" color="rgba(255,255,255,0.75)">
              {rangeLabel(trip.startDate, trip.endDate)}
            </Fixed>
            <View style={styles.storyStats}>
              {storyStats.map(([value, label]) => (
                <View key={label} style={styles.storyStat}>
                  <Fixed variant="headline" color={colors.accent} numberOfLines={1} style={{ fontSize: 24, lineHeight: 28 }}>
                    {value}
                  </Fixed>
                  <Fixed variant="micro" color="rgba(255,255,255,0.8)">
                    {label}
                  </Fixed>
                </View>
              ))}
            </View>
            <View style={styles.storyBrand}>
              <View style={styles.storyLogo}>
                <Ionicons name="airplane" size={11} color={colors.onAccent} />
              </View>
              <Fixed variant="small" weight="extrabold" color={colors.white} style={{ letterSpacing: 0.3 }}>
                TogetherWeGo
              </Fixed>
            </View>
          </View>
        </View>
      </FadeIn>
      <View style={{ flexDirection: 'row', gap: 8, marginTop: 14 }}>
        {Platform.OS !== 'web' ? <Button label="Share picture" icon="image-outline" onPress={() => shareView(card, 'Share your trip reca
p', summary)} style={{ flex: 1 }} testID="share-picture" /> : null}
        <Button label={Platform.OS === 'web' ? 'Share story card' : 'Share text'} icon="share-social-outline" variant={Platform.OS ===
'web' ? 'primary' : 'secondary'} onPress={() => setShare(true)} style={{ flex: 1 }} />
      </View>

      <View style={styles.grid}>
        {stats.map(([icon, value, label]) => (
          <View key={label} style={styles.stat}>
            <View style={styles.statIcon}>
              <Ionicons name={icon} size={15} color={colors.onAccent} />
            </View>
            <T variant="h3">{value}</T>
            <T variant="caption" color={colors.textSecondary}>
              {label}
            </T>
          </View>
        ))}
      </View>
```

```
      <Card style={[styles.card, { marginTop: 14, gap: 12 }]}>
        <T variant="headline" accessibilityRole="header">
          Highlights
        </T>
        {topActivity ? <Highlight icon="trophy-outline" title="Group favourite" body={`${topActivity.title} — ${Object.values(topActivit
y.votes).filter((v) => v === 1).length}/${trip.members.length} said “I’m in”`} /> : null}
        {topPhoto ? <Highlight icon="heart-outline" title="Most-liked photo" body={`“${topPhoto.caption || 'Untitled'}” · ${topPhoto.lik
es.length} likes`} /> : null}
        {photographer && photographer.n ? <Highlight icon="camera-outline" title="Chief photographer" body={`${photographer.m.name.split
(' ')[0]} with ${photographer.n} photos`} /> : null}
        {topCat ? <Highlight icon="pie-chart-outline" title="Biggest spend" body={`${topCat[0]} · ${money(topCat[1])} (${Math.round((top
Cat[1] / Math.max(1, spent)) * 100)}%)`} /> : null}
        <Highlight icon="people-outline" title="The crew" body={trip.members.map((m) => `${m.name.split(' ')[0]} (${m.role})`).join(',
')} />
      </Card>

      <Card style={[styles.card, { marginTop: 14, gap: 10 }]}>
        <View style={{ flexDirection: 'row', justifyContent: 'space-between', alignItems: 'center' }}>
          <T variant="headline" accessibilityRole="header">
            Final balances
          </T>
          <Press onPress={() => router.navigate('/(tabs)/budget')} hitSlop={8}>
            <T variant="small" weight="semibold" color={colors.blue}>
              Settle up
            </T>
          </Press>
        </View>
        {owed.length ? (
          owed.slice(0, 4).map((p) => {
            const a = trip.members.find((m) => m.id === p.from);
            const b = trip.members.find((m) => m.id === p.to);
            return (
              <View key={`${p.from}${p.to}`} style={{ flexDirection: 'row', alignItems: 'center', gap: 8 }}>
                <Avatar name={a?.name ?? '?'} src={a?.avatar} size={24} />
                <T variant="small" style={{ flex: 1 }}>
                  {a?.name.split(' ')[0]} → {b?.name.split(' ')[0]}
                </T>
                <T variant="small" weight="semibold">
                  {money(p.amount)}
                </T>
              </View>
            );
          })
        ) : (
          <T variant="small" color={colors.textSecondary}>
            Everyone is square ✅
          </T>
        )}
      </Card>

      <Card style={[styles.card, { marginTop: 14, alignItems: 'center', gap: 8 }]}>
        <T variant="headline" accessibilityRole="header">
          How was the trip?
        </T>
        <View style={{ flexDirection: 'row', gap: 6 }}>
          {[1, 2, 3, 4, 5].map((n) => (
            <Press
              key={n}
              onPress={() => {
                setRating(n);
                toast(n >= 4 ? 'Saved — sounds like a great trip! 🥳' : 'Thanks for the feedback');
              }}
              accessibilityLabel={`${n} star${n > 1 ? 's' : ''}`}>
              <Ionicons name={n <= rating ? 'star' : 'star-outline'} size={32} color={colors.star} />
            </Press>
          ))}
        </View>
      </Card>

      <Button label="Open the photo dump" icon="images-outline" variant="secondary" onPress={() => router.push('/photos')} style={{ marg
inTop: 16 }} />
      <ShareSheet visible={share} onClose={() => setShare(false)} title="Share your trip recap" message={summary} preview={summary} />
    </Screen>
  );
}

/** Text for the story card: a fixed-size graphic, so it ignores the text-size setting. */
function Fixed(props: TProps) {
  return <T allowFontScaling={false} {...props} />;
}

function Highlight({ icon, title, body }: { icon: IconName; title: string; body: string }) {
  return (
    <View style={{ flexDirection: 'row', gap: 10 }}>
      <View style={styles.hlIcon}>
```

```
        <Ionicons name={icon} size={16} color={colors.onAccent} />
      </View>
      <View style={{ flex: 1 }}>
        <T variant="caption" color={colors.textSecondary}>
          {title}
        </T>
        <T variant="small" weight="semibold">
          {body}
        </T>
      </View>
    </View>
  );
}

const styles = StyleSheet.create({
  card: { borderRadius: radius.xl },
  story: { width: 270, height: 480, borderRadius: 28, overflow: 'hidden', backgroundColor: '#111111', justifyContent: 'space-between', p
adding: 20 },
  storyTag: { alignSelf: 'flex-start', backgroundColor: colors.accent, borderRadius: radius.pill, paddingHorizontal: 10, paddingVertica
l: 4 },
  storyBottom: { gap: 0 },
  storyStats: { flexDirection: 'row', flexWrap: 'wrap', rowGap: 10, columnGap: 8, marginTop: 16 },
  storyStat: { width: '47%', borderRadius: 14, backgroundColor: 'rgba(255,255,255,0.14)', paddingHorizontal: 12, paddingVertical: 8 },
  storyBrand: { flexDirection: 'row', alignItems: 'center', gap: 8, marginTop: 16 },
  storyLogo: { width: 22, height: 22, borderRadius: 11, backgroundColor: colors.accent, alignItems: 'center', justifyContent: 'center'
},
  grid: { flexDirection: 'row', flexWrap: 'wrap', gap: 8, marginTop: 14 },
  stat: { width: '31.8%', backgroundColor: colors.card, borderRadius: radius.lg, padding: 12, gap: 2, borderWidth: 1, borderColor: color
s.border, alignItems: 'flex-start', ...shadow },
  statIcon: { width: 28, height: 28, borderRadius: 14, backgroundColor: colors.accent, alignItems: 'center', justifyContent: 'center', m
arginBottom: 4 },
  hlIcon: { width: 32, height: 32, borderRadius: 16, backgroundColor: colors.accent, alignItems: 'center', justifyContent: 'center' },
});
```

src/app/settings.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { LinearGradient } from 'expo-linear-gradient';
import { router } from 'expo-router';
import { useState } from 'react';
import { StyleSheet, View } from 'react-native';

import { CurrencyPickerSheet } from '@/components/CurrencyPickerSheet';
import { AccessibilityCard } from '@/components/settings/AccessibilityCard';
import { AssistantCard } from '@/components/settings/AssistantCard';
import { DataCard } from '@/components/settings/DataCard';
import { DemoCard } from '@/components/settings/DemoCard';
import { NotificationsCard } from '@/components/settings/NotificationsCard';
import { ProfileCard } from '@/components/settings/ProfileCard';
import { SyncSettingsCard } from '@/components/sync/SyncSettingsCard';
import { Button } from '@/components/ui/Button';
import { Card, Chip, ListRow, ToggleRow } from '@/components/ui/bits';
import { confirmAction } from '@/components/ui/feedback';
import { Press } from '@/components/ui/Press';
import { Header, Screen } from '@/components/ui/Screen';
import { T } from '@/components/ui/T';
import { currencyByCode } from '@/data/reference';
import { LANGUAGES, useT } from '@/i18n';
import { useOsReduceMotion } from '@/hooks/useReduceMotion';
import { useAppStore } from '@/store/useAppStore';
import { appearance, colors, cornerStyles, radius, themes } from '@/theme';

/** Settings: how the app looks and behaves, your profile, sync, reminders, accessibility and your data. Each card is its own component.
*/
export default function Settings() {
  const t = useT();
  const settings = useAppStore((s) => s.settings);
  const update = useAppStore((s) => s.updateSettings);
  const signOut = useAppStore((s) => s.signOut);
  const personal = useAppStore((s) => s.personal);
  const osReduceMotion = useOsReduceMotion((s) => s.on);
  const updatePersonal = useAppStore((s) => s.updatePersonal);
  const [curOpen, setCurOpen] = useState(false);

  return (
    <Screen header={<Header title={t('settings.title')} subtitle="Make TogetherWeGo yours" />}>
      {/* Make it yours */}
      <Press onPress={() => router.push('/personalize')} style={styles.yours} accessibilityLabel="Make it yours — change theme, interest
s and home screen">
        <LinearGradient colors={[colors.heroA, colors.heroB, colors.heroC]} start={{ x: 0, y: 0 }} end={{ x: 1, y: 1 }} style={StyleShee
t.absoluteFill} />
        <View style={styles.yoursEmoji}>
          <T style={{ fontSize: 28, lineHeight: 34 }}>{personal.emoji}</T>
        </View>
        <View style={{ flex: 1 }}>
          <T variant="title" weight="bold" color={colors.white}>
```

```
            Make it yours 🎨
          </T>
          <T variant="caption" color="rgba(255,255,255,0.9)">
            {themes[appearance.theme].emoji} {themes[appearance.theme].name} theme · {cornerStyles[appearance.corners].name} corners ·
{personal.interests.length} interests
          </T>
        </View>
        <Ionicons name="chevron-forward" size={20} color={colors.white} />
      </Press>
      <Card style={{ marginBottom: 14 }}>
        <ToggleRow icon="sparkles-outline" label="Animations" sub={osReduceMotion ? 'Off for now: your phone’s Reduce Motion setting is
on' : 'Flying plane, floating cards and smooth entrances'} value={personal.motion && !osReduceMotion} onChange={(v) => updatePersonal({
motion: v })} />
      </Card>

      <SyncSettingsCard />

      <ProfileCard />

      {/* Language */}
      <Card style={{ gap: 10, marginTop: 14 }}>
        <T variant="title">{t('settings.language')}</T>
        <View style={styles.wrap}>
          {LANGUAGES.map((l) => (
            <Chip key={l.code} label={`${l.flag} ${l.native}`} active={settings.language === l.code} onPress={() => update({ language:
l.code })} />
          ))}
        </View>
      </Card>

      {/* Currency & time */}
      <Card style={{ marginTop: 14 }}>
        <ListRow icon="cash-outline" label={t('settings.currency')} sub={`${settings.homeCurrency} · ${currencyByCode(settings.homeCurre
ncy)?.name ?? ''}`} onPress={() => setCurOpen(true)} />
        <ToggleRow icon="globe-outline" label="Show itinerary times in trip time" sub="Off = convert to your phone’s time zone" value={s
ettings.showTripTime} onChange={(v) => update({ showTripTime: v })} />
      </Card>

      <NotificationsCard />
      <AccessibilityCard />
      <AssistantCard />
      <DemoCard />
      <DataCard />

      <Button
        label={t('settings.signOut')}
        variant="danger"
        icon="log-out-outline"
        style={{ marginTop: 16 }}
        onPress={() =>
          confirmAction('Sign out?', 'Your trips stay saved on this device.', () => {
            signOut();
            router.replace('/sign-in');
          }, { confirmLabel: 'Sign out' })
        }
      />
      <T variant="caption" color={colors.textMuted} center style={{ marginTop: 14 }}>
        TogetherWeGo v1.0 · Made for FBLA Mobile Application Development 2026–27
      </T>

      <CurrencyPickerSheet visible={curOpen} onClose={() => setCurOpen(false)} value={settings.homeCurrency} onPick={(c) => update({ hom
eCurrency: c })} title="Home currency" />
    </Screen>
  );
}

const styles = StyleSheet.create({
  yours: { flexDirection: 'row', alignItems: 'center', gap: 12, padding: 14, borderRadius: radius.lg, overflow: 'hidden', marginBottom:
14 },
  yoursEmoji: { width: 50, height: 50, borderRadius: 25, backgroundColor: 'rgba(255,255,255,0.25)', alignItems: 'center', justifyConten
t: 'center' },
  wrap: { flexDirection: 'row', flexWrap: 'wrap', gap: 8 },
});
```

src/app/sign-in.tsx

```
import { Ionicons, MaterialCommunityIcons } from '@expo/vector-icons';
import { Link, router } from 'expo-router';
import { useRef, useState } from 'react';
import { StyleSheet, View, type TextInput } from 'react-native';

import { GoogleSheet } from '@/components/auth/GoogleSheet';
import { showLegal } from '@/components/auth/legal';
import { Logo } from '@/components/Logo';
import { Button } from '@/components/ui/Button';
import { Divider } from '@/components/ui/bits';
```

```
import { toast } from '@/components/ui/feedback';
import { Input } from '@/components/ui/Input';
import { Float } from '@/components/ui/motion';
import { Press, tap } from '@/components/ui/Press';
import { Screen } from '@/components/ui/Screen';
import { T } from '@/components/ui/T';
import { useT } from '@/i18n';
import { copy } from '@/services/share';
import { DEMO_EMAIL, DEMO_PASSWORD } from '@/store/seed';
import { useAppStore } from '@/store/useAppStore';
import { colors, radius } from '@/theme';
import { validateEmail } from '@/utils/validation';

export default function SignIn() {
  const t = useT();
  const signIn = useAppStore((s) => s.signIn);
  const signInWithGoogle = useAppStore((s) => s.signInWithGoogle);
  const [email, setEmail] = useState('');
  const [password, setPassword] = useState('');
  const [remember, setRemember] = useState(true);
  const [errors, setErrors] = useState<{ email?: string | null; password?: string | null; form?: string }>({});
  const [loading, setLoading] = useState(false);
  const [googleOpen, setGoogleOpen] = useState(false);
  const pwRef = useRef<TextInput>(null);

  const submit = async () => {
    const e = validateEmail(email);
    const p = password ? null : 'Enter your password';
    setErrors({ email: e, password: p });
    if (e || p) return tap('warning');
    setLoading(true);
    const res = await signIn(email, password, remember);
    setLoading(false);
    if (!res.ok) {
      setErrors({ form: res.error });
      tap('warning');
      return;
    }
    tap('success');
    router.replace('/(tabs)/explore');
  };

  const fillDemo = () => {
    setEmail(DEMO_EMAIL);
    setPassword(DEMO_PASSWORD);
    setErrors({});
    toast('Demo account filled in — tap Sign In');
  };

  return (
    <Screen contentStyle={{ paddingTop: 28 }}>
      <Float amount={6} duration={2400}>
        <Logo />
      </Float>
      <T variant="display" style={{ marginTop: 28 }} accessibilityRole="header">
        {t('auth.welcome')}
      </T>
      <T variant="bodySm" color={colors.textSecondary} style={{ marginTop: 4, marginBottom: 22 }}>
        {t('auth.welcomeSub')}
      </T>

      <View style={{ gap: 12 }}>
        <Input
          inlineLabel
          label={t('auth.email')}
          icon="mail-outline"
          value={email}
          onChangeText={(v) => {
            setEmail(v);
            if (errors.email || errors.form) setErrors({});
          }}
          autoCapitalize="none"
          autoComplete="email"
          keyboardType="email-address"
          textContentType="emailAddress"
          returnKeyType="next"
          onSubmitEditing={() => pwRef.current?.focus()}
          error={errors.email}
          testID="email"
        />
        <Input
          ref={pwRef}
          inlineLabel
          label={t('auth.password')}
          icon="lock-closed-outline"
          value={password}
```

```
          onChangeText={(v) => {
            setPassword(v);
            if (errors.password || errors.form) setErrors({});
          }}
          secureToggle
          autoComplete="password"
          textContentType="password"
          returnKeyType="go"
          onSubmitEditing={submit}
          error={errors.password}
          testID="password"
        />
      </View>

      <View style={styles.row}>
        <Press onPress={() => setRemember((r) => !r)} style={styles.remember} accessibilityRole="checkbox" accessibilityState={{ checke
d: remember }} accessibilityLabel={t('auth.remember')}>
          <View style={[styles.box, remember && styles.boxOn]}>{remember ? <Ionicons name="checkmark" size={13} color={colors.white} />
: null}</View>
          <T variant="small" color={colors.textSecondary}>
            {t('auth.remember')}
          </T>
        </Press>
        <Link href="/forgot-password" asChild>
          <Press accessibilityRole="link">
            <T variant="small" weight="semibold" color={colors.primary}>
              {t('auth.forgot')}
            </T>
          </Press>
        </Link>
      </View>

      {errors.form ? (
        <View style={styles.formError} accessibilityLiveRegion="assertive">
          <Ionicons name="alert-circle" size={18} color={colors.red} />
          <T variant="small" color={colors.red} style={{ flex: 1 }}>
            {errors.form}
          </T>
        </View>
      ) : null}

      <Button label={t('auth.signIn')} onPress={submit} loading={loading} testID="sign-in" />

      <View style={styles.orRow}>
        <Divider style={{ flex: 1 }} />
        <T variant="caption" color={colors.textMuted}>
          {t('auth.or')}
        </T>
        <Divider style={{ flex: 1 }} />
      </View>

      <Press onPress={() => setGoogleOpen(true)} style={styles.google} accessibilityLabel={t('auth.google')}>
        <MaterialCommunityIcons name="google" size={18} color={colors.text} />
        <T variant="title" weight="medium">
          {t('auth.google')}
        </T>
      </Press>

      <View style={styles.signupRow}>
        <T variant="small" color={colors.textSecondary}>
          {t('auth.noAccount')}{' '}
        </T>
        <Link href="/sign-up" asChild>
          <Press accessibilityRole="link">
            <T variant="small" weight="bold">
              {t('auth.signUp')}
            </T>
          </Press>
        </Link>
      </View>

      {/* Tapping the card fills the form. The copy buttons sit beside the text, not inside a button: on the web a button inside a butto
n is invalid HTML. */}
      <View style={styles.demo}>
        <Press onPress={fillDemo} scaleTo={0.99} accessibilityLabel="Use the demo account" style={styles.demoHead}>
          <Ionicons name="information-circle-outline" size={16} color={colors.primary} />
          <T variant="small" weight="semibold" color={colors.primary}>
            {t('auth.demo')}
          </T>
          <T variant="caption" color={colors.textMuted} style={{ marginLeft: 'auto' }}>
            Tap to fill
          </T>
        </Press>
        {[
          ['Email', DEMO_EMAIL],
          ['Password', DEMO_PASSWORD],
```

```
        ].map(([k, v]) => (
          <View key={k} style={styles.demoRow}>
            <Press onPress={fillDemo} scaleTo={0.99} accessible={false} accessibilityRole="none" style={styles.demoFill}>
              <T variant="caption" color={colors.textSecondary} style={{ width: 62 }}>
                {k}
              </T>
              <T variant="caption" style={{ flex: 1 }} selectable>
                {v}
              </T>
            </Press>
            <Press onPress={() => copy(v, `${k} copied`)} accessibilityLabel={`Copy demo ${k.toLowerCase()}`} style={styles.copyBtn} hit
Slop={6}>
              <Ionicons name="copy-outline" size={13} color={colors.primary} />
            </Press>
          </View>
        ))}
      </View>

      <T variant="caption" color={colors.textMuted} center style={{ marginTop: 18 }}>
        By continuing, you agree to our{' '}
        <T variant="caption" weight="semibold" color={colors.textSecondary} onPress={() => showLegal('Terms of Service')}>
          Terms of Service
        </T>{' '}
        and{' '}
        <T variant="caption" weight="semibold" color={colors.textSecondary} onPress={() => showLegal('Privacy Policy')}>
          Privacy Policy
        </T>
      </T>

      <GoogleSheet
        visible={googleOpen}
        onClose={() => setGoogleOpen(false)}
        onPick={async (mail, name) => {
          setGoogleOpen(false);
          await signInWithGoogle(mail, name);
          tap('success');
          router.replace('/(tabs)/explore');
        }}
      />
    </Screen>
  );
}

const styles = StyleSheet.create({
  row: { flexDirection: 'row', justifyContent: 'space-between', alignItems: 'center', marginVertical: 16 },
  remember: { flexDirection: 'row', alignItems: 'center', gap: 8 },
  box: { width: 18, height: 18, borderRadius: 5, borderWidth: 1.5, borderColor: colors.borderStrong, alignItems: 'center', justifyConten
t: 'center' },
  boxOn: { backgroundColor: colors.primary, borderColor: colors.primary },
  formError: { flexDirection: 'row', gap: 8, alignItems: 'center', backgroundColor: colors.redSoft, padding: 12, borderRadius: radius.s
m, marginBottom: 12 },
  orRow: { flexDirection: 'row', alignItems: 'center', gap: 10, marginVertical: 18 },
  google: {
    height: 52,
    borderRadius: radius.md,
    borderWidth: 1,
    borderColor: colors.border,
    backgroundColor: colors.card,
    flexDirection: 'row',
    alignItems: 'center',
    justifyContent: 'center',
    gap: 10,
  },
  signupRow: { flexDirection: 'row', justifyContent: 'center', alignItems: 'center', marginTop: 20 },
  demo: { marginTop: 22, padding: 14, borderRadius: radius.md, backgroundColor: colors.card, borderWidth: 1, borderColor: colors.border
},
  demoHead: { flexDirection: 'row', alignItems: 'center', gap: 6, marginBottom: 10 },
  demoRow: { flexDirection: 'row', alignItems: 'center', paddingVertical: 4 },
  demoFill: { flex: 1, flexDirection: 'row', alignItems: 'center' },
  copyBtn: { width: 26, height: 26, borderRadius: 8, backgroundColor: colors.primarySofter, alignItems: 'center', justifyContent: 'cente
r' },
});
```

src/app/sign-up.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { router } from 'expo-router';
import { useState } from 'react';
import { StyleSheet, View } from 'react-native';

import { Button } from '@/components/ui/Button';
import { Input } from '@/components/ui/Input';
import { Press, tap } from '@/components/ui/Press';
import { Header, Screen } from '@/components/ui/Screen';
import { T } from '@/components/ui/T';
import { useAppStore } from '@/store/useAppStore';
```

```
import { colors, radius } from '@/theme';
import { passwordStrength, validateEmail, validateName, validatePassword } from '@/utils/validation';

export default function SignUp() {
  const signUp = useAppStore((s) => s.signUp);
  const [name, setName] = useState('');
  const [email, setEmail] = useState('');
  const [password, setPassword] = useState('');
  const [confirm, setConfirm] = useState('');
  const [agree, setAgree] = useState(false);
  const [errors, setErrors] = useState<Record<string, string | null | undefined>>({});
  const [loading, setLoading] = useState(false);
  const strength = passwordStrength(password);

  const submit = async () => {
    const e = {
      name: validateName(name),
      email: validateEmail(email),
      password: validatePassword(password),
      confirm: !confirm ? 're-enter your password' : confirm !== password ? 'Passwords don’t match' : null,
      agree: agree ? null : 'Please accept the terms to continue',
    };
    setErrors(e);
    if (Object.values(e).some(Boolean)) return tap('warning');
    setLoading(true);
    const res = await signUp(name, email, password);
    setLoading(false);
    if (!res.ok) {
      setErrors({ email: res.error });
      return tap('warning');
    }
    tap('success');
    router.replace('/(tabs)/explore');
    // Brand-new account: offer the personalization survey straight away.
    setTimeout(() => router.push('/personalize'), 350);
  };

  const bars = [colors.red, colors.orange, colors.yellow, '#6BAA75', colors.primary];

  return (
    <Screen header={<Header title="Create account" subtitle="Start planning trips with your crew" />}>
      <View style={{ gap: 14 }}>
        <Input label="Full name" icon="person-outline" value={name} onChangeText={setName} autoCapitalize="words" autoComplete="name" er
ror={errors.name} placeholder="e.g. Alex Rivera" />
        <Input label="Email address" icon="mail-outline" value={email} onChangeText={setEmail} autoCapitalize="none" keyboardType="email
-address" autoComplete="email" error={errors.email} placeholder="you@email.com" />
        <View style={{ gap: 8 }}>
          <Input label="Password" icon="lock-closed-outline" value={password} onChangeText={setPassword} secureToggle error={errors.pass
word} placeholder="At least 8 characters" textContentType="newPassword" />
          {password ? (
            <View style={{ gap: 6 }}>
              <View style={{ flexDirection: 'row', gap: 4 }}>
                {[0, 1, 2, 3].map((i) => (
                  <View key={i} style={[styles.bar, { backgroundColor: i < strength.score ? bars[strength.score] : colors.border }]} />
                ))}
              </View>
              <T variant="caption" color={colors.textSecondary}>
                Strength: {strength.label}
              </T>
            </View>
          ) : null}
        </View>
        <Input label="Confirm password" icon="lock-closed-outline" value={confirm} onChangeText={setConfirm} secureToggle error={errors.
confirm} placeholder="Re-enter password" />
        <Press onPress={() => setAgree((a) => !a)} style={styles.agree} accessibilityRole="checkbox" accessibilityState={{ checked: agre
e }}>
          <View style={[styles.box, agree && styles.boxOn]}>{agree ? <Ionicons name="checkmark" size={13} color={colors.white} /> : nul
l}</View>
          <T variant="small" color={colors.textSecondary} style={{ flex: 1 }}>
            I agree to the Terms of Service and Privacy Policy
          </T>
        </Press>
        {errors.agree ? (
          <T variant="caption" color={colors.red}>
            {errors.agree}
          </T>
        ) : null}
        <Button label="Create account" onPress={submit} loading={loading} />
        <View style={styles.note}>
          <Ionicons name="shield-checkmark-outline" size={16} color={colors.primary} />
          <T variant="caption" color={colors.textSecondary} style={{ flex: 1 }}>
            Your password is salted and hashed on-device — it’s never stored in plain text.
          </T>
        </View>
      </View>
    </Screen>
```

```
  );
}

const styles = StyleSheet.create({
  bar: { flex: 1, height: 5, borderRadius: 3 },
  agree: { flexDirection: 'row', alignItems: 'center', gap: 10 },
  box: { width: 20, height: 20, borderRadius: 6, borderWidth: 1.5, borderColor: colors.borderStrong, alignItems: 'center', justifyConten
t: 'center' },
  boxOn: { backgroundColor: colors.primary, borderColor: colors.primary },
  note: { flexDirection: 'row', gap: 8, alignItems: 'center', padding: 12, borderRadius: radius.sm, backgroundColor: colors.primarySofte
r },
});
```

src/app/survey.tsx

```
import { Image } from 'expo-image';
import { LinearGradient } from 'expo-linear-gradient';
import { router } from 'expo-router';
import { useState } from 'react';
import { StyleSheet, View } from 'react-native';

import { AddToTripSheet } from '@/components/explore';
import { Button } from '@/components/ui/Button';
import { Pill, ProgressBar } from '@/components/ui/bits';
import { Press, tap } from '@/components/ui/Press';
import { Header, Screen } from '@/components/ui/Screen';
import { T } from '@/components/ui/T';
import { destinationById, destinations } from '@/data/destinations';
import { imageSource } from '@/data/images';
import { surveyQuestions } from '@/data/reference';
import { useAppStore } from '@/store/useAppStore';
import { colors, radius } from '@/theme';
import { money } from '@/utils/format';

/** Scores every destination against the answers: tag matches + budget fit + rating. */
function recommend(answers: string[][]): { id: string; match: number }[] {
  const tags = answers.flat();
  const budgetTag = tags.find((t) => t.startsWith('$'));
  const budget = budgetTag ? Number(budgetTag.slice(1)) : 2;
  const scored = destinations.map((d) => {
    let s = 0;
    for (const t of tags) if (!t.startsWith('$') && d.tags.includes(t as never)) s += 2;
    s -= Math.abs(d.priceLevel - budget) * 1.5;
    s += (d.rating - 4.5) * 4;
    return { id: d.id, s };
  });
  const max = Math.max(...scored.map((x) => x.s));
  const min = Math.min(...scored.map((x) => x.s));
  return scored
    .sort((a, b) => b.s - a.s)
    .slice(0, 5)
    .map((x) => ({ id: x.id, match: Math.round(70 + ((x.s - min) / Math.max(1, max - min)) * 29) }));
}

export default function Survey() {
  const saved = useAppStore((s) => s.surveyResult);
  const save = useAppStore((s) => s.setSurveyResult);
  const [step, setStep] = useState(saved ? surveyQuestions.length : 0);
  const [answers, setAnswers] = useState<string[][]>([]);
  const [results, setResults] = useState<{ id: string; match: number }[]>(saved ? saved.map((id, i) => ({ id, match: 98 - i * 4 })) :
[]);
  const [adding, setAdding] = useState<string | null>(null);

  const pick = (tags: string[]) => {
    const next = [...answers.slice(0, step), tags];
    setAnswers(next);
    tap();
    if (step + 1 >= surveyQuestions.length) {
      const r = recommend(next);
      setResults(r);
      save(r.map((x) => x.id));
    }
    setStep(step + 1);
  };

  if (step >= surveyQuestions.length) {
    return (
      <Screen header={<Header title="Your matches" subtitle="Based on your travel style" />}>
        <T variant="small" color={colors.textSecondary} style={{ marginBottom: 14 }}>
          We matched your answers against 32 destinations — tap one to explore it, or add it to a trip.
        </T>
        <View style={{ gap: 14 }}>
          {results.map((r, i) => {
            const d = destinationById(r.id)!;
            return (
              // The Add button sits next to the card button, not inside it: on the web a button inside a button is invalid HTML.
```

```
              <View key={r.id} style={[styles.result, i === 0 && { height: 300 }]}>
                <Press onPress={() => router.push({ pathname: '/destination/[id]', params: { id: d.id } })} style={StyleSheet.absoluteFi
ll} scaleTo={0.985} accessibilityLabel={`${d.city}, ${r.match}% match`}>
                  <Image source={imageSource(d.image)} style={StyleSheet.absoluteFill} contentFit="cover" transition={250} />
                  <LinearGradient colors={['rgba(0,0,0,0)', 'rgba(0,0,0,0.72)']} locations={[0.35, 1]} style={StyleSheet.absoluteFill} /
>
                  <View style={styles.match}>
                    <T variant="micro" weight="bold" color={colors.primary}>
                      {r.match}% match
                    </T>
                  </View>
                  <View style={styles.resultText}>
                    {i === 0 ? (
                      <T variant="kicker" color="rgba(255,255,255,0.85)">
                        Top pick for you
                      </T>
                    ) : null}
                    <T variant={i === 0 ? 'display' : 'h2'} color={colors.white}>
                      {d.city}
                    </T>
                    <T variant="small" color="rgba(255,255,255,0.9)" numberOfLines={2}>
                      {d.blurb}
                    </T>
                    <View style={styles.resultPills}>
                      <Pill label={`~${money(d.dailyCost)}/day`} tone="yellow" />
                      <Pill label={`★ ${d.rating}`} />
                    </View>
                  </View>
                </Press>
                <View style={styles.resultAdd}>
                  <Button label="Add" size="sm" full={false} variant="light" icon="add" onPress={() => setAdding(d.id)} />
                </View>
              </View>
            );
          })}
        </View>
        <Button
          label="Retake survey"
          variant="secondary"
          icon="refresh-outline"
          style={{ marginTop: 16 }}
          onPress={() => {
            setAnswers([]);
            setStep(0);
          }}
        />
        <AddToTripSheet destinationId={adding} onClose={() => setAdding(null)} />
      </Screen>
    );
  }

  const q = surveyQuestions[step];
  return (
    <Screen header={<Header title="Find your next trip" subtitle={`Question ${step + 1} of ${surveyQuestions.length}`} />}>
      <ProgressBar value={step / surveyQuestions.length} height={6} />
      <T variant="h1" style={{ marginTop: 22, marginBottom: 18 }} accessibilityRole="header">
        {q.q}
      </T>
      <View style={styles.grid}>
        {q.options.map((o) => (
          <Press key={o.label} onPress={() => pick(o.tags)} style={styles.option} accessibilityLabel={o.label}>
            <T style={{ fontSize: 40, lineHeight: 48 }}>{o.emoji}</T>
            <T variant="small" weight="semibold" center>
              {o.label}
            </T>
          </Press>
        ))}
      </View>
      {step > 0 ? <Button label="Back" variant="ghost" icon="arrow-back" onPress={() => setStep(step - 1)} style={{ marginTop: 16 }} />
: null}
    </Screen>
  );
}

const styles = StyleSheet.create({
  grid: { flexDirection: 'row', flexWrap: 'wrap', gap: 12 },
  option: { width: '47.5%', aspectRatio: 1.05, borderRadius: radius.xl, backgroundColor: colors.card, borderWidth: 1, borderColor: color
s.border, alignItems: 'center', justifyContent: 'center', gap: 8, padding: 12 },
  result: { height: 220, borderRadius: radius.xl, overflow: 'hidden' },
  match: { position: 'absolute', top: 12, right: 12, backgroundColor: colors.white, paddingHorizontal: 10, paddingVertical: 5, borderRad
ius: 999 },
  resultText: { position: 'absolute', left: 16, right: 16, bottom: 14 },
  resultPills: { flexDirection: 'row', gap: 6, marginTop: 6, alignItems: 'center', minHeight: 36 },
  resultAdd: { position: 'absolute', right: 16, bottom: 14 },
});
```

src/app/trips.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { Image } from 'expo-image';
import { router } from 'expo-router';
import { StyleSheet, View } from 'react-native';

import { Button } from '@/components/ui/Button';
import { AvatarStack, EmptyState, Pill } from '@/components/ui/bits';
import { toast } from '@/components/ui/feedback';
import { Press } from '@/components/ui/Press';
import { Header, Screen } from '@/components/ui/Screen';
import { T } from '@/components/ui/T';
import { destinationById } from '@/data/destinations';
import { imageSource } from '@/data/images';
import { totalSpent, tripStatus, useAppStore, useMyTrips } from '@/store/useAppStore';
import type { Trip, TripStatus } from '@/store/types';
import { colors, radius, shadow } from '@/theme';
import { money } from '@/utils/format';
import { rangeLabel } from '@/utils/time';

const ORDER: { key: TripStatus; title: string }[] = [
  { key: 'active', title: 'In progress' },
  { key: 'planning', title: 'Upcoming' },
  { key: 'completed', title: 'Completed' },
];

export default function Trips() {
  const trips = useMyTrips();
  const activeId = useAppStore((s) => s.activeTripId);
  const setActive = useAppStore((s) => s.setActiveTrip);

  const open = (t: Trip) => {
    setActive(t.id);
    toast(`Switched to ${t.name}`);
    router.navigate('/(tabs)/itinerary');
  };

  return (
    <Screen header={<Header title="My trips" subtitle={`${trips.length} trip${trips.length === 1 ? '' : 's'} · plan, travel, remember`}
/>}>
      <View style={{ flexDirection: 'row', gap: 10, marginBottom: 18 }}>
        <Button label="New trip" icon="add" size="md" onPress={() => router.push('/new-trip')} style={{ flex: 1 }} />
        <Button label="Join with code" icon="people-outline" variant="secondary" size="md" onPress={() => router.push('/join')} style={{
flex: 1 }} />
      </View>
      {trips.length === 0 ? <EmptyState icon="airplane-outline" title="No trips yet" body="Create one or join a friend’s trip with their
invite code." /> : null}
      {ORDER.map(({ key, title }) => {
        const list = trips.filter((t) => tripStatus(t) === key);
        if (!list.length) return null;
        return (
          <View key={key} style={{ marginBottom: 18, gap: 10 }}>
            <T variant="kicker" color={colors.textSecondary}>
              {title}
            </T>
            {list.map((t) => {
              const d = destinationById(t.destinationId);
              const isActive = t.id === activeId;
              return (
                <Press key={t.id} onPress={() => open(t)} style={[styles.card, isActive && styles.cardActive]} accessibilityLabel={`${t.
name}${isActive ? ', current trip' : ''}`}>
                  <Image source={imageSource(t.cover)} style={styles.cover} contentFit="cover" />
                  <View style={{ flex: 1, gap: 4 }}>
                    <View style={{ flexDirection: 'row', alignItems: 'center', gap: 6 }}>
                      <T variant="title" numberOfLines={1} style={{ flex: 1 }}>
                        {t.name}
                      </T>
                      {isActive ? <Ionicons name="checkmark-circle" size={18} color={colors.primary} /> : null}
                    </View>
                    <T variant="caption" color={colors.textSecondary}>
                      {d?.city}, {d?.country}
                    </T>
                    <T variant="caption" color={colors.textSecondary}>
                      {rangeLabel(t.startDate, t.endDate)}
                    </T>
                    <View style={{ flexDirection: 'row', alignItems: 'center', gap: 8, marginTop: 2 }}>
                      <AvatarStack people={t.members} max={4} size={22} />
                      <Pill label={`${money(totalSpent(t))} / ${money(t.budget)}`} tone="gray" />
                    </View>
                  </View>
                </Press>
              );
            })}
          </View>
        );
```

```
      ))}
    </Screen>
  );
}

const styles = StyleSheet.create({
  card: { flexDirection: 'row', gap: 12, backgroundColor: colors.card, borderRadius: radius.lg, padding: 10, borderWidth: 1, borderColo
r: colors.border, ...shadow },
  cardActive: { borderColor: colors.primary, borderWidth: 2 },
  cover: { width: 88, height: 96, borderRadius: radius.md },
});
```

src/app/weather.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { useLocalSearchParams } from 'expo-router';
import { useState } from 'react';
import { ScrollView, StyleSheet, View } from 'react-native';

import { Button } from '@/components/ui/Button';
import { Card, Chip, Pill } from '@/components/ui/bits';
import { toast } from '@/components/ui/feedback';
import { Header, Screen } from '@/components/ui/Screen';
import { T } from '@/components/ui/T';
import { destinationById } from '@/data/destinations';
import { useWeather } from '@/hooks/useTrip';
import { cToF, fetchWeather, weatherInfo } from '@/services/weather';
import { useActiveTrip, useAppStore } from '@/store/useAppStore';
import { colors, radius } from '@/theme';
import { timeAgo } from '@/utils/format';
import { dayLabel, weekdayShort } from '@/utils/time';

export default function Weather() {
  const params = useLocalSearchParams<{ id?: string }>();
  const trip = useActiveTrip();
  const options = trip ? [trip.destinationId, ...trip.extraDestinationIds] : [];
  const [id, setId] = useState(params.id ?? trip?.destinationId ?? 'tokyo');
  const [unitF, setUnitF] = useState(false);
  const [loading, setLoading] = useState(false);
  const addPacking = useAppStore((s) => s.addPackingItems);
  const w = useWeather(id);
  const d = destinationById(id)!;
  const u = (c: number) => (unitF ? `${cToF(c)}°F` : `${c}°C`);
  const info = w ? weatherInfo(w.current.code) : null;
  const rainy = w?.daily.filter((x) => x.rain >= 50) ?? [];
  const best = w?.daily.slice().sort((a, b) => a.rain - b.rain || b.max - a.max)[0];
  const maxT = Math.max(...(w?.daily.map((x) => x.max) ?? [30]));
  const minT = Math.min(...(w?.daily.map((x) => x.min) ?? [0]));

  const tips: string[] = [];
  if (w) {
    if (rainy.length) tips.push('Compact umbrella or rain jacket');
    if (maxT >= 27) tips.push('Sunscreen, hat & refillable water bottle');
    if (minT <= 12) tips.push('Warm layer for evenings');
    if (minT <= 3) tips.push('Gloves & beanie');
    tips.push('Comfortable walking shoes');
  }

  return (
    <Screen
      header={<Header title={`Weather · ${d.city}`} subtitle={w ? `${w.live ? 'Live forecast' : 'Estimated (offline)'} · updated ${timeA
go(w.fetchedAt)}` : 'Loading…'} />}
      onRefresh={async () => {
        setLoading(true);
        const r = await fetchWeather(id, true);
        setLoading(false);
        toast(r?.live ? 'Forecast updated' : 'Offline — showing the last saved forecast', { tone: r?.live ? 'ok' : 'warn' });
      }}
      refreshing={loading}>
      {options.length > 1 ? (
        <ScrollView horizontal showsHorizontalScrollIndicator={false} contentContainerStyle={{ gap: 6 }} style={{ marginBottom: 12 }}>
          {options.map((o) => (
            <Chip key={o} small label={destinationById(o)?.city ?? o} active={o === id} onPress={() => setId(o)} />
          ))}
        </ScrollView>
      ) : null}

      {w && info ? (
        <>
          <View style={styles.hero}>
            <View style={{ flex: 1 }}>
              <T variant="kicker" color="rgba(255,255,255,0.75)">
                Right now
              </T>
              <T variant="display" color={colors.white} style={{ fontSize: 54, lineHeight: 60 }}>
                {u(w.current.temp)}
```

```
              </T>
              <T variant="title" color={colors.white}>
                {info.label}
              </T>
              <View style={{ flexDirection: 'row', gap: 14, marginTop: 8 }}>
                <T variant="caption" color="rgba(255,255,255,0.85)">
                  💧 {w.current.humidity}%
                </T>
                <T variant="caption" color="rgba(255,255,255,0.85)">
                  💨 {w.current.wind} km/h
                </T>
              </View>
            </View>
            <Ionicons name={info.icon} size={84} color="#FFD27A" />
          </View>

          <View style={styles.unitRow}>
            {!w.live ? <Pill label="Estimate — connect to refresh" tone="orange" icon="cloud-offline-outline" /> : <Pill label="Live · O
pen-Meteo" icon="radio-button-on" />}
            <View style={{ flexDirection: 'row', gap: 6 }}>
              <Chip small label="°C" active={!unitF} onPress={() => setUnitF(false)} />
              <Chip small label="°F" active={unitF} onPress={() => setUnitF(true)} />
            </View>
          </View>

          <Card padded={false} style={{ paddingHorizontal: 14, paddingVertical: 6 }}>
            {w.daily.map((x, i) => {
              const di = weatherInfo(x.code);
              const span = Math.max(1, maxT - minT);
              return (
                <View key={x.date} style={[styles.day, i === w.daily.length - 1 && { borderBottomWidth: 0 }]}>
                  <View style={{ width: 70 }}>
                    <T variant="small" weight="semibold">
                      {i === 0 ? 'Today' : weekdayShort(x.date)}
                    </T>
                    <T variant="micro" color={colors.textMuted}>
                      {dayLabel(x.date)}
                    </T>
                  </View>
                  <Ionicons name={di.icon} size={22} color={x.rain >= 50 ? colors.blue : '#E8A23A'} />
                  <T variant="caption" color={x.rain >= 50 ? colors.blue : colors.textMuted} style={{ width: 38, textAlign: 'right' }}>
                    {x.rain}%
                  </T>
                  <T variant="small" color={colors.textSecondary} style={{ width: 38, textAlign: 'right' }}>
                    {unitF ? cToF(x.min) : x.min}°
                  </T>
                  <View style={styles.bar}>
                    <View style={[styles.barFill, { left: `${((x.min - minT) / span) * 100}%`, right: `${((maxT - x.max) / span) * 100}%
` }]} />
                  </View>
                  <T variant="small" weight="semibold" style={{ width: 34 }}>
                    {unitF ? cToF(x.max) : x.max}°
                  </T>
                </View>
              );
            })}
          </Card>

          {best ? (
            <Card style={{ marginTop: 14, flexDirection: 'row', gap: 12, alignItems: 'center' }}>
              <Ionicons name="sunny" size={26} color="#E8A23A" />
              <View style={{ flex: 1 }}>
                <T variant="small" weight="semibold">
                  Best outdoor day: {weekdayShort(best.date)} {dayLabel(best.date)}
                </T>
                <T variant="caption" color={colors.textSecondary}>
                  {best.rain}% rain, up to {u(best.max)} — plan hikes and day trips here.
                </T>
              </View>
            </Card>
          ) : null}

          <Card style={{ marginTop: 14, gap: 8 }}>
            <T variant="title">What to pack</T>
            {tips.map((t) => (
              <View key={t} style={{ flexDirection: 'row', gap: 8, alignItems: 'center' }}>
                <Ionicons name="checkmark-circle-outline" size={16} color={colors.primary} />
                <T variant="small">{t}</T>
              </View>
            ))}
            {trip ? (
              <Button
                label="Add these to the packing list"
                variant="soft"
                size="md"
                icon="briefcase-outline"
```

```
                onPress={() => {
                  addPacking(trip.id, tips.map((text) => ({ text, category: 'Weather' })));
                  toast('Added to the shared packing list');
                }}
              />
            ) : null}
          </Card>
        </>
      ) : (
        <Card>
          <T variant="small" color={colors.textSecondary}>
            Fetching the forecast…
          </T>
        </Card>
      )}
    </Screen>
  );
}

const styles = StyleSheet.create({
  hero: { flexDirection: 'row', alignItems: 'center', backgroundColor: '#2E6B8F', borderRadius: radius.xl, padding: 20 },
  unitRow: { flexDirection: 'row', justifyContent: 'space-between', alignItems: 'center', marginVertical: 14 },
  day: { flexDirection: 'row', alignItems: 'center', gap: 10, paddingVertical: 10, borderBottomWidth: 1, borderBottomColor: colors.borde
r },
  bar: { flex: 1, height: 6, borderRadius: 3, backgroundColor: colors.bgAlt },
  barFill: { position: 'absolute', top: 0, bottom: 0, borderRadius: 3, backgroundColor: '#F2B33D' },
});
```

src/components/CurrencyConverter.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { useState } from 'react';
import { StyleSheet, TextInput, View } from 'react-native';

import { convert, useRates } from '@/services/currency';
import { colors, fonts, radius } from '@/theme';
import { money } from '@/utils/format';
import { parseAmount } from '@/utils/validation';

import { CurrencyPickerSheet } from './CurrencyPickerSheet';
import { Press } from './ui/Press';
import { T } from './ui/T';

/** Two-way converter used on the Budget tab and the Currency screen. */
export function CurrencyConverter({ from, to, onChange }: { from: string; to: string; onChange: (from: string, to: string) => void }) {
  const rates = useRates();
  const [amount, setAmount] = useState('50');
  const [picking, setPicking] = useState<'from' | 'to' | null>(null);
  const n = parseAmount(amount);
  const out = n === null ? null : convert(n, from, to, rates);
  const unit = convert(1, from, to, rates);

  const code = (which: 'from' | 'to', c: string) => (
    <Press onPress={() => setPicking(which)} style={styles.code} accessibilityLabel={`Change ${which} currency, currently ${c}`}>
      <T variant="small" weight="bold">
        {c}
      </T>
      <Ionicons name="chevron-down" size={16} color={colors.text} />
    </Press>
  );

  return (
    <View style={{ gap: 10 }}>
      <View style={styles.row}>
        <View style={[styles.box, n === null && !!amount && { borderColor: colors.red }]}>
          <TextInput
            value={amount}
            onChangeText={(v) => setAmount(v.replace(/[^0-9.,]/g, ''))}
            keyboardType="decimal-pad"
            style={styles.input}
            accessibilityLabel={`Amount in ${from}`}
            maxLength={12}
            placeholder="0"
            placeholderTextColor={colors.textMuted}
          />
        </View>
        {code('from', from)}
      </View>
      <View style={styles.swapRow}>
        <Press onPress={() => onChange(to, from)} style={styles.swap} accessibilityLabel="Swap currencies" hitSlop={8}>
          <Ionicons name="swap-vertical" size={16} color={colors.text} />
        </Press>
      </View>
      <View style={styles.row}>
        <View style={styles.box}>
          <T variant="h3" weight="bold" numberOfLines={1} adjustsFontSizeToFit style={{ flex: 1 }}>
```

```
            {out === null ? '—' : money(out, to, { decimals: true })}
          </T>
        </View>
        {code('to', to)}
      </View>
      {n === null && amount ? (
        <T variant="caption" color={colors.red}>
          Enter a number, e.g. 25 or 25.50
        </T>
      ) : null}
      <View style={styles.rateRow}>
        <View style={[styles.liveDot, { backgroundColor: rates.live ? colors.positive : colors.orange }]} />
        <T variant="caption" color={colors.textSecondary}>
          1 {from} = {unit < 0.01 ? unit.toFixed(5) : unit.toFixed(unit > 100 ? 1 : 4)} {to} · {rates.live ? 'live rate' : 'offline rat
e'}
        </T>
      </View>
      <CurrencyPickerSheet
        visible={!!picking}
        onClose={() => setPicking(null)}
        value={picking === 'from' ? from : to}
        onPick={(c) => (picking === 'from' ? onChange(c, to) : onChange(from, c))}
      />
    </View>
  );
}

const styles = StyleSheet.create({
  row: { flexDirection: 'row', alignItems: 'center', gap: 10 },
  box: { flex: 1, minWidth: 0, height: 50, borderRadius: radius.md, backgroundColor: colors.bgAlt, borderWidth: 1, borderColor: 'transpa
rent', flexDirection: 'row', alignItems: 'center', paddingHorizontal: 16 },
  input: { flex: 1, minWidth: 0, fontFamily: fonts.semibold, fontSize: 18, color: colors.text, ...({ outlineStyle: 'none' } as object)
},
  code: { width: 108, height: 50, borderRadius: radius.md, backgroundColor: colors.bgAlt, flexDirection: 'row', alignItems: 'center', ju
stifyContent: 'space-between', paddingHorizontal: 14 },
  swapRow: { alignItems: 'center' },
  swap: { width: 34, height: 34, borderRadius: 17, backgroundColor: colors.bgAlt, alignItems: 'center', justifyContent: 'center' },
  rateRow: { flexDirection: 'row', alignItems: 'center', gap: 6 },
  liveDot: { width: 7, height: 7, borderRadius: 4 },
});
```

src/components/CurrencyPickerSheet.tsx

```
import { useState } from 'react';
import { StyleSheet, TextInput, View } from 'react-native';

import { currencies } from '@/data/reference';
import { colors, fonts, radius } from '@/theme';

import { Press } from './ui/Press';
import { Sheet } from './ui/Sheet';
import { T } from './ui/T';

export function CurrencyPickerSheet({ visible, onClose, value, onPick, title = 'Choose currency' }: { visible: boolean; onClose: () => v
oid; value: string; onPick: (code: string) => void; title?: string }) {
  const [q, setQ] = useState('');
  const list = currencies.filter((c) => !q || c.code.toLowerCase().includes(q.toLowerCase()) || c.name.toLowerCase().includes(q.toLowerC
ase()));
  return (
    <Sheet visible={visible} onClose={onClose} title={title}>
      <TextInput value={q} onChangeText={setQ} placeholder="Search currency…" placeholderTextColor={colors.textMuted} style={styles.sear
ch} autoCapitalize="none" />
      <View style={{ gap: 6 }}>
        {list.map((c) => (
          <Press
            key={c.code}
            onPress={() => {
              onPick(c.code);
              onClose();
            }}
            style={[styles.row, c.code === value && styles.rowOn]}
            accessibilityState={{ selected: c.code === value }}>
            <View style={styles.sym}>
              <T variant="small" weight="bold" color={colors.primary}>
                {c.symbol.length > 3 ? c.code.slice(0, 1) : c.symbol}
              </T>
            </View>
            <T variant="bodySm" weight="semibold" style={{ width: 48 }}>
              {c.code}
            </T>
            <T variant="small" color={colors.textSecondary} style={{ flex: 1 }}>
              {c.name}
            </T>
          </Press>
        ))}
      </View>
```

```
    </Sheet>
  );
}

const styles = StyleSheet.create({
  search: { height: 44, borderRadius: radius.sm, borderWidth: 1, borderColor: colors.border, paddingHorizontal: 12, fontFamily: fonts.re
gular, color: colors.text, backgroundColor: colors.cardMuted },
  row: { flexDirection: 'row', alignItems: 'center', gap: 10, padding: 10, borderRadius: radius.sm },
  rowOn: { backgroundColor: colors.primarySofter },
  sym: { width: 32, height: 32, borderRadius: 16, backgroundColor: colors.primarySofter, alignItems: 'center', justifyContent: 'center'
},
});
```

src/components/GuidedTour.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { useRef, useState } from 'react';
import { Animated, Modal, ScrollView, StyleSheet, View, useWindowDimensions } from 'react-native';
import { useSafeAreaInsets } from 'react-native-safe-area-context';

import { useAppStore } from '@/store/useAppStore';
import { MAX_WIDTH, colors, radius, shadow } from '@/theme';

import { LogoMark } from './Logo';
import { Button } from './ui/Button';
import type { IconName } from './ui/bits';
import { FadeIn, nativeDriver, useMotion } from './ui/motion';
import { Press } from './ui/Press';
import { T } from './ui/T';

const STEPS: { icon: IconName; tab: string; title: string; body: string; points: string[] }[] = [
  { icon: 'home-outline', tab: 'Home', title: 'Your trip at a glance', body: 'The moment you open the app you see the hero photo, invite
code, weather and your group budget.', points: ['Up next and countdown', 'Weather and local time', 'Budget progress'] },
  { icon: 'calendar-outline', tab: 'Plan', title: 'Plan it together', body: 'Build the itinerary day by day. Everyone votes “I’m in”, th
en the group locks the winners in.', points: ['Itinerary by day', 'Vote on ideas', 'Lock in the plan'] },
  { icon: 'wallet-outline', tab: 'Budget', title: 'Split it fairly', body: 'Log expenses in any currency and see exactly who owes who. S
nap a receipt so nothing gets lost.', points: ['Who owes who', 'Receipts and categories', 'Settle up in fewer payments'] },
  { icon: 'chatbubbles-outline', tab: 'Chat', title: 'Talk it out', body: 'Message the crew in one place, or ask Scout, the built-in AI,
about weather, food and what’s next.', points: ['Group chat', 'Scout AI assistant', 'Works offline'] },
  { icon: 'grid-outline', tab: 'More', title: 'Everything else', body: 'Offline documents, the trip map, a packing list, the photo dump,
a trip quiz and more live here.', points: ['Offline docs', 'Trip map', 'Packing, photos and quiz'] },
];

/** First-run walkthrough: five swipeable cards, one per tab. Marks the tour as seen when it closes. */
export function GuidedTour({ visible, onClose }: { visible: boolean; onClose: () => void }) {
  const setToured = useAppStore((s) => s.setToured);
  const finish = () => {
    setToured(true);
    onClose();
  };
  return (
    <Modal visible={visible} animationType="fade" onRequestClose={finish} statusBarTranslucent>
      {visible ? <TourBody onFinish={finish} /> : null}
    </Modal>
  );
}

function TourBody({ onFinish }: { onFinish: () => void }) {
  const insets = useSafeAreaInsets();
  const { width } = useWindowDimensions();
  const motion = useMotion();
  const pageW = Math.min(width, MAX_WIDTH);
  const scroller = useRef<ScrollView>(null);
  const [index, setIndex] = useState(0);
  const [scrollX] = useState(() => new Animated.Value(0));
  const last = index === STEPS.length - 1;

  const goTo = (i: number) => {
    setIndex(i);
    scroller.current?.scrollTo({ x: i * pageW, animated: true });
  };

  return (
    <View style={[styles.root, { paddingTop: insets.top + 8, paddingBottom: Math.max(insets.bottom, 16) }]}>
      <View style={[styles.column, { width: pageW }]}>
        <View style={styles.top}>
          <View style={styles.brand}>
            <LogoMark size={30} />
            <T variant="small" weight="extrabold">
              TogetherWeGo
            </T>
          </View>
          {last ? null : (
            <Press onPress={onFinish} hitSlop={10} accessibilityLabel="Skip the tour">
              <T variant="small" weight="semibold" color={colors.textSecondary}>
                Skip
```

```
              </T>
            </Press>
          )}
        </View>

        <FadeIn style={{ flex: 1 }} duration={420} distance={16}>
          <Animated.ScrollView
            ref={scroller}
            horizontal
            pagingEnabled
            showsHorizontalScrollIndicator={false}
            scrollEventThrottle={16}
            onScroll={Animated.event([{ nativeEvent: { contentOffset: { x: scrollX } } }], {
              useNativeDriver: nativeDriver,
              listener: (e: { nativeEvent: { contentOffset: { x: number } } }) => setIndex(Math.max(0, Math.min(STEPS.length - 1, Math.r
ound(e.nativeEvent.contentOffset.x / pageW)))),
            })}
            style={{ flex: 1 }}>
            {STEPS.map((s, i) => {
              const range = [(i - 1) * pageW, i * pageW, (i + 1) * pageW];
              const pop = motion ? { opacity: scrollX.interpolate({ inputRange: range, outputRange: [0.45, 1, 0.45], extrapolate: 'clam
p' }), transform: [{ scale: scrollX.interpolate({ inputRange: range, outputRange: [0.92, 1, 0.92], extrapolate: 'clamp' }) }] } : null;
              return (
                <View key={s.tab} style={[styles.page, { width: pageW }]} accessible accessibilityLabel={`Step ${i + 1} of ${STEPS.lengt
h}. ${s.tab}. ${s.title}. ${s.body}`}>
                  <Animated.View style={[styles.card, pop]}>
                    <T variant="display" color={colors.bgAlt} style={styles.number}>
                      {`0${i + 1}`}
                    </T>
                    <View style={styles.icon}>
                      <Ionicons name={s.icon} size={30} color={colors.onAccent} />
                    </View>
                    <T variant="kicker" color={colors.textSecondary} style={{ marginTop: 22 }}>
                      {s.tab}
                    </T>
                    <T variant="display" style={{ fontSize: 34, lineHeight: 38, marginTop: 4 }}>
                      {s.title}
                    </T>
                    <T variant="body" color={colors.textSecondary} style={{ marginTop: 10 }}>
                      {s.body}
                    </T>
                    <View style={{ gap: 10, marginTop: 20 }}>
                      {s.points.map((p) => (
                        <View key={p} style={styles.point}>
                          <View style={styles.tick}>
                            <Ionicons name="checkmark" size={13} color={colors.onAccent} />
                          </View>
                          <T variant="bodySm" weight="semibold">
                            {p}
                          </T>
                        </View>
                      ))}
                    </View>
                  </Animated.View>
                </View>
              );
            })}
          </Animated.ScrollView>
        </FadeIn>

        <View style={styles.dots} accessibilityRole="tablist">
          {STEPS.map((s, i) => (
            <Press key={s.tab} onPress={() => goTo(i)} hitSlop={8} haptic={false} accessibilityRole="tab" accessibilityState={{ selecte
d: i === index }} accessibilityLabel={`Step ${i + 1} of ${STEPS.length}: ${s.tab}`}>
              <View style={[styles.dot, i === index && styles.dotOn]} />
            </Press>
          ))}
        </View>
        <View style={{ paddingHorizontal: 20 }}>
          <Button label={last ? 'Got it' : 'Next'} iconRight={last ? 'checkmark' : 'arrow-forward'} onPress={() => (last ? onFinish() :
goTo(index + 1))} />
        </View>
      </View>
    </View>
  );
}

const styles = StyleSheet.create({
  root: { flex: 1, backgroundColor: colors.bg, alignItems: 'center' },
  column: { flex: 1 },
  top: { flexDirection: 'row', alignItems: 'center', justifyContent: 'space-between', paddingHorizontal: 20, height: 44 },
  brand: { flexDirection: 'row', alignItems: 'center', gap: 8 },
  page: { paddingHorizontal: 20, justifyContent: 'center' },
  card: { backgroundColor: colors.card, borderRadius: 28, padding: 24, borderWidth: 1, borderColor: colors.border, ...shadow },
  number: { position: 'absolute', top: 14, right: 20, fontSize: 64, lineHeight: 68 },
  icon: { width: 64, height: 64, borderRadius: 32, backgroundColor: colors.accent, alignItems: 'center', justifyContent: 'center' },
```

```
  point: { flexDirection: 'row', alignItems: 'center', gap: 10 },
  tick: { width: 22, height: 22, borderRadius: 11, backgroundColor: colors.accent, alignItems: 'center', justifyContent: 'center' },
  dots: { flexDirection: 'row', alignSelf: 'center', alignItems: 'center', gap: 8, paddingVertical: 18 },
  dot: { width: 8, height: 8, borderRadius: radius.pill, backgroundColor: colors.borderStrong },
  dotOn: { width: 24, backgroundColor: colors.primary },
});
```

src/components/Logo.tsx

```
import { MaterialCommunityIcons } from '@expo/vector-icons';
import { View } from 'react-native';

import { colors } from '@/theme';

import { T } from './ui/T';

/** Brand mark: forest-green tile with a take-off plane (matches the app icon). */
export function LogoMark({ size = 40 }: { size?: number }) {
  return (
    <View
      accessibilityLabel="TogetherWeGo logo"
      style={{ width: size, height: size, borderRadius: size * 0.28, backgroundColor: colors.primary, alignItems: 'center', justifyConte
nt: 'center' }}>
      <MaterialCommunityIcons name="airplane-takeoff" size={size * 0.56} color={colors.white} />
    </View>
  );
}

export function Logo({ size = 40 }: { size?: number }) {
  return (
    <View style={{ flexDirection: 'row', alignItems: 'center', gap: 10 }}>
      <LogoMark size={size} />
      <T variant="h3" weight="bold">
        TogetherWeGo
      </T>
    </View>
  );
}
```

src/components/QrCode.tsx

```
import { useMemo } from 'react';
import Svg, { Path, Rect } from 'react-native-svg';

import { qrMatrix, qrPath } from '@/utils/qr';

const QUIET = 4; // squares of empty border all around; scanners need it to find the code

/** A QR code drawn with SVG. Always black on white, whatever the app theme, because that is what scanners read best. */
export function QrCode({ value, size = 220, accessibilityLabel }: { value: string; size?: number; accessibilityLabel?: string }) {
  const { total, path } = useMemo(() => {
    const matrix = qrMatrix(value);
    return { total: matrix.length + QUIET * 2, path: qrPath(matrix) };
  }, [value]);
  const px = Math.max(1, Math.floor(size / total)) * total; // a whole number of screen points per square keeps the edges sharp
  return (
    <Svg width={px} height={px} viewBox={`${-QUIET} ${-QUIET} ${total} ${total}`} accessibilityRole="image" accessibilityLabel={accessib
ilityLabel}>
      <Rect x={-QUIET} y={-QUIET} width={total} height={total} fill="#FFFFFF" />
      <Path d={path} fill="#000000" />
    </Svg>
  );
}
```

src/components/ReminderService.tsx

```
import { useEffect, useMemo } from 'react';
import { AppState } from 'react-native';

import { checkDueReminders, syncReminders } from '@/services/notifications';
import { useAppStore } from '@/store/useAppStore';

/**
 * Invisible component that keeps reminders live:
 * - re-schedules OS notifications whenever reminder-relevant data changes
 * - polls every 20 s while the app is open to surface due reminders in-app
 */
export function ReminderService() {
  const trips = useAppStore((s) => s.trips);
  const lead = useAppStore((s) => s.settings.reminderLeadMin);
  const enabled = useAppStore((s) => s.settings.notificationsEnabled);
  const userId = useAppStore((s) => s.session?.userId);

  const signature = useMemo(
    () =>
      [lead, enabled, userId]
```

```
        .concat(trips.flatMap((t) => t.activities.filter((a) => a.reminder).map((a) => `${a.id}|${a.date}|${a.time}`)))
        .join(','),
    [trips, lead, enabled, userId],
  );

  useEffect(() => {
    const timer = setTimeout(() => {
      syncReminders().catch(() => {});
    }, 1200);
    return () => clearTimeout(timer);
  }, [signature]);

  useEffect(() => {
    checkDueReminders();
    const id = setInterval(checkDueReminders, 20000);
    const sub = AppState.addEventListener('change', (st) => st === 'active' && checkDueReminders());
    return () => {
      clearInterval(id);
      sub.remove();
    };
  }, []);

  return null;
}
```

src/components/ShareSheet.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { StyleSheet, View } from 'react-native';

import { copy, shareText, shareTo, socialTargets } from '@/services/share';
import { colors, radius } from '@/theme';

import { Button } from './ui/Button';
import { Press } from './ui/Press';
import { Sheet } from './ui/Sheet';
import { T } from './ui/T';

/** Social share picker: direct deep links into social apps + the OS share sheet. */
export function ShareSheet({ visible, onClose, title, message, preview }: { visible: boolean; onClose: () => void; title: string; messag
e: string; preview?: string }) {
  return (
    <Sheet visible={visible} onClose={onClose} title={title} subtitle="Send it straight to your group’s favourite app">
      {preview ? (
        <View style={styles.preview}>
          <T variant="small" color={colors.textSecondary}>
            {preview}
          </T>
        </View>
      ) : null}
      <View style={styles.grid}>
        {socialTargets.map((s) => (
          <Press
            key={s.key}
            onPress={() => {
              onClose();
              shareTo(s.key, message);
            }}
            style={styles.item}
            accessibilityLabel={`Share via ${s.label}`}>
            <View style={[styles.icon, { backgroundColor: s.color }]}>
              <Ionicons name={s.icon} size={22} color={colors.white} />
            </View>
            <T variant="micro" weight="medium">
              {s.label}
            </T>
          </Press>
        ))}
      </View>
      <View style={{ flexDirection: 'row', gap: 8 }}>
        <Button label="Copy" variant="secondary" icon="copy-outline" size="md" onPress={() => copy(message)} style={{ flex: 1 }} />
        <Button
          label="More apps"
          icon="share-outline"
          size="md"
          onPress={() => {
            onClose();
            shareText(message);
          }}
          style={{ flex: 1 }}
        />
      </View>
    </Sheet>
  );
}
```

```
const styles = StyleSheet.create({
  preview: { backgroundColor: colors.cardMuted, borderRadius: radius.md, padding: 12, borderWidth: 1, borderColor: colors.border },
  grid: { flexDirection: 'row', flexWrap: 'wrap', rowGap: 16 },
  item: { width: '25%', alignItems: 'center', gap: 6 },
  icon: { width: 52, height: 52, borderRadius: 26, alignItems: 'center', justifyContent: 'center' },
});
```

src/components/SkylineHero.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { Image } from 'expo-image';
import { LinearGradient } from 'expo-linear-gradient';
import { useEffect, useState, type ReactNode } from 'react';
import { Animated, Easing, StyleSheet, View } from 'react-native';

import { nativeDriver, useLoop } from '@/components/ui/motion';
import { Press } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import { imageSource } from '@/data/images';
import { useSafeAreaInsets } from 'react-native-safe-area-context';

import { useNow } from '@/hooks/useTrip';
import { colors } from '@/theme';
import { partsInTz } from '@/utils/time';

export type TimeOfDay = 'day' | 'night';

/** Hours 19:00–05:59 at the destination count as night. */
export function timeOfDayAt(tz: string, now: number): TimeOfDay {
  const h = Number(partsInTz(now, tz).hhmm.slice(0, 2));
  return h >= 19 || h < 6 ? 'night' : 'day';
}

/** Deterministic pseudo-random list so stars sit in the same place every render. */
const STARS = Array.from({ length: 34 }, (_, i) => {
  const a = Math.sin(i * 91.7) * 43758.5453;
  const b = Math.sin(i * 12.9898 + 4.1) * 24634.6345;
  return { x: (a - Math.floor(a)) * 100, y: (b - Math.floor(b)) * 52, size: 1.5 + ((i * 7) % 3), group: i % 3 };
});

/**
 * Photo hero that follows the day/night cycle of the trip’s city: by day a warm sky glare, a sun and drifting
 * clouds; by night the photo is graded to blue hour with twinkling stars and a moon. The sun/moon chip lets you
 * preview the other half of the day. Children are drawn on top (header buttons, headline…).
 */
export function SkylineHero({ image, tz, height, children }: { image: string; tz: string; height: number; children?: ReactNode }) {
  const insets = useSafeAreaInsets();
  const now = useNow(60000);
  const auto = timeOfDayAt(tz, now);
  const [forced, setForced] = useState<TimeOfDay | null>(null);
  const mode = forced ?? auto;
  const [night] = useState(() => new Animated.Value(auto === 'night' ? 1 : 0));

  useEffect(() => {
    const anim = Animated.timing(night, { toValue: mode === 'night' ? 1 : 0, duration: 1100, easing: Easing.inOut(Easing.cubic), useNati
veDriver: nativeDriver });
    anim.start();
    return () => anim.stop();
  }, [mode, night]);

  const zoom = useLoop(18000, { pingPong: true });
  const twinkle = [useLoop(1700, { pingPong: true, rest: 0.6 }), useLoop(2300, { pingPong: true, delay: 400, rest: 0.6 }), useLoop(2900,
{ pingPong: true, delay: 900, rest: 0.6 })];
  const cloudA = useLoop(26000, { rest: 0.2 });
  const cloudB = useLoop(38000, { delay: 3000, rest: 0.7 });

  const dayOpacity = night.interpolate({ inputRange: [0, 1], outputRange: [1, 0] });

  return (
    <View style={{ height, overflow: 'hidden', backgroundColor: '#0B1020' }}>
      <Animated.View style={[StyleSheet.absoluteFill, { transform: [{ scale: zoom.interpolate({ inputRange: [0, 1], outputRange: [1.02,
1.14] }) }] }]}>
        <Image source={imageSource(image)} style={StyleSheet.absoluteFill} contentFit="cover" transition={300} accessibilityIgnoresInver
tColors />
      </Animated.View>

      {/* Day: warm light, sun, clouds */}
      <Animated.View pointerEvents="none" style={[StyleSheet.absoluteFill, { opacity: dayOpacity }]}>
        <LinearGradient colors={['rgba(255,214,120,0.30)', 'rgba(255,255,255,0)']} start={{ x: 1, y: 0 }} end={{ x: 0.2, y: 0.6 }} style
={StyleSheet.absoluteFill} />
        <View style={styles.sunGlow} />
        <View style={styles.sun} />
        <Animated.View style={[styles.cloud, { top: height * 0.16, transform: [{ translateX: cloudA.interpolate({ inputRange: [0, 1], ou
tputRange: [-120, 520] }) }] }]}>
          <Ionicons name="cloud" size={64} color="rgba(255,255,255,0.38)" />
        </Animated.View>
```

```
        <Animated.View style={[styles.cloud, { top: height * 0.3, transform: [{ translateX: cloudB.interpolate({ inputRange: [0, 1], out
putRange: [520, -140] }) }] }]}>
          <Ionicons name="cloud" size={44} color="rgba(255,255,255,0.28)" />
        </Animated.View>
      </Animated.View>

      {/* Night: blue-hour grade, stars, moon */}
      <Animated.View pointerEvents="none" style={[StyleSheet.absoluteFill, { opacity: night }]}>
        <LinearGradient colors={['rgba(5,10,45,0.82)', 'rgba(20,28,85,0.58)', 'rgba(8,10,35,0.72)']} locations={[0, 0.55, 1]} style={Sty
leSheet.absoluteFill} />
        <View style={[StyleSheet.absoluteFill, { backgroundColor: 'rgba(40,60,160,0.16)' }]} />
        {STARS.map((s, i) => (
          <Animated.View key={i} style={[styles.star, { left: `${s.x}%`, top: `${s.y}%`, width: s.size, height: s.size, borderRadius: s.
size, opacity: twinkle[s.group].interpolate({ inputRange: [0, 1], outputRange: [0.25, 1] }) }]} />
        ))}
        <View style={styles.moonGlow} />
        <View style={styles.moon}>
          <Ionicons name="moon" size={26} color="#FFF4C7" />
        </View>
      </Animated.View>

      <LinearGradient pointerEvents="none" colors={['rgba(0,0,0,0.35)', 'rgba(0,0,0,0)', 'rgba(0,0,0,0.72)']} locations={[0, 0.4, 1]} st
yle={StyleSheet.absoluteFill} />

      {children}

      <Press
        onPress={() => setForced(mode === 'night' ? 'day' : 'night')}
        onLongPress={() => setForced(null)}
        style={[styles.toggle, { top: insets.top + 62 }]}
        accessibilityLabel={mode === 'night' ? 'Showing night. Tap for day' : 'Showing day. Tap for night'}
        accessibilityHint="Long press to follow the local time again">
        <Ionicons name={mode === 'night' ? 'moon' : 'sunny'} size={13} color={colors.white} />
        <T variant="micro" weight="bold" color={colors.white}>
          {mode === 'night' ? 'Night' : 'Day'}
          {forced ? '' : ' · live'}
        </T>
      </Press>
    </View>
  );
}

const styles = StyleSheet.create({
  sun: { position: 'absolute', right: 34, top: 54, width: 46, height: 46, borderRadius: 23, backgroundColor: '#FFE08A', opacity: 0.9 },
  sunGlow: { position: 'absolute', right: 8, top: 28, width: 98, height: 98, borderRadius: 49, backgroundColor: '#FFD25E', opacity: 0.28
},
  cloud: { position: 'absolute', left: 0 },
  star: { position: 'absolute', backgroundColor: '#FFFFFF' },
  moon: { position: 'absolute', right: 32, top: 52, width: 44, height: 44, borderRadius: 22, alignItems: 'center', justifyContent: 'cent
er', backgroundColor: 'rgba(255,244,199,0.14)' },
  moonGlow: { position: 'absolute', right: 10, top: 30, width: 88, height: 88, borderRadius: 44, backgroundColor: 'rgba(255,244,199,0.1
4)' },
  toggle: { position: 'absolute', right: 16, flexDirection: 'row', alignItems: 'center', gap: 5, paddingHorizontal: 10, paddingVertical:
6, borderRadius: 999, backgroundColor: 'rgba(0,0,0,0.38)' },
});
```

src/components/TabBar.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { Tabs } from 'expo-router/js-tabs';
import { useEffect, useState, type ComponentProps } from 'react';
import { Animated, StyleSheet, View } from 'react-native';
import { useSafeAreaInsets } from 'react-native-safe-area-context';

import { useT } from '@/i18n';
import { useActiveTrip, useAppStore } from '@/store/useAppStore';
import { MAX_WIDTH, colors } from '@/theme';

import type { IconName } from './ui/bits';
import { nativeDriver, useMotion } from './ui/motion';
import { Press } from './ui/Press';
import { T } from './ui/T';

type BottomTabBarProps = Parameters<NonNullable<ComponentProps<typeof Tabs>['tabBar']>>[0];

const TABS: Record<string, { label: string; icon: IconName; active: IconName }> = {
  explore: { label: 'nav.explore', icon: 'home-outline', active: 'home' },
  itinerary: { label: 'nav.itinerary', icon: 'calendar-outline', active: 'calendar' },
  budget: { label: 'nav.budget', icon: 'wallet-outline', active: 'wallet' },
  chat: { label: 'nav.chat', icon: 'chatbubble-outline', active: 'chatbubble' },
  profile: { label: 'nav.profile', icon: 'grid-outline', active: 'grid' },
};

/** Bottom navigation matching the reference: soft-green pill behind the active tab. */
export function TabBar({ state, navigation }: BottomTabBarProps) {
  const t = useT();
```

```
  const insets = useSafeAreaInsets();
  const trip = useActiveTrip();
  const lastSeen = useAppStore((s) => s.notifications.filter((n) => !n.read && n.kind === 'chat').length);
  const unreadChat = trip ? lastSeen : 0;
  const motion = useMotion();
  const visible = state.routes.filter((r) => TABS[r.name]);
  const activeSlot = Math.max(0, visible.findIndex((r) => r.key === state.routes[state.index]?.key));
  const [innerW, setInnerW] = useState(0);
  const [slide] = useState(() => new Animated.Value(activeSlot));
  const [pop] = useState(() => new Animated.Value(1));

  // The highlight pill glides to the selected tab and the new icon does a little bounce.
  useEffect(() => {
    if (!motion) {
      slide.setValue(activeSlot);
      return;
    }
    Animated.spring(slide, { toValue: activeSlot, useNativeDriver: nativeDriver, friction: 8, tension: 90 }).start();
    pop.setValue(0.7);
    Animated.spring(pop, { toValue: 1, useNativeDriver: nativeDriver, friction: 4, tension: 160 }).start();
  }, [activeSlot, motion, slide, pop]);

  const slotW = innerW > 0 ? (innerW - 16) / Math.max(1, visible.length) : 0;
  const pillW = Math.min(76, slotW - 4);

  return (
    <View style={[styles.bar, { paddingBottom: Math.max(insets.bottom, 10) }]}>
      <View style={styles.inner} onLayout={(e) => setInnerW(e.nativeEvent.layout.width)}>
        {slotW > 0 ? (
          <Animated.View
            pointerEvents="none"
            style={[
              styles.indicator,
              { width: pillW, left: 8 + (slotW - pillW) / 2, transform: [{ translateX: slide.interpolate({ inputRange: [0, Math.max(1, v
isible.length - 1)], outputRange: [0, slotW * Math.max(1, visible.length - 1)] }) }] },
            ]}
          />
        ) : null}
        {state.routes.map((route, index) => {
          const cfg = TABS[route.name];
          if (!cfg) return null;
          const focused = state.index === index;
          const onPress = () => {
            const event = navigation.emit({ type: 'tabPress', target: route.key, canPreventDefault: true });
            if (!focused && !event.defaultPrevented) navigation.navigate(route.name);
          };
          return (
            <Press
              key={route.key}
              onPress={onPress}
              accessibilityRole="tab"
              accessibilityState={{ selected: focused }}
              accessibilityLabel={t(cfg.label)}
              style={styles.item}
              scaleTo={0.92}>
              <View style={[styles.pill, focused && slotW === 0 && styles.pillActive]}>
                <Animated.View style={focused ? { transform: [{ scale: pop }] } : undefined}>
                  <Ionicons name={focused ? cfg.active : cfg.icon} size={21} color={focused ? colors.primary : colors.textMuted} />
                  {route.name === 'chat' && unreadChat > 0 && !focused ? <View style={styles.dot} /> : null}
                </Animated.View>
                <T variant="micro" weight={focused ? 'semibold' : 'medium'} color={focused ? colors.primary : colors.textMuted} numberOf
Lines={1} style={{ fontSize: 10.5 }}>
                  {t(cfg.label)}
                </T>
              </View>
            </Press>
          );
        })}
      </View>
    </View>
  );
}

const styles = StyleSheet.create({
  bar: { backgroundColor: colors.card, borderTopWidth: 1, borderTopColor: colors.border, alignItems: 'center' },
  inner: { flexDirection: 'row', width: '100%', maxWidth: MAX_WIDTH, paddingHorizontal: 8, paddingTop: 8 },
  item: { flex: 1, alignItems: 'center' },
  pill: { alignItems: 'center', justifyContent: 'center', gap: 2, paddingHorizontal: 10, minWidth: 58, height: 50, borderRadius: 25 },
  pillActive: { backgroundColor: colors.navActive },
  indicator: { position: 'absolute', top: 8, height: 50, borderRadius: 25, backgroundColor: colors.navActive },
  dot: { position: 'absolute', top: -1, right: -3, width: 8, height: 8, borderRadius: 4, backgroundColor: colors.red, borderWidth: 1.5,
borderColor: colors.card },
});
```

src/components/auth/GoogleSheet.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { useState } from 'react';
import { StyleSheet, View } from 'react-native';

import { Button } from '@/components/ui/Button';
import { Input } from '@/components/ui/Input';
import { Press } from '@/components/ui/Press';
import { Sheet } from '@/components/ui/Sheet';
import { T } from '@/components/ui/T';
import { DEMO_EMAIL } from '@/store/seed';
import { colors, radius } from '@/theme';
import { validateEmail, validateName } from '@/utils/validation';

/** Simulated Google account chooser — a production build would use OAuth via expo-auth-session. */
export function GoogleSheet({ visible, onClose, onPick }: { visible: boolean; onClose: () => void; onPick: (email: string, name: string)
=> void }) {
  const [other, setOther] = useState(false);
  const [mail, setMail] = useState('');
  const [name, setName] = useState('');
  const [errs, setErrs] = useState<{ mail?: string | null; name?: string | null }>({});
  return (
    <Sheet visible={visible} onClose={onClose} title="Choose an account" subtitle="to continue to TogetherWeGo">
      <Press onPress={() => onPick(DEMO_EMAIL, 'Maya Chen')} style={styles.gAcct}>
        <View style={[styles.gAvatar, { backgroundColor: colors.primary }]}>
          <T variant="title" color={colors.white}>
            M
          </T>
        </View>
        <View style={{ flex: 1 }}>
          <T variant="bodySm" weight="semibold">
            Maya Chen
          </T>
          <T variant="caption" color={colors.textSecondary}>
            {DEMO_EMAIL}
          </T>
        </View>
        <Ionicons name="chevron-forward" size={16} color={colors.textMuted} />
      </Press>
      {other ? (
        <View style={{ gap: 12 }}>
          <Input label="Name" value={name} onChangeText={setName} error={errs.name} autoCapitalize="words" icon="person-outline" />
          <Input label="Google email" value={mail} onChangeText={setMail} error={errs.mail} autoCapitalize="none" keyboardType="email-ad
dress" icon="mail-outline" />
          <Button
            label="Continue"
            onPress={() => {
              const e = { mail: validateEmail(mail), name: validateName(name) };
              setErrs(e);
              if (!e.mail && !e.name) onPick(mail, name.trim());
            }}
          />
        </View>
      ) : (
        <Press onPress={() => setOther(true)} style={styles.gAcct}>
          <View style={[styles.gAvatar, { backgroundColor: colors.bgAlt }]}>
            <Ionicons name="person-add-outline" size={18} color={colors.textSecondary} />
          </View>
          <T variant="bodySm" weight="medium">
            Use another account
          </T>
        </Press>
      )}
      <T variant="caption" color={colors.textMuted}>
        Demo sign-in: in a store build this opens Google’s secure OAuth screen.
      </T>
    </Sheet>
  );
}

const styles = StyleSheet.create({
  gAcct: { flexDirection: 'row', alignItems: 'center', gap: 12, padding: 12, borderRadius: radius.md, borderWidth: 1, borderColor: color
s.border },
  gAvatar: { width: 40, height: 40, borderRadius: 20, alignItems: 'center', justifyContent: 'center' },
});
```

src/components/auth/legal.ts

```
import { showDialog } from '@/components/ui/feedback';

/** The short Terms / Privacy notes shown from the sign-in and sign-up screens. */
export function showLegal(title: 'Terms of Service' | 'Privacy Policy') {
  showDialog(
    title,
```

```
    title === 'Privacy Policy'
      ? 'TogetherWeGo stores your trips on this device. Passwords are salted and hashed (PBKDF2); API keys go in the secure keychain. We never sell data. Location is only used when you tap a nearby/airport feature.'
      : 'TogetherWeGo is a student project built for FBLA Mobile Application Development. Use it to plan trips with friends — be kind in chats and only upload documents you own.',
    [{ label: 'Got it' }],
    'shield-checkmark-outline',
  );
}
```

src/components/budget/AddExpenseSheet.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { Image } from 'expo-image';
import { useState } from 'react';
import { Platform, ScrollView, View } from 'react-native';

import { expenseCategories } from '@/components/trip/meta';
import { Button } from '@/components/ui/Button';
import { Input } from '@/components/ui/Input';
import { Press, tap } from '@/components/ui/Press';
import { Sheet } from '@/components/ui/Sheet';
import { T } from '@/components/ui/T';
import { Chip } from '@/components/ui/bits';
import { toast } from '@/components/ui/feedback';
import { destinationById } from '@/data/destinations';
import { useT } from '@/i18n';
import { convert, useRates } from '@/services/currency';
import { deleteLocalFile, pickImage } from '@/services/files';
import type { ExpenseCategory, Trip } from '@/store/types';
import { totalSpent, useAppStore } from '@/store/useAppStore';
import { colors } from '@/theme';
import { money } from '@/utils/format';
import { todayInTz } from '@/utils/time';
import { firstError, maxLen, parseAmount, required, validateAmount } from '@/utils/validation';

import { styles } from './styles';

/** The "Add expense" form in a bottom sheet: description, amount (any currency), category, who paid, split, receipt. */
export function AddExpenseSheet({ visible, onClose, trip, userId }: { visible: boolean; onClose: () => void; trip: Trip; userId: string
}) {
  const t = useT();
  const rates = useRates();
  const addExpense = useAppStore((s) => s.addExpense);
  const dest = destinationById(trip.destinationId);
  const local = dest?.currency ?? 'EUR';

  const [title, setTitle] = useState('');
  const [amount, setAmount] = useState('');
  const [entryCur, setEntryCur] = useState<'USD' | string>('USD');
  const [cat, setCat] = useState<ExpenseCategory>('Food');
  const [paidBy, setPaidBy] = useState(userId);
  const [split, setSplit] = useState<string[]>(trip.members.map((m) => m.id));
  const [receipt, setReceipt] = useState<string | null>(null);
  const [attaching, setAttaching] = useState(false);
  const [errors, setErrors] = useState<Record<string, string | null>>({});

  const attach = async (camera: boolean) => {
    setAttaching(true);
    try {
      const f = await pickImage(camera);
      if (f) {
        deleteLocalFile(receipt ?? undefined);
        setReceipt(f.uri);
      }
    } catch (e) {
      toast(e instanceof Error ? e.message : 'Couldn’t open photos', { tone: 'warn' });
    } finally {
      setAttaching(false);
    }
  };

  const removeReceipt = () => {
    deleteLocalFile(receipt ?? undefined);
    setReceipt(null);
  };

  const submit = () => {
    const e = {
      title: firstError(title, required('what it was for'), maxLen(40, 'Description')),
      amount: validateAmount(amount, { max: 100000 }),
      split: split.length ? null : 'Pick at least one person to split with',
    };
    setErrors(e);
    if (Object.values(e).some(Boolean)) return tap('warning');
    const n = parseAmount(amount)!;
```

```
    const usd = entryCur === 'USD' ? n : convert(n, entryCur, 'USD', rates);
    const spent = totalSpent(trip);
    addExpense(trip.id, {
      title: title.trim(),
      amount: Math.round(usd * 100) / 100,
      category: cat,
      paidBy,
      splitBetween: split,
      date: todayInTz(dest?.tz ?? 'UTC'),
      original: entryCur === 'USD' ? undefined : { amount: n, currency: entryCur },
      receipt: receipt ?? undefined,
    });
    tap('success');
    toast(`Added ${money(usd)} · ${money(usd / split.length)} each`);
    setTitle('');
    setAmount('');
    setReceipt(null);
    onClose();
    if (spent + usd > trip.budget && spent <= trip.budget) toast('Heads up: this puts the group over budget', { tone: 'warn' });
  };

  return (
    <Sheet
      visible={visible}
      onClose={onClose}
      title={t('budget.addExpense')}
      footer={<Button label={t('budget.add')} icon="add" onPress={submit} testID="add-expense" />}>
      <View style={{ gap: 12 }}>
        <Input placeholder={t('budget.whatFor')} value={title} onChangeText={setTitle} error={errors.title} maxLength={45} testID="expen
se-title" />
        <View style={{ flexDirection: 'row', gap: 8 }}>
          <Input placeholder="100" value={amount} onChangeText={setAmount} keyboardType="decimal-pad" error={errors.amount} containerSty
le={{ flex: 1 }} icon="cash-outline" testID="expense-amount" />
          <View style={styles.curToggle}>
            {['USD', local].filter((c, i, a) => a.indexOf(c) === i).map((c) => (
              <Press key={c} onPress={() => setEntryCur(c)} style={[styles.curBtn, entryCur === c && styles.curOn]} accessibilityLabel=
{`Enter amount in ${c}`}>
                <T variant="caption" weight="semibold" color={entryCur === c ? colors.onPrimary : colors.textSecondary}>
                  {c}
                </T>
              </Press>
            ))}
          </View>
        </View>
        {entryCur !== 'USD' && parseAmount(amount) ? (
          <T variant="caption" color={colors.textSecondary}>
            ≈ {money(convert(parseAmount(amount)!, entryCur, 'USD', rates), 'USD', { decimals: true })} — saved in USD so the group can
compare
          </T>
        ) : null}
        <ScrollView horizontal showsHorizontalScrollIndicator={false} contentContainerStyle={{ gap: 6 }}>
          {expenseCategories.map((c) => (
            <Chip key={c.key} small label={c.key} icon={c.icon} active={cat === c.key} onPress={() => setCat(c.key)} />
          ))}
        </ScrollView>
        <T variant="caption" weight="semibold" color={colors.textSecondary}>
          Paid by
        </T>
        <ScrollView horizontal showsHorizontalScrollIndicator={false} contentContainerStyle={{ gap: 6 }}>
          {trip.members.map((m) => (
            <Chip key={m.id} small label={m.name.split(' ')[0]} image={m.avatar} active={paidBy === m.id} onPress={() => setPaidBy(m.i
d)} />
          ))}
        </ScrollView>
        <View style={styles.rowBetween}>
          <T variant="caption" weight="semibold" color={colors.textSecondary}>
            Split between ({split.length})
          </T>
          <Press onPress={() => setSplit(split.length === trip.members.length ? [userId] : trip.members.map((m) => m.id))} hitSlop={8}>
            <T variant="caption" weight="bold">
              {split.length === trip.members.length ? 'Just me' : 'Everyone'}
            </T>
          </Press>
        </View>
        <View style={{ flexDirection: 'row', flexWrap: 'wrap', gap: 6 }}>
          {trip.members.map((m) => (
            <Chip
              key={m.id}
              small
              label={m.name.split(' ')[0]}
              icon={split.includes(m.id) ? 'checkmark' : undefined}
              active={split.includes(m.id)}
              onPress={() => setSplit((s) => (s.includes(m.id) ? s.filter((x) => x !== m.id) : [...s, m.id]))}
            />
          ))}
        </View>
```

```
        {errors.split ? (
          <T variant="caption" color={colors.red}>
            {errors.split}
          </T>
        ) : null}
        <View style={styles.receiptRow}>
          <Button label="Attach receipt" icon="camera-outline" variant="soft" size="md" loading={attaching} onPress={() => attach(fals
e)} style={{ flex: 1 }} />
          {Platform.OS !== 'web' ? <Button label="Take photo" icon="scan-outline" variant="soft" size="md" disabled={attaching} onPress=
{() => attach(true)} style={{ flex: 1 }} /> : null}
        </View>
        {receipt ? (
          <View style={styles.receiptRow}>
            <View>
              <Image source={{ uri: receipt }} style={styles.thumb} contentFit="cover" accessibilityLabel="Receipt photo" />
              <Press onPress={removeReceipt} style={styles.thumbX} hitSlop={8} accessibilityLabel="Remove receipt">
                <Ionicons name="close" size={13} color={colors.onPrimary} />
              </Press>
            </View>
            <T variant="small" color={colors.textSecondary} style={{ flex: 1 }}>
              Receipt attached. It is saved with the expense.
            </T>
          </View>
        ) : null}
      </View>
    </Sheet>
  );
}
```

src/components/budget/BalancesCard.tsx

```
import { View } from 'react-native';

import { Button } from '@/components/ui/Button';
import { T } from '@/components/ui/T';
import { Avatar, Card } from '@/components/ui/bits';
import { useT } from '@/i18n';
import type { Trip } from '@/store/types';
import { balances } from '@/store/useAppStore';
import { colors } from '@/theme';

import { styles } from './styles';

/** "Who owes who": every member's balance, plus the Settle up button. */
export function BalancesCard({ trip, userId, fmt, onSettle }: { trip: Trip; userId: string; fmt: (usd: number) => string; onSettle: () =
> void }) {
  const t = useT();
  const bal = balances(trip);
  const firstName = (id: string, name: string) => (id === userId ? 'You' : name.split(' ')[0]);
  return (
    <>
    <View style={styles.sectionRow}>
      <T variant="headline" accessibilityRole="header" style={{ flexShrink: 1 }}>
        Who owes who
      </T>
      <Button label={t('budget.settle')} size="sm" full={false} variant="soft" icon="swap-horizontal" onPress={onSettle} />
    </View>
    <Card padded={false} style={styles.card}>
      {trip.members.map((m, i) => {
        const v = bal[m.id] ?? 0;
        return (
          <View key={m.id} style={[styles.balRow, i > 0 && styles.balDivider]}>
            <Avatar name={m.name} src={m.avatar} size={34} />
            <View style={styles.nameWrap}>
              <T variant="bodySm" weight="semibold" numberOfLines={1} style={{ flexShrink: 1 }}>
                {firstName(m.id, m.name)}
              </T>
              {m.role === 'Treasurer' ? (
                <View style={styles.badge}>
                  <T variant="micro" weight="bold" color={colors.onAccent}>
                    Treasurer
                  </T>
                </View>
              ) : null}
            </View>
            <T variant="small" weight="bold" color={Math.abs(v) < 0.5 ? colors.textMuted : v > 0 ? colors.positive : colors.red}>
              {Math.abs(v) < 0.5 ? 'settled' : v > 0 ? `gets back ${fmt(v)}` : `owes ${fmt(-v)}`}
            </T>
          </View>
        );
      })}
    </Card>
    </>
  );
}
```

src/components/budget/DailyPlanner.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { router } from 'expo-router';
import { useState } from 'react';
import { TextInput, View } from 'react-native';

import { placeCategoryIcon } from '@/components/trip/meta';
import { Press } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import { Card } from '@/components/ui/bits';
import { destinationById } from '@/data/destinations';
import { placesFor } from '@/data/places';
import { useT } from '@/i18n';
import type { Trip } from '@/store/types';
import { totalSpent } from '@/store/useAppStore';
import { categoryColors, colors } from '@/theme';
import { money, priceTier } from '@/utils/format';
import { distanceKm, kmLabel } from '@/utils/geo';
import { daysBetween } from '@/utils/time';
import { parseAmount } from '@/utils/validation';

import { styles } from './styles';

/** Daily spending planner: how much per person per day, does it fit what is left, and budget finds nearby. */
export function DailyPlanner({ trip }: { trip: Trip }) {
  const t = useT();
  const dest = destinationById(trip.destinationId);
  const [daily, setDaily] = useState(String(dest?.dailyCost ?? 120));
  const tripLen = daysBetween(trip.startDate, trip.endDate) + 1;
  const dailyN = parseAmount(daily) ?? 0;
  const remaining = trip.budget - totalSpent(trip);
  const plannedTotal = dailyN * tripLen * trip.members.length;
  const activityAllowance = dailyN * 0.3;
  const finds = dest
    ? [trip.destinationId, ...trip.extraDestinationIds]
        .flatMap((id) => placesFor(id))
        .filter((p) => p.cost <= activityAllowance)
        .map((p) => ({ p, km: distanceKm(dest, p) }))
        .filter((x) => x.km <= 8)
        .sort((a, b) => b.p.rating - a.p.rating)
        .slice(0, 3)
    : [];
  return (
    <>
    <T variant="headline" accessibilityRole="header" style={styles.section}>
      {t('budget.planner')}
    </T>
    <Card style={[styles.card, { gap: 12 }]}>
      <T variant="small" color={colors.textSecondary}>
        How much does each person want to spend per day in {dest?.city}? (Typical: {money(dest?.dailyCost ?? 0)})
      </T>
      <View style={styles.dailyRow}>
        <T variant="h3">$</T>
        <TextInput value={daily} onChangeText={(v) => setDaily(v.replace(/[^0-9.]/g, ''))} keyboardType="decimal-pad" style={styles.dail
yInput} accessibilityLabel="Daily spend per person in USD" maxLength={6} />
        <T variant="small" color={colors.textSecondary}>
          / person / day
        </T>
      </View>
      {dailyN > 0 ? (
        <>
          <View style={styles.splitRow}>
            {[
              ['Food', 0.35],
              ['Activities', 0.3],
              ['Transport', 0.15],
              ['Shopping', 0.12],
              ['Other', 0.08],
            ].map(([k, f]) => (
              <View key={k as string} style={styles.splitCell}>
                <View style={[styles.dot, { backgroundColor: categoryColors[k as string] }]} />
                <T variant="micro" color={colors.textSecondary}>
                  {k as string}
                </T>
                <T variant="small" weight="bold">
                  {money(dailyN * (f as number))}
                </T>
              </View>
            ))}
          </View>
          <View style={[styles.fit, { backgroundColor: plannedTotal <= remaining ? colors.primarySofter : colors.orangeSoft }]}>
            <Ionicons name={plannedTotal <= remaining ? 'checkmark-circle' : 'alert-circle'} size={16} color={plannedTotal <= remaining
? colors.positive : colors.orange} />
            <T variant="caption" style={{ flex: 1 }}>
              {tripLen} days × {trip.members.length} people = {money(plannedTotal)} on the ground ·{' '}
```

```
              {plannedTotal <= remaining ? `fits the ${money(remaining)} left` : `${money(plannedTotal - remaining)} more than what's le
ft`}
            </T>
          </View>
          <T variant="small" weight="bold" style={{ marginTop: 2 }}>
            Budget finds near {dest?.city} (≤ {money(activityAllowance)} each)
          </T>
          {finds.map(({ p, km }) => (
            <Press key={p.id} onPress={() => router.push({ pathname: '/nearby', params: { focus: p.id, maxCost: String(Math.round(activi
tyAllowance)) } })} style={styles.findRow}>
              <View style={styles.findIcon}>
                <Ionicons name={placeCategoryIcon[p.category]} size={15} color={colors.text} />
              </View>
              <T variant="small" weight="medium" style={{ flex: 1 }} numberOfLines={1}>
                {p.name}
              </T>
              <T variant="caption" color={colors.textSecondary}>
                {priceTier(p.cost)} · {kmLabel(km)}
              </T>
            </Press>
          ))}
        </>
      ) : (
        <T variant="caption" color={colors.red}>
          Enter a daily amount greater than 0
        </T>
      )}
    </Card>
    </>
  );
}
```

src/components/budget/ExpenseDetailSheet.tsx

```
import { Image } from 'expo-image';
import { useState } from 'react';
import { ActivityIndicator, View } from 'react-native';

import { Button } from '@/components/ui/Button';
import { Press } from '@/components/ui/Press';
import { Sheet } from '@/components/ui/Sheet';
import { T } from '@/components/ui/T';
import { AvatarStack, Pill } from '@/components/ui/bits';
import { confirmAction, toast } from '@/components/ui/feedback';
import { deleteLocalFile } from '@/services/files';
import type { Expense, Trip } from '@/store/types';
import { memberById, useAppStore } from '@/store/useAppStore';
import { useSharedFile } from '@/sync/useFile';
import { colors } from '@/theme';
import { money } from '@/utils/format';

import { styles } from './styles';

/** The receipt photo: this phone's own, or the copy from the friend who paid. Tap to enlarge. */
function Receipt({ expense }: { expense: Expense }) {
  const [big, setBig] = useState(false);
  const { uri, loading } = useSharedFile(expense.receipt, expense.receiptBlob);
  if (!uri && !loading) return null;
  return (
    <View style={{ gap: 8 }}>
      <T variant="kicker" color={colors.textSecondary}>
        Receipt
      </T>
      {uri ? (
        <Press onPress={() => setBig((b) => !b)} scaleTo={0.99} accessibilityLabel={big ? 'Shrink receipt' : 'Enlarge receipt'} style={{
alignSelf: big ? 'stretch' : 'flex-start' }}>
          <Image source={{ uri }} style={big ? styles.receiptBig : styles.receiptSmall} contentFit={big ? 'contain' : 'cover'} accessibi
lityLabel="Receipt photo" />
        </Press>
      ) : (
        <View style={{ flexDirection: 'row', alignItems: 'center', gap: 8 }}>
          <ActivityIndicator color={colors.primary} />
          <T variant="small" color={colors.textSecondary}>
            Getting the receipt…
          </T>
        </View>
      )}
    </View>
  );
}

/** Details of one expense: amount, who paid, the split, the receipt photo, and a Delete button. */
export function ExpenseDetailSheet({ detail, onClose, trip, fmt }: { detail: Expense | null; onClose: () => void; trip: Trip; fmt: (usd:
number) => string }) {
  const deleteExpense = useAppStore((s) => s.deleteExpense);
  return (
```

```
    <Sheet visible={!!detail} onClose={onClose} title={detail?.title} subtitle={detail ? `${detail.category} · ${detail.date}` : undefin
ed}>
      {detail ? (
        <>
          <View style={styles.rowBetween}>
            <T variant="display" style={{ fontSize: 36, lineHeight: 40 }}>
              {fmt(detail.amount)}
            </T>
            {detail.original ? <Pill label={`Paid as ${money(detail.original.amount, detail.original.currency)}`} tone="blue" /> : null}
          </View>
          <View style={styles.rowBetween}>
            <T variant="small" color={colors.textSecondary}>
              Paid by {memberById(trip, detail.paidBy)?.name}
            </T>
            <T variant="small" weight="bold">
              {fmt(detail.amount / Math.max(1, detail.splitBetween.length))} each
            </T>
          </View>
          <AvatarStack people={detail.splitBetween.map((id) => memberById(trip, id)).filter((m): m is NonNullable<typeof m> => !!m)} max
={6} size={30} />
          <Receipt expense={detail} />
          <Button
            label="Delete expense"
            variant="danger"
            icon="trash-outline"
            onPress={() => {
              const d = detail;
              onClose();
              confirmAction('Delete expense?', `${d.title} (${money(d.amount)}) will be removed.`, () => {
                deleteExpense(trip.id, d.id);
                deleteLocalFile(d.receipt);
                toast('Expense deleted');
              }, { destructive: true, confirmLabel: 'Delete' });
            }}
          />
        </>
      ) : null}
    </Sheet>
  );
}
```

src/components/budget/ExpensesSection.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { useState } from 'react';
import { TextInput, View } from 'react-native';

import { T } from '@/components/ui/T';
import { Card } from '@/components/ui/bits';
import { toast } from '@/components/ui/feedback';
import { useT } from '@/i18n';
import type { Expense, Trip } from '@/store/types';
import { memberById, useAppStore } from '@/store/useAppStore';
import { categoryColors, colors } from '@/theme';
import { money } from '@/utils/format';
import { parseAmount, validateAmount } from '@/utils/validation';

import { styles } from './styles';

/** The expenses list with the group budget cap field. Tapping an expense opens its details. */
export function ExpensesSection({ trip, fmt, onOpen }: { trip: Trip; fmt: (usd: number) => string; onOpen: (e: Expense) => void }) {
  const t = useT();
  const updateTrip = useAppStore((s) => s.updateTrip);
  const [capText, setCapText] = useState(String(trip.budget));
  const [capErr, setCapErr] = useState<string | null>(null);

  const saveCap = () => {
    const e = validateAmount(capText, { min: 1, max: 1_000_000, label: 'a budget' });
    setCapErr(e);
    if (e) return;
    const v = parseAmount(capText)!;
    updateTrip(trip.id, { budget: v });
    toast(`Budget cap set to ${money(v)}`);
  };

  return (
    <>
    <View style={[styles.sectionRow, { marginTop: 26 }]}>
      <T variant="headline" accessibilityRole="header" style={{ flexShrink: 1 }}>
        {t('budget.expenses')}
      </T>
      <View style={styles.capBox}>
        <T variant="caption" weight="semibold" color={colors.textSecondary}>
          {t('budget.cap')}
        </T>
        <TextInput
```

```
          value={capText}
          onChangeText={(v) => setCapText(v.replace(/[^0-9.]/g, ''))}
          onBlur={saveCap}
          onSubmitEditing={saveCap}
          keyboardType="decimal-pad"
          style={[styles.capInput, capErr && { borderColor: colors.red }]}
          accessibilityLabel="Group budget cap in USD"
          maxLength={8}
        />
      </View>
    </View>
    {capErr ? (
      <T variant="caption" color={colors.red} style={{ marginBottom: 8, marginTop: -4 }}>
        {capErr}
      </T>
    ) : null}
    <View style={{ gap: 10 }}>
      {trip.expenses.map((e) => (
        <Card key={e.id} padded={false} onPress={() => onOpen(e)} style={styles.expense} accessibilityLabel={`${e.title}, ${fmt(e.amoun
t)}`}>
          <View style={[styles.expenseDot, { backgroundColor: e.settlement ? colors.textMuted : categoryColors[e.category] ?? colors.tex
tMuted }]} />
          <View style={{ flex: 1 }}>
            <View style={styles.titleRow}>
              <T variant="bodySm" weight="bold" numberOfLines={1} style={{ flexShrink: 1 }}>
                {e.title}
              </T>
              {e.receipt || e.receiptBlob ? <Ionicons name="attach" size={15} color={colors.textMuted} accessibilityLabel="Has receipt"
/> : null}
            </View>
            <T variant="caption" color={colors.textSecondary} numberOfLines={1}>
              {e.settlement ? 'Settlement' : e.category} · {t('budget.paidBy', { name: memberById(trip, e.paidBy)?.name.split(' ')[0] ??
'—' })}
              {e.original ? ` · ${money(e.original.amount, e.original.currency)}` : ''}
            </T>
          </View>
          <T variant="small" weight="extrabold">
            {fmt(e.amount)}
          </T>
        </Card>
      ))}
    </View>
    </>
  );
}
```

src/components/budget/SettleSheet.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { View } from 'react-native';

import { Button } from '@/components/ui/Button';
import { tap } from '@/components/ui/Press';
import { Sheet } from '@/components/ui/Sheet';
import { T } from '@/components/ui/T';
import { Avatar } from '@/components/ui/bits';
import { toast } from '@/components/ui/feedback';
import type { Trip } from '@/store/types';
import { memberById, settlePlan, useAppStore } from '@/store/useAppStore';
import { colors } from '@/theme';
import { money } from '@/utils/format';

import { styles } from './styles';

/** "Settle up": the fewest payments that make everyone even, each with a Mark paid button. */
export function SettleSheet({ visible, onClose, trip }: { visible: boolean; onClose: () => void; trip: Trip }) {
  const settleUp = useAppStore((s) => s.settleUp);
  const plan = settlePlan(trip);
  return (
    <Sheet visible={visible} onClose={onClose} title="Settle up" subtitle="Fewest payments to make everyone even">
      {plan.length ? (
        plan.map((p) => {
          const from = memberById(trip, p.from);
          const to = memberById(trip, p.to);
          return (
            <View key={`${p.from}-${p.to}`} style={styles.settleRow}>
              <Avatar name={from?.name ?? '?'} src={from?.avatar} size={30} />
              <Ionicons name="arrow-forward" size={14} color={colors.textMuted} />
              <Avatar name={to?.name ?? '?'} src={to?.avatar} size={30} />
              <View style={{ flex: 1 }}>
                <T variant="small" weight="semibold">
                  {from?.name.split(' ')[0]} pays {to?.name.split(' ')[0]}
                </T>
                <T variant="caption" color={colors.textSecondary}>
                  {money(p.amount, 'USD', { decimals: true })}
                </T>
```

```
              </View>
              <Button
                label="Mark paid"
                size="sm"
                full={false}
                onPress={() => {
                  settleUp(trip.id, p.from, p.to, p.amount);
                  tap('success');
                  toast('Payment recorded ✅');
                }}
              />
            </View>
          );
        })
      ) : (
        <View style={{ alignItems: 'center', gap: 8, paddingVertical: 20 }}>
          <Ionicons name="checkmark-circle" size={40} color={colors.positive} />
          <T variant="title">Everyone is square! 🎉</T>
        </View>
      )}
    </Sheet>
  );
}
```

src/components/budget/SpendSummary.tsx

```
import { View } from 'react-native';

import { Donut } from '@/components/ui/Donut';
import { T } from '@/components/ui/T';
import { Card } from '@/components/ui/bits';
import { categoryColors, colors } from '@/theme';

import { styles } from './styles';

/** The donut chart of spending by category, plus the per-person and daily-allowance tiles. */
export function SpendSummary({ byCat, spent, budget, remaining, per, daysLeft, fmt }: {
  byCat: [string, number][];
  spent: number;
  budget: number;
  remaining: number;
  per: number;
  daysLeft: number;
  /** Formats a US-dollar amount in the currency the user is viewing. */
  fmt: (usd: number) => string;
}) {
  return (
    <>
    <Card style={[styles.card, { marginTop: 14 }]}>
      <View style={styles.donutRow}>
        <Donut
          size={132}
          parts={byCat.map(([c, v]) => ({ value: v, color: categoryColors[c] ?? colors.textMuted }))}
          centerLabel={remaining < 0 ? 'OVER' : 'LEFT'}
          centerValue={fmt(Math.abs(remaining))}
        />
        <View style={{ flex: 1, gap: 8 }}>
          <T variant="kicker" weight="bold" color={colors.textSecondary} numberOfLines={1} adjustsFontSizeToFit style={styles.tinyKicke
r}>
            {`Spent ${fmt(spent)} / ${fmt(budget)}`}
          </T>
          {byCat.map(([c, v]) => (
            <View key={c} style={styles.legendRow}>
              <View style={[styles.dot, { backgroundColor: categoryColors[c] ?? colors.textMuted }]} />
              <T variant="small" weight="medium" style={{ flex: 1 }} numberOfLines={1}>
                {c}
              </T>
              <T variant="small" weight="bold">
                {fmt(v)}
              </T>
            </View>
          ))}
          {!byCat.length ? (
            <T variant="small" color={colors.textMuted}>
              No expenses yet. Tap + to add your first one.
            </T>
          ) : null}
        </View>
      </View>
    </Card>
    <View style={styles.tiles}>
      <Card style={[styles.card, styles.tile]}>
        <T variant="kicker" weight="bold" color={colors.textSecondary} style={styles.tinyKicker}>
          Per person
        </T>
        <T variant="h2" weight="extrabold" numberOfLines={1} adjustsFontSizeToFit>
```

```
          {fmt(per)}
        </T>
      </Card>
      <Card style={[styles.card, styles.tile]}>
        <T variant="kicker" weight="bold" color={colors.textSecondary} style={styles.tinyKicker}>
          Daily allowance
        </T>
        <T variant="h2" weight="extrabold" numberOfLines={1} adjustsFontSizeToFit>
          {fmt(Math.max(0, remaining) / daysLeft)}
        </T>
      </Card>
    </View>
    </>
  );
}
```

src/components/budget/styles.ts

```
import { StyleSheet } from 'react-native';

import { colors, fonts, radius } from '@/theme';

/** Styles shared by the Budget screen and its parts. */
export const styles = StyleSheet.create({
  head: { flexDirection: 'row', alignItems: 'flex-end', gap: 12, marginTop: 12, marginBottom: 14 },
  plus: { width: 44, height: 44, borderRadius: 22, backgroundColor: colors.primary, alignItems: 'center', justifyContent: 'center', marg
inBottom: 4 },
  card: { borderRadius: radius.xl },
  donutRow: { flexDirection: 'row', alignItems: 'center', gap: 18 },
  legendRow: { flexDirection: 'row', alignItems: 'center', gap: 8 },
  dot: { width: 9, height: 9, borderRadius: 5 },
  tinyKicker: { fontSize: 10.5, letterSpacing: 0.6 },
  tiles: { flexDirection: 'row', gap: 12, marginTop: 12 },
  tile: { flex: 1, gap: 4, paddingHorizontal: 16, paddingVertical: 14 },
  sectionRow: { flexDirection: 'row', justifyContent: 'space-between', alignItems: 'center', gap: 8, marginTop: 24, marginBottom: 12 },
  section: { marginTop: 24, marginBottom: 12 },
  rowBetween: { flexDirection: 'row', justifyContent: 'space-between', alignItems: 'center', gap: 8 },
  balRow: { flexDirection: 'row', alignItems: 'center', gap: 12, paddingHorizontal: 16, paddingVertical: 13 },
  balDivider: { borderTopWidth: 1, borderTopColor: colors.border },
  nameWrap: { flex: 1, flexDirection: 'row', alignItems: 'center', gap: 8 },
  badge: { backgroundColor: colors.accent, borderRadius: 999, paddingHorizontal: 8, paddingVertical: 2 },
  dailyRow: { flexDirection: 'row', alignItems: 'center', gap: 8 },
  dailyInput: { width: 96, height: 46, borderRadius: radius.md, backgroundColor: colors.bgAlt, paddingHorizontal: 14, fontFamily: fonts.
semibold, fontSize: 17, color: colors.text, ...({ outlineStyle: 'none' } as object) },
  splitRow: { flexDirection: 'row', justifyContent: 'space-between' },
  splitCell: { alignItems: 'center', gap: 2 },
  fit: { flexDirection: 'row', gap: 8, alignItems: 'center', padding: 12, borderRadius: radius.md },
  findRow: { flexDirection: 'row', alignItems: 'center', gap: 10, paddingVertical: 4 },
  findIcon: { width: 30, height: 30, borderRadius: 15, backgroundColor: colors.bgAlt, alignItems: 'center', justifyContent: 'center' },
  capBox: { flexDirection: 'row', alignItems: 'center', gap: 8 },
  capInput: { width: 80, height: 34, borderRadius: 999, borderWidth: 1, borderColor: colors.border, backgroundColor: colors.card, textAl
ign: 'center', fontFamily: fonts.bold, fontSize: 13, color: colors.text, ...({ outlineStyle: 'none' } as object) },
  expense: { flexDirection: 'row', alignItems: 'center', gap: 12, borderRadius: radius.lg, paddingHorizontal: 16, paddingVertical: 14 },
  expenseDot: { width: 11, height: 11, borderRadius: 6 },
  titleRow: { flexDirection: 'row', alignItems: 'center', gap: 5 },
  curToggle: { flexDirection: 'row', height: 50, borderRadius: radius.md, padding: 4, gap: 4, backgroundColor: colors.bgAlt },
  curBtn: { paddingHorizontal: 12, borderRadius: radius.sm, justifyContent: 'center' },
  curOn: { backgroundColor: colors.primary },
  receiptRow: { flexDirection: 'row', alignItems: 'center', gap: 10 },
  thumb: { width: 64, height: 64, borderRadius: radius.sm, backgroundColor: colors.bgAlt },
  thumbX: { position: 'absolute', top: -6, right: -6, width: 20, height: 20, borderRadius: 10, backgroundColor: colors.primary, alignIte
ms: 'center', justifyContent: 'center', borderWidth: 1.5, borderColor: colors.white },
  receiptSmall: { width: 96, height: 96, borderRadius: radius.md, backgroundColor: colors.bgAlt },
  receiptBig: { width: '100%', height: 380, borderRadius: radius.md, backgroundColor: colors.bgAlt },
  settleRow: { flexDirection: 'row', alignItems: 'center', gap: 8, padding: 12, borderRadius: radius.md, backgroundColor: colors.bgAlt
},
});
```

src/components/chat/ChatBubble.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { StyleSheet, View } from 'react-native';

import { T } from '@/components/ui/T';
import type { ChatMessage, Trip } from '@/store/types';
import { colors, shadow } from '@/theme';

const AVATAR_COLORS = ['#E8533F', '#1E73E8', '#2FB863', '#E0A100', '#7C5CD6'];

/** One chat message: a centred note for system messages, otherwise a bubble with the sender's avatar and name. */
export function ChatBubble({ item, prev, trip, userId }: { item: ChatMessage; prev: ChatMessage | undefined; trip: Trip; userId: string
}) {
  if (item.kind === 'system') {
    return (
      <View style={styles.system}>
        <T variant="caption" color={colors.textMuted} center>
```

```
          {item.text}
        </T>
      </View>
    );
  }
  const mine = item.authorId === userId;
  const grouped = !!prev && prev.kind !== 'system' && prev.authorId === item.authorId && item.ts - prev.ts < 5 * 60000;
  const m = trip.members.find((x) => x.id === item.authorId);
  const memberIndex = trip.members.findIndex((x) => x.id === item.authorId);
  const avatarBg = memberIndex >= 0 ? AVATAR_COLORS[memberIndex % AVATAR_COLORS.length] : colors.textMuted;
  const name = item.ai ? 'Scout AI' : (m?.name.split(' ')[0] ?? 'Former member');
  return (
    <View style={[styles.msgRow, mine && { flexDirection: 'row-reverse' }, grouped && { marginTop: 2 }]}>
      {!mine ? (
        <View style={{ width: 30 }}>
          {!grouped ? (
            item.ai ? (
              <View style={[styles.avatar, { backgroundColor: colors.black }]}>
                <Ionicons name="sparkles" size={14} color={colors.white} />
              </View>
            ) : (
              <View style={[styles.avatar, { backgroundColor: avatarBg }]}>
                <T variant="micro" weight="bold" color={colors.white} style={{ fontSize: 12 }}>
                  {(m?.name.trim()[0] ?? '?').toUpperCase()}
                </T>
              </View>
            )
          ) : null}
        </View>
      ) : null}
      <View style={{ maxWidth: '78%', alignItems: mine ? 'flex-end' : 'flex-start' }}>
        {!grouped && !mine ? (
          <View style={styles.nameRow}>
            <T variant="micro" weight="semibold" color={colors.textSecondary}>
              {name}
            </T>
            {item.ai && item.aiSource === 'claude' ? (
              <T variant="micro" color={colors.purple}>
                · Claude
              </T>
            ) : null}
          </View>
        ) : null}
        <View style={[styles.bubble, mine ? styles.mine : item.ai ? styles.ai : styles.theirs]}>
          <T variant="bodySm" color={mine ? colors.white : item.ai ? colors.onAccent : colors.text} selectable>
            {item.text}
          </T>
        </View>
      </View>
    </View>
  );
}

const styles = StyleSheet.create({
  system: { alignSelf: 'center', backgroundColor: colors.bgAlt, paddingHorizontal: 12, paddingVertical: 5, borderRadius: 999, marginVert
ical: 8 },
  msgRow: { flexDirection: 'row', alignItems: 'flex-end', gap: 8, marginTop: 12 },
  nameRow: { flexDirection: 'row', gap: 3, marginBottom: 3, marginHorizontal: 6 },
  avatar: { width: 30, height: 30, borderRadius: 15, alignItems: 'center', justifyContent: 'center' },
  bubble: { paddingHorizontal: 16, paddingVertical: 11, borderRadius: 20 },
  mine: { backgroundColor: colors.blue },
  theirs: { backgroundColor: colors.card, ...shadow },
  ai: { backgroundColor: colors.accentSoft, ...shadow },
});
```

src/components/chat/ChatHeader.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { router } from 'expo-router';
import { StyleSheet, View } from 'react-native';
import { useSafeAreaInsets } from 'react-native-safe-area-context';

import { SyncBadge } from '@/components/sync/SyncBadge';
import { IconButton } from '@/components/ui/bits';
import { Press } from '@/components/ui/Press';
import { Segmented } from '@/components/ui/Segmented';
import { T } from '@/components/ui/T';
import type { Trip } from '@/store/types';
import { colors } from '@/theme';

export type ChatTab = 'crew' | 'ai';
const TABS = [
  { key: 'crew' as const, label: 'Crew' },
  { key: 'ai' as const, label: 'Scout AI' },
];
```

```
/** Trip name, the Crew / Scout AI switch, whether the trip is live-synced, and (in Scout AI) which brain answers. */
export function ChatHeader({ trip, tab, onTab, synced, aiMode }: { trip: Trip; tab: ChatTab; onTab: (t: ChatTab) => void; synced: boolea
n; aiMode: 'offline' | 'claude' }) {
  const insets = useSafeAreaInsets();
  return (
    <View style={[styles.header, { paddingTop: insets.top + 12 }]}>
      <View style={styles.titleRow}>
        <T variant="display" numberOfLines={1} style={{ flex: 1 }}>
          {trip.name}
        </T>
        <IconButton icon="images-outline" label="Photo dump" onPress={() => router.push('/photos')} />
      </View>
      <Segmented options={TABS} value={tab} onChange={onTab} />
      {synced ? (
        <View style={styles.syncRow}>
          <SyncBadge tripId={trip.id} />
          <T variant="micro" color={colors.textSecondary}>
            Shared with your group
          </T>
        </View>
      ) : null}
      {tab === 'ai' ? (
        <Press onPress={() => router.push('/settings')} style={styles.modePill} accessibilityLabel="Assistant settings">
          <Ionicons name={aiMode === 'claude' ? 'sparkles' : 'cloud-offline-outline'} size={12} color={colors.text} />
          <T variant="micro" weight="semibold" color={colors.text}>
            {aiMode === 'claude' ? 'Answers by Claude · change in Settings' : 'Answers offline from your trip data · change in Setting
s'}
          </T>
        </Press>
      ) : null}
    </View>
  );
}

const styles = StyleSheet.create({
  header: { gap: 12, paddingHorizontal: 20, paddingBottom: 12, backgroundColor: colors.card, borderBottomWidth: 1, borderBottomColor: co
lors.border },
  titleRow: { flexDirection: 'row', alignItems: 'center', gap: 8 },
  syncRow: { flexDirection: 'row', alignItems: 'center', gap: 8 },
  modePill: { alignSelf: 'flex-start', flexDirection: 'row', alignItems: 'center', gap: 4, paddingHorizontal: 10, paddingVertical: 6, bo
rderRadius: 999, backgroundColor: colors.bgAlt },
});
```

src/components/chat/Composer.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { StyleSheet, TextInput, View } from 'react-native';

import { Press } from '@/components/ui/Press';
import { colors, fonts } from '@/theme';

/** The message box and send button at the bottom of the chat. */
export function Composer({ value, onChange, onSend, placeholder }: { value: string; onChange: (v: string) => void; onSend: () => void; p
laceholder: string }) {
  return (
    <View style={styles.inputBar}>
      <TextInput
        value={value}
        onChangeText={onChange}
        placeholder={placeholder}
        placeholderTextColor={colors.textMuted}
        style={styles.input}
        multiline
        maxLength={1000}
        accessibilityLabel="Message"
        onSubmitEditing={onSend}
        submitBehavior="submit"
        returnKeyType="send"
        testID="chat-input"
      />
      <Press onPress={onSend} style={[styles.send, !value.trim() && { opacity: 0.4 }]} disabled={!value.trim()} accessibilityLabel="Send
message" testID="chat-send">
        <Ionicons name="arrow-up" size={20} color={colors.white} />
      </Press>
    </View>
  );
}

const styles = StyleSheet.create({
  inputBar: { flexDirection: 'row', alignItems: 'flex-end', gap: 8, paddingHorizontal: 16, paddingTop: 6, paddingBottom: 10 },
  input: {
    flex: 1,
    minHeight: 44,
    maxHeight: 120,
    borderRadius: 22,
    backgroundColor: colors.card,
```

```
    borderWidth: 1,
    borderColor: colors.border,
    paddingHorizontal: 18,
    paddingTop: 12,
    paddingBottom: 12,
    fontFamily: fonts.regular,
    fontSize: 15,
    color: colors.text,
    ...({ outlineStyle: 'none' } as object),
  },
  send: { width: 44, height: 44, borderRadius: 22, backgroundColor: colors.black, alignItems: 'center', justifyContent: 'center' },
});
```

src/components/chat/EmptyChat.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { StyleSheet, View } from 'react-native';

import { Card } from '@/components/ui/bits';
import { T } from '@/components/ui/T';
import { colors } from '@/theme';

/** What an empty Crew chat or an empty Scout AI thread shows. */
export function EmptyChat({ ai }: { ai: boolean }) {
  return ai ? (
    <Card style={styles.intro}>
      <View style={styles.introIcon}>
        <Ionicons name="sparkles" size={20} color={colors.white} />
      </View>
      <T variant="headline">Scout AI</T>
      <T variant="small" color={colors.textSecondary} center>
        Scout AI knows your trip — budget, weather, food nearby, who owes who, flights… Ask away or tap a question below.
      </T>
    </Card>
  ) : (
    <View style={{ alignItems: 'center', padding: 30, gap: 6 }}>
      <Ionicons name="chatbubbles-outline" size={34} color={colors.textMuted} />
      <T variant="title">Say hi to your crew 👋</T>
      <T variant="small" color={colors.textSecondary} center>
        Messages here are shared with everyone on the trip. Switch to Scout AI to ask the assistant.
      </T>
    </View>
  );
}

const styles = StyleSheet.create({
  intro: { alignItems: 'center', gap: 8, paddingVertical: 24, marginTop: 8 },
  introIcon: { width: 48, height: 48, borderRadius: 24, backgroundColor: colors.black, alignItems: 'center', justifyContent: 'center' },
});
```

src/components/crew/AddMemberSheet.tsx

```
import { useState } from 'react';

import { Button } from '@/components/ui/Button';
import { toast } from '@/components/ui/feedback';
import { Input } from '@/components/ui/Input';
import { tap } from '@/components/ui/Press';
import { Sheet } from '@/components/ui/Sheet';
import { validateEmail, validateName } from '@/utils/validation';

/** Add someone by name (they can still join with the invite code later). */
export function AddMemberSheet({ visible, onClose, onAdd, onShare }: { visible: boolean; onClose: () => void; onAdd: (name: string, emai
l?: string) => void; onShare: () => void }) {
  const [name, setName] = useState('');
  const [email, setEmail] = useState('');
  const [errors, setErrors] = useState<{ name?: string | null; email?: string | null }>({});
  return (
    <Sheet visible={visible} onClose={onClose} title="Add a member" subtitle="Add someone by name, or share the invite code so they can
join themselves">
      <Button label="Share invite code instead" variant="soft" icon="share-social-outline" onPress={onShare} />
      <Input label="Name" value={name} onChangeText={setName} error={errors.name} icon="person-outline" autoCapitalize="words" />
      <Input label="Email (optional)" value={email} onChangeText={setEmail} error={errors.email} icon="mail-outline" autoCapitalize="non
e" keyboardType="email-address" />
      <Button
        label="Add member"
        icon="person-add-outline"
        onPress={() => {
          const e = { name: validateName(name), email: email.trim() ? validateEmail(email) : null };
          setErrors(e);
          if (e.name || e.email) return tap('warning');
          onAdd(name, email);
          toast(`${name.trim().split(' ')[0]} added to the trip`);
          setName('');
          setEmail('');
          onClose();
```

```
        }}
      />
    </Sheet>
  );
}
```

src/components/crew/ChecklistSection.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { useState } from 'react';
import { StyleSheet, TextInput, View } from 'react-native';

import { Press, tap } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import { Card, SectionHeader } from '@/components/ui/bits';
import { useT } from '@/i18n';
import { useAppStore } from '@/store/useAppStore';
import type { Trip } from '@/store/types';
import { colors, fonts, radius } from '@/theme';
import { firstError, maxLen, required } from '@/utils/validation';

/** The shared to-do list: tick things off, assign them, add new ones. */
export function ChecklistSection({ trip, onAssign }: { trip: Trip; onAssign: (taskId: string) => void }) {
  const t = useT();
  const s = useAppStore.getState();
  const [task, setTask] = useState('');
  const [taskErr, setTaskErr] = useState<string | null>(null);
  const done = trip.tasks.filter((k) => k.done).length;

  const addTask = () => {
    const e = firstError(task, required('a task'), maxLen(60, 'Task'));
    setTaskErr(e);
    if (e) return tap('warning');
    s.addTask(trip.id, task);
    setTask('');
    tap('success');
  };

  return (
    <>
      <SectionHeader title={t('hub.checklist')} action={t('hub.done', { a: done, b: trip.tasks.length })} style={{ marginTop: 26 }} />
      <Card padded={false} style={{ paddingHorizontal: 16, paddingVertical: 4 }}>
        {trip.tasks.map((k) => {
          const who = trip.members.find((m) => m.id === k.assignee);
          return (
            <View key={k.id} style={styles.task}>
              <Press onPress={() => s.toggleTask(trip.id, k.id)} accessibilityRole="checkbox" accessibilityState={{ checked: k.done }} a
ccessibilityLabel={k.text} hitSlop={6}>
                <Ionicons name={k.done ? 'checkmark-circle' : 'ellipse-outline'} size={22} color={k.done ? colors.primary : colors.borde
rStrong} />
              </Press>
              <Press onPress={() => s.toggleTask(trip.id, k.id)} style={{ flex: 1 }} haptic={false}>
                <T variant="small" color={k.done ? colors.textMuted : colors.text} style={k.done && { textDecorationLine: 'line-through'
}}>
                  {k.text}
                </T>
              </Press>
              <Press onPress={() => onAssign(k.id)} hitSlop={6} accessibilityLabel="Assign task">
                <T variant="caption" color={colors.textMuted}>
                  {who ? who.name.split(' ')[0] : 'Everyone'}
                </T>
              </Press>
              <Press onPress={() => s.deleteTask(trip.id, k.id)} hitSlop={6} accessibilityLabel={`Delete task ${k.text}`}>
                <Ionicons name="close" size={16} color={colors.textMuted} />
              </Press>
            </View>
          );
        })}
        <View style={styles.addTask}>
          <TextInput
            value={task}
            onChangeText={(v) => {
              setTask(v);
              if (taskErr) setTaskErr(null);
            }}
            placeholder={t('hub.addTask')}
            placeholderTextColor={colors.textMuted}
            style={[styles.taskInput, taskErr && { borderColor: colors.red }]}
            onSubmitEditing={addTask}
            returnKeyType="done"
            maxLength={70}
            accessibilityLabel="New task"
          />
          <Press onPress={addTask} style={styles.addBtn} accessibilityLabel="Add task">
            <Ionicons name="add" size={20} color={colors.white} />
          </Press>
```

```
        </View>
        {taskErr ? (
          <T variant="caption" color={colors.red} style={{ marginBottom: 8 }}>
            {taskErr}
          </T>
        ) : null}
      </Card>
    </>
  );
}

const styles = StyleSheet.create({
  task: { flexDirection: 'row', alignItems: 'center', gap: 10, paddingVertical: 11, borderBottomWidth: 1, borderBottomColor: colors.bord
er },
  addTask: { flexDirection: 'row', alignItems: 'center', gap: 8, paddingVertical: 10 },
  taskInput: { flex: 1, height: 42, borderRadius: radius.sm, borderWidth: 1, borderColor: colors.border, backgroundColor: colors.cardMut
ed, paddingHorizontal: 12, fontFamily: fonts.regular, color: colors.text },
  addBtn: { width: 40, height: 40, borderRadius: 20, backgroundColor: colors.primary, alignItems: 'center', justifyContent: 'center' },
});
```

src/components/crew/DocsSection.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { router } from 'expo-router';
import { StyleSheet, View } from 'react-native';

import { Press } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import { Card, Pill, SectionHeader } from '@/components/ui/bits';
import { useT } from '@/i18n';
import type { Trip } from '@/store/types';
import { colors } from '@/theme';

/** A short list of the trip's documents, with a button to open the full vault. */
export function DocsSection({ trip }: { trip: Trip }) {
  const t = useT();
  return (
    <>
      <SectionHeader title={t('hub.documents')} action="Open vault" onAction={() => router.push('/documents')} style={{ marginTop: 26, m
arginBottom: 2 }} />
      <T variant="caption" color={colors.textSecondary} style={{ marginBottom: 10 }}>
        {t('hub.documentsSub')}
      </T>
      {trip.docs.length ? (
        <Card padded={false}>
          {trip.docs.slice(0, 4).map((d, i) => (
            <Press key={d.id} onPress={() => router.push({ pathname: '/documents', params: { open: d.id } })} style={[styles.doc, i > 0
&& styles.line]} scaleTo={0.99}>
              <View style={styles.docIcon}>
                <Ionicons name={d.kind === 'image' ? 'image-outline' : d.kind === 'note' ? 'document-text-outline' : 'document-outline'}
size={18} color={colors.text} />
              </View>
              <View style={{ flex: 1 }}>
                <T variant="small" weight="bold" numberOfLines={1}>
                  {d.name}
                </T>
                <T variant="caption" color={colors.textMuted}>
                  {d.kind === 'note' ? 'Note' : d.kind.toUpperCase()}
                </T>
              </View>
              {d.offline ? <Pill label={t('hub.offline')} icon="cloud-done-outline" tone="gray" /> : null}
            </Press>
          ))}
        </Card>
      ) : null}
    </>
  );
}

const styles = StyleSheet.create({
  line: { borderTopWidth: 1, borderTopColor: colors.border },
  doc: { flexDirection: 'row', alignItems: 'center', gap: 12, paddingHorizontal: 16, paddingVertical: 12 },
  docIcon: { width: 36, height: 36, borderRadius: 18, backgroundColor: colors.bgAlt, alignItems: 'center', justifyContent: 'center' },
});
```

src/components/crew/InviteCard.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { useState } from 'react';
import { StyleSheet, View } from 'react-native';

import { Button } from '@/components/ui/Button';
import { Card } from '@/components/ui/bits';
import { confirmAction, toast } from '@/components/ui/feedback';
import { Press } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
```

```
import { useT } from '@/i18n';
import { copy } from '@/services/share';
import { useAppStore } from '@/store/useAppStore';
import type { Trip } from '@/store/types';
import { colors, radius } from '@/theme';

import { InviteQrSheet } from './InviteQrSheet';

/** The trip's invite code with Copy, Share, QR code and (for the organizer) New code. */
export function InviteCard({ trip, owner, onShare }: { trip: Trip; owner: boolean; onShare: () => void }) {
  const t = useT();
  const [qrOpen, setQrOpen] = useState(false);
  return (
    <Card style={styles.invite}>
      <T variant="kicker" color={colors.textSecondary}>
        {t('hub.invite')}
      </T>
      <View style={styles.codeRow}>
        <T variant="display" numberOfLines={1} adjustsFontSizeToFit style={styles.code} accessibilityLabel={`Invite code ${trip.inviteCo
de}`}>
          {trip.inviteCode}
        </T>
        <Press onPress={() => copy(trip.inviteCode, `Invite code ${trip.inviteCode} copied`)} style={styles.copyPill} accessibilityLabel
={`Copy invite code ${trip.inviteCode}`}>
          <Ionicons name="copy-outline" size={15} color={colors.onAccent} />
          <T variant="small" weight="bold" color={colors.onAccent}>
            Copy
          </T>
        </Press>
      </View>
      <T variant="caption" color={colors.textSecondary}>
        Friends enter this in “Join a trip”, or scan the QR code
      </T>
      <View style={styles.row}>
        <Button label={t('hub.share')} icon="share-social-outline" size="md" onPress={onShare} style={{ flex: 1 }} />
        <Button label="QR code" icon="qr-code-outline" variant="secondary" size="md" onPress={() => setQrOpen(true)} style={{ flex: 1 }}
testID="invite-qr" />
      </View>
      {owner ? (
        <Button
          label="New code"
          icon="refresh-outline"
          variant="ghost"
          size="sm"
          onPress={() =>
            confirmAction('New invite code?', 'The old code will stop working. Members already on the trip stay.', () => {
              useAppStore.getState().regenerateInvite(trip.id);
              toast('New invite code created');
            })
          }
        />
      ) : null}
      <InviteQrSheet visible={qrOpen} onClose={() => setQrOpen(false)} trip={trip} />
    </Card>
  );
}

const styles = StyleSheet.create({
  invite: { marginTop: 14, gap: 6, borderRadius: radius.xl, padding: 18 },
  codeRow: { flexDirection: 'row', alignItems: 'center', justifyContent: 'space-between', gap: 12 },
  code: { flexShrink: 1, fontSize: 44, lineHeight: 52, letterSpacing: 3 },
  copyPill: { flexDirection: 'row', alignItems: 'center', gap: 6, backgroundColor: colors.accent, paddingHorizontal: 16, height: 38, bor
derRadius: 999 },
  row: { flexDirection: 'row', alignItems: 'center', gap: 10 },
});
```

src/components/crew/InviteQrSheet.tsx

```
import { StyleSheet, View, useWindowDimensions } from 'react-native';

import { QrCode } from '@/components/QrCode';
import { Sheet } from '@/components/ui/Sheet';
import { T } from '@/components/ui/T';
import { inviteLink } from '@/services/share';
import type { Trip } from '@/store/types';
import { parseInviteUrl } from '@/sync/invite';
import { colors, radius } from '@/theme';

/** The invite as a QR code: a friend points their phone camera at it and the app opens on the join screen. */
export function InviteQrSheet({ visible, onClose, trip }: { visible: boolean; onClose: () => void; trip: Trip }) {
  const { width } = useWindowDimensions();
  const link = inviteLink(trip.inviteCode);
  const size = Math.min(280, width - 96);
  return (
    <Sheet visible={visible} onClose={onClose} title="Scan to join" subtitle={trip.name}>
      <View style={styles.center}>
```

```
        <View style={styles.box}>
          <QrCode value={link} size={size} accessibilityLabel={`QR code for invite ${trip.inviteCode}`} />
        </View>
        <T variant="display" style={{ letterSpacing: 3 }} accessibilityLabel={`Invite code ${trip.inviteCode}`}>
          {trip.inviteCode}
        </T>
        <T variant="small" color={colors.textSecondary} center>
          Friends point their phone camera here, or open TogetherWeGo → Join a trip → Scan, or type the code.
        </T>
        {parseInviteUrl(link).server ? (
          <T variant="caption" color={colors.textMuted} center>
            It also carries your sync server’s address, so their phone connects by itself.
          </T>
        ) : null}
      </View>
    </Sheet>
  );
}

const styles = StyleSheet.create({
  center: { alignItems: 'center', gap: 12, paddingBottom: 8 },
  box: { padding: 12, borderRadius: radius.lg, backgroundColor: '#FFFFFF', borderWidth: 1, borderColor: colors.border },
});
```

src/components/crew/MembersSection.tsx

```
import { StyleSheet, View } from 'react-native';

import { Press } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import { Avatar, Card, SectionHeader } from '@/components/ui/bits';
import { useT } from '@/i18n';
import type { Member, Trip } from '@/store/types';
import { colors } from '@/theme';

/** Everyone on the trip with their role. Tap a person to change their role. */
export function MembersSection({ trip, userId, onPick, onAdd }: { trip: Trip; userId: string; onPick: (m: Member) => void; onAdd: () =>
void }) {
  const t = useT();
  return (
    <>
      <SectionHeader title={t('hub.members')} action="+ Add" onAction={onAdd} style={{ marginTop: 26 }} />
      <Card padded={false}>
        {trip.members.map((m, i) => (
          <Press key={m.id} onPress={() => onPick(m)} style={[styles.member, i > 0 && styles.line]} scaleTo={0.99} accessibilityLabel={`
${m.name}, ${m.role}. Tap to change role`}>
            <Avatar name={m.name} src={m.avatar} size={40} />
            <View style={{ flex: 1 }}>
              <T variant="title" weight="bold" numberOfLines={1}>
                {m.name}
                {m.id === userId ? ' (you)' : ''}
                {m.id === trip.ownerId ? ' 👑' : ''}
              </T>
              <T variant="caption" color={colors.textSecondary} numberOfLines={1}>
                {m.invited ? 'Invited · hasn’t joined yet' : m.homeCity ? `From ${m.homeCity}` : 'Member'}
              </T>
            </View>
            <View style={styles.rolePill}>
              <T variant="micro" weight="bold" color={colors.onAccent}>
                {m.role}
              </T>
            </View>
          </Press>
        ))}
      </Card>
    </>
  );
}

const styles = StyleSheet.create({
  line: { borderTopWidth: 1, borderTopColor: colors.border },
  member: { flexDirection: 'row', alignItems: 'center', gap: 12, paddingHorizontal: 16, paddingVertical: 12 },
  rolePill: { paddingHorizontal: 10, paddingVertical: 4, borderRadius: 999, backgroundColor: colors.accent },
});
```

src/components/crew/NewPollSheet.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { useState } from 'react';
import { StyleSheet, TextInput, View } from 'react-native';

import { Button } from '@/components/ui/Button';
import { toast } from '@/components/ui/feedback';
import { Input } from '@/components/ui/Input';
import { Press } from '@/components/ui/Press';
import { Sheet } from '@/components/ui/Sheet';
```

```
import { T } from '@/components/ui/T';
import { colors, fonts, radius } from '@/theme';

/** Start a poll: a question and two to five options. */
export function NewPollSheet({ visible, onClose, onCreate }: { visible: boolean; onClose: () => void; onCreate: (q: string, opts: string
[]) => void }) {
  const [q, setQ] = useState('');
  const [opts, setOpts] = useState(['', '']);
  const [err, setErr] = useState<string | null>(null);
  return (
    <Sheet visible={visible} onClose={onClose} title="New poll" subtitle="Let the group decide">
      <Input label="Question" value={q} onChangeText={setQ} placeholder="e.g. Where should we eat on Day 3?" maxLength={80} />
      {opts.map((o, i) => (
        <View key={i} style={{ flexDirection: 'row', gap: 8, alignItems: 'center' }}>
          <TextInput
            value={o}
            onChangeText={(v) => setOpts((arr) => arr.map((x, j) => (j === i ? v : x)))}
            placeholder={`Option ${i + 1}`}
            placeholderTextColor={colors.textMuted}
            style={styles.optInput}
            maxLength={50}
            accessibilityLabel={`Option ${i + 1}`}
          />
          {opts.length > 2 ? (
            <Press onPress={() => setOpts((arr) => arr.filter((_, j) => j !== i))} accessibilityLabel={`Remove option ${i + 1}`}>
              <Ionicons name="close-circle" size={22} color={colors.textMuted} />
            </Press>
          ) : null}
        </View>
      ))}
      {opts.length < 5 ? <Button label="Add option" variant="ghost" size="sm" icon="add" onPress={() => setOpts((a) => [...a, ''])} /> :
null}
      {err ? (
        <T variant="caption" color={colors.red}>
          {err}
        </T>
      ) : null}
      <Button
        label="Start poll"
        icon="stats-chart-outline"
        onPress={() => {
          const clean = opts.map((o) => o.trim()).filter(Boolean);
          if (!q.trim()) return setErr('Write a question for the poll');
          if (clean.length < 2) return setErr('Add at least two options');
          if (new Set(clean.map((c) => c.toLowerCase())).size !== clean.length) return setErr('Options must be different');
          onCreate(q, clean);
          setQ('');
          setOpts(['', '']);
          setErr(null);
          toast('Poll started — the group has been notified');
          onClose();
        }}
      />
    </Sheet>
  );
}
const styles = StyleSheet.create({
  optInput: { flex: 1, height: 46, borderRadius: radius.md, borderWidth: 1, borderColor: colors.border, backgroundColor: colors.cardMute
d, paddingHorizontal: 12, fontFamily: fonts.regular, color: colors.text },
});
```

src/components/crew/PollsSection.tsx

```
import { StyleSheet, View } from 'react-native';

import { Press, tap } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import { Card, Pill, ProgressBar, SectionHeader } from '@/components/ui/bits';
import { confirmAction } from '@/components/ui/feedback';
import { useT } from '@/i18n';
import { useAppStore } from '@/store/useAppStore';
import type { Trip } from '@/store/types';
import { colors, radius } from '@/theme';

/** The group's polls: vote, close or delete, and start a new one. */
export function PollsSection({ trip, userId, owner, onNew }: { trip: Trip; userId: string; owner: boolean; onNew: () => void }) {
  const t = useT();
  const s = useAppStore.getState();
  return (
    <>
      <SectionHeader title={t('hub.vote')} action="+ New poll" onAction={onNew} style={{ marginTop: 26 }} />
      <View style={{ gap: 10 }}>
        {trip.polls.length ? (
          trip.polls.map((p) => {
            const total = p.options.reduce((n, o) => n + o.votes.length, 0);
            const voted = p.options.find((o) => o.votes.includes(userId));
```

```
            return (
              <Card key={p.id} style={{ gap: 10 }}>
                <View style={styles.rowBetween}>
                  <T variant="title" weight="bold" style={{ flex: 1 }}>
                    {p.question}
                  </T>
                  {p.closed ? <Pill label="Closed" tone="gray" /> : <Pill label={`${total} vote${total === 1 ? '' : 's'}`} tone="yellow"
/>}
                </View>
                {p.options.map((o) => {
                  const pctv = total ? Math.round((o.votes.length / total) * 100) : 0;
                  const mine = o.votes.includes(userId);
                  return (
                    <Press
                      key={o.id}
                      disabled={p.closed}
                      onPress={() => {
                        s.votePoll(trip.id, p.id, o.id);
                        tap();
                      }}
                      style={[styles.option, mine && styles.optionOn]}
                      accessibilityLabel={`${o.text}, ${pctv} percent${mine ? ', your vote' : ''}`}>
                      <View style={styles.rowBetween}>
                        <T variant="caption" weight={mine ? 'semibold' : 'regular'} style={{ flex: 1 }}>
                          {mine ? '✓ ' : ''}
                          {o.text}
                        </T>
                        <T variant="caption" color={colors.textSecondary}>
                          {pctv}%
                        </T>
                      </View>
                      <ProgressBar value={pctv / 100} height={5} />
                    </Press>
                  );
                })}
                <View style={styles.rowBetween}>
                  <T variant="caption" color={colors.textMuted} style={{ flex: 1 }}>
                    {voted ? 'Tap again to remove your vote' : p.closed ? 'Voting closed' : 'Tap an option to vote'}
                  </T>
                  {p.createdBy === userId || owner ? (
                    <View style={{ flexDirection: 'row', gap: 12 }}>
                      <Press onPress={() => s.closePoll(trip.id, p.id)} hitSlop={6}>
                        <T variant="caption" weight="semibold" color={colors.text}>
                          {p.closed ? 'Reopen' : 'Close'}
                        </T>
                      </Press>
                      <Press
                        onPress={() => confirmAction('Delete poll?', p.question, () => s.deletePoll(trip.id, p.id), { destructive: true,
confirmLabel: 'Delete' })}
                        hitSlop={6}>
                        <T variant="caption" weight="semibold" color={colors.red}>
                          Delete
                        </T>
                      </Press>
                    </View>
                  ) : null}
                </View>
              </Card>
            );
          })
        ) : (
          <Card>
            <T variant="small" color={colors.textSecondary}>
              No polls yet — start one to settle a decision fast.
            </T>
          </Card>
        )}
      </View>
    </>
  );
}

const styles = StyleSheet.create({
  rowBetween: { flexDirection: 'row', alignItems: 'center', justifyContent: 'space-between', gap: 8 },
  option: { gap: 6, padding: 10, borderRadius: radius.sm, borderWidth: 1, borderColor: colors.border },
  optionOn: { borderColor: colors.primaryLine, backgroundColor: colors.primarySofter },
});
```

src/components/crew/RoleSheet.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { StyleSheet, View } from 'react-native';

import { Button } from '@/components/ui/Button';
import { Press } from '@/components/ui/Press';
import { Sheet } from '@/components/ui/Sheet';
```

```
import { T } from '@/components/ui/T';
import { ROLE_OPTIONS, roleDescriptions, type Role } from '@/data/reference';
import type { Member } from '@/store/types';
import { colors, radius } from '@/theme';

/** Pick a role for a member (or remove them from the trip, if you are the organizer). */
export function RoleSheet({ member, canEdit, canRemove, onClose, onPick, onRemove }: { member: Member | null; canEdit: boolean; canRemov
e: boolean; onClose: () => void; onPick: (r: Role) => void; onRemove: () => void }) {
  return (
    <Sheet visible={!!member} onClose={onClose} title={member?.name} subtitle={canEdit ? 'Choose a role for this trip' : 'Only the organ
izer can change other people’s roles'}>
      {ROLE_OPTIONS.map((r) => (
        <Press key={r} disabled={!canEdit} onPress={() => onPick(r)} style={[styles.roleRow, member?.role === r && styles.optionOn, !can
Edit && { opacity: 0.6 }]} accessibilityState={{ selected: member?.role === r }}>
          <View style={{ flex: 1 }}>
            <T variant="small" weight="semibold">
              {r}
            </T>
            <T variant="caption" color={colors.textSecondary}>
              {roleDescriptions[r]}
            </T>
          </View>
          {member?.role === r ? <Ionicons name="checkmark-circle" size={20} color={colors.primary} /> : null}
        </Press>
      ))}
      {canRemove ? <Button label="Remove from trip" variant="danger" icon="person-remove-outline" onPress={onRemove} /> : null}
    </Sheet>
  );
}

const styles = StyleSheet.create({
  roleRow: { flexDirection: 'row', alignItems: 'center', gap: 10, padding: 12, borderRadius: radius.md, borderWidth: 1, borderColor: col
ors.border },
  optionOn: { borderColor: colors.primaryLine, backgroundColor: colors.primarySofter },
});
```

src/components/documents/DocFileBody.tsx

```
import { Image } from 'expo-image';
import { ActivityIndicator, View } from 'react-native';

import { Button } from '@/components/ui/Button';
import { T } from '@/components/ui/T';
import type { TripDoc } from '@/store/types';
import { useSharedFile } from '@/sync/useFile';
import { colors } from '@/theme';

/** The file part of an opened document: a preview for pictures, an "Open / share" button, or why there is no file here. */
export function DocFileBody({ doc, adderName, onOpen }: { doc: TripDoc; adderName: string; onOpen: (uri: string) => void }) {
  const { uri, loading } = useSharedFile(doc.uri, doc.blob);
  if (doc.kind === 'note') return null;
  if (!uri) {
    return loading ? (
      <View style={{ flexDirection: 'row', alignItems: 'center', gap: 10 }}>
        <ActivityIndicator color={colors.primary} />
        <T variant="small" color={colors.textSecondary}>
          Getting the file from {adderName}…
        </T>
      </View>
    ) : (
      <T variant="small" color={colors.textSecondary}>
        This file is on {adderName}’s phone. Its name and notes sync; the file arrives once their phone has sent it to the sync server.
      </T>
    );
  }
  return (
    <>
      {doc.kind === 'image' ? <Image source={{ uri }} style={{ width: '100%', height: 320, borderRadius: 14 }} contentFit="contain" /> :
null}
      <Button label="Open / share file" icon="open-outline" onPress={() => onOpen(uri)} />
    </>
  );
}
```

src/components/documents/DocRow.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { StyleSheet, View } from 'react-native';

import { Press } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import { Pill } from '@/components/ui/bits';
import { sizeLabel } from '@/services/files';
import type { Trip, TripDoc } from '@/store/types';
import { useSharedFile } from '@/sync/useFile';
import { colors, radius } from '@/theme';
```

```
import { timeAgo } from '@/utils/format';

const ICON = { image: 'image-outline', note: 'document-text-outline', pdf: 'document-outline', file: 'attach-outline' } as const;

/** One line in the document vault. The pill says whether the file can be opened with no connection. */
export function DocRow({ doc, trip, onPress }: { doc: TripDoc; trip: Trip; onPress: () => void }) {
  const { uri, loading } = useSharedFile(doc.uri, doc.blob);
  const adder = trip.members.find((m) => m.id === doc.addedBy)?.name.split(' ')[0];
  return (
    <Press onPress={onPress} style={styles.row} scaleTo={0.99} accessibilityLabel={`Open ${doc.name}`}>
      <View style={styles.icon}>
        <Ionicons name={ICON[doc.kind]} size={20} color={colors.primary} />
      </View>
      <View style={{ flex: 1 }}>
        <T variant="small" weight="semibold" numberOfLines={1}>
          {doc.name}
        </T>
        <T variant="caption" color={colors.textMuted}>
          {doc.kind === 'note' ? 'Note' : doc.kind.toUpperCase()} {doc.size ? `· ${sizeLabel(doc.size)}` : ''} · {timeAgo(doc.addedAt)}
· {adder}
        </T>
      </View>
      {doc.kind === 'note' || uri ? <Pill label="Offline" icon="cloud-done-outline" /> : loading ? <Pill label="Downloading" icon="cloud
-download-outline" tone="blue" /> : <Pill label={`On ${adder ?? 'a friend'}’s phone`} icon="phone-portrait-outline" tone="gray" />}
    </Press>
  );
}

const styles = StyleSheet.create({
  row: { flexDirection: 'row', alignItems: 'center', gap: 12, padding: 12, borderRadius: radius.md, backgroundColor: colors.card, border
Width: 1, borderColor: colors.border },
  icon: { width: 40, height: 40, borderRadius: 12, backgroundColor: colors.primarySofter, alignItems: 'center', justifyContent: 'center'
},
});
```

src/components/explore.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { Image } from 'expo-image';
import { LinearGradient } from 'expo-linear-gradient';
import { router } from 'expo-router';
import { StyleSheet, View, type StyleProp, type ViewStyle } from 'react-native';

import { destinationById, type Destination } from '@/data/destinations';
import { imageSource } from '@/data/images';
import { useActiveTrip, useAppStore } from '@/store/useAppStore';
import { colors, radius, shadow } from '@/theme';
import { money } from '@/utils/format';

import { Chip, ListRow } from './ui/bits';
import { toast } from './ui/feedback';
import { Press, tap } from './ui/Press';
import { Sheet } from './ui/Sheet';
import { T } from './ui/T';

export function FavStar({ id, size = 30 }: { id: string; size?: number }) {
  const fav = useAppStore((s) => s.favorites.includes(id));
  const toggle = useAppStore((s) => s.toggleFavorite);
  return (
    <Press
      onPress={() => {
        toggle(id);
        tap();
        toast(fav ? 'Removed from favourites' : 'Saved to favourites ⭐', { icon: fav ? 'star-outline' : 'star' });
      }}
      hitSlop={6}
      accessibilityLabel={fav ? 'Remove favourite' : 'Save as favourite'}
      style={[styles.star, { width: size, height: size, borderRadius: size / 2 }]}>
      <Ionicons name={fav ? 'star' : 'star-outline'} size={size * 0.5} color={fav ? colors.star : colors.white} />
    </Press>
  );
}

export function PlaceCard({ d, style, height = 190 }: { d: Destination; style?: StyleProp<ViewStyle>; height?: number }) {
  return (
    // The heart sits next to the card button, not inside it: on the web a button inside a button is invalid HTML.
    <View style={[styles.place, { height }, style]}>
      <Press onPress={() => router.push({ pathname: '/destination/[id]', params: { id: d.id } })} style={StyleSheet.absoluteFill} access
ibilityLabel={`${d.city}, ${d.country}, rated ${d.rating}`}>
        <Image source={imageSource(d.image)} style={StyleSheet.absoluteFill} contentFit="cover" transition={200} />
        <LinearGradient colors={['rgba(0,0,0,0)', 'rgba(0,0,0,0.65)']} locations={[0.45, 1]} style={StyleSheet.absoluteFill} />
        <View style={styles.placeText}>
          <T variant="title" weight="bold" color={colors.white} numberOfLines={1}>
            {d.city}
          </T>
          <View style={styles.row}>
```

```
            <Ionicons name="location-sharp" size={10} color="rgba(255,255,255,0.85)" />
            <T variant="micro" color="rgba(255,255,255,0.85)" numberOfLines={1} style={{ flex: 1 }}>
              {d.country}
            </T>
            <Ionicons name="star" size={10} color={colors.star} />
            <T variant="micro" weight="bold" color={colors.white}>
              {d.rating.toFixed(1)}
            </T>
          </View>
        </View>
      </Press>
      <View style={{ position: 'absolute', top: 10, right: 10 }}>
        <FavStar id={d.id} />
      </View>
    </View>
  );
}

export function AddToTripSheet({ destinationId, onClose }: { destinationId: string | null; onClose: () => void }) {
  const trip = useActiveTrip();
  const updateTrip = useAppStore((s) => s.updateTrip);
  const addBucket = useAppStore((s) => s.addBucket);
  const d = destinationId ? destinationById(destinationId) : undefined;
  if (!d) return <Sheet visible={false} onClose={onClose}>{null}</Sheet>;
  const already = trip && (trip.destinationId === d.id || trip.extraDestinationIds.includes(d.id));
  return (
    <Sheet visible={!!destinationId} onClose={onClose} title={`Add ${d.city}`} subtitle={`${d.country} · ★ ${d.rating} · ~${money(d.dail
yCost)}/day per person`}>
      {trip ? (
        <ListRow
          icon="add-circle-outline"
          label={already ? `Already part of ${trip.name}` : `Add as a stop on “${trip.name}”`}
          sub={already ? 'Its places show up in Nearby and the assistant' : 'Its places appear in Nearby, the map and assistant suggesti
ons'}
          onPress={() => {
            if (!already) {
              updateTrip(trip.id, { extraDestinationIds: [...trip.extraDestinationIds, d.id] });
              toast(`${d.city} added to ${trip.name}`);
            }
            onClose();
          }}
        />
      ) : null}
      <ListRow
        icon="airplane-outline"
        label="Start a new group trip here"
        sub="Pick dates, set a budget and invite friends"
        onPress={() => {
          onClose();
          setTimeout(() => router.push({ pathname: '/new-trip', params: { destination: d.id } }), 180);
        }}
      />
      <ListRow
        icon="flag-outline"
        label="Save to bucket list"
        onPress={() => {
          addBucket(`Visit ${d.city}, ${d.country}`, d.id);
          toast('Saved to your bucket list');
          onClose();
        }}
      />
      <View style={{ flexDirection: 'row', flexWrap: 'wrap', gap: 6 }}>
        {d.highlights.slice(0, 4).map((h) => (
          <Chip key={h} small tone="soft" label={h} />
        ))}
      </View>
    </Sheet>
  );
}

const styles = StyleSheet.create({
  star: { backgroundColor: 'rgba(0,0,0,0.32)', alignItems: 'center', justifyContent: 'center' },
  place: { borderRadius: radius.lg, overflow: 'hidden', backgroundColor: colors.bgAlt },
  placeText: { position: 'absolute', left: 12, right: 12, bottom: 12, gap: 2 },
  row: { flexDirection: 'row', alignItems: 'center', gap: 3 },
  featured: { backgroundColor: colors.card, borderRadius: radius.lg, overflow: 'hidden', borderWidth: 1, borderColor: colors.border, ...
shadow },
  featBadge: { position: 'absolute', top: 12, left: 12, backgroundColor: colors.primary, paddingHorizontal: 10, paddingVertical: 4, bord
erRadius: 999 },
  tag: { paddingHorizontal: 9, paddingVertical: 4, borderRadius: 999, borderWidth: 1, borderColor: colors.border },
  shareBtn: { width: 44, height: 44, borderRadius: 22, borderWidth: 1, borderColor: colors.border, alignItems: 'center', justifyContent:
'center' },
});
```

src/components/flights/AddFlightSheet.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { useState } from 'react';
import { ScrollView, StyleSheet, View } from 'react-native';

import { DatePickerSheet, TimePickerSheet } from '@/components/pickers';
import { Button } from '@/components/ui/Button';
import { Chip, ToggleRow } from '@/components/ui/bits';
import { toast } from '@/components/ui/feedback';
import { Input } from '@/components/ui/Input';
import { Press, tap } from '@/components/ui/Press';
import { Sheet } from '@/components/ui/Sheet';
import { T } from '@/components/ui/T';
import { airportByIata } from '@/data/airports';
import { destinationById } from '@/data/destinations';
import { FileTooLargeError, pickDocument, pickImage } from '@/services/files';
import { useActiveTrip, useAppStore } from '@/store/useAppStore';
import { colors, radius } from '@/theme';
import { distanceKm, kmLabel } from '@/utils/geo';
import { addDays, dayLabel, time12, zonedToUtc } from '@/utils/time';
import { maxLen, validateFlightNumber, validateIata } from '@/utils/validation';

/** Adds a flight (a friend's, or yours): who, which way, airports, times, seat and an optional ticket file. */
export function AddFlightSheet({ visible, onClose }: { visible: boolean; onClose: () => void }) {
  const trip = useActiveTrip()!;
  const dest = destinationById(trip.destinationId)!;
  const userId = useAppStore((s) => s.session?.userId ?? '');
  const addFlight = useAppStore((s) => s.addFlight);
  const addDoc = useAppStore((s) => s.addDoc);
  const [memberId, setMemberId] = useState(userId);
  const [direction, setDirection] = useState<'arrival' | 'departure'>('arrival');
  const [airline, setAirline] = useState('');
  const [flightNo, setFlightNo] = useState('');
  const [from, setFrom] = useState('');
  const [to, setTo] = useState(dest.airport);
  const [depDate, setDepDate] = useState(addDays(trip.startDate, -1));
  const [depTime, setDepTime] = useState('11:00');
  const [arrDate, setArrDate] = useState(trip.startDate);
  const [arrTime, setArrTime] = useState('14:30');
  const [seat, setSeat] = useState('');
  const [conf, setConf] = useState('');
  const [ticket, setTicket] = useState<{ uri: string; name: string; size?: number } | null>(null);
  const [saveDoc, setSaveDoc] = useState(true);
  const [picker, setPicker] = useState<null | 'depDate' | 'depTime' | 'arrDate' | 'arrTime'>(null);
  const [errors, setErrors] = useState<Record<string, string | null>>({});

  const setDir = (d: 'arrival' | 'departure') => {
    setDirection(d);
    if (d === 'arrival') {
      setTo(to || 'HND');
      setDepDate(addDays(trip.startDate, -1));
      setArrDate(trip.startDate);
    } else {
      setFrom(from || 'HND');
      setDepDate(trip.endDate);
      setArrDate(trip.endDate);
    }
  };

  const attach = async (kind: 'photo' | 'file') => {
    try {
      const f = kind === 'photo' ? await pickImage() : await pickDocument();
      if (f) {
        setTicket(f);
        toast('Ticket attached');
      }
    } catch (e) {
      toast(e instanceof FileTooLargeError ? e.message : 'Couldn’t attach that file', { tone: 'warn' });
    }
  };

  const submit = () => {
    const A = airportByIata(from);
    const B = airportByIata(to);
    const e: Record<string, string | null> = {
      airline: airline.trim() ? maxLen(30, 'Airline')(airline) : 'Enter the airline',
      flightNo: validateFlightNumber(flightNo),
      from: validateIata(from, 'departure'),
      to: validateIata(to, 'arrival') ?? (from.trim().toUpperCase() === to.trim().toUpperCase() ? 'Departure and arrival airports must d
iffer' : null),
      seat: seat && !/^\d{1,3}[A-K]$/i.test(seat.trim()) ? 'Seats look like 32A' : null,
    };
    let departUtc = 0;
    let arriveUtc = 0;
    if (A && B && !e.from && !e.to) {
```

```
      departUtc = zonedToUtc(depDate, depTime, A.tz);
      arriveUtc = zonedToUtc(arrDate, arrTime, B.tz);
      const hrs = (arriveUtc - departUtc) / 3600000;
      if (hrs <= 0) e.arr = 'Arrival must be after departure (times are local to each airport)';
      else if (hrs > 22) e.arr = `That's a ${Math.round(hrs)}h flight — double-check the dates`;
    }
    setErrors(e);
    if (Object.values(e).some(Boolean)) return tap('warning');
    const name = trip.members.find((m) => m.id === memberId)?.name.split(' ')[0] ?? '';
    addFlight(trip.id, {
      memberId,
      airline: airline.trim(),
      flightNo: flightNo.trim().toUpperCase().replace(/^([A-Z0-9]{2})\s?/, '$1 '),
      from: from.trim().toUpperCase(),
      to: to.trim().toUpperCase(),
      departUtc,
      arriveUtc,
      direction,
      seat: seat.trim().toUpperCase() || undefined,
      confirmation: conf.trim().toUpperCase() || undefined,
      ticketUri: ticket?.uri,
      ticketName: ticket?.name,
    });
    if (ticket && saveDoc) {
      addDoc(trip.id, { name: `${name}'s ticket · ${flightNo.toUpperCase()}`, kind: /\.pdf$/i.test(ticket.name) ? 'pdf' : 'image', uri:
ticket.uri, size: ticket.size, offline: true });
    }
    const near = B ? distanceKm(B, dest) : 0;
    tap('success');
    toast(direction === 'arrival' && near > 150 ? `Flight saved — note ${to.toUpperCase()} is ${kmLabel(near)} from ${dest.city}` : 'Fli
ght saved ✈️ The group can see it now');
    onClose();
  };

  const A = airportByIata(from);
  const B = airportByIata(to);
  return (
    <Sheet
      visible={visible}
      onClose={onClose}
      title="Add a flight"
      subtitle="Times are local to each airport — we convert them for everyone"
      footer={<Button label="Save flight" icon="airplane" onPress={submit} />}>
      <View style={{ flexDirection: 'row', gap: 8 }}>
        <Chip label="Arriving" icon="airplane" active={direction === 'arrival'} onPress={() => setDir('arrival')} />
        <Chip label="Departing" icon="airplane-outline" active={direction === 'departure'} onPress={() => setDir('departure')} />
      </View>
      <View style={{ gap: 6 }}>
        <T variant="small" weight="semibold">
          Traveller
        </T>
        <ScrollView horizontal showsHorizontalScrollIndicator={false} contentContainerStyle={{ gap: 6 }}>
          {trip.members.map((m) => (
            <Chip key={m.id} small label={m.name.split(' ')[0]} image={m.avatar} active={memberId === m.id} onPress={() => setMemberId
(m.id)} />
          ))}
        </ScrollView>
      </View>
      <View style={{ flexDirection: 'row', gap: 10 }}>
        <Input label="Airline" placeholder="ANA" value={airline} onChangeText={setAirline} error={errors.airline} containerStyle={{ fle
x: 1 }} />
        <Input label="Flight no." placeholder="NH 175" value={flightNo} onChangeText={setFlightNo} autoCapitalize="characters" error={er
rors.flightNo} containerStyle={{ flex: 1 }} />
      </View>
      <View style={{ flexDirection: 'row', gap: 10 }}>
        <Input label="From" placeholder="JFK" value={from} onChangeText={(v) => setFrom(v.toUpperCase().slice(0, 3))} autoCapitalize="ch
aracters" error={errors.from} hint={A ? A.city : undefined} containerStyle={{ flex: 1 }} />
        <Input label="To" placeholder={dest.airport} value={to} onChangeText={(v) => setTo(v.toUpperCase().slice(0, 3))} autoCapitalize
="characters" error={errors.to} hint={B ? B.city : undefined} containerStyle={{ flex: 1 }} />
      </View>
      <T variant="small" weight="semibold">
        Departs {A ? `(${A.city} time)` : ''}
      </T>
      <View style={{ flexDirection: 'row', gap: 10 }}>
        <Input value={`${dayLabel(depDate)}, ${depDate.slice(0, 4)}`} icon="calendar-outline" onPressField={() => setPicker('depDate')}
containerStyle={{ flex: 1.3 }} />
        <Input value={time12(depTime)} icon="time-outline" onPressField={() => setPicker('depTime')} containerStyle={{ flex: 1 }} />
      </View>
      <T variant="small" weight="semibold">
        Arrives {B ? `(${B.city} time)` : ''}
      </T>
      <View style={{ flexDirection: 'row', gap: 10 }}>
        <Input value={`${dayLabel(arrDate)}, ${arrDate.slice(0, 4)}`} icon="calendar-outline" onPressField={() => setPicker('arrDate')}
containerStyle={{ flex: 1.3 }} />
        <Input value={time12(arrTime)} icon="time-outline" onPressField={() => setPicker('arrTime')} containerStyle={{ flex: 1 }} />
      </View>
```

```
      {errors.arr ? (
        <T variant="caption" color={colors.red}>
          {errors.arr}
        </T>
      ) : null}
      <View style={{ flexDirection: 'row', gap: 10 }}>
        <Input label="Seat (optional)" placeholder="32A" value={seat} onChangeText={setSeat} autoCapitalize="characters" error={errors.s
eat} containerStyle={{ flex: 1 }} />
        <Input label="Confirmation (optional)" placeholder="ABC123" value={conf} onChangeText={setConf} autoCapitalize="characters" maxL
ength={10} containerStyle={{ flex: 1 }} />
      </View>
      <View style={{ gap: 8 }}>
        <T variant="small" weight="semibold">
          Ticket / boarding pass
        </T>
        {ticket ? (
          <View style={styles.attached}>
            <Ionicons name="document-attach-outline" size={18} color={colors.primary} />
            <T variant="caption" style={{ flex: 1 }} numberOfLines={1}>
              {ticket.name}
            </T>
            <Press onPress={() => setTicket(null)} accessibilityLabel="Remove attachment">
              <Ionicons name="close-circle" size={18} color={colors.textMuted} />
            </Press>
          </View>
        ) : (
          <View style={{ flexDirection: 'row', gap: 8 }}>
            <Button label="Photo" icon="image-outline" variant="secondary" size="md" onPress={() => attach('photo')} style={{ flex: 1 }}
/>
            <Button label="PDF / file" icon="document-outline" variant="secondary" size="md" onPress={() => attach('file')} style={{ fle
x: 1 }} />
          </View>
        )}
        {ticket ? <ToggleRow icon="cloud-done-outline" label="Also save to offline documents" value={saveDoc} onChange={setSaveDoc} /> :
null}
      </View>

      <DatePickerSheet visible={picker === 'depDate'} value={depDate} onClose={() => setPicker(null)} onChange={setDepDate} title="Depar
ture date" />
      <DatePickerSheet visible={picker === 'arrDate'} value={arrDate} onClose={() => setPicker(null)} onChange={setArrDate} title="Arriv
al date" />
      <TimePickerSheet visible={picker === 'depTime'} value={depTime} onClose={() => setPicker(null)} onChange={setDepTime} />
      <TimePickerSheet visible={picker === 'arrTime'} value={arrTime} onClose={() => setPicker(null)} onChange={setArrTime} />
    </Sheet>
  );
}

const styles = StyleSheet.create({
  attached: { flexDirection: 'row', alignItems: 'center', gap: 8, padding: 12, borderRadius: radius.md, backgroundColor: colors.primaryS
ofter },
});
```

src/components/home/BudgetTile.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { router } from 'expo-router';
import { StyleSheet, View } from 'react-native';

import { ProgressBar } from '@/components/ui/bits';
import { FadeIn } from '@/components/ui/motion';
import { Press } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import type { Trip } from '@/store/types';
import { colors, radius, shadow } from '@/theme';
import { money } from '@/utils/format';

/** How much of the group budget is spent, with a bar. Tap to open the Budget tab. */
export function BudgetTile({ trip, spent }: { trip: Trip; spent: number }) {
  return (
    <FadeIn delay={140}>
      <Press onPress={() => router.navigate('/(tabs)/budget')} style={styles.budget} accessibilityLabel="Group budget. Open budget">
        <View style={styles.rowBetween}>
          <T variant="kicker" color={colors.textSecondary}>
            Group budget
          </T>
          <Ionicons name="chevron-forward" size={16} color={colors.textMuted} />
        </View>
        <View style={{ flexDirection: 'row', alignItems: 'baseline', gap: 6, marginTop: 4 }}>
          <T variant="h1" weight="extrabold">
            {money(spent)}
          </T>
          <T variant="small" color={colors.textSecondary}>
            of {money(trip.budget)}
          </T>
        </View>
        <View style={{ marginTop: 12 }}>
```

```
          <ProgressBar value={spent / Math.max(1, trip.budget)} color={spent > trip.budget ? colors.red : colors.primary} height={9} />
        </View>
      </Press>
    </FadeIn>
  );
}

const styles = StyleSheet.create({
  budget: { marginTop: 12, backgroundColor: colors.card, borderRadius: radius.xl, padding: 16, ...shadow },
  rowBetween: { flexDirection: 'row', justifyContent: 'space-between', alignItems: 'center', gap: 8 },
});
```

src/components/home/DiscoverSection.tsx

```
import { router } from 'expo-router';
import { StyleSheet, View } from 'react-native';

import { PlaceCard } from '@/components/explore';
import { SectionHeader } from '@/components/ui/bits';
import { FadeIn } from '@/components/ui/motion';
import { destinations, type Destination } from '@/data/destinations';

/** Four other destinations to look at. */
export function DiscoverSection({ dest, colW }: { dest: Destination | undefined; colW: number }) {
  return (
    <FadeIn delay={380}>
      <SectionHeader title="Discover" action="See all" onAction={() => router.push('/destinations')} style={{ marginTop: 26 }} />
      <View style={styles.grid}>
        {destinations.filter((d) => d.id !== dest?.id).slice(0, 4).map((d) => (
          <PlaceCard key={d.id} d={d} style={{ width: (colW - 12) / 2 }} />
        ))}
      </View>
    </FadeIn>
  );
}

const styles = StyleSheet.create({
  grid: { flexDirection: 'row', flexWrap: 'wrap', gap: 12 },
});
```

src/components/home/GlanceTiles.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { router } from 'expo-router';
import { StyleSheet, View } from 'react-native';

import { FadeIn } from '@/components/ui/motion';
import { Press } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import type { Destination } from '@/data/destinations';
import type { useWeather } from '@/hooks/useTrip';
import { weatherInfo } from '@/services/weather';
import { colors, radius, shadow } from '@/theme';
import { partsInTz, time12 } from '@/utils/time';

/** Two small tiles: today's weather in the destination, and the local time there (with the time at home). */
export function GlanceTiles({ dest, weather, now, tz, homeTz }: { dest: Destination | undefined; weather: ReturnType<typeof useWeather>;
now: number; tz: string; homeTz: string }) {
  const todayWx = weather?.daily[0];
  return (
    <FadeIn delay={80}>
      <View style={styles.tiles}>
        <Press onPress={() => router.push('/weather')} style={styles.tile} accessibilityLabel={`Weather in ${dest?.city}`}>
          <T variant="kicker" color={colors.textSecondary}>
            Weather
          </T>
          <View style={styles.tileRow}>
            <Ionicons name={weather ? weatherInfo(weather.current.code).icon : 'partly-sunny'} size={34} color="#F5A524" />
            <T variant="h1" weight="extrabold" style={{ fontSize: 36, lineHeight: 40 }}>
              {weather ? `${weather.current.temp}°` : '—'}
            </T>
          </View>
          <T variant="micro" color={colors.textSecondary}>
            {todayWx ? `H ${Math.round(todayWx.max)}° · L ${Math.round(todayWx.min)}°` : 'Loading forecast…'}
          </T>
        </Press>
        <View style={styles.tile} accessible accessibilityLabel={`Local time in ${dest?.city}`}>
          <T variant="kicker" color={colors.textSecondary}>
            Local time
          </T>
          <T variant="h1" weight="extrabold" style={{ marginTop: 6 }}>
            {time12(partsInTz(now, tz).hhmm)}
          </T>
          <T variant="micro" color={colors.textSecondary} style={{ marginTop: 4 }}>
            Home {time12(partsInTz(now, homeTz).hhmm)}
          </T>
```

```
        </View>
      </View>
    </FadeIn>
  );
}

const styles = StyleSheet.create({
  tiles: { flexDirection: 'row', gap: 12 },
  tile: { flex: 1, backgroundColor: colors.card, borderRadius: radius.xl, padding: 16, minHeight: 112, ...shadow },
  tileRow: { flexDirection: 'row', alignItems: 'center', gap: 8, marginTop: 6 },
});
```

src/components/home/HomeHero.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { router } from 'expo-router';
import { StyleSheet, View } from 'react-native';
import { useSafeAreaInsets } from 'react-native-safe-area-context';

import { SkylineHero } from '@/components/SkylineHero';
import { AvatarStack, IconButton } from '@/components/ui/bits';
import { FadeIn } from '@/components/ui/motion';
import { Press } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import type { Destination } from '@/data/destinations';
import type { Trip } from '@/store/types';
import { colors } from '@/theme';

/** Big hero headline that follows the trip’s stage. */
export function headlineFor(status: 'planning' | 'active' | 'completed' | null, daysToGo: number) {
  if (status === 'active') return 'Living it.';
  if (status === 'completed') return 'That was epic.';
  if (status === 'planning') return daysToGo <= 1 ? 'Tomorrow.' : `${daysToGo} days to go.`;
  return 'Where to next?';
}

/** The picture at the top of Home: trip name, 360° button, notifications, a headline and the invite code. */
export function HomeHero({ trip, dest, tz, status, daysToGo, unread, onMenu, onCopyCode }: { trip: Trip | undefined; dest: Destination |
undefined; tz: string; status: 'planning' | 'active' | 'completed' | null; daysToGo: number; unread: number; onMenu: () => void; onCopyC
ode: () => void }) {
  const insets = useSafeAreaInsets();
  return (
    <SkylineHero image={dest?.image ?? 'amalfi'} tz={tz} height={400 + insets.top}>
      <View style={[styles.heroTop, { top: insets.top + 10 }]}>
        <Press onPress={() => (trip ? onMenu() : router.push('/trips'))} style={styles.tripPill} accessibilityLabel={trip ? `${trip.nam
e}. Trip options and switching` : 'Choose a trip'}>
          <T variant="small" weight="bold" color={colors.white} numberOfLines={1} style={{ maxWidth: 190 }}>
            {trip?.name ?? 'No trip yet'}
          </T>
          <Ionicons name="chevron-down" size={14} color={colors.white} />
        </Press>
        <View style={{ flex: 1 }} />
        <Press onPress={() => router.push({ pathname: '/panorama', params: { city: dest?.id } })} style={styles.pano} accessibilityLabel
="Open the 360 degree city view">
          <Ionicons name="globe-outline" size={15} color={colors.white} />
          <T variant="micro" weight="extrabold" color={colors.white}>
            360°
          </T>
        </Press>
        <IconButton icon="notifications-outline" label={`Notifications, ${unread} unread`} badge={unread} onPress={() => router.push('/n
otifications')} bg="rgba(255,255,255,0.92)" color="#111111" bordered={false} />
      </View>

      <View style={styles.heroBottom}>
        <View style={styles.place}>
          <Ionicons name="location-sharp" size={13} color="rgba(255,255,255,0.9)" />
          <T variant="kicker" color="rgba(255,255,255,0.92)">
            {dest ? `${dest.city}, ${dest.country}` : 'Somewhere great'}
          </T>
        </View>
        <FadeIn delay={120} distance={26}>
          <T variant="display" color={colors.white} style={{ fontSize: 60, lineHeight: 64 }} accessibilityRole="header">
            {headlineFor(status, daysToGo)}
          </T>
        </FadeIn>
        <View style={styles.heroRow}>
          {trip ? <AvatarStack people={trip.members} max={4} size={32} /> : <View />}
          {trip ? (
            <Press onPress={onCopyCode} style={styles.code} accessibilityLabel={`Invite code ${trip.inviteCode}. Tap to copy`}>
              <T variant="small" weight="extrabold" color={colors.onAccent}>
                Code {trip.inviteCode}
              </T>
              <Ionicons name="copy-outline" size={14} color={colors.onAccent} />
            </Press>
          ) : null}
        </View>
```

```
      </View>
    </SkylineHero>
  );
}

const styles = StyleSheet.create({
  heroTop: { position: 'absolute', left: 16, right: 16, flexDirection: 'row', alignItems: 'center', gap: 10 },
  pano: { flexDirection: 'row', alignItems: 'center', gap: 5, height: 40, paddingHorizontal: 12, borderRadius: 999, backgroundColor: 'rg
ba(20,20,24,0.55)' },
  tripPill: { flexDirection: 'row', alignItems: 'center', gap: 6, height: 40, paddingHorizontal: 14, borderRadius: 999, backgroundColor:
'rgba(20,20,24,0.55)' },
  heroBottom: { position: 'absolute', left: 20, right: 20, bottom: 44 },
  place: { flexDirection: 'row', alignItems: 'center', gap: 4, marginBottom: 2 },
  heroRow: { flexDirection: 'row', alignItems: 'center', justifyContent: 'space-between', marginTop: 12 },
  code: { flexDirection: 'row', alignItems: 'center', gap: 8, height: 38, paddingHorizontal: 14, borderRadius: 999, backgroundColor: col
ors.accent },
});
```

src/components/home/NearbySection.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { router } from 'expo-router';
import { useState } from 'react';
import { StyleSheet, View } from 'react-native';

import { placeCategoryIcon } from '@/components/trip/meta';
import { Card, SectionHeader } from '@/components/ui/bits';
import { FadeIn } from '@/components/ui/motion';
import { Press } from '@/components/ui/Press';
import { Slider } from '@/components/ui/Slider';
import { T } from '@/components/ui/T';
import type { Destination } from '@/data/destinations';
import { placesFor } from '@/data/places';
import type { Trip } from '@/store/types';
import { colors } from '@/theme';
import { lengthTag, money } from '@/utils/format';
import { distanceKm, kmLabel } from '@/utils/geo';

/** Places close to the trip's centre, with a radius slider. */
export function NearbySection({ trip, dest }: { trip: Trip; dest: Destination }) {
  const [radiusKm, setRadiusKm] = useState(5);
  const nearby = [trip.destinationId, ...trip.extraDestinationIds]
    .flatMap((id) => placesFor(id))
    .map((p) => ({ p, km: distanceKm(dest, p) }))
    .filter((x) => x.km <= radiusKm)
    .sort((a, b) => a.km - b.km);
  return (
    <FadeIn delay={340}>
      <SectionHeader title="Nearby" action="Filters" onAction={() => router.push('/nearby')} style={{ marginTop: 26 }} />
      <Card>
        <View style={styles.rowBetween}>
          <T variant="small" color={colors.textSecondary}>
            Search radius
          </T>
          <T variant="small" weight="extrabold">
            {radiusKm} km
          </T>
        </View>
        <Slider value={radiusKm} min={1} max={20} onChange={setRadiusKm} label="Search radius in kilometres" formatValue={(v) => `${v} k
m`} />
        <T variant="caption" color={colors.textMuted} style={{ marginBottom: 6 }}>
          {nearby.length} places within {radiusKm} km of central {dest.city}
        </T>
        {nearby.slice(0, 4).map(({ p, km }) => (
          <Press key={p.id} onPress={() => router.push({ pathname: '/nearby', params: { focus: p.id } })} style={styles.nearRow} scaleTo
={0.99}>
            <View style={styles.nearIcon}>
              <Ionicons name={placeCategoryIcon[p.category]} size={15} color={colors.text} />
            </View>
            <View style={{ flex: 1 }}>
              <T variant="small" weight="bold" numberOfLines={1}>
                {p.name}
              </T>
              <T variant="caption" color={colors.textSecondary} numberOfLines={1}>
                {p.category[0].toUpperCase() + p.category.slice(1)} · {lengthTag(p.durationMin)} · {p.cost ? money(p.cost) : 'Free'}
              </T>
            </View>
            <T variant="caption" weight="semibold" color={colors.textSecondary}>
              {kmLabel(km)}
            </T>
          </Press>
        ))}
      </Card>
    </FadeIn>
  );
}
```

```
const styles = StyleSheet.create({
  rowBetween: { flexDirection: 'row', justifyContent: 'space-between', alignItems: 'center', gap: 8 },
  nearRow: { flexDirection: 'row', alignItems: 'center', gap: 10, paddingVertical: 9, borderTopWidth: 1, borderTopColor: colors.border
},
  nearIcon: { width: 32, height: 32, borderRadius: 16, backgroundColor: colors.bgAlt, alignItems: 'center', justifyContent: 'center' },
});
```

src/components/home/PicksSection.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { Image } from 'expo-image';
import { LinearGradient } from 'expo-linear-gradient';
import { router } from 'expo-router';
import { ScrollView, StyleSheet, View } from 'react-native';

import { SectionHeader } from '@/components/ui/bits';
import { FadeIn } from '@/components/ui/motion';
import { Press } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import { imageSource } from '@/data/images';
import type { Personal } from '@/store/types';
import { colors, radius, shadow } from '@/theme';

import { pickedForYou } from './picks';

/** "Picked for you" cards ranked by the survey answers, or the invitation to take the survey. */
export function PicksSection({ personal }: { personal: Personal }) {
  const picks = pickedForYou(personal);
  return (
    personal.done && picks.length ? (
      <FadeIn delay={300}>
        <SectionHeader title="Picked for you" action="Edit" onAction={() => router.push('/personalize')} style={{ marginTop: 26 }} />
        <ScrollView horizontal showsHorizontalScrollIndicator={false} contentContainerStyle={{ gap: 12, paddingHorizontal: 20 }} style=
{{ marginHorizontal: -20 }}>
          {picks.map(({ d, match, why }) => (
            <Press key={d.id} onPress={() => router.push({ pathname: '/destination/[id]', params: { id: d.id } })} style={styles.pick} a
ccessibilityLabel={`${d.city}, ${match}% match`}>
              <Image source={imageSource(d.image)} style={StyleSheet.absoluteFill} contentFit="cover" transition={200} />
              <LinearGradient colors={['rgba(0,0,0,0)', 'rgba(0,0,0,0.75)']} locations={[0.35, 1]} style={StyleSheet.absoluteFill} />
              <View style={styles.match}>
                <T variant="micro" weight="bold" color={colors.onAccent}>
                  {match}% match
                </T>
              </View>
              <View style={{ position: 'absolute', left: 12, right: 12, bottom: 12 }}>
                <T variant="title" weight="bold" color={colors.white} numberOfLines={1}>
                  {d.city}
                </T>
                <T variant="micro" color="rgba(255,255,255,0.9)" numberOfLines={1}>
                  {why}
                </T>
              </View>
            </Press>
          ))}
        </ScrollView>
      </FadeIn>
    ) : !personal.done ? (
      <Press onPress={() => router.push('/personalize')} style={styles.cta} accessibilityLabel="Make TogetherWeGo yours — take the 60-se
cond survey">
        <LinearGradient colors={[colors.accent, '#F59E0B']} start={{ x: 0, y: 0 }} end={{ x: 1, y: 1 }} style={StyleSheet.absoluteFill}
/>
        <View style={styles.ctaEmoji}>
          <T style={{ fontSize: 24, lineHeight: 30 }}>🎨</T>
        </View>
        <View style={{ flex: 1 }}>
          <T variant="title" weight="extrabold" color={colors.onAccent}>
            Make it yours
          </T>
          <T variant="caption" color={colors.onAccent}>
            60-sec survey · colours, interests & your home screen
          </T>
        </View>
        <Ionicons name="arrow-forward-circle" size={28} color={colors.onAccent} />
      </Press>
    ) : null
  );
}

const styles = StyleSheet.create({
  pick: { width: 156, height: 196, borderRadius: radius.xl, overflow: 'hidden', ...shadow },
  match: { position: 'absolute', top: 10, left: 10, backgroundColor: colors.accent, paddingHorizontal: 8, paddingVertical: 4, borderRadi
us: 999 },
  cta: { marginTop: 22, flexDirection: 'row', alignItems: 'center', gap: 12, padding: 14, borderRadius: radius.xl, overflow: 'hidden',
...shadow },
  ctaEmoji: { width: 44, height: 44, borderRadius: 22, backgroundColor: 'rgba(255,255,255,0.4)', alignItems: 'center', justifyContent:
```

```
'center' },
});
```

src/components/home/QuickTiles.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { router } from 'expo-router';
import { StyleSheet, View } from 'react-native';

import { type IconName } from '@/components/ui/bits';
import { FadeIn } from '@/components/ui/motion';
import { Press } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import { colors, radius, shadow } from '@/theme';

/** Four shortcuts: vote, flights, documents, photos. */
export function QuickTiles({ votingCount }: { votingCount: number }) {
  const quick: { icon: IconName; label: string; go: () => void }[] = [
    { icon: 'checkbox-outline', label: `Vote (${votingCount})`, go: () => router.navigate({ pathname: '/(tabs)/itinerary', params: { ta
b: 'vote' } }) },
    { icon: 'airplane-outline', label: 'Flights', go: () => router.push('/flights') },
    { icon: 'folder-open-outline', label: 'Docs', go: () => router.push('/documents') },
    { icon: 'camera-outline', label: 'Photos', go: () => router.push('/photos') },
  ];
  return (
    <FadeIn delay={260}>
      <View style={styles.quickRow}>
        {quick.map((q) => (
          <Press key={q.label} onPress={q.go} style={styles.quick} accessibilityLabel={q.label}>
            <Ionicons name={q.icon} size={22} color={colors.text} />
            <T variant="micro" weight="bold" numberOfLines={1}>
              {q.label}
            </T>
          </Press>
        ))}
      </View>
    </FadeIn>
  );
}

const styles = StyleSheet.create({
  quickRow: { flexDirection: 'row', gap: 10, marginTop: 14 },
  quick: { flex: 1, height: 84, backgroundColor: colors.card, borderRadius: radius.xl, alignItems: 'center', justifyContent: 'center', g
ap: 8, ...shadow },
});
```

src/components/home/UpNextSection.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { router } from 'expo-router';
import { StyleSheet, View } from 'react-native';

import { typeEmoji } from '@/components/trip/meta';
import { Card, SectionHeader } from '@/components/ui/bits';
import { FadeIn, Pulse } from '@/components/ui/motion';
import { Press } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import type { nextUpcoming } from '@/services/notifications';
import type { Activity, Trip } from '@/store/types';
import { useAppStore } from '@/store/useAppStore';
import { colors, radius, shadow } from '@/theme';
import { countdownLabel, lengthTag } from '@/utils/format';
import { dayLabel, time12, weekdayShort } from '@/utils/time';

/** "Up next": today's checklist while the trip is on, and the next planned activity with a countdown. */
export function UpNextSection({ trip, status, todays, dayNo, tripLen, next, now }: { trip: Trip; status: 'planning' | 'active' | 'comple
ted' | null; todays: Activity[]; dayNo: number; tripLen: number; next: ReturnType<typeof nextUpcoming>; now: number }) {
  const updateActivity = useAppStore((s) => s.updateActivity);
  return (
    <FadeIn delay={200}>
      <SectionHeader title="Up next" action="See all" onAction={() => router.navigate('/(tabs)/itinerary')} style={{ marginTop: 24 }} />
      {status === 'active' ? (
        <Card style={{ marginBottom: 10, gap: 10 }}>
          <View style={styles.rowBetween}>
            <T variant="title" weight="bold">
              Today · Day {dayNo} of {tripLen}
            </T>
            <T variant="caption" color={colors.textSecondary}>
              {todays.filter((a) => a.done).length}/{todays.length} done
            </T>
          </View>
          {todays.length ? (
            todays.map((a) => (
              <Press key={a.id} onPress={() => updateActivity(trip.id, a.id, { done: !a.done })} style={styles.todayRow} accessibilityRo
le="checkbox" accessibilityState={{ checked: !!a.done }} accessibilityLabel={`${a.title} at ${a.time}`}>
                <Ionicons name={a.done ? 'checkmark-circle' : 'ellipse-outline'} size={22} color={a.done ? colors.positive : colors.bord
```

```
erStrong} />
                <T variant="caption" weight="bold" color={colors.textSecondary} style={{ width: 40 }}>
                  {a.time}
                </T>
                <T variant="small" weight="semibold" numberOfLines={1} style={[{ flex: 1 }, a.done && { textDecorationLine: 'line-throug
h', color: colors.textMuted }]}>
                  {a.title}
                </T>
              </Press>
            ))
          ) : (
            <T variant="small" color={colors.textSecondary}>
              A free day. Enjoy it!
            </T>
          )}
        </Card>
      ) : null}
      {next ? (
        <Press onPress={() => router.navigate('/(tabs)/itinerary')} style={styles.next} accessibilityLabel={`Next: ${next.a.title}`}>
          <View style={styles.nextBadge}>
            <T style={{ fontSize: 22, lineHeight: 28 }}>{typeEmoji[next.a.type]}</T>
          </View>
          <View style={{ flex: 1, gap: 2 }}>
            <T variant="small" weight="bold" numberOfLines={1}>
              {next.a.title}
            </T>
            <T variant="caption" color={colors.textSecondary} numberOfLines={1}>
              {weekdayShort(next.a.date)}, {dayLabel(next.a.date)} · {time12(next.a.time)} · {lengthTag(next.a.durationMin)}
            </T>
          </View>
          {next.a.reminder ? (
            <Pulse>
              <Ionicons name="notifications" size={16} color={colors.primary} />
            </Pulse>
          ) : null}
          <View style={styles.countdown}>
            <T variant="micro" weight="bold" color={colors.onPrimary}>
              {next.startUtc - now > 0 ? `in ${countdownLabel(next.startUtc - now)}` : 'Now'}
            </T>
          </View>
        </Press>
      ) : (
        <Card>
          <T variant="small" color={colors.textSecondary}>
            Nothing scheduled yet.
          </T>
        </Card>
      )}
    </FadeIn>
  );
}

const styles = StyleSheet.create({
  rowBetween: { flexDirection: 'row', justifyContent: 'space-between', alignItems: 'center', gap: 8 },
  todayRow: { flexDirection: 'row', alignItems: 'center', gap: 10, paddingVertical: 2 },
  next: { flexDirection: 'row', alignItems: 'center', gap: 12, backgroundColor: colors.card, borderRadius: radius.xl, padding: 14, ...sh
adow },
  nextBadge: { width: 46, height: 46, borderRadius: 23, backgroundColor: colors.bgAlt, alignItems: 'center', justifyContent: 'center' },
  countdown: { backgroundColor: colors.primary, paddingHorizontal: 10, paddingVertical: 6, borderRadius: 999 },
});
```

src/components/home/__tests__/picks.test.ts

```
import { destinations } from '@/data/destinations';
import type { Personal } from '@/store/types';

import { pickedForYou } from '../picks';

const personal = (over: Partial<Personal>): Personal => ({
  done: true,
  nickname: '',
  emoji: '✈️',
  interests: [],
  travelStyle: 'balanced',
  budgetStyle: 'mid',
  sections: { glance: true, budget: true, upnext: true, quick: true, forYou: true, nearby: true, discover: true },
  motion: true,
  ...over,
});

describe('picked for you', () => {
  it('shows nothing until an interest is chosen', () => {
    expect(pickedForYou(personal({ interests: [] }))).toEqual([]);
  });

  it('only suggests places that match a chosen interest, best match first', () => {
```

```
    const picks = pickedForYou(personal({ interests: ['beach'] }));
    expect(picks.length).toBeGreaterThan(0);
    expect(picks.length).toBeLessThanOrEqual(8);
    for (const { d } of picks) expect(d.tags).toContain('beach');
    const scores = picks.map((p) => p.match);
    expect([...scores].sort((a, b) => b - a)).toEqual(scores);
    for (const { match } of picks) {
      expect(match).toBeGreaterThanOrEqual(0);
      expect(match).toBeLessThanOrEqual(99);
    }
  });

  it('matching more of the chosen interests scores higher', () => {
    const [top] = pickedForYou(personal({ interests: ['beach', 'food', 'adventure'] }));
    const hits = (tags: string[]) => ['beach', 'food', 'adventure'].filter((i) => tags.includes(i)).length;
    const bestPossible = Math.max(...destinations.map((d) => hits(d.tags)));
    expect(hits(top.d.tags)).toBe(bestPossible);
  });

  it('prefers cheaper places for savers and pricier ones for treats', () => {
    const avgPrice = (style: Personal['budgetStyle']) => {
      const picks = pickedForYou(personal({ interests: ['food', 'beach', 'culture'], budgetStyle: style })).slice(0, 4);
      return picks.reduce((n, p) => n + p.d.priceLevel, 0) / picks.length;
    };
    expect(avgPrice('saver')).toBeLessThanOrEqual(avgPrice('treat'));
  });

  it('explains each pick with the interest emoji and the country', () => {
    const [first] = pickedForYou(personal({ interests: ['food'] }));
    expect(first.why).toContain('🍜');
    expect(first.why).toContain(first.d.country);
  });
});
```

src/components/home/picks.ts

```
import { destinations, type Destination } from '@/data/destinations';
import { INTERESTS } from '@/data/personal';
import type { Personal } from '@/store/types';

/** Ranks destinations against the “Make it yours” answers (interests + budget style). */
export function pickedForYou(p: Personal): { d: Destination; match: number; why: string }[] {
  if (!p.interests.length) return [];
  return destinations
    .map((d) => {
      const hits = INTERESTS.filter((i) => p.interests.includes(i.key) && d.tags.includes(i.key));
      const budgetFit = p.budgetStyle === 'saver' ? (d.priceLevel <= 2 ? 1 : 0) : p.budgetStyle === 'treat' ? (d.priceLevel >= 3 ? 1 :
0) : d.priceLevel <= 3 ? 1 : 0;
      const ratio = hits.length / Math.min(3, p.interests.length);
      const match = Math.min(99, Math.round(52 + 36 * Math.min(1, ratio) + 7 * budgetFit + (d.rating - 4.5) * 8));
      return { d, match, hits, why: `${hits.map((h) => h.emoji).join(' ')} · ${d.country}` };
    })
    .filter((x) => x.hits.length > 0)
    .sort((a, b) => b.match - a.match || b.d.rating - a.d.rating)
    .slice(0, 8);
}
```

src/components/join/QrScanner.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { CameraView, useCameraPermissions } from 'expo-camera';
import { useState } from 'react';
import { Modal, StyleSheet, View } from 'react-native';
import { useSafeAreaInsets } from 'react-native-safe-area-context';

import { Button } from '@/components/ui/Button';
import { Press } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import { colors } from '@/theme';

/** Full-screen camera that reads one QR code and hands its text back. */
export function QrScanner({ visible, onClose, onScan }: { visible: boolean; onClose: () => void; onScan: (text: string) => void }) {
  const insets = useSafeAreaInsets();
  const [permission, requestPermission] = useCameraPermissions();
  const [done, setDone] = useState(false); // the camera reports the same code many times a second

  return (
    <Modal visible={visible} animationType="slide" onRequestClose={onClose} onShow={() => setDone(false)}>
      <View style={styles.root}>
        {permission?.granted ? (
          <CameraView
            style={StyleSheet.absoluteFill}
            facing="back"
            barcodeScannerSettings={{ barcodeTypes: ['qr'] }}
            onBarcodeScanned={({ data }) => {
              if (done) return;
```

```
              setDone(true);
              onScan(data);
            }}
          />
        ) : (
          <View style={styles.ask}>
            <Ionicons name="camera-outline" size={44} color="rgba(255,255,255,0.8)" />
            <T variant="title" color={colors.white} center>
              {permission && !permission.canAskAgain ? 'Camera access is turned off' : 'Allow the camera to scan'}
            </T>
            <T variant="small" color="rgba(255,255,255,0.75)" center>
              {permission && !permission.canAskAgain ? 'Turn it on for TogetherWeGo in your phone’s settings, or type the invite code in
stead.' : 'It is only used to read the invite QR code your friend shows you.'}
            </T>
            {!permission || permission.canAskAgain ? <Button label="Allow camera" icon="camera-outline" onPress={() => requestPermission
()} /> : null}
          </View>
        )}
        <View style={[styles.top, { paddingTop: insets.top + 8 }]}>
          <Press onPress={onClose} style={styles.round} accessibilityLabel="Close scanner">
            <Ionicons name="close" size={22} color={colors.white} />
          </Press>
        </View>
        {permission?.granted ? (
          <>
            <View style={styles.middle} pointerEvents="none">
              <View style={styles.frame} />
            </View>
            <View style={[styles.hint, { paddingBottom: insets.bottom + 24 }]} pointerEvents="none">
              <T variant="small" weight="semibold" color={colors.white} center>
                Point the camera at your friend’s invite QR code
              </T>
            </View>
          </>
        ) : null}
      </View>
    </Modal>
  );
}

const styles = StyleSheet.create({
  root: { flex: 1, backgroundColor: '#0B0F0A' },
  ask: { flex: 1, alignItems: 'center', justifyContent: 'center', gap: 12, padding: 32 },
  top: { position: 'absolute', top: 0, left: 0, right: 0, paddingHorizontal: 16 },
  round: { width: 42, height: 42, borderRadius: 21, backgroundColor: 'rgba(0,0,0,0.45)', alignItems: 'center', justifyContent: 'center'
},
  middle: { position: 'absolute', top: 0, bottom: 0, left: 0, right: 0, alignItems: 'center', justifyContent: 'center' },
  hint: { position: 'absolute', bottom: 0, left: 0, right: 0, alignItems: 'center', paddingHorizontal: 24 },
  frame: { width: 240, height: 240, borderRadius: 24, borderWidth: 3, borderColor: 'rgba(255,255,255,0.85)' },
});
```

src/components/map/OfflineRadar.tsx

```
/**
 * Offline map: plots every stop around the trip centre using real distances
 * and bearings — no map tiles or internet needed. Handy on a plane or when
 * roaming data is off.
 */
import { useState } from 'react';
import { StyleSheet, View } from 'react-native';

import { Press } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import { colors } from '@/theme';
import { projectKm } from '@/utils/geo';

import type { MapPoint } from './types';

export function OfflineRadar({ center, points, radiusKm, selectedId, onSelect }: { center: { lat: number; lng: number }; points: MapPoin
t[]; radiusKm: number; selectedId?: string | null; onSelect: (id: string) => void }) {
  const [size, setSize] = useState(300);
  const half = size / 2;
  const scale = half / radiusKm;
  const rings = [0.25, 0.5, 1].map((f) => Math.max(0.5, Math.round(radiusKm * f * 10) / 10));

  return (
    <View style={styles.wrap} onLayout={(e) => setSize(Math.min(e.nativeEvent.layout.width, e.nativeEvent.layout.height))}>
      <View style={{ width: size, height: size }}>
        {rings.map((r) => (
          <View
            key={r}
            style={[styles.ring, { width: r * scale * 2, height: r * scale * 2, borderRadius: r * scale, left: half - r * scale, top: ha
lf - r * scale }]}>
            <T variant="micro" color={colors.textMuted} style={{ position: 'absolute', top: -2, alignSelf: 'center', backgroundColor: co
lors.bgAlt, paddingHorizontal: 3 }}>
              {r} km
```

```
            </T>
          </View>
        ))}
        <View style={[styles.axis, { left: half, top: 0, width: 1, height: size }]} />
        <View style={[styles.axis, { top: half, left: 0, height: 1, width: size }]} />
        <T variant="micro" weight="bold" color={colors.textSecondary} style={{ position: 'absolute', top: 4, left: half + 4 }}>
          N
        </T>
        {points.map((p) => {
          const { x, y } = projectKm(center, p);
          if (Math.hypot(x, y) > radiusKm * 1.02) return null;
          const sel = p.id === selectedId;
          return (
            <Press
              key={p.id}
              onPress={() => onSelect(p.id)}
              hitSlop={8}
              accessibilityLabel={`${p.title}, ${p.subtitle}`}
              style={[
                styles.dot,
                { left: half + x * scale - (sel ? 9 : 6), top: half - y * scale - (sel ? 9 : 6), backgroundColor: p.color, width: sel ?
18 : 12, height: sel ? 18 : 12, borderRadius: 9 },
              ]}
            />
          );
        })}
        <View style={[styles.center, { left: half - 7, top: half - 7 }]} />
      </View>
    </View>
  );
}

const styles = StyleSheet.create({
  wrap: { ...StyleSheet.absoluteFill, backgroundColor: colors.bgAlt, alignItems: 'center', justifyContent: 'center' },
  ring: { position: 'absolute', borderWidth: 1, borderColor: colors.borderStrong, borderStyle: 'dashed' },
  axis: { position: 'absolute', backgroundColor: colors.border },
  dot: { position: 'absolute', borderWidth: 2, borderColor: colors.white },
  center: { position: 'absolute', width: 14, height: 14, borderRadius: 7, backgroundColor: colors.primary, borderWidth: 3, borderColor:
colors.white },
});
```

src/components/map/TripMap.tsx

```
/** Native map (Apple Maps on iOS, Google Maps on Android) via react-native-maps. */
import { useEffect, useRef } from 'react';
import { StyleSheet } from 'react-native';
import MapView, { Circle, Marker } from 'react-native-maps';

import type { TripMapProps } from './types';

export function TripMap({ center, points, radiusKm, selectedId, onSelect, showsUser }: TripMapProps) {
  const ref = useRef<MapView>(null);
  const delta = Math.max(0.04, (radiusKm / 111) * 2.6);

  useEffect(() => {
    ref.current?.animateToRegion({ latitude: center.lat, longitude: center.lng, latitudeDelta: delta, longitudeDelta: delta }, 400);
  }, [center.lat, center.lng, delta]);

  useEffect(() => {
    const p = points.find((x) => x.id === selectedId);
    if (p) ref.current?.animateToRegion({ latitude: p.lat, longitude: p.lng, latitudeDelta: 0.03, longitudeDelta: 0.03 }, 350);
  }, [selectedId, points]);

  return (
    <MapView
      ref={ref}
      style={StyleSheet.absoluteFill}
      initialRegion={{ latitude: center.lat, longitude: center.lng, latitudeDelta: delta, longitudeDelta: delta }}
      showsUserLocation={showsUser}
      showsCompass
      toolbarEnabled={false}>
      <Circle center={{ latitude: center.lat, longitude: center.lng }} radius={radiusKm * 1000} strokeColor="rgba(31,77,37,0.6)" fillCol
or="rgba(31,77,37,0.08)" strokeWidth={1.5} />
      {points.map((p) => (
        <Marker
          key={p.id}
          coordinate={{ latitude: p.lat, longitude: p.lng }}
          title={p.title}
          description={p.subtitle}
          pinColor={p.color}
          onPress={() => onSelect(p.id)}
          tracksViewChanges={false}
        />
      ))}
    </MapView>
```

```
  );
}
```

src/components/map/TripMap.web.tsx

```
/** Web preview: OpenStreetMap embed (react-native-maps is native-only). */
import { createElement } from 'react';
import { StyleSheet, View } from 'react-native';

import type { TripMapProps } from './types';

export function TripMap({ center, points, radiusKm, selectedId }: TripMapProps) {
  const focus = points.find((p) => p.id === selectedId) ?? center;
  const d = Math.max(0.02, (radiusKm / 111) * 1.3);
  const bbox = [focus.lng - d, focus.lat - d * 0.7, focus.lng + d, focus.lat + d * 0.7].join(',');
  const src = `https://www.openstreetmap.org/export/embed.html?bbox=${bbox}&layer=mapnik&marker=${focus.lat},${focus.lng}`;
  return (
    <View style={StyleSheet.absoluteFill}>
      {createElement('iframe', { src, title: 'Trip map', style: { border: 0, width: '100%', height: '100%' }, loading: 'lazy' })}
    </View>
  );
}
```

src/components/map/types.ts

```
export interface MapPoint {
  id: string;
  lat: number;
  lng: number;
  title: string;
  subtitle: string;
  color: string;
  kind: 'activity' | 'place' | 'airport' | 'me';
}

export interface TripMapProps {
  center: { lat: number; lng: number };
  points: MapPoint[];
  radiusKm: number;
  selectedId?: string | null;
  onSelect: (id: string) => void;
  showsUser?: boolean;
}
```

src/components/panorama/Scene.tsx

```
import { LinearGradient } from 'expo-linear-gradient';
import { memo, useId, useState, type ReactNode } from 'react';
import { Animated, StyleSheet, View } from 'react-native';
import Svg, { Circle, Defs, G, Path, Pattern, RadialGradient, Rect, Stop, Use } from 'react-native-svg';

import { useLoop } from '@/components/ui/motion';
import type { Landmark, Pano, Roof } from '@/data/panoramas';

import { LOOKS, SHAPES, type Part, type Tone } from './shapes';

/** Layer speeds relative to the far layer. Wider layers scroll faster, which gives the parallax. */
export const K = { far: 1, mid: 1.4, near: 1.9, sky: 0.5 } as const;

export function layout(S: number) {
  const base = 4 * S;
  return { far: base * K.far, mid: base * K.mid, near: base * K.near, sky: base * K.sky };
}

/** Compass heading (0–360°) at the centre of the screen for a scroll position. */
export function headingOf(pos: number, S: number) {
  const W = layout(S).mid;
  return (((((pos * K.mid + S / 2) % W) + W) % W) / W) * 360;
}

/** Scroll position that centres a given compass heading. */
export function posForHeading(deg: number, S: number) {
  return ((deg / 360) * layout(S).mid - S / 2) / K.mid;
}

/** x position of something at compass heading `deg` inside a layer (so it lines up with the mid layer). */
function xFor(layerK: number, W: number, deg: number, S: number) {
  return (((posForHeading(deg, S) * layerK + S / 2) % W) + W) % W;
}

function rng(seed: number) {
  let a = seed >>> 0;
  return () => {
    a = (a + 0x6d2b79f5) >>> 0;
    let t = a;
    t = Math.imul(t ^ (t >>> 15), t | 1);
```

```
    t ^= t + Math.imul(t ^ (t >>> 7), t | 61);
    return ((t ^ (t >>> 14)) >>> 0) / 4294967296;
  };
}

function shade(hex: string, amt: number) {
  const n = parseInt(hex.slice(1), 16);
  const ch = (v: number) => Math.max(0, Math.min(255, Math.round(v + (amt >= 0 ? (255 - v) * amt : v * amt))));
  const r = ch((n >> 16) & 255);
  const g = ch((n >> 8) & 255);
  const b = ch(n & 255);
  return `#${((1 << 24) | (r << 16) | (g << 8) | b).toString(16).slice(1)}`;
}

interface Building {
  x: number;
  w: number;
  h: number;
  roof: Roof;
  v: number;
  amt: number;
}

function genBuildings(seed: number, W: number, wr: [number, number], hr: [number, number], gap: number, roofs: Roof[], towers: number,
H: number): Building[] {
  const rand = rng(seed);
  const out: Building[] = [];
  let x = 0;
  while (x < W) {
    let w = wr[0] + rand() * (wr[1] - wr[0]);
    if (W - x - w < wr[0]) w = W - x;
    const tall = rand() < towers;
    const h = H * (tall ? hr[1] * (0.6 + rand() * 0.4) : hr[0] + (hr[1] - hr[0]) * Math.pow(rand(), 1.8) * 0.7);
    out.push({ x, w: Math.max(4, w - gap), h, roof: roofs[Math.floor(rand() * roofs.length)], v: Math.floor(rand() * 4), amt: (rand() -
0.5) * 0.22 });
    x += w;
  }
  return out;
}

/** Four window grids with different lit patterns, so skylines don't look tiled. */
function WindowPatterns({ id, night, color, base }: { id: string; night: boolean; color: string; base: string }) {
  return (
    <>
      {[0, 1, 2, 3].map((v) => {
        const rand = rng(900 + v * 17);
        const cells: ReactNode[] = [];
        for (let i = 0; i < 4; i++)
          for (let j = 0; j < 4; j++) {
            const lit = rand() < 0.55;
            if (night && !lit) continue;
            cells.push(<Rect key={`${i}-${j}`} x={3 + i * 6} y={4 + j * 9} width={3} height={5} fill={night ? color : lit ? '#FFFFFF' :
base} opacity={night ? 0.95 : lit ? 0.4 : 0.55} />);
          }
        return (
          <Pattern key={v} id={`${id}${v}`} x={0} y={0} width={24} height={36} patternUnits="userSpaceOnUse">
            {cells}
          </Pattern>
        );
      })}
    </>
  );
}

function roofEl(b: Building, y: number, fill: string, night: boolean, key: string): ReactNode {
  const { x, w, h } = b;
  switch (b.roof) {
    case 'step':
      return <Rect key={key} x={x + w * 0.25} y={y - h * 0.08} width={w * 0.5} height={h * 0.08} fill={fill} />;
    case 'spire':
      return (
        <G key={key}>
          <Rect x={x + w / 2 - 0.8} y={y - h * 0.2} width={1.6} height={h * 0.2} fill={fill} />
          {night ? <Circle cx={x + w / 2} cy={y - h * 0.2} r={1.8} fill="#FF4B3E" /> : null}
        </G>
      );
    case 'slant':
      return <Path key={key} d={`M${x} ${y}L${x + w * 0.15} ${y - 4}L${x + w} ${y - 12}L${x + w} ${y}Z`} fill={fill} />;
    case 'gable':
      return <Path key={key} d={`M${x - 1} ${y}L${x + w / 2} ${y - w * 0.42}L${x + w + 1} ${y}Z`} fill={night ? shade('#8A4A3A', -0.6) :
'#B5614A'} />;
    case 'dome':
      return <Path key={key} d={`M${x + w * 0.1} ${y}A${w * 0.4} ${w * 0.4} 0 0 1 ${x + w * 0.9} ${y}Z`} fill={shade(fill, night ? 0.1 :
-0.1)} />;
    default:
      return null;
```

```
  }
}

function Landmarks({ list, layer, W, base, H, night, S }: { list: Landmark[]; layer: 'mid' | 'far'; W: number; base: number; H: number;
night: boolean; S: number }) {
  return (
    <>
      {list
        .filter((lm) => !!lm.far === (layer === 'far'))
        .map((lm) => {
          const look = LOOKS[lm.kind];
          const idx = night ? 1 : 0;
          const x = xFor(K[layer], W, lm.heading, S);
          const colour = (k: Tone) => look[k][idx];
          return (
            <G key={lm.id} transform={`translate(${x} ${base}) scale(${H * lm.size})`}>
              {night && look.glowR > 0 ? <Circle cx={0} cy={-0.5} r={look.glowR} fill={`url(#glow-${lm.kind})`} /> : null}
              {SHAPES[lm.kind].map((pt: Part, i) => {
                const f = colour(pt.k);
                if (pt.t === 'p') return <Path key={i} d={pt.d} fill={f} />;
                if (pt.t === 'r') return <Rect key={i} x={pt.x} y={pt.y} width={pt.w} height={pt.h} fill={f} />;
                if (pt.t === 'c') return <Circle key={i} cx={pt.x} cy={pt.y} r={pt.r} fill={f} />;
                if (pt.t === 'l') return <Path key={i} d={`M${pt.x1} ${pt.y1}L${pt.x2} ${pt.y2}`} stroke={f} strokeWidth={pt.w} />;
                return <Path key={i} d={pt.d} stroke={f} strokeWidth={pt.w} fill="none" />;
              })}
            </G>
          );
        })}
    </>
  );
}

function GlowDefs({ list }: { list: Landmark[] }) {
  return (
    <>
      {list.map((lm) => (
        <RadialGradient key={lm.id} id={`glow-${lm.kind}`} cx="50%" cy="50%" r="50%">
          <Stop offset="0" stopColor={LOOKS[lm.kind].glow} stopOpacity={0.55} />
          <Stop offset="1" stopColor={LOOKS[lm.kind].glow} stopOpacity={0} />
        </RadialGradient>
      ))}
    </>
  );
}

/** One scrolling layer: content is drawn once and repeated twice side by side so it can wrap around 360°. */
function Wrap({ W, H, defs, children }: { W: number; H: number; defs: ReactNode; children: ReactNode }) {
  const id = `strip${useId().replace(/\W/g, '')}`;
  return (
    <Svg width={W * 2} height={H} style={{ position: 'absolute', left: 0, top: 0 }}>
      <Defs>{defs}</Defs>
      <G id={id}>{children}</G>
      <Use href={`#${id}`} x={W} />
    </Svg>
  );
}

function skylineLayers(pano: Pano, S: number, H: number, night: boolean) {
  const W = layout(S);
  const i = night ? 1 : 0;
  const rand = rng(pano.seed * 7);
  const midBase = H * 0.58;
  const nearBase = H * 0.67;
  const farBase = H * 0.5;

  // ---- far layer: mountains or distant blocks
  const far: ReactNode[] = [];
  if (pano.mountains) {
    const f = rng(pano.seed + 3);
    const ph = [f() * 6, f() * 6, f() * 6];
    let d = `M0 ${farBase}`;
    for (let x = 0; x <= W.far; x += 12) {
      const t = (x / W.far) * Math.PI * 2;
      const amp =
        pano.mountains === 'hills'
          ? 0.05 + 0.03 * Math.sin(t * 3 + ph[0]) + 0.02 * Math.sin(t * 7 + ph[1])
          : pano.mountains === 'granite'
            ? 0.07 + 0.1 * Math.pow(Math.abs(Math.sin(t * 3 + ph[0])), 1.6) + 0.03 * Math.sin(t * 9 + ph[1])
            : 0.08 + 0.1 * Math.abs(Math.sin(t * 4 + ph[0])) + 0.04 * Math.sin(t * 11 + ph[2]);
      d += `L${x} ${farBase - H * amp}`;
    }
    d += `L${W.far} ${farBase}Z`;
    far.push(<Path key="mount" d={d} fill={pano.tone.far[i]} />);
  } else {
    genBuildings(pano.seed + 5, W.far, [12, 30], [0.03, 0.15], 2, ['flat', 'step', 'flat'], 0.1, H).forEach((b, k) =>
      far.push(<Rect key={`f${k}`} x={b.x} y={farBase - b.h} width={b.w} height={b.h} fill={shade(pano.tone.far[i], b.amt)} />),
```

```
    );
  }

  // ---- mid layer: the skyline
  const mid: ReactNode[] = [];
  const midDefs: ReactNode[] = [];
  if (pano.ground === 'slope') {
    const f = rng(pano.seed + 9);
    const ph = [f() * 6, f() * 6];
    const top = (x: number) => midBase - H * (0.13 + 0.06 * Math.sin((x / W.mid) * Math.PI * 2 * 2 + ph[0]) + 0.04 * Math.sin((x / W.mi
d) * Math.PI * 2 * 5 + ph[1]));
    let d = `M0 ${midBase}`;
    for (let x = 0; x <= W.mid; x += 12) d += `L${x} ${top(x)}`;
    d += `L${W.mid} ${midBase}Z`;
    mid.push(<Path key="hill" d={d} fill={pano.tone.mid[i]} />);
    const colours = pano.houses ?? ['#F2B8A2'];
    for (let row = 0; row < 2; row++) {
      let x = row * 11;
      let n = 0;
      while (x < W.mid - 40) {
        const w = 24 + rand() * 16;
        const h = H * (0.035 + rand() * 0.02);
        const y = Math.min(midBase - h, top(x + w / 2) + row * H * 0.035 + rand() * 6);
        const c = colours[Math.floor(rand() * colours.length)];
        const fill = night ? shade(c, -0.62) : c;
        mid.push(<Path key={`h${row}-${n}`} d={`M${x} ${y + h}L${x} ${y + 6}L${x + w / 2} ${y}L${x + w} ${y + 6}L${x + w} ${y + h}Z`} fi
ll={fill} />);
        if (night) mid.push(<Rect key={`w${row}-${n}`} x={x + w * 0.3} y={y + h * 0.45} width={4} height={5} fill={pano.lights} />);
        x += w + 3;
        n++;
      }
    }
  } else {
    midDefs.push(<WindowPatterns key="wp" id={night ? 'wmn' : 'wmd'} night={night} color={pano.lights} base={shade(pano.tone.mid[i], 0.2
5)} />);
    genBuildings(pano.seed, W.mid, pano.mid.w, pano.mid.h, pano.mid.gap, pano.mid.roofs, pano.mid.towers, H).forEach((b, k) => {
      const fill = shade(pano.tone.mid[i], b.amt);
      const y = midBase - b.h;
      mid.push(<Rect key={`m${k}`} x={b.x} y={y} width={b.w} height={b.h} fill={fill} />);
      if (b.w > 8 && b.h > 16) mid.push(<Rect key={`mw${k}`} x={b.x + 1} y={y + 2} width={b.w - 2} height={b.h - 2} fill={`url(#${night
? 'wmn' : 'wmd'}${b.v})`} />);
      mid.push(roofEl(b, y, fill, night, `mr${k}`));
      if (pano.ground === 'water')
        mid.push(<Rect key={`mr${k}f`} x={b.x} y={midBase} width={b.w} height={Math.min(H * 0.1, b.h * 0.45)} fill={night ? pano.lights
: '#FFFFFF'} opacity={night ? 0.12 : 0.1} />);
    });
  }

  // ---- near layer: low rooftops and street lamps
  const near: ReactNode[] = [];
  const nearDefs: ReactNode[] = [<WindowPatterns key="wp" id={night ? 'wnn' : 'wnd'} night={night} color={pano.lights} base={shade(pano.
tone.near[i], 0.2)} />];
  genBuildings(pano.seed + 1, W.near, [34, 78], [pano.near.h[0], pano.near.h[1]], 6, pano.houses ? ['gable', 'flat'] : ['flat', 'gable',
'step'], 0, H).forEach((b, k) => {
    const c = pano.houses ? pano.houses[k % pano.houses.length] : pano.tone.near[i];
    const fill = pano.houses ? (night ? shade(c, -0.68) : c) : shade(c, b.amt);
    const y = nearBase - b.h;
    near.push(
      <Rect key={`n${k}`} x={b.x} y={y} width={b.w} height={b.h} fill={fill} />,
      <Rect key={`nw${k}`} x={b.x + 2} y={y + 3} width={b.w - 4} height={b.h - 4} fill={`url(#${night ? 'wnn' : 'wnd'}${b.v})`} />,
      roofEl({ ...b, roof: b.roof === 'step' ? 'flat' : b.roof }, y, fill, night, `nr${k}`),
    );
  });
  for (let x = 40; x < W.near; x += 130 + rand() * 40) {
    near.push(<Rect key={`lp${x}`} x={x} y={nearBase - H * 0.075} width={2} height={H * 0.075} fill={night ? '#20243A' : '#4A505C'} />);
    near.push(night ? <Circle key={`lg${x}`} cx={x + 1} cy={nearBase - H * 0.078} r={13} fill={pano.lights} opacity={0.3} /> : null);
    near.push(<Circle key={`lb${x}`} cx={x + 1} cy={nearBase - H * 0.078} r={2.6} fill={night ? '#FFF1C9' : '#9AA3B0'} />);
  }
  // Foreground details: lane markings on streets, shimmer on water.
  if (pano.ground === 'road') {
    for (let x = 0; x < W.near; x += 64) near.push(<Rect key={`ln${x}`} x={x} y={H * 0.79} width={30} height={3} fill={night ? '#C9C3A0'
: '#E8E3C8'} opacity={night ? 0.3 : 0.7} />);
    near.push(<Rect key="edge" x={0} y={H * 0.72} width={W.near} height={2} fill={night ? '#2A2E48' : '#6B7280'} opacity={0.8} />);
  } else {
    for (let k = 0; k < 140; k++) near.push(<Rect key={`sh${k}`} x={rand() * W.near} y={H * (0.69 + rand() * 0.27)} width={14 + rand() *
40} height={1.6} rx={1} fill={night ? pano.lights : '#FFFFFF'} opacity={night ? 0.14 + rand() * 0.14 : 0.18 + rand() * 0.2} />);
  }
  return { far, mid, near, midDefs, nearDefs, W, farBase, midBase, nearBase };
}

function Skyline({ pano, S, H, night, pos }: { pano: Pano; S: number; H: number; night: boolean; pos: Animated.Value }) {
  const [built] = useState(() => skylineLayers(pano, S, H, night));
  const { W } = built;
  const glowDefs = <GlowDefs key="glow" list={pano.landmarks} />; // goes into a list of defs below, so it needs a key
  return (
    <>
```

```
      <Layer pos={pos} k={K.far} W={W.far} H={H} defs={null}>
        {built.far}
        <Landmarks list={pano.landmarks} layer="far" W={W.far} base={built.farBase} H={H} night={night} S={S} />
      </Layer>
      <Layer pos={pos} k={K.mid} W={W.mid} H={H} defs={[...built.midDefs, glowDefs]}>
        {built.mid}
        <Landmarks list={pano.landmarks} layer="mid" W={W.mid} base={built.midBase} H={H} night={night} S={S} />
      </Layer>
      <Layer pos={pos} k={K.near} W={W.near} H={H} defs={built.nearDefs}>
        {built.near}
      </Layer>
    </>
  );
}

/** A layer wrapper that scrolls with the shared position value. */
function Layer({ pos, k, W, H, defs, children }: { pos: Animated.Value; k: number; W: number; H: number; defs: ReactNode; children: Reac
tNode }) {
  const tx = Animated.multiply(Animated.modulo(Animated.multiply(pos, k), W), -1);
  return (
    <Animated.View pointerEvents="none" style={[StyleSheet.absoluteFill, { transform: [{ translateX: tx }] }]}>
      <Wrap W={W} H={H} defs={defs}>
        {children}
      </Wrap>
    </Animated.View>
  );
}

function SkyDecor({ pano, S, H, night }: { pano: Pano; S: number; H: number; night: boolean }) {
  const Ws = layout(S).sky;
  const [items] = useState(() => {
    const rand = rng(pano.seed * 13);
    const stars = Array.from({ length: 70 }, () => ({ x: rand() * Ws, y: rand() * H * 0.42, r: 0.6 + rand() * 1.4, o: 0.5 + rand() * 0.5
}));
    const clouds = Array.from({ length: 5 }, () => ({ x: rand() * Ws, y: H * (0.1 + rand() * 0.22), w: 70 + rand() * 90 }));
    return { stars, clouds };
  });
  const sunX = xFor(K.sky, Ws, 205, S);
  const moonX = xFor(K.sky, Ws, 150, S);
  return (
    <Wrap
      W={Ws}
      H={H}
      defs={
        <>
          <RadialGradient id="sun" cx="50%" cy="50%" r="50%">
            <Stop offset="0" stopColor="#FFE9A8" stopOpacity={0.9} />
            <Stop offset="1" stopColor="#FFE9A8" stopOpacity={0} />
          </RadialGradient>
          <RadialGradient id="moon" cx="50%" cy="50%" r="50%">
            <Stop offset="0" stopColor="#FFF4C7" stopOpacity={0.5} />
            <Stop offset="1" stopColor="#FFF4C7" stopOpacity={0} />
          </RadialGradient>
        </>
      }>
      {night ? (
        <>
          {items.stars.map((st, i) => (
            <Circle key={i} cx={st.x} cy={st.y} r={st.r} fill="#FFFFFF" opacity={st.o} />
          ))}
          <Circle cx={moonX} cy={H * 0.17} r={70} fill="url(#moon)" />
          <Circle cx={moonX} cy={H * 0.17} r={20} fill="#FFF4C7" />
          <Circle cx={moonX + 8} cy={H * 0.17 - 4} r={18} fill={pano.sky.nightTop} opacity={0.9} />
        </>
      ) : (
        <>
          <Circle cx={sunX} cy={H * 0.2} r={150} fill="url(#sun)" />
          <Circle cx={sunX} cy={H * 0.2} r={26} fill="#FFF3C4" />
          {items.clouds.map((c, i) => (
            <G key={i} opacity={0.75}>
              <Rect x={c.x} y={c.y} width={c.w} height={14} rx={7} fill="#FFFFFF" />
              <Rect x={c.x + c.w * 0.2} y={c.y - 9} width={c.w * 0.5} height={16} rx={8} fill="#FFFFFF" />
            </G>
          ))}
        </>
      )}
    </Wrap>
  );
}

function CarLights({ H, S, color }: { H: number; S: number; color: string }) {
  const a = useLoop(9000, { rest: 0 });
  const b = useLoop(12000, { rest: 0 });
  const dots = Array.from({ length: 18 }, (_, i) => i);
  const row = (key: string, v: Animated.Value, y: number, c: string, dir: 1 | -1) => (
    <Animated.View
```

```
      key={key}
      pointerEvents="none"
      style={{ position: 'absolute', top: y, left: 0, width: S * 2, height: 6, transform: [{ translateX: v.interpolate({ inputRange: [0,
1], outputRange: dir === 1 ? [-S, 0] : [0, -S] }) }] }}>
      {dots.map((i) => (
        <View key={i} style={{ position: 'absolute', left: (i * S * 2) / 18 + ((i * 37) % 23), width: 11, height: 4, borderRadius: 2, ba
ckgroundColor: c, opacity: 0.9 }} />
      ))}
    </Animated.View>
  );
  return (
    <>
      {row('w', a, H * 0.755, '#FFF6D8', -1)}
      {row('r', b, H * 0.8, '#FF5A4D', 1)}
      <View pointerEvents="none" style={{ position: 'absolute', left: 0, right: 0, top: H * 0.745, height: H * 0.08, backgroundColor: co
lor, opacity: 0.05 }} />
    </>
  );
}

function PanoramaSceneImpl({ pano, night, pos, S, H }: { pano: Pano; night: Animated.Value; pos: Animated.Value; S: number; H: number })
{
  const flicker = useLoop(2600, { pingPong: true, rest: 0.5 });
  const nightOp = Animated.multiply(night, flicker.interpolate({ inputRange: [0, 1], outputRange: [0.93, 1] }));
  const dayOp = night.interpolate({ inputRange: [0, 1], outputRange: [1, 0] });
  const water = pano.ground !== 'road';
  const ground = water ? { day: ['#5B97C9', '#2F6A9E'], night: ['#0B1235', '#04061A'] } : { day: ['#454C59', '#2B3039'], night: ['#0E112
2', '#05060F'] };
  const skyTx = Animated.multiply(Animated.modulo(Animated.multiply(pos, K.sky), layout(S).sky), -1);
  const sparkle = useLoop(3200, { pingPong: true, rest: 0.5 });
  return (
    <View style={StyleSheet.absoluteFill} pointerEvents="none">
      {/* Sky */}
      <Animated.View style={[StyleSheet.absoluteFill, { opacity: dayOp }]}>
        <LinearGradient colors={[pano.sky.dayTop, pano.sky.dayHorizon]} locations={[0.05, 0.55]} style={StyleSheet.absoluteFill} />
      </Animated.View>
      <Animated.View style={[StyleSheet.absoluteFill, { opacity: night }]}>
        <LinearGradient colors={[pano.sky.nightTop, pano.sky.nightHorizon]} locations={[0.05, 0.58]} style={StyleSheet.absoluteFill} />
      </Animated.View>
      <Animated.View style={[StyleSheet.absoluteFill, { opacity: dayOp, transform: [{ translateX: skyTx }] }]}>
        <SkyDecor pano={pano} S={S} H={H} night={false} />
      </Animated.View>
      <Animated.View style={[StyleSheet.absoluteFill, { opacity: Animated.multiply(night, sparkle.interpolate({ inputRange: [0, 1], outp
utRange: [0.8, 1] })), transform: [{ translateX: skyTx }] }]}>
        <SkyDecor pano={pano} S={S} H={H} night />
      </Animated.View>

      {/* Ground / water */}
      <Animated.View style={[styles.ground, { top: H * 0.5, opacity: dayOp }]}>
        <LinearGradient colors={ground.day as [string, string]} style={StyleSheet.absoluteFill} />
      </Animated.View>
      <Animated.View style={[styles.ground, { top: H * 0.5, opacity: night }]}>
        <LinearGradient colors={ground.night as [string, string]} style={StyleSheet.absoluteFill} />
      </Animated.View>

      {/* Skyline: day set and night set cross-fade */}
      <Animated.View style={[StyleSheet.absoluteFill, { opacity: dayOp }]}>
        <Skyline pano={pano} S={S} H={H} night={false} pos={pos} />
      </Animated.View>
      <Animated.View style={[StyleSheet.absoluteFill, { opacity: nightOp }]}>
        <Skyline pano={pano} S={S} H={H} night pos={pos} />
      </Animated.View>

      {/* Warm city glow above the skyline at night */}
      <Animated.View style={[styles.glow, { top: H * 0.3, height: H * 0.3, opacity: Animated.multiply(night, 0.5) }]}>
        <LinearGradient colors={['rgba(255,150,70,0)', pano.lights]} style={StyleSheet.absoluteFill} />
      </Animated.View>
      <Animated.View style={[styles.glow, { top: H * 0.3, height: H * 0.3, opacity: Animated.multiply(dayOp, 0.28) }]}>
        <LinearGradient colors={['rgba(255,255,255,0)', '#FFFFFF']} style={StyleSheet.absoluteFill} />
      </Animated.View>

      {!water ? (
        <Animated.View style={[StyleSheet.absoluteFill, { opacity: night }]}>
          <CarLights S={S} H={H} color={pano.lights} />
        </Animated.View>
      ) : null}
    </View>
  );
}

export const PanoramaScene = memo(PanoramaSceneImpl);

const styles = StyleSheet.create({
  ground: { position: 'absolute', left: 0, right: 0, bottom: 0 },
  glow: { position: 'absolute', left: 0, right: 0 },
});
```

src/components/panorama/shapes.ts

```
import type { LandmarkKind } from '@/data/panoramas';

/**
 * Landmark silhouettes in unit coordinates: x is centred, y is negative upwards, 1 = full height.
 * Tones: m = main body, a = accent, l = light (windows / lamps — glows at night).
 */
export type Tone = 'm' | 'a' | 'l';
export type Part =
  | { t: 'p'; d: string; k: Tone }
  | { t: 'r'; x: number; y: number; w: number; h: number; k: Tone }
  | { t: 'c'; x: number; y: number; r: number; k: Tone }
  | { t: 'l'; x1: number; y1: number; x2: number; y2: number; w: number; k: Tone }
  | { t: 's'; d: string; w: number; k: Tone };

const p = (d: string, k: Tone): Part => ({ t: 'p', d, k });
const r = (x: number, y: number, w: number, h: number, k: Tone): Part => ({ t: 'r', x, y, w, h, k });
const c = (x: number, y: number, rad: number, k: Tone): Part => ({ t: 'c', x, y, r: rad, k });
const l = (x1: number, y1: number, x2: number, y2: number, w: number, k: Tone): Part => ({ t: 'l', x1, y1, x2, y2, w, k });
const s = (d: string, w: number, k: Tone): Part => ({ t: 's', d, w, k });

export interface Look {
  /** [day, night] colour for each tone. */
  m: [string, string];
  a: [string, string];
  l: [string, string];
  /** Night glow colour. */
  glow: string;
  /** Glow radius in units. */
  glowR: number;
}

const shell = (x: number, w: number, hh: number) =>
  `M${x} -.06C${x} ${-0.06 - hh * 0.4} ${x + w * 0.4} ${-0.06 - hh} ${x + w * 0.8} ${-0.06 - hh}C${x + w * 0.9} ${-0.06 - hh * 0.6} ${x
+ w} ${-0.06 - hh * 0.2} ${x + w} -.06Z`;

function eye(): Part[] {
  const out: Part[] = [s('M-.5 -.5A.5 .5 0 1 0 .5 -.5A.5 .5 0 1 0 -.5 -.5', 0.012, 'a'), l(0, -0.5, -0.2, 0, 0.016, 'm'), l(0, -0.5, 0.
2, 0, 0.016, 'm')];
  for (let i = 0; i < 16; i++) {
    const a = (i / 16) * Math.PI * 2;
    const x = Math.cos(a) * 0.5;
    const y = -0.5 + Math.sin(a) * 0.5;
    out.push(l(0, -0.5, x, y, 0.003, 'm'), c(x, y, 0.017, 'l'));
  }
  return out;
}

function harbour(): Part[] {
  const out: Part[] = [s('M-1.1 0Q0 -1.5 1.1 0', 0.04, 'm'), r(-1.3, -0.12, 2.6, 0.03, 'm'), r(-1.3, -0.12, 2.6, 0.01, 'l'), r(-1.16, -
0.45, 0.12, 0.45, 'a'), r(1.04, -0.45, 0.12, 0.45, 'a')];
  for (let i = 1; i < 20; i++) {
    const t = i / 20;
    out.push(l(-1.1 + 2.2 * t, -3 * t * (1 - t), -1.1 + 2.2 * t, -0.12, 0.004, 'm'));
  }
  return out;
}

function operaParts(): Part[] {
  const shells: [number, number, number][] = [
    [-0.66, 0.3, 0.58],
    [-0.4, 0.28, 0.48],
    [-0.16, 0.26, 0.38],
    [0.08, 0.34, 0.64],
    [0.36, 0.32, 0.52],
    [0.62, 0.2, 0.3],
  ];
  return [r(-0.75, -0.06, 1.55, 0.06, 'a'), ...shells.map(([x, w, hh]) => p(shell(x, w, hh), 'm'))];
}

function colosseum(): Part[] {
  const out: Part[] = [p('M-.8 0L-.8-.4L-.62-.5L.5-.5L.8-.4L.8 0Z', 'm')];
  for (let row = 0; row < 3; row++) for (let i = 0; i < 15; i++) out.push(r(-0.76 + i * 0.105, -0.13 - row * 0.12, 0.05, 0.08, 'a'));
  return out;
}

function supertrees(): Part[] {
  const out: Part[] = [];
  for (const [cx, hh] of [
    [-0.5, 0.7],
    [0, 1],
    [0.5, 0.8],
  ] as const) {
    out.push(
      p(`M${cx - 0.03} 0L${cx - 0.008} ${-hh}L${cx + 0.008} ${-hh}L${cx + 0.03} 0Z`, 'm'),
```

```
      p(`M${cx - 0.24} ${-hh}Q${cx} ${-hh - 0.3} ${cx + 0.24} ${-hh}Z`, 'a'),
      c(cx, -hh - 0.07, 0.018, 'l'),
    );
  }
  return out;
}

export const SHAPES: Record<LandmarkKind, Part[]> = {
  eiffel: [
    p('M-.19 0L-.015-1L.015-1L.19 0L.13 0Q0 -.34 -.13 0Z', 'm'),
    r(-0.13, -0.405, 0.26, 0.016, 'm'),
    r(-0.075, -0.7, 0.15, 0.014, 'm'),
    r(-0.004, -1.07, 0.008, 0.07, 'm'),
    c(0, -0.41, 0.012, 'l'),
    c(0, -0.71, 0.01, 'l'),
    r(-0.004, -0.98, 0.008, 0.2, 'l'),
  ],
  tokyotower: [
    p('M-.14 0L-.03-.66L-.016-.78L-.006-1L.006-1L.016-.78L.03-.66L.14 0L.1 0Q0 -.22 -.1 0Z', 'm'),
    r(-0.065, -0.34, 0.13, 0.03, 'a'),
    r(-0.04, -0.62, 0.08, 0.025, 'a'),
    r(-0.12, -0.15, 0.24, 0.02, 'a'),
    c(0, -0.33, 0.012, 'l'),
  ],
  skytree: [p('M-.055 0L-.014-.6L.014-.6L.055 0Z', 'm'), r(-0.05, -0.64, 0.1, 0.03, 'a'), r(-0.032, -0.8, 0.064, 0.025, 'a'), r(-0.004,
-1, 0.008, 0.2, 'm'), r(-0.003, -0.6, 0.006, 0.4, 'l')],
  fuji: [p('M-1 0L-.12-.5L-.06-.52L0-.55L.06-.52L.12-.5L1 0Z', 'm'), p('M-.16-.43L-.12-.5L-.06-.52L0-.55L.06-.52L.12-.5L.16-.43L.09-.46
L.04-.4L-.02-.45L-.08-.39Z', 'a')],
  esb: [
    r(-0.15, -0.4, 0.3, 0.4, 'm'),
    r(-0.11, -0.62, 0.22, 0.22, 'm'),
    r(-0.075, -0.78, 0.15, 0.16, 'm'),
    r(-0.045, -0.88, 0.09, 0.1, 'a'),
    r(-0.022, -0.93, 0.044, 0.05, 'a'),
    r(-0.004, -1, 0.008, 0.07, 'a'),
    r(-0.03, -0.86, 0.06, 0.02, 'l'),
  ],
  wtc: [p('M-.075 0L-.075-.7L-.003-.86L.003-.86L.075-.7L.075 0Z', 'm'), r(-0.003, -1, 0.006, 0.15, 'a'), r(-0.003, -0.86, 0.006, 0.86,
'l')],
  chrysler: [r(-0.07, -0.66, 0.14, 0.66, 'm'), p('M-.07-.66Q-.07-.9 0-.95Q.07-.9 .07-.66Z', 'a'), r(-0.004, -1, 0.008, 0.06, 'a'), c(0,
-0.82, 0.014, 'l')],
  bigben: [
    r(-0.05, -0.78, 0.1, 0.78, 'm'),
    p('M-.065-.78L0-1L.065-.78Z', 'a'),
    r(-0.065, -0.8, 0.13, 0.03, 'a'),
    c(0, -0.62, 0.032, 'l'),
    r(0.05, -0.2, 0.7, 0.2, 'm'),
    r(0.2, -0.3, 0.05, 0.1, 'm'),
    r(0.45, -0.28, 0.04, 0.08, 'm'),
    p('M.2-.3L.225-.38L.25-.3Z', 'a'),
    r(0.1, -0.14, 0.6, 0.03, 'l'),
  ],
  eye: eye(),
  shard: [p('M-.07 0L0-1L.07 0Z', 'm'), l(0, -1, -0.03, 0, 0.004, 'a'), l(0, -1, 0.03, 0, 0.004, 'a'), l(0, -1, 0, 0, 0.006, 'l')],
  burj: [
    p('M-.07 0L-.05-.35L-.035-.5L-.03-.62L-.02-.78L-.012-.9L-.003-1L.003-1L.012-.9L.02-.78L.03-.62L.035-.5L.05-.35L.07 0Z', 'm'),
    l(0, 0, 0, -0.95, 0.005, 'l'),
    l(-0.03, -0.2, -0.015, -0.6, 0.003, 'a'),
    l(0.03, -0.2, 0.015, -0.6, 0.003, 'a'),
  ],
  burjalarab: [p('M-.2 0L-.2-.75Q-.02-1 .02-1Q.2-.45 .2 0Z', 'm'), r(-0.45, -0.03, 0.9, 0.03, 'a'), r(-0.012, -1.1, 0.012, 0.12, 'a'), l
(-0.2, -0.75, 0.02, -1, 0.012, 'l')],
  opera: operaParts(),
  harbour: harbour(),
  sydtower: [r(-0.012, -0.7, 0.024, 0.7, 'm'), p('M-.06-.7L.06-.7L.045-.8L-.045-.8Z', 'a'), r(-0.05, -0.83, 0.1, 0.05, 'a'), r(-0.004, -
1, 0.008, 0.17, 'm'), r(-0.045, -0.8, 0.09, 0.012, 'l')],
  mbs: [
    p('M-.52 0L-.34 0L-.22-1L-.4-1Z', 'm'),
    p('M-.09 0L.09 0L.09-1L-.09-1Z', 'm'),
    p('M.34 0L.52 0L.4-1L.22-1Z', 'm'),
    r(-0.66, -1.05, 1.32, 0.05, 'a'),
    p('M.66-1.05L.84-1.03L.66-1Z', 'a'),
    r(-0.66, -1, 1.32, 0.012, 'l'),
  ],
  supertrees: supertrees(),
  christ: [
    p('M-1 0Q-.4 -.15 -.2 -.6Q-.1 -.95 0 -.95Q.1 -.95 .2 -.6Q.5 -.15 1 0Z', 'm'),
    r(-0.016, -1, 0.032, 0.03, 'a'),
    r(-0.007, -1.12, 0.014, 0.12, 'a'),
    r(-0.06, -1.1, 0.12, 0.014, 'a'),
    c(0, -1.135, 0.009, 'a'),
  ],
  sugarloaf: [p('M-.5 0Q-.52 -.6 -.15 -.78Q.2 -.9 .4 -.5Q.52 -.2 .55 0Z', 'm'), p('M.45 0Q.6 -.35 .9 -.35Q1.3 -.3 1.5 0Z', 'm'), l(0.28,
-0.7, 1.0, -0.4, 0.004, 'a'), r(0.6, -0.6, 0.05, 0.04, 'l')],
  colosseum: colosseum(),
  stpeters: [
```

```
    r(-0.4, -0.28, 0.8, 0.28, 'm'),
    r(-0.16, -0.6, 0.32, 0.32, 'm'),
    p('M-.16-.6Q-.16-1 0-1Q.16-1 .16-.6Z', 'a'),
    r(-0.03, -1.12, 0.06, 0.12, 'm'),
    r(-0.004, -1.25, 0.008, 0.13, 'a'),
    r(-0.3, -0.2, 0.6, 0.02, 'l'),
  ],
  duomo: [
    r(-0.3, -0.5, 0.6, 0.5, 'm'),
    p('M-.22-.5Q-.22-.95 0-.95Q.22-.95 .22-.5Z', 'a'),
    l(-0.1, -0.5, -0.06, -0.93, 0.02, 'l'),
    l(0.1, -0.5, 0.06, -0.93, 0.02, 'l'),
    r(0.32, -0.78, 0.12, 0.78, 'm'),
    p('M.3-.78L.38-.92L.46-.78Z', 'a'),
    r(-0.24, -0.28, 0.12, 0.14, 'l'),
    r(0.12, -0.28, 0.12, 0.14, 'l'),
  ],
  arc: [p('M-.45 0L-.45-1L.45-1L.45 0L.18 0L.18-.55Q0 -.8 -.18-.55L-.18 0Z', 'm'), r(-0.45, -0.9, 0.9, 0.02, 'a'), r(-0.45, -1, 0.9, 0.0
5, 'a')],
  sacre: [
    r(-0.4, -0.35, 0.8, 0.35, 'm'),
    p('M-.15-.35Q-.15-.85 0-.9Q.15-.85 .15-.35Z', 'm'),
    p('M-.38-.35Q-.38-.6 -.28-.62Q-.18-.6 -.18-.35Z', 'm'),
    p('M.18-.35Q.18-.6 .28-.62Q.38-.6 .38-.35Z', 'm'),
    r(-0.012, -1, 0.024, 0.1, 'a'),
    r(-0.3, -0.2, 0.6, 0.02, 'l'),
  ],
};

const warm = (day: string, night: string): [string, string] => [day, night];

export const LOOKS: Record<LandmarkKind, Look> = {
  eiffel: { m: warm('#5B4B40', '#C98F2E'), a: warm('#6F5E50', '#E5B04C'), l: warm('#8C7A6A', '#FFE08A'), glow: '#FFC14D', glowR: 0.6 },
  tokyotower: { m: warm('#D9432B', '#FF7A3A'), a: warm('#F3F3F3', '#FFF1C9'), l: warm('#F3F3F3', '#FFF1C9'), glow: '#FF8A4A', glowR: 0.5
},
  skytree: { m: warm('#C9D2DC', '#6FB6FF'), a: warm('#EEF3F8', '#BFE4FF'), l: warm('#E8EEF5', '#CFEAFF'), glow: '#6FB6FF', glowR: 0.4 },
  fuji: { m: warm('#8C9DB8', '#1B2150'), a: warm('#FFFFFF', '#8FA4E0'), l: warm('#FFFFFF', '#8FA4E0'), glow: '#000000', glowR: 0 },
  esb: { m: warm('#B9B2A6', '#EAD9A8'), a: warm('#9C9488', '#FFD36B'), l: warm('#D7D1C6', '#FFF0B8'), glow: '#FFD36B', glowR: 0.5 },
  wtc: { m: warm('#8FA6C0', '#BFD7F5'), a: warm('#C9D5E2', '#FFFFFF'), l: warm('#DCE6F0', '#FFFFFF'), glow: '#BFD7F5', glowR: 0.45 },
  chrysler: { m: warm('#A9A39C', '#E6DFC8'), a: warm('#D5DAE0', '#F4F8FF'), l: warm('#F5F5F5', '#FFFFFF'), glow: '#F4F8FF', glowR: 0.45
},
  bigben: { m: warm('#B3A58E', '#D9B76E'), a: warm('#7D8B6E', '#F0D8A0'), l: warm('#E8E2D0', '#FFF6C9'), glow: '#FFD98A', glowR: 0.45 },
  eye: { m: warm('#8FA3B8', '#5EB6FF'), a: warm('#B5C7DA', '#8AD0FF'), l: warm('#FFFFFF', '#CFEFFF'), glow: '#5EB6FF', glowR: 0.7 },
  shard: { m: warm('#9DB6CE', '#CFE3FF'), a: warm('#C5D6E6', '#FFFFFF'), l: warm('#E6EEF6', '#FFFFFF'), glow: '#CFE3FF', glowR: 0.4 },
  burj: { m: warm('#B9C6D4', '#DDEBFF'), a: warm('#8EA2B8', '#7DB8FF'), l: warm('#E8F0F8', '#FFFFFF'), glow: '#9CC8FF', glowR: 0.4 },
  burjalarab: { m: warm('#F4F1EA', '#5A9BFF'), a: warm('#D9D5CB', '#B58BFF'), l: warm('#FFFFFF', '#E1D0FF'), glow: '#7B8CFF', glowR: 0.6
},
  opera: { m: warm('#F2EFE8', '#FFD9A0'), a: warm('#C9C3B6', '#C97F4A'), l: warm('#FFFFFF', '#FFE9C0'), glow: '#FFC878', glowR: 0.7 },
  harbour: { m: warm('#59616B', '#8EB8E8'), a: warm('#B59A7B', '#F4C27A'), l: warm('#FFFFFF', '#FFE08A'), glow: '#8EB8E8', glowR: 0.5 },
  sydtower: { m: warm('#9AA3AD', '#F4C95D'), a: warm('#C9A24B', '#FFE08A'), l: warm('#FFFFFF', '#FFF0B0'), glow: '#F4C95D', glowR: 0.4
},
  mbs: { m: warm('#B5BFC9', '#E6EEF8'), a: warm('#8FA0B2', '#FFB36B'), l: warm('#E8F2FF', '#FFE7A8'), glow: '#8FC8FF', glowR: 0.5 },
  supertrees: { m: warm('#4B6B52', '#2E4A3A'), a: warm('#6FA37A', '#C25BFF'), l: warm('#D0F0D8', '#FF8CF0'), glow: '#C25BFF', glowR: 0.7
},
  christ: { m: warm('#4D6B5A', '#10202A'), a: warm('#F2F2F2', '#FFF3C9'), l: warm('#FFFFFF', '#FFF3C9'), glow: '#FFF3C9', glowR: 0.1 },
  sugarloaf: { m: warm('#5E7467', '#16222B'), a: warm('#C9D2CC', '#FFD07A'), l: warm('#FFFFFF', '#FFD07A'), glow: '#FFD07A', glowR: 0 },
  colosseum: { m: warm('#C7A57C', '#6D4B33'), a: warm('#8E6E4A', '#FFC873'), l: warm('#E8D2B0', '#FFD89A'), glow: '#FFB45E', glowR: 0.5
},
  stpeters: { m: warm('#D9CDB6', '#F0CF8F'), a: warm('#B8A98B', '#FFE3A8'), l: warm('#F2EAD8', '#FFF0C8'), glow: '#FFD98A', glowR: 0.5
},
  duomo: { m: warm('#F2D5B0', '#FFCF8A'), a: warm('#E0A93C', '#FFE08A'), l: warm('#4E9E6B', '#7DFFB0'), glow: '#FFC878', glowR: 0.5 },
  arc: { m: warm('#C9BFA9', '#EBCB86'), a: warm('#A89E88', '#FFE2A0'), l: warm('#FFFFFF', '#FFF0C8'), glow: '#FFD98A', glowR: 0.5 },
  sacre: { m: warm('#F4EFE6', '#FFE9B8'), a: warm('#D9D2C4', '#FFF6D9'), l: warm('#FFFFFF', '#FFF6D9'), glow: '#FFE9B8', glowR: 0.5 },
};
```

src/components/personalize/options.ts

```
import type { IconName } from '@/components/ui/bits';
import type { HomeSection, Personal } from '@/store/types';

export const EMOJIS = ['✈️', '🌴', '🏔️', '🍜', '📸', '🎒', '🌸', '🏄', '🚂', '🗺️', '🌙', '🦋'];

export const SECTIONS: { key: HomeSection; label: string; sub: string; icon: IconName }[] = [
  { key: 'glance', label: 'Weather & local time', sub: 'Two quick tiles under the hero', icon: 'partly-sunny-outline' },
  { key: 'budget', label: 'Group budget', sub: 'Spent so far with a progress bar', icon: 'wallet-outline' },
  { key: 'upnext', label: 'Up next & today', sub: 'Countdown and today’s checklist', icon: 'alarm-outline' },
  { key: 'quick', label: 'Quick tiles', sub: 'Vote, flights, docs and photos', icon: 'grid-outline' },
  { key: 'forYou', label: 'Picked for you', sub: 'Destinations matching your interests', icon: 'sparkles-outline' },
  { key: 'nearby', label: 'Nearby radius', sub: 'Places around your trip', icon: 'navigate-outline' },
  { key: 'discover', label: 'Discover destinations', sub: 'Photo cards to dream about', icon: 'images-outline' },
];

export const PACE: {
  key: Personal['travelStyle'];
  label: string;
  emoji: string;
```

```
  sub: string;
}[] = [
  {
    key: 'chill',
    label: 'Chill',
    emoji: '🧘',
    sub: '2–3 things a day, long lunches',
  },
  {
    key: 'balanced',
    label: 'Balanced',
    emoji: '⚖️',
    sub: 'A plan, with room to wander',
  },
  {
    key: 'packed',
    label: 'Packed',
    emoji: '⚡',
    sub: 'See everything, sleep later',
  },
];
export const BUDGET: {
  key: Personal['budgetStyle'];
  label: string;
  emoji: string;
  sub: string;
}[] = [
  { key: 'saver', label: 'Saver', emoji: '🪙', sub: 'Street food & hostels' },
  { key: 'mid', label: 'Mid-range', emoji: '💳', sub: 'Comfy, not crazy' },
  {
    key: 'treat',
    label: 'Treat yo self',
    emoji: '💎',
    sub: 'Splurge on the good stuff',
  },
];

export const STEPS = ['Hello', 'Vibe', 'Interests', 'Style', 'Home', 'You', 'Done'] as const;
```

src/components/personalize/palette.ts

```
import { baseColors, themes, type ThemeKey } from '@/theme';

/** Colours of a palette (merged over Forest) — used for the live previews. */
export function paletteOf(k: ThemeKey) {
  return { ...baseColors, ...themes[k].palette };
}

export type Pal = ReturnType<typeof paletteOf>;
```

src/components/personalize/parts.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { LinearGradient } from 'expo-linear-gradient';
import { useEffect, useState } from 'react';
import { Animated, View } from 'react-native';

import type { IconName } from '@/components/ui/bits';
import { nativeDriver, useMotion } from '@/components/ui/motion';
import { Press } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import { cornerStyles, type CornerStyle } from '@/theme';

import type { Personal } from '@/store/types';

import { styles } from './styles';
import type { Pal } from './palette';

export function StepTitle({ pal, kicker, title, sub }: { pal: Pal; kicker: string; title: string; sub: string }) {
  return (
    <View style={{ marginBottom: 16 }}>
      <T variant="kicker" color={pal.primary}>
        {kicker}
      </T>
      <T variant="h1" color={pal.text} style={{ marginTop: 4 }} accessibilityRole="header">
        {title}
      </T>
      <T variant="bodySm" color={pal.textSecondary} style={{ marginTop: 6 }}>
        {sub}
      </T>
    </View>
  );
}

export function OptionRow({ pal, on, emoji, label, sub, onPress }: { pal: Pal; on: boolean; emoji: string; label: string; sub: string; o
nPress: () => void }) {
```

```
  return (
    <Press
      onPress={onPress}
      accessibilityRole="radio"
      accessibilityState={{ checked: on }}
      accessibilityLabel={label}
      style={[
        styles.option,
        {
          backgroundColor: on ? pal.primarySofter : pal.card,
          borderColor: on ? pal.primary : pal.border,
        },
      ]}>
      <T style={{ fontSize: 26, lineHeight: 32 }}>{emoji}</T>
      <View style={{ flex: 1 }}>
        <T variant="title" color={pal.text}>
          {label}
        </T>
        <T variant="caption" color={pal.textSecondary}>
          {sub}
        </T>
      </View>
      <Ionicons name={on ? 'radio-button-on' : 'radio-button-off'} size={22} color={on ? pal.primary : pal.textMuted} />
    </Press>
  );
}

export function SmallChip({ pal, on, label, onPress }: { pal: Pal; on: boolean; label: string; onPress: () => void }) {
  return (
    <Press
      onPress={onPress}
      accessibilityRole="radio"
      accessibilityState={{ checked: on }}
      accessibilityLabel={label}
      style={[
        styles.smallChip,
        {
          backgroundColor: on ? pal.primary : pal.card,
          borderColor: on ? pal.primary : pal.border,
        },
      ]}>
      <T variant="small" weight="semibold" color={on ? '#FFFFFF' : pal.text}>
        {label}
      </T>
    </Press>
  );
}

/** One answer on the last page. Tapping the row (icon, words or pencil) opens the step where it can be changed. */
export function Summary({ pal, icon, label, value, onPress }: { pal: Pal; icon: IconName; label: string; value: string; onPress: () => v
oid }) {
  return (
    <Press onPress={onPress} scaleTo={0.99} accessibilityLabel={`Change ${label.toLowerCase()}, now ${value}`} style={styles.secRow}>
      <View style={[styles.secIcon, { backgroundColor: pal.primarySofter }]}>
        <Ionicons name={icon} size={16} color={pal.primary} />
      </View>
      <T variant="small" color={pal.textSecondary} style={{ width: 120 }}>
        {label}
      </T>
      <T variant="small" weight="semibold" color={pal.text} style={{ flex: 1, textAlign: 'right' }}>
        {value}
      </T>
      <Ionicons name="create-outline" size={16} color={pal.textMuted} />
    </Press>
  );
}

/** A tiny phone-style preview of the Explore page in the chosen palette + corners. */
export function MiniPreview({ pal, corners, name, emoji }: { pal: Pal; corners: CornerStyle; name: string; emoji: string }) {
  const r = (n: number) => Math.round(n * cornerStyles[corners].scale);
  return (
    <View style={[styles.preview, { backgroundColor: pal.bg, borderColor: pal.borderStrong }]} accessibilityLabel="Live preview">
      <T variant="kicker" color={pal.textMuted} style={{ marginBottom: 8 }}>
        Live preview
      </T>
      <LinearGradient colors={[pal.heroA, pal.heroB, pal.heroC]} start={{ x: 0, y: 0 }} end={{ x: 1, y: 1 }} style={{ borderRadius: r(1
8), padding: 12, height: 86 }}>
        <T variant="caption" color="rgba(255,255,255,0.85)">
          Hi {name} {emoji}
        </T>
        <T variant="h3" color="#FFFFFF">
          Where to next?
        </T>
        <View style={[styles.previewSun, { backgroundColor: pal.accent }]} />
      </LinearGradient>
      <View style={{ flexDirection: 'row', gap: 8, marginTop: 8 }}>
```

```
        <View
          style={{
            flex: 1,
            height: 44,
            borderRadius: r(14),
            backgroundColor: pal.primary,
          }}
        />
        <View
          style={{
            flex: 1,
            height: 44,
            borderRadius: r(14),
            backgroundColor: pal.card,
            borderWidth: 1,
            borderColor: pal.border,
          }}
        />
      </View>
      <View style={{ flexDirection: 'row', gap: 6, marginTop: 8 }}>
        {[0, 1, 2].map((i) => (
          <View
            key={i}
            style={{
              paddingHorizontal: 12,
              height: 22,
              borderRadius: 11,
              backgroundColor: i === 0 ? pal.primary : pal.primarySoft,
            }}
          />
        ))}
      </View>
    </View>
  );
}

const CONFETTI = ['#F2B33D', '#E8833A', '#3D7DD8', '#7C5CD6', '#2A9D8F', '#D94F86', '#8CC97A', '#F5C451', '#5CC8D6', '#FF8A5B', '#B8326
8', '#2FAE66'];

/** One-shot confetti burst for the final step. */
export function Confetti({ pal }: { pal: Pal }) {
  const [v] = useState(() => new Animated.Value(0));
  const motion = useMotion();
  useEffect(() => {
    if (!motion) return;
    Animated.timing(v, {
      toValue: 1,
      duration: 1600,
      useNativeDriver: nativeDriver,
    }).start();
  }, [v, motion]);
  if (!motion) return null;
  return (
    <View pointerEvents="none" style={styles.confetti}>
      {CONFETTI.concat(CONFETTI).map((c, i) => {
        const angle = (i / 24) * Math.PI * 2;
        const dist = 90 + (i % 4) * 30;
        return (
          <Animated.View
            key={i}
            style={[
              styles.piece,
              {
                backgroundColor: i % 5 === 0 ? pal.accent : c,
                borderRadius: i % 3 === 0 ? 5 : 2,
                opacity: v.interpolate({
                  inputRange: [0, 0.1, 0.8, 1],
                  outputRange: [0, 1, 1, 0],
                }),
                transform: [
                  {
                    translateX: v.interpolate({
                      inputRange: [0, 1],
                      outputRange: [0, Math.cos(angle) * dist],
                    }),
                  },
                  {
                    translateY: v.interpolate({
                      inputRange: [0, 0.6, 1],
                      outputRange: [0, Math.sin(angle) * dist - 20, Math.sin(angle) * dist + 40],
                    }),
                  },
                  {
                    rotate: v.interpolate({
                      inputRange: [0, 1],
                      outputRange: ['0deg', `${(i % 2 ? 1 : -1) * 540}deg`],
```

```
                  }),
                },
              ],
            },
          ]}
        />
      );
    })}
  </View>
 );
}

/** Updates part of the survey answers. */
export type SetPersonal = (patch: Partial<Personal>) => void;
```

src/components/personalize/steps/StepDone.tsx

```
import { LinearGradient } from 'expo-linear-gradient';
import { View } from 'react-native';

import { T } from '@/components/ui/T';
import { INTERESTS } from '@/data/personal';
import { LANGUAGES, type LangCode } from '@/i18n';
import type { Personal } from '@/store/types';
import { cornerStyles, themes, type CornerStyle, type ThemeKey } from '@/theme';

import type { Pal } from '../palette';
import { styles } from '../styles';
import { SECTIONS, PACE, BUDGET, STEPS } from '../options';
import { MiniPreview, Summary, Confetti } from '../parts';

interface Props {
  pal: Pal;
  p: Personal;
  theme: ThemeKey;
  corners: CornerStyle;
  lang: LangCode;
  currency: string;
  shownName: string;
  /** Opens the step with this number so an answer can be changed. */
  onEdit: (step: number) => void;
}

export function StepDone({ pal, p, theme, corners, lang, currency, shownName, onEdit }: Props) {
  const edit = (step: (typeof STEPS)[number]) => () => onEdit(STEPS.indexOf(step));
  return (
    <View style={{ alignItems: 'center' }}>
      <Confetti pal={pal} />
      <LinearGradient colors={[pal.heroA, pal.heroB, pal.heroC]} start={{ x: 0, y: 0 }} end={{ x: 1, y: 1 }} style={styles.heroCircle}>
        <T style={{ fontSize: 54, lineHeight: 64 }}>{p.emoji}</T>
      </LinearGradient>
      <T variant="display" center color={pal.text} style={{ marginTop: 14 }}>
        Looking good, {shownName}!
      </T>
      <T variant="body" center color={pal.textSecondary} style={{ marginTop: 6 }}>
        Here’s your TogetherWeGo:
      </T>
      <T variant="caption" center color={pal.textMuted} style={{ marginTop: 2 }}>
        Tap any row to change it
      </T>
      <View
        style={[
          styles.card,
          {
            backgroundColor: pal.card,
            borderColor: pal.border,
            alignSelf: 'stretch',
            marginTop: 16,
          },
        ]}>
        <Summary pal={pal} icon="color-palette-outline" label="Theme" value={`${themes[theme].emoji} ${themes[theme].name}`} onPress={ed
it('Vibe')} />
        <Summary
          pal={pal}
          icon="heart-outline"
          label="Interests"
          value={
            p.interests.length
              ? INTERESTS.filter((i) => p.interests.includes(i.key))
                  .map((i) => i.emoji)
                  .join(' ')
              : 'Everything'
          }
          onPress={edit('Interests')}
        />
        <Summary pal={pal} icon="walk-outline" label="Style" value={`${PACE.find((x) => x.key === p.travelStyle)?.label} pace · ${BUDGE
```

```
T.find((x) => x.key === p.budgetStyle)?.label}`} onPress={edit('Style')} />
        <Summary pal={pal} icon="home-outline" label="Home sections" value={`${Object.values(p.sections).filter(Boolean).length} of ${SE
CTIONS.length} on`} onPress={edit('Home')} />
        <Summary pal={pal} icon="shapes-outline" label="Card shape" value={`${cornerStyles[corners].name} corners`} onPress={edit('Hom
e')} />
        <Summary pal={pal} icon="globe-outline" label="Language & currency" value={`${LANGUAGES.find((l) => l.code === lang)?.native} ·
${currency}`} onPress={edit('You')} />
      </View>
      <MiniPreview pal={pal} corners={corners} name={shownName} emoji={p.emoji} />
    </View>
  );
}
```

src/components/personalize/steps/StepHello.tsx

```
import { LinearGradient } from 'expo-linear-gradient';
import { View } from 'react-native';

import { Float } from '@/components/ui/motion';
import { T } from '@/components/ui/T';
import type { Personal } from '@/store/types';

import type { Pal } from '../palette';
import { styles } from '../styles';

const bubblePos = [
  { left: 0, top: 30 },
  { right: 4, top: 10 },
  { left: 22, bottom: 0 },
  { right: 0, bottom: 20 },
  { left: 92, top: -6 },
  { right: 86, bottom: -12 },
] as const;

export function StepHello({ pal, p, firstName }: { pal: Pal; p: Personal; firstName: string }) {
  return (
    <View style={{ alignItems: 'center', paddingTop: 10 }}>
      <View style={styles.bubbles}>
        {['🌍', '✈️', '🎒', '🏝️', '🍜', '📸'].map((e, i) => (
          <Float key={e} amount={10} duration={1800 + i * 260} delay={i * 120} style={[styles.bubble, bubblePos[i], { backgroundColor: p
al.card, borderColor: pal.border }]}>
            <T style={{ fontSize: 26, lineHeight: 32 }}>{e}</T>
          </Float>
        ))}
        <LinearGradient colors={[pal.heroA, pal.heroB, pal.heroC]} start={{ x: 0, y: 0 }} end={{ x: 1, y: 1 }} style={styles.heroCircle}
>
          <T style={{ fontSize: 54, lineHeight: 64 }}>{p.emoji}</T>
        </LinearGradient>
      </View>
      <T variant="kicker" color={pal.primary} style={{ marginTop: 18 }}>
        60-second survey
      </T>
      <T variant="display" center color={pal.text} style={{ marginTop: 6 }}>
        Let’s make it{'\n'}yours, {firstName}
      </T>
      <T variant="body" center color={pal.textSecondary} style={{ marginTop: 10, maxWidth: 320 }}>
        Pick a colour vibe, your travel interests and what shows up on your home screen. You can change any of it later in Settings.
      </T>
      <View
        style={{
          flexDirection: 'row',
          flexWrap: 'wrap',
          gap: 8,
          justifyContent: 'center',
          marginTop: 18,
        }}>
        {['🎨 6 colour themes', '❤️ Your interests', '🏠 Your home screen', '😎 Your avatar'].map((x) => (
          <View
            key={x}
            style={[
              styles.tag,
              {
                backgroundColor: pal.primarySofter,
                borderColor: pal.primaryLine,
              },
            ]}>
            <T variant="small" weight="medium" color={pal.primary}>
              {x}
            </T>
          </View>
        ))}
      </View>
    </View>
  );
}
```

src/components/personalize/steps/StepHome.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { Switch, View } from 'react-native';

import { Press } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import type { Personal } from '@/store/types';
import { cornerStyles, type CornerStyle } from '@/theme';

import type { Pal } from '../palette';
import { styles } from '../styles';
import { SECTIONS } from '../options';
import { StepTitle } from '../parts';
import type { SetPersonal } from '../parts';

export function StepHome({ pal, p, set, corners, setCorners }: { pal: Pal; p: Personal; set: SetPersonal; corners: CornerStyle; setCorne
rs: (c: CornerStyle) => void }) {
  return (
    <View>
      <StepTitle pal={pal} kicker="Your home screen" title="Build your Explore page" sub="Switch sections on or off, pick a card shape a
nd decide how lively it should feel." />
      <View style={[styles.card, { backgroundColor: pal.card, borderColor: pal.border }]}>
        {SECTIONS.map((s, i) => (
          <View
            key={s.key}
            style={[
              styles.secRow,
              i > 0 && {
                borderTopWidth: 1,
                borderTopColor: pal.border,
              },
            ]}>
            <View style={[styles.secIcon, { backgroundColor: pal.primarySofter }]}>
              <Ionicons name={s.icon} size={17} color={pal.primary} />
            </View>
            <View style={{ flex: 1 }}>
              <T variant="small" weight="semibold" color={pal.text}>
                {s.label}
              </T>
              <T variant="caption" color={pal.textSecondary}>
                {s.sub}
              </T>
            </View>
            <Switch
              value={p.sections[s.key]}
              onValueChange={(v) => set({ sections: { ...p.sections, [s.key]: v } })}
              trackColor={{
                true: pal.primary,
                false: pal.borderStrong,
              }}
              thumbColor="#FFFFFF"
              accessibilityLabel={`Show ${s.label}`}
            />
          </View>
        ))}
      </View>

      <T variant="title" color={pal.text} style={{ marginTop: 18, marginBottom: 8 }}>
        Card shape
      </T>
      <View style={{ flexDirection: 'row', gap: 8 }}>
        {(Object.keys(cornerStyles) as CornerStyle[]).map((c) => {
          const on = corners === c;
          return (
            <Press
              key={c}
              onPress={() => setCorners(c)}
              accessibilityRole="radio"
              accessibilityState={{ checked: on }}
              accessibilityLabel={`${cornerStyles[c].name} corners`}
              style={[
                styles.cornerOpt,
                {
                  backgroundColor: on ? pal.primarySoft : pal.card,
                  borderColor: on ? pal.primary : pal.border,
                },
              ]}>
              <View
                style={[
                  styles.cornerDemo,
                  {
                    borderRadius: Math.round(14 * cornerStyles[c].scale),
                    backgroundColor: pal.primary,
                  },
                ]}
```

```
              />
              <T variant="small" weight="semibold" color={on ? pal.primary : pal.text}>
                {cornerStyles[c].name}
              </T>
            </Press>
          );
        })}
      </View>

      <View
        style={[
          styles.card,
          styles.secRow,
          {
            backgroundColor: pal.card,
            borderColor: pal.border,
            marginTop: 14,
          },
        ]}>
        <View style={[styles.secIcon, { backgroundColor: pal.accentSoft }]}>
          <Ionicons name="sparkles" size={17} color={pal.accent} />
        </View>
        <View style={{ flex: 1 }}>
          <T variant="small" weight="semibold" color={pal.text}>
            Animations
          </T>
          <T variant="caption" color={pal.textSecondary}>
            Flying plane, floating cards, smooth entrances
          </T>
        </View>
        <Switch value={p.motion} onValueChange={(v) => set({ motion: v })} trackColor={{ true: pal.primary, false: pal.borderStrong }} t
humbColor="#FFFFFF" accessibilityLabel="Animations" />
      </View>
    </View>
  );
}
```

src/components/personalize/steps/StepInterests.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { Image } from 'expo-image';
import { LinearGradient } from 'expo-linear-gradient';
import { StyleSheet, View } from 'react-native';

import { FadeIn } from '@/components/ui/motion';
import { Press } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import { photos } from '@/data/images';
import { INTERESTS } from '@/data/personal';
import type { Personal } from '@/store/types';

import type { Pal } from '../palette';
import { styles } from '../styles';
import { StepTitle } from '../parts';
import type { SetPersonal } from '../parts';

export function StepInterests({ pal, p, set }: { pal: Pal; p: Personal; set: SetPersonal }) {
  const toggleInterest = (k: string) => set({ interests: p.interests.includes(k) ? p.interests.filter((x) => x !== k) : [...p.interests,
k] });
  return (
    <View>
      <StepTitle
        pal={pal}
        kicker="Your interests"
        title="What gets you excited to travel?"
        sub={`Pick as many as you like — “Picked for you” on Explore uses these. ${p.interests.length ? `${p.interests.length} selected`
: ''}`}
      />
      <View style={styles.grid}>
        {INTERESTS.map((it, i) => {
          const on = p.interests.includes(it.key);
          return (
            <FadeIn key={it.key} delay={i * 50} from="scale" style={{ width: '48%' }}>
              <Press
                onPress={() => toggleInterest(it.key)}
                accessibilityRole="checkbox"
                accessibilityState={{ checked: on }}
                accessibilityLabel={it.label}
                style={[styles.interest, { borderColor: on ? pal.primary : 'transparent' }]}>
                <Image source={photos[it.image]} style={StyleSheet.absoluteFill} contentFit="cover" />
                <LinearGradient colors={on ? [`${pal.primary}55`, `${pal.primary}EE`] : ['rgba(0,0,0,0)', 'rgba(0,0,0,0.72)']} style={St
yleSheet.absoluteFill} />
                <View
                  style={[
                    styles.check,
                    {
```

```
                    backgroundColor: on ? pal.accent : 'rgba(255,255,255,0.35)',
                  },
                ]}>
                {on ? <Ionicons name="checkmark" size={14} color="#1A1C17" /> : null}
              </View>
              <T style={{ fontSize: 24 }}>{it.emoji}</T>
              <T variant="small" weight="bold" color="#FFFFFF">
                {it.label}
              </T>
            </Press>
          </FadeIn>
        );
      })}
    </View>
  </View>
);
}
```

src/components/personalize/steps/StepStyle.tsx

```
import { View } from 'react-native';

import { T } from '@/components/ui/T';
import type { Personal } from '@/store/types';

import type { Pal } from '../palette';
import { PACE, BUDGET } from '../options';
import { OptionRow, StepTitle } from '../parts';
import type { SetPersonal } from '../parts';

export function StepStyle({ pal, p, set }: { pal: Pal; p: Personal; set: SetPersonal }) {
  return (
    <View>
      <StepTitle pal={pal} kicker="Travel style" title="How do you like to travel?" sub="Helps the Trip Assistant and recommendations ma
tch your pace and wallet." />
      <T variant="title" color={pal.text} style={{ marginBottom: 8 }}>
        Pace
      </T>
      <View style={{ gap: 8 }}>
        {PACE.map((o) => (
          <OptionRow key={o.key} pal={pal} on={p.travelStyle === o.key} emoji={o.emoji} label={o.label} sub={o.sub} onPress={() => set({
travelStyle: o.key })} />
        ))}
      </View>
      <T variant="title" color={pal.text} style={{ marginTop: 18, marginBottom: 8 }}>
        Budget
      </T>
      <View style={{ gap: 8 }}>
        {BUDGET.map((o) => (
          <OptionRow key={o.key} pal={pal} on={p.budgetStyle === o.key} emoji={o.emoji} label={o.label} sub={o.sub} onPress={() => set({
budgetStyle: o.key })} />
        ))}
      </View>
    </View>
  );
}
```

src/components/personalize/steps/StepVibe.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { LinearGradient } from 'expo-linear-gradient';
import { View } from 'react-native';

import { FadeIn } from '@/components/ui/motion';
import { Press } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import type { Personal } from '@/store/types';
import { themes, type CornerStyle, type ThemeKey } from '@/theme';

import type { Pal } from '../palette';
import { paletteOf } from '../palette';
import { styles } from '../styles';
import { MiniPreview, StepTitle } from '../parts';

export function StepVibe({ pal, p, theme, setTheme, corners, shownName }: { pal: Pal; p: Personal; theme: ThemeKey; setTheme: (k: ThemeK
ey) => void; corners: CornerStyle; shownName: string }) {
  return (
    <View>
      <StepTitle pal={pal} kicker="Pick your vibe" title="Which colours feel like you?" sub="The whole app re-paints itself — buttons, c
ards, the animated home banner, everything." />
      <View style={styles.grid}>
        {(Object.keys(themes) as ThemeKey[]).map((k, i) => {
          const tp = paletteOf(k);
          const on = theme === k;
          return (
            <FadeIn key={k} delay={i * 60} from="scale" style={{ width: '48%' }}>
```

```
              <Press
                onPress={() => setTheme(k)}
                accessibilityRole="radio"
                accessibilityState={{ checked: on }}
                accessibilityLabel={`${themes[k].name} theme`}
                style={[
                  styles.themeCard,
                  {
                    backgroundColor: tp.bg,
                    borderColor: on ? pal.primary : pal.border,
                    borderWidth: on ? 2.5 : 1,
                  },
                ]}>
                <LinearGradient colors={[tp.heroA, tp.heroB, tp.heroC]} start={{ x: 0, y: 0 }} end={{ x: 1, y: 1 }} style={styles.themeH
ero}>
                  <T style={{ fontSize: 22 }}>{themes[k].emoji}</T>
                  <View style={[styles.themeSun, { backgroundColor: tp.accent }]} />
                </LinearGradient>
                <View style={{ padding: 10, gap: 6 }}>
                  <View
                    style={{
                      flexDirection: 'row',
                      alignItems: 'center',
                      justifyContent: 'space-between',
                    }}>
                    <T variant="title" weight="bold" color={tp.text}>
                      {themes[k].name}
                    </T>
                    {on ? <Ionicons name="checkmark-circle" size={20} color={pal.primary} /> : null}
                  </View>
                  <T variant="caption" color={tp.textSecondary} numberOfLines={1}>
                    {themes[k].vibe}
                  </T>
                  <View style={{ flexDirection: 'row', gap: 5 }}>
                    {[tp.primary, tp.primarySoft, tp.accent, tp.card].map((c, j) => (
                      <View
                        key={j}
                        style={[
                          styles.swatch,
                          {
                            backgroundColor: c,
                            borderColor: tp.border,
                          },
                        ]}
                      />
                    ))}
                  </View>
                </View>
              </Press>
            </FadeIn>
          );
        })}
      </View>
      <MiniPreview pal={pal} corners={corners} name={shownName} emoji={p.emoji} />
    </View>
  );
}
```

src/components/personalize/steps/StepYou.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { TextInput, View } from 'react-native';

import { Press } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import { currencies } from '@/data/reference';
import { LANGUAGES, type LangCode } from '@/i18n';
import type { Personal } from '@/store/types';

import type { Pal } from '../palette';
import { styles } from '../styles';
import { EMOJIS } from '../options';
import { SmallChip, StepTitle } from '../parts';
import type { SetPersonal } from '../parts';

export function StepYou({ pal, p, set, firstName, lang, setLang, currency, setCurrency }: { pal: Pal; p: Personal; set: SetPersonal; fir
stName: string; lang: LangCode; setLang: (l: LangCode) => void; currency: string; setCurrency: (c: string) => void }) {
  return (
    <View>
      <StepTitle pal={pal} kicker="About you" title="How should we greet you?" sub="Choose a nickname and a travel avatar. They show up
on your home screen." />
      <View style={[styles.input, { backgroundColor: pal.card, borderColor: pal.border }]}>
        <Ionicons name="happy-outline" size={18} color={pal.textMuted} />
        <TextInput
          value={p.nickname}
          onChangeText={(v) => set({ nickname: v })}
```

```
          placeholder={`Nickname (default: ${firstName})`}
          placeholderTextColor={pal.textMuted}
          maxLength={24}
          style={[styles.inputText, { color: pal.text }]}
          accessibilityLabel="Nickname"
        />
      </View>
      <T variant="title" color={pal.text} style={{ marginTop: 18, marginBottom: 8 }}>
        Travel avatar
      </T>
      <View style={{ flexDirection: 'row', flexWrap: 'wrap', gap: 8 }}>
        {EMOJIS.map((e) => {
          const on = p.emoji === e;
          return (
            <Press
              key={e}
              onPress={() => set({ emoji: e })}
              accessibilityRole="radio"
              accessibilityState={{ checked: on }}
              accessibilityLabel={`Avatar ${e}`}
              scaleTo={0.85}
              style={[
                styles.emoji,
                {
                  backgroundColor: on ? pal.primarySoft : pal.card,
                  borderColor: on ? pal.primary : pal.border,
                },
              ]}>
              <T style={{ fontSize: 24, lineHeight: 30 }}>{e}</T>
            </Press>
          );
        })}
      </View>
      <T variant="title" color={pal.text} style={{ marginTop: 18, marginBottom: 8 }}>
        Language
      </T>
      <View style={{ flexDirection: 'row', flexWrap: 'wrap', gap: 8 }}>
        {LANGUAGES.map((l) => (
          <SmallChip key={l.code} pal={pal} on={lang === l.code} label={`${l.flag} ${l.native}`} onPress={() => setLang(l.code)} />
        ))}
      </View>
      <T variant="title" color={pal.text} style={{ marginTop: 18, marginBottom: 8 }}>
        Home currency
      </T>
      <View style={{ flexDirection: 'row', flexWrap: 'wrap', gap: 8 }}>
        {currencies.slice(0, 10).map((c) => (
          <SmallChip key={c.code} pal={pal} on={currency === c.code} label={`${c.symbol} ${c.code}`} onPress={() => setCurrency(c.code)}
/>
        ))}
      </View>
    </View>
  );
}
```

src/components/personalize/styles.ts

```
import { StyleSheet } from 'react-native';

import { fonts, radius, shadow } from '@/theme';

/** Styles shared by every survey step. */
export const styles = StyleSheet.create({
  bubbles: {
    width: 260,
    height: 190,
    alignItems: 'center',
    justifyContent: 'center',
  },
  bubble: {
    position: 'absolute',
    width: 54,
    height: 54,
    borderRadius: 27,
    borderWidth: 1,
    alignItems: 'center',
    justifyContent: 'center',
    ...shadow,
  },
  heroCircle: {
    width: 116,
    height: 116,
    borderRadius: 58,
    alignItems: 'center',
    justifyContent: 'center',
    ...shadow,
  },
```

```
tag: {
  paddingHorizontal: 12,
  paddingVertical: 7,
  borderRadius: 999,
  borderWidth: 1,
},
grid: {
  flexDirection: 'row',
  flexWrap: 'wrap',
  justifyContent: 'space-between',
  rowGap: 12,
},
themeCard: { borderRadius: radius.lg, overflow: 'hidden' },
themeHero: { height: 64, padding: 10, justifyContent: 'flex-end' },
themeSun: {
  position: 'absolute',
  right: 12,
  top: 10,
  width: 22,
  height: 22,
  borderRadius: 11,
},
swatch: { width: 18, height: 18, borderRadius: 9, borderWidth: 1 },
interest: {
  height: 128,
  borderRadius: radius.lg,
  overflow: 'hidden',
  padding: 12,
  justifyContent: 'flex-end',
  borderWidth: 3,
},
check: {
  position: 'absolute',
  top: 10,
  right: 10,
  width: 24,
  height: 24,
  borderRadius: 12,
  alignItems: 'center',
  justifyContent: 'center',
},
option: {
  flexDirection: 'row',
  alignItems: 'center',
  gap: 12,
  padding: 14,
  borderRadius: radius.lg,
  borderWidth: 1.5,
},
card: {
  borderRadius: radius.lg,
  borderWidth: 1,
  paddingHorizontal: 14,
  paddingVertical: 4,
},
secRow: {
  flexDirection: 'row',
  alignItems: 'center',
  gap: 10,
  paddingVertical: 10,
},
secIcon: {
  width: 34,
  height: 34,
  borderRadius: 11,
  alignItems: 'center',
  justifyContent: 'center',
},
cornerOpt: {
  flex: 1,
  alignItems: 'center',
  gap: 8,
  paddingVertical: 14,
  borderRadius: radius.md,
  borderWidth: 1.5,
},
cornerDemo: { width: 44, height: 30 },
input: {
  flexDirection: 'row',
  alignItems: 'center',
  gap: 10,
  borderWidth: 1,
  borderRadius: radius.md,
  paddingHorizontal: 14,
  height: 52,
},
```

```
  inputText: {
    flex: 1,
    fontFamily: fonts.regular,
    fontSize: 15,
    height: '100%',
    ...({ outlineStyle: 'none' } as object),
  },
  emoji: {
    width: 52,
    height: 52,
    borderRadius: 16,
    borderWidth: 1.5,
    alignItems: 'center',
    justifyContent: 'center',
  },
  smallChip: {
    paddingHorizontal: 13,
    paddingVertical: 8,
    borderRadius: 999,
    borderWidth: 1,
  },
  preview: {
    alignSelf: 'stretch',
    marginTop: 18,
    borderWidth: 1,
    borderRadius: 22,
    padding: 12,
  },
  previewSun: {
    position: 'absolute',
    right: 14,
    top: 12,
    width: 26,
    height: 26,
    borderRadius: 13,
  },
  confetti: {
    position: 'absolute',
    top: 58,
    left: 0,
    right: 0,
    alignItems: 'center',
    zIndex: 2,
  },
  piece: { position: 'absolute', width: 9, height: 14 },
});
```

src/components/photos/PhotoTile.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { Image } from 'expo-image';
import { ActivityIndicator, StyleSheet, View } from 'react-native';

import { Press } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import { imageSource } from '@/data/images';
import type { Photo } from '@/store/types';
import { useSharedFile } from '@/sync/useFile';
import { colors } from '@/theme';

/** One square in the photo grid: the picture, a spinner while a friend's copy downloads, or a caption card. */
export function PhotoTile({ photo, size, who, onPress }: { photo: Photo; size: number; who: string; onPress: () => void }) {
  const { uri, loading } = useSharedFile(photo.uri, photo.blob);
  return (
    <Press onPress={onPress} style={{ width: size, height: size }} scaleTo={0.96} accessibilityLabel={photo.caption || 'Photo'}>
      {uri ? (
        <Image source={imageSource(uri)} style={styles.img} contentFit="cover" transition={150} />
      ) : (
        <View style={[styles.img, styles.missing]}>
          {loading ? <ActivityIndicator color={colors.textMuted} /> : <Ionicons name="image-outline" size={26} color={colors.textMuted}
/>}
          <T variant="micro" color={colors.textMuted} center numberOfLines={2}>
            {photo.caption || `Photo from ${who}`}
          </T>
        </View>
      )}
      {photo.likes.length ? (
        <View style={styles.likes}>
          <Ionicons name="heart" size={10} color={colors.white} />
          <T variant="micro" color={colors.white}>
            {photo.likes.length}
          </T>
        </View>
      ) : null}
    </Press>
  );
```

```
}

const styles = StyleSheet.create({
  img: { width: '100%', height: '100%', borderRadius: 8 },
  missing: { backgroundColor: colors.bgAlt, alignItems: 'center', justifyContent: 'center', gap: 4, padding: 8 },
  likes: { position: 'absolute', bottom: 5, left: 5, flexDirection: 'row', alignItems: 'center', gap: 2, backgroundColor: 'rgba(0,0,0,0.
45)', paddingHorizontal: 5, paddingVertical: 2, borderRadius: 999 },
});
```

src/components/photos/PhotoViewer.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { Image } from 'expo-image';
import { ActivityIndicator, StyleSheet, View } from 'react-native';
import { useSafeAreaInsets } from 'react-native-safe-area-context';

import { Button } from '@/components/ui/Button';
import { Press } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import { Avatar } from '@/components/ui/bits';
import { confirmAction } from '@/components/ui/feedback';
import { imageSource } from '@/data/images';
import { deleteLocalFile } from '@/services/files';
import { sharePhoto } from '@/services/share';
import type { Photo, Trip } from '@/store/types';
import { useSharedFile } from '@/sync/useFile';
import { colors } from '@/theme';
import { timeAgo } from '@/utils/format';
import { dayLabel } from '@/utils/time';

/** The full-screen view of one photo: the picture, who posted it, likes, share and (for your own photos) delete. */
export function PhotoViewer({ photo, trip, userId, onClose, onShare, onLike, onDelete }: { photo: Photo; trip: Trip; userId: string; onC
lose: () => void; onShare: () => void; onLike: () => void; onDelete: () => void }) {
  const insets = useSafeAreaInsets();
  const { uri, loading } = useSharedFile(photo.uri, photo.blob);
  const author = trip.members.find((x) => x.id === photo.by);
  const liked = photo.likes.includes(userId);
  return (
    <View style={[styles.viewer, { paddingTop: insets.top + 8, paddingBottom: insets.bottom + 16 }]}>
      <View style={styles.viewerTop}>
        <Press onPress={onClose} style={styles.round} accessibilityLabel="Close photo">
          <Ionicons name="close" size={22} color={colors.white} />
        </Press>
        <View style={{ flexDirection: 'row', gap: 10 }}>
          <Press onPress={onShare} style={styles.round} accessibilityLabel="Share photo to social media">
            <Ionicons name="share-social-outline" size={20} color={colors.white} />
          </Press>
          {photo.by === userId ? (
            <Press
              onPress={() =>
                confirmAction('Delete photo?', 'It will be removed from the photo dump for everyone.', () => {
                  deleteLocalFile(photo.uri);
                  onDelete();
                }, { destructive: true, confirmLabel: 'Delete' })
              }
              style={styles.round}
              accessibilityLabel="Delete photo">
              <Ionicons name="trash-outline" size={20} color={colors.white} />
            </Press>
          ) : null}
        </View>
      </View>
      {uri ? (
        <Image source={imageSource(uri)} style={{ flex: 1, width: '100%' }} contentFit="contain" />
      ) : (
        <View style={{ flex: 1, alignItems: 'center', justifyContent: 'center', gap: 8, padding: 30 }}>
          {loading ? <ActivityIndicator color="rgba(255,255,255,0.8)" size="large" /> : <Ionicons name="phone-portrait-outline" size={4
0} color="rgba(255,255,255,0.6)" />}
          <T variant="small" color="rgba(255,255,255,0.8)" center>
            {loading ? `Getting ${author?.name.split(' ')[0] ?? 'your friend'}’s photo…` : `This photo is on ${author?.name.split(' ')
[0] ?? 'a friend'}’s phone. Likes and captions still sync.`}
          </T>
        </View>
      )}
      <View style={styles.viewerBottom}>
        <View style={{ flexDirection: 'row', alignItems: 'center', gap: 10 }}>
          <Avatar name={author?.name ?? '?'} src={author?.avatar} size={32} />
          <View style={{ flex: 1 }}>
            <T variant="small" weight="semibold" color={colors.white}>
              {photo.caption || 'Untitled'}
            </T>
            <T variant="caption" color="rgba(255,255,255,0.7)">
              {author?.name.split(' ')[0]} · {photo.day ? dayLabel(photo.day) : ''} · {timeAgo(photo.ts)}
            </T>
          </View>
          <Press onPress={onLike} style={styles.likeBtn} accessibilityLabel={liked ? 'Unlike' : 'Like'}>
```

```
            <Ionicons name={liked ? 'heart' : 'heart-outline'} size={22} color={liked ? '#FF6B81' : colors.white} />
            <T variant="small" weight="semibold" color={colors.white}>
              {photo.likes.length}
            </T>
          </Press>
        </View>
        {uri ? <Button label="Save & share" variant="light" size="md" icon="download-outline" onPress={() => sharePhoto(uri, photo.capti
on)} /> : null}
      </View>
    </View>
  );
}

const styles = StyleSheet.create({
  viewer: { flex: 1, backgroundColor: '#0B0F0A' },
  viewerTop: { flexDirection: 'row', justifyContent: 'space-between', paddingHorizontal: 16, marginBottom: 8 },
  round: { width: 42, height: 42, borderRadius: 21, backgroundColor: 'rgba(255,255,255,0.14)', alignItems: 'center', justifyContent: 'ce
nter' },
  viewerBottom: { paddingHorizontal: 16, paddingTop: 12, gap: 12 },
  likeBtn: { flexDirection: 'row', alignItems: 'center', gap: 4, paddingHorizontal: 12, paddingVertical: 8, borderRadius: 999, backgroun
dColor: 'rgba(255,255,255,0.14)' },
});
```

src/components/pickers.tsx

```
/** Cross-platform time & date pickers (identical on iOS, Android and web). */
import { Ionicons } from '@expo/vector-icons';
import { useMemo, useState } from 'react';
import { StyleSheet, View } from 'react-native';

import { colors, radius } from '@/theme';
import { addDays, daysBetween, parseYMD, time12, ymdFromUtcDate } from '@/utils/time';

import { Button } from './ui/Button';
import { Chip } from './ui/bits';
import { Press } from './ui/Press';
import { Sheet } from './ui/Sheet';
import { T } from './ui/T';

function Stepper({ label, value, onUp, onDown }: { label: string; value: string; onUp: () => void; onDown: () => void }) {
  return (
    <View style={styles.stepper}>
      <Press onPress={onUp} accessibilityLabel={`Increase ${label}`} style={styles.stepBtn}>
        <Ionicons name="chevron-up" size={22} color={colors.primary} />
      </Press>
      <T variant="display" style={{ fontSize: 40, lineHeight: 48 }} accessibilityLabel={`${label} ${value}`}>
        {value}
      </T>
      <Press onPress={onDown} accessibilityLabel={`Decrease ${label}`} style={styles.stepBtn}>
        <Ionicons name="chevron-down" size={22} color={colors.primary} />
      </Press>
      <T variant="caption" color={colors.textMuted}>
        {label}
      </T>
    </View>
  );
}

export function TimePickerSheet({ visible, value, onClose, onChange }: { visible: boolean; value: string; onClose: () => void; onChange:
(v: string) => void }) {
  const [h, setH] = useState(10);
  const [m, setM] = useState(0);
  // Re-sync with the field each time the sheet opens (derived during render, no effect needed).
  const [openedWith, setOpenedWith] = useState<string | null>(null);
  const openKey = visible ? value : null;
  if (openKey !== openedWith) {
    setOpenedWith(openKey);
    if (visible) {
      const [hh, mm] = value.split(':').map(Number);
      setH(hh || 0);
      setM(mm || 0);
    }
  }
  const hhmm = `${String(h).padStart(2, '0')}:${String(m).padStart(2, '0')}`;
  const pm = h >= 12;
  const presets = ['07:00', '09:00', '12:30', '15:00', '18:00', '19:30', '21:00'];

  return (
    <Sheet
      visible={visible}
      onClose={onClose}
      title="Pick a time"
      subtitle={`Trip local time · ${time12(hhmm)}`}
      footer={
        <Button
          label="Set time"
```

```
          onPress={() => {
            onChange(hhmm);
            onClose();
          }}
        />
      }>
      <View style={styles.timeRow}>
        <Stepper label="hour" value={String(h % 12 === 0 ? 12 : h % 12)} onUp={() => setH((x) => (x + 1) % 24)} onDown={() => setH((x) =
> (x + 23) % 24)} />
        <T variant="display" style={{ fontSize: 40, lineHeight: 48, marginTop: -18 }}>
          :
        </T>
        <Stepper label="minute" value={String(m).padStart(2, '0')} onUp={() => setM((x) => (x + 5) % 60)} onDown={() => setM((x) => (x +
55) % 60)} />
        <View style={{ gap: 8, marginLeft: 6 }}>
          <Chip label="AM" active={!pm} onPress={() => pm && setH((x) => x - 12)} small />
          <Chip label="PM" active={pm} onPress={() => !pm && setH((x) => x + 12)} small />
        </View>
      </View>
      <View style={styles.wrap}>
        {presets.map((p) => (
          <Chip
            key={p}
            small
            label={time12(p)}
            active={p === hhmm}
            onPress={() => {
              const [a, b] = p.split(':').map(Number);
              setH(a);
              setM(b);
            }}
          />
        ))}
      </View>
    </Sheet>
  );
}

const WD = ['S', 'M', 'T', 'W', 'T', 'F', 'S'];
const MONTHS = ['January', 'February', 'March', 'April', 'May', 'June', 'July', 'August', 'September', 'October', 'November', 'Decembe
r'];

export function DatePickerSheet({
  visible,
  value,
  onClose,
  onChange,
  min,
  max,
  title = 'Pick a date',
}: {
  visible: boolean;
  value: string;
  onClose: () => void;
  onChange: (v: string) => void;
  min?: string;
  max?: string;
  title?: string;
}) {
  const [cursor, setCursor] = useState(value);
  const [openedWith, setOpenedWith] = useState<string | null>(null);
  const openKey = visible ? value : null;
  if (openKey !== openedWith) {
    setOpenedWith(openKey);
    if (visible) setCursor(value);
  }
  const { y, m } = parseYMD(cursor);
  const cells = useMemo(() => {
    const first = new Date(Date.UTC(y, m - 1, 1));
    const offset = first.getUTCDay();
    const daysInMonth = new Date(Date.UTC(y, m, 0)).getUTCDate();
    const out: (string | null)[] = Array(offset).fill(null);
    for (let d = 1; d <= daysInMonth; d++) out.push(ymdFromUtcDate(new Date(Date.UTC(y, m - 1, d))));
    while (out.length % 7) out.push(null);
    return out;
  }, [y, m]);

  const shiftMonth = (n: number) => setCursor(ymdFromUtcDate(new Date(Date.UTC(y, m - 1 + n, 1))));
  const disabled = (d: string) => (min && daysBetween(min, d) < 0) || (max && daysBetween(d, max) < 0);

  return (
    <Sheet visible={visible} onClose={onClose} title={title}>
      <View style={styles.monthRow}>
        <Press onPress={() => shiftMonth(-1)} accessibilityLabel="Previous month" style={styles.monthBtn}>
          <Ionicons name="chevron-back" size={18} color={colors.text} />
        </Press>
```

```
        <T variant="title">
          {MONTHS[m - 1]} {y}
        </T>
        <Press onPress={() => shiftMonth(1)} accessibilityLabel="Next month" style={styles.monthBtn}>
          <Ionicons name="chevron-forward" size={18} color={colors.text} />
        </Press>
      </View>
      <View style={styles.grid}>
        {WD.map((w, i) => (
          <View key={`w${i}`} style={styles.cell}>
            <T variant="caption" color={colors.textMuted} weight="semibold">
              {w}
            </T>
          </View>
        ))}
        {cells.map((d, i) =>
          d ? (
            <Press
              key={d}
              disabled={!!disabled(d)}
              onPress={() => {
                onChange(d);
                onClose();
              }}
              accessibilityLabel={d}
              accessibilityState={{ selected: d === value, disabled: !!disabled(d) }}
              style={[styles.cell, styles.day, d === value && styles.daySel, disabled(d) && { opacity: 0.3 }]}>
              <T variant="bodySm" weight={d === value ? 'bold' : 'regular'} color={d === value ? colors.white : colors.text}>
                {parseYMD(d).d}
              </T>
            </Press>
          ) : (
            <View key={`e${i}`} style={styles.cell} />
          ),
        )}
      </View>
      <View style={styles.wrap}>
        <Chip small label="Today" onPress={() => setCursor(ymdFromUtcDate(new Date()))} />
        <Chip small label="+1 week" onPress={() => setCursor(addDays(cursor, 7))} />
        <Chip small label="+1 month" onPress={() => shiftMonth(1)} />
      </View>
    </Sheet>
  );
}

const styles = StyleSheet.create({
  timeRow: { flexDirection: 'row', alignItems: 'center', justifyContent: 'center', gap: 6, paddingVertical: 8 },
  stepper: { alignItems: 'center', width: 76 },
  stepBtn: { width: 44, height: 36, borderRadius: radius.sm, backgroundColor: colors.primarySofter, alignItems: 'center', justifyConten
t: 'center' },
  wrap: { flexDirection: 'row', flexWrap: 'wrap', gap: 8 },
  monthRow: { flexDirection: 'row', alignItems: 'center', justifyContent: 'space-between' },
  monthBtn: { width: 36, height: 36, borderRadius: 18, borderWidth: 1, borderColor: colors.border, alignItems: 'center', justifyContent:
'center' },
  grid: { flexDirection: 'row', flexWrap: 'wrap' },
  cell: { width: `${100 / 7}%`, aspectRatio: 1, alignItems: 'center', justifyContent: 'center' },
  day: { borderRadius: 999 },
  daySel: { backgroundColor: colors.primary },
});
```

src/components/settings/AccessibilityCard.tsx

```
import { StyleSheet, View } from 'react-native';

import { Card, Chip, ToggleRow } from '@/components/ui/bits';
import { T } from '@/components/ui/T';
import { useT } from '@/i18n';
import { useAppStore } from '@/store/useAppStore';
import type { TextScale } from '@/store/types';
import { colors } from '@/theme';

const SCALES: { v: TextScale; l: string }[] = [
  { v: 1, l: 'Default' },
  { v: 1.12, l: 'Large' },
  { v: 1.25, l: 'Extra large' },
];

/** Bigger text and haptics. */
export function AccessibilityCard() {
  const t = useT();
  const settings = useAppStore((s) => s.settings);
  const update = useAppStore((s) => s.updateSettings);

  return (
    <Card style={{ marginTop: 14, gap: 8 }}>
      <T variant="title">Accessibility</T>
```

```
      <T variant="small" weight="semibold">
        {t('settings.textSize')}
      </T>
      <View style={styles.wrap}>
        {SCALES.map((s) => (
          <Chip key={s.v} label={s.l} active={settings.textScale === s.v} onPress={() => update({ textScale: s.v })} />
        ))}
      </View>
      <ToggleRow icon="phone-portrait-outline" label="Haptic feedback" value={settings.haptics} onChange={(v) => update({ haptics: v })}
/>
      <T variant="caption" color={colors.textSecondary}>
        Every button has a screen-reader label, colours meet WCAG AA contrast, and the app follows your phone’s font size too.
      </T>
    </Card>
  );
}

const styles = StyleSheet.create({
  wrap: { flexDirection: 'row', flexWrap: 'wrap', gap: 8 },
});
```

src/components/settings/AssistantCard.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { useEffect, useState } from 'react';
import { StyleSheet, View } from 'react-native';

import { Button } from '@/components/ui/Button';
import { Card, Chip } from '@/components/ui/bits';
import { toast } from '@/components/ui/feedback';
import { Input } from '@/components/ui/Input';
import { Press } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import { CLAUDE_MODEL } from '@/services/assistant';
import { getApiKey, looksLikeApiKey, setApiKey } from '@/services/secrets';
import { useAppStore } from '@/store/useAppStore';
import { colors, radius } from '@/theme';

/** Offline answers or Claude, and the optional API key (kept in the phone's secure keychain). */
export function AssistantCard() {
  const settings = useAppStore((s) => s.settings);
  const update = useAppStore((s) => s.updateSettings);
  const [key, setKey] = useState('');
  const [hasKey, setHasKey] = useState(false);
  const [keyErr, setKeyErr] = useState<string | null>(null);

  useEffect(() => {
    getApiKey().then((k) => setHasKey(!!k));
  }, []);

  const saveKey = async () => {
    if (!looksLikeApiKey(key)) return setKeyErr('Anthropic keys start with “sk-ant-”. Check you copied the whole key.');
    await setApiKey(key.trim());
    setKey('');
    setKeyErr(null);
    setHasKey(true);
    update({ aiMode: 'claude' });
    toast('Key saved securely — Claude mode on');
  };

  return (
    <Card style={{ marginTop: 14, gap: 10 }}>
      <View style={{ flexDirection: 'row', alignItems: 'center', gap: 8 }}>
        <Ionicons name="sparkles" size={18} color={colors.primary} />
        <T variant="title">Trip Assistant</T>
      </View>
      <View style={styles.wrap}>
        <Chip icon="cloud-offline-outline" label="Offline (built-in)" active={settings.aiMode === 'offline'} onPress={() => update({ aiM
ode: 'offline' })} />
        <Chip
          icon="sparkles-outline"
          label="Claude AI"
          active={settings.aiMode === 'claude'}
          onPress={() => {
            if (!hasKey) return toast('Add your Anthropic API key first', { tone: 'warn' });
            update({ aiMode: 'claude' });
          }}
        />
      </View>
      <T variant="caption" color={colors.textSecondary}>
        Offline mode answers from your trip data with no internet. Claude mode sends questions plus trip details to Anthropic’s {CLAUDE_
MODEL} model using your own key, which is stored in the phone’s secure keychain.
      </T>
      {hasKey ? (
        <View style={styles.keyRow}>
          <Ionicons name="key" size={16} color={colors.primary} />
```

```
          <T variant="small" style={{ flex: 1 }}>
            API key saved securely
          </T>
          <Press
            onPress={async () => {
              await setApiKey(null);
              setHasKey(false);
              update({ aiMode: 'offline' });
              toast('Key removed');
            }}
            hitSlop={8}>
            <T variant="small" weight="semibold" color={colors.red}>
              Remove
            </T>
          </Press>
        </View>
      ) : (
        <>
          <Input placeholder="sk-ant-…" value={key} onChangeText={(v) => { setKey(v); setKeyErr(null); }} secureToggle autoCapitalize="n
one" autoCorrect={false} error={keyErr} icon="key-outline" />
          <Button label="Save key" size="md" variant="secondary" onPress={saveKey} disabled={!key} />
        </>
      )}
    </Card>
  );
}

const styles = StyleSheet.create({
  wrap: { flexDirection: 'row', flexWrap: 'wrap', gap: 8 },
  keyRow: { flexDirection: 'row', alignItems: 'center', gap: 8, padding: 12, borderRadius: radius.md, backgroundColor: colors.primarySof
ter },
});
```

src/components/settings/DataCard.tsx

```
import { router } from 'expo-router';

import { Card, ListRow } from '@/components/ui/bits';
import { confirmAction, toast } from '@/components/ui/feedback';
import { shareText } from '@/services/share';
import { resetSync } from '@/sync/actions';
import { useActiveTrip, useAppStore } from '@/store/useAppStore';
import { colors } from '@/theme';

/** Help, about, export, clearing the sample trips and resetting the app. */
export function DataCard() {
  const trip = useActiveTrip();
  const resetEverything = useAppStore((s) => s.resetEverything);
  const clearDemoData = useAppStore((s) => s.clearDemoData);
  const ensureSeed = useAppStore((s) => s.ensureSeed);

  return (
    <Card style={{ marginTop: 14 }}>
      <ListRow icon="help-circle-outline" label="How to use TogetherWeGo" onPress={() => router.push('/help')} />
      <ListRow icon="information-circle-outline" label="About, credits & sources" onPress={() => router.push('/about')} />
      {trip ? (
        <ListRow
          icon="download-outline"
          label="Export this trip"
          sub="Share a JSON backup of the itinerary, budget and checklist"
          onPress={() => {
            const { messages, photos, docs, ...rest } = trip;
            void messages;
            void photos;
            void docs;
            shareText(JSON.stringify(rest, null, 1), `${trip.name} backup`);
          }}
        />
      ) : null}
      <ListRow
        icon="leaf-outline"
        label="Clear the sample trips"
        sub="Start with a blank app. Your own trips and accounts stay"
        onPress={() =>
          confirmAction('Clear the sample trips?', 'Amalfi Squad and the other example trips are removed from this phone. Trips you made
yourself are not touched.', () => {
            clearDemoData();
            toast('Sample trips cleared');
          }, { destructive: true, confirmLabel: 'Clear' })
        }
      />
      <ListRow
        icon="refresh-circle-outline"
        tint={colors.red}
        label="Reset app data"
        sub="Deletes all trips and accounts on this device"
```

```
        onPress={() =>
          confirmAction('Reset everything?', 'All trips, chats and accounts on this device will be erased and the demo data restored.',
async () => {
            await resetSync(); // also forget which trips were shared with friends
            resetEverything();
            await ensureSeed();
            router.replace('/sign-in');
          }, { destructive: true, confirmLabel: 'Reset' })
        }
      />
    </Card>
  );
}
```

src/components/settings/DemoCard.tsx

```
import { StyleSheet, View } from 'react-native';

import { Card, Chip } from '@/components/ui/bits';
import { confirmAction, toast } from '@/components/ui/feedback';
import { T } from '@/components/ui/T';
import { useAppStore } from '@/store/useAppStore';
import { colors } from '@/theme';

/** Re-creates the sample trip as upcoming, in progress or finished. */
export function DemoCard() {
  const reseed = useAppStore((s) => s.reseedDemo);
  const timeline = useAppStore((s) => s.demoTimeline);

  return (
    <Card style={{ marginTop: 14, gap: 10 }}>
      <T variant="title">Demo timeline</T>
      <T variant="caption" color={colors.textSecondary}>
        Re-create the sample Amalfi Squad trip as upcoming, in progress or finished to show every stage from planning through completio
n.
      </T>
      <View style={styles.wrap}>
        {(
          [
            ['upcoming', 'Starts in 3 days'],
            ['active', 'Day 2 in progress'],
            ['completed', 'Just finished'],
          ] as const
        ).map(([k, l]) => (
          <Chip
            key={k}
            small
            label={l}
            active={timeline === k}
            onPress={() =>
              confirmAction('Reset the demo trip?', `“Amalfi Squad” will be rebuilt as “${l}”. Your other trips are untouched.`, () => {
                reseed(k);
                toast('Demo trip updated');
              })
            }
          />
        ))}
      </View>
    </Card>
  );
}

const styles = StyleSheet.create({
  wrap: { flexDirection: 'row', flexWrap: 'wrap', gap: 8 },
});
```

src/components/settings/NotificationsCard.tsx

```
import { StyleSheet, View } from 'react-native';

import { Button } from '@/components/ui/Button';
import { Card, Chip, ToggleRow } from '@/components/ui/bits';
import { toast } from '@/components/ui/feedback';
import { T } from '@/components/ui/T';
import { useT } from '@/i18n';
import { ensurePermission, sendTestReminder } from '@/services/notifications';
import { useAppStore } from '@/store/useAppStore';

/** Reminder switch, how early to remind, and a test button. */
export function NotificationsCard() {
  const t = useT();
  const settings = useAppStore((s) => s.settings);
  const update = useAppStore((s) => s.updateSettings);

  return (
```

```
    <Card style={{ marginTop: 14, gap: 6 }}>
      <ToggleRow
        icon="notifications-outline"
        label={t('settings.notifications')}
        sub="Live alerts before activities with the 🔔 on"
        value={settings.notificationsEnabled}
        onChange={async (v) => {
          update({ notificationsEnabled: v });
          if (v) {
            const ok = await ensurePermission();
            toast(ok ? 'Reminders on' : 'Allow notifications in your phone settings', { tone: ok ? 'ok' : 'warn' });
          }
        }}
      />
      <T variant="small" weight="semibold">
        {t('settings.lead')}
      </T>
      <View style={styles.wrap}>
        {[10, 30, 60, 120].map((m) => (
          <Chip key={m} small label={m < 60 ? `${m} min` : `${m / 60} hr`} active={settings.reminderLeadMin === m} onPress={() => update
({ reminderLeadMin: m })} />
        ))}
      </View>
      <Button label="Send a test reminder" variant="soft" size="md" icon="alarm-outline" onPress={() => { sendTestReminder(5); toast('Ar
riving in 5 seconds…', { icon: 'alarm-outline' }); }} style={{ marginTop: 6 }} />
    </Card>
  );
}

const styles = StyleSheet.create({
  wrap: { flexDirection: 'row', flexWrap: 'wrap', gap: 8 },
});
```

src/components/settings/ProfileCard.tsx

```
import { useState } from 'react';
import { ScrollView, StyleSheet, View } from 'react-native';

import { Button } from '@/components/ui/Button';
import { Avatar, Card, Chip } from '@/components/ui/bits';
import { toast } from '@/components/ui/feedback';
import { Input } from '@/components/ui/Input';
import { Press, tap } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import { pickImage } from '@/services/files';
import { useAppStore, useCurrentUser } from '@/store/useAppStore';
import { colors } from '@/theme';
import { maxLen, validateName } from '@/utils/validation';

const AVATARS = ['avatar-maya', 'avatar-diego', 'avatar-priya', 'avatar-sam', 'avatar-ana', 'avatar-noah'];

/** Name, home city, avatar (a built-in one, a photo, or just initials). */
export function ProfileCard() {
  const user = useCurrentUser();
  const updateProfile = useAppStore((s) => s.updateProfile);
  const [name, setName] = useState(user?.name ?? '');
  const [city, setCity] = useState(user?.homeCity ?? '');
  const [errors, setErrors] = useState<Record<string, string | null>>({});

  const saveProfile = () => {
    const e = { name: validateName(name), city: maxLen(40, 'City')(city) };
    setErrors(e);
    if (e.name || e.city) return tap('warning');
    updateProfile({ name: name.trim(), homeCity: city.trim() || undefined });
    toast('Profile saved');
  };

  const pickAvatarPhoto = async () => {
    try {
      const f = await pickImage();
      if (f) {
        updateProfile({ avatar: f.uri });
        toast('Profile photo updated');
      }
    } catch {
      toast('Couldn’t open your photos', { tone: 'warn' });
    }
  };

  return (
    <Card style={{ gap: 12 }}>
      <T variant="title">Profile</T>
      <View style={{ flexDirection: 'row', alignItems: 'center', gap: 12 }}>
        <Avatar name={user?.name ?? 'You'} src={user?.avatar} size={60} />
        <View style={{ flex: 1, gap: 6 }}>
          <ScrollView horizontal showsHorizontalScrollIndicator={false} contentContainerStyle={{ gap: 6 }}>
```

```
            {AVATARS.map((a) => (
              <Press key={a} onPress={() => updateProfile({ avatar: a })} accessibilityLabel="Choose this avatar" style={[styles.avatarO
pt, user?.avatar === a && { borderColor: colors.primary }]}>
                <Avatar name="Avatar" src={a} size={36} />
              </Press>
            ))}
          </ScrollView>
          <View style={{ flexDirection: 'row', gap: 8 }}>
            <Chip small icon="image-outline" label="Upload photo" onPress={pickAvatarPhoto} />
            <Chip small icon="text-outline" label="Initials" onPress={() => updateProfile({ avatar: undefined })} />
          </View>
        </View>
      </View>
      <Input label="Name" value={name} onChangeText={setName} error={errors.name} icon="person-outline" />
      <Input label="Home city" value={city} onChangeText={setCity} error={errors.city} icon="home-outline" placeholder="e.g. Middletown,
DE" />
      <T variant="caption" color={colors.textSecondary}>
        Signed in as {user?.email} · {user?.provider === 'google' ? 'Google' : 'email & password'}
      </T>
      <Button label="Save profile" size="md" onPress={saveProfile} />
    </Card>
  );
}

const styles = StyleSheet.create({
  avatarOpt: { borderRadius: 22, borderWidth: 2, borderColor: 'transparent', padding: 1 },
});
```

src/components/sync/SyncBadge.tsx

```
import { Pill } from '@/components/ui/bits';
import { useSyncStatus } from '@/sync/status';

/** A small pill that says whether a shared trip is up to date: "Live · 2 phones", "Offline · 3 waiting"... */
export function SyncBadge({ tripId }: { tripId: string }) {
  const shared = useSyncStatus((s) => s.shared.includes(tripId));
  const online = useSyncStatus((s) => s.online);
  const pending = useSyncStatus((s) => s.pending);
  const peers = useSyncStatus((s) => s.peers);
  if (!shared) return null;
  if (online === false) return <Pill label={pending ? `Offline · ${pending} waiting` : 'Offline'} tone="yellow" />;
  if (online === null) return <Pill label="Connecting…" tone="gray" />;
  return <Pill label={peers > 1 ? `Live · ${peers} phones` : 'Live'} tone="green" />;
}
```

src/components/sync/SyncSettingsCard.tsx

```
import { useState } from 'react';
import { StyleSheet, TextInput, View } from 'react-native';

import { Button } from '@/components/ui/Button';
import { Card } from '@/components/ui/bits';
import { toast } from '@/components/ui/feedback';
import { T } from '@/components/ui/T';
import { testServer, type ConnectionTest } from '@/sync/actions';
import { SERVER_PORT, guessServerUrl, useSyncPrefs } from '@/sync/prefs';
import { useSyncStatus } from '@/sync/status';
import { normalizeServerUrl } from '@/sync/transport';
import { colors, fonts, radius } from '@/theme';

/** In Settings: where the sync server is, a test button, and how syncing is going. */
export function SyncSettingsCard() {
  const saved = useSyncPrefs((s) => s.serverUrl);
  const setServerUrl = useSyncPrefs((s) => s.setServerUrl);
  const status = useSyncStatus();
  const guess = guessServerUrl();
  const [text, setText] = useState(saved);
  const [result, setResult] = useState<ConnectionTest | null>(null);
  const [testing, setTesting] = useState(false);

  const address = text.trim() || guess || '';

  const save = () => {
    const url = text.trim();
    if (url && !normalizeServerUrl(url)) return toast('That doesn’t look like an address', { tone: 'warn' });
    setServerUrl(url ? normalizeServerUrl(url)! : '');
    toast(url ? 'Sync address saved' : 'Using the automatic address');
  };

  const test = async () => {
    setTesting(true);
    setResult(await testServer(address));
    setTesting(false);
  };

  return (
```

```
    <Card style={{ gap: 10 }}>
      <T variant="title">Sync across phones</T>
      <T variant="small" color={colors.textSecondary}>
        To plan together on different phones: on the computer that runs the app, start the sync server with <T variant="small" weight="b
old">npm run server</T>, then make sure every phone has its address.
      </T>
      <TextInput
        value={text}
        onChangeText={(v) => {
          setText(v);
          setResult(null);
        }}
        onBlur={save}
        placeholder={guess ? guess.replace(/^https?:\/\//, '') : `192.168.1.20:${SERVER_PORT}`}
        placeholderTextColor={colors.textMuted}
        autoCapitalize="none"
        autoCorrect={false}
        keyboardType="url"
        style={styles.input}
        accessibilityLabel="Sync server address"
        testID="sync-address"
      />
      <T variant="caption" color={colors.textMuted}>
        {saved || !guess ? 'The address printed by the server.' : `Found automatically: ${guess}. Leave this blank to use it.`}
      </T>
      <View style={{ flexDirection: 'row', gap: 8 }}>
        <Button label="Test connection" icon="pulse-outline" variant="secondary" size="md" loading={testing} onPress={test} style={{ fle
x: 1 }} testID="sync-test" />
        <Button label="Save" icon="checkmark" size="md" onPress={save} style={{ flex: 1 }} />
      </View>
      {result ? (
        <T variant="small" weight="semibold" color={result.ok ? colors.positive : colors.red} accessibilityLiveRegion="polite" testID="s
ync-result">
          {result.ok ? `Connected ✓ ${result.trips} shared trip${result.trips === 1 ? '' : 's'} on this server` : result.message}
        </T>
      ) : null}
      <T variant="caption" color={colors.textSecondary}>
        {status.shared.length
          ? `${status.shared.length} trip${status.shared.length === 1 ? ' is' : 's are'} syncing · ${status.online === false ? `offline,
${status.pending} change${status.pending === 1 ? '' : 's'} waiting` : status.online ? `online${status.peers > 1 ? `, ${status.peers} pho
nes` : ''}` : 'connecting…'}`
          : 'No trips are shared yet. Share one from More → Crew & roles.'}
      </T>
    </Card>
  );
}

const styles = StyleSheet.create({
  input: {
    height: 48,
    borderRadius: radius.md,
    borderWidth: 1,
    borderColor: colors.border,
    backgroundColor: colors.bgAlt,
    paddingHorizontal: 14,
    fontFamily: fonts.semibold,
    fontSize: 15,
    color: colors.text,
    ...({ outlineStyle: 'none' } as object),
  },
});
```

src/components/sync/SyncTripCard.tsx

```
import { router } from 'expo-router';
import { useState } from 'react';
import { View } from 'react-native';

import { Button } from '@/components/ui/Button';
import { Card } from '@/components/ui/bits';
import { confirmAction, toast } from '@/components/ui/feedback';
import { T } from '@/components/ui/T';
import { shareTrip, stopSharing } from '@/sync/actions';
import { effectiveServerUrl, useSyncPrefs } from '@/sync/prefs';
import { useSyncStatus } from '@/sync/status';
import { useAppStore } from '@/store/useAppStore';
import type { Trip } from '@/store/types';
import { colors } from '@/theme';

import { SyncBadge } from './SyncBadge';

/** On the Crew screen: share this trip with friends' phones, or see that it is already live. */
export function SyncTripCard({ trip }: { trip: Trip }) {
  const shared = useSyncStatus((s) => s.shared.includes(trip.id));
  const error = useSyncStatus((s) => s.error);
  const [busy, setBusy] = useState(false);
```

```
  useSyncPrefs((s) => s.serverUrl); // re-check when the address changes
  const hasServer = !!effectiveServerUrl();

  const start = async () => {
    setBusy(true);
    let r = await shareTrip(trip.id);
    setBusy(false);
    if (r.ok) return toast('Sharing! Friends can now join with the invite code 🎉');
    if (r.reason === 'code-taken') {
      return confirmAction(
        'Invite code already in use',
        `Another shared trip already uses ${trip.inviteCode}. To join that trip, use “Join a trip” instead. To share this one, give it a
new code.`,
        async () => {
          useAppStore.getState().regenerateInvite(trip.id);
          setBusy(true);
          r = await shareTrip(trip.id);
          setBusy(false);
          toast(r.ok ? 'Sharing with a new invite code' : r.message, r.ok ? undefined : { tone: 'warn' });
        },
        { confirmLabel: 'New code and share' },
      );
    }
    toast(r.message, { tone: 'warn' });
  };

  return (
    <Card style={{ gap: 10, marginTop: 14 }}>
      <View style={{ flexDirection: 'row', alignItems: 'center', justifyContent: 'space-between', gap: 8 }}>
        <T variant="title" weight="bold" style={{ flex: 1 }}>
          Sync across phones
        </T>
        <SyncBadge tripId={trip.id} />
      </View>
      {shared ? (
        <>
          <T variant="small" color={colors.textSecondary}>
            Chat, votes, the plan and expenses now update live on every phone in this trip. Changes made offline are sent when you are b
ack online.
          </T>
          {error ? (
            <T variant="caption" color={colors.orange}>
              {error}. Your changes are saved here and will sync when the server is back.
            </T>
          ) : null}
          <Button
            label="Stop syncing on this phone"
            variant="secondary"
            size="md"
            onPress={() =>
              confirmAction('Stop syncing?', 'This trip stays on this phone, but it will no longer update with your friends’ phones.', a
sync () => {
                await stopSharing(trip.id);
                toast('Syncing stopped for this trip');
              })
            }
          />
        </>
      ) : hasServer ? (
        <>
          <T variant="small" color={colors.textSecondary}>
            Share this trip so friends who join with the invite code see the same plan, chat and budget on their own phones.
          </T>
          <Button label="Share this trip across phones" icon="phone-portrait-outline" loading={busy} onPress={start} />
        </>
      ) : (
        <>
          <T variant="small" color={colors.textSecondary}>
            Phones sync through a small server that runs on your computer. Set its address once and this trip can be shared.
          </T>
          <Button label="Set up sync" icon="settings-outline" variant="secondary" size="md" onPress={() => router.push('/settings')} />
        </>
      )}
    </Card>
  );
}
```

src/components/trip/ActivityCard.tsx

```
import { Ionicons } from '@expo/vector-icons';
import * as Linking from 'expo-linking';
import { useState } from 'react';
import { LayoutAnimation, Platform, StyleSheet, View } from 'react-native';

import { AvatarStack, type IconName } from '@/components/ui/bits';
import { confirmAction, toast } from '@/components/ui/feedback';
```

```
import { Press, tap } from '@/components/ui/Press';
import { T } from '@/components/ui/T';
import { destinationById } from '@/data/destinations';
import { useT } from '@/i18n';
import { convert, useRates } from '@/services/currency';
import { ensurePermission } from '@/services/notifications';
import { useAppStore } from '@/store/useAppStore';
import type { Activity, Trip } from '@/store/types';
import { colors, radius, shadow } from '@/theme';
import { lengthTag, money } from '@/utils/format';
import { dayLabel, weekdayShort } from '@/utils/time';

import { typeEmoji } from './meta';

interface Props {
  trip: Trip;
  activity: Activity;
  onEdit: (a: Activity) => void;
  onAssign: (a: Activity) => void;
  /** 'vote' adds the progress bar + “Lock in” button used on the Vote tab. */
  mode?: 'plan' | 'vote';
}

/** One activity: emoji badge, when/where, tags, and a yellow vote pill with the people who said “I’m in”. */
export function ActivityCard({ trip, activity: a, onEdit, onAssign, mode = 'plan' }: Props) {
  const t = useT();
  const rates = useRates();
  const [open, setOpen] = useState(false);
  const userId = useAppStore((s) => s.session?.userId ?? '');
  const vote = useAppStore((s) => s.voteActivity);
  const update = useAppStore((s) => s.updateActivity);
  const remove = useAppStore((s) => s.deleteActivity);
  const currency = destinationById(trip.destinationId)?.currency ?? 'USD';
  const ups = Object.entries(a.votes).filter(([, v]) => v === 1).map(([id]) => id);
  const myVote = a.votes[userId];
  const total = trip.members.length;
  const needed = Math.floor(total / 2) + 1;
  const yesPeople = ups.map((id) => trip.members.find((m) => m.id === id)).filter(Boolean) as Trip['members'];
  const cost = a.costPerPerson > 0 ? `${money(convert(a.costPerPerson, 'USD', currency, rates), currency)} pp` : 'Free';

  const toggle = () => {
    if (Platform.OS !== 'web') LayoutAnimation.configureNext(LayoutAnimation.Presets.easeInEaseOut);
    setOpen((o) => !o);
  };

  const directions = () => {
    const q = a.lat && a.lng ? `${a.lat},${a.lng}` : encodeURIComponent(`${a.title} ${a.location}`);
    Linking.openURL(`https://www.google.com/maps/search/?api=1&query=${q}`).catch(() => toast('Couldn’t open maps', { tone: 'warn' }));
  };

  const toggleReminder = async () => {
    const next = !a.reminder;
    if (next) {
      const ok = await ensurePermission();
      if (!ok && Platform.OS !== 'web') toast('Enable notifications in Settings to get alerts', { tone: 'warn', icon: 'notifications-off
-outline' });
    }
    update(trip.id, a.id, { reminder: next });
    tap();
    toast(next ? `Reminder on — we’ll ping the group before ${a.title}` : 'Reminder turned off', { icon: next ? 'notifications' : 'notif
ications-off-outline' });
  };

  const lockIn = () => {
    if (trip.ownerId !== userId) {
      const boss = trip.members.find((m) => m.id === trip.ownerId)?.name.split(' ')[0] ?? 'the organizer';
      return toast(`Only ${boss} (the organizer) can lock this in`, { tone: 'warn', icon: 'lock-closed-outline' });
    }
    update(trip.id, a.id, { status: 'confirmed', reminder: true });
    tap('success');
    toast(`Locked in: ${a.title} 🔒`, { icon: 'lock-closed' });
  };

  return (
    <View style={[styles.card, a.done && { opacity: 0.6 }]}>
      {/* The delete button sits next to the card button, not inside it: on the web a button inside a button is invalid HTML. */}
      <View style={styles.top}>
        <Press onPress={toggle} scaleTo={0.99} style={styles.topMain} accessibilityLabel={`${a.title}, ${weekdayShort(a.date)} ${a.tim
e}, ${ups.length} of ${total} in`} accessibilityHint="Shows details and actions">
          <View style={styles.badge}>
            <T style={{ fontSize: 22, lineHeight: 28 }}>{typeEmoji[a.type]}</T>
          </View>
          <View style={{ flex: 1, gap: 5 }}>
            <T variant="title" weight="bold" numberOfLines={open ? undefined : 2} style={[{ fontSize: 17, lineHeight: 22 }, a.done && {
textDecorationLine: 'line-through' }]}>
              {a.title}
```

```
            </T>
            <View style={styles.meta}>
              <View style={styles.inline}>
                <Ionicons name="time-outline" size={13} color={colors.textMuted} />
                <T variant="caption" color={colors.textSecondary}>
                  {weekdayShort(a.date)}, {dayLabel(a.date)} · {a.time}
                </T>
              </View>
              <View style={[styles.inline, { flexShrink: 1 }]}>
                <Ionicons name="location-outline" size={13} color={colors.textMuted} />
                <T variant="caption" color={colors.textSecondary} numberOfLines={1} style={{ flexShrink: 1 }}>
                  {a.location}
                </T>
              </View>
            </View>
            <View style={styles.tags}>
              <Tag label={lengthTag(a.durationMin)} />
              <Tag label={cost} />
              {a.status !== 'confirmed' && mode === 'plan' ? <Tag label={t(a.status === 'voting' ? 'itin.voting' : 'itin.pending')} tone
="warn" /> : null}
            </View>
          </View>
        </Press>
        <Press
          onPress={() =>
            confirmAction('Delete activity?', `“${a.title}” will be removed for everyone.`, () => {
              remove(trip.id, a.id);
              toast('Activity deleted');
            }, { confirmLabel: 'Delete', destructive: true })
          }
          hitSlop={10}
          accessibilityLabel={`Delete ${a.title}`}>
          <Ionicons name="trash-outline" size={17} color={colors.textMuted} />
        </Press>
      </View>

      <View style={styles.bottom}>
        <Press
          onPress={() => {
            vote(trip.id, a.id, 1);
            tap();
          }}
          style={[styles.votePill, myVote === 1 ? styles.votePillOn : styles.votePillOff]}
          accessibilityRole="checkbox"
          accessibilityState={{ checked: myVote === 1 }}
          accessibilityLabel={`I’m in. ${ups.length} of ${total} are in`}>
          <Ionicons name={myVote === 1 ? 'thumbs-up' : 'thumbs-up-outline'} size={14} color={myVote === 1 ? colors.onAccent : colors.tex
tSecondary} />
          <T variant="small" weight="bold" color={myVote === 1 ? colors.onAccent : colors.textSecondary}>
            {ups.length}/{total}
          </T>
        </Press>
        {yesPeople.length ? <AvatarStack people={yesPeople} max={4} size={24} /> : null}
        {mode === 'vote' ? (
          <>
            <View style={styles.track} accessibilityRole="progressbar" accessibilityLabel={`${ups.length} of ${needed} votes needed`}>
              <View style={[styles.fill, { width: `${Math.min(1, ups.length / needed) * 100}%` }]} />
            </View>
            <Press onPress={lockIn} style={styles.lock} accessibilityLabel={`Lock in ${a.title}`}>
              <Ionicons name="checkmark" size={14} color={colors.onPrimary} />
              <T variant="caption" weight="bold" color={colors.onPrimary}>
                Lock in
              </T>
            </Press>
          </>
        ) : (
          <>
            <View style={{ flex: 1 }} />
            <Press onPress={toggleReminder} hitSlop={8} accessibilityLabel={a.reminder ? 'Turn reminder off' : 'Turn reminder on'} style
={[styles.bell, a.reminder && styles.bellOn]}>
              <Ionicons name={a.reminder ? 'notifications' : 'notifications-outline'} size={15} color={a.reminder ? colors.onPrimary : c
olors.textSecondary} />
            </Press>
          </>
        )}
      </View>

      {open ? (
        <View style={styles.details}>
          {a.notes ? (
            <View style={styles.notes}>
              <Ionicons name="document-text-outline" size={14} color={colors.textSecondary} />
              <T variant="small" color={colors.textSecondary} style={{ flex: 1 }}>
                {a.notes}
              </T>
            </View>
```

```
          ) : null}
          {a.status === 'voting' ? (
            <T variant="caption" color={colors.textMuted}>
              Confirms automatically when more than half the crew votes “I’m in”, or the organizer can lock it in from the Vote tab.
            </T>
          ) : null}
          <View style={styles.actions}>
            <Action icon={myVote === -1 ? 'thumbs-down' : 'thumbs-down-outline'} label={myVote === -1 ? 'Skipping' : 'Skip'} onPress={()
=> vote(trip.id, a.id, -1)} />
            <Action icon="people-outline" label="Assign" onPress={() => onAssign(a)} />
            <Action icon="navigate-outline" label="Directions" onPress={directions} />
            <Action icon={a.done ? 'refresh-outline' : 'checkmark-done-outline'} label={a.done ? 'Undo' : 'Done'} onPress={() => update
(trip.id, a.id, { done: !a.done })} />
            <Action
              icon="swap-horizontal"
              label="Status"
              onPress={() => {
                const next = a.status === 'confirmed' ? 'voting' : a.status === 'voting' ? 'pending' : 'confirmed';
                update(trip.id, a.id, { status: next });
                toast(`Marked as ${next}`);
              }}
            />
            <Action icon="create-outline" label="Edit" onPress={() => onEdit(a)} />
          </View>
        </View>
      ) : null}
    </View>
  );
}

function Tag({ label, tone }: { label: string; tone?: 'warn' }) {
  return (
    <View style={[styles.tag, tone === 'warn' && { backgroundColor: colors.accentSoft }]}>
      <T variant="micro" weight="semibold" color={tone === 'warn' ? colors.onAccent : colors.textSecondary}>
        {label}
      </T>
    </View>
  );
}

function Action({ icon, label, onPress }: { icon: IconName; label: string; onPress: () => void }) {
  return (
    <Press onPress={onPress} style={styles.action} accessibilityLabel={label}>
      <View style={styles.actionIcon}>
        <Ionicons name={icon} size={16} color={colors.text} />
      </View>
      <T variant="micro" color={colors.textSecondary}>
        {label}
      </T>
    </Press>
  );
}

const styles = StyleSheet.create({
  card: { backgroundColor: colors.card, borderRadius: radius.xl, padding: 16, marginBottom: 12, ...shadow },
  top: { flexDirection: 'row', alignItems: 'flex-start', gap: 12 },
  topMain: { flex: 1, flexDirection: 'row', alignItems: 'flex-start', gap: 12 },
  badge: { width: 46, height: 46, borderRadius: 23, backgroundColor: colors.bgAlt, alignItems: 'center', justifyContent: 'center' },
  meta: { flexDirection: 'row', alignItems: 'center', flexWrap: 'wrap', columnGap: 10, rowGap: 2 },
  inline: { flexDirection: 'row', alignItems: 'center', gap: 4 },
  tags: { flexDirection: 'row', flexWrap: 'wrap', gap: 6, marginTop: 2 },
  tag: { backgroundColor: colors.bgAlt, borderRadius: 999, paddingHorizontal: 9, paddingVertical: 4 },
  bottom: { flexDirection: 'row', alignItems: 'center', gap: 8, marginTop: 14 },
  votePill: { flexDirection: 'row', alignItems: 'center', gap: 6, height: 34, paddingHorizontal: 12, borderRadius: 999 },
  votePillOn: { backgroundColor: colors.accent },
  votePillOff: { backgroundColor: colors.bgAlt },
  track: { flex: 1, height: 6, borderRadius: 3, backgroundColor: colors.bgAlt, overflow: 'hidden' },
  fill: { height: 6, borderRadius: 3, backgroundColor: colors.primary },
  lock: { flexDirection: 'row', alignItems: 'center', gap: 4, height: 34, paddingHorizontal: 14, borderRadius: 999, backgroundColor: col
ors.primary },
  bell: { width: 34, height: 34, borderRadius: 17, alignItems: 'center', justifyContent: 'center', backgroundColor: colors.bgAlt },
  bellOn: { backgroundColor: colors.primary },
  details: { marginTop: 14, paddingTop: 14, borderTopWidth: 1, borderTopColor: colors.border, gap: 12 },
  notes: { flexDirection: 'row', gap: 8, backgroundColor: colors.cardMuted, padding: 10, borderRadius: radius.sm },
  actions: { flexDirection: 'row', justifyContent: 'space-between' },
  action: { alignItems: 'center', gap: 4, minWidth: 44 },
  actionIcon: { width: 36, height: 36, borderRadius: 18, backgroundColor: colors.bgAlt, alignItems: 'center', justifyContent: 'center'
},
});
```

src/components/trip/AddActivitySheet.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { useMemo, useState } from 'react';
import { ScrollView, StyleSheet, View } from 'react-native';
```

```
import { TimePickerSheet } from '@/components/pickers';
import { Button } from '@/components/ui/Button';
import { Chip, ToggleRow } from '@/components/ui/bits';
import { toast } from '@/components/ui/feedback';
import { Input } from '@/components/ui/Input';
import { tap } from '@/components/ui/Press';
import { Sheet } from '@/components/ui/Sheet';
import { T } from '@/components/ui/T';
import { useT } from '@/i18n';
import { ensurePermission } from '@/services/notifications';
import { useAppStore } from '@/store/useAppStore';
import type { Activity, ActivityType, Trip } from '@/store/types';
import { colors } from '@/theme';
import { durationLabel } from '@/utils/format';
import { dayLabel, time12, tripDays, weekdayShort } from '@/utils/time';
import { maxLen, parseAmount, required, validateAmount, firstError } from '@/utils/validation';

import { activityTypes } from './meta';

export interface ActivityDraft {
  title?: string;
  location?: string;
  type?: ActivityType;
  durationMin?: number;
  costPerPerson?: number;
  lat?: number;
  lng?: number;
  placeId?: string;
  notes?: string;
}

const DURATIONS = [30, 60, 90, 120, 180, 300, 480];

interface Props {
  visible: boolean;
  onClose: () => void;
  trip: Trip;
  date: string;
  editing?: Activity | null;
  draft?: ActivityDraft | null;
}

export function AddActivitySheet({ visible, onClose, trip, date, editing, draft }: Props) {
  const t = useT();
  const add = useAppStore((s) => s.addActivity);
  const update = useAppStore((s) => s.updateActivity);
  const userId = useAppStore((s) => s.session?.userId ?? '');
  const [type, setType] = useState<ActivityType>('sightseeing');
  const [day, setDay] = useState(date);
  const [time, setTime] = useState('10:00');
  const [title, setTitle] = useState('');
  const [location, setLocation] = useState('');
  const [duration, setDuration] = useState(90);
  const [cost, setCost] = useState('');
  const [notes, setNotes] = useState('');
  const [vote, setVote] = useState(false);
  const [remind, setRemind] = useState(true);
  const [timeOpen, setTimeOpen] = useState(false);
  const [errors, setErrors] = useState<Record<string, string | null>>({});

  // Reset the form whenever the sheet opens for a new activity / edit / draft.
  const [openedFor, setOpenedFor] = useState<string | object | null>(null);
  const openKey = visible ? (editing?.id ?? draft ?? 'new') : null;
  if (openKey !== openedFor) {
    setOpenedFor(openKey);
    if (visible) resetForm();
  }

  function resetForm() {
    const src = editing ?? draft ?? {};
    setType(src.type ?? 'sightseeing');
    setDay(editing?.date ?? date);
    setTime(editing?.time ?? '10:00');
    setTitle(src.title ?? '');
    setLocation(src.location ?? '');
    setDuration(src.durationMin ?? 90);
    setCost(src.costPerPerson ? String(src.costPerPerson) : '');
    setNotes(src.notes ?? '');
    setVote(editing ? editing.status === 'voting' : false);
    setRemind(editing ? editing.reminder : true);
    setErrors({});
  }

  const days = tripDays(trip.startDate, trip.endDate);

  // Semantic check: warn about overlapping plans on the same day.
```

```
  const clash = useMemo(() => {
    const [h, m] = time.split(':').map(Number);
    const start = h * 60 + m;
    const end = start + duration;
    return trip.activities.find((a) => {
      if (a.date !== day || a.id === editing?.id) return false;
      const [ah, am] = a.time.split(':').map(Number);
      const s = ah * 60 + am;
      return start < s + a.durationMin && s < end;
    });
  }, [trip.activities, day, time, duration, editing?.id]);

  const submit = async () => {
    const e = {
      title: firstError(title, required('an activity name'), maxLen(60, 'Activity name')),
      location: firstError(location, required('a location'), maxLen(60, 'Location')),
      cost: cost.trim() ? validateAmount(cost, { min: 0, max: 10000 }) : null,
      notes: maxLen(280, 'Notes')(notes),
    };
    setErrors(e);
    if (Object.values(e).some(Boolean)) return tap('warning');
    const payload = {
      date: day,
      time,
      title: title.trim(),
      location: location.trim(),
      type,
      durationMin: duration,
      costPerPerson: cost.trim() ? parseAmount(cost) ?? 0 : 0,
      notes: notes.trim() || undefined,
      status: vote ? ('voting' as const) : editing && editing.status !== 'voting' ? editing.status : ('confirmed' as const),
      reminder: remind,
    };
    if (remind) ensurePermission().catch(() => {});
    if (editing) {
      update(trip.id, editing.id, payload);
      toast('Activity updated');
    } else {
      add(trip.id, { ...payload, assigned: [userId], lat: draft?.lat, lng: draft?.lng, placeId: draft?.placeId });
      toast(`Added to ${dayLabel(day)}${vote ? ' — group vote started' : ''}`);
    }
    tap('success');
    onClose();
  };

  return (
    <Sheet
      visible={visible}
      onClose={onClose}
      title={editing ? 'Edit Activity' : t('add.title')}
      subtitle={t('add.subtitle')}
      footer={<Button label={editing ? t('add.save') : t('add.submit')} onPress={submit} testID="add-activity-submit" />}>
      <View style={{ gap: 8 }}>
        <T variant="small" weight="semibold">
          {t('add.type')}
        </T>
        <ScrollView horizontal showsHorizontalScrollIndicator={false} contentContainerStyle={{ gap: 8 }}>
          {activityTypes.map((a) => (
            <Chip key={a.key} label={a.label} icon={a.icon} active={type === a.key} onPress={() => setType(a.key)} />
          ))}
        </ScrollView>
      </View>

      <View style={{ gap: 8 }}>
        <T variant="small" weight="semibold">
          Day
        </T>
        <ScrollView horizontal showsHorizontalScrollIndicator={false} contentContainerStyle={{ gap: 8 }}>
          {days.map((d, i) => (
            <Chip key={d} small label={`Day ${i + 1} · ${weekdayShort(d)} ${dayLabel(d)}`} active={day === d} onPress={() => setDay(d)}
/>
          ))}
        </ScrollView>
      </View>

      <Input
        label={t('add.time')}
        icon="time-outline"
        value={time12(time)}
        onPressField={() => setTimeOpen(true)}
        right={<Ionicons name="pencil" size={15} color={colors.textMuted} />}
      />

      <Input label={t('add.name')} placeholder={t('add.namePh')} value={title} onChangeText={setTitle} error={errors.title} maxLength={7
0} testID="activity-name" />
      <Input label={t('add.location')} placeholder={t('add.locationPh')} icon="location-sharp" value={location} onChangeText={setLocatio
```

```
n} error={errors.location} maxLength={70} testID="activity-location" />

      <View style={{ gap: 8 }}>
        <T variant="small" weight="semibold">
          {t('add.duration')}
        </T>
        <View style={styles.wrap}>
          {DURATIONS.map((d) => (
            <Chip key={d} small label={d === 300 ? 'Half day' : d === 480 ? 'Full day' : durationLabel(d)} active={duration === d} onPre
ss={() => setDuration(d)} />
          ))}
        </View>
      </View>

      {clash ? (
        <View style={styles.warn} accessibilityLiveRegion="polite">
          <Ionicons name="warning-outline" size={16} color="#B45A1C" />
          <T variant="caption" color="#B45A1C" style={{ flex: 1 }}>
            Overlaps with “{clash.title}” at {time12(clash.time)}. You can still add it.
          </T>
        </View>
      ) : null}

      <Input label={t('add.cost')} placeholder="0 = free" icon="cash-outline" value={cost} onChangeText={setCost} keyboardType="decimal-
pad" error={errors.cost} />
      <Input label={t('add.notes')} placeholder={t('add.notesPh')} value={notes} onChangeText={setNotes} multiline error={errors.notes}
maxLength={300} />

      <View>
        <ToggleRow icon="people-outline" label={t('add.vote')} sub="Everyone gets a vote; majority confirms it" value={vote} onChange={s
etVote} />
        <ToggleRow icon="notifications-outline" label={t('add.remind')} value={remind} onChange={setRemind} />
      </View>

      <TimePickerSheet visible={timeOpen} value={time} onClose={() => setTimeOpen(false)} onChange={setTime} />
    </Sheet>
  );
}

const styles = StyleSheet.create({
  wrap: { flexDirection: 'row', flexWrap: 'wrap', gap: 8 },
  warn: { flexDirection: 'row', gap: 8, alignItems: 'center', backgroundColor: colors.orangeSoft, padding: 10, borderRadius: 10 },
});
```

src/components/trip/MemberPickerSheet.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { StyleSheet, View } from 'react-native';

import { Avatar } from '@/components/ui/bits';
import { Press } from '@/components/ui/Press';
import { Sheet } from '@/components/ui/Sheet';
import { T } from '@/components/ui/T';
import type { Member } from '@/store/types';
import { colors, radius } from '@/theme';

/** Multi- or single-select list of trip members. */
export function MemberPickerSheet({
  visible,
  onClose,
  members,
  selected,
  onToggle,
  title = 'Assign people',
  single,
}: {
  visible: boolean;
  onClose: () => void;
  members: Member[];
  selected: string[];
  onToggle: (id: string) => void;
  title?: string;
  single?: boolean;
}) {
  return (
    <Sheet visible={visible} onClose={onClose} title={title} subtitle={single ? 'Choose one person' : 'Tap to add or remove'}>
      {members.map((m) => {
        const on = selected.includes(m.id);
        return (
          <Press
            key={m.id}
            onPress={() => {
              onToggle(m.id);
              if (single) onClose();
            }}
            style={[styles.row, on && styles.rowOn]}
```

```
            accessibilityRole="checkbox"
            accessibilityState={{ checked: on }}>
            <Avatar name={m.name} src={m.avatar} size={36} />
            <View style={{ flex: 1 }}>
              <T variant="bodySm" weight="semibold">
                {m.name}
              </T>
              <T variant="caption" color={colors.textSecondary}>
                {m.role}
              </T>
            </View>
            <Ionicons name={on ? 'checkmark-circle' : 'ellipse-outline'} size={22} color={on ? colors.primary : colors.borderStrong} />
          </Press>
        );
      })}
    </Sheet>
  );
}

const styles = StyleSheet.create({
  row: { flexDirection: 'row', alignItems: 'center', gap: 12, padding: 10, borderRadius: radius.md, borderWidth: 1, borderColor: colors.
border },
  rowOn: { borderColor: colors.primaryLine, backgroundColor: colors.primarySofter },
});
```

src/components/trip/NoTrip.tsx

```
import { router } from 'expo-router';
import { View } from 'react-native';

import { Button } from '@/components/ui/Button';
import { EmptyState } from '@/components/ui/bits';
import { Screen } from '@/components/ui/Screen';

/** Shown on trip tabs when the signed-in user isn't part of any trip yet. */
export function NoTrip({ title = 'No trip selected' }: { title?: string }) {
  return (
    <Screen contentStyle={{ flexGrow: 1, justifyContent: 'center' }}>
      <EmptyState icon="airplane-outline" title={title} body="Create a trip and invite your group, or join a friend’s trip with their in
vite code.">
        <View style={{ gap: 10, alignSelf: 'stretch', marginTop: 12 }}>
          <Button label="Create a trip" icon="add" onPress={() => router.push('/new-trip')} />
          <Button label="Join with a code" icon="people-outline" variant="secondary" onPress={() => router.push('/join')} />
        </View>
      </EmptyState>
    </Screen>
  );
}
```

src/components/trip/TripMenuSheet.tsx

```
import { router } from 'expo-router';

import { ListRow } from '@/components/ui/bits';
import { confirmAction, toast } from '@/components/ui/feedback';
import { Sheet } from '@/components/ui/Sheet';
import { printPlanner, sharePlannerPdf } from '@/services/print';
import { inviteMessage, shareText } from '@/services/share';
import { tripStatus, useAppStore } from '@/store/useAppStore';
import type { Trip } from '@/store/types';
import { colors } from '@/theme';

export function TripMenuSheet({ visible, onClose, trip }: { visible: boolean; onClose: () => void; trip: Trip }) {
  const userId = useAppStore((s) => s.session?.userId);
  const updateTrip = useAppStore((s) => s.updateTrip);
  const leaveTrip = useAppStore((s) => s.leaveTrip);
  const deleteTrip = useAppStore((s) => s.deleteTrip);
  const owner = trip.ownerId === userId;
  const status = tripStatus(trip);
  const go = (href: Parameters<typeof router.push>[0]) => {
    onClose();
    setTimeout(() => router.push(href), 180);
  };

  return (
    <Sheet visible={visible} onClose={onClose} title={trip.name} subtitle="Trip options">
      <ListRow icon="swap-horizontal" label="Switch or create trip" sub="All your group trips in one place" onPress={() => go('/trips')}
/>
      <ListRow icon="create-outline" label="Edit trip details" sub="Name, dates and group budget" onPress={() => go({ pathname: '/new-tr
ip', params: { edit: trip.id } })} />
      <ListRow icon="map-outline" label="Trip map" onPress={() => go('/map')} />
      <ListRow icon="airplane-outline" label="Flights & arrivals" onPress={() => go('/flights')} />
      <ListRow
        icon="print-outline"
        label="Print planner"
        sub="A clean day-by-day printout with checkboxes"
```

```
        onPress={async () => {
          onClose();
          try {
            await printPlanner(trip);
          } catch {
            toast('Allow pop-ups to print the planner', { tone: 'warn' });
          }
        }}
      />
      <ListRow
        icon="document-text-outline"
        label="Save planner as PDF"
        sub="Share it to Files, email or the group chat"
        onPress={async () => {
          onClose();
          await sharePlannerPdf(trip).catch(() => toast('Couldn’t create the PDF', { tone: 'warn' }));
        }}
      />
      <ListRow icon="share-social-outline" label="Share invite" sub={`Code ${trip.inviteCode}`} onPress={() => shareText(inviteMessage(t
rip.name, trip.inviteCode))} />
      <ListRow icon="trophy-outline" label="Trip recap" sub="Stats, highlights and memories" onPress={() => go('/recap')} />
      <ListRow
        icon={status === 'completed' ? 'refresh-outline' : 'flag-outline'}
        label={status === 'completed' ? 'Reopen trip' : 'Mark trip as complete'}
        sub={status === 'completed' ? 'Move it back to planning' : 'Wrap up, settle balances and see the recap'}
        onPress={() => {
          const done = status !== 'completed';
          updateTrip(trip.id, { completed: done });
          onClose();
          toast(done ? 'Trip completed 🎉 — here’s your recap' : 'Trip reopened');
          if (done) setTimeout(() => router.push('/recap'), 250);
        }}
      />
      {owner ? (
        <ListRow
          icon="trash-outline"
          tint={colors.red}
          label="Delete trip"
          sub="Removes it for everyone on this device"
          onPress={() => {
            onClose();
            confirmAction('Delete this trip?', `“${trip.name}” and all its plans, expenses and chats will be deleted. This can’t be undo
ne.`, () => {
              deleteTrip(trip.id);
              toast('Trip deleted');
              router.navigate('/(tabs)/explore');
            }, { confirmLabel: 'Delete trip', destructive: true });
          }}
        />
      ) : (
        <ListRow
          icon="exit-outline"
          tint={colors.red}
          label="Leave trip"
          onPress={() => {
            onClose();
            confirmAction('Leave this trip?', 'You can rejoin later with the invite code.', () => {
              leaveTrip(trip.id);
              toast('You left the trip');
              router.navigate('/(tabs)/explore');
            }, { confirmLabel: 'Leave', destructive: true });
          }}
        />
      )}
    </Sheet>
  );
}
```

src/components/trip/meta.ts

```
import type { IconName } from '@/components/ui/bits';
import type { PlaceCategory } from '@/data/places';
import type { ActivityStatus, ActivityType, ExpenseCategory } from '@/store/types';
import { colors } from '@/theme';

export const activityTypes: { key: ActivityType; label: string; icon: IconName; color: string; bg: string }[] = [
  { key: 'transport', label: 'Transport', icon: 'train-outline', color: '#2B5FB0', bg: colors.blueSoft },
  { key: 'stay', label: 'Stay', icon: 'bed-outline', color: colors.purple, bg: colors.purpleSoft },
  { key: 'food', label: 'Food', icon: 'restaurant-outline', color: '#C2410C', bg: colors.orangeSoft },
  { key: 'sightseeing', label: 'Sightseeing', icon: 'camera-outline', color: colors.teal, bg: colors.tealSoft },
  { key: 'culture', label: 'Culture', icon: 'business-outline', color: '#8B5E0F', bg: colors.yellowSoft },
  { key: 'adventure', label: 'Adventure', icon: 'bonfire-outline', color: colors.primary, bg: colors.primarySoft },
  { key: 'nightlife', label: 'Nightlife', icon: 'musical-notes-outline', color: '#B4235A', bg: '#FCE4EE' },
  { key: 'shopping', label: 'Shopping', icon: 'bag-handle-outline', color: '#5B4AB8', bg: colors.purpleSoft },
];
```

```
export const typeMeta = (t: ActivityType) => activityTypes.find((x) => x.key === t) ?? activityTypes[3];

export const statusMeta: Record<ActivityStatus, { tone: 'green' | 'orange' | 'blue'; key: string; dot: string }> = {
  confirmed: { tone: 'green', key: 'itin.confirmed', dot: colors.primary },
  voting: { tone: 'orange', key: 'itin.voting', dot: colors.orange },
  pending: { tone: 'blue', key: 'itin.pending', dot: colors.blue },
};

/** Map place categories (Nearby) onto itinerary activity types. */
export const placeToActivityType: Record<PlaceCategory, ActivityType> = {
  food: 'food',
  sightseeing: 'sightseeing',
  adventure: 'adventure',
  culture: 'culture',
  nightlife: 'nightlife',
  shopping: 'shopping',
  relax: 'sightseeing',
};

export const placeCategoryIcon: Record<PlaceCategory, IconName> = {
  food: 'restaurant-outline',
  sightseeing: 'camera-outline',
  adventure: 'bonfire-outline',
  culture: 'business-outline',
  nightlife: 'musical-notes-outline',
  shopping: 'bag-handle-outline',
  relax: 'leaf-outline',
};

export const expenseCategories: { key: ExpenseCategory; icon: IconName }[] = [
  { key: 'Food', icon: 'restaurant-outline' },
  { key: 'Activities', icon: 'ticket-outline' },
  { key: 'Transport', icon: 'train-outline' },
  { key: 'Stays', icon: 'bed-outline' },
  { key: 'Flights', icon: 'airplane-outline' },
  { key: 'Shopping', icon: 'bag-handle-outline' },
  { key: 'Other', icon: 'ellipsis-horizontal' },
];

/** Filter chips for itinerary/nearby (the brief's "adventures, sightseeing, etc."). */
export const activityFilters: { key: 'all' | ActivityType; labelKey: string }[] = [
  { key: 'all', labelKey: 'cat.all' },
  { key: 'adventure', labelKey: 'cat.adventure' },
  { key: 'sightseeing', labelKey: 'cat.sightseeing' },
  { key: 'food', labelKey: 'cat.food' },
  { key: 'culture', labelKey: 'cat.culture' },
  { key: 'nightlife', labelKey: 'cat.nightlife' },
  { key: 'shopping', labelKey: 'cat.shopping' },
];

/** Emoji shown in the round icon of an activity card and in the filter chips. */
export const typeEmoji: Record<ActivityType, string> = {
  transport: '🚐',
  stay: '🏡',
  food: '🍝',
  sightseeing: '📷',
  culture: '🏛️',
  adventure: '🥾',
  nightlife: '🍸',
  shopping: '🛍️',
};
```

src/components/ui/Button.tsx

```
import { Ionicons } from '@expo/vector-icons';
import type { ComponentProps } from 'react';
import { LinearGradient } from 'expo-linear-gradient';
import { ActivityIndicator, StyleSheet, View, type StyleProp, type ViewStyle } from 'react-native';

import { colors, radius, shadow } from '@/theme';

import { Press } from './Press';
import { T } from './T';

type IconName = ComponentProps<typeof Ionicons>['name'];

export interface ButtonProps {
  label: string;
  onPress?: () => void;
  variant?: 'primary' | 'secondary' | 'soft' | 'ghost' | 'danger' | 'light';
  size?: 'lg' | 'md' | 'sm';
  icon?: IconName;
  iconRight?: IconName;
  loading?: boolean;
  disabled?: boolean;
  full?: boolean;
```

```
  style?: StyleProp<ViewStyle>;
  accessibilityHint?: string;
  testID?: string;
}

const palette = {
  primary: { bg: colors.primary, fg: colors.onPrimary, border: colors.primary },
  secondary: { bg: colors.card, fg: colors.text, border: colors.border },
  soft: { bg: colors.primarySoft, fg: colors.primary, border: colors.primarySoft },
  ghost: { bg: 'transparent', fg: colors.primary, border: 'transparent' },
  danger: { bg: colors.redSoft, fg: colors.red, border: colors.redSoft },
  light: { bg: 'rgba(255,255,255,0.18)', fg: colors.white, border: 'rgba(255,255,255,0.35)' },
};

export function Button({
  label,
  onPress,
  variant = 'primary',
  size = 'lg',
  icon,
  iconRight,
  loading,
  disabled,
  full = true,
  style,
  accessibilityHint,
  testID,
}: ButtonProps) {
  const p = palette[variant];
  const h = size === 'lg' ? 52 : size === 'md' ? 44 : 36;
  return (
    <Press
      testID={testID}
      accessibilityLabel={label}
      accessibilityHint={accessibilityHint}
      accessibilityState={{ disabled: !!disabled || !!loading, busy: !!loading }}
      disabled={disabled || loading}
      onPress={onPress}
      style={[
        styles.base,
        { height: h, backgroundColor: p.bg, borderColor: p.border, paddingHorizontal: size === 'sm' ? 14 : 20 },
        full && { alignSelf: 'stretch' },
        variant === 'primary' && [shadow, { overflow: 'hidden' as const }],
        (disabled || loading) && { opacity: 0.55 },
        style,
      ]}>
      {variant === 'primary' ? <LinearGradient colors={[colors.primary, colors.heroB]} start={{ x: 0, y: 0 }} end={{ x: 1, y: 1 }} style
={StyleSheet.absoluteFill} /> : null}
      {loading ? (
        <ActivityIndicator color={p.fg} />
      ) : (
        <View style={styles.row}>
          {icon && <Ionicons name={icon} size={size === 'sm' ? 15 : 18} color={p.fg} />}
          <T variant={size === 'sm' ? 'small' : 'title'} weight="semibold" color={p.fg}>
            {label}
          </T>
          {iconRight && <Ionicons name={iconRight} size={size === 'sm' ? 15 : 18} color={p.fg} />}
        </View>
      )}
    </Press>
  );
}

const styles = StyleSheet.create({
  base: {
    borderRadius: radius.pill,
    borderWidth: 1,
    alignItems: 'center',
    justifyContent: 'center',
  },
  row: { flexDirection: 'row', alignItems: 'center', gap: 8 },
});
```

src/components/ui/Donut.tsx

```
import { StyleSheet, View } from 'react-native';
import Svg, { Circle } from 'react-native-svg';

import { colors } from '@/theme';

import { T } from './T';

/** Ring chart of spending by category with the remaining budget in the middle. */
export function Donut({ parts, size = 132, stroke = 20, centerLabel, centerValue }: { parts: { value: number; color: string }[]; size?:
number; stroke?: number; centerLabel: string; centerValue: string }) {
  const r = (size - stroke) / 2;
```

```
  const c = 2 * Math.PI * r;
  const total = parts.reduce((s, p) => s + p.value, 0);
  let offset = 0;
  return (
    <View style={{ width: size, height: size }} accessibilityRole="image" accessibilityLabel={`${centerLabel} ${centerValue}`}>
      <Svg width={size} height={size} viewBox={`0 0 ${size} ${size}`} style={{ transform: [{ rotate: '-90deg' }] }}>
        <Circle cx={size / 2} cy={size / 2} r={r} stroke={colors.bgAlt} strokeWidth={stroke} fill="none" />
        {total > 0
          ? parts.map((p, i) => {
              const len = (p.value / total) * c;
              const el = <Circle key={i} cx={size / 2} cy={size / 2} r={r} stroke={p.color} strokeWidth={stroke} fill="none" strokeDasha
rray={`${Math.max(0, len - 2)} ${c}`} strokeDashoffset={-offset} />;
              offset += len;
              return el;
            })
          : null}
      </Svg>
      <View style={[StyleSheet.absoluteFill, { alignItems: 'center', justifyContent: 'center' }]}>
        <T variant="micro" color={colors.textSecondary} style={{ letterSpacing: 1 }}>
          {centerLabel}
        </T>
        <T variant="h3" weight="extrabold">
          {centerValue}
        </T>
      </View>
    </View>
  );
}
```

src/components/ui/Input.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { forwardRef, useState, type ComponentProps, type ReactNode } from 'react';
import { Platform, StyleSheet, TextInput, View, type StyleProp, type TextInputProps, type ViewStyle } from 'react-native';

import { useAppStore } from '@/store/useAppStore';
import { colors, fonts, radius } from '@/theme';

import { Press } from './Press';
import { T } from './T';

type IconName = ComponentProps<typeof Ionicons>['name'];

export interface InputProps extends TextInputProps {
  label?: string;
  /** Show the label inside the field (floating style from the sign-in screen). */
  inlineLabel?: boolean;
  icon?: IconName;
  error?: string | null;
  hint?: string;
  right?: ReactNode;
  secureToggle?: boolean;
  containerStyle?: StyleProp<ViewStyle>;
  onPressField?: () => void;
}

export const Input = forwardRef<TextInput, InputProps>(function Input(
  { label, inlineLabel, icon, error, hint, right, secureToggle, containerStyle, multiline, onPressField, style, secureTextEntry, onFocu
s, onBlur, ...rest },
  ref,
) {
  const [focused, setFocused] = useState(false);
  const [hidden, setHidden] = useState(true);
  const scale = useAppStore((s) => s.settings.textScale);
  const hasValue = !!rest.value;
  const borderColor = error ? colors.red : focused ? colors.primary : colors.border;

  const field = (
    <View
      style={[
        styles.field,
        multiline && styles.multiline,
        { borderColor },
        focused && !error && styles.focused,
        error && styles.errored,
      ]}>
      {icon && <Ionicons name={icon} size={18} color={error ? colors.red : colors.textMuted} style={{ marginRight: 10 }} />}
      <View style={{ flex: 1 }}>
        {inlineLabel && label && (hasValue || focused) ? (
          <T variant="micro" color={colors.textSecondary} style={{ marginBottom: 1 }}>
            {label}
          </T>
        ) : null}
        <TextInput
          ref={ref}
          placeholderTextColor={colors.textMuted}
```

```
          multiline={multiline}
          editable={!onPressField}
          pointerEvents={onPressField ? 'none' : 'auto'}
          secureTextEntry={secureToggle ? hidden : secureTextEntry}
          accessibilityLabel={label}
          {...rest}
          placeholder={inlineLabel && !hasValue && !focused ? label : rest.placeholder}
          onFocus={(e) => {
            setFocused(true);
            onFocus?.(e);
          }}
          onBlur={(e) => {
            setFocused(false);
            onBlur?.(e);
          }}
          style={[
            styles.input,
            { fontSize: 15 * scale },
            multiline && { minHeight: 64, textAlignVertical: 'top' },
            Platform.OS === 'web' && ({ outlineStyle: 'none' } as object),
            style,
          ]}
        />
      </View>
      {secureToggle && (
        <Press accessibilityLabel={hidden ? 'Show password' : 'Hide password'} onPress={() => setHidden((h) => !h)} hitSlop={10}>
          <Ionicons name={hidden ? 'eye-outline' : 'eye-off-outline'} size={19} color={colors.textMuted} />
        </Press>
      )}
      {right}
    </View>
  );

  return (
    <View style={[{ gap: 6 }, containerStyle]}>
      {label && !inlineLabel ? (
        <T variant="small" weight="semibold">
          {label}
        </T>
      ) : null}
      {onPressField ? (
        <Press onPress={onPressField} scaleTo={0.99} accessibilityLabel={label}>
          {field}
        </Press>
      ) : (
        field
      )}
      {error ? (
        <T variant="caption" color={colors.red} accessibilityLiveRegion="polite">
          {error}
        </T>
      ) : hint ? (
        <T variant="caption" color={colors.textMuted}>
          {hint}
        </T>
      ) : null}
    </View>
  );
});

const styles = StyleSheet.create({
  field: {
    minHeight: 50,
    borderRadius: radius.md,
    borderWidth: 1.2,
    backgroundColor: colors.cardMuted,
    paddingHorizontal: 14,
    flexDirection: 'row',
    alignItems: 'center',
  },
  multiline: { alignItems: 'flex-start', paddingVertical: 12 },
  focused: { backgroundColor: colors.white },
  errored: { backgroundColor: '#FFF8F8' },
  input: {
    fontFamily: fonts.regular,
    color: colors.text,
    paddingVertical: Platform.OS === 'ios' ? 12 : 8,
  },
});
```

src/components/ui/Press.tsx

```
import * as Haptics from 'expo-haptics';
import { Platform, Pressable, type PressableProps, type StyleProp, type ViewStyle } from 'react-native';

import { useAppStore } from '@/store/useAppStore';
```

```
export function tap(kind: 'light' | 'medium' | 'success' | 'warning' = 'light') {
  if (Platform.OS === 'web' || !useAppStore.getState().settings.haptics) return;
  if (kind === 'success') Haptics.notificationAsync(Haptics.NotificationFeedbackType.Success).catch(() => {});
  else if (kind === 'warning') Haptics.notificationAsync(Haptics.NotificationFeedbackType.Warning).catch(() => {});
  else Haptics.impactAsync(kind === 'medium' ? Haptics.ImpactFeedbackStyle.Medium : Haptics.ImpactFeedbackStyle.Light).catch(() => {});
}

export interface PressProps extends Omit<PressableProps, 'style'> {
  style?: StyleProp<ViewStyle>;
  /** Scale applied while pressed. */
  scaleTo?: number;
  haptic?: boolean;
}

/** Pressable with a subtle press-in scale + optional haptic tick. */
export function Press({ style, scaleTo = 0.97, haptic = true, onPress, accessibilityRole = 'button', ...rest }: PressProps) {
  const state = rest.accessibilityState;
  return (
    <Pressable
      accessibilityRole={accessibilityRole}
      // Mirror the state as ARIA props so web screen readers announce selection too.
      aria-selected={state?.selected}
      aria-checked={state?.checked}
      {...rest}
      onPress={(e) => {
        if (haptic) tap();
        onPress?.(e);
      }}
      style={({ pressed }) => [style, pressed && !rest.disabled && { transform: [{ scale: scaleTo }], opacity: 0.92 }]}
    />
  );
}
```

src/components/ui/Screen.tsx

```
import { Ionicons } from '@expo/vector-icons';
import { LinearGradient } from 'expo-linear-gradient';
import { router } from 'expo-router';
import type { ReactNode } from 'react';
import { RefreshControl, ScrollView, StyleSheet, View, type StyleProp, type ViewStyle } from 'react-native';
import { useSafeAreaInsets } from 'react-native-safe-area-context';

import { MAX_WIDTH, appearance, colors } from '@/theme';

import { Press } from './Press';
import { FadeIn } from './motion';
import { T } from './T';

/** Page container: cream background, safe-area padding and a phone-width column on big screens. */
export function Screen({
  children,
  scroll = true,
  padded = true,
  contentStyle,
  header,
  footer,
  bottomInset = 24,
  onRefresh,
  refreshing,
}: {
  children: ReactNode;
  scroll?: boolean;
  padded?: boolean;
  contentStyle?: StyleProp<ViewStyle>;
  header?: ReactNode;
  footer?: ReactNode;
  bottomInset?: number;
  onRefresh?: () => void;
  refreshing?: boolean;
}) {
  const insets = useSafeAreaInsets();
  const inner = [padded && { paddingHorizontal: 20 }, { paddingBottom: bottomInset + (footer ? 0 : insets.bottom) }, contentStyle];
  return (
    <View style={[styles.root, { paddingTop: insets.top }]}>
      {/* Soft wash of the theme colour behind the top of every page (the flat Amalfi look skips it). */}
      {appearance.theme !== 'amalfi' ? (
        <>
          <LinearGradient pointerEvents="none" colors={[colors.primarySoft, colors.primarySofter, colors.bg]} locations={[0, 0.45, 1]} s
tyle={styles.wash} />
          <View pointerEvents="none" style={styles.glow} />
        </>
      ) : null}
      <View style={styles.column}>
        {header}
        {scroll ? (
```

```
          <ScrollView
            style={{ flex: 1 }}
            contentContainerStyle={inner}
            keyboardShouldPersistTaps="handled"
            showsVerticalScrollIndicator={false}
            refreshControl={onRefresh ? <RefreshControl refreshing={!!refreshing} onRefresh={onRefresh} tintColor={colors.primary} /> :
undefined}>
            <FadeIn duration={360} distance={12}>
              {children}
            </FadeIn>
          </ScrollView>
        ) : (
          <View style={[{ flex: 1 }, inner]}>{children}</View>
        )}
        {footer}
      </View>
    </View>
  );
}

/** Stack-screen header: blue back link, big condensed title + subtitle, optional right actions. */
export function Header({ title, subtitle, kicker, right, onBack, back = true, backLabel = 'Back' }: { title: string; subtitle?: string;
kicker?: string; right?: ReactNode; onBack?: () => void; back?: boolean; backLabel?: string }) {
  return (
    <View style={styles.header}>
      {back ? (
        <Press
          hitSlop={10}
          accessibilityLabel={`Go back to ${backLabel}`}
          style={styles.back}
          onPress={() => {
            if (onBack) onBack();
            else if (router.canGoBack()) router.back();
            else router.replace('/(tabs)/explore');
          }}>
          <Ionicons name="chevron-back" size={18} color={colors.blue} />
          <T variant="small" weight="semibold" color={colors.blue}>
            {backLabel}
          </T>
        </Press>
      ) : null}
      <View style={{ flexDirection: 'row', alignItems: 'flex-end', gap: 12 }}>
        <View style={{ flex: 1 }}>
          {kicker ? (
            <T variant="kicker" color={colors.textSecondary}>
              {kicker}
            </T>
          ) : null}
          <T variant="display" style={{ fontSize: 36, lineHeight: 40 }} numberOfLines={2} accessibilityRole="header">
            {title}
          </T>
          {subtitle ? (
            <T variant="small" color={colors.textSecondary} numberOfLines={2} style={{ marginTop: 2 }}>
              {subtitle}
            </T>
          ) : null}
        </View>
        {right}
      </View>
    </View>
  );
}

const styles = StyleSheet.create({
  root: { flex: 1, backgroundColor: colors.bg, alignItems: 'center', overflow: 'hidden' },
  wash: { position: 'absolute', top: 0, left: 0, right: 0, height: 300 },
  glow: { position: 'absolute', top: -90, right: -70, width: 240, height: 240, borderRadius: 120, backgroundColor: colors.accent, opacit
y: 0.12 },
  column: { flex: 1, width: '100%', maxWidth: MAX_WIDTH },
  header: { paddingHorizontal: 20, paddingTop: 8, paddingBottom: 14, gap: 4 },
  back: { flexDirection: 'row', alignItems: 'center', alignSelf: 'flex-start', marginLeft: -4 },
});
```

src/components/ui/Segmented.tsx

```
import { StyleSheet, View } from 'react-native';

import { colors } from '@/theme';

import { Press } from './Press';
import { T } from './T';

/** Pill-shaped tab switcher: grey track, white selected pill (Itinerary | Vote, Crew | Scout AI…). */
export function Segmented<K extends string>({ options, value, onChange }: { options: { key: K; label: string }[]; value: K; onChange:
(k: K) => void }) {
  return (
```

```
    <View style={styles.track} accessibilityRole="tablist">
      {options.map((o) => {
        const on = o.key === value;
        return (
          <Press key={o.key} onPress={() => onChange(o.key)} style={[styles.item, on && styles.on]} accessibilityRole="tab" accessibilit
yState={{ selected: on }} accessibilityLabel={o.label} scaleTo={0.98}>
            <T variant="small" weight={on ? 'bold' : 'semibold'} color={on ? colors.text : colors.textSecondary}>
              {o.label}
            </T>
          </Press>
        );
      })}
    </View>
  );
}

const styles = StyleSheet.create({
  track: { flexDirection: 'row', backgroundColor: colors.bgAlt, borderRadius: 999, padding: 4 },
  item: { flex: 1, height: 38, alignItems: 'center', justifyContent: 'center', borderRadius: 999, flexDirection: 'row', gap: 6 },
  on: { backgroundColor: colors.card, boxShadow: '0px 1px 4px rgba(0,0,0,0.12)' } as object,
});
```

src/components/ui/Sheet.tsx

```
import { useEffect, useState, type ReactNode } from 'react';
import {
  Animated,
  Easing,
  KeyboardAvoidingView,
  Modal,
  Platform,
  Pressable,
  ScrollView,
  StyleSheet,
  View,
  useWindowDimensions,
} from 'react-native';
import { useSafeAreaInsets } from 'react-native-safe-area-context';

import { MAX_WIDTH, colors, radius } from '@/theme';

import { T } from './T';

export interface SheetProps {
  visible: boolean;
  onClose: () => void;
  title?: string;
  subtitle?: string;
  children: ReactNode;
  footer?: ReactNode;
  /** Fraction of screen height the sheet may use. */
  maxHeight?: number;
  scroll?: boolean;
}

/** Bottom sheet with a drag-handle look, slide-up animation and a tappable backdrop. */
export function Sheet({ visible, onClose, title, subtitle, children, footer, maxHeight = 0.9, scroll = true }: SheetProps) {
  const { height } = useWindowDimensions();
  const insets = useSafeAreaInsets();
  const [mounted, setMounted] = useState(visible);
  const [anim] = useState(() => new Animated.Value(0));
  // Mount as soon as we're asked to show; unmount only after the close animation.
  if (visible && !mounted) setMounted(true);

  useEffect(() => {
    if (visible) {
      Animated.timing(anim, { toValue: 1, duration: 260, easing: Easing.out(Easing.cubic), useNativeDriver: Platform.OS !== 'web' }).sta
rt();
    } else if (mounted) {
      Animated.timing(anim, { toValue: 0, duration: 200, easing: Easing.in(Easing.cubic), useNativeDriver: Platform.OS !== 'web' }).star
t(() =>
        setMounted(false),
      );
    }
  }, [visible, anim, mounted]);

  if (!mounted) return null;

  const translateY = anim.interpolate({ inputRange: [0, 1], outputRange: [height * 0.6, 0] });

  return (
    <Modal transparent visible={mounted} onRequestClose={onClose} statusBarTranslucent animationType="none">
      <KeyboardAvoidingView behavior={Platform.OS === 'ios' ? 'padding' : undefined} style={{ flex: 1 }}>
        <Animated.View style={[StyleSheet.absoluteFill, { backgroundColor: colors.overlay, opacity: anim }]}>
          <Pressable style={{ flex: 1 }} onPress={onClose} accessibilityLabel="Close" accessibilityRole="button" />
        </Animated.View>
```

```
        <View style={styles.wrap} pointerEvents="box-none">
          <Animated.View
            accessibilityViewIsModal
            style={[styles.sheet, { maxHeight: height * maxHeight, paddingBottom: Math.max(insets.bottom, 16), transform: [{ translateY
}] }]}>
            <View style={styles.handle} />
            {title ? (
              <View style={{ paddingHorizontal: 22, paddingTop: 6, paddingBottom: 10 }}>
                <T variant="h2" accessibilityRole="header">
                  {title}
                </T>
                {subtitle ? (
                  <T variant="small" color={colors.textSecondary} style={{ marginTop: 2 }}>
                    {subtitle}
                  </T>
                ) : null}
              </View>
            ) : null}
            {scroll ? (
              <ScrollView
                keyboardShouldPersistTaps="handled"
                contentContainerStyle={{ paddingHorizontal: 22, paddingBottom: 12, gap: 16 }}
                showsVerticalScrollIndicator={false}>
                {children}
              </ScrollView>
            ) : (
              <View style={{ paddingHorizontal: 22, flexShrink: 1 }}>{children}</View>
            )}
            {footer ? <View style={{ paddingHorizontal: 22, paddingTop: 8 }}>{footer}</View> : null}
          </Animated.View>
        </View>
      </KeyboardAvoidingView>
    </Modal>
  );
}

const styles = StyleSheet.create({
  wrap: { flex: 1, justifyContent: 'flex-end', alignItems: 'center' },
  sheet: {
    width: '100%',
    maxWidth: MAX_WIDTH,
    backgroundColor: colors.card,
    borderTopLeftRadius: radius.xl + 4,
    borderTopRightRadius: radius.xl + 4,
  },
  handle: {
    alignSelf: 'center',
    width: 42,
    height: 5,
    borderRadius: 3,
    backgroundColor: colors.borderStrong,
    marginTop: 10,
    marginBottom: 10,
  },
});
```

src/components/ui/Slider.tsx

```
import { useState } from 'react';
import { StyleSheet, View, type GestureResponderEvent } from 'react-native';

import { colors } from '@/theme';

export interface SliderProps {
  value: number;
  min: number;
  max: number;
  step?: number;
  onChange: (v: number) => void;
  label: string;
  formatValue?: (v: number) => string;
}

/** Lightweight cross-platform slider (track + thumb) styled like the reference radius control. */
export function Slider({ value, min, max, step = 1, onChange, label, formatValue }: SliderProps) {
  const [width, setWidth] = useState(1);

  // Children ignore touches, so locationX is always relative to the track container.
  const update = (e: GestureResponderEvent) => {
    const ratio = Math.max(0, Math.min(1, e.nativeEvent.locationX / width));
    const next = Math.round((min + ratio * (max - min)) / step) * step;
    if (next !== value) onChange(next);
  };

  const pct = (value - min) / (max - min);
  return (
```

```
    <View
      accessible
      accessibilityRole="adjustable"
      accessibilityLabel={label}
      accessibilityValue={{ min, max, now: value, text: formatValue?.(value) }}
      accessibilityActions={[{ name: 'increment' }, { name: 'decrement' }]}
      onAccessibilityAction={(e) => {
        if (e.nativeEvent.actionName === 'increment') onChange(Math.min(max, value + step));
        if (e.nativeEvent.actionName === 'decrement') onChange(Math.max(min, value - step));
      }}
      style={styles.hit}
      onLayout={(e) => setWidth(e.nativeEvent.layout.width)}
      onStartShouldSetResponder={() => true}
      onMoveShouldSetResponder={() => true}
      onResponderTerminationRequest={() => false}
      onResponderGrant={update}
      onResponderMove={update}>
      <View pointerEvents="none" style={styles.track}>
        <View style={[styles.fill, { width: `${pct * 100}%` }]} />
      </View>
      <View pointerEvents="none" style={[styles.thumb, { left: pct * width - 11 }]} />
    </View>
  );
}

const styles = StyleSheet.create({
  hit: { height: 32, justifyContent: 'center' },
  track: { height: 6, borderRadius: 3, backgroundColor: colors.border, overflow: 'hidden' },
  fill: { height: '100%', backgroundColor: colors.primary, borderRadius: 3 },
  thumb: {
    position: 'absolute',
    width: 22,
    height: 22,
    borderRadius: 11,
    backgroundColor: colors.primary,
    borderWidth: 4,
    borderColor: colors.white,
    top: 5,
    shadowColor: '#000',
    shadowOpacity: 0.18,
    shadowRadius: 4,
    shadowOffset: { width: 0, height: 2 },
    elevation: 3,
  },
});
```

src/components/ui/T.tsx

```
import { StyleSheet, Text, type TextProps, type TextStyle } from 'react-native';

import { useAppStore } from '@/store/useAppStore';
import { colors, fonts } from '@/theme';

type Variant = 'display' | 'headline' | 'h1' | 'h2' | 'h3' | 'title' | 'body' | 'bodySm' | 'small' | 'caption' | 'kicker' | 'micro';
type Weight = 'regular' | 'medium' | 'semibold' | 'bold' | 'extrabold';

const variants: Record<Variant, TextStyle> = {
  display: { fontSize: 42, lineHeight: 46, fontFamily: fonts.display, letterSpacing: 0.3, textTransform: 'uppercase' },
  headline: { fontSize: 24, lineHeight: 28, fontFamily: fonts.display, letterSpacing: 0.3, textTransform: 'uppercase' },
  h1: { fontSize: 26, lineHeight: 32, fontFamily: fonts.bold, letterSpacing: -0.4 },
  h2: { fontSize: 22, lineHeight: 28, fontFamily: fonts.bold, letterSpacing: -0.3 },
  h3: { fontSize: 18, lineHeight: 24, fontFamily: fonts.semibold, letterSpacing: -0.2 },
  title: { fontSize: 15.5, lineHeight: 21, fontFamily: fonts.semibold },
  body: { fontSize: 15, lineHeight: 21, fontFamily: fonts.regular },
  bodySm: { fontSize: 14, lineHeight: 19, fontFamily: fonts.regular },
  small: { fontSize: 13, lineHeight: 18, fontFamily: fonts.regular },
  caption: { fontSize: 12, lineHeight: 16, fontFamily: fonts.regular },
  kicker: { fontSize: 11, lineHeight: 14, fontFamily: fonts.semibold, letterSpacing: 1.6, textTransform: 'uppercase' },
  micro: { fontSize: 10.5, lineHeight: 13, fontFamily: fonts.medium },
};

export interface TProps extends TextProps {
  variant?: Variant;
  weight?: Weight;
  color?: string;
  center?: boolean;
}

/** App text: applies the brand font, type scale and the user's text-size setting. */
export function T({ variant = 'body', weight, color, center, style, ...rest }: TProps) {
  const scale = useAppStore((s) => s.settings.textScale);
  const v = variants[variant];
  const scaled: TextStyle =
    scale === 1
      ? {}
      : { fontSize: (v.fontSize ?? 15) * scale, lineHeight: v.lineHeight ? v.lineHeight * scale : undefined };
```

```
  return (
    <Text
      maxFontSizeMultiplier={1.4}
      {...rest}
      style={[
        styles.base,
        v,
        scaled,
        weight && { fontFamily: fonts[weight] },
        color ? { color } : null,
        center && { textAlign: 'center' },
        style,
      ]}
    />
  );
}

const styles = StyleSheet.create({
  base: { color: colors.text },
});
```

src/components/ui/bits.tsx

```
/** Small presentational building blocks shared by every screen. */
import { Ionicons } from '@expo/vector-icons';
import { Image } from 'expo-image';
import type { ComponentProps, ReactNode } from 'react';
import { StyleSheet, Switch, View, type StyleProp, type ViewStyle } from 'react-native';

import { imageSource } from '@/data/images';
import { colors, isDark, radius, shadow } from '@/theme';
import { initials } from '@/utils/format';

import { Press } from './Press';
import { T } from './T';

export type IconName = ComponentProps<typeof Ionicons>['name'];

export function Card({ children, style, padded = true, onPress, accessibilityLabel }: { children: ReactNode; style?: StyleProp<ViewStyle
>; padded?: boolean; onPress?: () => void; accessibilityLabel?: string }) {
  const s = [styles.card, padded && { padding: 16 }, style];
  if (onPress)
    return (
      <Press onPress={onPress} style={s} scaleTo={0.985} accessibilityLabel={accessibilityLabel}>
        {children}
      </Press>
    );
  return <View style={s}>{children}</View>;
}

export function Chip({
  label,
  active,
  onPress,
  icon,
  image,
  small,
  tone,
  style,
}: {
  label: string;
  active?: boolean;
  onPress?: () => void;
  icon?: IconName;
  image?: string;
  small?: boolean;
  tone?: 'soft';
  style?: StyleProp<ViewStyle>;
}) {
  const bg = active ? colors.primary : tone === 'soft' ? colors.primarySofter : colors.card;
  const fg = active ? colors.white : colors.text;
  return (
    <Press
      onPress={onPress}
      accessibilityRole="button"
      accessibilityState={{ selected: !!active }}
      accessibilityLabel={label}
      style={[
        styles.chip,
        small && styles.chipSmall,
        { backgroundColor: bg, borderColor: active ? colors.primary : colors.border },
        !!image && { paddingLeft: 5 },
        style,
      ]}>
      {image ? <Image source={imageSource(image)} style={styles.chipImg} contentFit="cover" /> : null}
      {icon ? <Ionicons name={icon} size={small ? 13 : 15} color={fg} /> : null}
```

```
      <T variant={small ? 'caption' : 'small'} weight={active ? 'semibold' : 'medium'} color={fg} numberOfLines={1}>
        {label}
      </T>
    </Press>
  );
}

export function Avatar({ name, src, size = 34, ring }: { name: string; src?: string; size?: number; ring?: string }) {
  const palette = ['#DCEFD3', '#FDEBDD', '#E3ECFB', '#EEE8FB', '#FDF3DC', '#DDF3F0'];
  const bg = palette[(name.charCodeAt(0) + name.length) % palette.length];
  return (
    <View
      accessibilityLabel={name}
      style={{
        width: size,
        height: size,
        borderRadius: size / 2,
        backgroundColor: bg,
        alignItems: 'center',
        justifyContent: 'center',
        overflow: 'hidden',
        borderWidth: ring ? 2 : 0,
        borderColor: ring,
      }}>
      {src ? (
        <Image source={imageSource(src)} style={{ width: '100%', height: '100%' }} contentFit="cover" />
      ) : (
        <T variant={size > 40 ? 'title' : 'micro'} weight="bold" color={isDark ? '#1F4D25' : colors.primary} style={size <= 28 ? { fontS
ize: 9.5 } : null}>
          {initials(name)}
        </T>
      )}
    </View>
  );
}

export function AvatarStack({ people, max = 3, size = 28 }: { people: { name: string; avatar?: string }[]; max?: number; size?: number
}) {
  const shown = people.slice(0, max);
  const extra = people.length - shown.length;
  return (
    <View style={{ flexDirection: 'row', alignItems: 'center' }}>
      {shown.map((p, i) => (
        <View key={`${p.name}${i}`} style={{ marginLeft: i === 0 ? 0 : -size * 0.32 }}>
          <Avatar name={p.name} src={p.avatar} size={size} ring={colors.white} />
        </View>
      ))}
      {extra > 0 && (
        <View
          style={{
            marginLeft: -size * 0.32,
            width: size,
            height: size,
            borderRadius: size / 2,
            backgroundColor: colors.primarySoft,
            borderWidth: 2,
            borderColor: colors.white,
            alignItems: 'center',
            justifyContent: 'center',
          }}>
          <T variant="micro" weight="bold" color={colors.primary}>
            +{extra}
          </T>
        </View>
      )}
    </View>
  );
}

export function ProgressBar({ value, color = colors.primary, track = colors.border, height = 6 }: { value: number; color?: string; trac
k?: string; height?: number }) {
  const pct = Math.max(0, Math.min(1, value));
  return (
    <View
      accessibilityRole="progressbar"
      accessibilityValue={{ min: 0, max: 100, now: Math.round(pct * 100) }}
      style={{ height, borderRadius: height, backgroundColor: track, overflow: 'hidden' }}>
      <View style={{ width: `${pct * 100}%`, height: '100%', backgroundColor: color, borderRadius: height }} />
    </View>
  );
}

export function IconButton({
  icon,
  onPress,
  label,
```

```
  size = 40,
  bg = colors.card,
  color = colors.text,
  badge,
  bordered = true,
}: {
  icon: IconName;
  onPress?: () => void;
  label: string;
  size?: number;
  bg?: string;
  color?: string;
  badge?: number | boolean;
  bordered?: boolean;
}) {
  return (
    <Press
      onPress={onPress}
      accessibilityLabel={label}
      hitSlop={6}
      style={{
        width: size,
        height: size,
        borderRadius: size / 2,
        backgroundColor: bg,
        alignItems: 'center',
        justifyContent: 'center',
        borderWidth: bordered ? 1 : 0,
        borderColor: colors.border,
      }}>
      <Ionicons name={icon} size={size * 0.46} color={color} />
      {badge ? (
        <View style={styles.badge}>
          {typeof badge === 'number' ? (
            <T variant="micro" weight="bold" color={colors.white} style={{ fontSize: 9 }}>
              {badge > 9 ? '9+' : badge}
            </T>
          ) : null}
        </View>
      ) : null}
    </Press>
  );
}

export function SectionHeader({ title, action, onAction, style }: { title: string; action?: string; onAction?: () => void; style?: StyleProp<ViewStyle> }) {
  return (
    <View style={[styles.sectionHeader, style]}>
      <T variant="headline" accessibilityRole="header" style={{ flexShrink: 1 }}>
        {title}
      </T>
      {action ? (
        <Press onPress={onAction} hitSlop={8} accessibilityLabel={action}>
          <T variant="small" weight="semibold" color={colors.textSecondary}>
            {action}
          </T>
        </Press>
      ) : null}
    </View>
  );
}

export function Pill({ label, tone = 'green', icon }: { label: string; tone?: 'green' | 'orange' | 'blue' | 'red' | 'gray' | 'yellow' | 'purple'; icon?: IconName }) {
  const map = {
    green: [colors.primarySoft, isDark ? '#6FD39A' : colors.primary],
    orange: [colors.orangeSoft, isDark ? colors.orange : '#B45A1C'],
    blue: [colors.blueSoft, isDark ? '#7FB0F0' : '#2B5FB0'],
    red: [colors.redSoft, colors.red],
    gray: [colors.bgAlt, colors.textSecondary],
    yellow: [colors.yellowSoft, isDark ? colors.yellow : '#94660A'],
    purple: [colors.purpleSoft, colors.purple],
  } as const;
  const [bg, fg] = map[tone];
  return (
    <View style={[styles.pill, { backgroundColor: bg }]}>
      {icon ? <Ionicons name={icon} size={11} color={fg} /> : null}
      <T variant="micro" weight="semibold" color={fg}>
        {label}
      </T>
    </View>
  );
}

export function EmptyState({ icon, title, body, children }: { icon: IconName; title: string; body?: string; children?: ReactNode }) {
  return (
```

```
    <View style={styles.empty}>
      <View style={styles.emptyIcon}>
        <Ionicons name={icon} size={26} color={colors.primary} />
      </View>
      <T variant="title" center>
        {title}
      </T>
      {body ? (
        <T variant="small" color={colors.textSecondary} center>
          {body}
        </T>
      ) : null}
      {children}
    </View>
  );
}

export function ToggleRow({ label, sub, value, onChange, icon }: { label: string; sub?: string; value: boolean; onChange: (v: boolean) =
> void; icon?: IconName }) {
  return (
    <View style={styles.toggleRow}>
      {icon ? (
        <View style={styles.rowIcon}>
          <Ionicons name={icon} size={17} color={colors.primary} />
        </View>
      ) : null}
      <View style={{ flex: 1 }}>
        <T variant="bodySm" weight="medium">
          {label}
        </T>
        {sub ? (
          <T variant="caption" color={colors.textSecondary}>
            {sub}
          </T>
        ) : null}
      </View>
      <Switch
        accessibilityLabel={label}
        value={value}
        onValueChange={onChange}
        trackColor={{ false: colors.borderStrong, true: colors.primary }}
        thumbColor={colors.white}
        {...({ activeThumbColor: colors.white } as object)}
      />
    </View>
  );
}

export function ListRow({ icon, label, sub, onPress, right, tint = colors.primary }: { icon: IconName; label: string; sub?: string; onPr
ess?: () => void; right?: ReactNode; tint?: string }) {
  return (
    <Press onPress={onPress} style={styles.listRow} scaleTo={0.99} accessibilityLabel={label}>
      <View style={[styles.rowIcon, { backgroundColor: tint === colors.primary ? colors.primarySofter : `${tint}1F` }]}>
        <Ionicons name={icon} size={17} color={tint} />
      </View>
      <View style={{ flex: 1 }}>
        <T variant="bodySm" weight="medium">
          {label}
        </T>
        {sub ? (
          <T variant="caption" color={colors.textSecondary} numberOfLines={2}>
            {sub}
          </T>
        ) : null}
      </View>
      {right ?? <Ionicons name="chevron-forward" size={16} color={colors.textMuted} />}
    </Press>
  );
}

export function Divider({ style }: { style?: StyleProp<ViewStyle> }) {
  return <View style={[{ height: 1, backgroundColor: colors.border }, style]} />;
}

const styles = StyleSheet.create({
  card: {
    backgroundColor: colors.card,
    borderRadius: radius.lg,
    borderWidth: 1,
    borderColor: colors.border,
    ...shadow,
  },
  chip: {
    flexDirection: 'row',
    alignItems: 'center',
    gap: 6,
```

```
    height: 38,
    paddingHorizontal: 14,
    borderRadius: radius.pill,
    borderWidth: 1,
  },
  chipSmall: { height: 30, paddingHorizontal: 11, gap: 4 },
  chipImg: { width: 28, height: 28, borderRadius: 14 },
  badge: {
    position: 'absolute',
    top: 5,
    right: 5,
    minWidth: 15,
    height: 15,
    paddingHorizontal: 3,
    borderRadius: 8,
    backgroundColor: colors.red,
    alignItems: 'center',
    justifyContent: 'center',
    borderWidth: 1.5,
    borderColor: colors.white,
  },
  sectionHeader: { flexDirection: 'row', alignItems: 'center', justifyContent: 'space-between', marginBottom: 12 },
  pill: { flexDirection: 'row', alignItems: 'center', gap: 3, paddingHorizontal: 8, paddingVertical: 3, borderRadius: radius.pill, align
Self: 'flex-start' },
  empty: { alignItems: 'center', gap: 8, paddingVertical: 36, paddingHorizontal: 24 },
  emptyIcon: {
    width: 60,
    height: 60,
    borderRadius: 30,
    backgroundColor: colors.primarySofter,
    alignItems: 'center',
    justifyContent: 'center',
    marginBottom: 6,
  },
  toggleRow: { flexDirection: 'row', alignItems: 'center', gap: 12, paddingVertical: 10 },
  rowIcon: { width: 34, height: 34, borderRadius: 10, backgroundColor: colors.primarySofter, alignItems: 'center', justifyContent: 'cent
er' },
  listRow: { flexDirection: 'row', alignItems: 'center', gap: 12, paddingVertical: 11 },
});
```

src/components/ui/feedback.tsx

```
/**
 * Global toast + dialog system. Works the same on iOS, Android and web
 * (React Native's Alert is a no-op on web, so we render our own).
 */
import { Ionicons } from '@expo/vector-icons';
import { useEffect, useState } from 'react';
import { Animated, Modal, Platform, Pressable, StyleSheet, View } from 'react-native';
import { useSafeAreaInsets } from 'react-native-safe-area-context';
import { create } from 'zustand';

import { colors, radius, shadowStrong } from '@/theme';

import { Button } from './Button';
import type { IconName } from './bits';
import { tap } from './Press';
import { T } from './T';

interface ToastState {
  toast: { id: number; text: string; icon: IconName; tone: 'ok' | 'info' | 'warn' } | null;
  show: (text: string, opts?: { icon?: IconName; tone?: 'ok' | 'info' | 'warn' }) => void;
  hide: () => void;
}

export const useToast = create<ToastState>((set) => ({
  toast: null,
  show: (text, opts) => set({ toast: { id: Date.now(), text, icon: opts?.icon ?? 'checkmark-circle', tone: opts?.tone ?? 'ok' } }),
  hide: () => set({ toast: null }),
}));

export const toast = (text: string, opts?: { icon?: IconName; tone?: 'ok' | 'info' | 'warn' }) => {
  if (opts?.tone === 'warn') tap('warning');
  useToast.getState().show(text, opts);
};

export function ToastHost() {
  const t = useToast((s) => s.toast);
  const hide = useToast((s) => s.hide);
  const insets = useSafeAreaInsets();
  const [anim] = useState(() => new Animated.Value(0));

  useEffect(() => {
    if (!t) return;
    anim.setValue(0);
    Animated.spring(anim, { toValue: 1, useNativeDriver: Platform.OS !== 'web', friction: 8 }).start();
```

```
    const timer = setTimeout(() => {
      Animated.timing(anim, { toValue: 0, duration: 180, useNativeDriver: Platform.OS !== 'web' }).start(() => hide());
    }, 2600);
    return () => clearTimeout(timer);
  }, [t, anim, hide]);

  if (!t) return null;
  const bg = t.tone === 'warn' ? '#3A1F1F' : colors.primaryDark;
  return (
    <Animated.View
      pointerEvents="box-none"
      style={[
        styles.toastWrap,
        { top: insets.top + 10, opacity: anim, transform: [{ translateY: anim.interpolate({ inputRange: [0, 1], outputRange: [-30, 0] })
}] },
      ]}>
      <Pressable onPress={hide} style={[styles.toast, { backgroundColor: bg }]} accessibilityLiveRegion="polite" accessibilityRole="aler
t">
        <Ionicons name={t.icon} size={18} color={t.tone === 'warn' ? '#FFB4B4' : '#BFE5AE'} />
        <T variant="small" weight="medium" color={colors.white} style={{ flex: 1 }}>
          {t.text}
        </T>
      </Pressable>
    </Animated.View>
  );
}

export interface DialogAction {
  label: string;
  style?: 'default' | 'cancel' | 'destructive';
  onPress?: () => void;
}

interface DialogState {
  dialog: { title: string; message?: string; actions: DialogAction[]; icon?: IconName } | null;
  open: (d: DialogState['dialog']) => void;
  close: () => void;
}

export const useDialog = create<DialogState>((set) => ({
  dialog: null,
  open: (d) => set({ dialog: d }),
  close: () => set({ dialog: null }),
}));

export function showDialog(title: string, message?: string, actions: DialogAction[] = [{ label: 'OK' }], icon?: IconName) {
  useDialog.getState().open({ title, message, actions, icon });
}

export function confirmAction(
  title: string,
  message: string,
  onConfirm: () => void,
  opts: { confirmLabel?: string; destructive?: boolean } = {},
) {
  showDialog(
    title,
    message,
    [
      { label: 'Cancel', style: 'cancel' },
      { label: opts.confirmLabel ?? 'Confirm', style: opts.destructive ? 'destructive' : 'default', onPress: onConfirm },
    ],
    opts.destructive ? 'warning-outline' : 'help-circle-outline',
  );
}

export function DialogHost() {
  const d = useDialog((s) => s.dialog);
  const close = useDialog((s) => s.close);
  if (!d) return null;
  return (
    <Modal transparent visible animationType="fade" onRequestClose={close} statusBarTranslucent>
      <View style={styles.dialogBackdrop}>
        <View style={styles.dialog} accessibilityViewIsModal accessibilityRole="alert">
          {d.icon ? (
            <View style={styles.dialogIcon}>
              <Ionicons name={d.icon} size={24} color={colors.primary} />
            </View>
          ) : null}
          <T variant="h3" center>
            {d.title}
          </T>
          {d.message ? (
            <T variant="bodySm" color={colors.textSecondary} center style={{ marginTop: 6 }}>
              {d.message}
            </T>
```

```
          ) : null}
          <View style={{ gap: 8, marginTop: 18, alignSelf: 'stretch' }}>
            {d.actions.map((a) => (
              <Button
                key={a.label}
                label={a.label}
                size="md"
                variant={a.style === 'cancel' ? 'secondary' : a.style === 'destructive' ? 'danger' : 'primary'}
                onPress={() => {
                  close();
                  a.onPress?.();
                }}
              />
            ))}
          </View>
        </View>
      </View>
    </Modal>
  );
}

const styles = StyleSheet.create({
  toastWrap: { position: 'absolute', left: 16, right: 16, alignItems: 'center', zIndex: 1000 },
  toast: {
    flexDirection: 'row',
    alignItems: 'center',
    gap: 10,
    paddingHorizontal: 16,
    paddingVertical: 13,
    borderRadius: radius.md,
    maxWidth: 480,
    width: '100%',
    ...shadowStrong,
  },
  dialogBackdrop: { flex: 1, backgroundColor: colors.overlay, alignItems: 'center', justifyContent: 'center', padding: 28 },
  dialog: {
    width: '100%',
    maxWidth: 360,
    backgroundColor: colors.card,
    borderRadius: radius.xl,
    padding: 22,
    alignItems: 'center',
    ...shadowStrong,
  },
  dialogIcon: {
    width: 48,
    height: 48,
    borderRadius: 24,
    backgroundColor: colors.primarySofter,
    alignItems: 'center',
    justifyContent: 'center',
    marginBottom: 10,
  },
});
```

src/components/ui/motion.tsx

```
/**
 * Small motion kit built on React Native's Animated API.
 * Every animation respects the "Animations" switch in Make it yours / Settings
 * (people who get motion-sick, or just want a calmer app, can turn it off).
 */
import { useEffect, useState, type ReactNode } from 'react';
import { Animated, Easing, Platform, type StyleProp, type ViewStyle } from 'react-native';

import { effectiveMotion, useOsReduceMotion } from '@/hooks/useReduceMotion';
import { useAppStore } from '@/store/useAppStore';

export const nativeDriver = Platform.OS !== 'web';

/** True when animations should run: the "Animations" switch is on and the phone's Reduce Motion setting is off. */
export function useMotion() {
  const wanted = useAppStore((s) => s.personal.motion);
  const osReduce = useOsReduceMotion((s) => s.on);
  return effectiveMotion(wanted, osReduce);
}

/** Fades + slides its children in once, after `delay` ms. Use for staggered entrances. */
export function FadeIn({
  children,
  delay = 0,
  from = 'up',
  distance = 18,
  duration = 480,
  style,
}: {
```

```
  children: ReactNode;
  delay?: number;
  from?: 'up' | 'down' | 'left' | 'right' | 'scale';
  distance?: number;
  duration?: number;
  style?: StyleProp<ViewStyle>;
}) {
  const motion = useMotion();
  const [v] = useState(() => new Animated.Value(motion ? 0 : 1));
  useEffect(() => {
    if (!motion) {
      v.setValue(1); // motion was switched off (or the phone's setting arrived) after the first frame: show the content now
      return;
    }
    const anim = Animated.timing(v, {
      toValue: 1,
      duration,
      delay,
      easing: Easing.out(Easing.cubic),
      useNativeDriver: nativeDriver,
    });
    anim.start();
    return () => anim.stop();
  }, [v, delay, duration, motion]);

  const offset = v.interpolate({
    inputRange: [0, 1],
    outputRange: [from === 'down' || from === 'right' ? -distance : distance, 0],
  });
  const transform =
    from === 'scale'
      ? [
          {
            scale: v.interpolate({
              inputRange: [0, 1],
              outputRange: [0.86, 1],
            }),
          },
        ]
      : from === 'left' || from === 'right'
        ? [{ translateX: offset }]
        : [{ translateY: offset }];
  return <Animated.View style={[style, { opacity: v, transform }]}>{children}</Animated.View>;
}

/** A 0 → 1 value that loops forever (or sits at `rest` when motion is off). */
export function useLoop(duration: number, { delay = 0, rest = 0.5, pingPong = false }: { delay?: number; rest?: number; pingPong?: boole
an } = {}) {
  const motion = useMotion();
  const [v] = useState(() => new Animated.Value(motion ? 0 : rest));
  useEffect(() => {
    if (!motion) {
      v.setValue(rest);
      return;
    }
    v.setValue(0);
    const one = Animated.timing(v, {
      toValue: 1,
      duration,
      easing: pingPong ? Easing.inOut(Easing.sin) : Easing.linear,
      useNativeDriver: nativeDriver,
    });
    const seq = pingPong
      ? Animated.sequence([
          one,
          Animated.timing(v, {
            toValue: 0,
            duration,
            easing: Easing.inOut(Easing.sin),
            useNativeDriver: nativeDriver,
          }),
        ])
      : Animated.sequence([
          one,
          Animated.timing(v, {
            toValue: 0,
            duration: 0,
            useNativeDriver: nativeDriver,
          }),
        ]);
    const loop = Animated.loop(seq);
    const timer = setTimeout(() => loop.start(), delay);
    return () => {
      clearTimeout(timer);
      loop.stop();
    };
```

```
  }, [v, duration, delay, rest, pingPong, motion]);
  return v;
}

/** Gently bobs its children up and down. */
export function Float({ children, amount = 6, duration = 2200, delay = 0, style }: { children: ReactNode; amount?: number; duration?: nu
mber; delay?: number; style?: StyleProp<ViewStyle> }) {
  const v = useLoop(duration, { delay, pingPong: true });
  const translateY = v.interpolate({
    inputRange: [0, 1],
    outputRange: [-amount / 2, amount / 2],
  });
  return <Animated.View style={[style, { transform: [{ translateY }] }]}>{children}</Animated.View>;
}

/** Soft pulsing glow (scale + opacity) — great behind icons and "live" dots. */
export function Pulse({ children, scale = 1.12, duration = 1400, style }: { children?: ReactNode; scale?: number; duration?: number; sty
le?: StyleProp<ViewStyle> }) {
  const v = useLoop(duration, { pingPong: true, rest: 0 });
  return (
    <Animated.View
      style={[
        style,
        {
          opacity: v.interpolate({
            inputRange: [0, 1],
            outputRange: [1, 0.75],
          }),
          transform: [
            {
              scale: v.interpolate({
                inputRange: [0, 1],
                outputRange: [1, scale],
              }),
            },
          ],
        },
      ]}>
      {children}
    </Animated.View>
  );
}
```

src/hooks/__tests__/useReduceMotion.test.ts

```
import { effectiveMotion } from '../useReduceMotion';

describe('animations', () => {
  it('run only when wanted and the phone does not ask for reduced motion', () => {
    expect(effectiveMotion(true, false)).toBe(true);
    expect(effectiveMotion(true, true)).toBe(false); // the phone's setting wins
    expect(effectiveMotion(false, false)).toBe(false);
    expect(effectiveMotion(false, true)).toBe(false);
  });
});
```

src/hooks/useReduceMotion.ts

```
import { useEffect } from 'react';
import { AccessibilityInfo } from 'react-native';
import { create } from 'zustand';

/** Whether the phone's own "Reduce Motion" accessibility setting is on. */
export const useOsReduceMotion = create<{ on: boolean }>(() => ({ on: false }));

/** Animations run only when the person wants them AND the phone is not asking apps to calm down. */
export const effectiveMotion = (wanted: boolean, osReduce: boolean) => wanted && !osReduce;

/** Call once near the root: reads the setting and follows changes while the app is open. */
export function useWatchOsReduceMotion() {
  useEffect(() => {
    let alive = true;
    AccessibilityInfo.isReduceMotionEnabled()
      .then((on) => alive && useOsReduceMotion.setState({ on }))
      .catch(() => {});
    const subscription = AccessibilityInfo.addEventListener('reduceMotionChanged', (on) => useOsReduceMotion.setState({ on }));
    return () => {
      alive = false;
      subscription.remove();
    };
  }, []);
}
```

src/hooks/useTrip.ts

```
import { useEffect, useState } from 'react';

import { destinationById } from '@/data/destinations';
import { fetchRates } from '@/services/currency';
import { fetchWeather } from '@/services/weather';
import { useAppStore } from '@/store/useAppStore';
import type { Trip, WeatherSnapshot } from '@/store/types';

/** Cached weather for a destination; refreshes in the background when online. */
export function useWeather(destinationId: string | undefined): WeatherSnapshot | undefined {
  const snap = useAppStore((s) => (destinationId ? s.weather[destinationId] : undefined));
  useEffect(() => {
    if (destinationId) fetchWeather(destinationId).catch(() => {});
  }, [destinationId]);
  return snap;
}

export function useTripDestination(trip: Trip | undefined) {
  return trip ? destinationById(trip.destinationId) : undefined;
}

/** Re-renders every `ms` so countdowns and "time ago" labels stay live. */
export function useNow(ms = 30000) {
  const [now, setNow] = useState(() => Date.now());
  useEffect(() => {
    const id = setInterval(() => setNow(Date.now()), ms);
    return () => clearInterval(id);
  }, [ms]);
  return now;
}

export function useRatesLoader() {
  useEffect(() => {
    fetchRates().catch(() => {});
  }, []);
}
```

src/i18n/index.ts

```
import { useCallback } from 'react';

import { useAppStore } from '@/store/useAppStore';

import { dictionaries, english, type LangCode, type StringKey } from './strings';

export { LANGUAGES } from './strings';
export type { LangCode, StringKey } from './strings';

export function translate(lang: LangCode, key: StringKey | string, vars?: Record<string, string | number>): string {
  const dict = dictionaries[lang] ?? {};
  let s: string = (dict as Record<string, string>)[key] ?? (english as Record<string, string>)[key] ?? key;
  if (vars) for (const [k, v] of Object.entries(vars)) s = s.replace(new RegExp(`\\{${k}\\}`, 'g'), String(v));
  return s;
}

/** `const t = useT(); t('nav.explore')` — re-renders when the language changes. */
export function useT() {
  const lang = useAppStore((s) => s.settings.language);
  return useCallback((key: StringKey | string, vars?: Record<string, string | number>) => translate(lang, key, vars), [lang]);
}
```

src/i18n/strings.ts

```
/* UI strings for the six supported languages. English is the source of truth;
   missing keys in other languages fall back to English. */

export const LANGUAGES = [
  { code: 'en', label: 'English', native: 'English', flag: '🇺🇸' },
  { code: 'es', label: 'Spanish', native: 'Español', flag: '🇪🇸' },
  { code: 'fr', label: 'French', native: 'Français', flag: '🇫🇷' },
  { code: 'ja', label: 'Japanese', native: '日本語', flag: '🇯🇵' },
  { code: 'hi', label: 'Hindi', native: 'हिन्दी', flag: '🇮🇳' },
  { code: 'zh', label: 'Chinese', native: '中文', flag: '🇨🇳' },
] as const;

export type LangCode = (typeof LANGUAGES)[number]['code'];

const en = {
  'nav.explore': 'Home',
  'nav.itinerary': 'Plan',
  'nav.budget': 'Budget',
  'nav.chat': 'Chat',
  'nav.profile': 'More',
```

```
'greet.morning': 'Good morning',
'greet.afternoon': 'Good afternoon',
'greet.evening': 'Good evening',
'greet.night': 'Hello night owl',

'explore.title': 'Find Your Perfect\nDestination',
'explore.subtitle': 'Plan, explore, and travel together.',
'explore.search': 'Search a city or destination…',
'explore.popular': 'Popular Places',
'explore.featured': 'Featured Destination',
'explore.seeAll': 'See all',
'explore.matches': '{n} matches',
'explore.nearby': 'Nearby within radius',
'explore.activeTrip': 'Active trip',
'explore.upcomingTrip': 'Upcoming trip',
'explore.pastTrip': 'Completed trip',
'explore.left': '{amount} left',
'explore.members': '{n} members',
'explore.reminder': 'Reminder',
'explore.reminderSub': 'We'll ping everyone {n} min before.',
'explore.nearestAirport': 'Nearest airport',
'explore.addToTrip': 'Add to Trip',
'explore.noResults': 'No destinations match "{q}"',
'explore.tools': 'Trip tools',

'cat.all': 'All',
'cat.best': 'Best place',
'cat.adventure': 'Adventures',
'cat.food': 'Food',
'cat.culture': 'Culture',
'cat.beach': 'Beaches',
'cat.nature': 'Nature',
'cat.nightlife': 'Nightlife',
'cat.sightseeing': 'Sightseeing',
'cat.shopping': 'Shopping',
'cat.relax': 'Relax',

'len.any': 'Any length',
'len.u1': 'Under 1 hour',
'len.1-2': '1–2 hours',
'len.2-4': '2–4 hours',
'len.half': 'Half day',
'len.full': 'Full day',

'itin.members': '{n} members',
'itin.noActivities': 'No activities yet',
'itin.noActivitiesSub': 'Tap the + button to add your\nfirst activity for this day.',
'itin.noMatch': 'No activities match these filters',
'itin.addActivity': 'Add Activity',
'itin.flights': 'Flights & arrivals',
'itin.flightsSub': 'Times shown in {zone}. Tap to switch time zone.',
'itin.addFlight': 'Add my flight',
'itin.assigned': '{n} assigned',
'itin.confirmed': 'Confirmed',
'itin.voting': 'Voting',
'itin.pending': 'Pending',
'itin.perPerson': '/person',

'add.title': 'Add Activity',
'add.subtitle': 'Fill in the details for your new activity.',
'add.type': 'Activity Type',
'add.time': 'Time',
'add.name': 'Activity Name',
'add.namePh': 'e.g. Fushimi Inari Shrine visit',
'add.location': 'Location',
'add.locationPh': 'e.g. Fushimi, Kyoto',
'add.duration': 'Duration',
'add.cost': 'Cost per person (USD)',
'add.notes': 'Notes (optional)',
'add.notesPh': 'Any notes for the group…',
'add.vote': 'Put it to a group vote',
'add.remind': 'Remind everyone before it starts',
'add.submit': 'Add to Itinerary',
'add.save': 'Save changes',

'budget.kicker': 'Group spending',
'budget.title': 'Budget',
'budget.spent': 'Spent so far',
'budget.of': 'of {total} · {per} per person',
'budget.used': '{pct}% of the group budget used',
'budget.breakdown': 'Spending breakdown',
'budget.currency': 'Currency',
'budget.expenses': 'Expenses',
'budget.cap': 'Budget cap',
'budget.addExpense': 'Add an expense',
```

```
  'budget.whatFor': 'What was it for?',
  'budget.add': 'Add expense',
  'budget.settle': 'Settle up',
  'budget.planner': 'Daily spending planner',
  'budget.paidBy': 'paid by {name}',

  'chat.kicker': '{n} travellers',
  'chat.title': 'Trip chat',
  'chat.placeholder': 'Message your trip pals…',
  'chat.askAi': 'Ask the Trip Assistant…',
  'chat.assistant': 'Trip Assistant',
  'chat.typing': '{name} is typing…',
  'chat.q.budget': 'How’s the budget?',
  'chat.q.food': 'Best food nearby?',
  'chat.q.weather': 'What’s the weather?',
  'chat.q.next': 'What’s next?',

  'hub.kicker': 'Trip hub',
  'hub.title': 'Your group',
  'hub.invite': 'Invite code',
  'hub.share': 'Share',
  'hub.members': 'Members & roles',
  'hub.vote': 'Vote on activities',
  'hub.checklist': 'Checklist',
  'hub.done': '{a}/{b} done',
  'hub.addTask': 'Add a task',
  'hub.documents': 'Documents',
  'hub.documentsSub': 'Marked items stay readable without wifi or data.',
  'hub.offline': 'Offline',
  'hub.tools': 'More for this trip',

  'auth.welcome': 'Welcome back!',
  'auth.welcomeSub': 'Sign in to continue planning your adventures.',
  'auth.email': 'Email Address',
  'auth.password': 'Password',
  'auth.remember': 'Remember me',
  'auth.forgot': 'Forgot password?',
  'auth.signIn': 'Sign In',
  'auth.or': 'or continue with',
  'auth.google': 'Continue with Google',
  'auth.noAccount': 'Don’t have an account?',
  'auth.signUp': 'Sign Up',
  'auth.demo': 'Demo Account',
  'auth.terms': 'By continuing, you agree to our Terms of Service and Privacy Policy',

  'common.cancel': 'Cancel',
  'common.save': 'Save',
  'common.delete': 'Delete',
  'common.done': 'Done',
  'common.next': 'Next',
  'common.getStarted': 'Get Started',
  'common.skip': 'Skip',
  'common.back': 'Back',
  'common.edit': 'Edit',
  'common.close': 'Close',

  'settings.title': 'Settings',
  'settings.language': 'Language',
  'settings.currency': 'Home currency',
  'settings.notifications': 'Activity reminders',
  'settings.lead': 'Remind me before',
  'settings.textSize': 'Text size',
  'settings.signOut': 'Sign out',
};

export type StringKey = keyof typeof en;
type Dict = Partial<Record<StringKey, string>>;

const es: Dict = {
  'nav.explore': 'Inicio', 'nav.itinerary': 'Plan', 'nav.budget': 'Presupuesto', 'nav.chat': 'Chat', 'nav.profile': 'Más',
  'greet.morning': 'Buenos días', 'greet.afternoon': 'Buenas tardes', 'greet.evening': 'Buenas noches', 'greet.night': 'Hola, noctámbul
o',
  'explore.title': 'Encuentra tu\ndestino perfecto', 'explore.subtitle': 'Planea, explora y viaja en grupo.',
  'explore.search': 'Busca una ciudad o destino…', 'explore.popular': 'Lugares populares', 'explore.featured': 'Destino destacado',
  'explore.seeAll': 'Ver todo', 'explore.matches': '{n} resultados', 'explore.nearby': 'Cerca dentro del radio',
  'explore.activeTrip': 'Viaje activo', 'explore.upcomingTrip': 'Próximo viaje', 'explore.pastTrip': 'Viaje completado',
  'explore.left': 'Quedan {amount}', 'explore.members': '{n} miembros', 'explore.reminder': 'Recordatorio',
  'explore.reminderSub': 'Avisaremos a todos {n} min antes.', 'explore.nearestAirport': 'Aeropuerto más cercano',
  'explore.addToTrip': 'Añadir al viaje', 'explore.noResults': 'Ningún destino coincide con “{q}”', 'explore.tools': 'Herramientas',
  'cat.all': 'Todo', 'cat.best': 'Lo mejor', 'cat.adventure': 'Aventuras', 'cat.food': 'Comida', 'cat.culture': 'Cultura',
  'cat.beach': 'Playas', 'cat.nature': 'Naturaleza', 'cat.nightlife': 'Vida nocturna', 'cat.sightseeing': 'Turismo', 'cat.shopping': 'Co
mpras', 'cat.relax': 'Relax',
  'len.any': 'Cualquier duración', 'len.u1': 'Menos de 1 hora', 'len.1-2': '1–2 horas', 'len.2-4': '2–4 horas', 'len.half': 'Medio día',
'len.full': 'Día completo',
  'itin.members': '{n} miembros', 'itin.noActivities': 'Aún no hay actividades', 'itin.noActivitiesSub': 'Toca + para añadir la primera
\nactividad del día.',
```

```
  'itin.noMatch': 'Ninguna actividad coincide', 'itin.addActivity': 'Añadir actividad', 'itin.flights': 'Vuelos y llegadas',
  'itin.flightsSub': 'Horas en {zone}. Toca para cambiar la zona horaria.', 'itin.addFlight': 'Añadir mi vuelo', 'itin.assigned': '{n} a
signados',
  'itin.confirmed': 'Confirmado', 'itin.voting': 'Votación', 'itin.pending': 'Pendiente', 'itin.perPerson': '/persona',
  'add.title': 'Añadir actividad', 'add.subtitle': 'Completa los detalles de tu actividad.', 'add.type': 'Tipo de actividad', 'add.tim
e': 'Hora',
  'add.name': 'Nombre de la actividad', 'add.location': 'Ubicación', 'add.duration': 'Duración', 'add.cost': 'Coste por persona (USD)',
  'add.notes': 'Notas (opcional)', 'add.notesPh': 'Notas para el grupo…', 'add.vote': 'Someter a votación', 'add.remind': 'Recordar a to
dos antes',
  'add.submit': 'Añadir al itinerario', 'add.save': 'Guardar cambios',
  'budget.kicker': 'Gastos del grupo', 'budget.title': 'Presupuesto', 'budget.spent': 'Gastado hasta ahora', 'budget.of': 'de {total} ·
{per} por persona',
  'budget.used': '{pct}% del presupuesto usado', 'budget.breakdown': 'Desglose de gastos', 'budget.currency': 'Moneda', 'budget.expense
s': 'Gastos',
  'budget.cap': 'Límite', 'budget.addExpense': 'Añadir un gasto', 'budget.whatFor': '¿Para qué fue?', 'budget.add': 'Añadir gasto',
  'budget.settle': 'Saldar cuentas', 'budget.planner': 'Planificador diario', 'budget.paidBy': 'pagado por {name}',
  'chat.kicker': '{n} viajeros', 'chat.title': 'Chat del viaje', 'chat.placeholder': 'Escribe a tu grupo…', 'chat.askAi': 'Pregunta al a
sistente…',
  'chat.assistant': 'Asistente de viaje', 'chat.typing': '{name} está escribiendo…', 'chat.q.budget': '¿Cómo va el presupuesto?',
  'chat.q.food': '¿Buena comida cerca?', 'chat.q.weather': '¿Qué tiempo hace?', 'chat.q.next': '¿Qué sigue?',
  'hub.kicker': 'Centro del viaje', 'hub.title': 'Tu grupo', 'hub.invite': 'Código de invitación', 'hub.share': 'Compartir',
  'hub.members': 'Miembros y roles', 'hub.vote': 'Vota actividades', 'hub.checklist': 'Lista de tareas', 'hub.done': '{a}/{b} hechas',
  'hub.addTask': 'Añadir tarea', 'hub.documents': 'Documentos', 'hub.documentsSub': 'Disponibles sin wifi ni datos.', 'hub.offline': 'Si
n conexión', 'hub.tools': 'Más para este viaje',
  'auth.welcome': '¡Bienvenido de nuevo!', 'auth.welcomeSub': 'Inicia sesión para seguir planeando.', 'auth.email': 'Correo electrónic
o',
  'auth.password': 'Contraseña', 'auth.remember': 'Recordarme', 'auth.forgot': '¿Olvidaste tu contraseña?', 'auth.signIn': 'Iniciar sesi
ón',
  'auth.or': 'o continúa con', 'auth.google': 'Continuar con Google', 'auth.noAccount': '¿No tienes cuenta?', 'auth.signUp': 'Regístrat
e',
  'auth.demo': 'Cuenta de demostración', 'auth.terms': 'Al continuar, aceptas los Términos y la Política de privacidad',
  'common.cancel': 'Cancelar', 'common.save': 'Guardar', 'common.delete': 'Eliminar', 'common.done': 'Listo', 'common.next': 'Siguient
e',
  'common.getStarted': 'Empezar', 'common.skip': 'Omitir', 'common.back': 'Atrás', 'common.edit': 'Editar', 'common.close': 'Cerrar',
  'settings.title': 'Ajustes', 'settings.language': 'Idioma', 'settings.currency': 'Moneda local', 'settings.notifications': 'Recordator
ios',
  'settings.lead': 'Avisarme antes', 'settings.textSize': 'Tamaño del texto', 'settings.signOut': 'Cerrar sesión',
};

const fr: Dict = {
  'nav.explore': 'Accueil', 'nav.itinerary': 'Plan', 'nav.budget': 'Budget', 'nav.chat': 'Discussion', 'nav.profile': 'Plus',
  'greet.morning': 'Bonjour', 'greet.afternoon': 'Bon après-midi', 'greet.evening': 'Bonsoir', 'greet.night': 'Bonsoir, oiseau de nuit',
  'explore.title': 'Trouvez votre\ndestination idéale', 'explore.subtitle': 'Planifiez, explorez et voyagez ensemble.',
  'explore.search': 'Rechercher une ville ou destination…', 'explore.popular': 'Lieux populaires', 'explore.featured': 'Destination à la
une',
  'explore.seeAll': 'Tout voir', 'explore.matches': '{n} résultats', 'explore.nearby': 'À proximité dans le rayon',
  'explore.activeTrip': 'Voyage en cours', 'explore.upcomingTrip': 'Voyage à venir', 'explore.pastTrip': 'Voyage terminé',
  'explore.left': '{amount} restants', 'explore.members': '{n} membres', 'explore.reminder': 'Rappel', 'explore.reminderSub': 'On prévie
nt tout le monde {n} min avant.',
  'explore.nearestAirport': 'Aéroport le plus proche', 'explore.addToTrip': 'Ajouter au voyage', 'explore.noResults': 'Aucune destinatio
n pour « {q} »', 'explore.tools': 'Outils du voyage',
  'cat.all': 'Tout', 'cat.best': 'Incontournables', 'cat.adventure': 'Aventures', 'cat.food': 'Gastronomie', 'cat.culture': 'Culture',
  'cat.beach': 'Plages', 'cat.nature': 'Nature', 'cat.nightlife': 'Vie nocturne', 'cat.sightseeing': 'Visites', 'cat.shopping': 'Shoppin
g', 'cat.relax': 'Détente',
  'len.any': 'Toute durée', 'len.u1': 'Moins d’1 heure', 'len.1-2': '1–2 heures', 'len.2-4': '2–4 heures', 'len.half': 'Demi-journée',
'len.full': 'Journée entière',
  'itin.members': '{n} membres', 'itin.noActivities': 'Aucune activité', 'itin.noActivitiesSub': 'Touchez + pour ajouter la première\nac
tivité de la journée.',
  'itin.noMatch': 'Aucune activité ne correspond', 'itin.addActivity': 'Ajouter', 'itin.flights': 'Vols et arrivées',
  'itin.flightsSub': 'Heures affichées en {zone}. Touchez pour changer.', 'itin.addFlight': 'Ajouter mon vol', 'itin.assigned': '{n} ass
ignés',
  'itin.confirmed': 'Confirmé', 'itin.voting': 'Vote', 'itin.pending': 'En attente', 'itin.perPerson': '/pers.',
  'add.title': 'Ajouter une activité', 'add.subtitle': 'Renseignez les détails de l’activité.', 'add.type': 'Type d’activité', 'add.tim
e': 'Heure',
  'add.name': 'Nom de l’activité', 'add.location': 'Lieu', 'add.duration': 'Durée', 'add.cost': 'Coût par personne (USD)', 'add.notes':
'Notes (facultatif)',
  'add.notesPh': 'Des notes pour le groupe…', 'add.vote': 'Soumettre au vote', 'add.remind': 'Rappeler à tous avant le début', 'add.subm
it': 'Ajouter à l’itinéraire', 'add.save': 'Enregistrer',
  'budget.kicker': 'Dépenses du groupe', 'budget.title': 'Budget', 'budget.spent': 'Dépensé', 'budget.of': 'sur {total} · {per} par pers
onne',
  'budget.used': '{pct} % du budget utilisé', 'budget.breakdown': 'Répartition', 'budget.currency': 'Devise', 'budget.expenses': 'Dépens
es', 'budget.cap': 'Plafond',
  'budget.addExpense': 'Ajouter une dépense', 'budget.whatFor': 'Pour quoi ?', 'budget.add': 'Ajouter', 'budget.settle': 'Régler les com
ptes',
  'budget.planner': 'Planificateur quotidien', 'budget.paidBy': 'payé par {name}',
  'chat.kicker': '{n} voyageurs', 'chat.title': 'Discussion', 'chat.placeholder': 'Écrivez au groupe…', 'chat.askAi': 'Demandez à l’assi
stant…',
  'chat.assistant': 'Assistant voyage', 'chat.typing': '{name} écrit…', 'chat.q.budget': 'Où en est le budget ?', 'chat.q.food': 'Bonne
cuisine près d’ici ?',
  'chat.q.weather': 'Quel temps fait-il ?', 'chat.q.next': 'Et ensuite ?',
  'hub.kicker': 'Espace voyage', 'hub.title': 'Votre groupe', 'hub.invite': 'Code d’invitation', 'hub.share': 'Partager', 'hub.members':
'Membres et rôles',
  'hub.vote': 'Votez pour les activités', 'hub.checklist': 'Liste de tâches', 'hub.done': '{a}/{b} faites', 'hub.addTask': 'Ajouter une
tâche',
  'hub.documents': 'Documents', 'hub.documentsSub': 'Lisibles sans wifi ni données.', 'hub.offline': 'Hors ligne', 'hub.tools': 'Plus po
ur ce voyage',
```

```
  'auth.welcome': 'Bon retour !', 'auth.welcomeSub': 'Connectez-vous pour continuer à planifier.', 'auth.email': 'Adresse e-mail', 'auth.password': 'Mot de passe',
  'auth.remember': 'Se souvenir de moi', 'auth.forgot': 'Mot de passe oublié ?', 'auth.signIn': 'Se connecter', 'auth.or': 'ou continuer avec',
  'auth.google': 'Continuer avec Google', 'auth.noAccount': 'Pas de compte ?', 'auth.signUp': 'S’inscrire', 'auth.demo': 'Compte de démo',
  'auth.terms': 'En continuant, vous acceptez nos Conditions et notre Politique de confidentialité',
  'common.cancel': 'Annuler', 'common.save': 'Enregistrer', 'common.delete': 'Supprimer', 'common.done': 'Terminé', 'common.next': 'Suivant',
  'common.getStarted': 'Commencer', 'common.skip': 'Passer', 'common.back': 'Retour', 'common.edit': 'Modifier', 'common.close': 'Fermer',
  'settings.title': 'Réglages', 'settings.language': 'Langue', 'settings.currency': 'Devise principale', 'settings.notifications': 'Rappels d’activité',
  'settings.lead': 'Me rappeler avant', 'settings.textSize': 'Taille du texte', 'settings.signOut': 'Se déconnecter',
};

const ja: Dict = {
  'nav.explore': 'ホーム', 'nav.itinerary': '計画', 'nav.budget': '予算', 'nav.chat': 'チャット', 'nav.profile': 'その他',
  'greet.morning': 'おはようございます', 'greet.afternoon': 'こんにちは', 'greet.evening': 'こんばんは', 'greet.night': 'こんばんは',
  'explore.title': 'ぴったりの\n旅先を見つけよう', 'explore.subtitle': 'みんなで計画して、探して、旅しよう。', 'explore.search': '都市や旅先を検索…',
  'explore.popular': '人気の場所', 'explore.featured': '注目の旅先', 'explore.seeAll': 'すべて見る', 'explore.matches': '{n}件',
  'explore.nearby': '半径内のスポット', 'explore.activeTrip': '旅行中', 'explore.upcomingTrip': '次の旅行', 'explore.pastTrip': '完了した旅行',
  'explore.left': '残り{amount}', 'explore.members': '{n}人', 'explore.reminder': 'リマインダー', 'explore.reminderSub': '{n}分前に全員に通知します。',
  'explore.nearestAirport': '最寄りの空港', 'explore.addToTrip': '旅行に追加', 'explore.noResults': '「{q}」に一致する旅先はありません', 'explore.tools': '旅のツール',
  'cat.all': 'すべて', 'cat.best': 'おすすめ', 'cat.adventure': 'アドベンチャー', 'cat.food': 'グルメ', 'cat.culture': '文化', 'cat.beach': 'ビーチ',
  'cat.nature': '自然', 'cat.nightlife': 'ナイトライフ', 'cat.sightseeing': '観光', 'cat.shopping': 'ショッピング', 'cat.relax': 'リラックス',
  'len.any': 'すべての長さ', 'len.u1': '1時間未満', 'len.1-2': '1〜2時間', 'len.2-4': '2〜4時間', 'len.half': '半日', 'len.full': '1日',
  'itin.members': '{n}人', 'itin.noActivities': 'まだ予定はありません', 'itin.noActivitiesSub': '＋ボタンでこの日の\n最初の予定を追加しましょう。',
  'itin.noMatch': '条件に合う予定はありません', 'itin.addActivity': '予定を追加', 'itin.flights': 'フライトと到着', 'itin.flightsSub': '{zone}で表示中。タップで切替。',
  'itin.addFlight': 'フライトを追加', 'itin.assigned': '{n}人担当', 'itin.confirmed': '確定', 'itin.voting': '投票中', 'itin.pending': '保留', 'itin.perPerson': '/人',
  'add.title': '予定を追加', 'add.subtitle': '新しい予定の詳細を入力してください。', 'add.type': '種類', 'add.time': '時間', 'add.name': '予定の名前',
  'add.location': '場所', 'add.duration': '所要時間', 'add.cost': '1人あたりの費用 (USD)', 'add.notes': 'メモ（任意）', 'add.notesPh': 'グループへのメモ…',
  'add.vote': 'グループ投票にする', 'add.remind': '開始前に全員に通知', 'add.submit': '旅程に追加', 'add.save': '変更を保存',
  'budget.kicker': 'グループの支出', 'budget.title': '予算', 'budget.spent': 'これまでの支出', 'budget.of': '{total}中 · 1人あたり{per}',
  'budget.used': 'グループ予算の{pct}%を使用', 'budget.breakdown': '支出の内訳', 'budget.currency': '通貨', 'budget.expenses': '支出', 'budget.cap': '予算上限',
  'budget.addExpense': '支出を追加', 'budget.whatFor': '何の支出？', 'budget.add': '支出を追加', 'budget.settle': '精算', 'budget.planner': '1日の予算プランナー', 'budget.paidBy': '{name}が支払い',
  'chat.kicker': '旅行者{n}人', 'chat.title': '旅のチャット', 'chat.placeholder': '旅の仲間にメッセージ…', 'chat.askAi': 'アシスタントに質問…',
  'chat.assistant': '旅のアシスタント', 'chat.typing': '{name}が入力中…', 'chat.q.budget': '予算はどう？', 'chat.q.food': '近くのおすすめグルメは？',
  'chat.q.weather': '天気は？', 'chat.q.next': '次の予定は？',
  'hub.kicker': '旅のハブ', 'hub.title': 'あなたのグループ', 'hub.invite': '招待コード', 'hub.share': '共有', 'hub.members': 'メンバーと役割',
  'hub.vote': 'アクティビティに投票', 'hub.checklist': 'チェックリスト', 'hub.done': '{a}/{b} 完了', 'hub.addTask': 'タスクを追加',
  'hub.documents': '書類', 'hub.documentsSub': 'Wi-Fiや通信なしでも閲覧できます。', 'hub.offline': 'オフライン', 'hub.tools': 'この旅のツール',
  'auth.welcome': 'おかえりなさい！', 'auth.welcomeSub': 'ログインして旅の計画を続けましょう。', 'auth.email': 'メールアドレス', 'auth.password': 'パスワード',
  'auth.remember': 'ログイン状態を保持', 'auth.forgot': 'パスワードをお忘れですか？', 'auth.signIn': 'ログイン', 'auth.or': 'または',
  'auth.google': 'Googleで続行', 'auth.noAccount': 'アカウントをお持ちでないですか？', 'auth.signUp': '新規登録', 'auth.demo': 'デモアカウント',
  'auth.terms': '続行すると、利用規約とプライバシーポリシーに同意したことになります',
  'common.cancel': 'キャンセル', 'common.save': '保存', 'common.delete': '削除', 'common.done': '完了', 'common.next': '次へ', 'common.getStarted': 'はじめる',
  'common.skip': 'スキップ', 'common.back': '戻る', 'common.edit': '編集', 'common.close': '閉じる',
  'settings.title': '設定', 'settings.language': '言語', 'settings.currency': '自国通貨', 'settings.notifications': '予定のリマインダー',
  'settings.lead': '通知のタイミング', 'settings.textSize': '文字サイズ', 'settings.signOut': 'ログアウト',
};

const hi: Dict = {
  'nav.explore': 'होम', 'nav.itinerary': 'योजना', 'nav.budget': 'बजट', 'nav.chat': 'चैट', 'nav.profile': 'और',
  'greet.morning': 'सुप्रभात', 'greet.afternoon': 'नमस्ते', 'greet.evening': 'शुभ संध्या', 'greet.night': 'नमस्ते',
  'explore.title': 'अपनी परफ़ेक्ट\nमंज़िल खोजें', 'explore.subtitle': 'साथ मिलकर योजना बनाएँ, घूमें और यात्रा करें।', 'explore.search': 'शहर या जगह खोजें…',
  'explore.popular': 'लोकप्रिय जगहें', 'explore.featured': 'विशेष मंज़िल', 'explore.seeAll': 'सभी देखें', 'explore.matches': '{n} परिणाम',
  'explore.nearby': 'दायरे में आस-पास', 'explore.activeTrip': 'चालू यात्रा', 'explore.upcomingTrip': 'आगामी यात्रा', 'explore.pastTrip': 'पूरी हुई यात्रा',
  'explore.left': '{amount} बाकी', 'explore.members': '{n} सदस्य', 'explore.reminder': 'रिमाइंडर', 'explore.reminderSub': 'हम सबको {n} मिनट पहले याद दिलाएँगे।',
  'explore.nearestAirport': 'नज़दीकी हवाई अड्डा', 'explore.addToTrip': 'यात्रा में जोड़ें', 'explore.noResults': '“{q}” से कोई मंज़िल नहीं मिली', 'explore.tools': 'यात्रा टूल',
  'cat.all': 'सभी', 'cat.best': 'सर्वश्रेष्ठ', 'cat.adventure': 'रोमांच', 'cat.food': 'खाना', 'cat.culture': 'संस्कृति', 'cat.beach': 'समुद्र तट',
  'cat.nature': 'प्रकृति', 'cat.nightlife': 'नाइटलाइफ़', 'cat.sightseeing': 'दर्शनीय स्थल', 'cat.shopping': 'खरीदारी', 'cat.relax': 'आराम',
  'len.any': 'कोई भी अवधि', 'len.u1': '1 घंटे से कम', 'len.1-2': '1–2 घंटे', 'len.2-4': '2–4 घंटे', 'len.half': 'आधा दिन', 'len.full': 'पूरा दिन',
  'itin.members': '{n} सदस्य', 'itin.noActivities': 'अभी कोई गतिविधि नहीं', 'itin.noActivitiesSub': 'इस दिन की पहली गतिविधि जोड़ने के लिए\n+ बटन दबाएँ।',
  'itin.noMatch': 'इन फ़िल्टर से कोई गतिविधि नहीं मिली', 'itin.addActivity': 'गतिविधि जोड़ें', 'itin.flights': 'उड़ानें और आगमन',
  'itin.flightsSub': 'समय {zone} में दिखाया गया है। बदलने के लिए टैप करें।', 'itin.addFlight': 'मेरी उड़ान जोड़ें', 'itin.assigned': '{n} ज़िम्मेदार',
  'itin.confirmed': 'पक्का', 'itin.voting': 'वोटिंग', 'itin.pending': 'बाकी', 'itin.perPerson': '/व्यक्ति',
  'add.title': 'गतिविधि जोड़ें', 'add.subtitle': 'नई गतिविधि का विवरण भरें।', 'add.type': 'गतिविधि का प्रकार', 'add.time': 'समय', 'add.name': 'गतिविधि का नाम',
  'add.location': 'स्थान', 'add.duration': 'अवधि', 'add.cost': 'प्रति व्यक्ति खर्च (USD)', 'add.notes': 'नोट्स (वैकल्पिक)', 'add.notesPh': 'समूह के लिए नोट्स…',
  'add.vote': 'समूह वोट के लिए रखें', 'add.remind': 'शुरू होने से पहले सबको याद दिलाएँ', 'add.submit': 'यात्रा योजना में जोड़ें', 'add.save': 'बदलाव सहेजें',
```

```
  'budget.kicker': 'समूह खर्च', 'budget.title': 'बजट', 'budget.spent': 'अब तक खर्च', 'budget.of': '{total} में से · {per} प्रति व्यक्ति',
  'budget.used': 'समूह बजट का {pct}% इस्तेमाल', 'budget.breakdown': 'खर्च का ब्यौरा', 'budget.currency': 'मुद्रा', 'budget.expenses': 'खर्च', 'budget.cap': 'बजट सीमा',
  'budget.addExpense': 'खर्च जोड़ें', 'budget.whatFor': 'किस लिए?', 'budget.add': 'खर्च जोड़ें', 'budget.settle': 'हिसाब बराबर करें', 'budget.planner': 'दैनिक खर्च योजना', 'budget.paidBy': '{name} ने भुगतान किया',
  'chat.kicker': '{n} यात्री', 'chat.title': 'यात्रा चैट', 'chat.placeholder': 'अपने साथियों को संदेश भेजें…', 'chat.askAi': 'ट्रिप असिस्टेंट से पूछें…',
  'chat.assistant': 'ट्रिप असिस्टेंट', 'chat.typing': '{name} लिख रहे हैं…', 'chat.q.budget': 'बजट कैसा है?', 'chat.q.food': 'आस-पास अच्छा खाना?',
  'chat.q.weather': 'मौसम कैसा है?', 'chat.q.next': 'आगे क्या है?',
  'hub.kicker': 'ट्रिप हब', 'hub.title': 'आपका समूह', 'hub.invite': 'आमंत्रण कोड', 'hub.share': 'शेयर', 'hub.members': 'सदस्य और भूमिकाएँ',
  'hub.vote': 'गतिविधियों पर वोट करें', 'hub.checklist': 'चेकलिस्ट', 'hub.done': '{a}/{b} पूरे', 'hub.addTask': 'काम जोड़ें', 'hub.documents': 'दस्तावेज़',
  'hub.documentsSub': 'बिना वाई-फ़ाई या डेटा के भी पढ़ सकते हैं।', 'hub.offline': 'ऑफ़लाइन', 'hub.tools': 'इस यात्रा के लिए और',
  'auth.welcome': 'फिर से स्वागत है!', 'auth.welcomeSub': 'अपनी यात्रा की योजना जारी रखने के लिए साइन इन करें।', 'auth.email': 'ईमेल पता', 'auth.password': 'पासवर्ड',
  'auth.remember': 'मुझे याद रखें', 'auth.forgot': 'पासवर्ड भूल गए?', 'auth.signIn': 'साइन इन', 'auth.or': 'या इसके साथ जारी रखें',
  'auth.google': 'Google के साथ जारी रखें', 'auth.noAccount': 'खाता नहीं है?', 'auth.signUp': 'साइन अप', 'auth.demo': 'डेमो खाता',
  'auth.terms': 'जारी रखकर आप हमारी सेवा की शर्तों और गोपनीयता नीति से सहमत होते हैं',
  'common.cancel': 'रद्द करें', 'common.save': 'सहेजें', 'common.delete': 'हटाएँ', 'common.done': 'हो गया', 'common.next': 'आगे', 'common.getStarted': 'शुरू करें',
  'common.skip': 'छोड़ें', 'common.back': 'वापस', 'common.edit': 'बदलें', 'common.close': 'बंद करें',
  'settings.title': 'सेटिंग्स', 'settings.language': 'भाषा', 'settings.currency': 'घरेलू मुद्रा', 'settings.notifications': 'गतिविधि रिमाइंडर',
  'settings.lead': 'पहले याद दिलाएँ', 'settings.textSize': 'टेक्स्ट का आकार', 'settings.signOut': 'साइन आउट',
};

const zh: Dict = {
  'nav.explore': '首页', 'nav.itinerary': '计划', 'nav.budget': '预算', 'nav.chat': '聊天', 'nav.profile': '更多',
  'greet.morning': '早上好', 'greet.afternoon': '下午好', 'greet.evening': '晚上好', 'greet.night': '夜深了',
  'explore.title': '发现你的\n理想目的地', 'explore.subtitle': '一起计划、探索、出发。', 'explore.search': '搜索城市或目的地…',
  'explore.popular': '热门地点', 'explore.featured': '精选目的地', 'explore.seeAll': '查看全部', 'explore.matches': '{n} 个结果',
  'explore.nearby': '半径范围内的地点', 'explore.activeTrip': '进行中的旅行', 'explore.upcomingTrip': '即将出发', 'explore.pastTrip': '已完成的旅行',
  'explore.left': '剩余 {amount}', 'explore.members': '{n} 位成员', 'explore.reminder': '提醒', 'explore.reminderSub': '我们会提前 {n} 分钟通知大家。',
  'explore.nearestAirport': '最近的机场', 'explore.addToTrip': '加入旅行', 'explore.noResults': '没有与“{q}”匹配的目的地', 'explore.tools': '旅行工具',
  'cat.all': '全部', 'cat.best': '最佳', 'cat.adventure': '探险', 'cat.food': '美食', 'cat.culture': '文化', 'cat.beach': '海滩', 'cat.nature': '自然',
  'cat.nightlife': '夜生活', 'cat.sightseeing': '观光', 'cat.shopping': '购物', 'cat.relax': '放松',
  'len.any': '不限时长', 'len.u1': '1 小时以内', 'len.1-2': '1–2 小时', 'len.2-4': '2–4 小时', 'len.half': '半天', 'len.full': '全天',
  'itin.members': '{n} 位成员', 'itin.noActivities': '还没有活动', 'itin.noActivitiesSub': '点击 + 按钮添加\n当天的第一个活动。', 'itin.noMatch': '没有符合筛选的活动',
  'itin.addActivity': '添加活动', 'itin.flights': '航班与抵达', 'itin.flightsSub': '时间以 {zone} 显示，点击切换时区。', 'itin.addFlight': '添加我的航班',
  'itin.assigned': '{n} 人负责', 'itin.confirmed': '已确认', 'itin.voting': '投票中', 'itin.pending': '待定', 'itin.perPerson': '/人',
  'add.title': '添加活动', 'add.subtitle': '填写新活动的详细信息。', 'add.type': '活动类型', 'add.time': '时间', 'add.name': '活动名称', 'add.location': '地点',
  'add.duration': '时长', 'add.cost': '人均费用 (USD)', 'add.notes': '备注（可选）', 'add.notesPh': '给大家的备注…', 'add.vote': '发起小组投票',
  'add.remind': '开始前提醒所有人', 'add.submit': '加入行程', 'add.save': '保存修改',
  'budget.kicker': '小组开销', 'budget.title': '预算', 'budget.spent': '目前已花费', 'budget.of': '共 {total} · 人均 {per}', 'budget.used': '已使用小组预算的 {pct}%',
  'budget.breakdown': '支出明细', 'budget.currency': '货币', 'budget.expenses': '支出', 'budget.cap': '预算上限', 'budget.addExpense': '添加支出',
  'budget.whatFor': '用途是什么？', 'budget.add': '添加支出', 'budget.settle': '结算', 'budget.planner': '每日花费规划', 'budget.paidBy': '{name} 支付',
  'chat.kicker': '{n} 位旅伴', 'chat.title': '旅行群聊', 'chat.placeholder': '给旅伴发消息…', 'chat.askAi': '问问旅行助手…', 'chat.assistant': '旅行助手',
  'chat.typing': '{name} 正在输入…', 'chat.q.budget': '预算怎么样？', 'chat.q.food': '附近有什么好吃的？', 'chat.q.weather': '天气如何？', 'chat.q.next': '接下来做什么？',
  'hub.kicker': '旅行中心', 'hub.title': '你的小组', 'hub.invite': '邀请码', 'hub.share': '分享', 'hub.members': '成员与分工', 'hub.vote': '为活动投票',
  'hub.checklist': '清单', 'hub.done': '已完成 {a}/{b}', 'hub.addTask': '添加任务', 'hub.documents': '文件', 'hub.documentsSub': '无网络时也能查看。',
  'hub.offline': '离线可用', 'hub.tools': '更多旅行工具',
  'auth.welcome': '欢迎回来！', 'auth.welcomeSub': '登录以继续规划你的旅程。', 'auth.email': '电子邮箱', 'auth.password': '密码', 'auth.remember': '记住我',
  'auth.forgot': '忘记密码？', 'auth.signIn': '登录', 'auth.or': '或使用以下方式', 'auth.google': '使用 Google 继续', 'auth.noAccount': '还没有账号？',
  'auth.signUp': '注册', 'auth.demo': '演示账号', 'auth.terms': '继续即表示你同意我们的服务条款和隐私政策',
  'common.cancel': '取消', 'common.save': '保存', 'common.delete': '删除', 'common.done': '完成', 'common.next': '下一步', 'common.getStarted': '开始使用',
  'common.skip': '跳过', 'common.back': '返回', 'common.edit': '编辑', 'common.close': '关闭',
  'settings.title': '设置', 'settings.language': '语言', 'settings.currency': '本国货币', 'settings.notifications': '活动提醒', 'settings.lead': '提前提醒',
  'settings.textSize': '文字大小', 'settings.signOut': '退出登录',
};

export const dictionaries: Record<LangCode, Dict> = { en, es, fr, ja, hi, zh };
export const english = en;
```

src/services/__tests__/assistant.test.ts

```
import { CLAUDE_MODEL, answerLocally, askClaude, type AssistantContext } from '../assistant';
import type { ChatMessage } from '@/store/types';
import { buildDemoTrip, DEMO_USER_ID } from '@/store/seed';
import { settlePlan } from '@/utils/money';

const trip = buildDemoTrip('upcoming');
```

```
const ctx: AssistantContext = { trip, userId: DEMO_USER_ID, rates: null, homeCurrency: 'USD' };
const ask = (q: string) => answerLocally(q, ctx);

describe('offline Trip Assistant', () => {
  it('answers budget questions from the real trip data', () => {
    const a = ask('How’s the budget?');
    expect(a).toMatch(/\$2,086 of \$4,000/);
    expect(a).toMatch(/per person/);
  });

  it('lists the fewest payments to settle up and matches the app’s own plan', () => {
    const plan = settlePlan(trip);
    const a = ask('who owes who?');
    expect(a).toContain(`${plan.length} transfer`);
    expect(a).toMatch(/→/);
  });

  it('converts currency with an honest label when offline', () => {
    const a = ask('convert 50 usd to eur');
    expect(a).toMatch(/\$50\.00 ≈ /);
    expect(a).toMatch(/offline reference rate/);
  });

  it('knows local emergency numbers', () => {
    expect(ask('emergency, I need an ambulance')).toMatch(/Police \d+/);
  });

  it('suggests places that are not already on the itinerary', () => {
    expect(ask('what should we do?')).toMatch(/Ideas not on your itinerary yet|already planned/);
  });

  it('says what it can do when it does not understand', () => {
    expect(ask('zzzz qqqq')).toMatch(/I can help with/);
  });

  it('greets by trip name', () => {
    expect(ask('hello')).toContain(trip.name);
  });
});

describe('Claude mode', () => {
  const realFetch = global.fetch;
  afterEach(() => {
    global.fetch = realFetch;
    jest.useRealTimers();
  });

  const reply = (body: unknown, status = 200) => jest.fn(async (_url: string, _init?: RequestInit) => ({ ok: status < 400, status, json:
async () => body }) as Response);
  const sentBody = (fetchMock: ReturnType<typeof reply>) => JSON.parse(fetchMock.mock.calls[0][1]?.body as string);

  it('sends the question, the recent chat and the trip to the Messages API', async () => {
    const fetchMock = reply({ stop_reason: 'end_turn', content: [{ type: 'text', text: 'Take the ferry from Amalfi.' }] });
    global.fetch = fetchMock as unknown as typeof fetch;
    const history: ChatMessage[] = [
      { id: 'a', authorId: 'assistant', text: 'old answer that starts the thread', ts: 1, ai: true },
      { id: 'b', authorId: DEMO_USER_ID, text: 'Where do we eat?', ts: 2 },
      { id: 'c', authorId: 'assistant', text: 'Try Da Vincenzo.', ts: 3, ai: true },
      { id: 'd', authorId: 'system', text: 'someone joined', ts: 4, kind: 'system' },
    ];
    const text = await askClaude('sk-ant-test', 'How do we get to Capri?', history, ctx);
    expect(text).toBe('Take the ferry from Amalfi.');

    const [url, init] = fetchMock.mock.calls[0];
    expect(url).toBe('https://api.anthropic.com/v1/messages');
    expect(init?.method).toBe('POST');
    expect(init?.headers).toMatchObject({ 'x-api-key': 'sk-ant-test', 'anthropic-version': '2023-06-01' });
    const body = sentBody(fetchMock);
    expect(body.model).toBe(CLAUDE_MODEL);
    expect(body.system).toContain(trip.name);
    // the thread must start with a user message, system notes are left out, and the question comes last
    expect(body.messages[0].role).toBe('user');
    expect(body.messages.map((m: { role: string }) => m.role)).toEqual(['user', 'assistant', 'user']);
    expect(body.messages[0].content).toMatch(/Maya Chen: Where do we eat\?/);
    expect(body.messages.at(-1)).toEqual({ role: 'user', content: 'How do we get to Capri?' });
    expect(JSON.stringify(body.messages)).not.toContain('someone joined');
  });

  it('only sends the last 12 messages of the chat', async () => {
    const fetchMock = reply({ content: [{ type: 'text', text: 'ok' }] });
    global.fetch = fetchMock as unknown as typeof fetch;
    const history: ChatMessage[] = Array.from({ length: 30 }, (_, i) => ({ id: `m${i}`, authorId: DEMO_USER_ID, text: `message ${i}`, t
s: i }));
    await askClaude('sk-ant-test', 'And now?', history, ctx);
    expect(sentBody(fetchMock).messages).toHaveLength(13);
    expect(sentBody(fetchMock).messages[0].content).toContain('message 18');
```

```
  });

  it('joins several text blocks and ignores the rest', async () => {
    global.fetch = reply({ content: [{ type: 'text', text: 'First.' }, { type: 'tool_use' }, { type: 'text', text: 'Second.' }] }) as un
known as typeof fetch;
    expect(await askClaude('k', 'q', [], ctx)).toBe('First.\nSecond.');
  });

  it('fails clearly when the key is wrong, the model refuses or the answer is empty', async () => {
    global.fetch = reply({ error: { message: 'invalid x-api-key' } }, 401) as unknown as typeof fetch;
    await expect(askClaude('bad', 'q', [], ctx)).rejects.toThrow('Claude 401');
    global.fetch = reply({ stop_reason: 'refusal', content: [] }) as unknown as typeof fetch;
    await expect(askClaude('k', 'q', [], ctx)).rejects.toThrow('refused');
    global.fetch = reply({ content: [{ type: 'text', text: '   ' }] }) as unknown as typeof fetch;
    await expect(askClaude('k', 'q', [], ctx)).rejects.toThrow('empty');
  });

  it('gives up after 45 seconds instead of waiting forever', async () => {
    jest.useFakeTimers();
    global.fetch = jest.fn((_url, init) => new Promise((_resolve, reject) => (init as RequestInit).signal?.addEventListener('abort', ()
=> reject(new Error('aborted'))))) as unknown as typeof fetch;
    const pending = askClaude('k', 'q', [], ctx);
    const outcome = expect(pending).rejects.toThrow('aborted');
    await jest.advanceTimersByTimeAsync(45_000);
    await outcome;
  });
});
```

src/services/__tests__/pretendFriends.test.ts

```
import { friendReply, planFriendReply } from '../pretendFriends';

describe('friendReply', () => {
  it('answers by topic', () => {
    expect(friendReply('anyone hungry for dinner?')).toMatch(/food/);
    expect(friendReply('what time is the ferry?')).toMatch(/assistant|vote/);
    expect(friendReply('running a bit late')).toMatch(/save you a seat/);
    expect(friendReply('sent you the pics')).toMatch(/photo dump/);
    expect(friendReply('I paid $40 for the taxi')).toMatch(/Budget/);
  });

  it('falls back to a short cheer, chosen by the random number', () => {
    expect(friendReply('see you there', () => 0)).toBe('Love it 🔥');
    expect(friendReply('see you there', () => 0.999)).toBe('Let’s goooo 🙌');
  });
});

describe('planFriendReply', () => {
  it('stays quiet when nobody is there to answer', () => {
    expect(planFriendReply([], () => 0)).toBeNull();
  });

  it('replies about 70% of the time, with a delay of a few seconds', () => {
    expect(planFriendReply(['a', 'b'], () => 0.71)).toBeNull();
    const plan = planFriendReply(['a', 'b'], () => 0.5);
    expect(plan?.friend).toBe('b');
    expect(plan?.delay).toBeGreaterThanOrEqual(2600);
    expect(plan?.delay).toBeLessThanOrEqual(3800);
  });
});
```

src/services/assistant.ts

```
/**
 * Trip Assistant.
 * 1. Offline engine (default): intent matching over the live trip data, so it
 *    answers accurately with no internet — important at venues with bad wifi.
 * 2. Claude mode (optional): when a user adds their own Anthropic API key in
 *    Settings, questions go to Claude with the trip as context. Any failure
 *    falls back to the offline engine.
 */
import { destinationById } from '@/data/destinations';
import { placesFor } from '@/data/places';
import { currencies, emergencyByCountry, phrasebook } from '@/data/reference';
import { balances, settlePlan, totalSpent } from '@/utils/money';
import type { ChatMessage, RatesSnapshot, Trip, WeatherSnapshot } from '@/store/types';
import { durationLabel, money } from '@/utils/format';
import { distanceKm, kmLabel, nearestAirports } from '@/utils/geo';
import { addDays, dayLabel, deviceTimeZone, formatInTz, hoursAhead, partsInTz, time12, todayInTz, tzAbbrev } from '@/utils/time';

import { convert } from './currency';
import { cToF, weatherInfo } from './weather';

export const CLAUDE_MODEL = 'claude-opus-5-5';

export interface AssistantContext {
```

```
  trip: Trip;
  userId: string;
  weather?: WeatherSnapshot;
  rates: RatesSnapshot | null;
  homeCurrency: string;
}

const first = (t: Trip, id: string) => t.members.find((m) => m.id === id)?.name.split(' ')[0] ?? 'Someone';

const CURRENCY_WORDS: Record<string, string> = {
  dollar: 'USD', dollars: 'USD', usd: 'USD', $: 'USD', yen: 'JPY', jpy: 'JPY', '¥': 'JPY', euro: 'EUR', euros: 'EUR', eur: 'EUR', '€':
'EUR',
  pound: 'GBP', pounds: 'GBP', gbp: 'GBP', '£': 'GBP', won: 'KRW', krw: 'KRW', baht: 'THB', thb: 'THB', rupee: 'INR', rupees: 'INR', in
r: 'INR',
  peso: 'MXN', pesos: 'MXN', mxn: 'MXN', franc: 'CHF', francs: 'CHF', chf: 'CHF', cad: 'CAD', aud: 'AUD', sgd: 'SGD', yuan: 'CNY', cny:
'CNY',
};

function currencyFrom(word: string | undefined): string | undefined {
  if (!word) return undefined;
  const w = word.toLowerCase();
  if (CURRENCY_WORDS[w]) return CURRENCY_WORDS[w];
  const up = w.toUpperCase();
  return currencies.some((c) => c.code === up) ? up : undefined;
}

export function answerLocally(question: string, ctx: AssistantContext): string {
  const q = question.toLowerCase().trim();
  const { trip } = ctx;
  const dest = destinationById(trip.destinationId);
  const tz = dest?.tz ?? 'UTC';
  const spent = totalSpent(trip);
  const per = spent / Math.max(1, trip.members.length);
  const has = (re: RegExp) => re.test(q);

  // Currency conversion: "convert 50 usd to jpy", "how much is 3000 yen"
  const conv = q.match(/([\d][\d,.]*)\s*([a-z$€£¥]{1,8})?\s*(?:to|in|into)?\s*([a-z$€£¥]{3,8})?/);
  if (has(/convert|exchange|how much is|in (yen|dollars|euros|usd|jpy|eur)|→/) && conv) {
    const amount = Number(conv[1].replace(/,/g, ''));
    const from = currencyFrom(conv[2]) ?? (has(/yen|jpy/) && !has(/to (yen|jpy)/) ? 'JPY' : 'USD');
    let to = currencyFrom(conv[3]) ?? (from === (dest?.currency ?? 'JPY') ? ctx.homeCurrency : dest?.currency ?? 'JPY');
    if (to === from) to = from === 'USD' ? dest?.currency ?? 'EUR' : 'USD';
    const out = convert(amount, from, to, ctx.rates);
    return `${money(amount, from, { decimals: true })} ≈ ${money(out, to, { decimals: true })} ${ctx.rates?.live ? '(live rate)' : '(off
line reference rate)'}. Tip: the Budget tab has a full converter with home/local view.`;
  }

  if (has(/owe|settle|split|pay (back|me)|who pays|balance/)) {
    const plan = settlePlan(trip);
    if (!plan.length) return 'Everyone is square right now — no one owes anything. 🎉';
    const lines = plan.slice(0, 5).map((p) => `• ${first(trip, p.from)} → ${first(trip, p.to)}: ${money(p.amount, 'USD', { decimals: tru
e })}`);
    const mine = balances(trip)[ctx.userId] ?? 0;
    return `Here's the simplest way to settle up (${plan.length} transfer${plan.length > 1 ? 's' : ''}):\n${lines.join('\n')}\n\nYou are
${mine >= 0 ? `owed ${money(mine)}` : `owing ${money(-mine)}`}. Tap "Settle up" in Budget to record payments.`;
  }

  if (has(/budget|spent|spend|money|afford|left|expens|cost so far/)) {
    const pct = Math.round((spent / Math.max(1, trip.budget)) * 100);
    const byCat: Record<string, number> = {};
    trip.expenses.filter((e) => !e.settlement).forEach((e) => (byCat[e.category] = (byCat[e.category] ?? 0) + e.amount));
    const top = Object.entries(byCat).sort((a, b) => b[1] - a[1])[0];
    const remaining = trip.budget - spent;
    const days = Math.max(1, Math.round((Date.parse(trip.endDate) - Date.parse(trip.startDate)) / 86400000) + 1);
    return `You've committed ${money(spent)} of ${money(trip.budget)} — about ${pct}% of the group budget (${money(per)} per person).
${top ? `Biggest category: ${top[0]} (${money(top[1])}). ` : ''}${remaining >= 0 ? `${money(remaining)} left ≈ ${money(remaining / days
/ trip.members.length)} per person per day.` : `You're ${money(-remaining)} over — Food is usually the easiest place to trim.`}`;
  }

  if (has(/weather|rain|temperature|forecast|umbrella|cold|hot|sunny|snow|degrees/)) {
    const w = ctx.weather;
    if (!w) return `I don't have a forecast for ${dest?.city ?? 'the destination'} yet — open the Weather screen while online and I'll r
emember it for offline use.`;
    const now = weatherInfo(w.current.code);
    const tomorrow = w.daily[1];
    const rainy = w.daily.slice(0, 3).some((d) => d.rain >= 50);
    return `${dest?.city}: ${w.current.temp}°C (${cToF(w.current.temp)}°F), ${now.label.toLowerCase()}. ${tomorrow ? `Tomorrow ${tomorro
w.min}–${tomorrow.max}°C, ${tomorrow.rain}% chance of rain. ` : ''}${rainy ? 'Pack an umbrella ☔ — rain is likely in the next few days.'
: 'Looks good for walking days. 👟'}${w.live ? '' : ' (estimated — connect to refresh)'}`;
  }

  if (has(/food|eat|restaurant|dinner|lunch|breakfast|ramen|sushi|hungry|snack|cafe|coffee/)) {
    const cheap = has(/cheap|budget|afford|under/);
    const center = dest ? { lat: dest.lat, lng: dest.lng } : null;
    const list = placesFor(trip.destinationId)
      .concat(trip.extraDestinationIds.flatMap((id) => placesFor(id)))
```

```
      .filter((p) => p.category === 'food' && (!cheap || p.cost <= 20))
      .map((p) => ({ p, km: center ? distanceKm(center, p) : 0 }))
      .sort((a, b) => b.p.rating - a.p.rating)
      .slice(0, 3);
    if (!list.length) return 'I couldn’t find food spots for this destination yet — try the Nearby screen to widen the radius.';
    return `Top-rated food${cheap ? ' on a budget' : ''} near ${dest?.city}:\n${list
      .map(({ p, km }) => `• ${p.name} (${p.area}) — ★${p.rating}, ~${p.cost ? money(p.cost) : 'free'}/person, ${kmLabel(km)} from centr
e`)
      .join('\n')}\nWant one added to the itinerary? Open Nearby and tap “Add”.`;
  }

  if (has(/flight|land|arriv|depart|airport|plane/)) {
    const arrivals = trip.flights.filter((f) => f.direction === 'arrival').sort((a, b) => a.arriveUtc - b.arriveUtc);
    const missing = trip.members.filter((m) => !trip.flights.some((f) => f.memberId === m.id)).map((m) => m.name.split(' ')[0]);
    const air = dest ? nearestAirports(dest, 1)[0] : undefined;
    const lines = arrivals.map((f) => `• ${first(trip, f.memberId)} — ${f.flightNo}, lands ${formatInTz(f.arriveUtc, tz)} ${tzAbbrev(t
z)}`);
    return `${lines.length ? `Arrivals:\n${lines.join('\n')}` : 'No flights added yet.'}${missing.length ? `\n${missing.join(', ')} stil
l need to add flights.` : ''}${air ? `\nNearest airport to ${dest?.city} centre: ${air.iata} (${air.name}).` : ''}`;
  }

  if (has(/time zone|timezone|what time|jet ?lag|time difference/)) {
    const home = deviceTimeZone();
    const diff = hoursAhead(tz, home);
    const p = partsInTz(Date.now(), tz);
    return `It’s ${time12(p.hhmm)} in ${dest?.city} (${tzAbbrev(tz)}). That’s ${diff === 0 ? 'the same as' : `${Math.abs(diff)}h ${diff
> 0 ? 'ahead of' : 'behind'}`} your phone’s time zone. Itinerary times are always shown in local trip time.${Math.abs(diff) >= 6 ? ' Jet
-lag tip: get daylight on arrival day and avoid naps over 30 min.' : ''}`;
  }

  if (has(/how (do|to) (i |you |we )?say|translate|phrase|in japanese|in french|in spanish/)) {
    const book = dest ? phrasebook[dest.language] : undefined;
    if (!book) return 'Open the Phrasebook from the menu for common phrases with audio.';
    const hit = book.phrases.find((p) => q.includes(p.en.toLowerCase().replace(/[?!]/g, '').split(' / ')[0]));
    if (hit) return `“${hit.en}” in ${book.lang}: ${hit.local}${hit.roman ? ` (${hit.roman})` : ''}. Tap the speaker in the Phrasebook t
o hear it.`;
    return `Handy ${book.lang} phrases:\n${book.phrases.slice(0, 4).map((p) => `• ${p.en} — ${p.local}${p.roman ? ` (${p.roman})` : ''}
`).join('\n')}`;
  }

  if (has(/emergency|police|hospital|ambulance|lost passport|help me|doctor|stolen/)) {
    const em = emergencyByCountry[dest?.countryCode ?? ''] ?? emergencyByCountry.DEFAULT;
    return `In ${dest?.country ?? 'this country'}: Police ${em.police} · Ambulance ${em.ambulance} · Fire ${em.fire}.\n${em.tips[0] ??
''}\nThe Emergency screen works offline and can share your location by text.`;
  }

  if (has(/pack|bring|luggage|suitcase/)) {
    const left = trip.packing.filter((p) => !p.packed);
    const rainy = ctx.weather?.daily.slice(0, 5).some((d) => d.rain >= 50);
    return `${left.length ? `Still to pack (${left.length}): ${left.slice(0, 5).map((p) => p.text).join(', ')}${left.length > 5 ? '…' :
''}.` : 'Everything on the packing list is packed! 🎒'}${rainy ? ' Rain is forecast — add an umbrella.' : ''} ${dest?.countryCode === 'J
P' ? 'Japan uses type A plugs (100V) and is still cash-friendly — bring a coin purse.' : ''}`.trim();
  }

  if (has(/vote|poll|decide|undecided/)) {
    const open = trip.polls.filter((p) => !p.closed);
    const voting = trip.activities.filter((a) => a.status === 'voting');
    const pollLines = open.map((p) => {
      const leader = p.options.slice().sort((a, b) => b.votes.length - a.votes.length)[0];
      const voted = p.options.some((o) => o.votes.includes(ctx.userId));
      return `• “${p.question}” — leading: ${leader?.text ?? '—'}${voted ? '' : ' (you haven’t voted)'}`;
    });
    return `${pollLines.length ? `Open polls:\n${pollLines.join('\n')}` : 'No open polls.'}${voting.length ? `\nActivities waiting on vo
tes: ${voting.map((a) => a.title).join(', ')}.` : ''}`;
  }

  if (has(/task|checklist|to-?do|remaining|left to do/)) {
    const open = trip.tasks.filter((k) => !k.done);
    if (!open.length) return 'The checklist is all done ✅';
    return `${open.length} task${open.length > 1 ? 's' : ''} left:\n${open.map((k) => `• ${k.text}${k.assignee ? ` — ${first(trip, k.ass
ignee)}` : ''}`).join('\n')}`;
  }

  if (has(/who('s| is)|role|treasurer|navigator|photographer|members?/)) {
    return `Roles for ${trip.name}:\n${trip.members.map((m) => `• ${m.name.split(' ')[0]} — ${m.role}`).join('\n')}\nChange roles in Pro
file → Members & roles.`;
  }

  const dayMatch = q.match(/day\s*(\d{1,2})/);
  if (has(/next|today|tomorrow|schedule|itinerary|plan for|what('s| is) on/) || dayMatch) {
    const today = todayInTz(tz);
    let target = today;
    if (has(/tomorrow/)) target = addDays(today, 1);
    if (dayMatch) target = addDays(trip.startDate, Number(dayMatch[1]) - 1);
    if (has(/next/) && !dayMatch) {
      const nowP = partsInTz(Date.now(), tz);
```

```
      const upcoming = trip.activities
        .filter((a) => a.date > nowP.ymd || (a.date === nowP.ymd && a.time >= nowP.hhmm))
        .sort((a, b) => (a.date + a.time).localeCompare(b.date + b.time))[0];
      if (!upcoming) return 'Nothing else is scheduled — time to plan something fun! Try “suggest something to do”.';
      return `Next up: ${upcoming.title} — ${dayLabel(upcoming.date)} at ${time12(upcoming.time)}, ${upcoming.location} (${durationLabel
(upcoming.durationMin)}). ${upcoming.reminder ? 'Reminder is on 🔔' : 'Turn on the bell to get a reminder.'}`;
    }
    if (target < trip.startDate) target = trip.startDate;
    const acts = trip.activities.filter((a) => a.date === target).sort((a, b) => a.time.localeCompare(b.time));
    if (!acts.length) return `${dayLabel(target)} is a free day so far. Want ideas? Ask “what should we do?”`;
    return `${dayLabel(target)}:\n${acts.map((a) => `• ${time12(a.time)} ${a.title}${a.status !== 'confirmed' ? ` (${a.status})` : ''}
`).join('\n')}`;
  }

  if (has(/what (should|can|could) we do|idea|suggest|recommend|bored|activity|things to do|adventure|museum|culture|nightlife|shoppin
g/)) {
    const planned = new Set(trip.activities.map((a) => a.placeId));
    const cat = has(/adventure|hike|outdoor/) ? 'adventure' : has(/museum|culture|temple|shrine/) ? 'culture' : has(/night|bar|karaoke/)
? 'nightlife' : has(/shop/) ? 'shopping' : undefined;
    const ideas = placesFor(trip.destinationId)
      .concat(trip.extraDestinationIds.flatMap((id) => placesFor(id)))
      .filter((p) => !planned.has(p.id) && (!cat || p.category === cat))
      .sort((a, b) => b.rating - a.rating)
      .slice(0, 3);
    if (!ideas.length) return 'You’ve already planned the top spots! Try the Recommendations survey for new destinations.';
    return `Ideas not on your itinerary yet:\n${ideas.map((p) => `• ${p.name} — ${p.category}, ${durationLabel(p.durationMin)}, ${p.cost
? `${money(p.cost)}/person` : 'free'} (★${p.rating})`).join('\n')}\nAdd one from the Nearby screen or put it to a vote.`;
  }

  if (has(/^(hi|hey|hello|yo|sup|good (morning|evening|afternoon))\b/)) {
    return `Hey! 👋 I’m your Trip Assistant for ${trip.name}. Ask me about the budget, weather, food nearby, flights, what’s next, or “c
onvert 50 USD to ${dest?.currency ?? 'EUR'}”.`;
  }

  return `I can help with: budget & who owes who, weather, food nearby, today’s/tomorrow’s plan, flights & arrivals, currency (“convert
20 USD”), time zones, phrases, packing and emergencies. Try one of the quick questions below!`;
}

function systemPrompt(ctx: AssistantContext): string {
  const { trip } = ctx;
  const dest = destinationById(trip.destinationId);
  const tz = dest?.tz ?? 'UTC';
  const name = (id: string) => trip.members.find((m) => m.id === id)?.name ?? id;
  const acts = trip.activities
    .slice()
    .sort((a, b) => (a.date + a.time).localeCompare(b.date + b.time))
    .map((a) => `${a.date} ${a.time} | ${a.title} | ${a.location} | ${a.status} | $${a.costPerPerson}/pp | ${durationLabel(a.durationMi
n)}`)
    .join('\n');
  const flights = trip.flights.map((f) => `${name(f.memberId)} ${f.direction} ${f.flightNo} ${f.from}->${f.to} ${formatInTz(f.direction
=== 'arrival' ? f.arriveUtc : f.departUtc, tz)}`).join('\n');
  const w = ctx.weather;
  return [
    'You are the Trip Assistant inside TogetherWeGo, a group trip-planning app, replying in the group chat.',
    'Be friendly and concise: 1–4 short sentences or a short bullet list. Use the trip data below; do not invent bookings.',
    'You cannot edit the trip yourself — when someone wants a change, suggest it and point them to the right screen (Itinerary → Add Act
ivity, Budget → Add expense, Profile → polls/checklist).',
    '',
    `Trip: ${trip.name} — ${dest?.city}, ${dest?.country}; ${trip.startDate} to ${trip.endDate}; local time zone ${tz}; today (local)
${todayInTz(tz)}.`,
    `Members: ${trip.members.map((m) => `${m.name} (${m.role})`).join(', ')}.`,
    `Budget: spent $${Math.round(totalSpent(trip))} of $${trip.budget} cap. Expenses: ${trip.expenses.map((e) => `${e.title} $${e.amoun
t} (${e.category}, paid by ${name(e.paidBy)})`).join('; ')}.`,
    w ? `Weather now: ${w.current.temp}°C ${weatherInfo(w.current.code).label}; next days: ${w.daily.slice(0, 4).map((d) => `${d.date}
${d.min}-${d.max}°C rain ${d.rain}%`).join(', ')}.` : '',
    `Itinerary (local times):\n${acts || 'none yet'}`,
    flights ? `Flights:\n${flights}` : '',
    `Open tasks: ${trip.tasks.filter((k) => !k.done).map((k) => k.text).join(', ') || 'none'}.`,
  ]
    .filter(Boolean)
    .join('\n');
}

export async function askClaude(apiKey: string, question: string, history: ChatMessage[], ctx: AssistantContext): Promise<string> {
  // Plain fetch to the Messages API: the official SDK does not bundle for React Native.
  const messages: { role: 'user' | 'assistant'; content: string }[] = [];
  for (const m of history.filter((x) => x.kind !== 'system').slice(-12)) {
    if (m.ai) messages.push({ role: 'assistant', content: m.text });
    else messages.push({ role: 'user', content: `${ctx.trip.members.find((x) => x.id === m.authorId)?.name ?? 'Member'}: ${m.text}` });
  }
  while (messages.length && messages[0].role !== 'user') messages.shift();
  messages.push({ role: 'user', content: question });

  const controller = new AbortController();
  const timer = setTimeout(() => controller.abort(), 45_000);
  try {
```

```
    const res = await fetch('https://api.anthropic.com/v1/messages', {
      method: 'POST',
      headers: { 'content-type': 'application/json', 'x-api-key': apiKey, 'anthropic-version': '2023-06-01' },
      body: JSON.stringify({ model: CLAUDE_MODEL, max_tokens: 1024, system: systemPrompt(ctx), messages }),
      signal: controller.signal,
    });
    if (!res.ok) throw new Error(`Claude ${res.status}`);
    const data = (await res.json()) as { stop_reason?: string; content?: { type: string; text?: string }[] };
    if (data.stop_reason === 'refusal') throw new Error('refused');
    const text = (data.content ?? []).flatMap((b) => (b.type === 'text' && b.text ? [b.text] : [])).join('\n').trim();
    if (!text) throw new Error('empty');
    return text;
  } finally {
    clearTimeout(timer);
  }
}
```

src/services/currency.ts

```
/**
 * Exchange rates from open.er-api.com (free, no key), cached for 6 hours and
 * persisted for offline use. Falls back to bundled reference rates.
 */
import { currencies } from '@/data/reference';
import { useAppStore } from '@/store/useAppStore';
import type { RatesSnapshot } from '@/store/types';

const CACHE_MS = 6 * 60 * 60 * 1000;

export function fallbackRates(): RatesSnapshot {
  const rates: Record<string, number> = {};
  currencies.forEach((c) => (rates[c.code] = c.perUsd));
  return { base: 'USD', rates, fetchedAt: 0, live: false };
}

export async function fetchRates(force = false): Promise<RatesSnapshot> {
  const cached = useAppStore.getState().rates;
  if (!force && cached?.live && Date.now() - cached.fetchedAt < CACHE_MS) return cached;
  try {
    const ctrl = new AbortController();
    const timer = setTimeout(() => ctrl.abort(), 8000);
    const res = await fetch('https://open.er-api.com/v6/latest/USD', { signal: ctrl.signal });
    clearTimeout(timer);
    const j = await res.json();
    if (j.result !== 'success' || !j.rates) throw new Error('bad response');
    const snap: RatesSnapshot = { base: 'USD', rates: j.rates, fetchedAt: Date.now(), live: true };
    useAppStore.getState().setRates(snap);
    return snap;
  } catch {
    return cached ?? fallbackRates();
  }
}

export function convert(amount: number, from: string, to: string, rates: RatesSnapshot | null): number {
  const r = rates?.rates ?? fallbackRates().rates;
  const f = r[from] ?? fallbackRates().rates[from] ?? 1;
  const t = r[to] ?? fallbackRates().rates[to] ?? 1;
  return (amount / f) * t;
}

export function useRates(): RatesSnapshot {
  return useAppStore((s) => s.rates) ?? fallbackRates();
}
```

src/services/files.ts

```
/**
 * Offline document vault. Picked files are copied into the app's private
 * document directory (native) so they stay readable with no connection.
 * On web, small files are kept as data URLs in local storage.
 */
import * as DocumentPicker from 'expo-document-picker';
import { Directory, File, Paths } from 'expo-file-system';
import * as ImagePicker from 'expo-image-picker';
import { Platform } from 'react-native';

export const MAX_WEB_BYTES = 2 * 1024 * 1024;
export const MAX_NATIVE_BYTES = 25 * 1024 * 1024;

export interface SavedFile {
  uri: string;
  name: string;
  size?: number;
  kind: 'pdf' | 'image' | 'file';
}
```

```
function kindOf(name: string, mime?: string | null): SavedFile['kind'] {
  if (mime?.startsWith('image/') || /\.(png|jpe?g|heic|webp|gif)$/i.test(name)) return 'image';
  if (mime === 'application/pdf' || /\.pdf$/i.test(name)) return 'pdf';
  return 'file';
}

async function toDataUrl(file: Blob): Promise<string> {
  return new Promise((resolve, reject) => {
    const r = new FileReader();
    r.onload = () => resolve(String(r.result));
    r.onerror = () => reject(r.error);
    r.readAsDataURL(file);
  });
}

async function persist(uri: string, name: string, webFile?: Blob | null): Promise<string> {
  if (Platform.OS === 'web') {
    if (webFile) return toDataUrl(webFile);
    return uri;
  }
  const dir = new Directory(Paths.document, 'vault');
  if (!dir.exists) dir.create({ intermediates: true, idempotent: true });
  const safe = `${Date.now()}-${name.replace(/[^\w.-]+/g, '_')}`;
  const dest = new File(dir, safe);
  await new File(uri).copy(dest);
  return dest.uri;
}

export class FileTooLargeError extends Error {}

/** Lets the user pick a PDF/image/any file and saves an offline copy. */
export async function pickDocument(): Promise<SavedFile | null> {
  const res = await DocumentPicker.getDocumentAsync({ copyToCacheDirectory: true, multiple: false, type: ['application/pdf', 'image/*',
'*/*'] });
  if (res.canceled || !res.assets?.[0]) return null;
  const a = res.assets[0];
  const limit = Platform.OS === 'web' ? MAX_WEB_BYTES : MAX_NATIVE_BYTES;
  if (a.size && a.size > limit) throw new FileTooLargeError(`Files must be under ${Math.round(limit / 1024 / 1024)} MB`);
  const uri = await persist(a.uri, a.name, (a as { file?: Blob }).file ?? null);
  return { uri, name: a.name, size: a.size, kind: kindOf(a.name, a.mimeType) };
}

/** Picks a photo from the library (or camera) and keeps an offline copy. */
export async function pickImage(fromCamera = false): Promise<SavedFile | null> {
  if (fromCamera) {
    const perm = await ImagePicker.requestCameraPermissionsAsync();
    if (!perm.granted) throw new Error('Camera permission is needed to take photos.');
  }
  const opts: ImagePicker.ImagePickerOptions = { mediaTypes: ['images'], quality: 0.7, allowsEditing: false };
  const res = fromCamera ? await ImagePicker.launchCameraAsync(opts) : await ImagePicker.launchImageLibraryAsync(opts);
  if (res.canceled || !res.assets?.[0]) return null;
  const a = res.assets[0];
  const name = a.fileName ?? `photo-${Date.now()}.jpg`;
  if (Platform.OS === 'web') {
    if (a.fileSize && a.fileSize > MAX_WEB_BYTES) throw new FileTooLargeError('Photos must be under 2 MB in the web preview');
    return { uri: a.uri, name, size: a.fileSize, kind: 'image' };
  }
  const uri = await persist(a.uri, name);
  return { uri, name, size: a.fileSize, kind: 'image' };
}

export function deleteLocalFile(uri?: string) {
  if (!uri || Platform.OS === 'web' || !uri.startsWith('file:')) return;
  try {
    const f = new File(uri);
    if (f.exists) f.delete();
  } catch {
    // already gone
  }
}

export function sizeLabel(bytes?: number) {
  if (!bytes) return '';
  if (bytes < 1024) return `${bytes} B`;
  if (bytes < 1024 * 1024) return `${Math.round(bytes / 1024)} KB`;
  return `${(bytes / 1024 / 1024).toFixed(1)} MB`;
}
```

src/services/location.ts

```
/** Device location with graceful fallbacks (permission denied, web, timeouts). */
import * as Location from 'expo-location';

import type { LatLng } from '@/utils/geo';

export interface LocationResult extends LatLng {
```

```
  source: 'device' | 'fallback';
}

export async function getLocation(fallback: LatLng): Promise<LocationResult> {
  try {
    const perm = await Location.requestForegroundPermissionsAsync();
    if (perm.status !== 'granted') return { ...fallback, source: 'fallback' };
    const last = await Location.getLastKnownPositionAsync().catch(() => null);
    const pos =
      last ??
      (await Promise.race([
        Location.getCurrentPositionAsync({ accuracy: Location.Accuracy.Balanced }),
        new Promise<null>((r) => setTimeout(() => r(null), 8000)),
      ]));
    if (!pos) return { ...fallback, source: 'fallback' };
    return { lat: pos.coords.latitude, lng: pos.coords.longitude, source: 'device' };
  } catch {
    return { ...fallback, source: 'fallback' };
  }
}
```

src/services/notifications.ts

```
/**
 * Live activity reminders.
 * - iOS / Android: real local notifications scheduled with expo-notifications
 *   (they fire even when the app is closed).
 * - Every platform: an in-app watcher adds due reminders to the notification
 *   centre and, on web, shows a toast + browser notification.
 */
import * as Notifications from 'expo-notifications';
import { router } from 'expo-router';
import { Platform } from 'react-native';

import { toast } from '@/components/ui/feedback';
import { destinationById } from '@/data/destinations';
import { tripStatus, useAppStore } from '@/store/useAppStore';
import type { Activity, Trip } from '@/store/types';
import { time12, zonedToUtc } from '@/utils/time';

const isNative = Platform.OS !== 'web';
let initialised = false;

export function initNotifications() {
  if (initialised || !isNative) return;
  initialised = true;
  Notifications.setNotificationHandler({
    handleNotification: async () => ({ shouldShowBanner: true, shouldShowList: true, shouldPlaySound: true, shouldSetBadge: false }),
  });
  if (Platform.OS === 'android') {
    Notifications.setNotificationChannelAsync('reminders', {
      name: 'Activity reminders',
      importance: Notifications.AndroidImportance.HIGH,
      vibrationPattern: [0, 200, 120, 200],
      lightColor: '#FBC91B',
    }).catch(() => {});
  }
  Notifications.addNotificationResponseReceivedListener((resp) => {
    const route = resp.notification.request.content.data?.route;
    if (typeof route === 'string') setTimeout(() => router.push(route as never), 300);
  });
}

export async function ensurePermission(): Promise<boolean> {
  if (!isNative) {
    // Browsers may leave the prompt unanswered forever, so never block on it:
    // ask in the background and rely on in-app toasts until it's granted.
    if (typeof window !== 'undefined' && 'Notification' in window) {
      if (Notification.permission === 'granted') return true;
      if (Notification.permission === 'default') Notification.requestPermission().catch(() => {});
    }
    return false;
  }
  try {
    const cur = await Notifications.getPermissionsAsync();
    if (cur.granted) return true;
    const req = await Notifications.requestPermissionsAsync();
    return req.granted;
  } catch {
    return false;
  }
}

export function activityStartUtc(trip: Trip, a: Activity): number {
  const tz = destinationById(trip.destinationId)?.tz ?? 'UTC';
  return zonedToUtc(a.date, a.time, tz);
```

```
}

export function reminderBody(trip: Trip, a: Activity, lead: number) {
  return {
    title: `⏰ ${a.title} in ${lead} min`,
    body: `${time12(a.time)} · ${a.location} — ${trip.name}`,
  };
}

/** Reschedules every reminder for the signed-in user's upcoming activities. */
export async function syncReminders(): Promise<number> {
  const s = useAppStore.getState();
  if (!isNative) return 0;
  try {
    await Notifications.cancelAllScheduledNotificationsAsync();
  } catch {
    return 0;
  }
  if (!s.settings.notificationsEnabled || !s.session) return 0;
  const granted = (await Notifications.getPermissionsAsync().catch(() => null))?.granted;
  if (!granted) return 0;
  const lead = s.settings.reminderLeadMin;
  const now = Date.now();
  const due: { at: number; trip: Trip; a: Activity }[] = [];
  for (const trip of s.trips) {
    if (!trip.members.some((m) => m.id === s.session?.userId) || tripStatus(trip) === 'completed') continue;
    for (const a of trip.activities) {
      if (!a.reminder) continue;
      const at = activityStartUtc(trip, a) - lead * 60000;
      if (at > now + 5000 && at < now + 45 * 86400000) due.push({ at, trip, a });
    }
  }
  due.sort((x, y) => x.at - y.at);
  for (const { at, trip, a } of due.slice(0, 48)) {
    const { title, body } = reminderBody(trip, a, lead);
    await Notifications.scheduleNotificationAsync({
      content: { title, body, data: { route: '/(tabs)/itinerary', tripId: trip.id, activityId: a.id } },
      trigger: { type: Notifications.SchedulableTriggerInputTypes.DATE, date: new Date(at), channelId: 'reminders' },
    }).catch(() => {});
  }
  return Math.min(due.length, 48);
}

function browserNotify(title: string, body: string) {
  try {
    if (!isNative && typeof window !== 'undefined' && 'Notification' in window && Notification.permission === 'granted') {
      new Notification(title, { body, icon: '/favicon.ico' });
    }
  } catch {
    // ignore
  }
}

/** Sends a sample reminder in a few seconds so the feature can be demoed live. */
export async function sendTestReminder(seconds = 5) {
  const s = useAppStore.getState();
  const trip = s.trips.find((x) => x.id === s.activeTripId);
  const a = trip?.activities.find((x) => x.reminder) ?? trip?.activities[0];
  const title = a ? `⏰ ${a.title} starts soon` : '⏰ Test reminder';
  const body = a && trip ? `${time12(a.time)} · ${a.location} — ${trip.name}` : 'Reminders are working!';
  await ensurePermission();
  if (isNative) {
    await Notifications.scheduleNotificationAsync({
      content: { title, body, data: { route: '/(tabs)/itinerary' } },
      trigger: { type: Notifications.SchedulableTriggerInputTypes.TIME_INTERVAL, seconds, channelId: 'reminders' },
    }).catch(() => {});
  }
  setTimeout(() => {
    useAppStore.getState().pushNotification({ kind: 'reminder', title, body, tripId: trip?.id, route: '/(tabs)/itinerary' });
    if (!isNative) {
      toast(`${title} — ${body}`, { icon: 'alarm-outline', tone: 'info' });
      browserNotify(title, body);
    }
  }, seconds * 1000);
}

/** Called on an interval while the app is open: surfaces reminders that just came due. */
export function checkDueReminders() {
  const s = useAppStore.getState();
  if (!s.session || !s.settings.notificationsEnabled) return;
  const lead = s.settings.reminderLeadMin;
  const now = Date.now();
  for (const trip of s.trips) {
    if (!trip.members.some((m) => m.id === s.session?.userId)) continue;
    for (const a of trip.activities) {
      if (!a.reminder) continue;
```

```
      const start = activityStartUtc(trip, a);
      const fireAt = start - lead * 60000;
      const key = `${a.id}@${a.date}T${a.time}-${lead}`;
      if (now >= fireAt && now < start && !s.firedReminders[key]) {
        s.markReminderFired(key);
        const { title, body } = reminderBody(trip, a, Math.max(1, Math.round((start - now) / 60000)));
        s.pushNotification({ kind: 'reminder', title, body, tripId: trip.id, route: '/(tabs)/itinerary' });
        if (!isNative) {
          toast(`${title}`, { icon: 'alarm-outline', tone: 'info' });
          browserNotify(title, body);
        }
      }
    }
  }
}

/** The next upcoming reminder-enabled activity across the active trip (for the "live" banner). */
export function nextUpcoming(trip: Trip | undefined): { a: Activity; startUtc: number } | null {
  if (!trip) return null;
  const now = Date.now();
  let best: { a: Activity; startUtc: number } | null = null;
  for (const a of trip.activities) {
    const st = activityStartUtc(trip, a);
    if (st > now - 30 * 60000 && (!best || st < best.startUtc)) best = { a, startUtc: st };
  }
  return best;
}
```

src/services/pretendFriends.ts

```
/**
 * The demo crew: friendly canned replies so a trip with no real friends on the other end still feels alive.
 * Never used in a trip that is shared across phones (the real friends answer there).
 */
export function friendReply(text: string, random: () => number = Math.random): string {
  const q = text.toLowerCase();
  if (/food|eat|dinner|lunch|ramen|sushi|hungry/.test(q)) return 'I’m always down for food 😋 count me in!';
  if (/\?$/.test(q.trim())) return 'Good question — let’s ask the assistant or put it to a vote 🗳️';
  if (/late|delay|wait/.test(q)) return 'No stress, we’ll save you a seat 🙌';
  if (/photo|pic|camera/.test(q)) return 'Drop them in the photo dump! 📸';
  if (/money|pay|owe|\$/.test(q)) return 'I’ll log it in Budget so we can settle up later 💸';
  const pool = ['Love it 🔥', 'Works for me 👍', 'Yesss can’t wait!! ✈️', 'Adding it to my calendar 📅', 'Haha same 😂', 'Let’s goooo
🙌'];
  return pool[Math.floor(random() * pool.length)];
}

/** Picks a random group member to reply (70% of the time) and a natural-feeling delay in milliseconds. */
export function planFriendReply<M>(others: M[], random: () => number = Math.random): { friend: M; delay: number } | null {
  if (!others.length || random() >= 0.7) return null;
  return { friend: others[Math.floor(random() * others.length)], delay: 2600 + random() * 1200 };
}
```

src/services/print.ts

```
/** Printable trip planner: builds a styled HTML document and prints / saves it as PDF. */
import * as Print from 'expo-print';
import * as Sharing from 'expo-sharing';
import { Platform } from 'react-native';

import { destinationById } from '@/data/destinations';
import { emergencyByCountry } from '@/data/reference';
import { totalSpent } from '@/store/useAppStore';
import type { Trip } from '@/store/types';
import { durationLabel, money } from '@/utils/format';
import { dayLabel, formatInTz, rangeLabel, time12, tripDays, tzAbbrev, weekdayShort } from '@/utils/time';

const esc = (s: string) => s.replace(/[&<>"']/g, (c) => ({ '&': '&amp;', '<': '&lt;', '>': '&gt;', '"': '&quot;', "'": '&#39;' })[c]!);

export function plannerHtml(trip: Trip): string {
  const dest = destinationById(trip.destinationId);
  const tz = dest?.tz ?? 'UTC';
  const name = (id: string) => trip.members.find((m) => m.id === id)?.name ?? 'Member';
  const days = tripDays(trip.startDate, trip.endDate)
    .map((d, i) => {
      const acts = trip.activities.filter((a) => a.date === d).sort((a, b) => a.time.localeCompare(b.time));
      const rows = acts.length
        ? acts
            .map(
              (a) => `<tr><td class="time">${time12(a.time)}</td><td><b>${esc(a.title)}</b><div class="muted">${esc(a.location)} · ${dur
ationLabel(a.durationMin)}${a.costPerPerson ? ` · ${money(a.costPerPerson)}/person` : ''}</div>${a.notes ? `<div class="note">${esc(a.no
tes)}</div>` : ''}</td><td class="who">${a.assigned.map((x) => esc(name(x).split(' ')[0])).join(', ')}</td><td class="box">□</td></tr>`,
            )
            .join('')
        : '<tr><td colspan="4" class="muted">Free day — explore at your own pace ✨</td></tr>';
      return `<section class="day"><h3>Day ${i + 1} · ${weekdayShort(d)} ${dayLabel(d)}</h3><table>${rows}</table></section>`;
    })
```

```
    .join('');
  const flights = trip.flights
    .slice()
    .sort((a, b) => a.arriveUtc - b.arriveUtc)
    .map((f) => `<tr><td>${esc(name(f.memberId))}</td><td>${esc(f.flightNo)} · ${f.from}→${f.to}</td><td>${f.direction === 'arrival' ?
'Arrives' : 'Departs'} ${formatInTz(f.direction === 'arrival' ? f.arriveUtc : f.departUtc, tz)}</td></tr>`)
    .join('');
  const em = emergencyByCountry[dest?.countryCode ?? ''] ?? emergencyByCountry.DEFAULT;
  const spent = totalSpent(trip);
  return `<!doctype html><html><head><meta charset="utf-8"><title>${esc(trip.name)} — Planner</title>
<style>
  *{box-sizing:border-box} body{font-family:-apple-system,Segoe UI,Roboto,Helvetica,Arial,sans-serif;color:#1A1C17;margin:28px;font-siz
e:12.5px}
  header{display:flex;justify-content:space-between;align-items:flex-end;border-bottom:3px solid #1F4D25;padding-bottom:10px;margin-bott
om:16px}
  h1{margin:0;font-size:24px;color:#1F4D25} h2{font-size:15px;margin:18px 0 6px;color:#1F4D25} h3{font-size:13px;margin:14px 0 6px;backg
round:#EAF5E4;padding:6px 10px;border-radius:6px}
  table{width:100%;border-collapse:collapse} td{padding:6px 8px;border-bottom:1px solid #E7E9DF;vertical-align:top}
  .time{width:70px;font-weight:600;white-space:nowrap} .who{width:110px;color:#5F645A} .box{width:20px;font-size:15px}
  .muted{color:#7A7F73;font-size:11.5px} .note{margin-top:3px;font-style:italic;color:#5F645A}
  .grid{display:grid;grid-template-columns:repeat(3,1fr);gap:8px} .stat{background:#F4F4EE;border-radius:8px;padding:8px 10px}
  .stat b{display:block;font-size:16px;color:#1F4D25} .day{page-break-inside:avoid} footer{margin-top:18px;color:#959A8E;font-size:10.5p
x;text-align:center}
</style></head><body>
<header><div><h1>${esc(trip.name)}</h1><div class="muted">${esc(dest ? `${dest.city}, ${dest.country}` : '')} · ${rangeLabel(trip.startD
ate, trip.endDate)} · times in ${tzAbbrev(tz)}</div></div><div class="muted">Invite code <b>${trip.inviteCode}</b></div></header>
<div class="grid">
  <div class="stat">Travellers<b>${trip.members.length}</b>${trip.members.map((m) => esc(m.name.split(' ')[0]) + ' (' + esc(m.role) +
')').join(', ')}</div>
  <div class="stat">Budget<b>${money(spent)} / ${money(trip.budget)}</b>${money(trip.budget / Math.max(1, trip.members.length))} per per
son</div>
  <div class="stat">Emergency<b>Police ${em.police} · Ambulance ${em.ambulance}</b>${esc(em.tips[0] ?? '')}</div>
</div>
<h2>Day-by-day itinerary</h2>${days}
${flights ? `<h2>Flights & arrivals</h2><table>${flights}</table>` : ''}
<h2>Checklist</h2><table>${trip.tasks.map((k) => `<tr><td class="box">${k.done ? '☑' : '☐'}</td><td>${esc(k.text)}</td><td class="who">
${k.assignee ? esc(name(k.assignee).split(' ')[0]) : 'Everyone'}</td></tr>`).join('')}</table>
<footer>Printed from TogetherWeGo · Plan together, travel better</footer>
</body></html>`;
}

export async function printPlanner(trip: Trip) {
  const html = plannerHtml(trip);
  if (Platform.OS === 'web') {
    const w = window.open('', '_blank');
    if (!w) throw new Error('Pop-up blocked');
    w.document.write(html);
    w.document.close();
    w.focus();
    setTimeout(() => w.print(), 350);
    return;
  }
  await Print.printAsync({ html });
}

export async function sharePlannerPdf(trip: Trip) {
  if (Platform.OS === 'web') return printPlanner(trip);
  const { uri } = await Print.printToFileAsync({ html: plannerHtml(trip) });
  if (await Sharing.isAvailableAsync()) {
    await Sharing.shareAsync(uri, { mimeType: 'application/pdf', dialogTitle: `${trip.name} planner`, UTI: 'com.adobe.pdf' });
  }
}
```

src/services/secrets.ts

```
/**
 * Sensitive values (the optional Anthropic API key) live in the OS keychain /
 * keystore via expo-secure-store — never in the regular app database.
 * Web has no secure enclave, so the preview keeps it in session storage only.
 */
import * as SecureStore from 'expo-secure-store';
import { Platform } from 'react-native';

const KEY = 'twg_anthropic_key';

export async function getApiKey(): Promise<string | null> {
  try {
    if (Platform.OS === 'web') return typeof sessionStorage !== 'undefined' ? sessionStorage.getItem(KEY) : null;
    return await SecureStore.getItemAsync(KEY);
  } catch {
    return null;
  }
}

export async function setApiKey(value: string | null): Promise<void> {
  if (Platform.OS === 'web') {
```

```
    if (typeof sessionStorage === 'undefined') return;
    if (value) sessionStorage.setItem(KEY, value);
    else sessionStorage.removeItem(KEY);
    return;
  }
  if (value) await SecureStore.setItemAsync(KEY, value);
  else await SecureStore.deleteItemAsync(KEY);
}

export const looksLikeApiKey = (v: string) => /^sk-ant-[A-Za-z0-9_-]{20,}$/.test(v.trim());
```

src/services/share.ts

```
/**
 * Social media integration: native share sheet plus direct deep links into
 * WhatsApp, X, Facebook, Messenger, Telegram, Instagram, SMS and email.
 */
import { Asset } from 'expo-asset';
import * as Clipboard from 'expo-clipboard';
import * as Linking from 'expo-linking';
import * as Sharing from 'expo-sharing';
import type { RefObject } from 'react';
import { Platform, Share, type View } from 'react-native';
import { captureRef } from 'react-native-view-shot';

import { toast } from '@/components/ui/feedback';
import type { IconName } from '@/components/ui/bits';
import { photos, type PhotoKey } from '@/data/images';
import { buildInviteUrl, inviteText } from '@/sync/invite';
import { serverForInvites } from '@/sync/prefs';
import { useSyncStatus } from '@/sync/status';
import { useAppStore } from '@/store/useAppStore';

export type SocialTarget = 'whatsapp' | 'x' | 'facebook' | 'messenger' | 'telegram' | 'instagram' | 'sms' | 'email';

export const socialTargets: { key: SocialTarget; label: string; icon: IconName; color: string }[] = [
  { key: 'whatsapp', label: 'WhatsApp', icon: 'logo-whatsapp', color: '#25D366' },
  { key: 'instagram', label: 'Instagram', icon: 'logo-instagram', color: '#E1306C' },
  { key: 'x', label: 'X', icon: 'logo-twitter', color: '#111111' },
  { key: 'facebook', label: 'Facebook', icon: 'logo-facebook', color: '#1877F2' },
  { key: 'messenger', label: 'Messenger', icon: 'chatbubble-ellipses-outline', color: '#0084FF' },
  { key: 'telegram', label: 'Telegram', icon: 'paper-plane', color: '#229ED9' },
  { key: 'sms', label: 'Messages', icon: 'chatbox-ellipses-outline', color: '#34C759' },
  { key: 'email', label: 'Email', icon: 'mail-outline', color: '#5F645A' },
];

/** The sync server only goes into an invite when that trip is actually shared across phones. */
function serverFor(code: string): string | null {
  const trip = useAppStore.getState().trips.find((t) => t.inviteCode === code);
  return trip && useSyncStatus.getState().shared.includes(trip.id) ? serverForInvites() : null;
}

/** The link that opens TogetherWeGo on the join screen with the code (and the sync server, if the trip is shared) filled in. */
export function inviteLink(code: string): string {
  return buildInviteUrl(Linking.createURL('/join'), code, serverFor(code));
}

export function inviteMessage(tripName: string, code: string): string {
  return inviteText(tripName, code, inviteLink(code));
}

export async function copy(text: string, label = 'Copied to clipboard') {
  await Clipboard.setStringAsync(text);
  toast(label, { icon: 'copy-outline' });
}

/** Opens the OS share sheet (Instagram, Snapchat, iMessage, Slack… whatever is installed). */
export async function shareText(message: string, title = 'TogetherWeGo') {
  try {
    if (Platform.OS === 'web') {
      const nav = typeof navigator !== 'undefined' ? (navigator as Navigator & { share?: (d: object) => Promise<void> }) : undefined;
      if (nav?.share) {
        await nav.share({ title, text: message });
        return;
      }
      await copy(message, 'Share text copied — paste it anywhere');
      return;
    }
    await Share.share({ message, title });
  } catch {
    // user cancelled
  }
}

async function openFirst(urls: string[]) {
  for (const url of urls) {
```

```
    try {
      if (Platform.OS === 'web') {
        if (url.startsWith('http') || url.startsWith('mailto') || url.startsWith('sms')) {
          window.open(url, '_blank');
          return true;
        }
        continue;
      }
      if (url.startsWith('http') || (await Linking.canOpenURL(url))) {
        await Linking.openURL(url);
        return true;
      }
    } catch {
      // try next
    }
  }
  return false;
}

/** Shares straight into a specific social app, falling back to its web intent. */
export async function shareTo(target: SocialTarget, text: string, url?: string) {
  const t = encodeURIComponent(text);
  const u = encodeURIComponent(url ?? '');
  const map: Record<SocialTarget, string[]> = {
    whatsapp: [`whatsapp://send?text=${t}`, `https://wa.me/?text=${t}`],
    x: [`twitter://post?message=${t}`, `https://x.com/intent/tweet?text=${t}`],
    facebook: [`https://www.facebook.com/sharer/sharer.php?u=${u || encodeURIComponent('https://expo.dev')}&quote=${t}`],
    messenger: [`fb-messenger://share?link=${u}`, `https://www.messenger.com/`],
    telegram: [`tg://msg?text=${t}`, `https://t.me/share/url?url=${u}&text=${t}`],
    instagram: ['instagram://story-camera', 'instagram://app', 'https://www.instagram.com/'],
    sms: [Platform.OS === 'ios' ? `sms:&body=${t}` : `sms:?body=${t}`],
    email: [`mailto:?subject=${encodeURIComponent('Join our trip on TogetherWeGo')}&body=${t}`],
  };
  if (target === 'instagram') {
    // Instagram has no text-share URL: copy the caption first, then open the app.
    await Clipboard.setStringAsync(text);
    toast('Caption copied — paste it in Instagram', { icon: 'logo-instagram' });
  }
  const ok = await openFirst(map[target]);
  if (!ok) await shareText(text);
}

/** Shares a photo file (bundled demo photo or one from the camera roll). */
export async function sharePhoto(uri: string, caption: string) {
  try {
    let fileUri = uri;
    if (uri in photos) {
      const asset = Asset.fromModule(photos[uri as PhotoKey]);
      await asset.downloadAsync();
      fileUri = asset.localUri ?? asset.uri;
    }
    if (Platform.OS !== 'web' && (await Sharing.isAvailableAsync())) {
      await Sharing.shareAsync(fileUri, { dialogTitle: caption, mimeType: 'image/jpeg' });
      return;
    }
    await shareText(`${caption} 📸 — shared from TogetherWeGo`);
  } catch {
    await shareText(`${caption} 📸 — shared from TogetherWeGo`);
  }
}

/**
 * Turns something on screen (like the recap story card) into a picture and opens the phone's share sheet with it, so it
 * can go to Instagram, Snapchat, Messages or anywhere else that takes images. Falls back to plain text.
 */
export async function shareView(view: RefObject<View | null>, title: string, fallbackText: string) {
  try {
    if (Platform.OS === 'web' || !(await Sharing.isAvailableAsync())) return await shareText(fallbackText);
    const uri = await captureRef(view, { format: 'png', quality: 1, result: 'tmpfile' });
    await Sharing.shareAsync(uri, { mimeType: 'image/png', UTI: 'public.png', dialogTitle: title });
  } catch {
    await shareText(fallbackText);
  }
}
```

src/services/weather.ts

```
/**
 * Weather via Open-Meteo (free, no API key). Results are cached in the store
 * so the last forecast stays visible offline; if the network is down on first
 * load we fall back to a typical-climate estimate and label it as such.
 */
import { destinationById, type Destination } from '@/data/destinations';
import type { IconName } from '@/components/ui/bits';
import { useAppStore } from '@/store/useAppStore';
import type { WeatherSnapshot } from '@/store/types';
```

```
import { addDays, todayInTz } from '@/utils/time';

const CACHE_MS = 30 * 60 * 1000;

export function weatherInfo(code: number): { label: string; icon: IconName } {
  if (code === 0) return { label: 'Clear sky', icon: 'sunny' };
  if (code <= 2) return { label: 'Partly cloudy', icon: 'partly-sunny' };
  if (code === 3) return { label: 'Overcast', icon: 'cloud-outline' };
  if (code <= 48) return { label: 'Foggy', icon: 'cloud-outline' };
  if (code <= 57) return { label: 'Drizzle', icon: 'rainy-outline' };
  if (code <= 67) return { label: 'Rain', icon: 'rainy-outline' };
  if (code <= 77) return { label: 'Snow', icon: 'snow-outline' };
  if (code <= 82) return { label: 'Rain showers', icon: 'rainy-outline' };
  if (code <= 86) return { label: 'Snow showers', icon: 'snow-outline' };
  return { label: 'Thunderstorms', icon: 'thunderstorm-outline' };
}

function estimate(d: Destination): WeatherSnapshot {
  // Rough climate model: warmer near the equator, seasonal swing by hemisphere.
  const month = new Date().getUTCMonth();
  const north = d.lat >= 0;
  const season = Math.cos(((month - (north ? 6.5 : 0.5)) / 12) * 2 * Math.PI);
  const base = 27 - Math.abs(d.lat) * 0.42 + season * Math.min(12, Math.abs(d.lat) * 0.3);
  const today = todayInTz(d.tz);
  const codes = [1, 2, 3, 61, 2, 0, 80];
  return {
    destinationId: d.id,
    fetchedAt: Date.now(),
    live: false,
    current: { temp: Math.round(base), code: 2, wind: 11, humidity: 64 },
    daily: codes.map((code, i) => ({
      date: addDays(today, i),
      max: Math.round(base + 3 + ((i * 7) % 3)),
      min: Math.round(base - 5 + ((i * 5) % 3)),
      code,
      rain: code >= 61 ? 60 : code === 3 ? 25 : 10,
    })),
  };
}

export async function fetchWeather(destinationId: string, force = false): Promise<WeatherSnapshot | undefined> {
  const d = destinationById(destinationId);
  if (!d) return undefined;
  const cached = useAppStore.getState().weather[destinationId];
  if (!force && cached?.live && Date.now() - cached.fetchedAt < CACHE_MS) return cached;
  try {
    const url =
      `https://api.open-meteo.com/v1/forecast?latitude=${d.lat}&longitude=${d.lng}` +
      '&current=temperature_2m,relative_humidity_2m,weather_code,wind_speed_10m' +
      '&daily=weather_code,temperature_2m_max,temperature_2m_min,precipitation_probability_max&timezone=auto&forecast_days=7';
    const ctrl = new AbortController();
    const timer = setTimeout(() => ctrl.abort(), 8000);
    const res = await fetch(url, { signal: ctrl.signal });
    clearTimeout(timer);
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const j = await res.json();
    const snap: WeatherSnapshot = {
      destinationId,
      fetchedAt: Date.now(),
      live: true,
      current: {
        temp: Math.round(j.current.temperature_2m),
        code: j.current.weather_code,
        wind: Math.round(j.current.wind_speed_10m),
        humidity: Math.round(j.current.relative_humidity_2m),
      },
      daily: (j.daily.time as string[]).map((date, i) => ({
        date,
        max: Math.round(j.daily.temperature_2m_max[i]),
        min: Math.round(j.daily.temperature_2m_min[i]),
        code: j.daily.weather_code[i],
        rain: j.daily.precipitation_probability_max?.[i] ?? 0,
      })),
    };
    useAppStore.getState().setWeather(snap);
    return snap;
  } catch {
    if (cached) return cached;
    const est = estimate(d);
    useAppStore.getState().setWeather(est);
    return est;
  }
}

export function weatherSummary(w: WeatherSnapshot | undefined): string {
  if (!w) return 'Loading forecast…';
```

```
  const today = w.daily[0];
  const later = today && today.rain >= 50 ? ' · rain likely later' : today && today.rain >= 30 ? ' · chance of showers' : '';
  return `${weatherInfo(w.current.code).label.toLowerCase()}${later}`;
}

export const cToF = (c: number) => Math.round((c * 9) / 5 + 32);
```

src/store/__tests__/auth.test.ts

```
import { createHash } from 'crypto';
import { useAppStore } from '../useAppStore';

const reset = () => useAppStore.getState().resetEverything();

describe('accounts and sign-in', () => {
  beforeEach(reset);

  it('signs up, signs out and signs back in with the same password', async () => {
    const s = useAppStore.getState();
    expect(await s.signUp('Sam Okafor', 'sam@example.com', 'Passw0rd!x')).toEqual({ ok: true });
    const account = useAppStore.getState().accounts[0];
    expect(account.passwordHash).toMatch(/^pbkdf2-sha256\$/);
    expect(JSON.stringify(useAppStore.getState())).not.toContain('Passw0rd!x');

    useAppStore.getState().signOut();
    expect(useAppStore.getState().session).toBeNull();
    expect((await useAppStore.getState().signIn('SAM@example.com ', 'wrong-pass1', true)).ok).toBe(false);
    expect(await useAppStore.getState().signIn('SAM@example.com ', 'Passw0rd!x', true)).toEqual({ ok: true });
    expect(useAppStore.getState().session?.userId).toBe(account.id);
  });

  it('refuses a second account with the same email and a malformed email', async () => {
    await useAppStore.getState().signUp('A', 'a@example.com', 'Passw0rd!x');
    expect((await useAppStore.getState().signUp('B', 'A@example.com', 'Passw0rd!y')).ok).toBe(false);
    expect((await useAppStore.getState().signUp('B', 'not-an-email', 'Passw0rd!y')).ok).toBe(false);
    expect(useAppStore.getState().accounts).toHaveLength(1);
  });

  it('upgrades an old single-round SHA-256 account the first time it signs in', async () => {
    const salt = 'legacy-salt';
    const passwordHash = createHash('sha256').update(`${salt}:Passw0rd!x`).digest('hex');
    useAppStore.setState({ accounts: [{ id: 'u-old', name: 'Old User', email: 'old@example.com', passwordHash, salt, provider: 'passwor
d' }] });
    expect((await useAppStore.getState().signIn('old@example.com', 'bad-pass-1', false)).ok).toBe(false);
    expect(useAppStore.getState().accounts[0].passwordHash).toBe(passwordHash); // a failed sign-in changes nothing
    expect((await useAppStore.getState().signIn('old@example.com', 'Passw0rd!x', false)).ok).toBe(true);
    expect(useAppStore.getState().accounts[0].passwordHash).toMatch(/^pbkdf2-sha256\$/);
    useAppStore.getState().signOut();
    expect((await useAppStore.getState().signIn('old@example.com', 'Passw0rd!x', false)).ok).toBe(true);
  });

  it('resets a forgotten password', async () => {
    await useAppStore.getState().signUp('A', 'a@example.com', 'Passw0rd!x');
    useAppStore.getState().signOut();
    expect((await useAppStore.getState().resetPassword('nobody@example.com', 'Newpass1!x')).ok).toBe(false);
    expect((await useAppStore.getState().resetPassword('a@example.com', 'Newpass1!x')).ok).toBe(true);
    expect((await useAppStore.getState().signIn('a@example.com', 'Passw0rd!x', true)).ok).toBe(false);
    expect((await useAppStore.getState().signIn('a@example.com', 'Newpass1!x', true)).ok).toBe(true);
  });

  it('seeds the demo account and lets it sign in', async () => {
    await useAppStore.getState().ensureSeed();
    const s = useAppStore.getState();
    expect(s.trips.map((t) => t.name)).toContain('Amalfi Squad');
    expect(await s.signIn('maya.chen@togetherwego.app', 'TravelTogether2026!', true)).toEqual({ ok: true });
  });
});
```

src/store/__tests__/trips.test.ts

```
import { balances, settlePlan, useAppStore } from '../useAppStore';

const s = () => useAppStore.getState();
const tripOf = (id: string) => s().trips.find((t) => t.id === id)!;

async function signedInUser(name = 'Test User', email = 't@example.com') {
  s().resetEverything();
  await s().signUp(name, email, 'Passw0rd!x');
  return s().session!.userId;
}

const newTrip = () => s().createTrip({ name: 'Rome Crew', destinationId: 'rome', startDate: '2026-09-10', endDate: '2026-09-16', budget:
3000 });

describe('trips', () => {
  it('creates a trip owned by the signed-in user', async () => {
```

```
    const me = await signedInUser();
    const id = newTrip();
    const t = tripOf(id);
    expect(s().activeTripId).toBe(id);
    expect(t.ownerId).toBe(me);
    expect(t.members).toEqual([expect.objectContaining({ id: me, role: 'Organizer' })]);
    expect(t.inviteCode).toMatch(/^[A-Z]{3}-\d{4}$/);
    expect(t.tasks).toHaveLength(3);
    expect(t.messages[0]).toMatchObject({ ai: true });
  });

  it('lets a second person join by invite code, once', async () => {
    await signedInUser('Maya', 'maya@example.com');
    const id = newTrip();
    const code = tripOf(id).inviteCode;
    s().signOut();
    await s().signUp('Sam', 'sam@example.com', 'Passw0rd!x');
    const sam = s().session!.userId;
    expect(s().joinTrip(code)).toEqual({ ok: true, tripId: id });
    expect(tripOf(id).members.map((m) => m.id)).toContain(sam);
    expect(tripOf(id).messages.at(-1)).toMatchObject({ kind: 'system' });
    const again = s().joinTrip(code);
    expect(again.ok).toBe(false);
    expect(s().joinTrip('ZZZ-0000')).toMatchObject({ ok: false });
  });

  it('can join a trip from the built-in directory', async () => {
    await signedInUser();
    const r = s().joinTrip('SEO-5521');
    expect(r.ok).toBe(true);
    expect(s().trips.some((t) => t.inviteCode === 'SEO-5521')).toBe(true);
  });

  it('edits, leaves and deletes trips, moving the active trip sensibly', async () => {
    const me = await signedInUser();
    const a = newTrip();
    const b = s().createTrip({ name: 'Second', destinationId: 'tokyo', startDate: '2026-10-01', endDate: '2026-10-05', budget: 1000 });
    expect(s().activeTripId).toBe(b);
    s().updateTrip(a, { name: 'Renamed', budget: 4200 });
    expect(tripOf(a)).toMatchObject({ name: 'Renamed', budget: 4200 });
    s().leaveTrip(b);
    expect(tripOf(b).members.some((m) => m.id === me)).toBe(false);
    expect(s().activeTripId).toBe(a);
    s().deleteTrip(a);
    expect(s().trips.find((t) => t.id === a)).toBeUndefined();
    expect(s().activeTripId).toBeNull();
  });

  it('regenerates the invite code', async () => {
    await signedInUser();
    const id = newTrip();
    const before = tripOf(id).inviteCode;
    s().regenerateInvite(id);
    expect(tripOf(id).inviteCode).toMatch(/^[A-Z]{3}-\d{4}$/);
    expect(tripOf(id).inviteCode).not.toBe(undefined);
    void before;
  });
});

describe('itinerary and voting', () => {
  const base = { date: '2026-09-11', time: '10:00', title: 'Colosseum', location: 'Rome', type: 'sightseeing' as const, durationMin: 12
0, costPerPerson: 20, status: 'voting' as const, assigned: [], reminder: false };

  it('adds, edits and deletes activities', async () => {
    const me = await signedInUser();
    const t = newTrip();
    const id = s().addActivity(t, { ...base });
    expect(tripOf(t).activities[0]).toMatchObject({ id, createdBy: me, votes: { [me]: 1 } });
    s().updateActivity(t, id, { title: 'Colosseum tour', time: '09:30' });
    expect(tripOf(t).activities[0]).toMatchObject({ title: 'Colosseum tour', time: '09:30' });
    s().toggleAssign(t, id, me);
    expect(tripOf(t).activities[0].assigned).toEqual([me]);
    s().toggleAssign(t, id, me);
    expect(tripOf(t).activities[0].assigned).toEqual([]);
    s().deleteActivity(t, id);
    expect(tripOf(t).activities).toHaveLength(0);
  });

  it('confirms an activity once more than half the group votes yes', async () => {
    const me = await signedInUser();
    const t = newTrip();
    s().addMember(t, 'Friend One');
    s().addMember(t, 'Friend Two');
    const friend = tripOf(t).members[1].id;
    const id = s().addActivity(t, { ...base, votes: { [friend]: 1 } });
    expect(tripOf(t).activities[0].status).toBe('voting'); // 1 of 3
```

```
    s().voteActivity(t, id, 1);
    expect(tripOf(t).activities[0].votes[me]).toBe(1);
    expect(tripOf(t).activities[0].status).toBe('confirmed'); // 2 of 3
    s().voteActivity(t, id, 1); // same vote again takes it back
    expect(tripOf(t).activities[0].votes[me]).toBeUndefined();
  });

  it('removing a member cleans up their assignments and votes', async () => {
    await signedInUser();
    const t = newTrip();
    s().addMember(t, 'Friend');
    const friend = tripOf(t).members[1].id;
    const id = s().addActivity(t, { ...base, assigned: [friend], votes: { [friend]: -1 } });
    s().addTask(t, 'Book tickets', friend);
    s().removeMember(t, friend);
    const a = tripOf(t).activities.find((x) => x.id === id)!;
    expect(a.assigned).toEqual([]);
    expect(a.votes[friend]).toBeUndefined();
    expect(tripOf(t).tasks.find((k) => k.text === 'Book tickets')!.assignee).toBeUndefined();
  });
});

describe('budget', () => {
  it('tracks expenses and settles up to zero', async () => {
    const me = await signedInUser();
    const t = newTrip();
    s().addMember(t, 'Friend');
    const friend = tripOf(t).members[1].id;
    s().addExpense(t, { title: 'Hotel', amount: 300, category: 'Stays', paidBy: me, splitBetween: [me, friend], date: '2026-09-10' });
    s().addExpense(t, { title: 'Dinner', amount: 80, category: 'Food', paidBy: friend, splitBetween: [me, friend], date: '2026-09-11'
});
    expect(tripOf(t).expenses[0].title).toBe('Dinner'); // newest first
    expect(balances(tripOf(t))).toEqual({ [me]: 110, [friend]: -110 });
    const [payment] = settlePlan(tripOf(t));
    expect(payment).toEqual({ from: friend, to: me, amount: 110 });
    s().settleUp(t, payment.from, payment.to, payment.amount);
    expect(tripOf(t).expenses[0]).toMatchObject({ settlement: true, paidBy: friend });
    expect(settlePlan(tripOf(t))).toEqual([]);
    const spend = tripOf(t).expenses.filter((e) => !e.settlement).length;
    expect(spend).toBe(2);
    s().deleteExpense(t, tripOf(t).expenses[0].id);
    expect(tripOf(t).expenses).toHaveLength(2);
  });
});

describe('collaboration features', () => {
  it('polls: vote, change vote, close, delete', async () => {
    const me = await signedInUser();
    const t = newTrip();
    s().createPoll(t, 'Which beach?', ['Ostia', 'Sperlonga']);
    const poll = tripOf(t).polls[0];
    s().votePoll(t, poll.id, poll.options[0].id);
    expect(tripOf(t).polls[0].options[0].votes).toEqual([me]);
    s().votePoll(t, poll.id, poll.options[1].id); // switching moves the vote
    expect(tripOf(t).polls[0].options.map((o) => o.votes.length)).toEqual([0, 1]);
    s().closePoll(t, poll.id);
    s().votePoll(t, poll.id, poll.options[0].id); // closed polls ignore votes
    expect(tripOf(t).polls[0].options[0].votes).toEqual([]);
    s().deletePoll(t, poll.id);
    expect(tripOf(t).polls).toHaveLength(0);
  });

  it('checklist, packing, docs and photos', async () => {
    const me = await signedInUser();
    const t = newTrip();
    const first = tripOf(t).tasks[0];
    s().toggleTask(t, first.id);
    expect(tripOf(t).tasks[0].done).toBe(true);
    s().assignTask(t, first.id, me);
    expect(tripOf(t).tasks[0].assignee).toBe(me);
    s().deleteTask(t, first.id);
    expect(tripOf(t).tasks).toHaveLength(2);

    s().addPackingItems(t, [{ text: 'Passport', category: 'Documents' }, { text: 'passport', category: 'Documents' }, { text: 'Charger',
category: 'Tech' }]);
    expect(tripOf(t).packing.map((p) => p.text)).toEqual(['Passport', 'Charger']); // duplicates are skipped
    s().togglePacked(t, tripOf(t).packing[0].id);
    s().assignPacking(t, tripOf(t).packing[1].id, me);
    expect(tripOf(t).packing[0].packed).toBe(true);
    expect(tripOf(t).packing[1].broughtBy).toBe(me);

    s().addDoc(t, { name: 'Ticket.pdf', kind: 'pdf', offline: true });
    expect(tripOf(t).docs[0]).toMatchObject({ name: 'Ticket.pdf', addedBy: me });
    s().addPhoto(t, { uri: 'file:///p.jpg', caption: 'Pasta' });
    const photo = tripOf(t).photos[0];
    s().toggleLikePhoto(t, photo.id);
```

```
    expect(tripOf(t).photos[0].likes).toEqual([me]);
    s().toggleLikePhoto(t, photo.id);
    expect(tripOf(t).photos[0].likes).toEqual([]);
    s().deletePhoto(t, photo.id);
    s().deleteDoc(t, tripOf(t).docs[0].id);
    expect(tripOf(t).photos).toHaveLength(0);
    expect(tripOf(t).docs).toHaveLength(0);
  });

  it('records which server file belongs to a photo, a document and a receipt', async () => {
    const me = await signedInUser();
    const t = newTrip();
    s().addPhoto(t, { uri: 'file:///p.jpg', caption: 'Pasta' });
    s().addDoc(t, { name: 'Ticket', kind: 'pdf', uri: 'file:///t.pdf', offline: true });
    s().addExpense(t, { title: 'Taxi', amount: 20, category: 'Transport', paidBy: me, splitBetween: [me], date: '2026-09-10', receipt:
'file:///r.jpg' });
    const [photo, doc, expense] = [tripOf(t).photos[0], tripOf(t).docs[0], tripOf(t).expenses[0]];
    s().attachBlob(t, 'photos', photo.id, 'aaaa1111_jpg');
    s().attachBlob(t, 'docs', doc.id, 'bbbb2222_pdf');
    s().attachBlob(t, 'expenses', expense.id, 'cccc3333_jpg');
    expect(tripOf(t).photos[0]).toMatchObject({ uri: 'file:///p.jpg', blob: 'aaaa1111_jpg' });
    expect(tripOf(t).docs[0]).toMatchObject({ uri: 'file:///t.pdf', blob: 'bbbb2222_pdf' });
    expect(tripOf(t).expenses[0]).toMatchObject({ receipt: 'file:///r.jpg', receiptBlob: 'cccc3333_jpg' });
    s().attachBlob(t, 'photos', 'no-such-photo', 'dddd4444_jpg'); // an item deleted meanwhile is ignored
    s().attachBlob('no-such-trip', 'photos', photo.id, 'eeee5555_jpg');
    expect(tripOf(t).photos).toHaveLength(1);
  });

  it('chat keeps only the latest 400 messages', async () => {
    const me = await signedInUser();
    const t = newTrip();
    for (let i = 0; i < 405; i++) s().sendMessage(t, { authorId: me, text: `m${i}` });
    const msgs = tripOf(t).messages;
    expect(msgs).toHaveLength(400);
    expect(msgs.at(-1)!.text).toBe('m404');
    s().clearChat(t);
    expect(tripOf(t).messages).toEqual([]);
  });

  it('changes roles', async () => {
    const me = await signedInUser();
    const t = newTrip();
    s().setRole(t, me, 'Treasurer');
    expect(tripOf(t).members[0].role).toBe('Treasurer');
  });
});

describe('sample data', () => {
  it('can be cleared without touching the person’s own trips, and does not come back', async () => {
    s().resetEverything();
    await s().ensureSeed();
    expect(s().trips.some((t) => t.id === 'trip-amalfi')).toBe(true);
    await s().signUp('Own Trip', 'own@example.com', 'Passw0rd!x');
    const mine = newTrip();
    s().clearDemoData();
    expect(s().trips.map((t) => t.id)).toEqual([mine]);
    expect(s().activeTripId).toBe(mine);
    expect(s().notifications.every((n) => !n.tripId || n.tripId === mine)).toBe(true);
    expect(s().bucket).toEqual([]);
    expect(s().seeded).toBe(true);
    await s().ensureSeed(); // already seeded: nothing returns
    expect(s().trips.map((t) => t.id)).toEqual([mine]);
    expect(s().accounts.some((a) => a.email === 'own@example.com')).toBe(true);
  });

  it('keeps bucket-list items the person added themselves', async () => {
    s().resetEverything();
    await s().ensureSeed();
    s().addBucket('Learn to surf in Portugal');
    s().clearDemoData();
    expect(s().bucket.map((b) => b.text)).toEqual(['Learn to surf in Portugal']);
  });
});

describe('app-level state', () => {
  it('bucket list, favourites, notifications and settings', async () => {
    await signedInUser();
    s().addBucket('See the aurora', 'iceland');
    expect(s().bucket[0]).toMatchObject({ text: 'See the aurora', done: false });
    s().toggleBucket(s().bucket[0].id);
    expect(s().bucket[0].done).toBe(true);
    s().deleteBucket(s().bucket[0].id);
    expect(s().bucket).toEqual([]);

    s().toggleFavorite('rome');
    expect(s().favorites).toContain('rome');
```

```
    s().toggleFavorite('rome');
    expect(s().favorites).not.toContain('rome');

    for (let i = 0; i < 70; i++) s().pushNotification({ kind: 'system', title: `n${i}`, body: '' });
    expect(s().notifications).toHaveLength(60);
    s().markKindRead('system');
    expect(s().notifications.every((n) => n.read)).toBe(true);
    s().clearNotifications();
    expect(s().notifications).toEqual([]);

    s().updateSettings({ language: 'es', textScale: 1.25 });
    expect(s().settings).toMatchObject({ language: 'es', textScale: 1.25, haptics: true });
    s().updatePersonal({ nickname: 'Mimi', interests: ['food'] });
    expect(s().personal).toMatchObject({ nickname: 'Mimi', interests: ['food'], travelStyle: 'balanced' });
  });

  it('keeps the session only when "remember me" was ticked', async () => {
    await signedInUser();
    const opts = useAppStore.persist.getOptions();
    const saved = opts.partialize!({ ...s(), session: { userId: 'u', remember: false } });
    expect((saved as { session: unknown }).session).toBeNull();
    const kept = opts.partialize!({ ...s(), session: { userId: 'u', remember: true } });
    expect((kept as { session: unknown }).session).toEqual({ userId: 'u', remember: true });
  });

  it('re-seeds the demo trip for a different stage of the trip', async () => {
    await s().ensureSeed();
    s().reseedDemo('active');
    expect(s().demoTimeline).toBe('active');
    expect(s().trips.filter((t) => t.id === 'trip-amalfi')).toHaveLength(1);
  });
});
```

src/store/seed.ts

```
/**
 * Demo content. Dates are generated relative to "today" so the sample trip is
 * always fresh — whether the app is opened today or at the competition months
 * later. Settings → Demo timeline can re-seed it as upcoming, in-progress or
 * completed to show the whole trip lifecycle.
 */
import { packingBase } from '@/data/reference';
import { uid } from '@/utils/format';
import { addDays, todayInTz, zonedToUtc } from '@/utils/time';

import type { Activity, Expense, Flight, Member, PackingItem, Photo, Trip } from './types';

export const DEMO_EMAIL = 'maya.chen@togetherwego.app';
export const DEMO_PASSWORD = 'TravelTogether2026!';
export const DEMO_USER_ID = 'u-maya';
/** The sample trips that come with the app (the fourth is from an older version). "Clear sample trips" removes exactly these. */
export const DEMO_TRIP_IDS = ['trip-amalfi', 'trip-bali', 'trip-bcn', 'trip-tokyo'];

export type DemoTimeline = 'upcoming' | 'active' | 'completed';

const TOKYO_TZ = 'Asia/Tokyo';
const AMALFI_TZ = 'Europe/Rome';

/** The 6 members of the Tokyo & Kyoto trip. */
export const demoMembers: Member[] = [
  { id: DEMO_USER_ID, name: 'Maya Chen', email: DEMO_EMAIL, avatar: 'avatar-maya', role: 'Navigator', homeCity: 'New York' },
  { id: 'm-diego', name: 'Diego Alvarez', avatar: 'avatar-diego', role: 'Food Lead', homeCity: 'New York' },
  { id: 'm-priya', name: 'Priya Patel', avatar: 'avatar-priya', role: 'Treasurer', homeCity: 'San Francisco' },
  { id: 'm-sam', name: 'Sam Okafor', avatar: 'avatar-sam', role: 'Photographer', homeCity: 'New York' },
  { id: 'm-ana', name: 'Ana Silva', avatar: 'avatar-ana', role: 'Planner', homeCity: 'Boston' },
  { id: 'm-noah', name: 'Noah Kim', avatar: 'avatar-noah', role: 'Activities Lead', homeCity: 'Chicago' },
];

const ALL = demoMembers.map((m) => m.id);
const votesFrom = (ids: string[], down: string[] = []): Record<string, 1 | -1> => {
  const v: Record<string, 1 | -1> = {};
  ids.forEach((id) => (v[id] = 1));
  down.forEach((id) => (v[id] = -1));
  return v;
};

function act(start: string, day: number, a: Omit<Activity, 'id' | 'date' | 'createdBy' | 'reminder'> & { reminder?: boolean }): Activity
{
  return { id: uid('a-'), date: addDays(start, day), createdBy: DEMO_USER_ID, reminder: a.reminder ?? a.status === 'confirmed', ...a };
}

function demoActivities(start: string): Activity[] {
  return [
    // Day 1
    act(start, 0, { time: '14:30', title: 'Arrive at Haneda Airport (HND)', location: 'Ōta, Tokyo', type: 'transport', durationMin: 150,
costPerPerson: 0, status: 'confirmed', assigned: [DEMO_USER_ID, 'm-sam'], votes: votesFrom(ALL), lat: 35.5494, lng: 139.7798, notes: 'Me
```

```
et at the Keikyu Line ticket gates, Terminal 3 arrivals.' }),
    act(start, 0, { time: '17:30', title: 'Check-in — Shinjuku Grand Hotel', location: 'Shinjuku, Tokyo', type: 'stay', durationMin: 30,
costPerPerson: 180, status: 'confirmed', assigned: [DEMO_USER_ID], votes: votesFrom(ALL), lat: 35.6938, lng: 139.7034 }),
    act(start, 0, { time: '19:30', title: 'Welcome Dinner — Omoide Yokocho', location: 'Shinjuku, Tokyo', type: 'food', durationMin: 12
0, costPerPerson: 30, status: 'confirmed', assigned: ['m-diego'], votes: votesFrom(ALL.slice(0, 5)), lat: 35.6929, lng: 139.6996, placeI
d: 'omoide' }),
    // Day 2
    act(start, 1, { time: '08:00', title: 'Tsukiji Outer Market Breakfast', location: 'Tsukiji, Tokyo', type: 'food', durationMin: 90, c
ostPerPerson: 25, status: 'confirmed', assigned: ['m-diego'], votes: votesFrom(ALL.slice(0, 4)), lat: 35.6655, lng: 139.7707, placeId:
'tsukiji' }),
    act(start, 1, { time: '10:30', title: 'Senso-ji Temple & Asakusa', location: 'Asakusa, Tokyo', type: 'culture', durationMin: 120, co
stPerPerson: 0, status: 'voting', assigned: ['m-ana', 'm-noah'], votes: votesFrom(['m-ana', 'm-noah', 'm-priya'], ['m-sam']), lat: 35.71
48, lng: 139.7967, placeId: 'sensoji' }),
    act(start, 1, { time: '14:00', title: 'teamLab Planets Digital Art Museum', location: 'Toyosu, Tokyo', type: 'culture', durationMin:
150, costPerPerson: 30, status: 'pending', assigned: ['m-ana', 'm-priya', 'm-noah'], votes: votesFrom(ALL.slice(0, 5)), lat: 35.6492, ln
g: 139.7898, placeId: 'teamlab', notes: 'Wear shorts — you walk through knee-deep water!' }),
    act(start, 1, { time: '20:00', title: 'Shibuya Crossing & Rooftop Bar', location: 'Shibuya, Tokyo', type: 'nightlife', durationMin:
180, costPerPerson: 40, status: 'confirmed', assigned: ['m-sam'], votes: votesFrom(ALL), lat: 35.6595, lng: 139.7005, placeId: 'shibuya-
crossing' }),
    // Day 3
    act(start, 2, { time: '09:00', title: 'Meiji Jingu & Harajuku stroll', location: 'Harajuku, Tokyo', type: 'culture', durationMin: 15
0, costPerPerson: 0, status: 'confirmed', assigned: ['m-ana'], votes: votesFrom(ALL), lat: 35.6764, lng: 139.6993, placeId: 'meiji' }),
    act(start, 2, { time: '13:30', title: 'Akihabara arcades & retro games', location: 'Akihabara, Tokyo', type: 'shopping', durationMi
n: 120, costPerPerson: 15, status: 'confirmed', assigned: ['m-noah'], votes: votesFrom(ALL.slice(1)), lat: 35.6984, lng: 139.7731, place
Id: 'akihabara' }),
    act(start, 2, { time: '19:00', title: 'Karaoke night in Shibuya', location: 'Shibuya, Tokyo', type: 'nightlife', durationMin: 120, c
ostPerPerson: 20, status: 'voting', assigned: ['m-noah'], votes: votesFrom(['m-noah', 'm-diego', 'm-sam']), lat: 35.66, lng: 139.6985, p
laceId: 'karaoke' }),
    // Day 4
    act(start, 3, { time: '07:30', title: 'Day trip: Mt Fuji & Chureito Pagoda', location: 'Fujiyoshida, Yamanashi', type: 'adventure',
durationMin: 540, costPerPerson: 45, status: 'confirmed', assigned: [DEMO_USER_ID, 'm-sam'], votes: votesFrom(ALL), lat: 35.5016, lng: 1
38.8013, notes: 'Bus from Shinjuku Expressway Bus Terminal, gate B.' }),
    // Day 5
    act(start, 4, { time: '08:30', title: 'Shinkansen to Kyoto', location: 'Tokyo Station', type: 'transport', durationMin: 135, costPer
Person: 0, status: 'confirmed', assigned: [DEMO_USER_ID], votes: votesFrom(ALL), lat: 35.6812, lng: 139.7671, notes: 'Covered by JR Pas
s. Sit on the right (seat E) for Fuji views.' }),
    act(start, 4, { time: '13:00', title: 'Check-in — Gion ryokan', location: 'Gion, Kyoto', type: 'stay', durationMin: 30, costPerPerso
n: 140, status: 'confirmed', assigned: ['m-priya'], votes: votesFrom(ALL), lat: 35.0037, lng: 135.7788 }),
    act(start, 4, { time: '17:00', title: 'Gion evening walk', location: 'Gion, Kyoto', type: 'culture', durationMin: 90, costPerPerson:
0, status: 'confirmed', assigned: ['m-ana'], votes: votesFrom(ALL.slice(0, 5)), lat: 35.0037, lng: 135.7788, placeId: 'gion' }),
    // Day 6
    act(start, 5, { time: '06:30', title: 'Fushimi Inari sunrise hike', location: 'Fushimi, Kyoto', type: 'culture', durationMin: 150, c
ostPerPerson: 0, status: 'confirmed', assigned: ['m-sam'], votes: votesFrom(ALL.slice(0, 4)), lat: 34.9671, lng: 135.7727, placeId: 'ina
ri' }),
    act(start, 5, { time: '11:00', title: 'Nishiki Market lunch crawl', location: 'Nakagyo, Kyoto', type: 'food', durationMin: 90, costP
erPerson: 20, status: 'confirmed', assigned: ['m-diego'], votes: votesFrom(ALL), lat: 35.005, lng: 135.7649, placeId: 'nishiki' }),
    act(start, 5, { time: '15:00', title: 'Tea ceremony experience', location: 'Higashiyama, Kyoto', type: 'culture', durationMin: 60, c
ostPerPerson: 40, status: 'voting', assigned: ['m-priya'], votes: votesFrom(['m-priya', 'm-ana'], ['m-noah']), lat: 34.9986, lng: 135.78
09, placeId: 'tea' }),
    // Day 7
    act(start, 6, { time: '07:30', title: 'Arashiyama Bamboo Grove', location: 'Arashiyama, Kyoto', type: 'sightseeing', durationMin: 6
0, costPerPerson: 0, status: 'confirmed', assigned: ['m-sam'], votes: votesFrom(ALL), lat: 35.017, lng: 135.6713, placeId: 'bamboo' }),
    act(start, 6, { time: '13:00', title: 'Hozugawa river boat ride', location: 'Kameoka, Kyoto', type: 'adventure', durationMin: 120, c
ostPerPerson: 35, status: 'pending', assigned: ['m-noah'], votes: votesFrom(['m-noah', 'm-diego', 'm-sam', DEMO_USER_ID]), lat: 35.0128,
lng: 135.6744, placeId: 'hozugawa' }),
    // Day 8 left empty on purpose (free day)
    // Day 9
    act(start, 8, { time: '10:00', title: 'Shinkansen back to Tokyo', location: 'Kyoto Station', type: 'transport', durationMin: 135, co
stPerPerson: 0, status: 'confirmed', assigned: [DEMO_USER_ID], votes: votesFrom(ALL), lat: 34.9858, lng: 135.7588 }),
    act(start, 8, { time: '19:00', title: 'Farewell dinner — Gonpachi', location: 'Nishi-Azabu, Tokyo', type: 'food', durationMin: 120,
costPerPerson: 35, status: 'voting', assigned: ['m-diego'], votes: votesFrom(['m-diego', 'm-ana', 'm-priya']), lat: 35.6563, lng: 139.72
4, placeId: 'gonpachi' }),
    // Day 10
    act(start, 9, { time: '06:30', title: 'Head to Haneda for departures', location: 'Haneda Airport, Tokyo', type: 'transport', duratio
nMin: 60, costPerPerson: 6, status: 'confirmed', assigned: [DEMO_USER_ID], votes: votesFrom(ALL), lat: 35.5494, lng: 139.7798 }),
  ];
}

interface Airfield {
  tz: string;
  code: string;
}
const HND: Airfield = { tz: TOKYO_TZ, code: 'HND' };
const NAP: Airfield = { tz: AMALFI_TZ, code: 'NAP' };

function flightPair(at: Airfield, memberId: string, airline: string, outNo: string, inNo: string, from: string, start: string, end: stri
ng, arr: string, dep: string, hoursOut: number, hoursBack: number, arrDayOffset = 0): Flight[] {
  const arriveUtc = zonedToUtc(addDays(start, arrDayOffset), arr, at.tz);
  const departUtc = zonedToUtc(end, dep, at.tz);
  const confirmation = 'TWG' + memberId.slice(-3).toUpperCase();
  return [
    { id: uid('f-'), memberId, airline, flightNo: outNo, from, to: at.code, departUtc: arriveUtc - hoursOut * 3600000, arriveUtc, direct
ion: 'arrival', seat: '32A', confirmation },
    { id: uid('f-'), memberId, airline, flightNo: inNo, from: at.code, to: from, departUtc, arriveUtc: departUtc + hoursBack * 3600000,
direction: 'departure', seat: '41C', confirmation },
  ];
```

```
}

function demoExpenses(start: string): Expense[] {
  const before = addDays(start, -20);
  return [
    { id: uid('e-'), title: 'Round-trip flights', amount: 1840, category: 'Flights', paidBy: 'm-priya', splitBetween: ALL, date: before
},
    { id: uid('e-'), title: 'Shinjuku apartment', amount: 960, category: 'Stays', paidBy: DEMO_USER_ID, splitBetween: ALL, date: addDays
(before, 2) },
    { id: uid('e-'), title: 'JR Rail passes', amount: 420, category: 'Transport', paidBy: 'm-sam', splitBetween: ALL, date: addDays(befo
re, 5) },
    { id: uid('e-'), title: 'Group dinners', amount: 265, category: 'Food', paidBy: 'm-diego', splitBetween: ALL, date: addDays(before,
8), original: { amount: 40386, currency: 'JPY' } },
    { id: uid('e-'), title: 'teamLab tickets', amount: 132, category: 'Activities', paidBy: 'm-ana', splitBetween: ALL, date: addDays(be
fore, 9) },
  ];
}

function demoPacking(): PackingItem[] {
  const done = new Set(['Passport', 'Phone charger', 'Universal adapter', 'Comfortable walking shoes']);
  return packingBase.flatMap((g) => g.items.map((text) => ({ id: uid('p-'), text, category: g.category, packed: done.has(text) })));
}

function demoPhotos(start: string, now: number): Photo[] {
  const list: [string, string, string, number][] = [
    ['shibuya', 'Shibuya after dark 🌃', 'm-sam', 1],
    ['ramen', 'Diego's 2am ramen find 🍜', 'm-diego', 0],
    ['fushimi-inari', '10,000 gates and counting ⛩️', 'm-sam', 5],
    ['lantern-alley', 'Lantern alley vibes', 'm-ana', 2],
    ['sushi', 'Omakase at the market', 'm-priya', 1],
    ['tokyo-skyline', 'View from the tower', DEMO_USER_ID, 2],
    ['fuji-blossom', 'Fuji finally showed up 🗻', 'm-noah', 3],
    ['omoide-yokocho', 'Yakitori alley', 'm-diego', 0],
    ['akihabara', 'Akiba arcade crew', 'm-noah', 2],
    ['kyoto-skyline', 'Kyoto from Kiyomizu', 'm-ana', 5],
  ];
  return list.map(([uri, caption, by, day], i) => ({
    id: uid('ph-'), uri, caption, by, day: addDays(start, day), ts: now - i * 3600000,
    likes: ALL.filter((_, j) => (i + j) % 3 !== 0),
  }));
}

export function inviteCodeFor(name: string): string {
  const letters = name.replace(/[^A-Za-z]/g, '').toUpperCase().padEnd(3, 'X').slice(0, 3);
  return `${letters}-${Math.floor(1000 + Math.random() * 9000)}`;
}

/** The big 10-day Tokyo & Kyoto trip — kept as a second trip so multi-trip management has something to show. */
export function buildTokyoTrip(now = Date.now()): Trip {
  const today = todayInTz(TOKYO_TZ);
  const start = addDays(today, 45);
  const end = addDays(start, 9);
  const flights = [
    ...flightPair(HND, DEMO_USER_ID, 'ANA', 'NH 175', 'NH 176', 'JFK', start, end, '14:20', '09:05', 14, 13),
    ...flightPair(HND, 'm-diego', 'Japan Airlines', 'JL 006', 'JL 005', 'JFK', start, end, '16:40', '11:30', 14, 13),
    ...flightPair(HND, 'm-priya', 'United', 'UA 837', 'UA 838', 'SFO', start, end, '07:15', '09:05', 11, 10, 1),
    ...flightPair(HND, 'm-sam', 'ANA', 'NH 175', 'NH 176', 'JFK', start, addDays(end, -1), '14:20', '18:00', 14, 13),
  ];
  const t = (mins: number) => now - mins * 60000;
  return {
    id: 'trip-tokyo',
    name: 'Tokyo & Kyoto Explorer',
    destinationId: 'tokyo',
    extraDestinationIds: ['kyoto', 'mount-fuji'],
    startDate: start,
    endDate: end,
    cover: 'tokyo',
    budget: 4200,
    inviteCode: 'TKY-2481',
    ownerId: DEMO_USER_ID,
    members: demoMembers.map((m) => ({ ...m })),
    activities: demoActivities(start),
    expenses: demoExpenses(start),
    flights,
    polls: [
      {
        id: uid('poll-'), question: 'Day 4 evening — what are we doing?', createdBy: 'm-noah', createdAt: t(600),
        options: [
          { id: 'o1', text: 'Robot show in Shinjuku', votes: ['m-noah'] },
          { id: 'o2', text: 'Sumida river cruise', votes: ['m-priya'] },
          { id: 'o3', text: 'Karaoke night', votes: ['m-diego', 'm-ana', 'm-sam'] },
        ],
      },
      {
        id: uid('poll-'), question: 'Farewell dinner: where should we go?', createdBy: 'm-diego', createdAt: t(300),
        options: [
```

```
          { id: 'o1', text: 'Gonpachi izakaya (Kill Bill vibes)', votes: ['m-diego', 'm-ana'] },
          { id: 'o2', text: 'Sushi omakase splurge', votes: ['m-priya'] },
          { id: 'o3', text: 'Ramen crawl in Ebisu', votes: ['m-noah'] },
        ],
      },
    ],
    tasks: [
      { id: uid('t-'), text: 'Passports valid 6+ months', done: true },
      { id: uid('t-'), text: 'Book airport transfer', done: false, assignee: DEMO_USER_ID },
      { id: uid('t-'), text: 'Travel insurance', done: false, assignee: 'm-priya' },
      { id: uid('t-'), text: 'Download offline maps', done: true, assignee: 'm-sam' },
      { id: uid('t-'), text: 'Pocket wifi pickup code', done: false, assignee: 'm-diego' },
    ],
    docs: [
      {
        id: uid('d-'), name: 'Flight tickets (all members)', kind: 'note', addedBy: 'm-priya', addedAt: t(4000), offline: true,
        note: 'Maya — ANA NH 175 · JFK → HND · Conf. TWGAYA\nSam — ANA NH 175 · JFK → HND · Conf. TWGSAM\nDiego — JAL JL 006 · JFK → HND · Conf. TWGEGO\nPriya — United UA 837 · SFO → HND · Conf. TWGIYA\n\nCheck-in opens 24h before departure. Bring the passport used for booking.',
      },
      {
        id: uid('d-'), name: 'Apartment booking', kind: 'note', addedBy: DEMO_USER_ID, addedAt: t(3800), offline: true,
        note: 'Shinjuku Grand Hotel — 3 rooms, 4 nights\nCheck-in from 15:00 · Check-out 11:00\nBooking ref: SGH-55120\n\nShow the taxi driver:\n新宿駅までお願いします (To Shinjuku Station, please)',
      },
      {
        id: uid('d-'), name: 'Passport scans', kind: 'note', addedBy: DEMO_USER_ID, addedAt: t(3600), offline: true,
        note: 'Encrypted copies stored on each member’s phone.\nIf a passport is lost: report at the nearest koban (police box), then contact your embassy.\nUS Embassy Tokyo: +81 3-3224-5000',
      },
      {
        id: uid('d-'), name: 'Emergency contacts', kind: 'note', addedBy: 'm-ana', addedAt: t(3000), offline: true,
        note: 'Police 110 · Ambulance/Fire 119\nJapan Visitor Hotline (24h, English): 050-3816-2787\nTravel insurance: TravelSafe +1 800 555 0199, policy TS-88213',
      },
    ],
    messages: [
      { id: uid('c-'), authorId: 'm-priya', text: 'JR passes are booked 🎉 picking them up at Haneda', ts: t(320) },
      { id: uid('c-'), authorId: 'm-ana', text: 'Can we lock in teamLab for Day 2 afternoon? Tickets sell out', ts: t(260) },
      { id: uid('c-'), authorId: 'm-noah', text: 'Voted! Also started a poll for Day 4 evening 👀', ts: t(200) },
      { id: uid('c-'), authorId: 'm-diego', text: 'Found a ramen spot 4 min from the apartment, open till 2am 🍜', ts: t(42) },
      { id: uid('c-'), authorId: DEMO_USER_ID, text: 'Adding it to Day 1 after Shibuya!', ts: t(40) },
      { id: uid('c-'), authorId: 'assistant', ai: true, aiSource: 'local', text: 'Heads up: Priya lands at 07:15 on Day 2 — the morning after everyone else. Want me to move Tsukiji breakfast to 09:30 so she can join?', ts: t(38) },
    ],
    photos: demoPhotos(start, now),
    packing: demoPacking(),
    completed: false,
    createdAt: t(60 * 24 * 30),
  };
}


// ---------------------------------------------------------------------------
// Amalfi Squad — the main demo trip (4 friends, 6 days, Positano)
// ---------------------------------------------------------------------------
const squad: Member[] = [
  { id: DEMO_USER_ID, name: 'Maya Chen', email: DEMO_EMAIL, avatar: 'avatar-maya', role: 'Navigator', homeCity: 'New York' },
  { id: 'm-diego', name: 'Diego Alvarez', avatar: 'avatar-diego', role: 'Food Lead', homeCity: 'New York' },
  { id: 'm-priya', name: 'Priya Patel', avatar: 'avatar-priya', role: 'Treasurer', homeCity: 'San Francisco' },
  { id: 'm-sam', name: 'Sam Okafor', avatar: 'avatar-sam', role: 'Photographer', homeCity: 'New York' },
];
const SQUAD = squad.map((m) => m.id);

function amalfiActivities(start: string): Activity[] {
  return [
    act(start, 0, { time: '15:40', title: 'Arrive at Naples Airport (NAP)', location: 'Naples Airport', type: 'transport', durationMin: 90, costPerPerson: 0, status: 'confirmed', assigned: [DEMO_USER_ID], votes: votesFrom(SQUAD), lat: 40.886, lng: 14.2908, notes: 'Private transfer to Positano, about 1h 30m. Driver holds a TogetherWeGo sign.' }),
    act(start, 0, { time: '17:45', title: 'Check-in — Villa Margherita', location: 'Positano', type: 'stay', durationMin: 30, costPerPerson: 410, status: 'confirmed', assigned: ['m-priya'], votes: votesFrom(SQUAD), lat: 40.6283, lng: 14.4853, notes: 'Keys in the lockbox, code in the Docs folder.' }),
    act(start, 0, { time: '19:00', title: 'Dinner at Da Vincenzo', location: 'Viale Pasitea', type: 'food', durationMin: 120, costPerPerson: 64, status: 'confirmed', assigned: ['m-diego'], votes: votesFrom(SQUAD), lat: 40.6295, lng: 14.488, placeId: 'da-vincenzo' }),
    act(start, 1, { time: '08:00', title: 'Path of the Gods hike', location: 'Bomerano', type: 'adventure', durationMin: 300, costPerPerson: 0, status: 'confirmed', assigned: [DEMO_USER_ID, 'm-sam'], votes: votesFrom([DEMO_USER_ID, 'm-diego', 'm-sam']), lat: 40.619, lng: 14.5176, placeId: 'path-of-gods', notes: 'Bring water, sun hat and sturdy shoes. Bus from Amalfi to Bomerano at 07:15.' }),
    act(start, 1, { time: '20:00', title: 'Sunset aperitivo at Le Sirenuse', location: 'Positano', type: 'nightlife', durationMin: 90, costPerPerson: 40, status: 'confirmed', assigned: ['m-sam'], votes: votesFrom(SQUAD.slice(0, 3)), lat: 40.6293, lng: 14.4868, placeId: 'sunset-bar' }),
    act(start, 2, { time: '10:00', title: 'Boat tour to Capri', location: 'Positano Marina', type: 'adventure', durationMin: 540, costPerPerson: 140, status: 'confirmed', assigned: [DEMO_USER_ID, 'm-priya'], votes: votesFrom([DEMO_USER_ID, 'm-priya']), lat: 40.6272, lng: 14.4868, placeId: 'capri-boat' }),
    act(start, 3, { time: '11:00', title: 'Beach day at Fornillo Beach', location: 'Fornillo Beach', type: 'sightseeing', durationMin: 180, costPerPerson: 35, status: 'voting', assigned: [], votes: {}, reminder: false, lat: 40.6255, lng: 14.4795, placeId: 'fornillo' }),
    act(start, 3, { time: '16:00', title: 'Limoncello tasting', location: 'Amalfi', type: 'food', durationMin: 45, costPerPerson: 23, st
```

```
atus: 'voting', assigned: ['m-priya'], votes: votesFrom(['m-priya']), reminder: false, lat: 40.634, lng: 14.6027, placeId: 'limoncello'
}),
    act(start, 4, { time: '10:00', title: 'Villa Rufolo gardens in Ravello', location: 'Ravello', type: 'sightseeing', durationMin: 150,
costPerPerson: 8, status: 'confirmed', assigned: ['m-sam'], votes: votesFrom(SQUAD), lat: 40.649, lng: 14.6118, placeId: 'villa-rufolo'
}),
    act(start, 4, { time: '19:30', title: 'Farewell dinner — Trattoria da Adolfo', location: 'Laurito Beach', type: 'food', durationMin:
150, costPerPerson: 55, status: 'confirmed', assigned: ['m-diego'], votes: votesFrom(SQUAD.slice(0, 3)), lat: 40.6224, lng: 14.4731, pla
ceId: 'trattoria-chitarrella' }),
    act(start, 5, { time: '14:30', title: 'Transfer to Naples Airport', location: 'Positano', type: 'transport', durationMin: 100, costP
erPerson: 35, status: 'confirmed', assigned: [DEMO_USER_ID], votes: votesFrom(SQUAD), lat: 40.628, lng: 14.485 }),
  ];
}

export function buildDemoTrip(timeline: DemoTimeline = 'upcoming', now = Date.now()): Trip {
  const today = todayInTz(AMALFI_TZ);
  const offset = timeline === 'upcoming' ? 3 : timeline === 'active' ? -1 : -14;
  const start = addDays(today, offset);
  const end = addDays(start, 5);
  const t = (mins: number) => now - mins * 60000;
  const before = addDays(start, -20);
  return {
    id: 'trip-amalfi',
    name: 'Amalfi Squad',
    destinationId: 'positano',
    extraDestinationIds: [],
    startDate: start,
    endDate: end,
    cover: 'amalfi',
    budget: 4000,
    inviteCode: 'AML-2026',
    ownerId: DEMO_USER_ID,
    members: squad.map((m) => ({ ...m })),
    activities: amalfiActivities(start),
    expenses: [
      { id: uid('e-'), title: 'Villa Margherita (6 nights)', amount: 1650, category: 'Stays', paidBy: 'm-priya', splitBetween: SQUAD, da
te: before },
      { id: uid('e-'), title: 'Airport transfers', amount: 140, category: 'Transport', paidBy: DEMO_USER_ID, splitBetween: SQUAD, date:
addDays(before, 4) },
      { id: uid('e-'), title: 'Scooters & ferry tickets', amount: 210, category: 'Transport', paidBy: 'm-sam', splitBetween: SQUAD, dat
e: addDays(before, 6) },
      { id: uid('e-'), title: 'Dinner deposits & groceries', amount: 86, category: 'Food', paidBy: 'm-diego', splitBetween: SQUAD, date:
addDays(before, 8), original: { amount: 74, currency: 'EUR' } },
    ],
    flights: [
      ...flightPair(NAP, DEMO_USER_ID, 'Delta', 'DL 184', 'DL 185', 'JFK', start, end, '15:40', '19:15', 9.5, 10),
      ...flightPair(NAP, 'm-priya', 'ITA Airways', 'AZ 609', 'AZ 610', 'JFK', start, end, '17:05', '21:30', 9.5, 10),
    ],
    polls: [
      {
        id: uid('poll-'), question: 'Beach day — which beach?', createdBy: 'm-sam', createdAt: t(600),
        options: [
          { id: 'o1', text: 'Fornillo (quiet, pebbles)', votes: ['m-sam', 'm-diego'] },
          { id: 'o2', text: 'Spiaggia Grande (lively)', votes: ['m-priya'] },
          { id: 'o3', text: 'Fiordo di Furore (jump in!)', votes: [DEMO_USER_ID] },
        ],
      },
    ],
    tasks: [
      { id: uid('t-'), text: 'Passports valid 6+ months', done: true },
      { id: uid('t-'), text: 'Book airport transfers', done: true, assignee: DEMO_USER_ID },
      { id: uid('t-'), text: 'Travel insurance', done: false, assignee: 'm-priya' },
      { id: uid('t-'), text: 'Rent scooters (licences!)', done: false, assignee: 'm-sam' },
      { id: uid('t-'), text: 'Reserve Da Vincenzo for 4', done: true, assignee: 'm-diego' },
      { id: uid('t-'), text: 'Download offline maps', done: false, assignee: DEMO_USER_ID },
    ],
    docs: [
      {
        id: uid('d-'), name: 'Flight tickets', kind: 'note', addedBy: 'm-priya', addedAt: t(4000), offline: true,
        note: 'Maya — Delta DL 184 · JFK → NAP · Conf. TWGAYA\nPriya — ITA Airways AZ 609 · JFK → NAP · Conf. TWGIYA\n\nCheck-in opens 2
4h before departure. Bring the passport used for booking.',
      },
      {
        id: uid('d-'), name: 'Villa Margherita booking', kind: 'note', addedBy: 'm-priya', addedAt: t(3800), offline: true,
        note: 'Villa Margherita, Via Chiesa Nuova, Positano\nCheck-in from 15:00 · Check-out 10:00\nBooking ref: VMG-48213\nDoor code ar
rives by message on the day.\n\nTaxi phrase:\nVilla Margherita, per favore (Villa Margherita, please)',
      },
      {
        id: uid('d-'), name: 'Passport scans', kind: 'note', addedBy: DEMO_USER_ID, addedAt: t(3600), offline: true,
        note: 'Copies are stored on each member's phone.\nIf a passport is lost: report it to the Carabinieri, then contact your embass
y.\nUS Consulate Naples: +39 081 583 8111',
      },
      {
        id: uid('d-'), name: 'Emergency contacts', kind: 'note', addedBy: 'm-diego', addedAt: t(3000), offline: true,
        note: 'EU emergency 112 · Ambulance 118 · Fire 115\nTravel insurance: TravelSafe +1 800 555 0199, policy TS-88213\nNearest hospi
tal: Ospedale Costa d'Amalfi, Castiglione di Ravello',
      },
```

```
    ],
    messages: [
      { id: uid('c-'), authorId: 'm-priya', text: 'Villa is booked! 🏡 Paid the deposit.', ts: t(320) },
      { id: uid('c-'), authorId: 'm-sam', text: 'I'll handle maps & the scooter rental.', ts: t(260) },
      { id: uid('c-'), authorId: 'm-diego', text: 'Dinner tonight at 7, I reserved for 4 🍝', ts: t(200) },
    ],
    photos: [
      { id: uid('ph-'), uri: 'amalfi', caption: 'First look at the coast 😍', by: 'm-sam', day: start, ts: t(120), likes: [DEMO_USER_ID,
'm-priya'] },
      { id: uid('ph-'), uri: 'cinque-terre', caption: 'Cinque Terre next time?', by: 'm-priya', day: start, ts: t(90), likes: ['m-dieg
o'] },
    ],
    packing: demoPacking(),
    completed: timeline === 'completed',
    createdAt: t(60 * 24 * 30),
  };
}

export function buildSecondaryTrips(now = Date.now()): Trip[] {
  const today = todayInTz('UTC');
  const baliStart = addDays(today, 120);
  const bcnStart = addDays(today, -160);
  const four = demoMembers.slice(0, 4).map((m) => ({ ...m }));
  return [
    buildTokyoTrip(now),
    {
      id: 'trip-bali', name: 'Bali Surf & Chill', destinationId: 'bali', extraDestinationIds: [], startDate: baliStart, endDate: addDays
(baliStart, 6),
      cover: 'bali', budget: 3000, inviteCode: 'BAL-7316', ownerId: 'm-priya', members: four,
      activities: [
        { id: uid('a-'), date: addDays(baliStart, 1), time: '04:00', title: 'Mount Batur sunrise trek', location: 'Kintamani, Bali', typ
e: 'adventure', durationMin: 360, costPerPerson: 45, status: 'voting', assigned: ['m-priya'], votes: votesFrom(['m-priya', DEMO_USER_I
D]), reminder: false, createdBy: 'm-priya' },
        { id: uid('a-'), date: addDays(baliStart, 2), time: '09:00', title: 'Surf lesson at Kuta', location: 'Kuta, Bali', type: 'advent
ure', durationMin: 150, costPerPerson: 30, status: 'pending', assigned: ['m-sam'], votes: votesFrom(['m-sam']), reminder: false, created
By: 'm-sam' },
      ],
      expenses: [{ id: uid('e-'), title: 'Villa deposit', amount: 600, category: 'Stays', paidBy: 'm-priya', splitBetween: four.map((m)
=> m.id), date: today }],
      flights: [], polls: [], tasks: [{ id: uid('t-'), text: 'Decide villa vs. hostel', done: false, assignee: 'm-priya' }], docs: [],
      messages: [{ id: uid('c-'), authorId: 'm-priya', text: 'Started planning Bali for next year 🌴 who's in?', ts: now - 86400000 }],
      photos: [], packing: [], createdAt: now - 86400000 * 2,
    },
    {
      id: 'trip-bcn', name: 'Barcelona Spring Break', destinationId: 'barcelona', extraDestinationIds: [], startDate: bcnStart, endDate:
addDays(bcnStart, 5),
      cover: 'barcelona', budget: 2400, inviteCode: 'BCN-4402', ownerId: DEMO_USER_ID, members: four, completed: true,
      activities: [
        { id: uid('a-'), date: addDays(bcnStart, 1), time: '10:00', title: 'Sagrada Família tour', location: 'Eixample', type: 'cultur
e', durationMin: 120, costPerPerson: 34, status: 'confirmed', assigned: [DEMO_USER_ID], votes: votesFrom(four.map((m) => m.id)), reminde
r: false, createdBy: DEMO_USER_ID, done: true },
        { id: uid('a-'), date: addDays(bcnStart, 2), time: '13:00', title: 'Tapas crawl in El Born', location: 'El Born', type: 'food',
durationMin: 180, costPerPerson: 40, status: 'confirmed', assigned: ['m-diego'], votes: votesFrom(four.map((m) => m.id)), reminder: fals
e, createdBy: 'm-diego', done: true },
        { id: uid('a-'), date: addDays(bcnStart, 3), time: '11:00', title: 'Barceloneta beach day', location: 'Barceloneta', type: 'sigh
tseeing', durationMin: 300, costPerPerson: 0, status: 'confirmed', assigned: ['m-sam'], votes: votesFrom(four.map((m) => m.id)), reminde
r: false, createdBy: 'm-sam', done: true },
      ],
      expenses: [
        { id: uid('e-'), title: 'Flights', amount: 1320, category: 'Flights', paidBy: DEMO_USER_ID, splitBetween: four.map((m) => m.id),
date: bcnStart },
        { id: uid('e-'), title: 'Airbnb Gothic Quarter', amount: 640, category: 'Stays', paidBy: 'm-priya', splitBetween: four.map((m) =
> m.id), date: bcnStart },
        { id: uid('e-'), title: 'Tapas & dinners', amount: 310, category: 'Food', paidBy: 'm-diego', splitBetween: four.map((m) => m.i
d), date: addDays(bcnStart, 2) },
      ],
      flights: [], polls: [], tasks: [], docs: [], messages: [], packing: [],
      photos: [
        { id: uid('ph-'), uri: 'barcelona', caption: 'Rooftop views', by: 'm-sam', likes: [DEMO_USER_ID], ts: now - 86400000 * 150, day:
addDays(bcnStart, 1) },
      ],
      createdAt: now - 86400000 * 200,
    },
  ];
}

/** Trips that exist "on the server" and can be joined with an invite code. */
export function directoryTrips(now = Date.now()): Trip[] {
  const start = addDays(todayInTz('UTC'), 45);
  const jordan: Member = { id: 'm-jordan', name: 'Jordan Lee', role: 'Organizer', homeCity: 'Philadelphia' };
  const kai: Member = { id: 'm-kai', name: 'Kai Thompson', role: 'Food Lead', homeCity: 'Philadelphia' };
  return [
    {
      id: 'trip-seoul', name: 'Seoul Food Crawl', destinationId: 'seoul', extraDestinationIds: [], startDate: start, endDate: addDays(st
art, 5),
      cover: 'seoul', budget: 2600, inviteCode: 'SEO-5521', ownerId: jordan.id, members: [jordan, kai],
```

```
      activities: [
        { id: uid('a-'), date: addDays(start, 1), time: '18:00', title: 'Gwangjang Market night food tour', location: 'Jongno, Seoul', t
ype: 'food', durationMin: 150, costPerPerson: 35, status: 'confirmed', assigned: [kai.id], votes: votesFrom([jordan.id, kai.id]), remind
er: true, createdBy: kai.id },
      ],
      expenses: [], flights: [], polls: [], tasks: [], docs: [], photos: [], packing: [],
      messages: [{ id: uid('c-'), authorId: jordan.id, text: 'Welcome to the Seoul crew! Drop your flight times when you book ✈️', ts: n
ow - 3600000 }],
      createdAt: now - 86400000,
    },
  ];
}

export { AMALFI_TZ as DEMO_TZ };
```

src/store/selectors.ts

```
/** Selectors and derived data (the "view-model" helpers screens use to read the store). */
import { destinationById } from '@/data/destinations';
import { daysBetween, todayInTz } from '@/utils/time';

import { useAppStore } from './store';
import type { Account, Member, Trip, TripStatus } from './types';

export function useCurrentUser(): Account | undefined {
  return useAppStore((s) => s.accounts.find((a) => a.id === s.session?.userId));
}

export function useActiveTrip(): Trip | undefined {
  return useAppStore((s) => {
    const uidNow = s.session?.userId;
    const t = s.trips.find((x) => x.id === s.activeTripId);
    if (t && (!uidNow || t.members.some((m) => m.id === uidNow))) return t;
    return undefined;
  });
}

export function useMyTrips(): Trip[] {
  const userId = useAppStore((s) => s.session?.userId);
  const trips = useAppStore((s) => s.trips);
  return trips.filter((t) => t.members.some((m) => m.id === userId));
}

export function tripStatus(t: Trip): TripStatus {
  if (t.completed) return 'completed';
  const dest = destinationById(t.destinationId);
  const today = todayInTz(dest?.tz ?? 'UTC');
  if (daysBetween(today, t.startDate) > 0) return 'planning';
  if (daysBetween(t.endDate, today) > 0) return 'completed';
  return 'active';
}

export function memberById(t: Trip | undefined, id: string): Member | undefined {
  return t?.members.find((m) => m.id === id);
}
```

src/store/slices/app.ts

```
/**
 * App-wide state: onboarding, settings, personalization, bucket list, favourites, notifications and cached weather/rates.
 */
import { uid } from '@/utils/format';

import type { AppNotification, BucketItem, RatesSnapshot, Settings, WeatherSnapshot, Personal } from '../types';
import { initialAuth } from './auth';
import { initialDemo } from './demo';
import { initialTrips } from './trips';
import type { Slice } from './types';

export const defaultSettings: Settings = {
  language: 'en',
  homeCurrency: 'USD',
  notificationsEnabled: true,
  reminderLeadMin: 30,
  textScale: 1,
  showTripTime: true,
  aiMode: 'offline',
  haptics: true,
};

export const defaultPersonal: Personal = {
  done: false,
  nickname: '',
  emoji: '✈️',
  interests: [],
  travelStyle: 'balanced',
```

```
  budgetStyle: 'mid',
  sections: { glance: true, budget: true, upnext: true, quick: true, forYou: true, nearby: true, discover: true },
  motion: true,
};

export interface AppSlice {
  hydrated: boolean;
  hasOnboarded: boolean;
  bucket: BucketItem[];
  favorites: string[];
  notifications: AppNotification[];
  settings: Settings;
  weather: Record<string, WeatherSnapshot>;
  rates: RatesSnapshot | null;
  quizBest: number;
  surveyResult: string[] | null;
  personal: Personal;
  restyled: boolean;
  firedReminders: Record<string, number>;
  setHydrated: () => void;
  completeOnboarding: () => void;
  addBucket: (text: string, destinationId?: string) => void;
  toggleBucket: (id: string) => void;
  deleteBucket: (id: string) => void;
  toggleFavorite: (destinationId: string) => void;
  pushNotification: (n: Omit<AppNotification, 'id' | 'ts' | 'read'>) => void;
  markAllRead: () => void;
  markKindRead: (kind: AppNotification['kind']) => void;
  clearNotifications: () => void;
  updateSettings: (patch: Partial<Settings>) => void;
  setWeather: (w: WeatherSnapshot) => void;
  setRates: (r: RatesSnapshot) => void;
  setQuizBest: (n: number) => void;
  setSurveyResult: (ids: string[] | null) => void;
  updatePersonal: (patch: Partial<Personal>) => void;
  setRestyled: (v: boolean) => void;
  setToured: (v: boolean) => void;
  markReminderFired: (key: string) => void;
  resetEverything: () => void;
}

/** Starting values, also used by "reset everything". */
export const initialApp = {
  hydrated: false,
  hasOnboarded: false,
  bucket: [],
  favorites: ['tokyo', 'kyoto'],
  notifications: [],
  settings: defaultSettings,
  weather: {},
  rates: null,
  quizBest: 0,
  surveyResult: null,
  personal: defaultPersonal,
  restyled: false,
  firedReminders: {},
};

export const createAppSlice: Slice<AppSlice> = (set) => {
  return {
    ...initialApp,
    setHydrated: () => set({ hydrated: true }),

    completeOnboarding: () => set({ hasOnboarded: true }),

    addBucket: (text, destinationId) =>
      set((s) => {
        s.bucket.unshift({ id: uid('b-'), text: text.trim(), destinationId, done: false, addedAt: Date.now() });
      }),

    toggleBucket: (id) =>
      set((s) => {
        const b = s.bucket.find((x) => x.id === id);
        if (b) b.done = !b.done;
      }),

    deleteBucket: (id) =>
      set((s) => {
        s.bucket = s.bucket.filter((x) => x.id !== id);
      }),

    toggleFavorite: (destinationId) =>
      set((s) => {
        s.favorites = s.favorites.includes(destinationId)
          ? s.favorites.filter((x) => x !== destinationId)
          : [...s.favorites, destinationId];
```

```
      }),

    pushNotification: (n) =>
      set((s) => {
        s.notifications.unshift({ ...n, id: uid('n-'), ts: Date.now(), read: false });
        if (s.notifications.length > 60) s.notifications.length = 60;
      }),

    markAllRead: () =>
      set((s) => {
        s.notifications.forEach((n) => (n.read = true));
      }),

    markKindRead: (kind) =>
      set((s) => {
        s.notifications.forEach((n) => {
          if (n.kind === kind) n.read = true;
        });
      }),

    clearNotifications: () => set({ notifications: [] }),

    updateSettings: (patch) =>
      set((s) => {
        Object.assign(s.settings, patch);
      }),

    setWeather: (w) =>
      set((s) => {
        s.weather[w.destinationId] = w;
      }),

    setRates: (r) => set({ rates: r }),

    setQuizBest: (n) => set((s) => ({ quizBest: Math.max(s.quizBest, n) })),

    setSurveyResult: (ids) => set({ surveyResult: ids }),

    updatePersonal: (patch) =>
      set((s) => {
        Object.assign(s.personal, patch);
      }),

    setRestyled: (v) => set({ restyled: v }),

    setToured: (v) =>
      set((s) => {
        s.personal.toured = v;
      }),

    markReminderFired: (key) =>
      set((s) => {
        s.firedReminders[key] = Date.now();
      }),

    resetEverything: () =>
      set(() => ({ ...initialAuth,
        ...initialTrips,
        ...initialDemo,
        ...initialApp, hydrated: true, hasOnboarded: true })),
  };
};
```

src/store/slices/auth.ts

```
/**
 * Accounts and sign-in: who is using this phone.
 */
import * as Crypto from 'expo-crypto';
import { uid } from '@/utils/format';
import { EMAIL_RE } from '@/utils/validation';
import { hashPassword, verifyPassword } from '@/utils/password';

import type { Account } from '../types';
import type { Slice } from './types';

interface Session {
  userId: string;
  remember: boolean;
}

export type Result = { ok: true } | { ok: false; error: string };

export interface AuthSlice {
  session: Session | null;
  accounts: Account[];
```

```
  signIn: (email: string, password: string, remember: boolean) => Promise<Result>;
  signUp: (name: string, email: string, password: string) => Promise<Result>;
  signInWithGoogle: (email: string, name: string) => Promise<Result>;
  signOut: () => void;
  accountExists: (email: string) => boolean;
  resetPassword: (email: string, newPassword: string) => Promise<Result>;
  updateProfile: (patch: Partial<Pick<Account, 'name' | 'homeCity' | 'avatar'>>) => void;
}

/** Starting values, also used by "reset everything". */
export const initialAuth = {
  session: null,
  accounts: [],
};

export const createAuthSlice: Slice<AuthSlice> = (set, get) => {
  return {
    ...initialAuth,
    signIn: async (email, password, remember) => {
      const acct = get().accounts.find((a) => a.email.toLowerCase() === email.trim().toLowerCase());
      if (!acct) return { ok: false, error: 'No account found with that email. Try Sign Up instead.' };
      if (acct.provider === 'google' && !acct.passwordHash) return { ok: false, error: 'This account uses Google sign-in.' };
      const check = await verifyPassword(password, acct);
      if (!check.ok) return { ok: false, error: 'Incorrect password. Please try again.' };
      set((s) => {
        // Old accounts used a single SHA-256 round: quietly move them to the stronger hash.
        if (check.upgrade) {
          const mine = s.accounts.find((a) => a.id === acct.id);
          if (mine) mine.passwordHash = check.upgrade;
        }
        s.session = { userId: acct.id, remember };
        const mine = s.trips.find((t) => t.members.some((m) => m.id === acct.id));
        if (!s.trips.find((t) => t.id === s.activeTripId && t.members.some((m) => m.id === acct.id))) {
          s.activeTripId = mine?.id ?? null;
        }
      });
      return { ok: true };
    },

    signUp: async (name, email, password) => {
      const e = email.trim().toLowerCase();
      if (!EMAIL_RE.test(e)) return { ok: false, error: 'Enter a valid email address.' };
      if (get().accounts.some((a) => a.email.toLowerCase() === e)) {
        return { ok: false, error: 'An account with this email already exists. Sign in instead.' };
      }
      const salt = Crypto.randomUUID();
      const passwordHash = await hashPassword(password, salt);
      const id = uid('u-');
      set((s) => {
        s.accounts.push({ id, name: name.trim(), email: e, passwordHash, salt, provider: 'password' });
        s.session = { userId: id, remember: true };
        s.activeTripId = null;
      });
      return { ok: true };
    },

    signInWithGoogle: async (email, name) => {
      const e = email.trim().toLowerCase();
      let acct = get().accounts.find((a) => a.email.toLowerCase() === e);
      if (!acct) {
        acct = { id: uid('u-'), name, email: e, passwordHash: '', salt: '', provider: 'google' };
        const created = acct;
        set((s) => {
          s.accounts.push(created);
        });
      }
      const id = acct.id;
      set((s) => {
        s.session = { userId: id, remember: true };
        const mine = s.trips.find((t) => t.members.some((m) => m.id === id));
        s.activeTripId = mine?.id ?? null;
      });
      return { ok: true };
    },

    signOut: () => set({ session: null }),

    accountExists: (email) => get().accounts.some((a) => a.email.toLowerCase() === email.trim().toLowerCase()),

    resetPassword: async (email, newPassword) => {
      const e = email.trim().toLowerCase();
      if (!get().accounts.some((a) => a.email.toLowerCase() === e)) return { ok: false, error: 'No account found with that email.' };
      const salt = Crypto.randomUUID();
      const passwordHash = await hashPassword(newPassword, salt);
      set((s) => {
        const a = s.accounts.find((x) => x.email.toLowerCase() === e);
```

```
        if (a) {
          a.salt = salt;
          a.passwordHash = passwordHash;
          a.provider = 'password';
        }
      });
      return { ok: true };
    },

    updateProfile: (patch) =>
      set((s) => {
        const a = s.accounts.find((x) => x.id === s.session?.userId);
        if (!a) return;
        Object.assign(a, patch);
        for (const t of s.trips) {
          const m = t.members.find((x) => x.id === a.id);
          if (m) {
            if (patch.name) m.name = patch.name;
            if (patch.avatar !== undefined) m.avatar = patch.avatar;
            if (patch.homeCity !== undefined) m.homeCity = patch.homeCity;
          }
        }
      }),
  };
};
```

src/store/slices/budget.ts

```
/**
 * Money: expenses and settle-up payments. The maths itself lives in utils/money.ts.
 */
import { uid } from '@/utils/format';
import { todayInTz } from '@/utils/time';

import type { Expense } from '../types';
import { tripHelpers } from './helpers';
import type { Slice } from './types';

export interface BudgetSlice {
  addExpense: (tripId: string, e: Omit<Expense, 'id'>) => void;
  deleteExpense: (tripId: string, id: string) => void;
  settleUp: (tripId: string, from: string, to: string, amount: number) => void;
}

export const createBudgetSlice: Slice<BudgetSlice> = (set, get) => {
  const { withTrip } = tripHelpers(set, get);
  return {
    addExpense: (tripId, e) =>
      withTrip(tripId, (t) => {
        t.expenses.unshift({ ...e, id: uid('e-') });
      }),

    deleteExpense: (tripId, id) =>
      withTrip(tripId, (t) => {
        t.expenses = t.expenses.filter((x) => x.id !== id);
      }),

    settleUp: (tripId, from, to, amount) =>
      withTrip(tripId, (t) => {
        const toName = t.members.find((m) => m.id === to)?.name.split(' ')[0] ?? 'member';
        t.expenses.unshift({
          id: uid('e-'), title: `Settle-up to ${toName}`, amount: Math.round(amount * 100) / 100, category: 'Other',
          paidBy: from, splitBetween: [to], date: todayInTz('UTC'), settlement: true,
        });
      }),
  };
};
```

src/store/slices/demo.ts

```
/**
 * Demo content: the sample trips and the "Demo timeline" switch (upcoming / in progress / finished).
 */
import * as Crypto from 'expo-crypto';

import { uid } from '@/utils/format';
import { hashPassword } from '@/utils/password';

import { DEMO_EMAIL, DEMO_PASSWORD, DEMO_TRIP_IDS, DEMO_USER_ID, buildDemoTrip, buildSecondaryTrips, type DemoTimeline } from '../seed';
import type { Account, AppNotification, BucketItem } from '../types';
import type { Slice } from './types';

/** The sample bucket-list items: text, destination, done. */
const BUCKET_SEED: [string, string, boolean][] = [
  ['See the northern lights in Iceland', 'iceland', false],
  ['Sunrise at Machu Picchu', 'machu-picchu', false],
```

```
  ['Eat street tacos in Mexico City', 'mexico-city', false],
  ['Watch the sunset in Oia, Santorini', 'santorini', false],
  ['Tapas crawl in Barcelona', 'barcelona', true],
];

function seedBucket(): BucketItem[] {
  const now = Date.now();
  return BUCKET_SEED.map(([text, destinationId, done], i) => ({ id: uid('b-'), text, destinationId, done, addedAt: now - (9 - i) * 1e7
}));
}

function seedNotifications(): AppNotification[] {
  const now = Date.now();
  return [
    { id: uid('n-'), kind: 'vote', title: 'New poll from Sam', body: 'Beach day — which beach? Cast your vote.', ts: now - 600 * 60000,
read: false, tripId: 'trip-amalfi', route: '/crew' },
    { id: uid('n-'), kind: 'expense', title: 'Diego added an expense', body: 'Dinner deposits & groceries · $86 split 4 ways', ts: now -
420 * 60000, read: false, tripId: 'trip-amalfi', route: '/(tabs)/budget' },
    { id: uid('n-'), kind: 'chat', title: 'Diego in Amalfi Squad', body: 'Dinner tonight at 7, I reserved for 4 🍝', ts: now - 42 * 6000
0, read: false, tripId: 'trip-amalfi', route: '/(tabs)/chat' },
  ];
}

export interface DemoSlice {
  demoTimeline: DemoTimeline;
  seeded: boolean;
  ensureSeed: () => Promise<void>;
  reseedDemo: (timeline: DemoTimeline) => void;
  /** Removes the sample trips, their notifications and the sample bucket list; accounts and your own trips stay. */
  clearDemoData: () => void;
}

/** Starting values, also used by "reset everything". */
export const initialDemo = {
  demoTimeline: 'upcoming' as DemoTimeline,
  seeded: false,
};

export const createDemoSlice: Slice<DemoSlice> = (set, get) => {
  return {
    ...initialDemo,
    ensureSeed: async () => {
      if (get().seeded) return;
      const salt = Crypto.randomUUID();
      const passwordHash = await hashPassword(DEMO_PASSWORD, salt);
      const demo: Account = { id: DEMO_USER_ID, name: 'Maya Chen', email: DEMO_EMAIL, passwordHash, salt, avatar: 'avatar-maya', homeCit
y: 'New York', provider: 'password' };
      set((s) => {
        if (s.seeded) return;
        s.accounts = [demo, ...s.accounts.filter((a) => a.email !== DEMO_EMAIL)];
        s.trips = [buildDemoTrip('upcoming'), ...buildSecondaryTrips(), ...s.trips];
        s.activeTripId = 'trip-amalfi';
        s.bucket = seedBucket();
        s.notifications = seedNotifications();
        s.seeded = true;
      });
    },

    reseedDemo: (timeline) =>
      set((s) => {
        const fresh = buildDemoTrip(timeline);
        s.trips = [fresh, ...s.trips.filter((t) => t.id !== fresh.id)];
        s.activeTripId = fresh.id;
        s.demoTimeline = timeline;
        s.firedReminders = {};
      }),

    clearDemoData: () =>
      set((s) => {
        s.trips = s.trips.filter((t) => !DEMO_TRIP_IDS.includes(t.id));
        s.activeTripId = s.trips.find((t) => t.members.some((m) => m.id === s.session?.userId))?.id ?? null;
        s.notifications = s.notifications.filter((n) => !n.tripId || !DEMO_TRIP_IDS.includes(n.tripId));
        const sample = new Set(BUCKET_SEED.map(([text]) => text));
        s.bucket = s.bucket.filter((b) => !sample.has(b.text));
        s.firedReminders = {};
      }),
  };
};
```

src/store/slices/files.ts

```
/**
 * Lets the sync code record which file on the sync server belongs to a photo, a document or a receipt.
 */
import { tripHelpers } from './helpers';
import type { Slice } from './types';
```

```
export type FileList = 'photos' | 'docs' | 'expenses';

export interface FilesSlice {
  attachBlob: (tripId: string, list: FileList, itemId: string, blobId: string) => void;
}

export const createFilesSlice: Slice<FilesSlice> = (set, get) => {
  const { withTrip } = tripHelpers(set, get);
  return {
    attachBlob: (tripId, list, itemId, blobId) =>
      withTrip(tripId, (t) => {
        if (list === 'photos') {
          const photo = t.photos.find((x) => x.id === itemId);
          if (photo) photo.blob = blobId;
        } else if (list === 'docs') {
          const doc = t.docs.find((x) => x.id === itemId);
          if (doc) doc.blob = blobId;
        } else {
          const expense = t.expenses.find((x) => x.id === itemId);
          if (expense) expense.receiptBlob = blobId;
        }
      }),
  };
};
```

src/store/slices/groupTools.ts

```
/**
 * Shared group tools: polls, the checklist, the packing list and the document vault.
 */
import { uid } from '@/utils/format';

import type { PackingItem, Poll, TripDoc } from '../types';
import { tripHelpers } from './helpers';
import type { Slice } from './types';

export interface GroupToolsSlice {
  createPoll: (tripId: string, question: string, options: string[]) => void;
  votePoll: (tripId: string, pollId: string, optionId: string) => void;
  closePoll: (tripId: string, pollId: string) => void;
  deletePoll: (tripId: string, pollId: string) => void;
  addTask: (tripId: string, text: string, assignee?: string) => void;
  toggleTask: (tripId: string, id: string) => void;
  deleteTask: (tripId: string, id: string) => void;
  assignTask: (tripId: string, id: string, memberId?: string) => void;
  addDoc: (tripId: string, d: Omit<TripDoc, 'id' | 'addedAt' | 'addedBy'>) => void;
  deleteDoc: (tripId: string, id: string) => void;
  togglePacked: (tripId: string, id: string) => void;
  assignPacking: (tripId: string, id: string, memberId?: string) => void;
  addPackingItems: (tripId: string, items: Omit<PackingItem, 'id' | 'packed'>[]) => void;
  deletePackingItem: (tripId: string, id: string) => void;
}

export const createGroupToolsSlice: Slice<GroupToolsSlice> = (set, get) => {
  const { withTrip, me } = tripHelpers(set, get);
  return {
    createPoll: (tripId, question, options) =>
      withTrip(tripId, (t) => {
        t.polls.unshift({
          id: uid('poll-'), question: question.trim(), createdBy: me(), createdAt: Date.now(),
          options: options.map((text, i) => ({ id: `o${i + 1}`, text: text.trim(), votes: [] })),
        } satisfies Poll);
        t.messages.push({ id: uid('c-'), authorId: 'system', kind: 'system', text: `New poll: ${question.trim()}`, ts: Date.now() });
      }),

    votePoll: (tripId, pollId, optionId) =>
      withTrip(tripId, (t) => {
        const p = t.polls.find((x) => x.id === pollId);
        if (!p || p.closed) return;
        const user = me();
        const already = p.options.find((o) => o.id === optionId)?.votes.includes(user);
        p.options.forEach((o) => (o.votes = o.votes.filter((v) => v !== user)));
        if (!already) p.options.find((o) => o.id === optionId)?.votes.push(user);
      }),

    closePoll: (tripId, pollId) =>
      withTrip(tripId, (t) => {
        const p = t.polls.find((x) => x.id === pollId);
        if (p) p.closed = !p.closed;
      }),

    deletePoll: (tripId, pollId) =>
      withTrip(tripId, (t) => {
        t.polls = t.polls.filter((x) => x.id !== pollId);
      }),
```

```
    addTask: (tripId, text, assignee) =>
      withTrip(tripId, (t) => {
        t.tasks.push({ id: uid('t-'), text: text.trim(), done: false, assignee });
      }),

    toggleTask: (tripId, id) =>
      withTrip(tripId, (t) => {
        const k = t.tasks.find((x) => x.id === id);
        if (k) k.done = !k.done;
      }),

    deleteTask: (tripId, id) =>
      withTrip(tripId, (t) => {
        t.tasks = t.tasks.filter((x) => x.id !== id);
      }),

    assignTask: (tripId, id, memberId) =>
      withTrip(tripId, (t) => {
        const k = t.tasks.find((x) => x.id === id);
        if (k) k.assignee = memberId;
      }),

    addDoc: (tripId, d) =>
      withTrip(tripId, (t) => {
        t.docs.unshift({ ...d, id: uid('d-'), addedAt: Date.now(), addedBy: me() });
      }),

    deleteDoc: (tripId, id) =>
      withTrip(tripId, (t) => {
        t.docs = t.docs.filter((x) => x.id !== id);
      }),

    togglePacked: (tripId, id) =>
      withTrip(tripId, (t) => {
        const p = t.packing.find((x) => x.id === id);
        if (p) p.packed = !p.packed;
      }),

    assignPacking: (tripId, id, memberId) =>
      withTrip(tripId, (t) => {
        const p = t.packing.find((x) => x.id === id);
        if (p) p.broughtBy = memberId;
      }),

    addPackingItems: (tripId, items) =>
      withTrip(tripId, (t) => {
        const have = new Set(t.packing.map((p) => p.text.toLowerCase()));
        items.forEach((i) => {
          const key = i.text.toLowerCase();
          if (have.has(key)) return;
          have.add(key); // also skips duplicates inside this same batch
          t.packing.push({ ...i, id: uid('p-'), packed: false });
        });
      }),

    deletePackingItem: (tripId, id) =>
      withTrip(tripId, (t) => {
        t.packing = t.packing.filter((x) => x.id !== id);
      }),
  };
};
```

src/store/slices/helpers.ts

```
import { DEMO_USER_ID } from '../seed';
import type { Trip } from '../types';
import type { AppStore, SliceGet, SliceSet } from './types';

/** Shortcuts for every slice that edits a trip. */
export function tripHelpers(set: SliceSet, get: SliceGet) {
  /** Runs `fn` on the trip with this id. The trip is an immer draft, so it can be changed in place. */
  const withTrip = (tripId: string, fn: (t: Trip, s: AppStore) => void) =>
    set((s) => {
      const t = s.trips.find((x) => x.id === tripId);
      if (t) fn(t, s);
    });
  /** The signed-in user's id (the demo user when nobody is signed in, so the demo screens always work). */
  const me = () => get().session?.userId ?? DEMO_USER_ID;
  return { withTrip, me };
}
```

src/store/slices/itinerary.ts

```
/**
 * The day-by-day plan: activities (with voting) and flights.
```

```
 */
import { uid } from '@/utils/format';

import type { Activity, Flight } from '../types';
import { tripHelpers } from './helpers';
import type { Slice } from './types';

export interface ItinerarySlice {
  addActivity: (tripId: string, a: Omit<Activity, 'id' | 'createdBy' | 'votes'> & { votes?: Activity['votes'] }) => string;
  updateActivity: (tripId: string, id: string, patch: Partial<Activity>) => void;
  deleteActivity: (tripId: string, id: string) => void;
  voteActivity: (tripId: string, id: string, value: 1 | -1) => void;
  toggleAssign: (tripId: string, id: string, memberId: string) => void;
  addFlight: (tripId: string, f: Omit<Flight, 'id'>) => void;
  deleteFlight: (tripId: string, id: string) => void;
}

export const createItinerarySlice: Slice<ItinerarySlice> = (set, get) => {
  const { withTrip, me } = tripHelpers(set, get);
  return {
    addActivity: (tripId, a) => {
      const id = uid('a-');
      withTrip(tripId, (t) => {
        t.activities.push({ ...a, votes: a.votes ?? { [me()]: 1 }, id, createdBy: me() });
      });
      return id;
    },

    updateActivity: (tripId, id, patch) =>
      withTrip(tripId, (t) => {
        const a = t.activities.find((x) => x.id === id);
        if (a) Object.assign(a, patch);
      }),

    deleteActivity: (tripId, id) =>
      withTrip(tripId, (t) => {
        t.activities = t.activities.filter((x) => x.id !== id);
      }),

    voteActivity: (tripId, id, value) =>
      withTrip(tripId, (t) => {
        const a = t.activities.find((x) => x.id === id);
        if (!a) return;
        const user = me();
        if (a.votes[user] === value) delete a.votes[user];
        else a.votes[user] = value;
        // Majority rule: a voting item is confirmed once more than half the group upvotes it.
        const ups = Object.values(a.votes).filter((v) => v === 1).length;
        if (a.status === 'voting' && ups > t.members.length / 2) a.status = 'confirmed';
      }),

    toggleAssign: (tripId, id, memberId) =>
      withTrip(tripId, (t) => {
        const a = t.activities.find((x) => x.id === id);
        if (!a) return;
        a.assigned = a.assigned.includes(memberId) ? a.assigned.filter((x) => x !== memberId) : [...a.assigned, memberId];
      }),

    addFlight: (tripId, f) =>
      withTrip(tripId, (t) => {
        t.flights.push({ ...f, id: uid('f-') });
      }),

    deleteFlight: (tripId, id) =>
      withTrip(tripId, (t) => {
        t.flights = t.flights.filter((x) => x.id !== id);
      }),
  };
};
```

src/store/slices/social.ts

```
/**
 * Chat messages and the photo dump.
 */
import { uid } from '@/utils/format';

import type { ChatMessage, Photo } from '../types';
import { tripHelpers } from './helpers';
import type { Slice } from './types';

export interface SocialSlice {
  sendMessage: (tripId: string, msg: Omit<ChatMessage, 'id' | 'ts'> & { ts?: number }) => void;
  clearChat: (tripId: string) => void;
  addPhoto: (tripId: string, p: Omit<Photo, 'id' | 'ts' | 'likes' | 'by'>) => void;
  toggleLikePhoto: (tripId: string, id: string) => void;
```

```
  deletePhoto: (tripId: string, id: string) => void;
}

export const createSocialSlice: Slice<SocialSlice> = (set, get) => {
  const { withTrip, me } = tripHelpers(set, get);
  return {
    sendMessage: (tripId, msg) =>
      withTrip(tripId, (t) => {
        t.messages.push({ ...msg, id: uid('c-'), ts: msg.ts ?? Date.now() });
        if (t.messages.length > 400) t.messages.splice(0, t.messages.length - 400);
      }),

    clearChat: (tripId) =>
      withTrip(tripId, (t) => {
        t.messages = [];
      }),

    addPhoto: (tripId, p) =>
      withTrip(tripId, (t) => {
        t.photos.unshift({ ...p, id: uid('ph-'), ts: Date.now(), likes: [], by: me() });
      }),

    toggleLikePhoto: (tripId, id) =>
      withTrip(tripId, (t) => {
        const p = t.photos.find((x) => x.id === id);
        if (!p) return;
        const user = me();
        p.likes = p.likes.includes(user) ? p.likes.filter((x) => x !== user) : [...p.likes, user];
      }),

    deletePhoto: (tripId, id) =>
      withTrip(tripId, (t) => {
        t.photos = t.photos.filter((x) => x.id !== id);
      }),
  };
};
```

src/store/slices/trips.ts

```
/**
 * The list of trips: create, join with an invite code, leave, delete, edit, and who is in each trip.
 */
import { destinationById } from '@/data/destinations';
import { uid } from '@/utils/format';
import type { Role } from '@/data/reference';

import { directoryTrips, inviteCodeFor } from '../seed';
import type { Trip } from '../types';
import { tripHelpers } from './helpers';
import type { Slice } from './types';

export interface TripsSlice {
  trips: Trip[];
  activeTripId: string | null;
  setActiveTrip: (id: string) => void;
  createTrip: (t: { name: string; destinationId: string; startDate: string; endDate: string; budget: number }) => string;
  joinTrip: (code: string) => { ok: true; tripId: string } | { ok: false; error: string };
  leaveTrip: (tripId: string) => void;
  deleteTrip: (tripId: string) => void;
  updateTrip: (tripId: string, patch: Partial<Pick<Trip, 'name' | 'startDate' | 'endDate' | 'budget' | 'completed' | 'cover' | 'destinat
ionId' | 'extraDestinationIds'>>) => void;
  regenerateInvite: (tripId: string) => void;
  addMember: (tripId: string, name: string, email?: string) => void;
  removeMember: (tripId: string, memberId: string) => void;
  setRole: (tripId: string, memberId: string, role: Role) => void;
}

/** Starting values, also used by "reset everything". */
export const initialTrips = {
  trips: [],
  activeTripId: null,
};

export const createTripsSlice: Slice<TripsSlice> = (set, get) => {
  const { withTrip, me } = tripHelpers(set, get);
  return {
    ...initialTrips,
    setActiveTrip: (id) => set({ activeTripId: id }),

    createTrip: ({ name, destinationId, startDate, endDate, budget }) => {
      const id = uid('trip-');
      const acct = get().accounts.find((a) => a.id === me());
      const dest = destinationById(destinationId);
      set((s) => {
        s.trips.unshift({
          id, name: name.trim(), destinationId, extraDestinationIds: [], startDate, endDate,
```

```
          cover: dest?.image ?? 'planning', budget, inviteCode: inviteCodeFor(dest?.city ?? name), ownerId: me(),
          members: [{ id: me(), name: acct?.name ?? 'You', email: acct?.email, avatar: acct?.avatar, role: 'Organizer', homeCity: acct?.
homeCity }],
          activities: [], expenses: [], flights: [], polls: [], tasks: [
            { id: uid('t-'), text: 'Invite your travel group', done: false, assignee: me() },
            { id: uid('t-'), text: 'Agree on a budget', done: false },
            { id: uid('t-'), text: 'Add everyone's flights', done: false },
          ], docs: [], photos: [], packing: [],
          messages: [{ id: uid('c-'), authorId: 'assistant', ai: true, aiSource: 'local', ts: Date.now(), text: `Welcome to ${name.trim
()}! Share invite code to bring your group in, then ask me anything — budget, weather, food nearby or what to do next.` }],
          createdAt: Date.now(),
        });
        s.activeTripId = id;
      });
      return id;
    },

    joinTrip: (code) => {
      const c = code.trim().toUpperCase();
      const user = me();
      const acct = get().accounts.find((a) => a.id === user);
      const local = get().trips.find((t) => t.inviteCode === c);
      if (local) {
        if (local.members.some((m) => m.id === user)) {
          set({ activeTripId: local.id });
          return { ok: false, error: `You're already in "${local.name}".` };
        }
        set((s) => {
          const t = s.trips.find((x) => x.id === local.id)!;
          t.members.push({ id: user, name: acct?.name ?? 'New member', email: acct?.email, avatar: acct?.avatar, role: 'Member', homeCit
y: acct?.homeCity });
          t.messages.push({ id: uid('c-'), authorId: 'system', kind: 'system', text: `${acct?.name ?? 'Someone'} joined the trip 👋`, t
s: Date.now() });
          s.activeTripId = t.id;
        });
        return { ok: true, tripId: local.id };
      }
      const remote = directoryTrips().find((t) => t.inviteCode === c);
      if (!remote) return { ok: false, error: 'No trip found with that code. Check with your friend and try again.' };
      set((s) => {
        remote.members.push({ id: user, name: acct?.name ?? 'New member', email: acct?.email, avatar: acct?.avatar, role: 'Member', home
City: acct?.homeCity });
        remote.messages.push({ id: uid('c-'), authorId: 'system', kind: 'system', text: `${acct?.name ?? 'Someone'} joined the trip 👋`,
ts: Date.now() });
        s.trips.unshift(remote);
        s.activeTripId = remote.id;
      });
      return { ok: true, tripId: remote.id };
    },

    leaveTrip: (tripId) =>
      set((s) => {
        const t = s.trips.find((x) => x.id === tripId);
        if (!t) return;
        t.members = t.members.filter((m) => m.id !== s.session?.userId);
        if (s.activeTripId === tripId) {
          s.activeTripId = s.trips.find((x) => x.id !== tripId && x.members.some((m) => m.id === s.session?.userId))?.id ?? null;
        }
      }),

    deleteTrip: (tripId) =>
      set((s) => {
        s.trips = s.trips.filter((t) => t.id !== tripId);
        if (s.activeTripId === tripId) {
          s.activeTripId = s.trips.find((x) => x.members.some((m) => m.id === s.session?.userId))?.id ?? null;
        }
      }),

    updateTrip: (tripId, patch) => withTrip(tripId, (t) => Object.assign(t, patch)),

    regenerateInvite: (tripId) =>
      withTrip(tripId, (t) => {
        t.inviteCode = inviteCodeFor(destinationById(t.destinationId)?.city ?? t.name);
      }),

    addMember: (tripId, name, email) =>
      withTrip(tripId, (t) => {
        t.members.push({ id: uid('m-'), name: name.trim(), email: email?.trim() || undefined, role: 'Member', invited: true });
        t.messages.push({ id: uid('c-'), authorId: 'system', kind: 'system', text: `${name.trim()} was added to the trip`, ts: Date.now
() });
      }),

    removeMember: (tripId, memberId) =>
      withTrip(tripId, (t) => {
        t.members = t.members.filter((m) => m.id !== memberId);
        t.activities.forEach((a) => {
```

```
          a.assigned = a.assigned.filter((x) => x !== memberId);
          delete a.votes[memberId];
        });
        t.tasks.forEach((k) => {
          if (k.assignee === memberId) k.assignee = undefined;
        });
      }),

    setRole: (tripId, memberId, role) =>
      withTrip(tripId, (t) => {
        const m = t.members.find((x) => x.id === memberId);
        if (m) m.role = role;
      }),
  };
};
```

src/store/slices/types.ts

```
import type { StateCreator } from 'zustand';

import type { AppSlice } from './app';
import type { AuthSlice } from './auth';
import type { BudgetSlice } from './budget';
import type { DemoSlice } from './demo';
import type { FilesSlice } from './files';
import type { GroupToolsSlice } from './groupTools';
import type { ItinerarySlice } from './itinerary';
import type { SocialSlice } from './social';
import type { TripsSlice } from './trips';

/** Everything in the store: each slice contributes its own state and actions. */
export type AppStore = AuthSlice & TripsSlice & ItinerarySlice & BudgetSlice & GroupToolsSlice & SocialSlice & FilesSlice & DemoSlice &
AppSlice;

/** A slice is a function that gets `set` / `get` and returns its piece of state plus the actions that change it. */
export type Slice<T> = StateCreator<AppStore, [['zustand/persist', unknown], ['zustand/immer', never]], [], T>;
export type SliceSet = Parameters<Slice<object>>[0];
export type SliceGet = Parameters<Slice<object>>[1];
```

src/store/store.ts

```
/**
 * The one store for the whole app (the "model" in MVVM): zustand for state + actions, immer for safe
 * nested updates, and AsyncStorage persistence so everything keeps working offline.
 * The actions are grouped into slices (see ./slices); this file only puts them together.
 */
import AsyncStorage from '@react-native-async-storage/async-storage';
import { create } from 'zustand';
import { createJSONStorage, persist } from 'zustand/middleware';
import { immer } from 'zustand/middleware/immer';

import { DEMO_TRIP_IDS } from './seed';
import { createAppSlice } from './slices/app';
import { createAuthSlice } from './slices/auth';
import { createBudgetSlice } from './slices/budget';
import { createDemoSlice } from './slices/demo';
import { createFilesSlice } from './slices/files';
import { createGroupToolsSlice } from './slices/groupTools';
import { createItinerarySlice } from './slices/itinerary';
import { createSocialSlice } from './slices/social';
import { createTripsSlice } from './slices/trips';
import type { AppStore } from './slices/types';

export type { AppStore };

export const useAppStore = create<AppStore>()(
  persist(
    immer((...a) => ({
      ...createAuthSlice(...a),
      ...createTripsSlice(...a),
      ...createItinerarySlice(...a),
      ...createBudgetSlice(...a),
      ...createGroupToolsSlice(...a),
      ...createSocialSlice(...a),
      ...createFilesSlice(...a),
      ...createDemoSlice(...a),
      ...createAppSlice(...a),
    })),
    {
      name: 'togetherwego-v1',
      version: 2,
      // v2: "Amalfi Squad" replaced the Tokyo trip as the main demo — re-seed the demo trips once, keep the user's own.
      migrate: (persisted, version) => {
        const old = persisted as Partial<AppStore>;
        if (version >= 2) return old as AppStore;
        const demoIds = DEMO_TRIP_IDS;
```

```
        return { ...old, seeded: false, activeTripId: null, trips: (old.trips ?? []).filter((t) => !demoIds.includes(t.id)) } as AppStor
e;
      },
      // New settings added in later versions fall back to their defaults instead of vanishing.
      merge: (persisted, current) => {
        const p = (persisted ?? {}) as Partial<AppStore>;
        return { ...current, ...p, personal: { ...current.personal, ...p.personal, sections: { ...current.personal.sections, ...p.person
al?.sections } } } as AppStore;
      },
      storage: createJSONStorage(() => AsyncStorage),
      partialize: (s) => {
        // Don't persist the session unless "Remember me" was ticked.
        const { hydrated, ...rest } = s;
        void hydrated;
        return { ...rest, session: s.session?.remember || s.restyled ? s.session : null };
      },
      onRehydrateStorage: () => (state) => {
        state?.setHydrated();
      },
    },
  ),
);
```

src/store/types.ts

```
import type { Role } from '@/data/reference';
import type { LangCode } from '@/i18n/strings';

export type ActivityType = 'transport' | 'stay' | 'food' | 'sightseeing' | 'adventure' | 'culture' | 'nightlife' | 'shopping';
export type ActivityStatus = 'confirmed' | 'voting' | 'pending';

export interface Member {
  id: string;
  name: string;
  email?: string;
  avatar?: string;
  role: Role;
  homeCity?: string;
  /** Members added by name only (no account yet). */
  invited?: boolean;
}

export interface Activity {
  id: string;
  date: string;
  time: string;
  title: string;
  location: string;
  type: ActivityType;
  durationMin: number;
  costPerPerson: number;
  status: ActivityStatus;
  assigned: string[];
  /** memberId → +1 (up) / -1 (down) */
  votes: Record<string, 1 | -1>;
  notes?: string;
  reminder: boolean;
  lat?: number;
  lng?: number;
  placeId?: string;
  createdBy: string;
  done?: boolean;
}

export type ExpenseCategory = 'Flights' | 'Stays' | 'Food' | 'Activities' | 'Transport' | 'Shopping' | 'Other';

export interface Expense {
  id: string;
  title: string;
  /** Always stored in the trip's base currency (USD by default). */
  amount: number;
  category: ExpenseCategory;
  paidBy: string;
  splitBetween: string[];
  date: string;
  /** Original amount when entered in a foreign currency. */
  original?: { amount: number; currency: string };
  settlement?: boolean;
  /** Photo uri of the receipt (on the phone that took it). */
  receipt?: string;
  /** The receipt's file on the sync server, so friends' phones can fetch it. */
  receiptBlob?: string;
}

export interface Flight {
  id: string;
```

```
  memberId: string;
  airline: string;
  flightNo: string;
  from: string;
  to: string;
  departUtc: number;
  arriveUtc: number;
  direction: 'arrival' | 'departure';
  seat?: string;
  confirmation?: string;
  ticketUri?: string;
  ticketName?: string;
}

export interface PollOption {
  id: string;
  text: string;
  votes: string[];
}

export interface Poll {
  id: string;
  question: string;
  options: PollOption[];
  createdBy: string;
  closed?: boolean;
  createdAt: number;
}

export interface Task {
  id: string;
  text: string;
  done: boolean;
  assignee?: string;
}

export interface TripDoc {
  id: string;
  name: string;
  kind: 'pdf' | 'image' | 'note' | 'file';
  uri?: string;
  /** The file on the sync server, so friends' phones can fetch it. */
  blob?: string;
  note?: string;
  size?: number;
  addedBy: string;
  addedAt: number;
  offline: boolean;
}

export interface ChatMessage {
  id: string;
  authorId: string;
  text: string;
  ts: number;
  kind?: 'text' | 'system';
  /** True when the message came from the AI trip assistant. */
  ai?: boolean;
  aiSource?: 'local' | 'claude';
  /** Which thread the message belongs to: the human crew chat (default) or the Scout AI thread. */
  channel?: 'crew' | 'ai';
}

export interface Photo {
  id: string;
  uri: string;
  /** The file on the sync server, so friends' phones can fetch it. */
  blob?: string;
  caption: string;
  by: string;
  likes: string[];
  ts: number;
  day?: string;
}

export interface PackingItem {
  id: string;
  text: string;
  category: string;
  packed: boolean;
  /** Member id of whoever is bringing it. */
  broughtBy?: string;
}

export type TripStatus = 'planning' | 'active' | 'completed';
```

```ts
export interface Trip {
  id: string;
  name: string;
  destinationId: string;
  extraDestinationIds: string[];
  startDate: string;
  endDate: string;
  cover: string;
  /** Group budget cap in USD. */
  budget: number;
  inviteCode: string;
  ownerId: string;
  members: Member[];
  activities: Activity[];
  expenses: Expense[];
  flights: Flight[];
  polls: Poll[];
  tasks: Task[];
  docs: TripDoc[];
  messages: ChatMessage[];
  photos: Photo[];
  packing: PackingItem[];
  /** Manual override: user marked the trip as finished. */
  completed?: boolean;
  createdAt: number;
}

export interface Account {
  id: string;
  name: string;
  email: string;
  passwordHash: string;
  salt: string;
  avatar?: string;
  homeCity?: string;
  provider: 'password' | 'google';
}

export interface BucketItem {
  id: string;
  text: string;
  destinationId?: string;
  done: boolean;
  addedAt: number;
}

export interface AppNotification {
  id: string;
  title: string;
  body: string;
  ts: number;
  read: boolean;
  kind: 'reminder' | 'chat' | 'vote' | 'expense' | 'system' | 'member' | 'photo';
  tripId?: string;
  route?: string;
}

export type TextScale = 1 | 1.12 | 1.25;

export interface Settings {
  language: LangCode;
  homeCurrency: string;
  notificationsEnabled: boolean;
  reminderLeadMin: number;
  textScale: TextScale;
  showTripTime: boolean;
  aiMode: 'offline' | 'claude';
  haptics: boolean;
}

/** Home-screen blocks the user can show or hide in "Make it yours". */
export type HomeSection = 'glance' | 'budget' | 'upnext' | 'quick' | 'forYou' | 'nearby' | 'discover';

/** Answers from the "Make it yours" personalization survey. */
export interface Personal {
  done: boolean;
  nickname: string;
  emoji: string;
  interests: string[];
  travelStyle: 'chill' | 'balanced' | 'packed';
  budgetStyle: 'saver' | 'mid' | 'treat';
  sections: Record<HomeSection, boolean>;
  motion: boolean;
  /** True once the first-run guided tour has been seen. */
  toured?: boolean;
}
```

```
export interface WeatherSnapshot {
  destinationId: string;
  fetchedAt: number;
  current: { temp: number; code: number; wind: number; humidity: number };
  daily: { date: string; max: number; min: number; code: number; rain: number }[];
  live: boolean;
}

export interface RatesSnapshot {
  base: 'USD';
  rates: Record<string, number>;
  fetchedAt: number;
  live: boolean;
}
```

src/store/useAppStore.ts

```
/**
 * Public entry point for app state. Screens import from here:
 *   import { useAppStore, useActiveTrip, balances } from '@/store/useAppStore';
 * The pieces live in ./store (the store itself), ./slices (the actions), ./selectors (reading helpers)
 * and utils/money (expense maths).
 */
export { useAppStore, type AppStore } from './store';
export { memberById, tripStatus, useActiveTrip, useCurrentUser, useMyTrips } from './selectors';
export { defaultPersonal } from './slices/app';
export { balances, settlePlan, totalSpent } from '@/utils/money';
```

src/sync/__tests__/blobs.test.ts

```
import type { Trip } from '@/store/types';

import { SyncStore, SyncError } from '../../../server/store';
import { BlobSync, FileTransferError, MAX_SHARED_FILE_BYTES, assetRef, extensionOf, fileNameOf, isAssetRef, isLocalFile, makeBlobId, res
olveFile, type BlobFiles } from '../blobs';
import { SyncEngine } from '../engine';
import { FakeStore, MemoryStorage, fakeNow, makeTrip, memoryTransport, seeded } from '../testing';

type Server = InstanceType<typeof SyncStore>;

/** A phone's storage and its connection to the server's files, in memory. */
function fakeFiles(server: Server, name: string) {
  const disk = new Map<string, Buffer>();
  const files: BlobFiles & { offline: boolean; uploads: string[]; downloads: string[]; disk: Map<string, Buffer> } = {
    offline: false,
    uploads: [],
    downloads: [],
    disk,
    exists: async (uri) => disk.has(uri),
    size: async (uri) => disk.get(uri)?.length ?? null,
    async upload(uri, url) {
      if (files.offline) throw new FileTransferError('network', 'offline');
      const [, tripId, fileId] = /\/v1\/trips\/([^/]+)\/files\/([^/]+)$/.exec(url)!;
      files.uploads.push(fileId);
      try {
        await server.putBlob(tripId, fileId, disk.get(uri)!);
      } catch (e) {
        const status = e instanceof SyncError ? e.status : 500;
        throw new FileTransferError(status === 404 ? 'missing' : status === 400 || status === 413 ? 'rejected' : 'network', String(e));
      }
    },
    async download(url, fileName) {
      if (files.offline) throw new FileTransferError('network', 'offline');
      const [, tripId, fileId] = /\/v1\/trips\/([^/]+)\/files\/([^/]+)$/.exec(url)!;
      files.downloads.push(fileId);
      const bytes = await server.getBlob(tripId, fileId);
      if (!bytes) throw new FileTransferError('missing', 'no such file');
      const uri = `file:///${name}/shared/${fileName}`;
      disk.set(uri, bytes);
      return uri;
    },
  };
  return files;
}

// a pass that keeps failing schedules its own retries: stop them so the test run can finish
const started: { blobs: BlobSync }[] = [];
afterEach(() => started.splice(0).forEach((p) => p.blobs.stop()));

function phone(server: Server, userId: string, name: string, seed: number) {
  const store = new FakeStore();
  store.userId = userId;
  const transport = memoryTransport(server);
  const engine = new SyncEngine({ store, storage: new MemoryStorage(), getTransport: () => transport, pollWaitMs: 0, now: fakeNow(), ran
dom: seeded(seed).next });
```

```
  const files = fakeFiles(server, name);
  const saved = new Map<string, string>();
  const rejected = new Set<string>();
  const notices: string[] = [];
  const pending: number[] = [];
  const random = seeded(seed + 100).next;
  const blobs = new BlobSync({
    trips: () => store.trips,
    remoteId: (id) => engine.remoteId(id),
    baseUrl: () => 'http://sync.test',
    files,
    cache: { get: (id) => saved.get(id), set: (id, uri) => void saved.set(id, uri), isRejected: (k) => rejected.has(k), reject: (k) => v
oid rejected.add(k) },
    attach: (tripId, list, itemId, blobId) =>
      store.edit(tripId, (d) => {
        if (list === 'expenses') d.expenses.find((x) => x.id === itemId)!.receiptBlob = blobId;
        else (d[list].find((x) => x.id === itemId) as { blob?: string }).blob = blobId;
      }),
    newId: () => Math.floor(random() * 36 ** 8).toString(36).padStart(8, '0'),
    onNotice: (m) => notices.push(m),
    onPending: (n) => pending.push(n),
  });
  const created = { store, engine, files, blobs, saved, rejected, notices, pending, name };
  started.push(created);
  return created;
}

type Phone = ReturnType<typeof phone>;

const bytesOf = (text: string) => Buffer.from(text);

/** Ana has a trip with a local photo, receipt and (optionally) more; Ben joins it. */
async function setup(options: { server?: Server; tripEdit?: (t: Trip) => void } = {}) {
  const server = options.server ?? new SyncStore();
  const ana = phone(server, 'u-a', 'ana', 1);
  const ben = phone(server, 'u-b', 'ben', 2);
  const trip = makeTrip();
  trip.expenses[0].receipt = 'file:///ana/receipt.jpg';
  trip.photos[0].uri = 'file:///ana/photo.jpg';
  options.tripEdit?.(trip);
  ana.store.trips = [trip];
  ana.files.disk.set('file:///ana/photo.jpg', bytesOf('PHOTO-BYTES'));
  ana.files.disk.set('file:///ana/receipt.jpg', bytesOf('RECEIPT-BYTES'));
  await ana.engine.init();
  await ben.engine.init();
  await ana.engine.share('trip-1');
  await ana.engine.syncNow();
  await ben.engine.join('ROM-1234');
  return { server, ana, ben };
}

/** Everyone sends and fetches, files move, and the news about the files travels once more. */
async function settle(...phones: Phone[]) {
  for (let i = 0; i < 2; i++) {
    for (const p of phones) await p.engine.syncNow();
    for (const p of phones) await p.blobs.run();
  }
  for (const p of phones) await p.engine.syncNow();
}

const tripOf = (p: Phone) => p.store.getTrip('trip-1')!;

describe('helpers', () => {
  it('works out a file extension from an address', () => {
    expect(extensionOf('file:///a/Photo.JPG')).toBe('jpg');
    expect(extensionOf('file:///a/photo.jpeg?x=1#y')).toBe('jpg');
    expect(extensionOf('file:///a/ticket.pdf')).toBe('pdf');
    expect(extensionOf('')).toBe('png');
    expect(extensionOf('')).toBe('svg');
    expect(extensionOf('blob:http://localhost/1234-5678', 'jpg')).toBe('jpg');
    expect(extensionOf('file:///a/no-extension')).toBe('bin');
  });

  it('makes server ids that turn back into a usable file name', () => {
    const id = makeBlobId('k3j9x2m1', 'file:///a/trip-plan.PDF');
    expect(id).toBe('k3j9x2m1_pdf');
    expect(fileNameOf(id)).toBe('k3j9x2m1.pdf');
    expect(fileNameOf('plainid')).toBe('plainid');
  });

  it('tells real files from the pictures that ship with the app', () => {
    expect(isLocalFile('file:///a/b.jpg')).toBe(true);
    expect(isLocalFile('blob:http://x/1')).toBe(true);
    expect(isLocalFile('')).toBe(true);
    expect(isLocalFile('amalfi')).toBe(false);
    expect(isLocalFile('')).toBe(false);
```

```
    expect(isLocalFile(undefined)).toBe(false);
  });
});

describe('what to show for a file', () => {
  it('prefers this phone’s own file, then a bundled picture, then the downloaded copy', () => {
    expect(resolveFile('file:///mine.jpg', 'abc_jpg', 'file:///copy.jpg')).toEqual({ uri: 'file:///mine.jpg', loading: false });
    expect(resolveFile('', 'asset:amalfi', undefined)).toEqual({ uri: 'amalfi', loading: false });
    expect(resolveFile('', 'abc12345_jpg', 'file:///copy.jpg')).toEqual({ uri: 'file:///copy.jpg', loading: false });
    expect(resolveFile('', 'abc12345_jpg', undefined)).toEqual({ uri: undefined, loading: true });
    expect(resolveFile(undefined, undefined, undefined)).toEqual({ uri: undefined, loading: false });
    expect(isAssetRef(assetRef('amalfi'))).toBe(true);
    expect(isAssetRef('abc12345_jpg')).toBe(false);
  });
});

describe('sending and fetching files', () => {
  it('shares the pictures that come with the app by name instead of sending them', async () => {
    const { ana, ben } = await setup({
      tripEdit: (t) => {
        t.photos.push({ id: 'ph-demo', uri: 'amalfi', caption: 'Bundled', by: 'u-a', likes: [], ts: 6 });
      },
    });
    await settle(ana, ben);
    expect(tripOf(ana).photos.find((p) => p.id === 'ph-demo')?.blob).toBe('asset:amalfi');
    expect(tripOf(ben).photos.find((p) => p.id === 'ph-demo')?.blob).toBe('asset:amalfi');
    expect(ben.files.downloads).toHaveLength(2); // only the real photo and receipt were fetched
    expect(ana.files.uploads).toHaveLength(2);
  });

  it('uploads your photo and receipt and writes the server ids on them', async () => {
    const { server, ana } = await setup();
    await ana.blobs.run();
    const trip = tripOf(ana);
    expect(trip.photos[0].blob).toMatch(/^[a-z0-9]{8}_jpg$/);
    expect(trip.expenses[0].receiptBlob).toMatch(/^[a-z0-9]{8}_jpg$/);
    const remote = ana.engine.remoteId('trip-1')!;
    expect((await server.getBlob(remote, trip.photos[0].blob!))?.toString()).toBe('PHOTO-BYTES');
    expect((await server.getBlob(remote, trip.expenses[0].receiptBlob!))?.toString()).toBe('RECEIPT-BYTES');
    expect(ana.files.uploads).toHaveLength(2);
  });

  it('a friend’s phone downloads the files and keeps its own copy', async () => {
    const { ana, ben } = await setup();
    await settle(ana, ben);
    const photo = tripOf(ben).photos[0];
    expect(photo.blob).toBe(tripOf(ana).photos[0].blob);
    expect(photo.uri).toBe(''); // the path on Ana's phone means nothing here
    const copy = ben.saved.get(photo.blob!)!;
    expect(copy).toMatch(/^file:\/\/\/ben\/shared\/[a-z0-9]{8}\.jpg$/);
    expect(ben.files.disk.get(copy)?.toString()).toBe('PHOTO-BYTES');
    expect(ben.files.disk.get(ben.saved.get(tripOf(ben).expenses[0].receiptBlob!)!)?.toString()).toBe('RECEIPT-BYTES');
  });

  it('does not send a file twice or fetch it again', async () => {
    const { ana, ben } = await setup();
    await settle(ana, ben);
    await settle(ana, ben);
    await ana.blobs.run();
    await ben.blobs.run();
    expect(ana.files.uploads).toHaveLength(2);
    expect(ben.files.downloads).toHaveLength(2);
    expect(ana.files.downloads).toHaveLength(0); // the owner already has the files
  });

  it('two passes at the same time still upload once', async () => {
    const { ana } = await setup();
    await Promise.all([ana.blobs.run(), ana.blobs.run(), ana.blobs.run()]);
    expect(ana.files.uploads).toHaveLength(2);
  });

  it('leaves trips that are not shared alone', async () => {
    const { ana } = await setup({
      tripEdit: (t) => {
        t.photos.push({ id: 'ph-demo', uri: 'amalfi', caption: 'Bundled', by: 'u-a', likes: [], ts: 6 });
      },
    });
    ana.store.trips = [...ana.store.trips, { ...makeTrip(), id: 'trip-private', inviteCode: 'PRV-0001', photos: [{ id: 'x', uri: 'fil
e:///ana/photo.jpg', caption: '', by: 'u-a', likes: [], ts: 1 }] }];
    await ana.blobs.run();
    expect(ana.files.uploads).toHaveLength(2); // only the two files in the shared trip
    expect(ana.store.getTrip('trip-private')!.photos[0].blob).toBeUndefined();
  });

  it('handles documents and keeps their file type', async () => {
    const { ana, ben } = await setup({
```

```
      tripEdit: (t) => {
        t.docs.push({ id: 'd1', name: 'Hotel booking', kind: 'pdf', uri: 'file:///ana/booking.pdf', addedBy: 'u-a', addedAt: 3, offline:
true });
        t.docs.push({ id: 'd2', name: 'Packing tips', kind: 'note', note: 'Bring a hat', addedBy: 'u-a', addedAt: 4, offline: false });
      },
    });
    ana.files.disk.set('file:///ana/booking.pdf', bytesOf('PDF-BYTES'));
    await settle(ana, ben);
    const doc = tripOf(ben).docs.find((d) => d.id === 'd1')!;
    expect(doc.blob).toMatch(/_pdf$/);
    expect(ben.saved.get(doc.blob!)).toMatch(/\.pdf$/);
    expect(tripOf(ben).docs.find((d) => d.id === 'd2')?.blob).toBeUndefined(); // a note has no file
  });

  it('fetches the file again when the saved copy has disappeared', async () => {
    const { ana, ben } = await setup();
    await settle(ana, ben);
    const id = tripOf(ben).photos[0].blob!;
    ben.files.disk.delete(ben.saved.get(id)!); // the phone cleaned its storage
    await ben.blobs.run();
    expect(ben.files.downloads.filter((d) => d === id)).toHaveLength(2);
    expect(ben.files.disk.get(ben.saved.get(id)!)?.toString()).toBe('PHOTO-BYTES');
  });

  it('a friend who joins later also gets files that were added before', async () => {
    const { server, ana, ben } = await setup();
    await settle(ana, ben);
    const cara = phone(server, 'u-c', 'cara', 3);
    await cara.engine.init();
    await cara.engine.join('ROM-1234');
    await cara.blobs.run();
    expect(cara.saved.size).toBe(2);
  });
});

describe('when things go wrong', () => {
  it('keeps the file for later when there is no connection, then sends it', async () => {
    const { ana, ben } = await setup();
    ana.files.offline = true;
    await ana.blobs.run();
    expect(tripOf(ana).photos[0].blob).toBeUndefined();
    expect(ana.notices).toEqual([]); // being offline is not worth a message
    ana.files.offline = false;
    await settle(ana, ben);
    expect(tripOf(ben).photos[0].blob).toBeDefined();
    expect(ben.saved.size).toBe(2);
  });

  it('tries again by itself after a pause', async () => {
    jest.useFakeTimers();
    try {
      const { ana } = await setup();
      ana.files.offline = true;
      await ana.blobs.run(); // fails and schedules a retry
      expect(ana.files.uploads).toHaveLength(0);
      ana.files.offline = false;
      await jest.advanceTimersByTimeAsync(2900);
      expect(ana.files.uploads).toHaveLength(0); // still waiting
      await jest.advanceTimersByTimeAsync(300);
      expect(ana.files.uploads).toHaveLength(2);
      ana.blobs.stop();
    } finally {
      jest.useRealTimers();
    }
  });

  it('does not share a file that is too big, and says so once', async () => {
    const { ana } = await setup();
    ana.files.disk.set('file:///ana/photo.jpg', Buffer.alloc(MAX_SHARED_FILE_BYTES + 1024 * 1024));
    await ana.blobs.run();
    await ana.blobs.run();
    expect(ana.notices).toHaveLength(1);
    expect(ana.notices[0]).toMatch(/9\.0 MB/);
    expect(tripOf(ana).photos[0].blob).toBeUndefined();
    expect(tripOf(ana).expenses[0].receiptBlob).toBeDefined(); // the receipt still went
  });

  it('remembers a file the server refused instead of trying it forever', async () => {
    const server = new SyncStore({ maxBlobBytes: 5 });
    const { ana } = await setup({ server });
    await ana.blobs.run();
    await ana.blobs.run();
    expect(ana.notices.length).toBe(2); // one per file, each only once
    expect(ana.files.uploads).toHaveLength(2);
    expect(ana.rejected.size).toBe(2);
  });
```

```
  it('stops quietly when the file was deleted from the phone', async () => {
    const { ana } = await setup();
    ana.files.disk.delete('file:///ana/photo.jpg');
    await ana.blobs.run();
    expect(tripOf(ana).photos[0].blob).toBeUndefined();
    expect(ana.notices).toEqual([]);
    expect(tripOf(ana).expenses[0].receiptBlob).toBeDefined();
  });

  it('keeps going with other files when one is missing on the server', async () => {
    const { ana, ben } = await setup();
    await settle(ana, ben);
    // a second photo whose file the server never received, plus the earlier ones gone from Ben's phone
    ana.store.edit('trip-1', (d) => void d.photos.unshift({ id: 'ph-lost', uri: '', blob: 'lost0000_jpg', caption: 'Lost', by: 'u-a', li
kes: [], ts: 9 }));
    await ana.engine.syncNow();
    await ben.engine.syncNow();
    ben.files.disk.delete(ben.saved.get(tripOf(ben).expenses[0].receiptBlob!)!); // this one can be fetched again
    await ben.blobs.run();
    expect(ben.files.disk.get(ben.saved.get(tripOf(ben).expenses[0].receiptBlob!)!)?.toString()).toBe('RECEIPT-BYTES');
  });

  it('reports how many files are waiting', async () => {
    const { ana } = await setup();
    await ana.blobs.run();
    expect(ana.pending[0]).toBe(2);
    expect(ana.pending.at(-1)).toBe(0);
  });

  it('does nothing without a server address', async () => {
    const { ana } = await setup();
    const quiet = new BlobSync({ trips: () => ana.store.trips, remoteId: (id) => ana.engine.remoteId(id), baseUrl: () => null, files: an
a.files, cache: { get: () => undefined, set: () => undefined, isRejected: () => false, reject: () => undefined }, attach: () => undefine
d, newId: () => 'abcdefgh' });
    await quiet.run();
    expect(ana.files.uploads).toHaveLength(0);
  });
});
```

src/sync/__tests__/crdt.test.ts

```
import { produce } from 'immer';

import { buildDemoTrip } from '@/store/seed';
import type { Trip } from '@/store/types';

import { applyOps, diffTrip, materializeTrip, resolve, type Op, type Registers } from '../crdt';
import { HybridClock } from '../hlc';
import { mergeLocalOnly } from '../view';
import { Replica, fakeNow, makeTrip, seeded, sharedPart } from '../testing';

const json = <T,>(x: T): T => JSON.parse(JSON.stringify(x));

describe('describing and rebuilding a trip', () => {
  it('a rebuilt trip equals the original (shared parts)', () => {
    const trip = makeTrip();
    const clock = new HybridClock('a', fakeNow());
    const regs: Registers = {};
    applyOps(regs, diffTrip(undefined, trip, () => clock.next()));
    const rebuilt = materializeTrip(regs)!;
    expect(rebuilt).toEqual(sharedPart(trip));
  });

  it('works for the full demo trip (hundreds of items) and puts phone-only parts back', () => {
    const trip = json(buildDemoTrip('upcoming'));
    const clock = new HybridClock('a', fakeNow());
    const regs: Registers = {};
    applyOps(regs, diffTrip(undefined, trip, () => clock.next()));
    const rebuilt = materializeTrip(regs)!;
    expect(rebuilt).toEqual(sharedPart(trip));
    expect(json(mergeLocalOnly(rebuilt, trip))).toEqual(trip);
  });

  it('keeps the order lists had: newest last for itineraries, newest first for expenses', () => {
    const trip = produce(makeTrip(), (d) => {
      d.activities.push({ ...d.activities[0], id: 'a2', title: 'Forum' }, { ...d.activities[0], id: 'a3', title: 'Pantheon' });
      d.expenses.unshift({ ...d.expenses[0], id: 'e2', title: 'Dinner' }, { ...d.expenses[0], id: 'e3', title: 'Taxi' });
    });
    const clock = new HybridClock('a', fakeNow());
    const regs: Registers = {};
    applyOps(regs, diffTrip(undefined, trip, () => clock.next()));
    const rebuilt = materializeTrip(regs)!;
    expect(rebuilt.activities.map((a) => a.id)).toEqual(['a1', 'a2', 'a3']);
    expect(rebuilt.expenses.map((e) => e.id)).toEqual(['e2', 'e3', 'e1']); // the same order the array had
  });
```

```
  it('never shares private chat, emails or file paths', () => {
    const ops = diffTrip(undefined, makeTrip(), new HybridClock('a', fakeNow()).next.bind(new HybridClock('a', fakeNow())));
    const text = JSON.stringify(ops);
    expect(text).not.toContain('ana@example.com');
    expect(text).not.toContain('private answer');
    expect(text).not.toContain('file:///');
  });
});

describe('merging is order-proof and repeat-proof', () => {
  it('applying ops twice, or shuffled, gives the same registers', () => {
    const clock = new HybridClock('a', fakeNow());
    const ops = diffTrip(undefined, makeTrip(), () => clock.next());
    const once: Registers = {};
    applyOps(once, ops);
    const twice: Registers = {};
    applyOps(twice, ops);
    expect(applyOps(twice, ops)).toBe(false);
    const rnd = seeded(5);
    const shuffled: Registers = {};
    applyOps(shuffled, rnd.shuffle(ops));
    expect(twice).toEqual(once);
    expect(shuffled).toEqual(once);
  });

  it('ignores paths it does not know (changes from a newer app version)', () => {
    const regs: Registers = {};
    expect(applyOps(regs, [{ p: ['hologram', 'x'], v: 1, t: '1800000000000-0000-zzzz' }, { p: ['activities', 'a1', 'teleport'], v: 1, t:
'1800000000001-0000-zzzz' }])).toBe(false);
    expect(regs).toEqual({});
  });

  it('validates paths against the schema', () => {
    expect(resolve(['name'])).toEqual({ items: [], isItem: false });
    expect(resolve(['activities', 'a1'])).toEqual({ items: [['activities', 'a1']], isItem: true });
    expect(resolve(['activities', 'a1', 'votes', 'u-a'])?.isItem).toBe(false);
    expect(resolve(['polls', 'p1', 'options', 'o1', 'votes', 'u-b'])?.items).toHaveLength(2);
    expect(resolve(['activities', 'a1', 'votes'])).toBeNull(); // a map node alone is not a register
    expect(resolve(['activities'])).toBeNull();
    expect(resolve(['name', 'extra'])).toBeNull();
  });

  it('hides items that arrive without their required fields', () => {
    const clock = new HybridClock('a', fakeNow());
    const regs: Registers = {};
    applyOps(regs, diffTrip(undefined, makeTrip(), () => clock.next()));
    const t = clock.next();
    applyOps(regs, [{ p: ['tasks', 'partial', '_c'], v: 1, t }, { p: ['tasks', 'partial', 'done'], v: true, t: clock.next() }]);
    expect(materializeTrip(regs)!.tasks.map((k) => k.id)).toEqual(['t1']);
  });
});

describe('two phones editing at the same time', () => {
  const setup = () => {
    const now = fakeNow();
    const a = new Replica('aaaaaaaa', now, makeTrip());
    const b = new Replica('bbbbbbbb', now);
    b.receive(a.publish());
    return { a, b, now };
  };
  const swap = (a: Replica, b: Replica, opsA: Op[], opsB: Op[]) => {
    b.receive(opsA);
    a.receive(opsB);
  };

  it('keeps both new activities', () => {
    const { a, b } = setup();
    const oa = a.edit((d) => void d.activities.push({ ...d.activities[0], id: 'a-ana', title: 'Forum' }));
    const ob = b.edit((d) => void d.activities.push({ ...d.activities[0], id: 'a-ben', title: 'Pantheon' }));
    swap(a, b, oa, ob);
    expect(a.trip!.activities.map((x) => x.id)).toEqual(['a1', 'a-ana', 'a-ben']);
    expect(sharedPart(a.trip)).toEqual(sharedPart(b.trip));
  });

  it('merges edits to different fields of one activity', () => {
    const { a, b } = setup();
    const oa = a.edit((d) => void (d.activities[0].title = 'Colosseum tour'));
    const ob = b.edit((d) => void (d.activities[0].time = '09:00'));
    swap(a, b, oa, ob);
    for (const r of [a, b]) expect(r.trip!.activities[0]).toMatchObject({ title: 'Colosseum tour', time: '09:00' });
  });

  it('the later edit wins when both change the same field, on both phones', () => {
    const { a, b } = setup();
    const oa = a.edit((d) => void (d.name = 'Rome Crew!'));
```

```
  const ob = b.edit((d) => void (d.name = 'Roma Squad')); // made a moment later
  swap(a, b, oa, ob);
  expect(a.trip!.name).toBe('Roma Squad');
  expect(b.trip!.name).toBe('Roma Squad');
});

it('a delete beats a concurrent edit of the same item', () => {
  const { a, b } = setup();
  const oa = a.edit((d) => void (d.activities = []));
  const ob = b.edit((d) => void (d.activities[0].title = 'Edited while deleted'));
  swap(a, b, oa, ob);
  expect(a.trip!.activities).toEqual([]);
  expect(b.trip!.activities).toEqual([]);
});

it('votes from different people both count', () => {
  const { a, b } = setup();
  const oa = a.edit((d) => void (d.activities[0].votes['u-a'] = -1));
  const ob = b.edit((d) => void (d.activities[0].votes['u-b'] = 1));
  swap(a, b, oa, ob);
  expect(a.trip!.activities[0].votes).toEqual({ 'u-a': -1, 'u-b': 1 });
  expect(b.trip!.activities[0].votes).toEqual({ 'u-a': -1, 'u-b': 1 });
});

it('adds and removes inside lists of names independently', () => {
  const { a, b } = setup();
  const oa = a.edit((d) => void d.expenses[0].splitBetween.push('u-c'));
  const ob = b.edit((d) => void (d.expenses[0].splitBetween = d.expenses[0].splitBetween.filter((x) => x !== 'u-a')));
  swap(a, b, oa, ob);
  expect([...a.trip!.expenses[0].splitBetween].sort()).toEqual(['u-b', 'u-c']);
  expect([...b.trip!.expenses[0].splitBetween].sort()).toEqual(['u-b', 'u-c']);
});

it('poll votes (nested items) merge too', () => {
  const { a, b } = setup();
  const oa = a.edit((d) => void d.polls[0].options[0].votes.push('u-a'));
  const ob = b.edit((d) => void d.polls[0].options[1].votes.push('u-c'));
  swap(a, b, oa, ob);
  for (const r of [a, b]) expect(r.trip!.polls[0].options.map((o) => [...o.votes].sort())).toEqual([['u-a'], ['u-b', 'u-c']]);
});

it('chat messages from both phones appear in time order', () => {
  const { a, b } = setup();
  const oa = a.edit((d) => void d.messages.push({ id: 'm-a', authorId: 'u-a', text: 'first', ts: 2000 }));
  const ob = b.edit((d) => void d.messages.push({ id: 'm-b', authorId: 'u-b', text: 'second', ts: 2001 }));
  swap(a, b, oa, ob);
  expect(a.trip!.messages.filter((m) => !m.ai).map((m) => m.id)).toEqual(['m1', 'm-a', 'm-b']);
  expect(b.trip!.messages.map((m) => m.id)).toEqual(['m1', 'm-a', 'm-b']);
});

it('keeps the private Scout chat on its own phone', () => {
  const { a, b } = setup();
  expect(a.trip!.messages.map((m) => m.id)).toContain('ai1');
  expect(b.trip!.messages.map((m) => m.id)).not.toContain('ai1');
  a.edit((d) => void d.messages.push({ id: 'ai2', authorId: 'assistant', text: 'another', ts: 3, ai: true, channel: 'ai' }));
  expect(b.trip!.messages.map((m) => m.id)).not.toContain('ai2');
});

it('a person who leaves and rejoins comes back with fresh details only', () => {
  const { a, b } = setup();
  // Ben leaves (his member entry is deleted) ...
  const leave = b.edit((d) => void (d.members = d.members.filter((m) => m.id !== 'u-b')));
  a.receive(leave);
  expect(a.trip!.members.map((m) => m.id)).toEqual(['u-a']);
  // ... then rejoins with a different role and no home city.
  const rejoin = b.edit((d) => void d.members.push({ id: 'u-b', name: 'Ben R.', role: 'Photographer' }));
  a.receive(rejoin);
  expect(a.trip!.members.find((m) => m.id === 'u-b')).toEqual({ id: 'u-b', name: 'Ben R.', role: 'Photographer' });
});

it('removing a field removes it everywhere', () => {
  const { a, b } = setup();
  b.receive(a.edit((d) => void (d.activities[0].notes = 'Bring water')));
  expect(b.trip!.activities[0].notes).toBe('Bring water');
  b.receive(a.edit((d) => void delete d.activities[0].notes));
  expect(b.trip!.activities[0].notes).toBeUndefined();
});

it('a phone that was offline catches up from everything it missed', () => {
  const { a, b } = setup();
  const missed: Op[] = [];
  missed.push(...a.edit((d) => void d.tasks.push({ id: 't2', text: 'Pack', done: false })));
  missed.push(...a.edit((d) => void (d.tasks[0].done = true)));
  missed.push(...a.edit((d) => void d.messages.push({ id: 'm2', authorId: 'u-a', text: 'back online?', ts: 9 })));
  b.receive(missed);
  expect(sharedPart(b.trip)).toEqual(sharedPart(a.trip));
```

```
  });
});

describe('many phones, many random edits: everyone ends up identical', () => {
  const ids = ['u-a', 'u-b', 'u-c'];

  const mutations: ((d: Trip, rnd: ReturnType<typeof seeded>, n: number) => void)[] = [
    (d, r, n) => void d.activities.push({ ...makeTrip().activities[0], id: `a-${n}`, title: `Act ${n}`, status: r.pick(['voting', 'confi
rmed', 'pending'] as const) }),
    (d, r) => void (d.activities.length && (d.activities[r.int(d.activities.length)].title = `Retitled ${r.int(99)}`)),
    (d, r) => void (d.activities.length && (d.activities[r.int(d.activities.length)].votes[r.pick(ids)] = r.pick([1, -1] as const))),
    (d, r) => {
      if (!d.activities.length) return;
      const a = d.activities[r.int(d.activities.length)];
      delete a.votes[r.pick(ids)];
    },
    (d, r) => void (d.activities.length && d.activities.splice(r.int(d.activities.length), 1)),
    (d, r) => {
      if (!d.activities.length) return;
      const a = d.activities[r.int(d.activities.length)];
      const who = r.pick(ids);
      a.assigned = a.assigned.includes(who) ? a.assigned.filter((x) => x !== who) : [...a.assigned, who];
    },
    (d, r, n) => void d.expenses.unshift({ id: `e-${n}`, title: `Exp ${n}`, amount: r.int(5000) / 100, category: 'Food', paidBy: r.pick
(ids), splitBetween: ids.filter(() => r.next() < 0.7), date: '2026-09-11' }),
    (d, r) => void (d.expenses.length && d.expenses.splice(r.int(d.expenses.length), 1)),
    (d, r, n) => void d.messages.push({ id: `m-${n}`, authorId: r.pick(ids), text: `msg ${n}`, ts: n }),
    (d, r, n) => void d.tasks.push({ id: `t-${n}`, text: `Task ${n}`, done: false }),
    (d, r) => void (d.tasks.length && (d.tasks[r.int(d.tasks.length)].done = r.next() < 0.5)),
    (d, r) => void (d.tasks.length && d.tasks.splice(r.int(d.tasks.length), 1)),
    (d, r) => void (d.name = `Trip ${r.int(100)}`),
    (d, r) => void (d.budget = 1000 + r.int(5000)),
    (d, r) => {
      const o = d.polls[0]?.options[r.int(2)];
      if (!o) return;
      const who = r.pick(ids);
      o.votes = o.votes.includes(who) ? o.votes.filter((x) => x !== who) : [...o.votes, who];
    },
    (d, r, n) => void d.members.push({ id: `m-${n}`, name: `Member ${n}`, role: 'Member' }),
    (d, r) => void (d.members.length > 1 && d.members.splice(1 + r.int(d.members.length - 1), 1)),
    (d, r) => void (d.photos.length && (d.photos[0].likes = d.photos[0].likes.includes('u-c') ? [] : ['u-c'])),
  ];

  // More seeds on demand:  SYNC_SEEDS=200 npx jest src/sync
  it.each(Array.from({ length: Number(process.env.SYNC_SEEDS ?? 6) }, (_, i) => i + 1))('seed %i: 3 phones, 90 edits, shuffled delivery
with duplicates', (seed) => {
    const rnd = seeded(seed * 7919);
    const now = fakeNow();
    const phones = [new Replica('aaaaaaaa', now, makeTrip()), new Replica('bbbbbbbb', now), new Replica('cccccccc', now)];
    const baseline = phones[0].publish();
    phones[1].receive(baseline);
    phones[2].receive(baseline);

    const inbox: Op[][][] = phones.map(() => []);
    for (let n = 0; n < 90; n++) {
      const i = rnd.int(3);
      const ops = phones[i].edit((d) => rnd.pick(mutations)(d, rnd, n));
      if (ops.length) phones.forEach((_, j) => j !== i && inbox[j].push(ops));
      if (rnd.next() < 0.35) {
        // some phone gets whatever is waiting for it, in random order, sometimes twice
        const j = rnd.int(3);
        const batches = rnd.shuffle(inbox[j]);
        inbox[j] = [];
        batches.forEach((b) => {
          phones[j].receive(b);
          if (rnd.next() < 0.2) phones[j].receive(b);
        });
      }
    }
    phones.forEach((p, j) => {
      rnd.shuffle(inbox[j]).forEach((b) => p.receive(b));
    });

    const [p0, p1, p2] = phones;
    expect(p1.regs).toEqual(p0.regs);
    expect(p2.regs).toEqual(p0.regs);
    expect(sharedPart(p1.trip)).toEqual(sharedPart(p0.trip));
    expect(sharedPart(p2.trip)).toEqual(sharedPart(p0.trip));
    // and what each phone shows is exactly the merge of everything
    expect(sharedPart(p0.trip)).toEqual(json(materializeTrip(p0.regs)));
  });
});
```

src/sync/__tests__/e2e.test.ts

```
/** @jest-environment node */
import http from 'http';
import type { AddressInfo } from 'net';

import { createServer } from '../../../server/index';
import { SyncStore } from '../../../server/store';
import { SyncEngine } from '../engine';
import { FakeStore, MemoryStorage, makeTrip, seeded, sharedPart } from '../testing';
import { createHttpTransport, normalizeServerUrl } from '../transport';

/** Jest's React Native setup replaces the global fetch, so these tests bring a tiny real one built on Node's http. */
const nodeFetch = ((input: string, init: RequestInit = {}) =>
  new Promise((resolve, reject) => {
    const req = http.request(input, { method: init.method ?? 'GET', headers: init.headers as Record<string, string>, agent: false }, (re
s) => {
      let text = '';
      res.on('data', (c) => (text += c));
      res.on('end', () => resolve({ status: res.statusCode, json: async () => JSON.parse(text) }));
    });
    req.on('error', reject);
    init.signal?.addEventListener('abort', () => {
      req.destroy();
      reject(Object.assign(new Error('aborted'), { name: 'AbortError' }));
    });
    if (init.body) req.write(init.body);
    req.end();
  })) as unknown as typeof fetch;

const until = async (check: () => boolean, ms = 3000) => {
  const start = Date.now();
  while (!check()) {
    if (Date.now() - start > ms) throw new Error('timed out waiting');
    await new Promise((r) => setTimeout(r, 15));
  }
};

describe('two phones talking to the real server over HTTP', () => {
  let server: ReturnType<typeof createServer>;
  let url: string;
  const engines: SyncEngine[] = [];

  beforeEach(async () => {
    server = createServer(new SyncStore());
    await new Promise<void>((r) => server.listen(0, '127.0.0.1', r));
    url = `http://127.0.0.1:${(server.address() as AddressInfo).port}`;
  });
  afterEach(async () => {
    engines.splice(0).forEach((e) => e.stop());
    await new Promise((r) => server.close(r));
  });

  function phone(userId: string, seed: number) {
    const store = new FakeStore();
    store.userId = userId;
    const engine = new SyncEngine({ store, storage: new MemoryStorage(), getTransport: () => createHttpTransport(url, nodeFetch), rando
m: seeded(seed).next, pollWaitMs: 5000 });
    engines.push(engine);
    return { store, engine };
  }

  it('shares, joins and keeps chatting live (long-polling)', async () => {
    const ana = phone('u-a', 1);
    const ben = phone('u-b', 2);
    ana.store.trips = [makeTrip()];
    await ana.engine.init();
    await ben.engine.init();
    await ana.engine.share('trip-1');
    ana.engine.start();
    await until(() => ana.engine.getStatus().pending === 0 && ana.engine.getStatus().online === true);

    expect((await ben.engine.join('ROM-1234'))?.name).toBe('Rome Crew');
    ben.engine.start();

    ana.store.edit('trip-1', (d) => void d.messages.push({ id: 'live1', authorId: 'u-a', text: 'Can you see this?', ts: 9000 }));
    await until(() => !!ben.store.getTrip('trip-1')?.messages.some((m) => m.id === 'live1'));

    ben.store.edit('trip-1', (d) => void (d.activities[0].votes['u-b'] = 1));
    await until(() => ana.store.getTrip('trip-1')?.activities[0].votes['u-b'] === 1);

    expect(sharedPart(ana.store.getTrip('trip-1'))).toEqual(sharedPart(ben.store.getTrip('trip-1')));
    await until(() => ana.engine.getStatus().peers === 2);
  });

  it('reports a clear error when the server is not there', async () => {
```

```
    const lonely = phone('u-a', 3);
    lonely.store.trips = [makeTrip()];
    await lonely.engine.init();
    await expect(lonely.engine.share('trip-1')).resolves.toBeUndefined();
    await new Promise((r) => server.close(r));
    server = createServer(new SyncStore()); // so afterEach can close something
    await new Promise<void>((r) => server.listen(0, '127.0.0.1', r));
    await lonely.engine.syncNow();
    expect(lonely.engine.getStatus()).toMatchObject({ online: false, error: 'Can’t reach the sync server' });
  });
});

describe('server addresses', () => {
  it('accepts the ways people type an address', () => {
    expect(normalizeServerUrl('192.168.1.20:8787')).toBe('http://192.168.1.20:8787');
    expect(normalizeServerUrl(' http://192.168.1.20:8787/ ')).toBe('http://192.168.1.20:8787');
    expect(normalizeServerUrl('https://sync.example.com')).toBe('https://sync.example.com');
    expect(normalizeServerUrl('localhost:8787/extra/path')).toBe('http://localhost:8787');
  });

  it('rejects things that are not addresses', () => {
    expect(normalizeServerUrl('')).toBeNull();
    expect(normalizeServerUrl('   ')).toBeNull();
    expect(normalizeServerUrl('http://')).toBeNull();
  });
});
```

src/sync/__tests__/engine.test.ts

```
import type { Trip } from '@/store/types';

import { SyncStore } from '../../../server/store';
import { CodeTakenError, SyncEngine } from '../engine';
import { FakeStore, MemoryStorage, fakeNow, makeTrip, memoryTransport, seeded, sharedPart } from '../testing';
import type { SyncTransport } from '../transport';

type Server = InstanceType<typeof SyncStore>;

function phone(server: Server, userId: string, seed = 1, now = fakeNow()) {
  const store = new FakeStore();
  store.userId = userId;
  const storage = new MemoryStorage();
  const transport = memoryTransport(server);
  const remote: { prev: Trip | undefined; next: Trip }[] = [];
  const engine = new SyncEngine({ store, storage, getTransport: () => transport, pollWaitMs: 0, now, random: seeded(seed).next, onRemote
Change: (prev, next) => remote.push({ prev, next }) });
  return { store, storage, transport, engine, remote, userId, now };
}

/** Everyone sends, then everyone fetches, twice, so changes travel all the way around. */
async function settle(...phones: ReturnType<typeof phone>[]) {
  for (let i = 0; i < 2; i++) for (const p of phones) await p.engine.syncNow();
}

async function sharedTrip() {
  const server = new SyncStore();
  const ana = phone(server, 'u-a', 1);
  const ben = phone(server, 'u-b', 2);
  ana.store.trips = [makeTrip()];
  await ana.engine.init();
  await ben.engine.init();
  await ana.engine.share('trip-1');
  await ana.engine.syncNow();
  const joined = await ben.engine.join('ROM-1234');
  return { server, ana, ben, joined };
}

describe('sharing and joining a trip', () => {
  it('a friend joins with the invite code and gets the same trip', async () => {
    const { ana, ben, joined } = await sharedTrip();
    expect(joined?.id).toBe('trip-1');
    expect(ben.store.trips).toHaveLength(1);
    expect(sharedPart(ben.store.getTrip('trip-1'))).toEqual(sharedPart(ana.store.getTrip('trip-1')));
    expect(ben.engine.isShared('trip-1')).toBe(true);
  });

  it('keeps each phone’s private parts private', async () => {
    const { ana, ben } = await sharedTrip();
    expect(ana.store.getTrip('trip-1')!.messages.map((m) => m.id)).toContain('ai1'); // Ana's Scout chat
    expect(ben.store.getTrip('trip-1')!.messages.map((m) => m.id)).not.toContain('ai1');
    expect(ben.store.getTrip('trip-1')!.members.find((m) => m.id === 'u-a')?.email).toBeUndefined();
    expect(ana.store.getTrip('trip-1')!.members.find((m) => m.id === 'u-a')?.email).toBe('ana@example.com');
  });

  it('an unknown code finds nothing', async () => {
    const { ben } = await sharedTrip();
```

```
    expect(await ben.engine.join('ZZZ-0000')).toBeNull();
  });

  it('refuses to share a second trip under a code another shared trip already uses', async () => {
    const { server } = await sharedTrip();
    const cara = phone(server, 'u-c', 3);
    cara.store.trips = [{ ...makeTrip(), id: 'trip-other' }];
    await cara.engine.init();
    await expect(cara.engine.share('trip-other')).rejects.toBeInstanceOf(CodeTakenError);
  });

  it('a big trip (700 chat messages) travels in pages', async () => {
    const server = new SyncStore();
    const ana = phone(server, 'u-a', 1);
    const ben = phone(server, 'u-b', 2);
    const trip = makeTrip();
    for (let i = 0; i < 700; i++) trip.messages.push({ id: `bulk${i}`, authorId: 'u-a', text: `message ${i}`, ts: 2000 + i });
    ana.store.trips = [trip];
    await ana.engine.init();
    await ben.engine.init();
    await ana.engine.share('trip-1');
    await ana.engine.syncNow();
    await ben.engine.join('ROM-1234');
    expect(ben.store.getTrip('trip-1')!.messages.filter((m) => m.id.startsWith('bulk'))).toHaveLength(700);
    expect(server.stats().ops).toBeGreaterThan(2500); // more than two pages of 1000 changes
  });
});

describe('working together', () => {
  it('chat, votes and expenses flow both ways', async () => {
    const { ana, ben } = await sharedTrip();
    ben.store.edit('trip-1', (d) => {
      d.messages.push({ id: 'm-ben', authorId: 'u-b', text: 'Hello from Ben', ts: 5000 });
      d.activities[0].votes['u-b'] = 1;
    });
    ana.store.edit('trip-1', (d) => {
      d.expenses.unshift({ id: 'e-ana', title: 'Pizza', amount: 40, category: 'Food', paidBy: 'u-a', splitBetween: ['u-a', 'u-b'], date:
'2026-09-11' });
    });
    await settle(ana, ben);
    for (const p of [ana, ben]) {
      const t = p.store.getTrip('trip-1')!;
      expect(t.messages.map((m) => m.id)).toContain('m-ben');
      expect(t.activities[0].votes).toEqual({ 'u-a': 1, 'u-b': 1 });
      expect(t.expenses[0].id).toBe('e-ana');
    }
    expect(sharedPart(ana.store.getTrip('trip-1'))).toEqual(sharedPart(ben.store.getTrip('trip-1')));
  });

  it('a new member joining shows up for everyone', async () => {
    const { server, ana, ben } = await sharedTrip();
    const cara = phone(server, 'u-c', 3);
    await cara.engine.init();
    await cara.engine.join('ROM-1234');
    cara.store.edit('trip-1', (d) => void d.members.push({ id: 'u-c', name: 'Cara', role: 'Member' }));
    await settle(ana, ben, cara);
    for (const p of [ana, ben, cara]) expect(p.store.getTrip('trip-1')!.members.map((m) => m.id)).toEqual(['u-a', 'u-b', 'u-c']);
  });

  it('does not send friends’ changes back to the server', async () => {
    const { ana, ben, server } = await sharedTrip();
    ben.store.edit('trip-1', (d) => void d.tasks.push({ id: 't9', text: 'Buy water', done: false }));
    await settle(ana, ben);
    const total = server.stats().ops;
    await settle(ana, ben);
    await settle(ana, ben);
    expect(server.stats().ops).toBe(total);
  });

  it('tells the app when friends changed something', async () => {
    const { ana, ben } = await sharedTrip();
    ben.store.edit('trip-1', (d) => void d.messages.push({ id: 'm-ben', authorId: 'u-b', text: 'Ping', ts: 6000 }));
    await settle(ana, ben);
    expect(ana.remote.length).toBeGreaterThan(0);
    const last = ana.remote.at(-1)!;
    expect(last.prev!.messages.some((m) => m.id === 'm-ben')).toBe(false);
    expect(last.next.messages.some((m) => m.id === 'm-ben')).toBe(true);
  });

  it('concurrent edits to different things both survive', async () => {
    const { ana, ben } = await sharedTrip();
    ana.store.edit('trip-1', (d) => void (d.activities[0].title = 'Colosseum at sunrise'));
    ben.store.edit('trip-1', (d) => void (d.activities[0].time = '07:30'));
    await settle(ana, ben);
    for (const p of [ana, ben]) expect(p.store.getTrip('trip-1')!.activities[0]).toMatchObject({ title: 'Colosseum at sunrise', time: '0
7:30' });
```

```
  });

  it('keeps private Scout messages off the server', async () => {
    const { ana, ben } = await sharedTrip();
    ana.store.edit('trip-1', (d) => void d.messages.push({ id: 'ai-new', authorId: 'assistant', text: 'secret', ts: 7000, ai: true, chan
nel: 'ai' }));
    await settle(ana, ben);
    expect(ben.store.getTrip('trip-1')!.messages.map((m) => m.id)).not.toContain('ai-new');
  });
});

describe('the live connection', () => {
  it('asks for a quick answer first, so a quiet connection shows as live right away', async () => {
    const server = new SyncStore();
    const ana = phone(server, 'u-a', 1);
    ana.store.trips = [makeTrip()];
    await ana.engine.init();
    await ana.engine.share('trip-1');
    await ana.engine.syncNow();

    const waits: number[] = [];
    const base = memoryTransport(server);
    const transport: SyncTransport = {
      ...base,
      // a long wait only ends when the engine stops (like a real long poll with nothing new to report)
      pull: (id, since, client, waitMs, signal) => {
        waits.push(waitMs);
        return waitMs > 0 ? new Promise((_resolve, reject) => signal?.addEventListener('abort', () => reject(new Error('aborted')))) : b
ase.pull(id, since, client, 0);
      },
    };
    const store = new FakeStore();
    store.userId = 'u-b';
    const engine = new SyncEngine({ store, storage: new MemoryStorage(), getTransport: () => transport, pollWaitMs: 5000, random: seeded
(2).next });
    await engine.init();
    engine.start();
    await engine.join('ROM-1234');
    await new Promise((resolve) => setTimeout(resolve, 20));
    expect(waits.slice(-2)).toEqual([0, 5000]);
    expect(engine.getStatus().online).toBe(true);
    engine.stop();
  });
});

describe('losing the connection and the app closing', () => {
  it('queues changes while offline and sends them when the connection returns', async () => {
    const { ana, ben } = await sharedTrip();
    ana.transport.offline = true;
    ana.store.edit('trip-1', (d) => void d.tasks.push({ id: 't-offline', text: 'Written offline', done: false }));
    await ana.engine.syncNow();
    expect(ana.engine.getStatus()).toMatchObject({ online: false, error: 'Can’t reach the sync server' });
    expect(ana.engine.getStatus().pending).toBeGreaterThan(0);
    await ben.engine.syncNow();
    expect(ben.store.getTrip('trip-1')!.tasks.map((t) => t.id)).not.toContain('t-offline');

    ana.transport.offline = false;
    await settle(ana, ben);
    expect(ana.engine.getStatus()).toMatchObject({ online: true, pending: 0, error: null });
    expect(ben.store.getTrip('trip-1')!.tasks.map((t) => t.id)).toContain('t-offline');
  });

  it('remembers where it was after the app restarts', async () => {
    const { server, ana, ben } = await sharedTrip();
    await ana.engine.saveNow();
    // "restart": a new engine with the same saved state and the same trips
    const again = phone(server, 'u-a', 9);
    again.store.trips = ana.store.trips;
    again.storage.saved = ana.storage.saved;
    await again.engine.init();
    expect(again.engine.isShared('trip-1')).toBe(true);
    expect(again.engine.nodeId).toBe(ana.engine.nodeId);
    const before = again.store.getTrip('trip-1');
    await again.engine.syncNow();
    expect(again.store.getTrip('trip-1')).toBe(before); // nothing new, nothing re-applied
    ben.store.edit('trip-1', (d) => void d.tasks.push({ id: 't-after', text: 'After restart', done: false }));
    await ben.engine.syncNow();
    await again.engine.syncNow();
    expect(again.store.getTrip('trip-1')!.tasks.map((t) => t.id)).toContain('t-after');
  });

  it('sends a change that was made while the app was closed', async () => {
    const { server, ana, ben } = await sharedTrip();
    await ana.engine.saveNow();
    const savedMeta = ana.storage.saved;
    // The change reaches the saved trips, but the app is killed before the sync engine hears about it.
```

```
    const edited = ana.store.trips.map((t) => ({ ...t, name: 'Renamed while closed' }));
    const again = phone(server, 'u-a', 9);
    again.store.trips = edited;
    again.storage.saved = savedMeta;
    await again.engine.init();
    await settle(again, ben);
    expect(ben.store.getTrip('trip-1')!.name).toBe('Renamed while closed');
  });

  it('stops syncing a trip that is deleted on this phone, and friends keep theirs', async () => {
    const { ana, ben, server } = await sharedTrip();
    ana.store.removeTrip('trip-1');
    await ana.engine.syncNow();
    expect(ana.engine.isShared('trip-1')).toBe(false);
    await ben.engine.syncNow();
    expect(ben.store.getTrip('trip-1')).toBeDefined();
    expect(server.stats().trips).toBe(1);
  });

  it('detaching leaves a normal local trip', async () => {
    const { ana, ben } = await sharedTrip();
    ana.engine.detach('trip-1');
    ana.store.edit('trip-1', (d) => void (d.name = 'Local only now'));
    await settle(ana, ben);
    expect(ben.store.getTrip('trip-1')!.name).toBe('Rome Crew');
    expect(ana.store.getTrip('trip-1')!.name).toBe('Local only now');
  });

  it('reports how many phones are in the trip', async () => {
    const { ana, ben } = await sharedTrip();
    await settle(ana, ben);
    expect(ana.engine.getStatus().peers).toBe(2);
    expect(ana.engine.getStatus().shared).toEqual(['trip-1']);
  });
});
```

src/sync/__tests__/invite.test.ts

```
import { buildInviteUrl, inviteText, parseInviteUrl } from '../invite';

describe('invite links', () => {
  it('survive the round trip for every kind of address the app can have', () => {
    for (const base of ['togetherwego://join', 'exp://192.168.1.20:8081/--/join', 'http://localhost:8081/join', 'https://example.org/ap
p/join']) {
      const link = buildInviteUrl(base, 'AML-2026', 'http://192.168.1.20:8787');
      expect(link.startsWith(base)).toBe(true);
      expect(parseInviteUrl(link)).toEqual({ code: 'AML-2026', server: 'http://192.168.1.20:8787' });
    }
  });

  it('leaves the server out when there is none', () => {
    const link = buildInviteUrl('togetherwego://join', 'SEO-5521');
    expect(link).toBe('togetherwego://join?code=SEO-5521');
    expect(parseInviteUrl(link)).toEqual({ code: 'SEO-5521', server: null });
  });

  it('adds to an address that already has a query', () => {
    expect(buildInviteUrl('https://x.org/join?lang=en', 'AML-2026')).toBe('https://x.org/join?lang=en&code=AML-2026');
    expect(parseInviteUrl('https://x.org/join?lang=en&code=AML-2026').code).toBe('AML-2026');
  });

  it('tidies codes typed in lower case or without the dash', () => {
    expect(parseInviteUrl('togetherwego://join?code=aml2026').code).toBe('AML-2026');
    expect(parseInviteUrl('togetherwego://join?code=aml-2026#section').code).toBe('AML-2026');
  });

  it('ignores anything that is not an invite', () => {
    expect(parseInviteUrl('')).toEqual({ code: null, server: null });
    expect(parseInviteUrl('https://example.org/')).toEqual({ code: null, server: null });
    expect(parseInviteUrl('togetherwego://join?code=hello')).toEqual({ code: null, server: null });
    expect(parseInviteUrl('togetherwego://join?code=AML-2026&server=not%20an%20address%20at%20all').server).toBeNull();
    expect(parseInviteUrl('togetherwego://join?code=%E0%A4%A').code).toBeNull(); // broken percent-encoding does not crash
  });

  it('writes a message that works even when the link does not open the app', () => {
    const text = inviteText('Amalfi Squad', 'AML-2026', 'togetherwego://join?code=AML-2026');
    expect(text).toContain('Amalfi Squad');
    expect(text).toContain('AML-2026');
    expect(text).toContain('togetherwego://join?code=AML-2026');
    expect(text).toContain('Join a trip');
  });
});
```

src/sync/__tests__/secureRandom.test.ts

```
import { newNodeId } from '../hlc';
import { secureRandom } from '../secureRandom';

describe('secureRandom', () => {
  it('gives numbers from 0 up to (not including) 1', () => {
    for (let i = 0; i < 1000; i++) {
      const n = secureRandom();
      expect(n).toBeGreaterThanOrEqual(0);
      expect(n).toBeLessThan(1);
    }
  });

  it('does not repeat itself, and makes distinct ids', () => {
    expect(new Set(Array.from({ length: 500 }, secureRandom)).size).toBe(500);
    const ids = new Set(Array.from({ length: 500 }, () => newNodeId(secureRandom)));
    expect(ids.size).toBe(500);
    for (const id of ids) expect(id).toMatch(/^[0-9a-z]{8}$/);
  });
});
```

src/sync/__tests__/transport.test.ts

```
import { NetworkError, ServerError, createHttpTransport, normalizeServerUrl } from '../transport';

/** A fetch that never answers on its own but stops when told to (like a wifi that drops packets). */
const hangingFetch: typeof fetch = (_url, init) =>
  new Promise((_resolve, reject) => {
    init?.signal?.addEventListener('abort', () => reject(Object.assign(new Error('aborted'), { name: 'AbortError' })));
  });

const jsonFetch = (status: number, body: unknown): typeof fetch => async () => ({ status, json: async () => body }) as Response;

// The phone's own URL class (React Native's) is a simple text matcher, much looser than Node's or a browser's.
// Run the address checks against both, so a typo is rejected on the phone too.
const PHONE_URL = jest.requireActual('react-native/Libraries/Blob/URL').URL as typeof URL;
describe.each([
  ['Node', URL],
  ['the phone', PHONE_URL],
])('normalizeServerUrl with the URL class of %s', (_name, UrlClass) => {
  const original = globalThis.URL;
  beforeAll(() => {
    globalThis.URL = UrlClass;
  });
  afterAll(() => {
    globalThis.URL = original;
  });

  it('adds http:// and drops paths and trailing slashes', () => {
    expect(normalizeServerUrl('192.168.1.20:8787')).toBe('http://192.168.1.20:8787');
    expect(normalizeServerUrl(' http://192.168.1.20:8787/ ')).toBe('http://192.168.1.20:8787');
    expect(normalizeServerUrl('https://sync.example.org/path?x=1')).toBe('https://sync.example.org');
    expect(normalizeServerUrl('HTTP://Laptop.local:8787')).toBe('http://laptop.local:8787');
    expect(normalizeServerUrl('localhost:8787')).toBe('http://localhost:8787');
  });

  it('rejects things that are not addresses', () => {
    for (const bad of ['', '   ', 'http://', 'not an address at all', 'my laptop', 'ftp://192.168.1.2', 'http://192.168.1.2 :8787', 'htt
p://-bad-.com', 'http://[::1]:8787', 'http://host:99999999']) {
      expect(normalizeServerUrl(bad)).toBeNull();
    }
  });
});

describe('normalizeServerUrl', () => {
  it('adds http:// and drops paths and trailing slashes', () => {
    expect(normalizeServerUrl('192.168.1.20:8787')).toBe('http://192.168.1.20:8787');
    expect(normalizeServerUrl(' http://192.168.1.20:8787/ ')).toBe('http://192.168.1.20:8787');
    expect(normalizeServerUrl('https://sync.example.org/path?x=1')).toBe('https://sync.example.org');
  });
  it('rejects empty and nonsense input', () => {
    expect(normalizeServerUrl('')).toBeNull();
    expect(normalizeServerUrl('   ')).toBeNull();
    expect(normalizeServerUrl('http://')).toBeNull();
  });
});

describe('http transport', () => {
  it('gives up on a server that never answers instead of waiting forever', async () => {
    const t = createHttpTransport('http://10.0.0.9:8787', hangingFetch, { timeoutMs: 40 });
    const started = Date.now();
    await expect(t.push('r1', 'AML-2026', 'phone001', [])).rejects.toThrow(NetworkError);
    await expect(t.pull('r1', 0, 'phone001', 0)).rejects.toThrow(/too long/);
    expect(Date.now() - started).toBeLessThan(1000);
```

```
  });

  it('looks up invite codes with a shorter limit than other requests', async () => {
    const t = createHttpTransport('http://10.0.0.9:8787', hangingFetch, { timeoutMs: 5000, lookupTimeoutMs: 40 });
    const started = Date.now();
    await expect(t.lookup('AML-2026')).rejects.toThrow(/too long/);
    expect(Date.now() - started).toBeLessThan(1000);
  });

  it('gives waiting requests extra time on top of the limit', async () => {
    const t = createHttpTransport('http://10.0.0.9:8787', hangingFetch, { timeoutMs: 20 });
    const started = Date.now();
    await expect(t.pull('r1', 0, 'me', 150)).rejects.toThrow(NetworkError);
    expect(Date.now() - started).toBeGreaterThanOrEqual(150);
  });

  it('lets the app cancel a request without calling it a network problem', async () => {
    const t = createHttpTransport('http://10.0.0.9:8787', hangingFetch, { timeoutMs: 5000 });
    const stop = new AbortController();
    const pending = t.pull('r1', 0, 'me', 1000, stop.signal);
    stop.abort();
    const error = await pending.catch((e: unknown) => e);
    expect(error).toBeInstanceOf(Error);
    expect(error).not.toBeInstanceOf(NetworkError);
  });

  it('turns a refused connection into a NetworkError', async () => {
    const refused: typeof fetch = async () => {
      throw new TypeError('Network request failed');
    };
    await expect(createHttpTransport('http://10.0.0.9:8787', refused).lookup('AML-2026')).rejects.toThrow(NetworkError);
  });

  it('treats 404 on a code lookup as "no such trip" and other errors as server errors', async () => {
    expect(await createHttpTransport('http://x:1', jsonFetch(404, { error: 'No trip with that code' })).lookup('ZZZ-0000')).toBeNull();
    expect(await createHttpTransport('http://x:1', jsonFetch(200, { tripId: 'rabc' })).lookup('AML-2026')).toBe('rabc');
    const err = await createHttpTransport('http://x:1', jsonFetch(429, { error: 'Too many tries. Wait a minute.' })).lookup('AML-2026').
catch((e: unknown) => e);
    expect(err).toBeInstanceOf(ServerError);
    expect((err as ServerError).status).toBe(429);
    expect((err as ServerError).message).toMatch(/Too many tries/);
  });
});
```

src/sync/actions.ts

```
/** The few things screens ask the sync engine to do: share a trip, join one, test the server. */
import { useAppStore } from '@/store/useAppStore';

import { useBlobCache } from './blobCache';
import { CodeTakenError } from './engine';
import { deleteSharedFiles } from './fileio';
import { SERVER_PORT, effectiveServerUrl } from './prefs';
import { syncReady } from './runtime';
import { NetworkError, ServerError, normalizeServerUrl } from './transport';

export type JoinOutcome = { ok: true; tripId: string; synced: boolean; offlineCopy: boolean } | { ok: false; error: string };

/**
 * Joins a trip by invite code. If a sync server is set, its shared trip is downloaded first, so everyone ends up
 * in the same trip. If the server is not reachable (or does not know the code) the code is tried on this phone,
 * which is how the demo trips and offline use keep working.
 */
export async function joinTripAnywhere(code: string): Promise<JoinOutcome> {
  let unreachable = false;
  if (effectiveServerUrl()) {
    try {
      const engine = await syncReady();
      if (await engine.join(code)) {
        const r = useAppStore.getState().joinTrip(code); // adds this person to the trip we just downloaded
        return r.ok ? { ok: true, tripId: r.tripId, synced: true, offlineCopy: false } : { ok: false, error: r.error };
      }
    } catch (e) {
      if (e instanceof ServerError && e.status === 429) return { ok: false, error: e.message };
      unreachable = e instanceof NetworkError;
    }
  }
  const r = useAppStore.getState().joinTrip(code);
  return r.ok ? { ok: true, tripId: r.tripId, synced: false, offlineCopy: unreachable } : { ok: false, error: r.error };
}

export type ShareOutcome = { ok: true } | { ok: false; reason: 'no-server' | 'code-taken' | 'network' | 'error'; message: string };

/** Starts sharing a trip with other phones. */
export async function shareTrip(tripId: string): Promise<ShareOutcome> {
  const url = effectiveServerUrl();
```

```
  if (!url) return { ok: false, reason: 'no-server', message: 'Set the sync server address first (Settings → Sync across phones).' };
  try {
    const engine = await syncReady();
    await engine.share(tripId);
    engine.start();
    return { ok: true };
  } catch (e) {
    if (e instanceof CodeTakenError) return { ok: false, reason: 'code-taken', message: e.message };
    if (e instanceof NetworkError) return { ok: false, reason: 'network', message: `Can’t reach the sync server at ${url}. Is it runnin
g, and are you on the same wifi?` };
    return { ok: false, reason: 'error', message: e instanceof Error ? e.message : 'Something went wrong' };
  }
}

/** Stops syncing a trip on this phone. It stays as a normal trip. */
export async function stopSharing(tripId: string): Promise<void> {
  (await syncReady()).detach(tripId);
}

/** Forgets all sync state (used when the app's data is reset). */
export async function resetSync(): Promise<void> {
  await (await syncReady()).reset();
  useBlobCache.getState().clear();
  deleteSharedFiles();
}

export type ConnectionTest = { ok: true; trips: number } | { ok: false; message: string };

/** Asks a server address "are you there?" with a short time limit. */
export async function testServer(input: string, fetchImpl: typeof fetch = fetch): Promise<ConnectionTest> {
  const url = normalizeServerUrl(input);
  if (!url) return { ok: false, message: `That doesn’t look like an address. Try something like 192.168.1.20:${SERVER_PORT}` };
  const controller = new AbortController();
  const timer = setTimeout(() => controller.abort(), 5000);
  try {
    const res = await fetchImpl(`${url}/health`, { signal: controller.signal });
    if (!res.ok) return { ok: false, message: `The address answered, but not like the sync server (error ${res.status}).` };
    const data = (await res.json()) as { ok?: boolean; trips?: number };
    return data.ok ? { ok: true, trips: data.trips ?? 0 } : { ok: false, message: 'The address answered, but it is not the TogetherWeGo
sync server.' };
  } catch {
    return { ok: false, message: `Couldn’t reach ${url}. Check that the server is running (npm run server) and that this phone is on the
same wifi.` };
  } finally {
    clearTimeout(timer);
  }
}
```

src/sync/adapter.ts

```
import { useAppStore } from '@/store/useAppStore';

import type { StoreAdapter } from './engine';

/** Lets the sync engine read and write trips in the app's store. */
export const storeAdapter: StoreAdapter = {
  getTrip: (id) => useAppStore.getState().trips.find((t) => t.id === id),
  getUserId: () => useAppStore.getState().session?.userId,
  setTrip: (trip) => {
    const { trips } = useAppStore.getState();
    useAppStore.setState({ trips: trips.map((t) => (t.id === trip.id ? trip : t)) });
  },
  addTrip: (trip) => {
    const { trips } = useAppStore.getState();
    useAppStore.setState({ trips: [trip, ...trips.filter((t) => t.id !== trip.id)], activeTripId: trip.id });
  },
  subscribe: (listener) => useAppStore.subscribe(() => listener()),
};
```

src/sync/blobCache.ts

```
import AsyncStorage from '@react-native-async-storage/async-storage';
import { Platform } from 'react-native';
import { create } from 'zustand';
import { createJSONStorage, persist } from 'zustand/middleware';

import type { BlobCache } from './blobs';

interface BlobCacheState {
  /** Server file id -> where this phone keeps its copy. */
  files: Record<string, string>;
  /** Items that cannot be shared, so the app does not complain about them every time it opens. */
  rejected: Record<string, true>;
  setFile: (blobId: string, uri: string) => void;
  reject: (key: string) => void;
  clear: () => void;
```

```
}

/**
 * What this phone has downloaded. Saved between runs, except in the web preview, where a copy is a temporary
 * address that stops working when the page closes (so it is fetched again next time).
 */
export const useBlobCache = create<BlobCacheState>()(
  persist(
    (set) => ({
      files: {},
      rejected: {},
      setFile: (blobId, uri) => set((s) => ({ files: { ...s.files, [blobId]: uri } })),
      reject: (key) => set((s) => ({ rejected: { ...s.rejected, [key]: true } })),
      clear: () => set({ files: {}, rejected: {} }),
    }),
    {
      name: 'togetherwego-files',
      storage: createJSONStorage(() => AsyncStorage),
      partialize: (s) => ({ files: Platform.OS === 'web' ? {} : s.files, rejected: s.rejected }),
    },
  ),
);

/** The same cache, in the shape the file syncing code asks for. */
export const blobCache: BlobCache = {
  get: (blobId) => useBlobCache.getState().files[blobId],
  set: (blobId, uri) => useBlobCache.getState().setFile(blobId, uri),
  isRejected: (key) => !!useBlobCache.getState().rejected[key],
  reject: (key) => useBlobCache.getState().reject(key),
};

/** How many files are still waiting to be sent or fetched (for a small "syncing files" hint). */
export const useFileStatus = create<{ pending: number }>(() => ({ pending: 0 }));
```

src/sync/blobs.ts

```
/**
 * Files (photos, documents, receipts) travel through the sync server on their own, next to the trip's text changes.
 *
 *  - Your own new file is uploaded, then the server's id for it (`blob`) is written on the item. That small change
 *    syncs like any other, so friends' phones hear that a file exists.
 *  - A friend's item that has a `blob` but no file on this phone is downloaded and remembered in a cache, so it also
 *    opens with no connection later.
 *
 * A pass is safe to run at any time. Failures are retried later with a growing pause and never block the screens.
 */
import type { FileList } from '@/store/slices/files';
import type { Trip } from '@/store/types';

import { quiet } from './timers';

/** Matches the server's limit (server/index.js). Bigger files stay on the phone that has them. */
export const MAX_SHARED_FILE_BYTES = 8 * 1024 * 1024;

export type TransferFailure = 'network' | 'rejected' | 'missing';

/** What went wrong with an upload or download: no connection, the server said no, or it does not have the file. */
export class FileTransferError extends Error {
  constructor(
    readonly kind: TransferFailure,
    message: string,
  ) {
    super(message);
  }
}

/** Where files live on this phone and how they reach the server. The app has a native and a web version; tests use a fake. */
export interface BlobFiles {
  exists(uri: string): Promise<boolean>;
  /** Size in bytes, or null when there is no such file. */
  size(uri: string): Promise<number | null>;
  /** Sends the file to `url` (HTTP PUT with the raw bytes). Throws FileTransferError. */
  upload(uri: string, url: string): Promise<void>;
  /** Fetches `url` and keeps it as a file called `name`; returns where it is now. Throws FileTransferError. */
  download(url: string, name: string): Promise<string>;
}

/** This phone's memory of which downloaded file belongs to which blob. */
export interface BlobCache {
  get(blobId: string): string | undefined;
  set(blobId: string, uri: string): void;
  /** Items that cannot be shared (too big, file gone): remembered so the app does not complain every time it opens. */
  isRejected(key: string): boolean;
  reject(key: string): void;
}
```

```
/** For each list that holds files: the field with this phone's file path and the field with the server's id. */
export const FILE_FIELDS: Record<FileList, { local: 'uri' | 'receipt'; blob: 'blob' | 'receiptBlob' }> = {
  photos: { local: 'uri', blob: 'blob' },
  docs: { local: 'uri', blob: 'blob' },
  expenses: { local: 'receipt', blob: 'receiptBlob' },
};

const LISTS = Object.keys(FILE_FIELDS) as FileList[];

/** A real file on this phone, as opposed to the names of the pictures that come with the app ("amalfi"). */
export const isLocalFile = (uri: string | undefined): uri is string => !!uri && /^(file|content|blob|data):/i.test(uri);

/**
 * The pictures that come with the app are not files to send: every phone has them already. A photo that uses one is
 * shared by name, as `asset:amalfi`, in the same field that holds a server file id for real files.
 */
const ASSET_PREFIX = 'asset:';
export const isAssetRef = (blob: string | undefined): blob is string => !!blob && blob.startsWith(ASSET_PREFIX);
export const assetRef = (key: string) => `${ASSET_PREFIX}${key}`;

/**
 * What to show for a photo, document or receipt: this phone's own file, a bundled picture shared by name, the copy
 * downloaded from a friend, or (while that is on its way) nothing yet.
 */
export function resolveFile(local: string | undefined, blob: string | undefined, downloaded: string | undefined): { uri: string | undefi
ned; loading: boolean } {
  if (local) return { uri: local, loading: false };
  if (isAssetRef(blob)) return { uri: blob.slice(ASSET_PREFIX.length), loading: false };
  if (downloaded) return { uri: downloaded, loading: false };
  return { uri: undefined, loading: !!blob };
}

const MIME_EXT: Record<string, string> = { jpeg: 'jpg', 'svg+xml': 'svg', 'x-icon': 'ico' };

/** "photo.JPG?x=1" -> "jpg". A data: address names its type instead of a file. */
export function extensionOf(uri: string, hint = 'bin'): string {
  const data = /^data:[\w.+-]+\/([\w.+-]+)/i.exec(uri);
  const raw = (data ? data[1] : (/\.([A-Za-z0-9]{1,5})$/.exec(uri.split(/[?#]/)[0])?.[1] ?? hint)).toLowerCase();
  return MIME_EXT[raw] ?? (raw.replace(/[^a-z0-9]/g, '').slice(0, 5) || hint);
}

/** The server's id for a file. It ends in `_ext` so the downloaded copy gets a name that apps can open. */
export const makeBlobId = (random: string, uri: string, hint?: string) => `${random}_${extensionOf(uri, hint)}`;

/** "k3j9x2m1_pdf" -> "k3j9x2m1.pdf" */
export const fileNameOf = (blobId: string) => blobId.replace(/_([a-z0-9]{1,5})$/, '.$1');

const hintFor = (list: FileList, item: { kind?: string }) => (list === 'docs' ? (item.kind === 'pdf' ? 'pdf' : item.kind === 'image' ?
'jpg' : 'bin') : 'jpg');

interface Job {
  kind: 'upload' | 'download' | 'reference';
  tripId: string;
  remoteId: string;
  list: FileList;
  itemId: string;
  /** For an upload: this phone's file. For a reference: the bundled picture's name. */
  uri?: string;
  /** For a download: the server's id. */
  blobId?: string;
  hint: string;
}

export interface BlobSyncOptions {
  /** The trips in the app. */
  trips: () => Trip[];
  /** The id the server uses for a trip, or undefined when the trip is not shared. */
  remoteId: (tripId: string) => string | undefined;
  /** The sync server's address, or null when none is set. */
  baseUrl: () => string | null;
  files: BlobFiles;
  cache: BlobCache;
  /** Writes the server's id onto the item (so it syncs like any other change). */
  attach: (tripId: string, list: FileList, itemId: string, blobId: string) => void;
  newId: () => string;
  /** Something the person should hear about (a file that is too big to share). */
  onNotice?: (message: string) => void;
  /** How many files are still waiting (for a "syncing…" hint). */
  onPending?: (count: number) => void;
}

export class BlobSync {
  private timer: ReturnType<typeof setTimeout> | null = null;
  private running = false;
  private again = false;
  private pause = 0;
```

```
  private stopped = false;

  constructor(private readonly o: BlobSyncOptions) {}

  /** Look for files to send or fetch soon. Cheap to call after every change; calls close together become one pass. */
  schedule(delayMs = 500): void {
    if (this.stopped || this.timer) return;
    this.timer = quiet(
      setTimeout(() => {
        this.timer = null;
        void this.run();
      }, delayMs),
    );
  }

  stop(): void {
    this.stopped = true;
    if (this.timer) clearTimeout(this.timer);
    this.timer = null;
  }

  /** Does one full pass (or asks the running one to go round again). Resolves when it is done. */
  async run(): Promise<void> {
    if (this.stopped) return;
    if (this.running) {
      this.again = true;
      return;
    }
    this.running = true;
    try {
      let failed: boolean;
      do {
        this.again = false;
        failed = await this.pass();
      } while (this.again && !this.stopped);
      this.pause = failed ? Math.min(this.pause ? this.pause * 2 : 3000, 60000) : 0;
      if (failed) this.schedule(this.pause);
    } finally {
      this.running = false;
    }
  }

  private collect(): Job[] {
    const jobs: Job[] = [];
    for (const trip of this.o.trips()) {
      const remoteId = this.o.remoteId(trip.id);
      if (!remoteId) continue;
      for (const list of LISTS) {
        const { local, blob } = FILE_FIELDS[list];
        for (const item of trip[list] as unknown as Record<string, string | undefined>[]) {
          const uri = item[local];
          const blobId = item[blob];
          const base = { tripId: trip.id, remoteId, list, itemId: item.id as string, hint: hintFor(list, item as { kind?: string }) };
          if (isLocalFile(uri) && !blobId) jobs.push({ ...base, kind: 'upload', uri });
          else if (list === 'photos' && uri && !isLocalFile(uri) && !blobId) jobs.push({ ...base, kind: 'reference', uri });
          else if (blobId && !isLocalFile(uri) && !isAssetRef(blobId)) jobs.push({ ...base, kind: 'download', blobId });
        }
      }
    }
    return jobs;
  }

  /** Returns true when something should be tried again later. */
  private async pass(): Promise<boolean> {
    const base = this.o.baseUrl();
    if (!base) {
      this.o.onPending?.(0);
      return false;
    }
    const jobs = this.collect();
    for (const job of jobs.filter((j) => j.kind === 'reference')) this.o.attach(job.tripId, job.list, job.itemId, assetRef(job.uri as st
ring)); // nothing to send: just say which picture it is
    jobs.splice(0, jobs.length, ...jobs.filter((j) => j.kind !== 'reference'));
    this.o.onPending?.(jobs.length);
    let retry = false;
    for (const job of jobs) {
      if (this.stopped) return false;
      try {
        if (job.kind === 'upload') await this.upload(base, job);
        else await this.download(base, job);
      } catch (e) {
        retry = true;
        if (e instanceof FileTransferError && e.kind === 'network') break; // the server is out of reach: no point trying the rest now
      }
    }
    if (!retry) this.o.onPending?.(0);
```

```
    return retry;
  }

  private async upload(base: string, job: Job): Promise<void> {
    const key = `${job.tripId}/${job.list}/${job.itemId}`;
    if (this.o.cache.isRejected(key)) return;
    const uri = job.uri as string;
    const size = await this.o.files.size(uri);
    if (size === null) return this.o.cache.reject(key); // the file is gone from this phone: nothing to send
    if (size > MAX_SHARED_FILE_BYTES) return this.refuse(key, size);
    const blobId = makeBlobId(this.o.newId(), uri, job.hint);
    try {
      await this.o.files.upload(uri, `${base}/v1/trips/${job.remoteId}/files/${blobId}`);
    } catch (e) {
      if (e instanceof FileTransferError && e.kind === 'rejected') return this.refuse(key, size);
      throw e; // no connection, or the server has not heard of this trip yet: try again later
    }
    this.o.attach(job.tripId, job.list, job.itemId, blobId);
  }

  private refuse(key: string, size: number): void {
    this.o.cache.reject(key);
    this.o.onNotice?.(`That file is ${(size / 1024 / 1024).toFixed(1)} MB. Files over ${MAX_SHARED_FILE_BYTES / 1024 / 1024} MB stay on
your phone and are not shared with friends.`);
  }

  private async download(base: string, job: Job): Promise<void> {
    const blobId = job.blobId as string;
    const have = this.o.cache.get(blobId);
    if (have && (await this.o.files.exists(have))) return;
    const uri = await this.o.files.download(`${base}/v1/trips/${job.remoteId}/files/${blobId}`, fileNameOf(blobId));
    this.o.cache.set(blobId, uri);
  }
}
```

src/sync/crdt.ts

```
/**
 * The merge engine behind phone-to-phone sync.
 *
 *   diffTrip      : "what changed?"  Compares two versions of a trip and returns the small changes (ops) between them.
 *   applyOps      : "merge these"    Folds ops from any phone into the registers. Order and repeats do not matter.
 *   materializeTrip: "show me the trip" Rebuilds a Trip object from the registers.
 *
 * Rules that make every phone end up identical:
 *  1. Every op carries a unique, sortable stamp (see hlc.ts). For one register the op with the biggest stamp wins.
 *  2. Deleting an item is remembered (a tombstone). Changes older than the tombstone are ignored, so a delete beats a
 *     concurrent edit. Creating the item again later (a newer creation stamp) brings it back.
 *  3. Applying the same op twice, or in a different order, gives the same result.
 */
import type { Trip } from '@/store/types';

import { compareStamps } from './hlc';
import { CREATED, PATH_SEP, TRIP_SCHEMA, pathKey, type ItemSchema, type ListSchema } from './schema';

export type Json = string | number | boolean | null | Json[] | { [k: string]: Json };

/** One change. `p` is the path of the register, `v` the new value, `d` marks a delete, `t` is the stamp. */
export interface Op {
  p: string[];
  v?: Json;
  d?: 1;
  t: string;
}

/** What we remember about one register: the stamp of the change that wrote it, and its value (or that it was deleted). */
export interface Reg {
  t: string;
  v?: Json;
  d?: 1;
}
export type Registers = Record<string, Reg>;


// ---------------------------------------------------------------- diff

/** The ops that turn `prev` into `next`. Pass `undefined` as `prev` to describe a whole trip from scratch. */
export function diffTrip(prev: Trip | undefined, next: Trip, stamp: () => string): Op[] {
  const out: Op[] = [];
  diffItem([], prev, next, TRIP_SCHEMA, out, stamp);
  return out;
}

const setOp = (p: string[], v: Json, t: string): Op => ({ p, v, t });
const delOp = (p: string[], t: string): Op => ({ p, d: 1, t });
```

```
function diffItem(path: string[], prev: any, next: any, S: ItemSchema, out: Op[], stamp: () => string): void {
  for (const f of S.scalars) {
    const a = prev?.[f];
    const b = next[f];
    if (a !== b) out.push(b === undefined ? delOp([...path, f], stamp()) : setOp([...path, f], b, stamp()));
  }
  for (const f of S.json ?? []) {
    const a = prev?.[f];
    const b = next[f];
    if (a === b || (a !== undefined && b !== undefined && deepEqual(a, b))) continue;
    out.push(b === undefined ? delOp([...path, f], stamp()) : setOp([...path, f], b, stamp()));
  }
  for (const f of S.sets ?? []) {
    if (prev?.[f] === next[f]) continue;
    const before = new Set<string>(prev?.[f] ?? []);
    const after = new Set<string>(next[f] ?? []);
    for (const x of after) if (!before.has(x)) out.push(setOp([...path, f, x], 1, stamp()));
    for (const x of before) if (!after.has(x)) out.push(delOp([...path, f, x], stamp()));
  }
  for (const f of S.maps ?? []) {
    if (prev?.[f] === next[f]) continue;
    const a: Record<string, Json> = prev?.[f] ?? {};
    const b: Record<string, Json> = next[f] ?? {};
    for (const k of Object.keys(b)) if (a[k] !== b[k]) out.push(setOp([...path, f, k], b[k], stamp()));
    for (const k of Object.keys(a)) if (!(k in b)) out.push(delOp([...path, f, k], stamp()));
  }
  for (const [name, L] of Object.entries(S.lists ?? {})) diffList([...path, name], prev?.[name], next[name], L, out, stamp);
}

function diffList(path: string[], prevArr: any[] | undefined, nextArr: any[] | undefined, L: ListSchema, out: Op[], stamp: () => strin
g): void {
  if (prevArr === nextArr) return;
  const shared = (arr: any[] | undefined) => (arr ?? []).filter((x) => !L.share || L.share(x));
  const prevItems = new Map<string, any>(shared(prevArr).map((x) => [x.id, x]));
  const nextItems = shared(nextArr);
  const nextIds = new Set<string>(nextItems.map((x) => x.id));
  for (const id of prevItems.keys()) if (!nextIds.has(id)) out.push(delOp([...path, id], stamp()));
  // Oldest first, so creation stamps rise in the same order the items were made.
  const oldestFirst = L.order === 'asc' ? nextItems : [...nextItems].reverse();
  for (const item of oldestFirst) {
    const old = prevItems.get(item.id);
    if (!old) {
      out.push(setOp([...path, item.id, CREATED], 1, stamp()));
      diffItem([...path, item.id], undefined, item, L.item, out, stamp);
    } else if (old !== item) {
      diffItem([...path, item.id], old, item, L.item, out, stamp);
    }
  }
}

// ---------------------------------------------------------------- merge

/** Where a path points: the chain of items it sits inside, and whether it is an item itself. */
interface Resolved {
  items: string[][];
  isItem: boolean;
}

/** Checks a path against the schema. Unknown paths (from a newer version of the app) return null and are ignored. */
export function resolve(path: readonly string[], schema: ItemSchema = TRIP_SCHEMA, items: string[][] = [], at = 0): Resolved | null {
  const seg = path[at];
  if (seg === undefined) return null;
  const last = at === path.length - 1;
  if (schema.scalars.includes(seg) || schema.json?.includes(seg)) return last ? { items, isItem: false } : null;
  if (seg === CREATED && items.length > 0) return last ? { items, isItem: false } : null;
  if (schema.sets?.includes(seg) || schema.maps?.includes(seg)) return at === path.length - 2 ? { items, isItem: false } : null;
  const list = schema.lists?.[seg];
  if (list) {
    if (path.length < at + 2) return null;
    const itemPath = path.slice(0, at + 2) as string[];
    const chain = [...items, itemPath];
    if (path.length === at + 2) return { items: chain, isItem: true };
    return resolve(path, list.item, chain, at + 2);
  }
  return null;
}

/** Folds ops into the registers. Returns true if anything actually changed. */
export function applyOps(regs: Registers, ops: readonly Op[]): boolean {
  let changed = false;
  for (const op of ops) if (applyOp(regs, op)) changed = true;
  return changed;
}

function applyOp(regs: Registers, op: Op): boolean {
  const info = resolve(op.p);
```

```
  if (!info) return false;
  if (info.isItem && !op.d) return false; // an item node can only be deleted; it is created by its creation stamp
  // A change older than a delete of the item it belongs to is void.
  const owners = info.isItem ? info.items.slice(0, -1) : info.items;
  for (const itemPath of owners) {
    const tomb = regs[pathKey(itemPath)];
    if (tomb?.d && compareStamps(op.t, tomb.t) <= 0) return false;
  }
  const key = pathKey(op.p);
  const cur = regs[key];
  if (cur && compareStamps(op.t, cur.t) <= 0) return false;
  regs[key] = op.d ? { t: op.t, d: 1 } : { t: op.t, v: op.v };
  // Everything older that was inside a deleted item is void; dropping it keeps every phone's registers identical.
  if (info.isItem) forgetOlderBelow(regs, key, op.t);
  return true;
}

function forgetOlderBelow(regs: Registers, itemKey: string, t: string): void {
  const prefix = itemKey + PATH_SEP;
  for (const key of Object.keys(regs)) if (key.startsWith(prefix) && compareStamps(regs[key].t, t) < 0) delete regs[key];
}

// ---------------------------------------------------------------- rebuild

interface Node {
  reg?: Reg;
  kids: Map<string, Node>;
}

function buildTree(regs: Registers): Node {
  const root: Node = { kids: new Map() };
  for (const [key, reg] of Object.entries(regs)) {
    let n = root;
    for (const seg of key.split(PATH_SEP)) {
      let c = n.kids.get(seg);
      if (!c) {
        c = { kids: new Map() };
        n.kids.set(seg, c);
      }
      n = c;
    }
    n.reg = reg;
  }
  return root;
}

/** Rebuilds the shared part of a trip from the registers. Returns null if the trip is not complete yet. */
export function materializeTrip(regs: Registers): Trip | null {
  return readItem(TRIP_SCHEMA, buildTree(regs), '', undefined) as Trip | null;
}

/** `floor` is the stamp of the newest delete above this item: anything written before it is ignored. */
function readItem(S: ItemSchema, node: Node, floor: string, id: string | undefined): any | null {
  const alive = (r?: Reg): r is Reg => !!r && !r.d && compareStamps(r.t, floor) > 0;
  const obj: any = id === undefined ? {} : { id };
  for (const f of [...S.scalars, ...(S.json ?? [])]) {
    const r = node.kids.get(f)?.reg;
    if (alive(r)) obj[f] = r.v;
  }
  for (const f of S.sets ?? []) {
    const kids = [...(node.kids.get(f)?.kids ?? [])].filter(([, k]) => alive(k.reg));
    kids.sort((a, b) => compareStamps(a[1].reg!.t, b[1].reg!.t) || (a[0] < b[0] ? -1 : 1));
    obj[f] = kids.map(([k]) => k);
  }
  for (const f of S.maps ?? []) {
    obj[f] = Object.fromEntries([...(node.kids.get(f)?.kids ?? [])].filter(([, k]) => alive(k.reg)).map(([k, v]) => [k, v.reg!.v]));
  }
  for (const [name, L] of Object.entries(S.lists ?? {})) obj[name] = readList(L, node.kids.get(name), floor);
  for (const r of S.required ?? []) if (obj[r] === undefined) return null;
  return obj;
}

function readList(L: ListSchema, node: Node | undefined, floor: string): any[] {
  const found: { order: string; id: string; obj: any }[] = [];
  for (const [id, child] of node?.kids ?? []) {
    const created = child.kids.get(CREATED)?.reg;
    const tomb = child.reg?.d ? child.reg.t : '';
    const itemFloor = compareStamps(tomb, floor) > 0 ? tomb : floor;
    if (!created || created.d || compareStamps(created.t, itemFloor) <= 0) continue;
    const obj = readItem(L.item, child, itemFloor, id);
    if (obj) found.push({ order: created.t, id, obj });
  }
  found.sort((a, b) => compareStamps(a.order, b.order) || (a.id < b.id ? -1 : 1));
  const items = found.map((f) => f.obj);
  return L.order === 'desc' ? items.reverse() : items;
}
```

```
// ---------------------------------------------------------------- helpers

export function deepEqual(a: any, b: any): boolean {
  if (a === b) return true;
  if (typeof a !== 'object' || typeof b !== 'object' || a === null || b === null) return false;
  if (Array.isArray(a) !== Array.isArray(b)) return false;
  const ka = Object.keys(a);
  const kb = Object.keys(b);
  if (ka.length !== kb.length) return false;
  return ka.every((k) => k in b && deepEqual(a[k], b[k]));
}
```

src/sync/engine.ts

```
/**
 * The sync engine: keeps a shared trip identical on every phone that has joined it.
 *
 *   phone edits the store ──► diff ──► ops ──► registers + outbox ──► server
 *   server ──► ops from friends ──► registers ──► rebuilt trip ──► store
 *
 * It knows nothing about React or the network library: the app store, the server connection and the place the
 * state is saved are all passed in, which is what lets the tests run two phones against one server in memory.
 */
import type { Trip } from '@/store/types';

import { applyOps, diffTrip, materializeTrip, type Op, type Registers } from './crdt';
import { HybridClock, newNodeId, type ClockState } from './hlc';
import { NetworkError, ServerError, type SyncTransport } from './transport';
import { quiet } from './timers';
import { toViewTrip } from './view';

/** What we remember about one shared trip. */
export interface TripSync {
  /** The trip's id on the server (long and random, so it works like a password). */
  remoteId: string;
  /** Everything we know about every shared field, with the stamp of its last change. */
  regs: Registers;
  /** How many of the server's changes we have already read. */
  cursor: number;
  /** Our own changes the server has not confirmed yet, in batches. */
  outbox: Op[][];
}

export interface SyncMeta {
  nodeId: string;
  clock?: ClockState;
  trips: Record<string, TripSync>;
}

/** How the engine reads and writes the app's trips. The real one wraps the zustand store. */
export interface StoreAdapter {
  getTrip(tripId: string): Trip | undefined;
  getUserId(): string | undefined;
  /** Replaces a trip that is already in the store. */
  setTrip(trip: Trip): void;
  /** Puts a trip that came from a friend into the store (replacing one with the same id) and makes it the active trip. */
  addTrip(trip: Trip): void;
  /** Calls the listener after any change to the trips. Returns a function that stops listening. */
  subscribe(listener: () => void): () => void;
}

export interface MetaStorage {
  load(): Promise<SyncMeta | null>;
  save(meta: SyncMeta): Promise<void>;
}

/** What the screens show about syncing. */
export interface SyncStatus {
  /** null until the first try, then whether the last conversation with the server worked. */
  online: boolean | null;
  /** Changes made here that the server has not confirmed. */
  pending: number;
  /** Phones talking to the server about our shared trips (including this one). */
  peers: number;
  lastSyncAt: number | null;
  error: string | null;
  /** Ids (in the app) of the trips being shared. */
  shared: string[];
}

export interface EngineOptions {
  store: StoreAdapter;
  storage: MetaStorage;
  /** The connection to the server, or null while no server address is set. Asked again every time it is needed. */
  getTransport: () => SyncTransport | null;
```

```
  /** Called after changes from friends were put into the store. */
  onRemoteChange?: (previous: Trip | undefined, next: Trip) => void;
  onStatus?: (status: SyncStatus) => void;
  now?: () => number;
  random?: () => number;
  /** How long to hold a request open waiting for news (ms). Tests use 0. */
  pollWaitMs?: number;
}

export class CodeTakenError extends Error {
  constructor(readonly code: string) {
    super(`The invite code ${code} is already used by another shared trip.`);
  }
}

interface Attached {
  sync: TripSync;
  /** The trip as it was when we last compared, so the next comparison finds only what changed since. */
  shadow: Trip | undefined;
  pushing: boolean;
  peers: number;
  retry: ReturnType<typeof setTimeout> | null;
  abort: AbortController | null;
  running: boolean;
  /** Ends a backoff wait early (used when stopping). */
  wake: (() => void) | null;
}

const BATCH = 400;
const chunk = <T>(items: T[], size: number): T[][] => Array.from({ length: Math.ceil(items.length / size) }, (_, i) => items.slice(i * s
ize, (i + 1) * size));

export class SyncEngine {
  private meta: SyncMeta = { nodeId: '', trips: {} };
  private clock!: HybridClock;
  private attached = new Map<string, Attached>();
  private unsubscribe: (() => void) | null = null;
  private applying = false;
  private running = false;
  private saveTimer: ReturnType<typeof setTimeout> | null = null;
  private status: SyncStatus = { online: null, pending: 0, peers: 0, lastSyncAt: null, error: null, shared: [] };

  constructor(private readonly o: EngineOptions) {}

  // ------------------------------------------------------------------ setup

  /** Loads what was saved, starts watching the store, and sends any change that was made while the app was closed. */
  async init(): Promise<void> {
    const saved = await this.o.storage.load();
    this.meta = saved ?? { nodeId: newNodeId(this.o.random), trips: {} };
    if (!this.meta.nodeId) this.meta.nodeId = newNodeId(this.o.random);
    this.clock = new HybridClock(this.meta.nodeId, this.o.now);
    this.clock.restore(this.meta.clock);
    for (const [tripId, sync] of Object.entries(this.meta.trips)) {
      this.attach(tripId, sync, materializeTrip(sync.regs) ? toViewTrip(materializeTrip(sync.regs)!, undefined) : undefined);
    }
    this.unsubscribe?.();
    this.unsubscribe = this.o.store.subscribe(() => this.onStoreChange());
    this.onStoreChange(); // catches changes made while the app was closed (they differ from the saved registers)
    this.publishStatus();
  }

  get nodeId(): string {
    return this.meta.nodeId;
  }

  isShared(tripId: string): boolean {
    return this.attached.has(tripId);
  }

  /** The id the server uses for a shared trip (undefined when the trip is not shared). */
  remoteId(tripId: string): string | undefined {
    return this.attached.get(tripId)?.sync.remoteId;
  }

  getStatus(): SyncStatus {
    return this.status;
  }

  /** Starts talking to the server: sending changes and listening for friends' changes. */
  start(): void {
    this.running = true;
    for (const [tripId, a] of this.attached) {
      this.schedulePush(tripId, 0);
      this.startPullLoop(tripId, a);
    }
```

```
  }

  stop(): void {
    this.running = false;
    for (const a of this.attached.values()) {
      a.abort?.abort();
      a.wake?.();
      if (a.retry) clearTimeout(a.retry);
      a.retry = null;
      a.running = false;
    }
    if (this.saveTimer) void this.saveNow(); // do not lose recent changes when the app goes to the background
  }

  /** Stops everything and forgets all sync state (used by "reset app data"). */
  async reset(): Promise<void> {
    this.stop();
    this.unsubscribe?.();
    this.unsubscribe = null;
    this.attached.clear();
    this.meta = { nodeId: newNodeId(this.o.random), trips: {} };
    await this.o.storage.save(this.meta);
    this.publishStatus();
  }

  // ---------------------------------------------------------------- sharing and joining

  /** Starts sharing a trip: describes the whole trip as changes and sends them. */
  async share(tripId: string): Promise<void> {
    if (this.attached.has(tripId)) return;
    const trip = this.o.store.getTrip(tripId);
    if (!trip) throw new Error('That trip no longer exists.');
    const transport = this.o.getTransport();
    if (!transport) throw new Error('Set the sync server address first.');
    const existing = await transport.lookup(trip.inviteCode);
    if (existing) throw new CodeTakenError(trip.inviteCode);
    const sync: TripSync = { remoteId: `r${newNodeId(this.o.random)}${newNodeId(this.o.random)}${newNodeId(this.o.random)}`, regs: {}, c
ursor: 0, outbox: [] };
    const ops = diffTrip(undefined, trip, () => this.clock.next());
    applyOps(sync.regs, ops);
    sync.outbox = chunk(ops, BATCH);
    this.meta.trips[tripId] = sync;
    this.attach(tripId, sync, trip);
    this.scheduleSave();
    this.publishStatus();
    if (this.running) {
      this.schedulePush(tripId, 0);
      this.startPullLoop(tripId, this.attached.get(tripId)!);
    }
  }

  /**
   * Joins a friend's shared trip using its invite code. Downloads the trip and puts it in the store.
   * Returns null when no trip has that code. Throws NetworkError when the server cannot be reached.
   */
  async join(code: string): Promise<Trip | null> {
    const transport = this.o.getTransport();
    if (!transport) throw new Error('Set the sync server address first.');
    const remoteId = await transport.lookup(code);
    if (!remoteId) return null;
    const already = [...this.attached.entries()].find(([, a]) => a.sync.remoteId === remoteId);
    if (already) return this.o.store.getTrip(already[0]) ?? null;
    const sync: TripSync = { remoteId, regs: {}, cursor: 0, outbox: [] };
    for (;;) {
      const page = await transport.pull(remoteId, sync.cursor, this.meta.nodeId, 0);
      page.ops.forEach((op) => this.clock.observe(op.t));
      applyOps(sync.regs, page.ops);
      sync.cursor = page.seq;
      if (sync.cursor >= page.total) break;
    }
    const shared = materializeTrip(sync.regs);
    if (!shared) throw new Error('That trip is not ready yet. Ask your friend to open the app and try again.');
    const trip = toViewTrip(shared, undefined);
    this.meta.trips[trip.id] = sync;
    this.attach(trip.id, sync, trip);
    this.applying = true;
    try {
      this.o.store.addTrip(trip);
    } finally {
      this.applying = false;
    }
    this.scheduleSave();
    this.publishStatus();
    if (this.running) {
      this.startPullLoop(trip.id, this.attached.get(trip.id)!);
    }
```

```
  return trip;
}

/** Stops syncing a trip on this phone. The trip stays here as a normal local trip. */
detach(tripId: string): void {
  const a = this.attached.get(tripId);
  if (!a) return;
  a.abort?.abort();
  a.wake?.();
  if (a.retry) clearTimeout(a.retry);
  a.running = false;
  this.attached.delete(tripId);
  delete this.meta.trips[tripId];
  this.scheduleSave();
  this.publishStatus();
}

// ---------------------------------------------------------------- local changes

private attach(tripId: string, sync: TripSync, shadow: Trip | undefined): void {
  this.attached.set(tripId, { sync, shadow, pushing: false, peers: 0, retry: null, abort: null, running: false, wake: null });
}

private onStoreChange(): void {
  if (this.applying) {
    // our own write of friends' changes: remember it as the new baseline instead of sending it back
    for (const [tripId, a] of this.attached) a.shadow = this.o.store.getTrip(tripId) ?? a.shadow;
    return;
  }
  let queued = false;
  for (const [tripId, a] of [...this.attached]) {
    const current = this.o.store.getTrip(tripId);
    if (!current) {
      this.detach(tripId); // the trip was deleted on this phone: stop syncing it, friends keep theirs
      continue;
    }
    if (current === a.shadow) continue;
    const ops = diffTrip(a.shadow, current, () => this.clock.next());
    a.shadow = current;
    if (!ops.length) continue;
    applyOps(a.sync.regs, ops);
    a.sync.outbox.push(...chunk(ops, BATCH));
    queued = true;
    this.schedulePush(tripId, 30);
  }
  if (queued) {
    this.scheduleSave();
    this.publishStatus();
  }
}

private schedulePush(tripId: string, delayMs: number): void {
  const a = this.attached.get(tripId);
  if (!a || !this.running || a.retry) return;
  a.retry = quiet(
    setTimeout(() => {
      a.retry = null;
      void this.push(tripId);
    }, delayMs),
  );
}

/** Sends our unconfirmed changes. Safe to call at any time; only one send per trip runs at once. */
async push(tripId: string): Promise<void> {
  const a = this.attached.get(tripId);
  const transport = this.o.getTransport();
  if (!a || a.pushing || !a.sync.outbox.length || !transport) return;
  a.pushing = true;
  try {
    while (a.sync.outbox.length) {
      const code = this.o.store.getTrip(tripId)?.inviteCode;
      await transport.push(a.sync.remoteId, code, this.meta.nodeId, a.sync.outbox[0]);
      a.sync.outbox.shift();
      this.scheduleSave();
    }
    this.setOnline(true, null);
  } catch (e) {
    this.setOnline(false, describe(e));
    if (this.running && !(e instanceof ServerError && e.status === 400)) this.schedulePush(tripId, 3000);
  } finally {
    a.pushing = false;
    this.publishStatus();
  }
}

// ---------------------------------------------------------------- changes from friends
```

```
/** Asks the server for new changes once and applies them. `waitMs` > 0 holds the request open for news. */
async pull(tripId: string, waitMs = 0, signal?: AbortSignal): Promise<number> {
  const a = this.attached.get(tripId);
  const transport = this.o.getTransport();
  if (!a || !transport) return 0;
  const page = await transport.pull(a.sync.remoteId, a.sync.cursor, this.meta.nodeId, waitMs, signal);
  if (!this.attached.has(tripId)) return 0;
  a.peers = page.peers;
  a.sync.cursor = page.seq;
  this.applyRemote(tripId, page.ops);
  this.scheduleSave();
  this.setOnline(true, null);
  this.publishStatus();
  return page.ops.length;
}

private applyRemote(tripId: string, ops: Op[]): void {
  const a = this.attached.get(tripId);
  if (!a || !ops.length) return;
  ops.forEach((op) => this.clock.observe(op.t));
  if (!applyOps(a.sync.regs, ops)) return; // nothing new (for example our own changes coming back)
  const shared = materializeTrip(a.sync.regs);
  if (!shared) return;
  const current = this.o.store.getTrip(tripId);
  const view = toViewTrip(shared, current);
  if (view === current) return;
  this.applying = true;
  try {
    this.o.store.setTrip(view);
  } finally {
    this.applying = false;
  }
  a.shadow = view;
  this.o.onRemoteChange?.(current, view);
}

private startPullLoop(tripId: string, a: Attached): void {
  if (a.running) return;
  a.running = true;
  void (async () => {
    let delay = 0;
    let quick = true; // the first answer after starting (or after a failure) is not held back, so "Live" shows at once
    while (this.running && a.running && this.attached.get(tripId) === a) {
      a.abort = new AbortController();
      try {
        await this.pull(tripId, quick ? 0 : (this.o.pollWaitMs ?? 20000), a.abort.signal);
        delay = 0;
        quick = false;
      } catch (e) {
        if (!this.running || !a.running || a.abort.signal.aborted) break;
        quick = true;
        this.setOnline(false, describe(e));
        this.publishStatus();
        delay = Math.min(delay ? delay * 2 : 1000, 15000);
        await new Promise<void>((resolve) => {
          const timer = quiet(setTimeout(resolve, delay));
          a.wake = () => {
            clearTimeout(timer);
            resolve();
          };
        });
        a.wake = null;
      }
    }
    a.running = false;
  })();
}

/** Sends and fetches once for every shared trip. Used by "Sync now" and by the tests. */
async syncNow(): Promise<void> {
  for (const tripId of [...this.attached.keys()]) {
    await this.push(tripId);
    try {
      await this.pull(tripId, 0);
    } catch (e) {
      this.setOnline(false, describe(e));
      this.publishStatus();
    }
  }
}

// ---------------------------------------------------------------- saving and status

private scheduleSave(): void {
  if (this.saveTimer) return;
```

```
    this.saveTimer = quiet(
      setTimeout(() => {
        this.saveTimer = null;
        void this.saveNow();
      }, 800), // saving writes everything, so batch bursts of changes (the app also saves when it goes to the background)
    );
  }

  async saveNow(): Promise<void> {
    if (this.saveTimer) {
      clearTimeout(this.saveTimer);
      this.saveTimer = null;
    }
    this.meta.clock = this.clock.snapshot();
    await this.o.storage.save(this.meta);
  }

  private setOnline(online: boolean, error: string | null): void {
    this.status = { ...this.status, online, error, lastSyncAt: online ? (this.o.now ?? Date.now)() : this.status.lastSyncAt };
  }

  private publishStatus(): void {
    const all = [...this.attached.values()];
    this.status = {
      ...this.status,
      pending: all.reduce((n, a) => n + a.sync.outbox.reduce((m, b) => m + b.length, 0), 0),
      peers: all.reduce((n, a) => Math.max(n, a.peers), 0),
      shared: [...this.attached.keys()],
    };
    this.o.onStatus?.(this.status);
  }
}

function describe(e: unknown): string {
  if (e instanceof NetworkError) return 'Can’t reach the sync server';
  if (e instanceof ServerError) return e.message;
  return e instanceof Error ? e.message : 'Sync problem';
}
```

src/sync/fileio.ts

```
/** Moves files between this phone and the sync server: native files on iOS and Android, browser files on the web. */
import { Directory, File, Paths, UploadType } from 'expo-file-system';
import { Platform } from 'react-native';

import { FileTransferError, type BlobFiles } from './blobs';

/** Which kind of failure an HTTP status means. 5xx and anything else odd is treated like a lost connection: try again later. */
function failureFor(status: number, what: string): FileTransferError {
  if (status === 400 || status === 413) return new FileTransferError('rejected', `The server did not accept the file (${status})`);
  if (status === 404) return new FileTransferError('missing', `${what} is not on the server`);
  return new FileTransferError('network', `The server answered ${status}`);
}

const nativeFiles: BlobFiles = {
  async exists(uri) {
    try {
      return new File(uri).exists;
    } catch {
      return false;
    }
  },
  async size(uri) {
    try {
      const file = new File(uri);
      return file.exists ? file.size : null;
    } catch {
      return null;
    }
  },
  async upload(uri, url) {
    let status: number;
    try {
      const res = await new File(uri).upload(url, {
        httpMethod: 'PUT',
        uploadType: UploadType.BINARY_CONTENT,
        headers: { 'Content-Type': 'application/octet-stream' },
        sessionType: 'foreground',
      });
      status = res.status;
    } catch (e) {
      throw new FileTransferError('network', e instanceof Error ? e.message : 'Upload failed');
    }
    if (status < 200 || status >= 300) throw failureFor(status, 'The trip');
  },
  async download(url, name) {
```

```
    const dir = new Directory(Paths.document, 'shared');
    if (!dir.exists) dir.create({ intermediates: true, idempotent: true });
    try {
      const file = await File.downloadFileAsync(url, new File(dir, name), { idempotent: true });
      return file.uri;
    } catch (e) {
      const message = e instanceof Error ? e.message : 'Download failed';
      throw /\b404\b/.test(message) ? new FileTransferError('missing', 'That file is not on the server') : new FileTransferError('networ
k', message);
    }
  },
};

const webFiles: BlobFiles = {
  async exists(uri) {
    try {
      return (await fetch(uri)).ok;
    } catch {
      return false;
    }
  },
  async size(uri) {
    try {
      return (await (await fetch(uri)).blob()).size;
    } catch {
      return null;
    }
  },
  async upload(uri, url) {
    let res: Response;
    try {
      res = await fetch(url, { method: 'PUT', body: await (await fetch(uri)).blob() });
    } catch (e) {
      throw new FileTransferError('network', e instanceof Error ? e.message : 'Upload failed');
    }
    if (!res.ok) throw failureFor(res.status, 'The trip');
  },
  async download(url) {
    let res: Response;
    try {
      res = await fetch(url);
    } catch (e) {
      throw new FileTransferError('network', e instanceof Error ? e.message : 'Download failed');
    }
    if (!res.ok) throw failureFor(res.status, 'That file');
    return URL.createObjectURL(await res.blob());
  },
};

export const deviceFiles: BlobFiles = Platform.OS === 'web' ? webFiles : nativeFiles;

/** Deletes everything this phone downloaded from friends (used by "reset app data"). */
export function deleteSharedFiles(): void {
  if (Platform.OS === 'web') return;
  try {
    const dir = new Directory(Paths.document, 'shared');
    if (dir.exists) dir.delete();
  } catch {
    // nothing to delete
  }
}
```

src/sync/hlc.ts

```
/**
 * A hybrid logical clock. Every change gets a stamp that is unique and sortable as plain text:
 *   "<13-digit milliseconds>-<4-char counter>-<device id>"
 * Two phones with slightly different clocks still agree on the order of changes, because a phone that sees a
 * newer stamp from a friend moves its own clock forward before it makes the next change.
 */
export interface ClockState {
  ms: number;
  counter: number;
}

export class HybridClock {
  private last: ClockState = { ms: 0, counter: 0 };

  constructor(
    private readonly node: string,
    private readonly now: () => number = Date.now,
  ) {}

  /** A new stamp, always bigger than every stamp this clock has made or seen. */
  next(): string {
    const ms = Math.max(this.now(), this.last.ms);
```

```
    const counter = ms === this.last.ms ? this.last.counter + 1 : 0;
    this.last = { ms, counter };
    return format(ms, counter, this.node);
  }

  /** Call with every stamp received from another phone so our next stamp is newer. */
  observe(stamp: string): void {
    const parsed = parseStamp(stamp);
    if (!parsed) return;
    if (parsed.ms > this.last.ms || (parsed.ms === this.last.ms && parsed.counter > this.last.counter)) {
      this.last = { ms: parsed.ms, counter: parsed.counter };
    }
  }

  /** The latest stamp state, saved so the clock never goes backwards after a restart. */
  snapshot(): ClockState {
    return { ...this.last };
  }

  restore(state: ClockState | undefined): void {
    if (state && state.ms > this.last.ms) this.last = { ...state };
  }
}

function format(ms: number, counter: number, node: string): string {
  return `${String(ms).padStart(13, '0')}-${counter.toString(36).padStart(4, '0')}-${node}`;
}

export function parseStamp(stamp: string): { ms: number; counter: number; node: string } | null {
  const m = /^(\d{13})-([0-9a-z]{4})-([\w]+)$/.exec(stamp);
  return m ? { ms: Number(m[1]), counter: parseInt(m[2], 36), node: m[3] } : null;
}

/** Stamp comparison: -1, 0 or 1. Plain string order is the right order by design. */
export const compareStamps = (a: string, b: string) => (a < b ? -1 : a > b ? 1 : 0);

/** A short random device id (letters and numbers only). */
export function newNodeId(random: () => number = Math.random): string {
  let id = '';
  while (id.length < 8) id += Math.floor(random() * 36).toString(36);
  return id;
}
```

src/sync/invite.ts

```
/** The link and the message a friend gets when you invite them. Opening the link fills in the join screen. */
import { INVITE_RE, normalizeInvite } from '@/utils/validation';

import { normalizeServerUrl } from './transport';

/**
 * `base` is the app's own address for the join screen (togetherwego://join, or exp://…/--/join while developing).
 * The sync server is included when there is one, so a friend's phone does not need it typed in.
 */
export function buildInviteUrl(base: string, code: string, server?: string | null): string {
  const query = [`code=${encodeURIComponent(code)}`];
  if (server) query.push(`server=${encodeURIComponent(server)}`);
  return `${base}${base.includes('?') ? '&' : '?'}${query.join('&')}`;
}

const decode = (value: string) => {
  try {
    return decodeURIComponent(value.replace(/\+/g, ' '));
  } catch {
    return value;
  }
};

/** What an invite link or a scanned QR code says: a valid invite code and (if it names one) a sync server. */
export function parseInviteUrl(url: string): { code: string | null; server: string | null } {
  const query = url.trim().split('#')[0].split('?')[1] ?? '';
  const params = new Map<string, string>();
  for (const pair of query.split('&')) {
    const [key, value = ''] = pair.split('=');
    if (key) params.set(key, decode(value));
  }
  const code = normalizeInvite(params.get('code') ?? '');
  const server = normalizeServerUrl(params.get('server') ?? '');
  return { code: INVITE_RE.test(code) ? code : null, server };
}

/** A plain-words message for chat apps: the code, the link, and what to do if the link does not open the app. */
export function inviteText(tripName: string, code: string, link: string): string {
  return `Join my trip “${tripName}” on TogetherWeGo ✈️\nInvite code: ${code}\nOpen: ${link}\nOr open the app, tap Join a trip and type
the code.`;
}
```

src/sync/notify.ts

```
import type { Trip } from '@/store/types';
import { useAppStore } from '@/store/useAppStore';

const nameOf = (trip: Trip, id: string) => trip.members.find((m) => m.id === id)?.name.split(' ')[0] ?? 'Someone';

/**
 * Turns changes that arrived from friends into entries in the in-app notification list (and the unread dot on the
 * Chat tab): new chat messages, polls, expenses, photos and people joining.
 */
export function notifyRemoteChange(prev: Trip | undefined, next: Trip, me: string | undefined): void {
  if (!prev) return;
  const { pushNotification } = useAppStore.getState();
  const known = (list: { id: string }[]) => new Set(list.map((x) => x.id));

  const seenMessages = known(prev.messages);
  const messages = next.messages.filter((m) => !seenMessages.has(m.id) && m.authorId !== me && m.kind !== 'system' && !m.ai);
  if (messages.length === 1) {
    pushNotification({ kind: 'chat', title: `${nameOf(next, messages[0].authorId)} in ${next.name}`, body: messages[0].text, tripId: nex
t.id, route: '/(tabs)/chat' });
  } else if (messages.length > 1) {
    pushNotification({ kind: 'chat', title: `${messages.length} new messages in ${next.name}`, body: messages.at(-1)!.text, tripId: nex
t.id, route: '/(tabs)/chat' });
  }

  const seenPolls = known(prev.polls);
  for (const p of next.polls.filter((x) => !seenPolls.has(x.id) && x.createdBy !== me).slice(0, 2)) {
    pushNotification({ kind: 'vote', title: `New poll from ${nameOf(next, p.createdBy)}`, body: `${p.question} Cast your vote.`, tripId:
next.id, route: '/crew' });
  }

  const seenExpenses = known(prev.expenses);
  for (const e of next.expenses.filter((x) => !seenExpenses.has(x.id) && x.paidBy !== me && !x.settlement).slice(0, 2)) {
    pushNotification({ kind: 'expense', title: `${nameOf(next, e.paidBy)} added an expense`, body: `${e.title} · $${Math.round(e.amoun
t)} split ${e.splitBetween.length} ways`, tripId: next.id, route: '/(tabs)/budget' });
  }

  const seenPhotos = known(prev.photos);
  const photos = next.photos.filter((x) => !seenPhotos.has(x.id) && x.by !== me);
  if (photos.length) {
    const who = nameOf(next, photos[0].by);
    pushNotification({ kind: 'photo', title: photos.length === 1 ? `${who} added a photo` : `${photos.length} new photos in ${next.name}
`, body: photos[0].caption || 'Have a look in the photo dump 📸', tripId: next.id, route: '/photos' });
  }

  const seenMembers = known(prev.members);
  for (const m of next.members.filter((x) => !seenMembers.has(x.id) && x.id !== me).slice(0, 2)) {
    pushNotification({ kind: 'member', title: `${m.name.split(' ')[0]} joined ${next.name}`, body: 'Say hi in the trip chat 👋', tripId:
next.id, route: '/(tabs)/chat' });
  }
}
```

src/sync/prefs.ts

```
import AsyncStorage from '@react-native-async-storage/async-storage';
import Constants from 'expo-constants';
import { Platform } from 'react-native';
import { create } from 'zustand';
import { createJSONStorage, persist } from 'zustand/middleware';

import { normalizeServerUrl } from './transport';

/** The port the sync server listens on (see server/index.js). */
export const SERVER_PORT = 8787;

interface SyncPrefs {
  /** What the user typed in Settings. Empty means "use the automatic guess". */
  serverUrl: string;
  setServerUrl: (url: string) => void;
}

export const useSyncPrefs = create<SyncPrefs>()(
  persist(
    (set) => ({
      serverUrl: '',
      setServerUrl: (serverUrl) => set({ serverUrl }),
    }),
    { name: 'togetherwego-sync-prefs', storage: createJSONStorage(() => AsyncStorage) },
  ),
);

const IPV4 = /^\d{1,3}(\.\d{1,3}){3}$/;

/**
```

```
 * While developing, the sync server runs on the same computer as Expo, so the phone can work out its address
 * from the Expo address it already used. Returns null when that is not possible (a published app, a tunnel...).
 */
export function guessServerUrl(): string | null {
  if (Platform.OS === 'web') {
    if (typeof window === 'undefined') return null;
    const { protocol, hostname } = window.location;
    return hostname === 'localhost' || IPV4.test(hostname) ? `${protocol}//${hostname}:${SERVER_PORT}` : null;
  }
  const host = Constants.expoConfig?.hostUri?.split(':')[0];
  return host && IPV4.test(host) ? `http://${host}:${SERVER_PORT}` : null;
}

/** The address the app will use: what the user typed, or else the automatic guess. */
export function effectiveServerUrl(): string | null {
  return normalizeServerUrl(useSyncPrefs.getState().serverUrl) ?? guessServerUrl();
}

/** The address to put in an invite. "localhost" only means something on the computer that runs the server, so it is left out. */
export function serverForInvites(): string | null {
  const url = effectiveServerUrl();
  return url && !/^https?:\/\/(localhost|127\.\d+\.\d+\.\d+)(:|$)/i.test(url) ? url : null;
}
```

src/sync/runtime.ts

```
/**
 * Starts the sync engine for the real app and wires it to the store, the saved state, the server address and
 * the app lifecycle (it pauses in the background and resumes when you come back).
 */
import { AppState } from 'react-native';

import { toast } from '@/components/ui/feedback';
import { useAppStore } from '@/store/useAppStore';

import { storeAdapter } from './adapter';
import { BlobSync } from './blobs';
import { blobCache, useFileStatus } from './blobCache';
import { SyncEngine } from './engine';
import { deviceFiles } from './fileio';
import { newNodeId } from './hlc';
import { notifyRemoteChange } from './notify';
import { effectiveServerUrl, useSyncPrefs } from './prefs';
import { secureRandom } from './secureRandom';
import { useSyncStatus } from './status';
import { asyncStorageMeta } from './storage';
import { createHttpTransport, type SyncTransport } from './transport';

let engine: SyncEngine | null = null;
let blobSync: BlobSync | null = null;
let ready: Promise<void> | null = null;
let cached: { url: string; transport: SyncTransport } | null = null;

function currentTransport(): SyncTransport | null {
  const url = effectiveServerUrl();
  if (!url) return null;
  if (cached?.url !== url) cached = { url, transport: createHttpTransport(url) };
  return cached.transport;
}

export function getEngine(): SyncEngine {
  engine ??= new SyncEngine({
    store: storeAdapter,
    storage: asyncStorageMeta,
    getTransport: currentTransport,
    random: secureRandom,
    onStatus: (status) => useSyncStatus.setState(status),
    onRemoteChange: (prev, next) => notifyRemoteChange(prev, next, useAppStore.getState().session?.userId),
  });
  return engine;
}

/** Sends your photos, documents and receipts to the server and fetches your friends' ones. */
export function getBlobSync(): BlobSync {
  blobSync ??= new BlobSync({
    trips: () => useAppStore.getState().trips,
    remoteId: (tripId) => getEngine().remoteId(tripId),
    baseUrl: effectiveServerUrl,
    files: deviceFiles,
    cache: blobCache,
    attach: (tripId, list, itemId, blobId) => useAppStore.getState().attachBlob(tripId, list, itemId, blobId),
    newId: () => newNodeId(secureRandom),
    onNotice: (message) => toast(message, { tone: 'warn', icon: 'alert-circle-outline' }),
    onPending: (pending) => useFileStatus.setState({ pending }),
  });
  return blobSync;
```

```
}

/** Call once when the app is ready. Safe to call again. */
export function startSync(): Promise<void> {
  ready ??= (async () => {
    const e = getEngine();
    await e.init();
    const files = getBlobSync();
    useAppStore.subscribe(() => files.schedule()); // a new photo here, or a friend's change that mentions a file
    useSyncPrefs.subscribe(() => files.schedule());
    useSyncStatus.subscribe((now, before) => {
      if (now.online && !before.online) files.schedule(0); // the connection came back
    });
    files.schedule(1000);
    if (AppState.currentState === 'active') e.start();
    AppState.addEventListener('change', (state) => {
      if (state === 'active') e.start();
      else if (state === 'background') e.stop();
    });
  })();
  return ready;
}

/** Waits for the engine to be loaded (used before sharing or joining). */
export async function syncReady(): Promise<SyncEngine> {
  await startSync();
  return getEngine();
}
```

src/sync/schema.ts

```
/**
 * What part of a trip is shared between phones, and how each part merges.
 *
 * A trip is treated as a tree of small "registers". Each register remembers the stamp of the change that last
 * wrote it, and the newest stamp always wins ("last writer wins"), so two phones that exchange their changes
 * end up identical no matter what order the changes arrive in.
 *
 *  - scalars : one value (title, date, amount...)
 *  - json    : one whole object that is replaced together (an expense's original currency amount)
 *  - sets    : lists of text values where only "is it in the list" matters (who is assigned, who votes)
 *  - maps    : name -> value (each member's vote on an activity)
 *  - lists   : lists of items with an id (activities, expenses, chat messages...), each an item with its own fields
 *
 * Left out on purpose: member emails (privacy), the private Scout AI chat, and the file paths of photos, documents,
 * receipts and tickets, because other phones cannot open a path into this phone. Photos, documents and receipts
 * instead share a `blob` id: the file itself travels through the server (see blobs.ts) and each phone keeps its
 * own copy. Flight tickets stay on the phone that added them (they hold personal details).
 */
export interface ItemSchema {
  /** Fields that hold one plain value. */
  scalars: readonly string[];
  /** Scalars that must be present for the item to count as complete. */
  required?: readonly string[];
  json?: readonly string[];
  sets?: readonly string[];
  maps?: readonly string[];
  lists?: Readonly<Record<string, ListSchema>>;
}

export interface ListSchema {
  item: ItemSchema;
  /** How the app keeps this list: 'asc' = newest last (push), 'desc' = newest first (unshift). */
  order: 'asc' | 'desc';
  /** Items this returns false for stay on this phone only. */
  share?: (item: any) => boolean;
}

/** Messages in the private Scout AI thread are never shared. */
export const isPrivateMessage = (m: { channel?: string }) => m.channel === 'ai';

const member: ItemSchema = { scalars: ['name', 'avatar', 'role', 'homeCity', 'invited'], required: ['name', 'role'] };

const activity: ItemSchema = {
  scalars: ['date', 'time', 'title', 'location', 'type', 'durationMin', 'costPerPerson', 'status', 'notes', 'reminder', 'lat', 'lng', 'p
laceId', 'createdBy', 'done'],
  required: ['date', 'time', 'title', 'type', 'status'],
  sets: ['assigned'],
  maps: ['votes'],
};

const expense: ItemSchema = {
  scalars: ['title', 'amount', 'category', 'paidBy', 'date', 'settlement', 'receiptBlob'],
  required: ['title', 'amount', 'paidBy'],
  json: ['original'],
  sets: ['splitBetween'],
```

```
};

const flight: ItemSchema = {
  scalars: ['memberId', 'airline', 'flightNo', 'from', 'to', 'departUtc', 'arriveUtc', 'direction', 'seat', 'confirmation'],
  required: ['memberId', 'flightNo', 'from', 'to', 'departUtc', 'arriveUtc', 'direction'],
};

const poll: ItemSchema = {
  scalars: ['question', 'createdBy', 'closed', 'createdAt'],
  required: ['question'],
  lists: { options: { item: { scalars: ['text'], required: ['text'], sets: ['votes'] }, order: 'asc' } },
};

const task: ItemSchema = { scalars: ['text', 'done', 'assignee'], required: ['text'] };
const doc: ItemSchema = { scalars: ['name', 'kind', 'note', 'size', 'addedBy', 'addedAt', 'offline', 'blob'], required: ['name', 'kind']
};
const message: ItemSchema = { scalars: ['authorId', 'text', 'ts', 'kind', 'ai', 'aiSource', 'channel'], required: ['authorId', 'text',
'ts'] };
const photo: ItemSchema = { scalars: ['caption', 'by', 'ts', 'day', 'blob'], required: ['by', 'ts'], sets: ['likes'] };
const packing: ItemSchema = { scalars: ['text', 'category', 'packed', 'broughtBy'], required: ['text'] };

export const TRIP_SCHEMA: ItemSchema = {
  scalars: ['id', 'name', 'destinationId', 'startDate', 'endDate', 'cover', 'budget', 'inviteCode', 'ownerId', 'completed', 'createdA
t'],
  required: ['id', 'name', 'destinationId', 'startDate', 'endDate', 'budget', 'inviteCode', 'ownerId'],
  sets: ['extraDestinationIds'],
  lists: {
    members: { item: member, order: 'asc' },
    activities: { item: activity, order: 'asc' },
    expenses: { item: expense, order: 'desc' },
    flights: { item: flight, order: 'asc' },
    polls: { item: poll, order: 'desc' },
    tasks: { item: task, order: 'asc' },
    docs: { item: doc, order: 'desc' },
    messages: { item: message, order: 'asc', share: (m) => !isPrivateMessage(m) },
    photos: { item: photo, order: 'desc' },
    packing: { item: packing, order: 'asc' },
  },
};

/** The register that records when an item was created (its stamp is also the item's place in the list). */
export const CREATED = '_c';
export const PATH_SEP = '\u0001';
export const pathKey = (p: readonly string[]) => p.join(PATH_SEP);
```

src/sync/secureRandom.ts

```
import * as Crypto from 'expo-crypto';

/**
 * A random number in [0, 1) from the phone's secure random generator. The ids that work like passwords (a trip's id on
 * the sync server, a file's id) come from this, not from Math.random, which is not meant to be unguessable.
 */
export function secureRandom(): number {
  const [a, b, c, d] = Crypto.getRandomBytes(4);
  return (((a << 24) | (b << 16) | (c << 8) | d) >>> 0) / 4294967296;
}
```

src/sync/status.ts

```
import { create } from 'zustand';

import type { SyncStatus } from './engine';

/** What the screens show about syncing. The engine writes to it; screens only read. */
export const useSyncStatus = create<SyncStatus>(() => ({ online: null, pending: 0, peers: 0, lastSyncAt: null, error: null, shared: []
}));
```

src/sync/storage.ts

```
import AsyncStorage from '@react-native-async-storage/async-storage';

import type { MetaStorage, SyncMeta } from './engine';

const KEY = 'togetherwego-sync-v1';

/** Keeps the sync engine's memory (what it has merged, what it has not sent yet) on the phone. */
export const asyncStorageMeta: MetaStorage = {
  async load() {
    try {
      const raw = await AsyncStorage.getItem(KEY);
      return raw ? (JSON.parse(raw) as SyncMeta) : null;
    } catch {
      return null;
    }
```

```
  },
  async save(meta) {
    await AsyncStorage.setItem(KEY, JSON.stringify(meta));
  },
};
```

src/sync/testing.ts

```
import { produce } from 'immer';

import type { Trip } from '@/store/types';

import { applyOps, diffTrip, materializeTrip, type Op, type Registers } from './crdt';
import { HybridClock } from './hlc';
import type { MetaStorage, StoreAdapter, SyncMeta } from './engine';
import { NetworkError, type SyncTransport } from './transport';
import { toViewTrip } from './view';

/** A tiny trip with a few of everything, for tests. */
export function makeTrip(): Trip {
  return {
    id: 'trip-1',
    name: 'Rome Crew',
    destinationId: 'rome',
    extraDestinationIds: [],
    startDate: '2026-09-10',
    endDate: '2026-09-16',
    cover: 'rome',
    budget: 3000,
    inviteCode: 'ROM-1234',
    ownerId: 'u-a',
    members: [
      { id: 'u-a', name: 'Ana', email: 'ana@example.com', role: 'Organizer' },
      { id: 'u-b', name: 'Ben', role: 'Member' },
    ],
    activities: [
      { id: 'a1', date: '2026-09-11', time: '10:00', title: 'Colosseum', location: 'Rome', type: 'sightseeing', durationMin: 120, costPe
rPerson: 20, status: 'voting', assigned: ['u-a'], votes: { 'u-a': 1 }, reminder: false, createdBy: 'u-a' },
    ],
    expenses: [{ id: 'e1', title: 'Hotel', amount: 300, category: 'Stays', paidBy: 'u-a', splitBetween: ['u-a', 'u-b'], date: '2026-09-1
0', receipt: 'file:///receipt.jpg' }],
    flights: [],
    polls: [{ id: 'p1', question: 'Beach?', createdBy: 'u-a', createdAt: 1, options: [{ id: 'o1', text: 'Ostia', votes: [] }, { id: 'o
2', text: 'Sperlonga', votes: ['u-b'] }] }],
    tasks: [{ id: 't1', text: 'Book tickets', done: false }],
    docs: [],
    messages: [
      { id: 'm1', authorId: 'u-a', text: 'Hi all', ts: 1000 },
      { id: 'ai1', authorId: 'assistant', text: 'private answer', ts: 1001, ai: true, channel: 'ai' },
    ],
    photos: [{ id: 'ph1', uri: 'file:///photo.jpg', caption: 'Pasta', by: 'u-a', likes: ['u-b'], ts: 5 }],
    packing: [],
    createdAt: 1,
  };
}

/** A fake clock shared by every phone in a test: each reading is one millisecond later. */
export function fakeNow(start = 1_800_000_000_000) {
  let t = start;
  return () => (t += 1);
}

/** One simulated phone: its own copy of the trip, its registers and its clock. */
export class Replica {
  regs: Registers = {};
  trip: Trip | undefined;
  clock: HybridClock;

  constructor(node: string, now: () => number, trip?: Trip) {
    this.clock = new HybridClock(node, now);
    this.trip = trip;
  }

  /** Makes a change the way a store action would, and returns the ops to send to the others. */
  edit(fn: (draft: Trip) => void): Op[] {
    const next = produce(this.trip!, fn);
    const ops = diffTrip(this.trip, next, () => this.clock.next());
    applyOps(this.regs, ops);
    this.trip = next;
    return ops;
  }

  /** Shares a trip for the first time: describes the whole trip as ops. */
  publish(): Op[] {
    const ops = diffTrip(undefined, this.trip!, () => this.clock.next());
    applyOps(this.regs, ops);
```

```
    return ops;
  }

  /** Receives ops from another phone. */
  receive(ops: readonly Op[]): void {
    ops.forEach((o) => this.clock.observe(o.t));
    if (applyOps(this.regs, ops)) {
      const shared = materializeTrip(this.regs);
      if (shared) this.trip = toViewTrip(shared, this.trip);
    }
  }
}

/** The trip without anything that stays on one phone, for comparing phones with each other. */
export function sharedPart(trip: Trip | undefined): unknown {
  if (!trip) return undefined;
  const t = JSON.parse(JSON.stringify(trip)) as Trip;
  t.messages = t.messages.filter((m) => m.channel !== 'ai');
  t.members.forEach((m) => delete m.email);
  t.photos.forEach((p) => delete (p as Partial<typeof p>).uri);
  t.docs.forEach((d) => delete d.uri);
  t.expenses.forEach((e) => delete e.receipt);
  t.flights.forEach((f) => {
    delete f.ticketUri;
    delete f.ticketName;
  });
  return t;
}

/** Small deterministic random numbers so a failing run can be repeated. */
export function seeded(seed: number) {
  let s = seed >>> 0;
  const next = () => {
    s = (Math.imul(s, 1664525) + 1013904223) >>> 0;
    return s / 4294967296;
  };
  return {
    next,
    int: (n: number) => Math.floor(next() * n),
    pick: <T,>(a: readonly T[]): T => a[Math.floor(next() * a.length)],
    shuffle: <T,>(a: readonly T[]): T[] => {
      const r = [...a];
      for (let i = r.length - 1; i > 0; i--) {
        const j = Math.floor(next() * (i + 1));
        [r[i], r[j]] = [r[j], r[i]];
      }
      return r;
    },
  };
}

// ---------------------------------------------------------------------- engine test doubles

/** An app store with just the trips: enough for the engine, and it notifies listeners like zustand does. */
export class FakeStore implements StoreAdapter {
  trips: Trip[] = [];
  userId: string | undefined = 'u-a';
  private listeners = new Set<() => void>();

  getTrip = (id: string) => this.trips.find((t) => t.id === id);
  getUserId = () => this.userId;
  setTrip = (trip: Trip) => {
    this.trips = this.trips.map((t) => (t.id === trip.id ? trip : t));
    this.emit();
  };
  addTrip = (trip: Trip) => {
    this.trips = [trip, ...this.trips.filter((t) => t.id !== trip.id)];
    this.emit();
  };
  removeTrip = (id: string) => {
    this.trips = this.trips.filter((t) => t.id !== id);
    this.emit();
  };
  subscribe = (l: () => void) => {
    this.listeners.add(l);
    return () => void this.listeners.delete(l);
  };
  private emit() {
    [...this.listeners].forEach((l) => l());
  }

  /** Changes a trip the way a store action would. */
  edit(tripId: string, fn: (d: Trip) => void) {
    this.trips = this.trips.map((t) => (t.id === tripId ? produce(t, fn) : t));
    this.emit();
  }
```

```
}

export class MemoryStorage implements MetaStorage {
  saved: string | null = null;
  async load(): Promise<SyncMeta | null> {
    return this.saved ? (JSON.parse(this.saved) as SyncMeta) : null;
  }
  async save(meta: SyncMeta): Promise<void> {
    this.saved = JSON.stringify(meta);
  }
}

/** Connects an engine straight to a server store in memory, and can be switched "offline". */
export function memoryTransport(server: {
  append: (...args: any[]) => { seq: number };
  since: (...args: any[]) => { ops: unknown[]; seq: number; total: number; peers: number };
  lookup: (code: string) => string | null;
}): SyncTransport & { offline: boolean; calls: { push: number; pull: number } } {
  const t = {
    offline: false,
    calls: { push: 0, pull: 0 },
    async lookup(code: string) {
      if (t.offline) throw new NetworkError('offline');
      return server.lookup(code);
    },
    async push(tripId: string, code: string | undefined, client: string, ops: Op[]) {
      if (t.offline) throw new NetworkError('offline');
      t.calls.push++;
      return server.append(tripId, code, client, JSON.parse(JSON.stringify(ops)));
    },
    async pull(tripId: string, since: number, client: string) {
      if (t.offline) throw new NetworkError('offline');
      t.calls.pull++;
      return server.since(tripId, since, client) as Awaited<ReturnType<SyncTransport['pull']>>;
    },
  };
  return t;
}
```

src/sync/timers.ts

```
/** In Node (the tests) a waiting timer would keep the program alive; in the app, timers have no such method. */
export function quiet<T>(timer: T): T {
  (timer as { unref?: () => void }).unref?.();
  return timer;
}
```

src/sync/transport.ts

```
/** Talking to the sync server over plain HTTP (see server/index.js). */
import type { Op } from './crdt';

export interface PullResult {
  ops: Op[];
  /** Position to ask from next time. */
  seq: number;
  /** How many changes the server holds in total (more than `seq` means there is another page). */
  total: number;
  /** How many phones talked to the server about this trip in the last minute. */
  peers: number;
}

export interface SyncTransport {
  /** Invite code to the trip's server id, or null when no trip has that code. */
  lookup(code: string): Promise<string | null>;
  push(tripId: string, code: string | undefined, client: string, ops: Op[]): Promise<{ seq: number }>;
  /** Fetches changes after `since`. With `waitMs` the server holds the request until something new arrives. */
  pull(tripId: string, since: number, client: string, waitMs: number, signal?: AbortSignal): Promise<PullResult>;
}

/** The server could not be reached (wrong address, wifi off, server not running). */
export class NetworkError extends Error {}
/** The server answered with an error. */
export class ServerError extends Error {
  constructor(
    readonly status: number,
    message: string,
  ) {
    super(message);
  }
}

const HOST = /^[A-Za-z0-9]([A-Za-z0-9.-]*[A-Za-z0-9])?$/;

/**
 * "192.168.1.5:8787" or "http://192.168.1.5:8787/" -> "http://192.168.1.5:8787". Returns null if it cannot be an address.
```

```
 * The phone's built-in URL class is much looser than a browser's (it accepts spaces in a host name), so the host is checked here too.
 */
export function normalizeServerUrl(input: string): string | null {
  let url = input.trim();
  if (!url) return null;
  url = /^[a-z][a-z0-9+.-]*:\/\//i.test(url) ? url.replace(/^[a-z][a-z0-9+.-]*:\/\//i, (scheme) => scheme.toLowerCase()) : `http://${ur
l}`;
  try {
    const parsed = new URL(url);
    if (!/^https?:$/i.test(parsed.protocol) || !HOST.test(parsed.hostname)) return null;
    if (!/^[A-Za-z0-9.-]+(:\d{1,5})?$/.test(parsed.host)) return null;
    return `${parsed.protocol.toLowerCase()}//${parsed.host.toLowerCase()}`;
  } catch {
    return null;
  }
}

export interface HttpTransportOptions {
  /** How long a request may take before we give up on the server (ms). Waiting requests get extra time on top. */
  timeoutMs?: number;
  /** Looking up an invite code is quick on a home network, so it gives up sooner (a server that is not there should not stall "Join").
*/
  lookupTimeoutMs?: number;
}

export function createHttpTransport(serverUrl: string, fetchImpl: typeof fetch = fetch, options: HttpTransportOptions = {}): SyncTranspo
rt {
  const base = serverUrl.replace(/\/+$/, '');
  const timeoutMs = options.timeoutMs ?? 8000;
  const lookupTimeoutMs = options.lookupTimeoutMs ?? 3000;

  /** One request with a time limit, so a wifi that silently drops packets cannot freeze "Join" for a minute. */
  async function request<T>(path: string, init?: RequestInit & { signal?: AbortSignal }, extraMs = 0, limitMs = timeoutMs): Promise<{ st
atus: number; data: T }> {
    const limit = new AbortController();
    const outer = init?.signal;
    const cancel = () => limit.abort();
    outer?.addEventListener('abort', cancel);
    const timer = setTimeout(cancel, limitMs + extraMs);
    try {
      let res: Response;
      try {
        res = await fetchImpl(base + path, { ...init, signal: limit.signal });
      } catch (e) {
        if (outer?.aborted) throw e; // the app cancelled this request on purpose
        throw new NetworkError(limit.signal.aborted ? 'The sync server took too long to answer' : e instanceof Error ? e.message : 'Netw
ork request failed');
      }
      let data: unknown = null;
      try {
        data = await res.json();
      } catch {
        // an empty or non-JSON body is handled below
      }
      return { status: res.status, data: data as T };
    } finally {
      clearTimeout(timer);
      outer?.removeEventListener('abort', cancel);
    }
  }

  const fail = (status: number, data: unknown): never => {
    const message = (data as { error?: string } | null)?.error ?? `Server answered ${status}`;
    throw new ServerError(status, message);
  };

  return {
    async lookup(code) {
      const { status, data } = await request<{ tripId: string }>(`/v1/codes/${encodeURIComponent(code)}`, undefined, 0, lookupTimeoutM
s);
      if (status === 404) return null;
      if (status !== 200) fail(status, data);
      return data.tripId;
    },
    async push(tripId, code, client, ops) {
      const { status, data } = await request<{ seq: number }>(`/v1/trips/${tripId}/ops`, {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({ code, client, ops }),
      });
      if (status !== 200) fail(status, data);
      return data;
    },
    async pull(tripId, since, client, waitMs, signal) {
      const { status, data } = await request<PullResult>(`/v1/trips/${tripId}/ops?since=${since}&wait=${waitMs}&client=${client}`, { sig
nal }, waitMs);
```

```
      if (status !== 200) fail(status, data);
      return data;
    },
  };
}
```

src/sync/useFile.ts

```
import { useBlobCache } from './blobCache';
import { resolveFile } from './blobs';

export interface SharedFile {
  /** Where to open the file from, when this phone has it. */
  uri: string | undefined;
  /** True while a copy from a friend is on its way. */
  loading: boolean;
}

/**
 * The file to show for a photo, document or receipt: this phone's own file if it has one, otherwise the copy
 * that was downloaded from a friend through the sync server.
 */
export function useSharedFile(local: string | undefined, blob: string | undefined): SharedFile {
  const downloaded = useBlobCache((s) => (blob ? s.files[blob] : undefined));
  return resolveFile(local, blob, downloaded);
}
```

src/sync/view.ts

```
/**
 * Puts a shared trip together with the parts that only live on this phone, so screens see one ordinary Trip.
 *
 * Phone-only parts: the private Scout AI chat, member emails, and the files behind photos, documents, receipts and
 * tickets (other phones cannot open a path that points into this phone's storage).
 */
import type { Trip } from '@/store/types';

import { deepEqual } from './crdt';


/** Fields that stay on this phone, per list, and what other phones show when they do not have them. */
const LOCAL_FIELDS: Record<string, { fields: string[]; fallback?: Record<string, unknown> }> = {
  members: { fields: ['email'] },
  photos: { fields: ['uri'], fallback: { uri: '' } },
  docs: { fields: ['uri'] },
  expenses: { fields: ['receipt'] },
  flights: { fields: ['ticketUri', 'ticketName'] },
};

export function mergeLocalOnly(shared: Trip, current: Trip | undefined): Trip {
  const out: any = { ...shared };
  for (const [list, { fields, fallback }] of Object.entries(LOCAL_FIELDS)) {
    const mine = new Map<string, any>(((current as any)?.[list] ?? []).map((x: any) => [x.id, x]));
    out[list] = (shared as any)[list].map((item: any) => {
      const own = mine.get(item.id);
      let next = item;
      for (const f of fields) {
        const value = own?.[f] ?? fallback?.[f];
        if (value !== undefined && next[f] !== value) next = { ...next, [f]: value };
      }
      return next;
    });
  }
  const privateMessages = (current?.messages ?? []).filter((m) => m.channel === 'ai').sort((a, b) => a.ts - b.ts);
  if (privateMessages.length) out.messages = placePrivate(shared.messages, privateMessages);
  return out as Trip;
}

/** Slots the private messages in by time without reordering the shared ones (their order is the order they were sent). */
function placePrivate(shared: Trip['messages'], mine: Trip['messages']): Trip['messages'] {
  const out = [...shared];
  for (const m of mine) {
    let i = out.length;
    while (i > 0 && out[i - 1].ts > m.ts) i--;
    out.splice(i, 0, m);
  }
  return out;
}

/**
 * Keeps the old object for every part that did not change, so React only re-renders what is new.
 * Returns `prev` itself when nothing changed at all.
 */
export function reuseUnchanged(prev: Trip | undefined, next: Trip): Trip {
  if (!prev) return next;
  const out: any = { ...next };
```

```
  let same = true;
  for (const key of Object.keys(next) as (keyof Trip)[]) {
    const a: any = prev[key];
    const b: any = next[key];
    if (Array.isArray(b) && Array.isArray(a) && b.length && typeof b[0] === 'object') {
      const old = new Map<string, any>(a.map((x: any) => [x.id, x]));
      const items = b.map((item: any) => {
        const o = old.get(item.id);
        return o && deepEqual(o, item) ? o : item;
      });
      out[key] = items.length === a.length && items.every((x: any, i: number) => x === a[i]) ? a : items;
    } else if (deepEqual(a, b)) {
      out[key] = a;
    }
    if (out[key] !== a) same = false;
  }
  return same ? prev : (out as Trip);
}

/** What the store should hold for a shared trip after the registers changed. */
export function toViewTrip(shared: Trip, current: Trip | undefined): Trip {
  return reuseUnchanged(current, mergeLocalOnly(shared, current));
}
```

src/theme/index.ts

```
/**
 * TogetherWeGo design tokens.
 * The default "Forest" palette is sampled from the reference design: warm cream
 * canvas, deep forest green for primary actions, soft sage for highlights.
 *
 * Users can pick another palette and corner style in "Make it yours"
 * (src/app/personalize.tsx). Styles are created once when each module loads,
 * so the saved choice is read synchronously here — before any screen module
 * evaluates — and applied by mutating these token objects in place. Changing
 * the theme then reloads the app (see applyAppearance).
 */
import { File, Paths } from 'expo-file-system';
import { DevSettings, Platform } from 'react-native';

export const colors = {
  bg: '#F4F4EE',
  bgAlt: '#EEF0E7',
  card: '#FFFFFF',
  cardMuted: '#F7F8F3',
  border: '#E7E9DF',
  borderStrong: '#D5D9CC',

  primary: '#1F4D25',
  primaryDark: '#163A1B',
  primaryPressed: '#173E1C',
  primarySoft: '#DCEFD3',
  primarySofter: '#EAF5E4',
  primaryLine: '#C3DFB6',
  navActive: '#CFE6C4',

  /** Pop colour for highlights, badges and the animated hero. */
  accent: '#F2B33D',
  accentSoft: '#FDF3DC',
  heroA: '#1F4D25',
  heroB: '#2F7A3A',
  heroC: '#8CC97A',

  text: '#1A1C17',
  textSecondary: '#5F645A',
  textMuted: '#959A8E',
  onPrimary: '#FFFFFF',
  /** Text on the yellow accent. */
  onAccent: '#111111',
  /** Money coming back / good news. */
  positive: '#1E9E57',

  orange: '#E8833A',
  orangeSoft: '#FDEBDD',
  blue: '#3D7DD8',
  blueSoft: '#E3ECFB',
  yellow: '#F2B33D',
  yellowSoft: '#FDF3DC',
  red: '#D64545',
  redSoft: '#FBE4E4',
  purple: '#7C5CD6',
  purpleSoft: '#EEE8FB',
  teal: '#2A9D8F',
  tealSoft: '#DDF3F0',
  star: '#F5B301',
```

```
  overlay: 'rgba(16, 24, 14, 0.45)',
  white: '#FFFFFF',
  black: '#000000',
};

type Palette = Partial<typeof colors>;
export type ThemeKey = 'amalfi' | 'forest' | 'ocean' | 'sunset' | 'lavender' | 'sakura' | 'midnight';
export type CornerStyle = 'rounded' | 'bubbly' | 'sharp';

/** Every palette the user can choose from in "Make it yours". */
export const themes: Record<
  ThemeKey,
  {
    name: string;
    emoji: string;
    vibe: string;
    dark?: boolean;
    palette: Palette;
  }
> = {
  amalfi: {
    name: 'Amalfi',
    emoji: '🍋',
    vibe: 'Bold black, sunny yellow',
    palette: {
      bg: '#F1F2F4', bgAlt: '#E8E9ED', cardMuted: '#F5F6F8', border: '#E6E7EC', borderStrong: '#D2D4DB',
      primary: '#0B0B0D', primaryDark: '#000000', primaryPressed: '#1B1B1F', primarySoft: '#E9EAEE', primarySofter: '#F3F4F6', primaryLi
ne: '#D5D7DE', navActive: 'rgba(0,0,0,0)',
      accent: '#FBC91B', accentSoft: '#FEF3C7', heroA: '#0B0B0D', heroB: '#0B0B0D', heroC: '#FBC91B',
      text: '#0D0D10', textSecondary: '#666973', textMuted: '#9A9DA8',
      blue: '#1E73E8', blueSoft: '#E1EDFD',
    },
  },
  forest: {
    name: 'Forest',
    emoji: '🌲',
    vibe: 'Calm, earthy, classic',
    palette: {},
  },
  ocean: {
    name: 'Ocean',
    emoji: '🌊',
    vibe: 'Fresh blues & sea breeze',
    palette: {
      bg: '#F1F6FA',
      bgAlt: '#E6EFF6',
      cardMuted: '#F5F9FC',
      border: '#DFE8EF',
      borderStrong: '#C9D7E3',
      primary: '#0F4C81',
      primaryDark: '#0A3A63',
      primaryPressed: '#0C406D',
      primarySoft: '#D5E6F5',
      primarySofter: '#E8F1FA',
      primaryLine: '#B3D0EA',
      navActive: '#CFE2F4',
      accent: '#FF8A5B',
      accentSoft: '#FFE9DF',
      heroA: '#0A3A63',
      heroB: '#1F78B4',
      heroC: '#5CC8D6',
      text: '#14202B',
      textSecondary: '#566573',
      textMuted: '#8F9CA8',
    },
  },
  sunset: {
    name: 'Sunset',
    emoji: '🌅',
    vibe: 'Warm oranges & golden hour',
    palette: {
      bg: '#FFF6EE',
      bgAlt: '#FCEBDD',
      cardMuted: '#FFF9F4',
      border: '#F3E2D3',
      borderStrong: '#E8CDB6',
      primary: '#C2410C',
      primaryDark: '#9A3412',
      primaryPressed: '#A8390B',
      primarySoft: '#FDDCC6',
      primarySofter: '#FFEDE1',
      primaryLine: '#F8C3A0',
      navActive: '#FCD6BD',
      accent: '#7C3AED',
      accentSoft: '#EDE4FD',
      heroA: '#9A3412',
```

```
    heroB: '#EA580C',
    heroC: '#FBBF24',
    text: '#2A1A12',
    textSecondary: '#6E5A4E',
    textMuted: '#A8968A',
  },
},
lavender: {
  name: 'Lavender',
  emoji: '💜',
  vibe: 'Dreamy purples & soft pastels',
  palette: {
    bg: '#F7F4FC',
    bgAlt: '#EEE9F8',
    cardMuted: '#FAF8FD',
    border: '#E7E1F2',
    borderStrong: '#D3C9E8',
    primary: '#5B3FA8',
    primaryDark: '#462F85',
    primaryPressed: '#4E3693',
    primarySoft: '#E4DBF7',
    primarySofter: '#F0EBFB',
    primaryLine: '#CDBDF0',
    navActive: '#DED2F6',
    accent: '#EC4899',
    accentSoft: '#FCE4F1',
    heroA: '#462F85',
    heroB: '#7C5CD6',
    heroC: '#F0A6CA',
    text: '#1E1A2B',
    textSecondary: '#625B73',
    textMuted: '#9A94A8',
  },
},
sakura: {
  name: 'Sakura',
  emoji: '🌸',
  vibe: 'Cherry-blossom pink & cream',
  palette: {
    bg: '#FFF5F7',
    bgAlt: '#FCE9EE',
    cardMuted: '#FFF9FA',
    border: '#F5DFE5',
    borderStrong: '#EBC8D2',
    primary: '#B83268',
    primaryDark: '#912451',
    primaryPressed: '#A12A5B',
    primarySoft: '#FAD7E4',
    primarySofter: '#FDEAF1',
    primaryLine: '#F2B6CC',
    navActive: '#F8D0DF',
    accent: '#14B8A6',
    accentSoft: '#D9F6F2',
    heroA: '#912451',
    heroB: '#D94F86',
    heroC: '#FFB4C8',
    text: '#2B1820',
    textSecondary: '#735A64',
    textMuted: '#A8939B',
  },
},
midnight: {
  name: 'Midnight',
  emoji: '🌙',
  vibe: 'Dark mode with neon green',
  dark: true,
  palette: {
    bg: '#0E1513',
    bgAlt: '#16201D',
    card: '#18221F',
    cardMuted: '#1C2724',
    border: '#26332F',
    borderStrong: '#34443F',
    primary: '#2FAE66',
    primaryDark: '#238A50',
    primaryPressed: '#28995A',
    primarySoft: '#1E3A2C',
    primarySofter: '#1A2E25',
    primaryLine: '#2D5240',
    navActive: '#21402F',
    accent: '#F5C451',
    accentSoft: '#3A3220',
    heroA: '#0B2E1E',
    heroB: '#16603C',
    heroC: '#2FAE66',
    text: '#ECF2EE',
```

```
      textSecondary: '#A9B8B1',
      textMuted: '#74857E',
      orangeSoft: '#3B2A1C',
      blueSoft: '#1C2A3D',
      yellowSoft: '#3A3220',
      redSoft: '#3D1F21',
      purpleSoft: '#2A2340',
      tealSoft: '#173431',
      overlay: 'rgba(0, 0, 0, 0.6)',
      positive: '#4ADE80', onAccent: '#111111',
    },
  },
};

export const cornerStyles: Record<CornerStyle, { name: string; scale: number }> = {
  rounded: { name: 'Rounded', scale: 1 },
  bubbly: { name: 'Bubbly', scale: 1.4 },
  sharp: { name: 'Crisp', scale: 0.45 },
};

export interface Appearance {
  theme: ThemeKey;
  corners: CornerStyle;
}

const PREFS_KEY = 'togetherwego-appearance';
const DEFAULT_APPEARANCE: Appearance = { theme: 'amalfi', corners: 'rounded' };

function prefsFile() {
  return new File(Paths.document, 'appearance.json');
}

function readAppearance(): Appearance {
  try {
    let raw: string | null = null;
    if (Platform.OS === 'web') raw = typeof localStorage !== 'undefined' ? localStorage.getItem(PREFS_KEY) : null;
    else {
      const f = prefsFile();
      raw = f.exists ? f.textSync() : null;
    }
    const parsed = raw ? (JSON.parse(raw) as Partial<Appearance>) : {};
    return {
      theme: parsed.theme && parsed.theme in themes ? parsed.theme : DEFAULT_APPEARANCE.theme,
      corners: parsed.corners && parsed.corners in cornerStyles ? parsed.corners : DEFAULT_APPEARANCE.corners,
    };
  } catch {
    return DEFAULT_APPEARANCE;
  }
}

function writeAppearance(a: Appearance) {
  try {
    const raw = JSON.stringify(a);
    if (Platform.OS === 'web') localStorage.setItem(PREFS_KEY, raw);
    else prefsFile().write(raw);
  } catch {
    // Storage unavailable (private browsing etc.) — the choice lasts for this session only.
  }
}

/** The appearance the app is currently drawn with (fixed until the next reload). */
export const appearance: Appearance = readAppearance();
export const isDark = !!themes[appearance.theme].dark;
/** The untouched Forest tokens (for previews of other palettes). */
export const baseColors = { ...colors };
Object.assign(colors, themes[appearance.theme].palette);

/** Saves a new look and restarts the UI so every screen picks it up. */
export function applyAppearance(next: Appearance): boolean {
  writeAppearance(next);
  if (next.theme === appearance.theme && next.corners === appearance.corners) return true;
  try {
    if (Platform.OS === 'web') {
      const home = (globalThis as { __TWG_HOME?: string }).__TWG_HOME;
      // Restart from the app's front door (not the survey page we're on).
      window.location.replace(home ?? `${window.location.origin}/`);
    } else DevSettings.reload('Applying your new look');
    return true;
  } catch {
    return false;
  }
}

/** Category colours used by the spending breakdown and expense dots. */
export const categoryColors: Record<string, string> = {
  Flights: '#6C4CE0',
```

```
  Stays: colors.text,
  Food: '#FBC91B',
  Activities: '#E8833A',
  Transport: '#1E73E8',
  Shopping: '#D94F86',
  Other: '#9A9DA8',
};

const cornerScale = cornerStyles[appearance.corners].scale;
const r = (n: number) => Math.round(n * cornerScale);
export const radius = {
  xs: r(8),
  sm: r(12),
  md: r(16),
  lg: r(20),
  xl: r(24),
  pill: 999,
};

export const space = {
  xxs: 2,
  xs: 4,
  sm: 8,
  md: 12,
  lg: 16,
  xl: 20,
  xxl: 24,
  xxxl: 32,
} as const;

export const fonts = {
  regular: 'Manrope_400Regular',
  medium: 'Manrope_500Medium',
  semibold: 'Manrope_600SemiBold',
  bold: 'Manrope_700Bold',
  extrabold: 'Manrope_800ExtraBold',
  /** Tall condensed headline face (THE PLAN, BUDGET…). */
  display: 'Anton_400Regular',
} as const;

export const shadow = Platform.select({
  web: { boxShadow: '0px 6px 18px rgba(30, 50, 20, 0.07)' },
  default: {
    shadowColor: '#1E3214',
    shadowOpacity: 0.07,
    shadowRadius: 12,
    shadowOffset: { width: 0, height: 6 },
    elevation: 2,
  },
}) as object;

export const shadowStrong = Platform.select({
  web: { boxShadow: '0px 10px 28px rgba(20, 45, 15, 0.18)' },
  default: {
    shadowColor: '#14280F',
    shadowOpacity: 0.18,
    shadowRadius: 18,
    shadowOffset: { width: 0, height: 10 },
    elevation: 6,
  },
}) as object;

/** Width of the app column on tablets / web so layouts stay phone-shaped. */
export const MAX_WIDTH = 520;
```

src/utils/__tests__/money.test.ts

```
import type { Expense, Member, Trip } from '@/store/types';
import { balances, balancesCents, settlePlan, totalSpent, zeroSumGroups } from '../money';

const member = (id: string): Member => ({ id, name: id, role: 'Member' });
let n = 0;
const exp = (paidBy: string, amount: number, splitBetween: string[], extra: Partial<Expense> = {}): Expense => ({
  id: `e${n++}`, title: 'x', amount, category: 'Food', paidBy, splitBetween, date: '2026-01-01', ...extra,
});
const ledger = (ids: string[], expenses: Expense[]): Pick<Trip, 'members' | 'expenses'> => ({ members: ids.map(member), expenses });
const sum = (o: Record<string, number>) => Object.values(o).reduce((a, b) => a + b, 0);

/** Applies a settle-up plan as settlement expenses, the way the app records payments. */
const applyPlan = (t: Pick<Trip, 'members' | 'expenses'>) =>
  ({ ...t, expenses: [...t.expenses, ...settlePlan(t).map((p) => exp(p.from, p.amount, [p.to], { settlement: true }))] });

describe('balances', () => {
  it('credits the payer and charges everyone in the split', () => {
    const b = balances(ledger(['a', 'b', 'c'], [exp('a', 90, ['a', 'b', 'c'])]));
    expect(b).toEqual({ a: 60, b: -30, c: -30 });
```

```
  });

  it('gives the odd cents to the first people so the total stays exactly zero', () => {
    const b = balancesCents(ledger(['a', 'b', 'c'], [exp('a', 10, ['a', 'b', 'c'])]));
    expect(b).toEqual({ a: 666, b: -333, c: -333 }); // $10.00 / 3 = 3.34 + 3.33 + 3.33: a's own share is the 3.34
    expect(sum(b)).toBe(0);
  });

  it('ignores split members who are no longer in the trip', () => {
    const b = balances(ledger(['a', 'b'], [exp('a', 100, ['a', 'b', 'ghost'])]));
    expect(b).toEqual({ a: 50, b: -50 });
  });

  it('is exact for amounts that break floating point (0.1 + 0.2)', () => {
    const b = balancesCents(ledger(['a', 'b'], [exp('a', 0.1, ['b']), exp('a', 0.2, ['b'])]));
    expect(b).toEqual({ a: 30, b: -30 });
  });
});

describe('totalSpent', () => {
  it('does not count settle-up payments as spending', () => {
    const t = ledger(['a', 'b'], [exp('a', 40, ['a', 'b']), exp('b', 25, ['a'], { settlement: true })]);
    expect(totalSpent(t)).toBe(40);
  });
});

describe('settlePlan', () => {
  it('is empty when everyone is square', () => {
    expect(settlePlan(ledger(['a', 'b'], [exp('a', 20, ['a', 'b']), exp('b', 20, ['a', 'b'])]))).toEqual([]);
  });

  it('pays the right person the right amount', () => {
    const t = ledger(['a', 'b', 'c'], [exp('a', 90, ['a', 'b', 'c'])]);
    expect(settlePlan(t)).toEqual([{ from: 'b', to: 'a', amount: 30 }, { from: 'c', to: 'a', amount: 30 }]);
  });

  it('zeroes every balance once the plan is paid', () => {
    const t = ledger(['a', 'b', 'c', 'd'], [exp('a', 120, ['a', 'b', 'c', 'd']), exp('b', 45.5, ['b', 'c']), exp('d', 10, ['a', 'd'])]);
    const after = balances(applyPlan(t));
    Object.values(after).forEach((v) => expect(Math.abs(v)).toBeLessThan(0.51));
    expect(settlePlan(applyPlan(t))).toEqual([]);
  });

  it('finds fewer payments than the simple largest-first method when the group splits into pairs', () => {
    // Balances: p +6, q +5 are owed; r -3, s -3, u -5 owe. {p,r,s} and {q,u} each add up to zero,
    // so 3 payments are enough, while largest-first needs 4.
    const t = ledger(['p', 'q', 'r', 's', 'u'], [
      exp('p', 6, ['r', 's']), // p paid 6 for r and s -> r -3, s -3, p +6
      exp('q', 5, ['u']), //      q paid 5 for u      -> u -5, q +5
    ]);
    const plan = settlePlan(t);
    expect(plan).toHaveLength(3);
    expect(plan).toContainEqual({ from: 'u', to: 'q', amount: 5 });
  });

  it('never needs more than (people - 1) payments and always settles exactly', () => {
    let seed = 7;
    const rnd = () => ((seed = (seed * 1103515245 + 12345) & 0x7fffffff) / 0x7fffffff);
    for (let round = 0; round < 60; round++) {
      const size = 2 + Math.floor(rnd() * 7);
      const ids = Array.from({ length: size }, (_, i) => `m${i}`);
      const expenses = Array.from({ length: 1 + Math.floor(rnd() * 8) }, () => {
        const split = ids.filter(() => rnd() < 0.6);
        return exp(ids[Math.floor(rnd() * size)], Math.round(rnd() * 20000) / 100 + 1, split.length ? split : [ids[0]]);
      });
      const t = ledger(ids, expenses);
      const plan = settlePlan(t);
      expect(plan.length).toBeLessThanOrEqual(size - 1);
      Object.values(balances(applyPlan(t))).forEach((v) => expect(Math.abs(v)).toBeLessThan(0.51));
    }
  });
});

describe('zeroSumGroups', () => {
  it('splits into the most groups that each add up to zero', () => {
    expect(zeroSumGroups([5, -5, 3, -3]).map((g) => g.length).sort()).toEqual([2, 2]);
    expect(zeroSumGroups([6, 5, -3, -3, -5])).toHaveLength(2);
    expect(zeroSumGroups([4, 3, -5, -2])).toHaveLength(1);
    expect(zeroSumGroups([])).toEqual([]);
  });
});
```

src/utils/__tests__/password.test.ts

```
import { createHash } from 'crypto';
import { PBKDF2_ROUNDS, hashPassword, isLegacyHash, safeEqual, verifyPassword } from '../password';
```

```
describe('hashPassword', () => {
  it('matches the official PBKDF2-HMAC-SHA256 test vector', async () => {
    // From RFC 7914 section 11: P = "passwd", S = "salt", c = 1 (first 32 bytes of the 64-byte result)
    expect(await hashPassword('passwd', 'salt', 1)).toBe('pbkdf2-sha256$1$55ac046e56e3089fec1691c22544b605f94185216dde0465e68b9d57c20dac
bc');
  });

  it('is repeatable, and the salt changes the result', async () => {
    const a = await hashPassword('TravelTogether2026!', 'salt-one');
    expect(await hashPassword('TravelTogether2026!', 'salt-one')).toBe(a);
    expect(await hashPassword('TravelTogether2026!', 'salt-two')).not.toBe(a);
    expect(a).toMatch(new RegExp(`^pbkdf2-sha256\\$${PBKDF2_ROUNDS}\\$[0-9a-f]{64}$`));
  });

  it('never contains the password', async () => {
    expect(await hashPassword('hunter2hunter2', 's')).not.toContain('hunter2');
  });
});

describe('verifyPassword', () => {
  it('accepts the right password and rejects the wrong one', async () => {
    const account = { salt: 'abc', passwordHash: await hashPassword('Correct-Horse-9', 'abc') };
    expect(await verifyPassword('Correct-Horse-9', account)).toEqual({ ok: true });
    expect((await verifyPassword('correct-horse-9', account)).ok).toBe(false);
    expect((await verifyPassword('', account)).ok).toBe(false);
  });

  it('upgrades a version-1 hash (single SHA-256) after a correct sign-in', async () => {
    const legacy = createHash('sha256').update('old-salt:Passw0rd!').digest('hex');
    expect(isLegacyHash(legacy)).toBe(true);
    const account = { salt: 'old-salt', passwordHash: legacy };
    expect((await verifyPassword('wrong', account)).ok).toBe(false);
    const ok = await verifyPassword('Passw0rd!', account);
    expect(ok.ok).toBe(true);
    expect(ok.upgrade).toMatch(/^pbkdf2-sha256\$/);
    expect(await verifyPassword('Passw0rd!', { salt: 'old-salt', passwordHash: ok.upgrade! })).toEqual({ ok: true });
  });

  it('upgrades a hash made with fewer rounds', async () => {
    const weak = await hashPassword('Passw0rd!', 's', 1000);
    const r = await verifyPassword('Passw0rd!', { salt: 's', passwordHash: weak });
    expect(r.ok).toBe(true);
    expect(r.upgrade).toContain(`$${PBKDF2_ROUNDS}$`);
  });

  it('rejects damaged or unknown hash formats instead of crashing', async () => {
    for (const bad of ['', 'nonsense', 'pbkdf2-sha256$abc$00', 'pbkdf2-sha256$99999999$00', 'md5$1$00']) {
      expect((await verifyPassword('x', { salt: 's', passwordHash: bad })).ok).toBe(false);
    }
  });

  it('supports passwords with accents and emoji', async () => {
    const account = { salt: 's', passwordHash: await hashPassword('Contraseña✈️2026', 's') };
    expect((await verifyPassword('Contraseña✈️2026', account)).ok).toBe(true);
  });
});

describe('safeEqual', () => {
  it('compares strings', () => {
    expect(safeEqual('abc', 'abc')).toBe(true);
    expect(safeEqual('abc', 'abd')).toBe(false);
    expect(safeEqual('abc', 'abcd')).toBe(false);
  });
});
```

src/utils/__tests__/qr.test.ts

```
import jsQR from 'jsqr';

import { qrMatrix, qrPath } from '../qr';

/** Draws the matrix the way a screen would (with the empty border a scanner needs) and reads it back with a QR decoder. */
function scan(matrix: boolean[][], scale = 6, quiet = 4): string | null {
  const size = (matrix.length + quiet * 2) * scale;
  const pixels = new Uint8ClampedArray(size * size * 4).fill(255);
  matrix.forEach((row, r) =>
    row.forEach((dark, c) => {
      if (!dark) return;
      for (let y = 0; y < scale; y++) {
        for (let x = 0; x < scale; x++) {
          const i = (((r + quiet) * scale + y) * size + (c + quiet) * scale + x) * 4;
          pixels[i] = pixels[i + 1] = pixels[i + 2] = 0;
        }
      }
    }),
```

```
  );
  return jsQR(pixels, size, size)?.data ?? null;
}

describe('QR codes', () => {
  it('can be read back by a scanner', () => {
    for (const text of [
      'AML-2026',
      'togetherwego://join?code=AML-2026',
      'exp://192.168.1.20:8081/--/join?code=AML-2026&server=http%3A%2F%2F192.168.1.20%3A8787',
      'Join my trip “Amalfi Squad” ✈️ — code AML-2026',
      'x'.repeat(180),
    ]) {
      expect(scan(qrMatrix(text))).toBe(text);
    }
  });

  it('is square, bigger for longer text, and has the three corner markers', () => {
    const short = qrMatrix('AML-2026');
    const long = qrMatrix('x'.repeat(180));
    expect(short.every((row) => row.length === short.length)).toBe(true);
    expect(long.length).toBeGreaterThan(short.length);
    const n = short.length;
    // the 7x7 finder pattern: dark ring, light ring, dark 3x3 centre — top-left, top-right, bottom-left
    for (const [r0, c0] of [[0, 0], [0, n - 7], [n - 7, 0]]) {
      expect(short[r0][c0]).toBe(true);
      expect(short[r0 + 1][c0 + 1]).toBe(false);
      expect(short[r0 + 3][c0 + 3]).toBe(true);
    }
  });

  it('draws the same squares as the grid says', () => {
    const matrix = qrMatrix('AML-2026');
    const path = qrPath(matrix);
    // every "M x y h len v1 h-len z" is a run of dark squares: counting their lengths gives the number of dark squares
    const dark = [...path.matchAll(/M(\d+) (\d+)h(\d+)v1h-\d+z/g)].reduce((sum, m) => sum + Number(m[3]), 0);
    expect(dark).toBe(matrix.flat().filter(Boolean).length);
    expect(qrPath([[false, false]])).toBe('');
    expect(qrPath([[true, true, false, true]])).toBe('M0 0h2v1h-2zM3 0h1v1h-1z');
  });
});
```

src/utils/__tests__/time.test.ts

```
import { addDays, daysBetween, formatInTz, hoursAhead, isValidYMD, partsInTz, tripDays, tzOffsetMinutes, zonedToUtc } from '../time';

const iso = (ms: number) => new Date(ms).toISOString();

describe('calendar dates', () => {
  it('adds days across months and leap years', () => {
    expect(addDays('2026-01-31', 1)).toBe('2026-02-01');
    expect(addDays('2028-02-28', 2)).toBe('2028-03-01');
    expect(addDays('2026-12-31', 1)).toBe('2027-01-01');
    expect(addDays('2026-03-01', -1)).toBe('2026-02-28');
  });

  it('counts days between two dates', () => {
    expect(daysBetween('2026-06-10', '2026-06-16')).toBe(6);
    expect(daysBetween('2026-06-16', '2026-06-10')).toBe(-6);
    expect(tripDays('2026-06-10', '2026-06-12')).toEqual(['2026-06-10', '2026-06-11', '2026-06-12']);
  });

  it('rejects dates that do not exist', () => {
    expect(isValidYMD('2026-02-30')).toBe(false);
    expect(isValidYMD('2026-13-01')).toBe(false);
    expect(isValidYMD('2026-6-1')).toBe(false);
    expect(isValidYMD('2028-02-29')).toBe(true);
  });
});

describe('time zones', () => {
  it('knows summer time and winter time in Rome', () => {
    expect(tzOffsetMinutes('Europe/Rome', Date.parse('2026-07-01T12:00:00Z'))).toBe(120);
    expect(tzOffsetMinutes('Europe/Rome', Date.parse('2026-01-15T12:00:00Z'))).toBe(60);
  });

  it('turns a wall-clock time in a city into the exact moment', () => {
    expect(iso(zonedToUtc('2026-07-01', '12:00', 'Europe/Rome'))).toBe('2026-07-01T10:00:00.000Z');
    expect(iso(zonedToUtc('2026-01-15', '12:00', 'Europe/Rome'))).toBe('2026-01-15T11:00:00.000Z');
    expect(iso(zonedToUtc('2026-06-10', '18:30', 'America/New_York'))).toBe('2026-06-10T22:30:00.000Z');
  });

  it('a New York to Rome flight converts to the right arrival day and duration', () => {
    const depart = zonedToUtc('2026-06-10', '18:30', 'America/New_York');
    const arrive = zonedToUtc('2026-06-11', '08:45', 'Europe/Rome');
    expect((arrive - depart) / 60000).toBe(8 * 60 + 15);
```

```
    expect(partsInTz(arrive, 'America/New_York')).toEqual({ ymd: '2026-06-11', hhmm: '02:45' });
    expect(formatInTz(arrive, 'Europe/Rome')).toMatch(/08:45$/);
  });

  it('round-trips wall time to UTC and back', () => {
    for (const tz of ['Asia/Tokyo', 'Australia/Sydney', 'America/Los_Angeles', 'Asia/Kolkata']) {
      const utc = zonedToUtc('2026-09-15', '07:05', tz);
      expect(partsInTz(utc, tz)).toEqual({ ymd: '2026-09-15', hhmm: '07:05' });
    }
  });

  it('hours ahead between cities', () => {
    const july = Date.parse('2026-07-01T12:00:00Z');
    expect(hoursAhead('Asia/Tokyo', 'America/New_York', july)).toBe(13);
    expect(hoursAhead('Europe/Rome', 'America/New_York', july)).toBe(6);
    expect(hoursAhead('Asia/Kolkata', 'UTC', july)).toBe(5.5);
  });
});
```

src/utils/__tests__/validation.test.ts

```
import {
  firstError, maxLen, normalizeInvite, parseAmount, passwordStrength, required, validateAmount, validateEmail, validateFlightNumber,
  validateIata, validateInvite, validateName, validatePassword, validateTime, validateTripDates,
} from '../validation';

describe('syntax checks (is it shaped right?)', () => {
  it('emails', () => {
    expect(validateEmail('maya@mail.com')).toBeNull();
    expect(validateEmail('  maya@mail.com  ')).toBeNull();
    expect(validateEmail('')).toMatch(/Enter/);
    expect(validateEmail('maya@mail')).toMatch(/doesn’t look right/);
    expect(validateEmail('maya mail@x.com')).not.toBeNull();
  });

  it('passwords need 8+ characters with a letter and a number', () => {
    expect(validatePassword('TravelTogether2026!')).toBeNull();
    expect(validatePassword('short1')).toMatch(/8 characters/);
    expect(validatePassword('onlyletters')).toMatch(/letter and one number/);
    expect(validatePassword('12345678')).toMatch(/letter and one number/);
  });

  it('password strength rises with length and variety', () => {
    expect(passwordStrength('abc').score).toBe(0);
    expect(passwordStrength('abcdefgh1').score).toBeLessThan(passwordStrength('Abcdefgh1234!').score);
    expect(passwordStrength('Abcdefgh1234!').label).toBe('Very strong');
  });

  it('names accept accents and apostrophes but not digits or symbols', () => {
    expect(validateName('José O’Brien-Müller')).toBeNull();
    expect(validateName('A')).toMatch(/too short/);
    expect(validateName('R2D2')).toMatch(/only contain letters/);
    expect(validateName('x'.repeat(41))).toMatch(/under 40/);
  });

  it('times are 24-hour HH:MM', () => {
    expect(validateTime('09:30')).toBeNull();
    expect(validateTime('23:59')).toBeNull();
    expect(validateTime('24:00')).not.toBeNull();
    expect(validateTime('9:30')).not.toBeNull();
    expect(validateTime('12:60')).not.toBeNull();
  });

  it('invite codes are normalised then checked', () => {
    expect(normalizeInvite('aml2026')).toBe('AML-2026');
    expect(normalizeInvite('  aml-2026 ')).toBe('AML-2026');
    expect(validateInvite('AML-2026')).toBeNull();
    expect(validateInvite('AML2026')).toMatch(/ABC-1234/);
    expect(validateInvite('')).toMatch(/Enter the code/);
  });

  it('flight numbers', () => {
    expect(validateFlightNumber('NH 175')).toBeNull();
    expect(validateFlightNumber('ua837')).toBeNull();
    expect(validateFlightNumber('FLIGHT')).not.toBeNull();
  });
});

describe('meaning checks (does it make sense for a trip?)', () => {
  it('amounts: positive, sensible, plain numbers', () => {
    expect(parseAmount('1,200.50')).toBe(1200.5);
    expect(parseAmount('12.345')).toBeNull();
    expect(parseAmount('-5')).toBeNull();
    expect(parseAmount('12abc')).toBeNull();
    expect(validateAmount('0')).toMatch(/more than 0/);
    expect(validateAmount('2000000')).toMatch(/double-check/);
```

```
    expect(validateAmount('25.50')).toBeNull();
  });

  it('a trip cannot end before it starts, or run for months', () => {
    expect(validateTripDates('2026-06-10', '2026-06-16')).toEqual({});
    expect(validateTripDates('2026-06-10', '2026-06-09').end).toMatch(/on or after/);
    expect(validateTripDates('2026-06-10', '2026-09-01').end).toMatch(/at most 60 days/);
  });

  it('dates must exist on the calendar', () => {
    expect(validateTripDates('2026-02-30', '2026-03-02').start).toMatch(/valid start/);
    expect(validateTripDates('2028-02-29', '2028-03-02')).toEqual({}); // leap year
    expect(validateTripDates('2026-02-29', '2026-03-02').start).toBeDefined(); // not a leap year
  });

  it('airport codes must be real airports', () => {
    expect(validateIata('jfk', 'departure')).toBeNull();
    expect(validateIata('FCO', 'arrival')).toBeNull();
    expect(validateIata('XXX', 'arrival')).toMatch(/don’t recognise/);
    expect(validateIata('JF', 'arrival')).toMatch(/3 letters/);
    expect(validateIata('', 'arrival')).toMatch(/Enter the arrival/);
  });
});

describe('helpers', () => {
  it('firstError returns the first failing validator only', () => {
    expect(firstError('', required('a name'), maxLen(5))).toBe('Enter a name');
    expect(firstError('toolong', required('a name'), maxLen(5, 'Name'))).toBe('Name must be 5 characters or fewer');
    expect(firstError('ok', required('a name'), maxLen(5))).toBeNull();
  });
});
```

src/utils/format.ts

```
import { currencyByCode } from '@/data/reference';

export function uid(prefix = ''): string {
  return `${prefix}${Date.now().toString(36)}${Math.random().toString(36).slice(2, 8)}`;
}

/** Formats money with the right symbol and sensible decimals for the currency. */
export function money(amount: number, code = 'USD', opts: { decimals?: boolean } = {}): string {
  const cur = currencyByCode(code);
  const bigUnit = ['JPY', 'KRW', 'IDR', 'VND', 'ISK'].includes(code);
  const decimals = opts.decimals && !bigUnit ? 2 : 0;
  const n = Math.abs(amount).toLocaleString('en-US', {
    minimumFractionDigits: decimals,
    maximumFractionDigits: decimals,
  });
  const sign = amount < 0 ? '−' : '';
  const sym = cur?.symbol ?? `${code} `;
  return sym.length > 2 ? `${sign}${n} ${code}` : `${sign}${sym}${n}`;
}

export function initials(name: string): string {
  const parts = name.trim().split(/\s+/).filter(Boolean);
  if (!parts.length) return '?';
  if (parts.length === 1) return parts[0].slice(0, 2).toUpperCase();
  return (parts[0][0] + parts[parts.length - 1][0]).toUpperCase();
}

export function pluralize(n: number, word: string, plural = `${word}s`) {
  return `${n} ${n === 1 ? word : plural}`;
}

export function durationLabel(min: number): string {
  if (min < 60) return `${min}m`;
  const h = Math.floor(min / 60);
  const m = min % 60;
  return m ? `${h}h ${m}m` : `${h}h`;
}

export type LengthFilter = 'any' | 'u1' | '1-2' | '2-4' | 'half' | 'full';

export const LENGTH_FILTERS: { key: LengthFilter; labelKey: string }[] = [
  { key: 'any', labelKey: 'len.any' },
  { key: 'u1', labelKey: 'len.u1' },
  { key: '1-2', labelKey: 'len.1-2' },
  { key: '2-4', labelKey: 'len.2-4' },
  { key: 'half', labelKey: 'len.half' },
  { key: 'full', labelKey: 'len.full' },
];

/** Buckets match the brief: under 1h, 1–2h, 2–4h, half day (4–6h), full day (6h+). */
export function matchesLength(min: number, f: LengthFilter): boolean {
  switch (f) {
```

```
    case 'any':
      return true;
    case 'u1':
      return min < 60;
    case '1-2':
      return min >= 60 && min <= 120;
    case '2-4':
      return min > 120 && min <= 240;
    case 'half':
      return min > 240 && min <= 360;
    case 'full':
      return min > 360;
  }
}

export function lengthBucketLabel(min: number): string {
  if (min < 60) return 'Under 1 hour';
  if (min <= 120) return '1–2 hours';
  if (min <= 240) return '2–4 hours';
  if (min <= 360) return 'Half day';
  return 'Full day';
}

/** Short tag for activity cards: <1h, 1-2h, 2-4h, Half day, Full day. */
export function lengthTag(min: number): string {
  if (min < 60) return '<1h';
  if (min <= 120) return '1-2h';
  if (min <= 240) return '2-4h';
  if (min <= 360) return 'Half day';
  return 'Full day';
}

export function priceTier(cost: number): string {
  if (cost === 0) return 'Free';
  if (cost < 15) return '$';
  if (cost < 40) return '$$';
  return '$$$';
}

export function timeAgo(ts: number, now = Date.now()): string {
  const s = Math.max(0, Math.round((now - ts) / 1000));
  if (s < 45) return 'just now';
  const m = Math.round(s / 60);
  if (m < 60) return `${m}m ago`;
  const h = Math.round(m / 60);
  if (h < 24) return `${h}h ago`;
  const d = Math.round(h / 24);
  return `${d}d ago`;
}

export function countdownLabel(ms: number): string {
  if (ms <= 0) return 'now';
  const totalMin = Math.round(ms / 60000);
  const d = Math.floor(totalMin / 1440);
  const h = Math.floor((totalMin % 1440) / 60);
  const m = totalMin % 60;
  if (d > 0) return `${d}d ${h}h`;
  if (h > 0) return `${h}h ${m}m`;
  return `${m}m`;
}
```

src/utils/geo.ts

```
import { airports, type Airport } from '@/data/airports';

export interface LatLng {
  lat: number;
  lng: number;
}

const R = 6371;
const rad = (d: number) => (d * Math.PI) / 180;

export function distanceKm(a: LatLng, b: LatLng): number {
  const dLat = rad(b.lat - a.lat);
  const dLng = rad(b.lng - a.lng);
  const h = Math.sin(dLat / 2) ** 2 + Math.cos(rad(a.lat)) * Math.cos(rad(b.lat)) * Math.sin(dLng / 2) ** 2;
  return 2 * R * Math.asin(Math.sqrt(h));
}

export function bearingDeg(a: LatLng, b: LatLng): number {
  const y = Math.sin(rad(b.lng - a.lng)) * Math.cos(rad(b.lat));
  const x = Math.cos(rad(a.lat)) * Math.sin(rad(b.lat)) - Math.sin(rad(a.lat)) * Math.cos(rad(b.lat)) * Math.cos(rad(b.lng - a.lng));
  return ((Math.atan2(y, x) * 180) / Math.PI + 360) % 360;
}
```

```
export function compassPoint(deg: number): string {
  const pts = ['N', 'NE', 'E', 'SE', 'S', 'SW', 'W', 'NW'];
  return pts[Math.round(deg / 45) % 8];
}

export function kmLabel(km: number): string {
  if (km < 1) return `${Math.round(km * 1000)} m`;
  if (km < 10) return `${km.toFixed(1)} km`;
  return `${Math.round(km)} km`;
}

/** Rough door-to-door estimate for city driving / airport transfer. */
export function driveMinutes(km: number): number {
  return Math.max(5, Math.round((km / 38) * 60 + 8));
}

export function nearestAirports(from: LatLng, count = 5): (Airport & { km: number })[] {
  return airports
    .map((a) => ({ ...a, km: distanceKm(from, a) }))
    .sort((x, y) => x.km - y.km)
    .slice(0, count);
}

/** Projects a point into x/y kilometres east/north of `center` (equirectangular, fine for < 50 km). */
export function projectKm(center: LatLng, p: LatLng) {
  return {
    x: (p.lng - center.lng) * 111.32 * Math.cos(rad(center.lat)),
    y: (p.lat - center.lat) * 110.57,
  };
}
```

src/utils/money.ts

```
/**
 * Group-expense maths. Pure functions with no React or storage imports, so they are easy to test.
 * Everything is computed in whole cents internally: floating-point dollars drift (0.1 + 0.2 !== 0.3).
 */
import type { Trip } from '@/store/types';

type Ledger = Pick<Trip, 'members' | 'expenses'>;
export interface Transfer {
  from: string;
  to: string;
  /** Dollars, rounded to cents. */
  amount: number;
}

export const toCents = (dollars: number) => Math.round(dollars * 100);

/** Total spent by the group (settlement payments are not spending). */
export function totalSpent(t: Pick<Trip, 'expenses'>): number {
  return t.expenses.filter((e) => !e.settlement).reduce((sum, e) => sum + e.amount, 0);
}

/**
 * Net balance per member in cents: positive = is owed money, negative = owes money.
 * An expense is split equally; when it does not divide evenly the extra cents go to the first people in the
 * split, so the balances always add up to exactly zero.
 */
export function balancesCents(t: Ledger): Record<string, number> {
  const bal: Record<string, number> = {};
  t.members.forEach((m) => (bal[m.id] = 0));
  for (const e of t.expenses) {
    const split = e.splitBetween.filter((id) => id in bal);
    if (!split.length) continue;
    const cents = toCents(e.amount);
    const base = Math.floor(cents / split.length);
    const extra = cents - base * split.length;
    bal[e.paidBy] = (bal[e.paidBy] ?? 0) + cents;
    split.forEach((id, i) => (bal[id] -= base + (i < extra ? 1 : 0)));
  }
  return bal;
}

/** Same as {@link balancesCents} but in dollars, for display. */
export function balances(t: Ledger): Record<string, number> {
  return Object.fromEntries(Object.entries(balancesCents(t)).map(([id, c]) => [id, c / 100]));
}

const MAX_EXACT = 12; // 2^12 subsets is instant; beyond that we use the simple greedy plan
const DUST_CENTS = 50; // a balance under 50 cents counts as settled

/**
 * Fewest payments that settle everyone up.
 * The minimum number of payments is (people with a balance) minus (the most separate groups whose balances add
 * up to zero): each such group can settle among itself with one fewer payment than it has people.
```

```
 * We find that maximum by dynamic programming over subsets, then settle each group greedily.
 */
export function settlePlan(t: Ledger): Transfer[] {
  const people = Object.entries(balancesCents(t))
    .filter(([, c]) => c !== 0)
    .map(([id, c]) => ({ id, c }))
    .sort((a, b) => b.c - a.c || a.id.localeCompare(b.id));
  const groups = people.length <= MAX_EXACT ? zeroSumGroups(people.map((p) => p.c)) : [people.map((_, i) => i)];
  const plan: Transfer[] = [];
  for (const g of groups) plan.push(...greedyGroup(g.map((i) => people[i])));
  return plan.filter((p) => p.amount * 100 >= DUST_CENTS);
}

/** Splits balances (summing to 0) into the maximum number of groups that each sum to 0. Returns index groups. */
export function zeroSumGroups(values: number[]): number[][] {
  const n = values.length;
  if (n === 0) return [];
  const full = (1 << n) - 1;
  const sum = new Array<number>(1 << n).fill(0);
  for (let m = 1; m <= full; m++) {
    const low = m & -m;
    sum[m] = sum[m ^ low] + values[31 - Math.clz32(low)];
  }
  const best = new Array<number>(1 << n).fill(0);
  const prev = new Array<number>(1 << n).fill(0);
  for (let m = 1; m <= full; m++) {
    let bestPrev = 0;
    let bestVal = -1;
    for (let i = 0; i < n; i++) {
      if (!(m & (1 << i))) continue;
      const p = m ^ (1 << i);
      if (best[p] > bestVal) {
        bestVal = best[p];
        bestPrev = p;
      }
    }
    best[m] = bestVal + (sum[m] === 0 ? 1 : 0);
    prev[m] = bestPrev;
  }
  // Walk back from the full set: every mask on the path with a zero sum closes one group.
  const closers: number[] = [];
  for (let m = full; m !== 0; m = prev[m]) if (sum[m] === 0) closers.push(m);
  closers.reverse();
  const groups: number[][] = [];
  let before = 0;
  for (const m of closers) {
    const g = m & ~before;
    groups.push(Array.from({ length: n }, (_, i) => i).filter((i) => g & (1 << i)));
    before = m;
  }
  return groups;
}

/** Pairs the biggest debtor with the biggest creditor until everyone in the group is square. */
function greedyGroup(people: { id: string; c: number }[]): Transfer[] {
  const debtors = people.filter((p) => p.c < 0).map((p) => ({ id: p.id, v: -p.c })).sort((a, b) => b.v - a.v || a.id.localeCompare(b.i
d));
  const creditors = people.filter((p) => p.c > 0).map((p) => ({ id: p.id, v: p.c })).sort((a, b) => b.v - a.v || a.id.localeCompare(b.i
d));
  const out: Transfer[] = [];
  let i = 0;
  let j = 0;
  while (i < debtors.length && j < creditors.length) {
    const amt = Math.min(debtors[i].v, creditors[j].v);
    out.push({ from: debtors[i].id, to: creditors[j].id, amount: amt / 100 });
    debtors[i].v -= amt;
    creditors[j].v -= amt;
    if (debtors[i].v === 0) i++;
    if (creditors[j].v === 0) j++;
  }
  return out;
}
```

src/utils/password.ts

```
/**
 * Password hashing. Stored as `pbkdf2-sha256$<rounds>$<hex>`; the salt lives next to it on the account.
 *
 * Why not a bigger number of rounds? This runs in the phone's JavaScript engine, which is far slower than a
 * server, and the async version below yields to the UI so the screen never freezes. The round count is stored
 * in the hash, so it can be raised later and old hashes are upgraded on the next successful sign-in.
 * shortcut: a real service should hash on the server with Argon2id or bcrypt; this protects the on-device copy.
 */
import { pbkdf2Async } from '@noble/hashes/pbkdf2.js';
import { sha256 } from '@noble/hashes/sha2.js';
import { bytesToHex, utf8ToBytes } from '@noble/hashes/utils.js';
```

```
export const PBKDF2_ROUNDS = 10_000;
const PREFIX = 'pbkdf2-sha256';
const MAX_ROUNDS = 1_000_000; // refuse absurd values so a corrupted record cannot hang the app

/** Version 1 of the app stored a single round of SHA-256 over `salt:password` (64 hex characters). */
export const isLegacyHash = (stored: string) => /^[0-9a-f]{64}$/.test(stored);

export async function hashPassword(password: string, salt: string, rounds = PBKDF2_ROUNDS): Promise<string> {
  const key = await pbkdf2Async(sha256, password.normalize('NFKC'), salt, { c: rounds, dkLen: 32, asyncTick: 8 });
  return `${PREFIX}$${rounds}$${bytesToHex(key)}`;
}

/** Compares two strings in time that does not depend on where they first differ. */
export function safeEqual(a: string, b: string): boolean {
  if (a.length !== b.length) return false;
  let diff = 0;
  for (let i = 0; i < a.length; i++) diff |= a.charCodeAt(i) ^ b.charCodeAt(i);
  return diff === 0;
}

/**
 * Checks a password against a stored hash. When it is right but the stored hash is old or weak, `upgrade`
 * holds the new hash the caller should save.
 */
export async function verifyPassword(password: string, account: { passwordHash: string; salt: string }): Promise<{ ok: boolean; upgrad
e?: string }> {
  const { passwordHash, salt } = account;
  if (isLegacyHash(passwordHash)) {
    const legacy = bytesToHex(sha256(utf8ToBytes(`${salt}:${password}`)));
    return safeEqual(legacy, passwordHash) ? { ok: true, upgrade: await hashPassword(password, salt) } : { ok: false };
  }
  const [prefix, roundsText] = passwordHash.split('$');
  const rounds = Number(roundsText);
  if (prefix !== PREFIX || !Number.isInteger(rounds) || rounds < 1 || rounds > MAX_ROUNDS) return { ok: false };
  const actual = await hashPassword(password, salt, rounds);
  if (!safeEqual(actual, passwordHash)) return { ok: false };
  return rounds < PBKDF2_ROUNDS ? { ok: true, upgrade: await hashPassword(password, salt) } : { ok: true };
}
```

src/utils/qr.ts

```
/** QR codes for invite links: friends point their phone camera at the screen and the link opens TogetherWeGo. */
import qrcode from 'qrcode-generator';

/** The library writes one byte per letter, which breaks accents and emoji; links with them need UTF-8. */
function utf8Bytes(text: string): number[] {
  const out: number[] = [];
  for (const ch of text) {
    const cp = ch.codePointAt(0) as number;
    if (cp < 0x80) out.push(cp);
    else if (cp < 0x800) out.push(0xc0 | (cp >> 6), 0x80 | (cp & 63));
    else if (cp < 0x10000) out.push(0xe0 | (cp >> 12), 0x80 | ((cp >> 6) & 63), 0x80 | (cp & 63));
    else out.push(0xf0 | (cp >> 18), 0x80 | ((cp >> 12) & 63), 0x80 | ((cp >> 6) & 63), 0x80 | (cp & 63));
  }
  return out;
}
qrcode.stringToBytes = utf8Bytes;

/** The code as a grid of squares: true is dark. Medium error correction keeps it readable with a smudged screen. */
export function qrMatrix(text: string): boolean[][] {
  const qr = qrcode(0, 'M'); // 0 = pick the smallest size that fits
  qr.addData(text);
  qr.make();
  const n = qr.getModuleCount();
  return Array.from({ length: n }, (_, row) => Array.from({ length: n }, (_, col) => qr.isDark(row, col)));
}

/** One SVG path that covers all dark squares (each row is drawn as a few long rectangles, not hundreds of tiny ones). */
export function qrPath(matrix: boolean[][]): string {
  let d = '';
  matrix.forEach((row, y) => {
    let x = 0;
    while (x < row.length) {
      if (!row[x]) {
        x++;
        continue;
      }
      let run = 1;
      while (x + run < row.length && row[x + run]) run++;
      d += `M${x} ${y}h${run}v1h-${run}z`;
      x += run;
    }
  });
  return d;
}
```

src/utils/time.ts

```
/**
 * Date & time-zone helpers. Trip dates are stored as plain 'YYYY-MM-DD' strings
 * in the destination's local time, flights as absolute UTC timestamps. Uses Intl
 * when available and falls back to a fixed-offset table.
 */

const FALLBACK_OFFSETS: Record<string, number> = {
  'Asia/Tokyo': 540, 'Asia/Seoul': 540, 'Asia/Shanghai': 480, 'Asia/Hong_Kong': 480, 'Asia/Taipei': 480,
  'Asia/Singapore': 480, 'Asia/Makassar': 480, 'Asia/Kuala_Lumpur': 480, 'Asia/Manila': 480,
  'Asia/Bangkok': 420, 'Asia/Jakarta': 420, 'Asia/Ho_Chi_Minh': 420, 'Asia/Kolkata': 330,
  'Indian/Maldives': 300, 'Asia/Dubai': 240, 'Asia/Qatar': 180, 'Europe/Istanbul': 180,
  'Europe/London': 0, 'Europe/Dublin': 0, 'Europe/Lisbon': 0, 'Atlantic/Reykjavik': 0,
  'Europe/Paris': 60, 'Europe/Rome': 60, 'Europe/Madrid': 60, 'Europe/Berlin': 60, 'Europe/Amsterdam': 60,
  'Europe/Zurich': 60, 'Europe/Prague': 60, 'Europe/Vienna': 60, 'Africa/Casablanca': 60,
  'Europe/Athens': 120, 'Africa/Johannesburg': 120, 'Africa/Cairo': 120, 'Africa/Nairobi': 180,
  'Australia/Sydney': 600, 'Australia/Melbourne': 600, 'Pacific/Auckland': 720,
  'America/New_York': -300, 'America/Toronto': -300, 'America/Detroit': -300, 'America/Chicago': -360,
  'America/Denver': -420, 'America/Edmonton': -420, 'America/Phoenix': -420,
  'America/Los_Angeles': -480, 'America/Vancouver': -480, 'Pacific/Honolulu': -600,
  'America/Mexico_City': -360, 'America/Cancun': -300, 'America/Lima': -300, 'America/Bogota': -300,
  'America/Sao_Paulo': -180, 'America/Argentina/Buenos_Aires': -180, 'America/Santiago': -240, UTC: 0,
};

const MONTHS = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec'];
const WEEKDAYS = ['Sun', 'Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat'];

export const YMD_RE = /^\d{4}-\d{2}-\d{2}$/;
export const HHMM_RE = /^([01]\d|2[0-3]):([0-5]\d)$/;

export function parseYMD(ymd: string) {
  const [y, m, d] = ymd.split('-').map(Number);
  return { y, m, d };
}

export function ymdFromUtcDate(date: Date): string {
  return `${date.getUTCFullYear()}-${String(date.getUTCMonth() + 1).padStart(2, '0')}-${String(date.getUTCDate()).padStart(2, '0')}`;
}

export function isValidYMD(ymd: string): boolean {
  if (!YMD_RE.test(ymd)) return false;
  const { y, m, d } = parseYMD(ymd);
  const dt = new Date(Date.UTC(y, m - 1, d));
  return dt.getUTCFullYear() === y && dt.getUTCMonth() === m - 1 && dt.getUTCDate() === d;
}

export function addDays(ymd: string, n: number): string {
  const { y, m, d } = parseYMD(ymd);
  return ymdFromUtcDate(new Date(Date.UTC(y, m - 1, d + n)));
}

export function daysBetween(a: string, b: string): number {
  const pa = parseYMD(a);
  const pb = parseYMD(b);
  return Math.round((Date.UTC(pb.y, pb.m - 1, pb.d) - Date.UTC(pa.y, pa.m - 1, pa.d)) / 86400000);
}

export function tripDays(start: string, end: string): string[] {
  const n = Math.max(0, daysBetween(start, end));
  return Array.from({ length: n + 1 }, (_, i) => addDays(start, i));
}

export function weekdayShort(ymd: string): string {
  const { y, m, d } = parseYMD(ymd);
  return WEEKDAYS[new Date(Date.UTC(y, m - 1, d)).getUTCDay()];
}

export function dayLabel(ymd: string): string {
  const { m, d } = parseYMD(ymd);
  return `${MONTHS[m - 1]} ${d}`;
}

export function rangeLabel(start: string, end: string): string {
  const a = parseYMD(start);
  const b = parseYMD(end);
  if (a.y === b.y) return `${MONTHS[a.m - 1]} ${a.d} – ${MONTHS[b.m - 1]} ${b.d}, ${b.y}`;
  return `${MONTHS[a.m - 1]} ${a.d}, ${a.y} – ${MONTHS[b.m - 1]} ${b.d}, ${b.y}`;
}

export function time12(hhmm: string): string {
  const [h, m] = hhmm.split(':').map(Number);
  const suffix = h >= 12 ? 'PM' : 'AM';
  const hr = h % 12 === 0 ? 12 : h % 12;
  return `${hr}:${String(m).padStart(2, '0')} ${suffix}`;
}
```

```
export function deviceTimeZone(): string {
  try {
    return Intl.DateTimeFormat().resolvedOptions().timeZone || 'UTC';
  } catch {
    return 'UTC';
  }
}

/** Minutes east of UTC for `tz` at the given instant. */
export function tzOffsetMinutes(tz: string, utcMs: number): number {
  try {
    const dtf = new Intl.DateTimeFormat('en-US', {
      timeZone: tz,
      hourCycle: 'h23',
      year: 'numeric',
      month: '2-digit',
      day: '2-digit',
      hour: '2-digit',
      minute: '2-digit',
      second: '2-digit',
    });
    const parts = dtf.formatToParts(new Date(utcMs));
    const get = (t: string) => Number(parts.find((p) => p.type === t)?.value);
    const asUtc = Date.UTC(get('year'), get('month') - 1, get('day'), get('hour') % 24, get('minute'), get('second'));
    const off = Math.round((asUtc - utcMs) / 60000);
    if (Number.isFinite(off)) return off;
  } catch {
    // fall through to table
  }
  if (tz === deviceTimeZone()) return -new Date(utcMs).getTimezoneOffset();
  return FALLBACK_OFFSETS[tz] ?? 0;
}

/** Converts a wall-clock time in `tz` to a UTC epoch (ms). */
export function zonedToUtc(ymd: string, hhmm: string, tz: string): number {
  const { y, m, d } = parseYMD(ymd);
  const [h, mi] = hhmm.split(':').map(Number);
  const guess = Date.UTC(y, m - 1, d, h, mi);
  const off1 = tzOffsetMinutes(tz, guess);
  let result = guess - off1 * 60000;
  const off2 = tzOffsetMinutes(tz, result);
  if (off2 !== off1) result = guess - off2 * 60000;
  return result;
}

/** Wall-clock parts of a UTC instant in a given zone. */
export function partsInTz(utcMs: number, tz: string) {
  const local = new Date(utcMs + tzOffsetMinutes(tz, utcMs) * 60000);
  return {
    ymd: ymdFromUtcDate(local),
    hhmm: `${String(local.getUTCHours()).padStart(2, '0')}:${String(local.getUTCMinutes()).padStart(2, '0')}`,
  };
}

export function formatInTz(utcMs: number, tz: string): string {
  const p = partsInTz(utcMs, tz);
  return `${dayLabel(p.ymd)} · ${p.hhmm}`;
}

export function utcOffsetLabel(tz: string, at = Date.now()): string {
  const off = tzOffsetMinutes(tz, at);
  const sign = off >= 0 ? '+' : '−';
  const h = Math.floor(Math.abs(off) / 60);
  const m = Math.abs(off) % 60;
  return `UTC${sign}${h}${m ? `:${String(m).padStart(2, '0')}` : ''}`;
}

const ABBREV: Record<string, string> = {
  'Asia/Tokyo': 'JST', 'Asia/Seoul': 'KST', 'Europe/London': 'GMT', 'Europe/Paris': 'CET',
  'America/New_York': 'ET', 'America/Chicago': 'CT', 'America/Denver': 'MT', 'America/Los_Angeles': 'PT',
  'Europe/Rome': 'CET', 'Europe/Madrid': 'CET', 'Europe/Berlin': 'CET', 'Europe/Amsterdam': 'CET', 'Europe/Prague': 'CET', 'Europe/Zuric
h': 'CET', 'Europe/Lisbon': 'WET',
  'Asia/Singapore': 'SGT', 'Asia/Bangkok': 'ICT', 'Australia/Sydney': 'AET', 'Asia/Kolkata': 'IST',
};

export function tzAbbrev(tz: string): string {
  const a = ABBREV[tz];
  if (!a) return tz.split('/').pop()?.replace(/_/g, ' ') ?? tz;
  // Summer time: Central European CET → CEST, UK GMT → BST, Western European WET → WEST.
  const summer = tzOffsetMinutes(tz, Date.now()) - tzOffsetMinutes('UTC', Date.now()) > ({ CET: 60, GMT: 0, WET: 0 }[a] ?? Infinity);
  return summer ? ({ CET: 'CEST', GMT: 'BST', WET: 'WEST' }[a] ?? a) : a;
}

/** Signed hour difference: positive when `tzA` is ahead of `tzB`. */
export function hoursAhead(tzA: string, tzB: string, at = Date.now()): number {
```

```
  return (tzOffsetMinutes(tzA, at) - tzOffsetMinutes(tzB, at)) / 60;
}

export function todayInTz(tz: string): string {
  return partsInTz(Date.now(), tz).ymd;
}

export function greetingFor(hour: number): string {
  if (hour < 5) return 'greet.night';
  if (hour < 12) return 'greet.morning';
  if (hour < 18) return 'greet.afternoon';
  return 'greet.evening';
}
```

src/utils/validation.ts

```
/**
 * Input validation used across forms. Each validator returns an error message
 * (string) or null when the value is valid. Checks are both syntactic
 * (format) and semantic (does the value make sense for a trip?).
 */
import { airportByIata } from '@/data/airports';
import { HHMM_RE, daysBetween, isValidYMD } from './time';

export type Validator = (value: string) => string | null;

export const required =
  (label: string): Validator =>
  (v) =>
    v.trim() ? null : `Enter ${label}`;

export const maxLen =
  (n: number, label = 'This'): Validator =>
  (v) =>
    v.trim().length > n ? `${label} must be ${n} characters or fewer` : null;

export const EMAIL_RE = /^[^\s@]+@[^\s@]+\.[^\s@]{2,}$/;

export function validateEmail(v: string): string | null {
  if (!v.trim()) return 'Enter your email address';
  if (!EMAIL_RE.test(v.trim())) return 'That email doesn’t look right (e.g. name@mail.com)';
  return null;
}

export function validatePassword(v: string): string | null {
  if (!v) return 'Enter a password';
  if (v.length < 8) return 'Use at least 8 characters';
  if (!/[A-Za-z]/.test(v) || !/\d/.test(v)) return 'Include at least one letter and one number';
  return null;
}

export function passwordStrength(v: string): { score: 0 | 1 | 2 | 3 | 4; label: string } {
  let score = 0;
  if (v.length >= 8) score++;
  if (v.length >= 12) score++;
  if (/[A-Z]/.test(v) && /[a-z]/.test(v)) score++;
  if (/\d/.test(v) && /[^A-Za-z0-9]/.test(v)) score++;
  const labels = ['Too weak', 'Weak', 'Okay', 'Strong', 'Very strong'];
  return { score: score as 0 | 1 | 2 | 3 | 4, label: labels[score] };
}

export function validateName(v: string): string | null {
  const t = v.trim();
  if (!t) return 'Enter a name';
  if (t.length < 2) return 'Name is too short';
  if (t.length > 40) return 'Keep names under 40 characters';
  if (!/^[\p{L}\p{M}'’ .-]+$/u.test(t)) return 'Names can only contain letters, spaces, hyphens and apostrophes';
  return null;
}

/**
 * Parses a money amount. Accepts "12", "12.50", "1,200" — rejects negatives,
 * letters and absurd values.
 */
export function parseAmount(v: string): number | null {
  const cleaned = v.replace(/,/g, '').trim();
  if (!/^\d+(\.\d{1,2})?$/.test(cleaned)) return null;
  const n = Number(cleaned);
  return Number.isFinite(n) ? n : null;
}

export function validateAmount(v: string, { min = 0.01, max = 1_000_000, label = 'an amount' } = {}): string | null {
  if (!v.trim()) return `Enter ${label}`;
  const n = parseAmount(v);
  if (n === null) return 'Use numbers only, e.g. 25 or 25.50';
  if (n < min) return min > 0 ? `Amount must be more than 0` : 'Amount can’t be negative';
```

```
  if (n > max) return `That’s more than ${max.toLocaleString()} — double-check the amount`;
  return null;
}

export function validateTime(v: string): string | null {
  return HHMM_RE.test(v) ? null : 'Use 24-hour time, e.g. 09:30';
}

export function validateTripDates(start: string, end: string): { start?: string; end?: string } {
  const errs: { start?: string; end?: string } = {};
  if (!isValidYMD(start)) errs.start = 'Pick a valid start date';
  if (!isValidYMD(end)) errs.end = 'Pick a valid end date';
  if (!errs.start && !errs.end) {
    const len = daysBetween(start, end);
    if (len < 0) errs.end = 'End date must be on or after the start date';
    else if (len > 60) errs.end = 'Trips can be at most 60 days long';
  }
  return errs;
}

export const INVITE_RE = /^[A-Z]{3}-\d{4}$/;

export function normalizeInvite(v: string): string {
  const raw = v.toUpperCase().replace(/[^A-Z0-9]/g, '');
  if (raw.length <= 3) return raw;
  return `${raw.slice(0, 3)}-${raw.slice(3, 7)}`;
}

export function validateInvite(v: string): string | null {
  if (!v.trim()) return 'Enter the code your friend shared';
  if (!INVITE_RE.test(v)) return 'Codes look like ABC-1234';
  return null;
}

export const FLIGHT_RE = /^[A-Z0-9]{2}\s?\d{1,4}[A-Z]?$/;

export function validateFlightNumber(v: string): string | null {
  const t = v.trim().toUpperCase();
  if (!t) return 'Enter the flight number';
  if (!FLIGHT_RE.test(t)) return 'Flight numbers look like NH 175 or UA837';
  return null;
}

export function validateIata(v: string, label: string): string | null {
  const t = v.trim().toUpperCase();
  if (!t) return `Enter the ${label} airport code`;
  if (!/^[A-Z]{3}$/.test(t)) return 'Airport codes are 3 letters, e.g. JFK';
  if (!airportByIata(t)) return `We don’t recognise “${t}” — try a major airport code`;
  return null;
}

/** Runs validators in order and returns the first error. */
export function firstError(value: string, ...validators: Validator[]): string | null {
  for (const v of validators) {
    const e = v(value);
    if (e) return e;
  }
  return null;
}
```

server/__tests__/server.test.js

```
/** @jest-environment node */
const fs = require('fs');
const http = require('http');
const os = require('os');
const path = require('path');

const { createServer } = require('../index');
const { SyncStore, SyncError } = require('../store');

const TRIP = 'trip_abcdefghijklmnop1234';
const stamp = (n, node = 'aaaa1111') => `${String(1800000000000 + n)}-0000-${node}`;
const op = (n, extra = {}) => ({ p: ['name'], v: `Name ${n}`, t: stamp(n), ...extra });

describe('SyncStore', () => {
  it('keeps an ordered log and hands out everything after a position', () => {
    const s = new SyncStore();
    expect(s.append(TRIP, 'ROM-1234', 'phoneA01', [op(1), op(2)])).toEqual({ seq: 2 });
    expect(s.append(TRIP, 'ROM-1234', 'phoneB01', [op(3)])).toEqual({ seq: 3 });
    const all = s.since(TRIP, 0, 'phoneA01');
    expect(all.ops.map((o) => o.v)).toEqual(['Name 1', 'Name 2', 'Name 3']);
    expect(s.since(TRIP, 2, 'phoneA01').ops.map((o) => o.v)).toEqual(['Name 3']);
    expect(s.since(TRIP, 3, 'phoneA01').ops).toEqual([]);
    expect(s.since(TRIP, 0, 'phoneA01', 2).ops).toHaveLength(2); // paging
  });
```

```
  it('looks up a trip by invite code and follows a changed code', () => {
    const s = new SyncStore();
    s.append(TRIP, 'ROM-1234', 'phoneA01', [op(1)]);
    expect(s.lookup('ROM-1234')).toBe(TRIP);
    expect(s.lookup('XXX-0000')).toBeNull();
    expect(s.lookup('not a code')).toBeNull();
    s.append(TRIP, 'ROM-9999', 'phoneA01', [op(2)]);
    expect(s.lookup('ROM-1234')).toBeNull();
    expect(s.lookup('ROM-9999')).toBe(TRIP);
  });

  it('rejects anything that is not a well-formed change', () => {
    const s = new SyncStore();
    const bad = [
      [],
      [{}],
      [{ p: [], v: 1, t: stamp(1) }],
      [{ p: [1], v: 1, t: stamp(1) }],
      [{ p: ['a'], v: 1, t: 'nope' }],
      [{ p: ['a'], t: stamp(1) }], // no value and no delete marker
      [{ p: ['a'], v: 1, d: 1, t: stamp(1) }], // both
      [{ p: ['a'], d: 2, t: stamp(1) }],
      [{ p: Array(9).fill('x'), v: 1, t: stamp(1) }],
      [{ p: ['a'], v: 'x'.repeat(5000), t: stamp(1) }],
    ];
    for (const ops of bad) expect(() => s.append(TRIP, null, 'phoneA01', ops)).toThrow(SyncError);
    expect(() => s.append('short', null, 'phoneA01', [op(1)])).toThrow(/trip id/);
    expect(() => s.append(TRIP, 'bad code', 'phoneA01', [op(1)])).toThrow(/invite code/);
    expect(() => s.append(TRIP, null, '!', [op(1)])).toThrow(/device id/);
    expect(() => s.append(TRIP, null, 'phoneA01', Array.from({ length: 501 }, (_, i) => op(i)))).toThrow(/Too many/);
    expect(s.trips.size).toBe(0);
  });

  it('stores deletes without a value', () => {
    const s = new SyncStore();
    s.append(TRIP, null, 'phoneA01', [{ p: ['tasks', 't1'], d: 1, t: stamp(1) }]);
    expect(s.since(TRIP, 0).ops[0]).toEqual({ p: ['tasks', 't1'], d: 1, t: stamp(1) });
  });

  it('wakes a waiting phone as soon as a change arrives, and gives up after the wait time', async () => {
    const s = new SyncStore();
    s.append(TRIP, null, 'phoneA01', [op(1)]);
    const started = Date.now();
    const waiting = s.wait(TRIP, 1, 5000);
    setTimeout(() => s.append(TRIP, null, 'phoneB01', [op(2)]), 30);
    await waiting;
    expect(Date.now() - started).toBeLessThan(1500);
    const t0 = Date.now();
    await s.wait(TRIP, 2, 60);
    expect(Date.now() - t0).toBeGreaterThanOrEqual(50);
  });

  it('counts the phones seen recently', () => {
    let clock = 1_000_000;
    const s = new SyncStore({ now: () => clock });
    s.append(TRIP, null, 'phoneA01', [op(1)]);
    s.append(TRIP, null, 'phoneB01', [op(2)]);
    expect(s.since(TRIP, 0, 'phoneC01').peers).toBe(3);
    clock += 60000; // a minute later only the phone that just asked is still "online"
    expect(s.since(TRIP, 0, 'phoneC01').peers).toBe(1);
  });

  it('survives a restart', () => {
    const file = path.join(fs.mkdtempSync(path.join(os.tmpdir(), 'twg-')), 'sync.json');
    const a = new SyncStore({ file });
    a.append(TRIP, 'ROM-1234', 'phoneA01', [op(1), op(2)]);
    a.flush();
    const b = new SyncStore({ file });
    expect(b.since(TRIP, 0).ops).toHaveLength(2);
    expect(b.lookup('ROM-1234')).toBe(TRIP);
  });
});

describe('files (photos, documents, receipts)', () => {
  const FILE = 'photo_abc12345';
  const fresh = (options) => {
    const s = new SyncStore(options);
    s.append(TRIP, null, 'phoneA01', [op(1)]); // a trip has to exist first
    return s;
  };

  it('keeps a file for a trip and gives the same bytes back', async () => {
    const s = fresh();
    const bytes = Buffer.from([0, 1, 2, 250, 251, 252, 255]);
    expect(await s.putBlob(TRIP, FILE, bytes)).toEqual({ size: 7 });
```

```
    expect(await s.getBlob(TRIP, FILE)).toEqual(bytes);
    expect(await s.getBlob(TRIP, 'someone_else_1')).toBeNull();
    expect(s.stats().files).toBe(1);
  });

  it('refuses unknown trips, bad ids, empty files and files that are too large', async () => {
    const s = fresh({ maxBlobBytes: 100 });
    await expect(s.putBlob('trip_zzzzzzzzzzzzzzzz9999', FILE, Buffer.from('x'))).rejects.toMatchObject({ status: 404 });
    await expect(s.putBlob(TRIP, '../escape', Buffer.from('x'))).rejects.toMatchObject({ status: 400 });
    await expect(s.putBlob(TRIP, 'a b', Buffer.from('x'))).rejects.toMatchObject({ status: 400 });
    await expect(s.putBlob(TRIP, FILE, Buffer.alloc(0))).rejects.toMatchObject({ status: 400 });
    await expect(s.putBlob(TRIP, FILE, 'not bytes')).rejects.toMatchObject({ status: 400 });
    await expect(s.putBlob(TRIP, FILE, Buffer.alloc(101))).rejects.toMatchObject({ status: 413 });
    expect(await s.getBlob(TRIP, '../escape')).toBeNull();
    expect(s.stats().files).toBe(0);
  });

  it('stops one trip from filling the server, and lets a file be replaced by a smaller one', async () => {
    const s = fresh({ maxBlobBytesPerTrip: 100 });
    await s.putBlob(TRIP, 'file_one_1', Buffer.alloc(60));
    await expect(s.putBlob(TRIP, 'file_two_2', Buffer.alloc(60))).rejects.toMatchObject({ status: 413 });
    await s.putBlob(TRIP, 'file_one_1', Buffer.alloc(30)); // same id: replaces, frees 30
    await s.putBlob(TRIP, 'file_two_2', Buffer.alloc(60));
    expect(s.trips.get(TRIP).blobBytes).toBe(90);
  });

  it('counts two uploads at the same moment against the limit', async () => {
    const s = fresh({ maxBlobBytesPerTrip: 100 });
    const results = await Promise.allSettled([s.putBlob(TRIP, 'file_one_1', Buffer.alloc(70)), s.putBlob(TRIP, 'file_two_2', Buffer.allo
c(70))]);
    expect(results.filter((r) => r.status === 'fulfilled')).toHaveLength(1);
    expect(s.trips.get(TRIP).blobBytes).toBe(70);
  });

  it('keeps files across a restart and ignores half-finished uploads', async () => {
    const dir = fs.mkdtempSync(path.join(os.tmpdir(), 'twg-'));
    const file = path.join(dir, 'sync.json');
    const a = fresh({ file });
    await a.putBlob(TRIP, FILE, Buffer.from('hello'));
    fs.writeFileSync(path.join(dir, 'files', TRIP, 'half_done_1.tmp'), 'junk'); // an interrupted upload
    a.flush();
    const b = new SyncStore({ file });
    expect(b.stats().files).toBe(1);
    expect(b.trips.get(TRIP).blobBytes).toBe(5);
    expect((await b.getBlob(TRIP, FILE)).toString()).toBe('hello');
  });
});

describe('HTTP API', () => {
  let server;
  let base;

  const call = (method, pathname, body) =>
    new Promise((resolve, reject) => {
      const data = body === undefined ? undefined : typeof body === 'string' ? body : JSON.stringify(body);
      const req = http.request(base + pathname, { method, headers: data ? { 'Content-Type': 'application/json', 'Content-Length': Buffe
r.byteLength(data) } : {} }, (res) => {
        let text = '';
        res.on('data', (c) => (text += c));
        res.on('end', () => resolve({ status: res.statusCode, headers: res.headers, json: text && res.headers['content-type']?.includes
('json') ? JSON.parse(text) : text }));
      });
      req.on('error', reject);
      if (data) req.write(data);
      req.end();
    });

  const callRaw = (method, pathname, body) =>
    new Promise((resolve, reject) => {
      const req = http.request(base + pathname, { method, headers: body ? { 'Content-Type': 'application/octet-stream', 'Content-Lengt
h': body.length } : {} }, (res) => {
        const chunks = [];
        res.on('data', (c) => chunks.push(c));
        res.on('end', () => resolve({ status: res.statusCode, headers: res.headers, body: Buffer.concat(chunks) }));
      });
      req.on('error', reject);
      if (body) req.write(body);
      req.end();
    });

  beforeEach(async () => {
    server = createServer(new SyncStore());
    await new Promise((r) => server.listen(0, '127.0.0.1', r));
    base = `http://127.0.0.1:${server.address().port}`;
  });
  afterEach(() => new Promise((r) => server.close(r)));
```

```
  it('says it is alive', async () => {
    const r = await call('GET', '/health');
    expect(r.status).toBe(200);
    expect(r.json).toMatchObject({ ok: true, trips: 0 });
    expect((await call('GET', '/')).json).toContain('sync server is running');
  });

  it('sends and receives changes, and looks up a code', async () => {
    const post = await call('POST', `/v1/trips/${TRIP}/ops`, { code: 'ROM-1234', client: 'phoneA01', ops: [op(1), op(2)] });
    expect(post).toMatchObject({ status: 200, json: { seq: 2 } });
    const get = await call('GET', `/v1/trips/${TRIP}/ops?since=1&client=phoneB01`);
    expect(get.status).toBe(200);
    expect(get.json.ops).toHaveLength(1);
    expect(get.json).toMatchObject({ seq: 2, peers: 2 });
    expect(get.headers['access-control-allow-origin']).toBe('*');
    expect((await call('GET', '/v1/codes/ROM-1234')).json).toEqual({ tripId: TRIP });
    expect((await call('GET', '/v1/codes/NOP-0000')).status).toBe(404);
  });

  it('answers the browser pre-flight request', async () => {
    const r = await call('OPTIONS', `/v1/trips/${TRIP}/ops`);
    expect(r.status).toBe(204);
    expect(r.headers['access-control-allow-methods']).toContain('POST');
    expect(r.headers['access-control-allow-methods']).toContain('PUT');
  });

  it('long-polling returns as soon as another phone posts', async () => {
    await call('POST', `/v1/trips/${TRIP}/ops`, { client: 'phoneA01', ops: [op(1)] });
    const started = Date.now();
    const waiting = call('GET', `/v1/trips/${TRIP}/ops?since=1&wait=8000&client=phoneB01`);
    setTimeout(() => call('POST', `/v1/trips/${TRIP}/ops`, { client: 'phoneA01', ops: [op(2)] }), 40);
    const r = await waiting;
    expect(Date.now() - started).toBeLessThan(2000);
    expect(r.json.ops.map((o) => o.v)).toEqual(['Name 2']);
  });

  it('returns clear errors', async () => {
    expect((await call('POST', `/v1/trips/${TRIP}/ops`, '{not json')).status).toBe(400);
    expect((await call('POST', `/v1/trips/${TRIP}/ops`, { client: 'phoneA01', ops: [{ p: [] }] })).status).toBe(400);
    expect((await call('GET', `/v1/trips/${TRIP}/ops?since=0`)).status).toBe(404);
    expect((await call('GET', '/v1/trips/short/ops')).status).toBe(400);
    expect((await call('GET', '/nothing')).status).toBe(404);
    expect((await call('POST', `/v1/trips/${TRIP}/ops`, { client: 'phoneA01', ops: [{ p: ['x'], v: 'y'.repeat(1_100_000), t: stamp(1) }]
})).status).toBe(413);
  });


  it('sends and receives files', async () => {
    await call('POST', `/v1/trips/${TRIP}/ops`, { client: 'phoneA01', ops: [op(1)] });
    const bytes = Buffer.from(Array.from({ length: 5000 }, (_, i) => i % 256));
    const put = await callRaw('PUT', `/v1/trips/${TRIP}/files/photo_abc12345`, bytes);
    expect(put.status).toBe(200);
    expect(JSON.parse(put.body.toString())).toEqual({ size: 5000 });
    const get = await callRaw('GET', `/v1/trips/${TRIP}/files/photo_abc12345`);
    expect(get.status).toBe(200);
    expect(get.body.equals(bytes)).toBe(true);
    expect(get.headers['cache-control']).toContain('immutable');
    expect(get.headers['access-control-allow-origin']).toBe('*');
    expect((await call('GET', '/health')).json.files).toBe(1);
  });

  it('answers files that are missing, misplaced or too large with clear errors', async () => {
    await call('POST', `/v1/trips/${TRIP}/ops`, { client: 'phoneA01', ops: [op(1)] });
    expect((await callRaw('GET', `/v1/trips/${TRIP}/files/nothing_here_1`)).status).toBe(404);
    expect((await callRaw('PUT', `/v1/trips/trip_unknownunknown123456/files/photo_abc12345`, Buffer.from('x'))).status).toBe(404);
    expect((await callRaw('PUT', `/v1/trips/${TRIP}/files/bad!id`, Buffer.from('x'))).status).toBe(400);
    expect((await callRaw('PUT', `/v1/trips/${TRIP}/files/photo_abc12345`, Buffer.alloc(9 * 1024 * 1024, 1))).status).toBe(413);
  });

  it('limits how fast invite codes can be guessed', async () => {
    let last;
    for (let i = 0; i < 61; i++) last = await call('GET', '/v1/codes/AAA-0000');
    expect(last.status).toBe(429);
  });
});
```

server/index.js

```
/**
 * TogetherWeGo sync server.  Run it with:  npm run server
 *
 * It lets friends' phones share one trip: chat, itinerary, votes, expenses... Phones send small changes here and
 * ask for everyone else's. There are no accounts. A trip is protected by its long random id, which a phone only
 * learns by entering the trip's invite code, so only run this on a network you trust (home wifi or a hotspot).
 *
```

```
 *   POST /v1/trips/:id/ops      { code, client, ops }        send changes
 *   GET  /v1/trips/:id/ops      ?since=N&wait=ms&client=ID   get changes (waits for new ones if there are none)
 *   PUT  /v1/trips/:id/files/:fileId   (raw bytes)           send a photo, document or receipt
 *   GET  /v1/trips/:id/files/:fileId                         get it back
 *   GET  /v1/codes/:code                                     invite code -> trip id
 *   GET  /health                                             is the server alive?
 */
const http = require('http');
const os = require('os');
const path = require('path');

const { SyncStore, SyncError } = require('./store');

const MAX_BODY = 1_000_000;
const MAX_FILE = 8 * 1024 * 1024;
const MAX_WAIT_MS = 25000;
const LOOKUPS_PER_MINUTE = 60;

function createServer(store, options = {}) {
  const lookups = new Map(); // ip -> { count, resetAt }
  const now = options.now || Date.now;

  function tooManyLookups(ip) {
    const entry = lookups.get(ip);
    if (!entry || entry.resetAt < now()) {
      lookups.set(ip, { count: 1, resetAt: now() + 60000 });
      return false;
    }
    entry.count += 1;
    return entry.count > LOOKUPS_PER_MINUTE;
  }

  function send(res, status, body) {
    const text = JSON.stringify(body);
    res.writeHead(status, {
      'Content-Type': 'application/json',
      'Content-Length': Buffer.byteLength(text),
      'Cache-Control': 'no-store',
      'Access-Control-Allow-Origin': '*',
      'Access-Control-Allow-Headers': 'content-type',
      'Access-Control-Allow-Methods': 'GET, POST, PUT, OPTIONS',
    });
    res.end(text);
  }

  function sendFile(res, bytes) {
    res.writeHead(200, {
      'Content-Type': 'application/octet-stream',
      'Content-Length': bytes.length,
      'Cache-Control': 'private, max-age=31536000, immutable', // a file never changes under its id
      'Access-Control-Allow-Origin': '*',
    });
    res.end(bytes);
  }

  /** The whole request body, or a 413 if it is bigger than `limit`. */
  function readBody(req, limit) {
    return new Promise((resolve, reject) => {
      let size = 0;
      let tooBig = false;
      const chunks = [];
      req.on('data', (c) => {
        size += c.length;
        if (size > limit * 4) return req.destroy(); // far too much: stop listening
        if (size > limit) tooBig = true; // keep reading (and ignoring) so the 413 answer can still reach the sender
        else chunks.push(c);
      });
      req.on('end', () => (tooBig ? reject(new SyncError(413, 'Request too large')) : resolve(Buffer.concat(chunks))));
      req.on('error', reject);
      req.on('close', () => reject(new SyncError(400, 'The request was cut off')));
    });
  }

  async function readJson(req) {
    const raw = await readBody(req, MAX_BODY);
    try {
      return JSON.parse(raw.toString('utf8') || '{}');
    } catch {
      throw new SyncError(400, 'Body must be JSON');
    }
  }

  return http.createServer(async (req, res) => {
    try {
      const url = new URL(req.url, 'http://localhost');
      const parts = url.pathname.split('/').filter(Boolean);
```

```
      if (req.method === 'OPTIONS') return send(res, 204, {});

      if (req.method === 'GET' && url.pathname === '/health') return send(res, 200, { ok: true, ...store.stats() });
      if (req.method === 'GET' && url.pathname === '/') {
        res.writeHead(200, { 'Content-Type': 'text/plain; charset=utf-8' });
        return res.end('TogetherWeGo sync server is running.\nIn the app: Settings, Sync across phones, enter this address.\n');
      }

      if (req.method === 'GET' && parts[0] === 'v1' && parts[1] === 'codes' && parts.length === 3) {
        const ip = req.socket.remoteAddress || '';
        if (tooManyLookups(ip)) throw new SyncError(429, 'Too many tries. Wait a minute.');
        const tripId = store.lookup(decodeURIComponent(parts[2]));
        if (!tripId) throw new SyncError(404, 'No trip with that code');
        return send(res, 200, { tripId });
      }

      if (parts[0] === 'v1' && parts[1] === 'trips' && parts[3] === 'ops' && parts.length === 4) {
        const tripId = parts[2];
        if (req.method === 'POST') {
          const body = await readJson(req);
          return send(res, 200, store.append(tripId, body.code, body.client, body.ops));
        }
        if (req.method === 'GET') {
          const since = Number(url.searchParams.get('since')) || 0;
          const wait = Math.min(Number(url.searchParams.get('wait')) || 0, MAX_WAIT_MS);
          const waiting = store.wait(tripId, since, wait);
          req.on('close', () => waiting.cancel && waiting.cancel()); // the phone hung up: stop waiting
          await waiting;
          return send(res, 200, store.since(tripId, since, url.searchParams.get('client')));
        }
      }
      if (parts[0] === 'v1' && parts[1] === 'trips' && parts[3] === 'files' && parts.length === 5) {
        const [tripId, fileId] = [parts[2], parts[4]];
        if (req.method === 'PUT') {
          const bytes = await readBody(req, MAX_FILE);
          return send(res, 200, await store.putBlob(tripId, fileId, bytes));
        }
        if (req.method === 'GET') {
          const bytes = await store.getBlob(tripId, fileId);
          if (!bytes) throw new SyncError(404, 'No such file');
          return sendFile(res, bytes);
        }
      }
      throw new SyncError(404, 'Not found');
    } catch (e) {
      if (e instanceof SyncError) return send(res, e.status, { error: e.message });
      console.error(e);
      return send(res, 500, { error: 'Server error' });
    }
  });
}

function lanAddresses() {
  return Object.values(os.networkInterfaces())
    .flat()
    .filter((i) => i && i.family === 'IPv4' && !i.internal)
    .map((i) => i.address);
}

if (require.main === module) {
  const port = Number(process.env.PORT) || 8787;
  const file = process.env.SYNC_DATA || path.join(__dirname, 'data', 'sync.json');
  const store = new SyncStore({ file });
  const server = createServer(store);
  server.listen(port, '0.0.0.0', () => {
    console.log('\nTogetherWeGo sync server is running.\n');
    console.log('Phones on the same wifi can use any of these addresses:');
    for (const ip of lanAddresses()) console.log(`   http://${ip}:${port}`);
    console.log(`   http://localhost:${port}   (this computer only)\n`);
    console.log(`Saving to ${file}\nPress Ctrl+C to stop.\n`);
  });
  const stop = () => {
    store.flush();
    process.exit(0);
  };
  process.on('SIGINT', stop);
  process.on('SIGTERM', stop);
}

module.exports = { createServer, lanAddresses };
```

server/store.js

```
/**
 * The sync server's memory: for every shared trip, an ordered log of changes ("ops") that phones send and
 * fetch. The server never needs to understand a change; phones merge them (see src/sync/crdt.ts), so this stays
```

```
 * small. Plain JavaScript with no dependencies so it runs anywhere Node runs.
 */
const fs = require('fs');
const path = require('path');

const TRIP_ID = /^[A-Za-z0-9_-]{16,64}$/;
const CODE = /^[A-Z]{3}-\d{4}$/;
const STAMP = /^\d{13}-[0-9a-z]{4}-\w{1,16}$/;
const CLIENT = /^\w{4,32}$/;
const MAX_OPS_PER_REQUEST = 500;
const MAX_OP_BYTES = 4000;
const MAX_OPS_PER_TRIP = 50000;
const PEER_WINDOW_MS = 45000;
const BLOB_ID = /^[A-Za-z0-9_-]{8,40}$/;
const MAX_BLOB_BYTES = 8 * 1024 * 1024;
const MAX_BLOB_BYTES_PER_TRIP = 150 * 1024 * 1024;

class SyncError extends Error {
  constructor(status, message) {
    super(message);
    this.status = status;
  }
}

/** Throws a SyncError for anything that is not a well-formed op. */
function checkOp(op) {
  if (!op || typeof op !== 'object') throw new SyncError(400, 'Each change must be an object');
  if (!Array.isArray(op.p) || op.p.length < 1 || op.p.length > 8 || op.p.some((s) => typeof s !== 'string' || s.length < 1 || s.length >
80)) {
    throw new SyncError(400, 'Bad change path');
  }
  if (typeof op.t !== 'string' || !STAMP.test(op.t)) throw new SyncError(400, 'Bad change stamp');
  if (op.d !== undefined && op.d !== 1) throw new SyncError(400, 'Bad delete marker');
  if (op.d === 1 && op.v !== undefined) throw new SyncError(400, 'A delete cannot carry a value');
  if (op.d !== 1 && op.v === undefined) throw new SyncError(400, 'A change needs a value');
  if (JSON.stringify(op).length > MAX_OP_BYTES) throw new SyncError(413, 'Change too large');
}

class SyncStore {
  /**
   * @param {{ file?: string, blobDir?: string, now?: () => number, maxBlobBytes?: number, maxBlobBytesPerTrip?: number }} [options]
   * Files (photos, documents, receipts) are kept next to the JSON file, or in memory when there is no file.
   */
  constructor(options = {}) {
    this.file = options.file;
    this.blobDir = options.blobDir || (options.file ? path.join(path.dirname(options.file), 'files') : null);
    this.limits = { blob: options.maxBlobBytes || MAX_BLOB_BYTES, perTrip: options.maxBlobBytesPerTrip || MAX_BLOB_BYTES_PER_TRIP };
    this.memBlobs = new Map();
    this.now = options.now || Date.now;
    /** @type {Map<string, { ops: object[], code: string | null, created: number, clients: Map<string, number>, blobs: Map<string, numbe
r>, blobBytes: number }>} */
    this.trips = new Map();
    /** @type {Map<string, string>} */
    this.codes = new Map();
    this.waiters = new Map();
    this.saveTimer = null;
    if (this.file) this.load();
  }

  trip(tripId, create) {
    if (!TRIP_ID.test(tripId)) throw new SyncError(400, 'Bad trip id');
    let t = this.trips.get(tripId);
    if (!t && create) {
      t = { ops: [], code: null, created: this.now(), clients: new Map(), blobs: new Map(), blobBytes: 0 };
      this.trips.set(tripId, t);
    }
    return t;
  }

  /** Adds changes from a phone. Returns the new log length (the "sequence number" phones remember). */
  append(tripId, code, client, ops) {
    if (!Array.isArray(ops) || ops.length === 0) throw new SyncError(400, 'Send at least one change');
    if (ops.length > MAX_OPS_PER_REQUEST) throw new SyncError(413, 'Too many changes in one request');
    if (!CLIENT.test(client || '')) throw new SyncError(400, 'Bad device id');
    if (code !== undefined && code !== null && !CODE.test(code)) throw new SyncError(400, 'Bad invite code');
    ops.forEach(checkOp);
    const t = this.trip(tripId, true);
    if (t.ops.length + ops.length > MAX_OPS_PER_TRIP) throw new SyncError(413, 'This trip has reached the server limit');
    for (const op of ops) t.ops.push({ p: op.p, t: op.t, ...(op.d ? { d: 1 } : { v: op.v }) });
    if (code && t.code !== code) {
      if (t.code) this.codes.delete(t.code);
      t.code = code;
      this.codes.set(code, tripId);
    }
    t.clients.set(client, this.now());
    this.wake(tripId);
```

```
    this.scheduleSave();
    return { seq: t.ops.length };
  }

  /** Changes after position `since`. */
  since(tripId, since, client, limit = 1000) {
    const t = this.trip(tripId, false);
    if (!t) throw new SyncError(404, 'Unknown trip');
    if (client && CLIENT.test(client)) t.clients.set(client, this.now());
    const from = Math.max(0, Math.min(Number(since) || 0, t.ops.length));
    return { ops: t.ops.slice(from, from + limit), seq: Math.min(t.ops.length, from + limit), total: t.ops.length, peers: this.peers(t)
};
  }

  /**
   * Waits until the trip has changes after `since` (or `ms` pass), so phones hear about changes almost instantly.
   * The returned promise has a `cancel()` for when the phone hangs up.
   */
  wait(tripId, since, ms) {
    const t = this.trip(tripId, false);
    if (!t) throw new SyncError(404, 'Unknown trip');
    if (t.ops.length > since || ms <= 0) return Promise.resolve();
    let finish;
    const promise = new Promise((resolve) => {
      const set = this.waiters.get(tripId) || new Set();
      this.waiters.set(tripId, set);
      const timer = setTimeout(() => finish(), ms);
      finish = () => {
        clearTimeout(timer);
        set.delete(finish);
        resolve();
      };
      set.add(finish);
    });
    promise.cancel = finish;
    return promise;
  }

  wake(tripId) {
    for (const done of [...(this.waiters.get(tripId) || [])]) done();
  }

  /** Keeps one file (a photo, document or receipt) for a trip. Sending the same id again replaces it. */
  async putBlob(tripId, blobId, bytes) {
    const t = this.trip(tripId, false);
    if (!t) throw new SyncError(404, 'Unknown trip');
    if (!BLOB_ID.test(blobId || '')) throw new SyncError(400, 'Bad file id');
    if (!Buffer.isBuffer(bytes) || bytes.length === 0) throw new SyncError(400, 'The file is empty');
    if (bytes.length > this.limits.blob) throw new SyncError(413, 'That file is too large to share');
    const before = t.blobs.get(blobId);
    if (t.blobBytes - (before || 0) + bytes.length > this.limits.perTrip) throw new SyncError(413, 'This trip has reached the server’s f
ile limit');
    t.blobBytes += bytes.length - (before || 0); // counted first, so two uploads at once cannot both slip under the limit
    t.blobs.set(blobId, bytes.length);
    try {
      if (this.blobDir) {
        const dir = path.join(this.blobDir, tripId);
        await fs.promises.mkdir(dir, { recursive: true });
        const tmp = path.join(dir, `${blobId}.tmp`);
        await fs.promises.writeFile(tmp, bytes);
        await fs.promises.rename(tmp, path.join(dir, blobId));
      } else {
        this.memBlobs.set(`${tripId}/${blobId}`, bytes);
      }
    } catch (e) {
      t.blobBytes += (before || 0) - bytes.length;
      if (before) t.blobs.set(blobId, before);
      else t.blobs.delete(blobId);
      throw e;
    }
    return { size: bytes.length };
  }

  /** A stored file, or null when there is none with that id. */
  async getBlob(tripId, blobId) {
    const t = this.trip(tripId, false);
    if (!t) throw new SyncError(404, 'Unknown trip');
    if (!BLOB_ID.test(blobId || '') || !t.blobs.has(blobId)) return null;
    if (!this.blobDir) return this.memBlobs.get(`${tripId}/${blobId}`) || null;
    try {
      return await fs.promises.readFile(path.join(this.blobDir, tripId, blobId));
    } catch (e) {
      if (e.code === 'ENOENT') return null;
      throw e;
    }
  }
```

```
  lookup(code) {
    return CODE.test(code || '') ? this.codes.get(code) || null : null;
  }

  peers(t) {
    const cutoff = this.now() - PEER_WINDOW_MS;
    let n = 0;
    for (const seen of t.clients.values()) if (seen >= cutoff) n++;
    return n;
  }

  stats() {
    const all = [...this.trips.values()];
    return { trips: this.trips.size, ops: all.reduce((a, t) => a + t.ops.length, 0), files: all.reduce((a, t) => a + t.blobs.size, 0) };
  }

  // ---- saving to a plain JSON file, so a restart loses nothing ----
  load() {
    try {
      const data = JSON.parse(fs.readFileSync(this.file, 'utf8'));
      for (const [id, t] of Object.entries(data.trips || {})) {
        this.trips.set(id, { ops: t.ops, code: t.code, created: t.created, clients: new Map(), ...this.indexBlobs(id) });
        if (t.code) this.codes.set(t.code, id);
      }
    } catch (e) {
      if (e.code !== 'ENOENT') console.error('Could not read', this.file, e.message);
    }
  }

  /** Which files are on disk for a trip (after a restart), and how big they are. */
  indexBlobs(tripId) {
    const blobs = new Map();
    let blobBytes = 0;
    if (this.blobDir) {
      try {
        for (const name of fs.readdirSync(path.join(this.blobDir, tripId))) {
          if (!BLOB_ID.test(name)) continue; // leftovers of an interrupted upload end in .tmp
          const size = fs.statSync(path.join(this.blobDir, tripId, name)).size;
          blobs.set(name, size);
          blobBytes += size;
        }
      } catch (e) {
        if (e.code !== 'ENOENT') console.error('Could not read files of', tripId, e.message);
      }
    }
    return { blobs, blobBytes };
  }

  scheduleSave() {
    if (!this.file || this.saveTimer) return;
    this.saveTimer = setTimeout(() => {
      this.saveTimer = null;
      this.flush();
    }, 300);
    if (this.saveTimer.unref) this.saveTimer.unref();
  }

  flush() {
    if (!this.file) return;
    const data = { trips: {} };
    for (const [id, t] of this.trips) data.trips[id] = { ops: t.ops, code: t.code, created: t.created };
    fs.mkdirSync(path.dirname(this.file), { recursive: true });
    const tmp = `${this.file}.tmp`;
    fs.writeFileSync(tmp, JSON.stringify(data));
    fs.renameSync(tmp, this.file);
  }
}

module.exports = { SyncStore, SyncError, TRIP_ID, CODE, BLOB_ID };
```

e2e/README.md

```
# Browser tests

These tests open the **web build** of the app in a phone-sized browser window and use it like a person would. They sit next to the unit
tests (`npm test`), which cover the logic without a screen.

```bash
npm install
npx playwright-core install chromium   # once: downloads a browser (or set E2E_CHROME to your Chrome)
npm run e2e                            # builds the web app, serves it, runs everything
npm run e2e -- --no-build              # reuse the last build
npm run e2e -- twophones overflow      # only some scripts
```
```

```
| Script | What it checks |
|---|---|
| `verify.js` | Home, Plan filters and Lock in, Budget (expenses, receipt, settle up), Chat and Scout AI, the More menu |
| `verify2.js` | Crew roles, polls and checklist, flights and validation, packing, recap, language switch, trip phases, joining a trip |
| `verify3.js` | Sign-up, the personalisation survey, themes, the first-run tour |
| `verify4.js` | The 360° city views: spin, day and night, landmarks, city picker |
| `motion.js` | The phone's *Reduce Motion* setting: nothing hidden, nothing moving, Settings says why |
| `overflow.js` | **Every screen** at 390 px, 320 px and with larger text: nothing may stick out sideways |
| `nesting.js` | **Every screen**: no button may sit inside another button (React's *"`<button>` cannot contain a nested `<button>`"* warning) |
| `twophones.js` | Two phones and the real sync server: share, join from a QR code, chat, expenses, offline catch-up, restart, photo, receipt and PDF |

Each script prints `PASS`/`FAIL` lines and `errors:` (browser console errors). `run.js` adds them up and exits with a non-zero code if anything failed.

The web build cannot show the native map, camera, notifications or file system; the on-phone parts of those features were built for iOS but not run on a physical phone.
```

e2e/lib.js

```
// Shared helpers for the browser tests: launch a browser, open the app as a phone-sized screen, sign in as the demo user.
const path = require('path');
const { chromium } = require('playwright-core');

/** Where the exported web app is being served (run.js starts it for you). */
exports.BASE = process.env.E2E_URL || 'http://localhost:8081';
exports.fixture = (name) => path.join(__dirname, 'fixtures', name);
exports.shot = (name) => path.join(__dirname, 'shots', name);
/** Set E2E_CHROME to a Chrome/Chromium path, or run `npx playwright-core install chromium` once. */
exports.launch = () => chromium.launch(process.env.E2E_CHROME ? { executablePath: process.env.E2E_CHROME } : {});

/** Every screen of the app (route paths). */
exports.ROUTES = ['', 'explore', 'itinerary', 'budget', 'chat', 'profile', 'about', 'airport', 'bucket-list', 'crew', 'currency', 'destinations', 'documents', 'emergency', 'flights', 'help', 'how-it-works', 'join', 'map', 'nearby', 'new-trip', 'notifications', 'packing', 'personalize', 'photos', 'phrasebook', 'quiz', 'recap', 'settings', 'trips', 'weather', 'destination/tokyo', 'sign-up', 'sign-in'];

/** Buttons and links that sit inside another button or link on the current screen, as `"outer" contains "inner"`. */
exports.nestedControls = (p) =>
  p.evaluate(() => {
    const SEL = 'button, a, [role=button], [role=link]';
    const name = (el) => (el.getAttribute('aria-label') || el.textContent || el.tagName).trim().slice(0, 40);
    const out = [];
    for (const inner of document.querySelectorAll(SEL)) {
      const outer = inner.parentElement && inner.parentElement.closest(SEL);
      if (outer) out.push(`"${name(outer)}" contains "${name(inner)}"`);
    }
    return [...new Set(out)];
  });

/** `beforeSignIn(page)` runs on the sign-in screen, before the demo account is used. */
exports.start = async ({ beforeSignIn } = {}) => {
  const b = await exports.launch();
  const ctx = await b.newContext({ viewport: { width: 390, height: 844 }, deviceScaleFactor: 2, isMobile: true, hasTouch: true });
  const p = await ctx.newPage();
  const errors = [];
  // a join that tries the (not running) sync server on :8787 and falls back to this phone logs one harmless "connection refused"
  p.on('console', (m) => { if (m.type() === 'error' && !m.text().includes('ERR_CERT') && !m.location().url.includes(':8787')) errors.push(m.text().slice(0, 400)); });
  p.on('pageerror', (e) => errors.push('PAGEERROR ' + e.message.slice(0, 400)));
  await p.goto(exports.BASE + '/', { waitUntil: 'networkidle' });
  await p.waitForTimeout(2000);
  if (await p.getByRole('button', { name: 'Skip' }).count()) { await p.getByRole('button', { name: 'Skip' }).click(); await p.waitForTimeout(1000); }
  if (beforeSignIn) await beforeSignIn(p);
  await p.getByLabel('Use the demo account').click(); await p.waitForTimeout(300);
  await p.getByTestId('sign-in').click(); await p.waitForTimeout(2500);
  if (!process.env.KEEP_TOUR && await p.getByRole('button', { name: 'Skip' }).count()) { await p.getByRole('button', { name: 'Skip' }).first().click(); await p.waitForTimeout(600); }
  return { b, p, errors };
};
```

e2e/motion.js

```
const { BASE, launch } = require('./lib');
// The phone's "Reduce Motion" setting is respected: nothing hidden, nothing animating, and Settings says why.
const results = [];
const check = (name, ok, extra = '') => results.push(`${ok ? 'PASS' : 'FAIL'}  ${name}${ok ? '' : '  ' + extra}`);
const vis = async (loc, t = 3000) => loc.filter({ visible: true }).first().waitFor({ state: 'visible', timeout: t }).then(() => true).catch(() => false);

async function open(b, reducedMotion) {
  const ctx = await b.newContext({ viewport: { width: 390, height: 844 }, isMobile: true, hasTouch: true, reducedMotion });
  const p = await ctx.newPage();
  await p.goto(BASE + '/', { waitUntil: 'networkidle' });
  await p.waitForTimeout(1500);
```

```
  if (await p.getByRole('button', { name: 'Skip' }).count()) await p.getByRole('button', { name: 'Skip' }).click();
  await p.waitForTimeout(500);
  await p.getByLabel('Use the demo account').click();
  await p.getByTestId('sign-in').click();
  await p.waitForTimeout(2500);
  if (await p.getByRole('button', { name: 'Skip' }).count()) await p.getByRole('button', { name: 'Skip' }).first().click().catch(() =>
{});
  return p;
}
const go = async (p, r) => { await p.goto(BASE + '/' + r, { waitUntil: 'networkidle' }); await p.waitForTimeout(1500); if (await p.getBy
Role('button', { name: 'Skip' }).count()) await p.getByRole('button', { name: 'Skip' }).first().click().catch(() => {}); };

(async () => {
  const b = await launch();
  // ---- reduce motion ON
  const calm = await open(b, 'reduce');
  await go(calm, '');
  const headlineOpacity = await calm.getByRole('heading', { name: /to go|Living|Tomorrow|epic/ }).first().evaluate((el) => getComputedSt
yle(el.closest('[style*="opacity"]') || el).opacity).catch(() => 'n/a');
  check('home headline is fully visible with Reduce Motion on', headlineOpacity === '1' || headlineOpacity === 'n/a', headlineOpacity);
  check('home content shows', await vis(calm.getByText(/to go|Living|Tomorrow/)));
  await go(calm, 'settings');
  check('settings says why animations are off', await vis(calm.getByText(/Reduce Motion setting is on/)));
  await go(calm, 'panorama?city=paris');
  check('360 view starts paused', await vis(calm.getByLabel('Auto rotate').getByText('Paused')));
  const before = await calm.getByText(/360° ·/).first().textContent();
  await calm.waitForTimeout(4000);
  const after = await calm.getByText(/360° ·/).first().textContent();
  check('360 view does not turn by itself', before === after, `${before} -> ${after}`);
  await calm.getByLabel('Auto rotate').click();
  check('tapping Auto explains why it stays off', await vis(calm.getByText(/Animations are off/)));
  await calm.getByLabel('Look right').click();
  await calm.waitForTimeout(150);
  const turned = await calm.getByText(/360° ·/).first().textContent();
  check('the arrows still turn the view at once', turned !== before, `${before} -> ${turned}`);

  // ---- reduce motion OFF: the same screens animate
  const lively = await open(b, 'no-preference');
  await go(lively, 'settings');
  check('settings offers animations normally', await vis(lively.getByText(/Flying plane, floating cards/)));
  await go(lively, 'panorama?city=paris');
  check('360 view auto-rotates by default', await vis(lively.getByLabel('Auto rotate').getByText('Auto')));
  console.log(results.join('\n'));
  await b.close();
})().catch((e) => { console.log(results.join('\n')); console.error('CRASH', e.message.slice(0, 500)); process.exit(1); });
```

e2e/nesting.js

```
// Looks for a button inside another button (or a link inside a link) on every screen. A browser lets you do it, React warns about it
// ("<button> cannot contain a nested <button>"), and keyboards and screen readers cannot use such a control properly.
const { BASE, ROUTES, start, nestedControls } = require('./lib');

(async () => {
  const failures = [];
  let screens = 0;
  const audit = async (page, where) => {
    screens++;
    for (const f of await nestedControls(page)) failures.push(`FAIL  ${where}: ${f}`);
  };

  const { b, p, errors } = await start({ beforeSignIn: (page) => audit(page, 'the sign-in screen') });
  const open = async (path) => {
    await p.goto(BASE + '/' + path, { waitUntil: 'load' });
    await p.waitForTimeout(1200);
  };

  for (const r of [...ROUTES, 'panorama']) {
    await open(r);
    await audit(p, '/' + r);
  }

  // screens that only show something after a tap (each step waits for what it expects, so a page that did not change is not audited by
mistake)
  const expect = (label) => p.getByLabel(label).first().waitFor({ state: 'visible', timeout: 5000 });
  await open('itinerary');
  await p.getByLabel(/, \d+ of \d+ in$/).first().click();
  await expect('Assign');
  await audit(p, 'an opened activity card');
  await p.getByRole('tab', { name: /Vote/ }).click();
  await expect(/^Lock in /);
  await audit(p, 'the Vote tab');
  await open('personalize');
  await p.getByLabel('Skip to the end').click();
  await expect('Apply my style');
  await audit(p, 'the last page of Make it yours');
  await open('survey');
```

```
  for (let i = 0; i < 5; i++) { await p.locator('button').filter({ hasText: /\p{Extended_Pictographic}/u }).first().click(); await p.wai
tForTimeout(250); }
  await p.getByText('Your matches').first().waitFor({ state: 'visible', timeout: 5000 });
  await audit(p, 'the survey matches');

  console.log(failures.length ? failures.join('\n') : `PASS  no button or link sits inside another one (${screens} screens and states ch
ecked)`);

  // React itself only complains in a development build (point E2E_URL at `npx expo start --web`), so this part is skipped on the export
ed app
  if (await p.evaluate(() => typeof __DEV__ !== 'undefined' && __DEV__)) {
    const warned = errors.filter((e) => /cannot contain a nested|cannot be a descendant of/.test(e));
    console.log(warned.length ? warned.map((w) => `FAIL  React warned: ${w.replace(/\s+/g, ' ').slice(0, 160)}`).join('\n') : 'PASS  Rea
ct printed no invalid-nesting warnings');
  }

  // prove the check can fail: plant a button inside a button and look again
  await p.evaluate(() => {
    const outer = document.createElement('button'); outer.setAttribute('aria-label', 'planted outer');
    const inner = document.createElement('button'); inner.setAttribute('aria-label', 'planted inner');
    outer.appendChild(inner); document.body.appendChild(outer);
  });
  const seen = await nestedControls(p);
  console.log(`${seen.some((s) => s.includes('planted outer') && s.includes('planted inner')) ? 'PASS' : 'FAIL'}  the check notices a bu
tton planted inside a button`);
  await b.close();
})().catch((e) => { console.error('CRASH', e.message.slice(0, 500)); process.exit(1); });
```

e2e/overflow.js

```
// Visits every screen at two phone widths and reports anything that sticks out sideways (the "page scrolls horizontally" bug).
const { BASE, start, ROUTES: routes } = require('./lib');
const passes = [{ w: 390, scale: 1 }, { w: 320, scale: 1 }, { w: 360, scale: 1.25 }]; // the last one is the "larger text" setting
const SKIP = new Set(['panorama']); // the 360° scene is meant to be wider than the screen

(async () => {
  const { b, p } = await start();
  let bad = 0;
  for (const { w, scale } of passes) {
    await p.setViewportSize({ width: w, height: 760 });
    await p.evaluate((sc) => { const j = JSON.parse(localStorage.getItem('togetherwego-v1')); j.state.settings.textScale = sc; localStor
age.setItem('togetherwego-v1', JSON.stringify(j)); }, scale);
    for (const r of routes) {
      if (SKIP.has(r)) continue;
      await p.goto(BASE + '/' + r, { waitUntil: 'load' });
      await p.waitForTimeout(1300);
      if (await p.getByRole('button', { name: 'Skip' }).count()) await p.getByRole('button', { name: 'Skip' }).first().click().catch(()
=> {});
      const res = await p.evaluate((vw) => {
        // not window.innerWidth: a mobile browser widens its layout viewport to fit whatever sticks out
        const doc = document.scrollingElement || document.documentElement;
        const scrollers = [...document.querySelectorAll('*')].filter((el) => /(auto|scroll)/.test(getComputedStyle(el).overflowX));
        const inScroller = (el) => scrollers.some((s) => s !== el && s.contains(el));
        const clipped = (el) => { for (let n = el.parentElement; n; n = n.parentElement) { const cs = getComputedStyle(n); if ((cs.overf
lowX === 'hidden' || cs.overflow === 'hidden') && n.getBoundingClientRect().right <= vw + 1) return true; } return false; };
        const out = [];
        for (const el of document.querySelectorAll('body *')) {
          const rc = el.getBoundingClientRect();
          if (rc.width < 2 || rc.right <= vw + 1 || getComputedStyle(el).visibility === 'hidden' || inScroller(el) || clipped(el)) conti
nue;
          out.push(`${el.tagName.toLowerCase()}[${(el.getAttribute('aria-label') || el.textContent || '').trim().slice(0, 30)}] w=${Mat
h.round(rc.width)} right=${Math.round(rc.right)}`);
          if (out.length >= 3) break;
        }
        return { pageScroll: doc.scrollWidth > vw + 1, scrollWidth: doc.scrollWidth, vw, out };
      }, w);
      if (res.pageScroll || res.out.length) { bad++; console.log(`FAIL  ${w}px x${scale} /${r}  page ${res.scrollWidth}>${res.vw}: ${re
s.pageScroll}  ${res.out.join(' | ')}`); }
    }
  }
  console.log(bad ? `${bad} screens overflow` : 'no screen overflows at 390 px, 320 px or with larger text');
  await b.close();
})().catch((e) => { console.error('CRASH', e.message.slice(0, 500)); process.exit(1); });
```

e2e/run.js

```
// Runs the browser tests: exports the web app, serves it, runs each script, prints a summary.
//   node e2e/run.js                 build + run everything
//   node e2e/run.js --no-build      reuse the last build (e2e/.dist)
//   node e2e/run.js verify twophones   only these scripts
// Needs a browser: run `npx playwright-core install chromium` once, or set E2E_CHROME to a Chrome path.
const { spawn, spawnSync } = require('child_process');
const fs = require('fs');
const path = require('path');
const { serve } = require('./serve');
```

```
const root = path.join(__dirname, '..');
const dist = path.join(__dirname, '.dist');
const port = Number(process.env.E2E_PORT) || 8081;
const args = process.argv.slice(2);
const wanted = args.filter((a) => !a.startsWith('--'));
const ALL = ['verify', 'verify2', 'verify3', 'verify4', 'motion', 'overflow', 'nesting', 'twophones'];
const LIMIT_MS = { overflow: 600000, twophones: 400000 }; // anything else gets 5 minutes

function run(script) {
  return new Promise((resolve) => {
    const child = spawn(process.execPath, [path.join(__dirname, `${script}.js`)], { cwd: __dirname, env: { ...process.env, E2E_URL: `htt
p://localhost:${port}` } });
    let out = '';
    child.stdout.on('data', (d) => (out += d));
    child.stderr.on('data', (d) => (out += d));
    let timedOut = false;
    const timer = setTimeout(() => {
      timedOut = true;
      child.kill();
    }, LIMIT_MS[script] ?? 300000);
    child.on('close', () => {
      clearTimeout(timer);
      resolve(timedOut ? `${out}\nCRASH ${script} did not finish in time` : out);
    });
  });
}

(async () => {
  fs.mkdirSync(path.join(__dirname, 'shots'), { recursive: true });
  if (!args.includes('--no-build')) {
    console.log('Exporting the web app…');
    const npx = process.platform === 'win32' ? 'npx.cmd' : 'npx';
    const r = spawnSync(npx, ['expo', 'export', '--platform', 'web', '--output-dir', dist], { cwd: root, stdio: 'inherit', shell: proces
s.platform === 'win32' });
    if (r.status !== 0) process.exit(1);
  }
  const server = await serve(dist, port);
  let pass = 0;
  let fail = 0;
  let crashed = 0;
  for (const script of ALL.filter((s) => !wanted.length || wanted.includes(s))) {
    console.log(`\n=== ${script}`);
    const out = await run(script);
    console.log(out.trim());
    pass += (out.match(/^PASS /gm) || []).length;
    fail += (out.match(/^FAIL /gm) || []).length;
    crashed += /^CRASH/m.test(out) ? 1 : 0;
  }
  server.close();
  console.log(`\n${pass} passed, ${fail} failed${crashed ? `, ${crashed} script(s) crashed` : ''}`);
  process.exit(fail || crashed ? 1 : 0);
})();
```

e2e/serve.js

```
// A tiny static file server for the exported web app (every unknown path falls back to index.html, like a single-page app host).
const http = require('http');
const fs = require('fs');
const path = require('path');

const types = { '.html': 'text/html', '.js': 'application/javascript', '.css': 'text/css', '.png': 'image/png', '.jpg': 'image/jpeg', '.
ttf': 'font/ttf', '.json': 'application/json', '.ico': 'image/x-icon', '.svg': 'image/svg+xml' };

exports.serve = (root, port) =>
  new Promise((resolve) => {
    const server = http.createServer((req, res) => {
      let file = path.join(root, decodeURIComponent(req.url.split('?')[0]));
      if (!file.startsWith(root) || !fs.existsSync(file) || fs.statSync(file).isDirectory()) file = path.join(root, 'index.html');
      res.writeHead(200, { 'Content-Type': types[path.extname(file)] || 'application/octet-stream' });
      fs.createReadStream(file).pipe(res);
    });
    server.listen(port, () => resolve(server));
  });
```

e2e/twophones.js

```
// Two separate "phones" (two browser profiles) + one real sync server on :8787.
const { spawn } = require('child_process');
const fs = require('fs');
const path = require('path');
const { BASE, fixture, shot, start } = require('./lib');
const jsQR = require('jsqr');

const results = [];
const check = (name, ok, extra = '') => results.push(`${ok ? 'PASS' : 'FAIL'}  ${name}${ok ? '' : '  ' + extra}`);
const vis = async (loc, t = 6000) => loc.filter({ visible: true }).first().waitFor({ state: 'visible', timeout: t }).then(() => true).ca
```

```
tch(() => false);
const gone = async (loc, t = 800) => !(await vis(loc, t));
const ROOT = path.join(__dirname, '..');
const DATA = '/tmp/twophones-sync.json';
let server;
const url = (r = '') => BASE + '/' + r;
/** With DOCS_SHOTS=1 the pictures used in the README are rewritten (docs/screenshots). */
const docShot = (page, name) => (process.env.DOCS_SHOTS ? page.screenshot({ path: path.join(ROOT, 'docs', 'screenshots', name), type: 'j
peg', quality: 80 }) : null);

const go = async (p, r) => {
  await p.goto(url(r), { waitUntil: 'load' }); // not 'networkidle': a synced trip always has one long-poll request open
  await p.waitForTimeout(2200);
  if (await p.getByRole('button', { name: 'Skip' }).count()) await p.getByRole('button', { name: 'Skip' }).first().click().catch(() =>
{});
};

async function health() {
  try {
    const r = await fetch('http://localhost:8787/health');
    return r.ok ? await r.json() : null;
  } catch {
    return null;
  }
}

/** Reads the QR code that is showing on a page the way a phone camera would: draw the SVG, decode the pixels. */
async function decodeQr(p) {
  const pixels = await p.evaluate(async () => {
    const svg = document.querySelector('svg[aria-label^="QR code"]');
    if (!svg) return null;
    const img = new Image();
    img.src = '' + btoa(new XMLSerializer().serializeToString(svg));
    await img.decode();
    const canvas = document.createElement('canvas');
    canvas.width = canvas.height = 600;
    const ctx = canvas.getContext('2d');
    ctx.fillStyle = '#fff';
    ctx.fillRect(0, 0, 600, 600);
    ctx.drawImage(img, 0, 0, 600, 600);
    return Array.from(ctx.getImageData(0, 0, 600, 600).data);
  });
  return pixels ? jsQR(Uint8ClampedArray.from(pixels), 600, 600)?.data ?? null : null;
}

async function say(p, text) {
  await p.getByTestId('chat-input').fill(text);
  await p.getByTestId('chat-send').click();
  await p.waitForTimeout(300);
}

(async () => {
  if (await health()) throw new Error('Something is already listening on :8787 - stop it first so the test uses a fresh server');
  try { fs.unlinkSync(DATA); } catch {}
  server = spawn('node', ['server/index.js'], { cwd: ROOT, env: { ...process.env, PORT: '8787', SYNC_DATA: DATA }, stdio: 'ignore' });
  for (let i = 0; i < 30 && !(await health()); i++) await new Promise((r) => setTimeout(r, 200));
  check('sync server is up', !!(await health()));

  // ---------------------------------------------------------------- phone A: the demo account, shares the trip
  const A = await start(); // signs in as the demo account
  const pa = A.p;
  const errorsA = A.errors;
  await go(pa, 'crew');
  check('A: crew has the sync card', await vis(pa.getByText('Sync across phones')));
  await pa.getByRole('button', { name: 'Share this trip across phones' }).click();
  check('A: sharing toast', await vis(pa.getByText(/Sharing! Friends can now join/)));
  check('A: badge goes Live', await vis(pa.getByText(/^Live/)));
  check('server now holds one trip', (await health())?.trips === 1);

  // ---------------------------------------------------------------- phone B: a brand-new account
  const ctxB = await A.b.newContext({ viewport: { width: 390, height: 844 }, deviceScaleFactor: 2, isMobile: true, hasTouch: true });
  const pb = await ctxB.newPage();
  const errorsB = [];
  pb.on('console', (m) => { if (m.type() === 'error') errorsB.push(m.text().slice(0, 300)); });
  pb.on('pageerror', (e) => errorsB.push('PAGEERROR ' + e.message.slice(0, 300)));
  await pb.goto(url(), { waitUntil: 'load' });
  await pb.waitForTimeout(2000);
  await pb.getByRole('button', { name: 'Skip' }).click();
  await pb.waitForTimeout(700);
  await pb.getByRole('link', { name: 'Sign Up' }).click();
  await pb.waitForTimeout(900);
  await pb.getByPlaceholder('e.g. Alex Rivera').fill('Jordan Smith');
  await pb.getByPlaceholder('you@email.com').fill('jordan.two@school.org');
  await pb.getByPlaceholder('At least 8 characters').fill('Trip2027go');
  await pb.getByPlaceholder('Re-enter password').fill('Trip2027go');
  await pb.getByRole('checkbox').click();
```

```
  await pb.getByRole('button', { name: 'Create account' }).click();
  await pb.waitForTimeout(2500);
  await pb.getByLabel('Skip to the end').click().catch(() => {});
  await pb.waitForTimeout(500);
  await pb.getByRole('button', { name: 'Apply my style' }).click().catch(() => {});
  await pb.waitForTimeout(2500);
  check('B: has no trip yet', await vis(pb.getByText(/Plan your first group trip/), 4000));

  // A shows the invite QR code; B "scans" it (same decoding a phone camera does) and opens the link
  await pa.getByTestId('invite-qr').click();
  await pa.waitForTimeout(800);
  const link = await decodeQr(pa);
  check('A: the QR code decodes to the invite link', link === `${BASE}/join?code=AML-2026`, String(link));
  await docShot(pa, '23-invite-qr.jpg');
  await pa.keyboard.press('Escape');

  // wrong code first
  await go(pb, 'join');
  await pb.getByTestId('invite-code').fill('ZZZ-9999');
  await pb.getByTestId('join-submit').click();
  check('B: unknown code says no', await vis(pb.getByText(/No trip found with that code/)));
  // an invite that names a sync server: the person is told before it is used
  await pb.goto(`${link}&server=${encodeURIComponent('http://192.168.1.20:8787')}`, { waitUntil: 'load' });
  await pb.waitForTimeout(1500);
  check('B: the link fills in the code', (await pb.getByTestId('invite-code').inputValue()) === 'AML-2026');
  check('B: a named sync server is announced', await vis(pb.getByText(/sync server at 192\.168\.1\.20:8787/)));
  await pb.goto(link, { waitUntil: 'load' });
  await pb.waitForTimeout(1500);
  check('B: opening the link fills in the code', (await pb.getByTestId('invite-code').inputValue()) === 'AML-2026');
  await pb.getByTestId('join-submit').click();
  check('B: join toast says it syncs live', await vis(pb.getByText(/now syncs live/)));
  await pb.waitForTimeout(1200);

  // B sees A's trip
  await go(pb, 'chat');
  check('B: sees A’s earlier chat messages', await vis(pb.getByText('Villa is booked! 🏡 Paid the deposit.')));
  check('B: its own join message/member is there', await vis(pb.getByText(/Jordan/)));
  if (process.env.DEBUG) {
    for (const t of [0, 4000, 12000]) {
      await pb.waitForTimeout(t);
      const badge = async (p) => (await p.locator('body').innerText()).split('\n').filter((l) => /Live|Offline|Connecting|phones/.test
(l)).join(' | ');
      console.log(`t+${t}: A[${await badge(pa)}] B[${await badge(pb)}]`);
    }
  }
  check('B: Live · 2 phones', await vis(pb.getByText(/Live · 2 phones/)));
  check('A: Live · 2 phones', await (async () => { await go(pa, 'chat'); return vis(pa.getByText(/Live · 2 phones/)); })());

  // ---------------------------------------------------------------- live chat both ways
  await say(pa, 'Hello from phone A');
  check('chat A→B', await vis(pb.getByText('Hello from phone A')));
  await say(pb, 'Hi from phone B');
  check('chat B→A', await vis(pa.getByText('Hi from phone B')));
  check('no fake friend replies in a synced trip', await gone(pa.getByText(/typing/i), 1500));

  // ---------------------------------------------------------------- expense B→A
  await go(pb, 'budget');
  await pb.getByTestId('add-expense-open').click();
  await pb.waitForTimeout(600);
  await pb.getByTestId('expense-title').fill('Taxi to the pier');
  await pb.getByTestId('expense-amount').fill('48');
  await pb.getByTestId('add-expense').click();
  await pb.waitForTimeout(800);
  await go(pa, 'budget');
  check('expense B→A', await vis(pa.getByText('Taxi to the pier')));

  // ---------------------------------------------------------------- files travel too: photo, receipt, document
  const choose = async (p, locator, file) => {
    const [fc] = await Promise.all([p.waitForEvent('filechooser', { timeout: 5000 }).catch(() => null), locator.click()]);
    if (fc) await fc.setFiles(file);
    return !!fc;
  };
  const until = async (fn, ms = 15000) => { const end = Date.now() + ms; while (Date.now() < end) { if (await fn().catch(() => false)) r
eturn true; await new Promise((r) => setTimeout(r, 300)); } return false; };

  await go(pa, 'photos');
  check('A: file chooser opens for photos', await choose(pa, pa.getByRole('button', { name: /Add photos/ }), fixture('receipt.jpg')));
  await pa.waitForTimeout(800);
  await pa.getByPlaceholder('Say something about it…').fill('Sunset from the villa');
  await pa.getByRole('button', { name: /Post to photo dump/ }).click();
  check('A: photo posted', await vis(pa.getByLabel('Sunset from the villa')));
  check('server received the photo file', await until(async () => (await health())?.files >= 1));
  await go(pb, 'photos');
  check('B: sees the photo tile', await vis(pb.getByLabel('Sunset from the villa')));
  check('B: the picture itself arrives', await until(async () => (await pb.locator('[aria-label="Sunset from the villa"] img').count())
> 0));
```

```
  await pb.screenshot({ path: shot('two-B-photos.png') });
  await docShot(pb, '24-sync-photos.jpg');

  await go(pb, 'budget');
  await pb.getByTestId('add-expense-open').click();
  await pb.waitForTimeout(600);
  await pb.getByTestId('expense-title').fill('Pier ferry');
  await pb.getByTestId('expense-amount').fill('36');
  check('B: file chooser opens for a receipt', await choose(pb, pb.getByRole('button', { name: /Attach receipt/ }), fixture('receipt.jp
g')));
  await pb.waitForTimeout(900);
  await pb.getByTestId('add-expense').click();
  await pb.waitForTimeout(800);
  check('server received the receipt too', await until(async () => (await health())?.files >= 2));
  await go(pa, 'budget');
  await pa.getByText('Pier ferry').first().click();
  const gotReceipt = await until(async () => (await pa.locator('[aria-label="Enlarge receipt"] img').count()) > 0);
  check('A: receipt photo arrives', gotReceipt);
  if (!gotReceipt) console.log('DEBUG has-receipt icons:', await pa.getByLabel('Has receipt').count(), '\n', (await pa.locator('body').i
nnerText()).replace(/\n+/g, ' | ').slice(-900));
  await pa.screenshot({ path: shot('two-A-receipt.png') });

  await go(pa, 'documents');
  check('A: file chooser opens for a document', await choose(pa, pa.getByRole('button', { name: 'File', exact: true }), fixture('tiny.pd
f')));
  check('A: document saved', await vis(pa.getByText(/Saved to the vault/)));
  check('server received the document', await until(async () => (await health())?.files >= 3));
  await go(pb, 'documents');
  check('B: sees the document', await vis(pb.getByLabel('Open tiny')));
  check('B: its pill says Offline once the file arrived', await until(async () => (await pb.getByLabel('Open tiny').getByText('Offlin
e').count()) > 0));
  await pb.getByLabel('Open tiny').click();
  check('B: can open or share the file', await vis(pb.getByRole('button', { name: 'Open / share file' })));
  await pb.keyboard.press('Escape');

  // ---------------------------------------------------------------- A goes offline, writes, restarts, comes back
  await go(pa, 'chat');
  await go(pb, 'chat');
  await A.b.contexts()[0].setOffline(true);
  await say(pa, 'Written with no signal');
  check('A: offline pill with 1 waiting', await vis(pa.getByText(/Offline · \d+ waiting|Offline/), 12000));
  check('B does not have it yet', await gone(pb.getByText('Written with no signal'), 1500));
  await pa.waitForTimeout(1200); // let the debounced save reach storage
  await A.b.contexts()[0].setOffline(false);
  check('B gets it after A is back', await vis(pb.getByText('Written with no signal'), 25000));
  check('A badge is Live again', await vis(pa.getByText(/^Live/), 15000));

  // ---------------------------------------------------------------- B closes the app (reload) and writes while closed state is restore
d
  await pb.reload({ waitUntil: 'load' });
  await pb.waitForTimeout(2500);
  await go(pb, 'chat');
  check('B after restart: still synced, same chat', await vis(pb.getByText('Hello from phone A')));
  await say(pa, 'Message while B restarted');
  check('B after restart receives live', await vis(pb.getByText('Message while B restarted'), 12000));

  // ---------------------------------------------------------------- screenshots
  await go(pa, 'chat');
  await pa.waitForTimeout(500);
  await pa.screenshot({ path: shot('two-A-chat.png') });
  await docShot(pa, '22-sync-chat.jpg');
  await pb.screenshot({ path: shot('two-B-chat.png') });
  await go(pa, 'crew');
  await pa.screenshot({ path: shot('two-A-crew.png') });

  console.log(results.join('\n'));
  const e = [...errorsA, ...errorsB].filter((m) => !/ERR_INTERNET_DISCONNECTED|ERR_NETWORK_CHANGED|Failed to load resource/.test(m));
  console.log('errors:', e.length ? e.join('\n') : 'none');
  await A.b.close();
  server.kill('SIGTERM');
})().catch((e) => {
  server?.kill('SIGTERM');
  console.log(results.join('\n'));
  console.error('CRASH', String(e.stack || e).slice(0, 900));
  process.exit(1);
});
```

e2e/verify.js

```
const { fixture, start } = require('./lib');
const results = [];
const check = (name, ok, extra = '') => { results.push(`${ok ? 'PASS' : 'FAIL'}  ${name}${ok ? '' : '  ' + extra}`); };
const vis = async (loc, t = 2500) => loc.filter({ visible: true }).first().waitFor({ state: 'visible', timeout: t }).then(() => true).ca
tch(() => false);
const tab = (p, n) => p.getByRole('tab', { name: n, exact: true }).click().then(() => p.waitForTimeout(1100));
(async () => {
```

```
  const { b, p, errors } = await start();
  // ---- Home
  check('home headline countdown', await vis(p.getByText(/days to go/i)));
  check('home weather tile', await vis(p.getByLabel(/Weather in Positano/)));
  check('home budget card', await vis(p.getByText('$2,086')));
  check('home quick tile Vote (2)', await vis(p.getByLabel('Vote (2)')));
  await p.getByLabel(/Showing (night|day)/).click(); await p.waitForTimeout(500);
  check('day/night toggle flips', await vis(p.getByLabel(/Showing (night|day)\. Tap for/)));
  await p.getByLabel(/Invite code AML-2026/).click(); await p.waitForTimeout(500);
  check('invite code copy toast', await vis(p.getByText(/Invite code AML-2026 copied/)));
  // ---- Plan
  await tab(p, 'Plan');
  check('plan lists confirmed dinner', await vis(p.getByText('Dinner at Da Vincenzo')));
  check('plan hides voting items', !(await vis(p.getByText('Limoncello tasting'), 600)));
  await p.getByRole('button', { name: /Food$/ }).first().click().catch(() => {}); await p.waitForTimeout(400);
  check('category filter hides hike', !(await vis(p.getByText('Path of the Gods hike'), 600)));
  await p.getByRole('button', { name: 'All', exact: true }).first().click(); await p.waitForTimeout(300);
  await p.getByRole('button', { name: /^Under 1 hour/ }).click(); await p.waitForTimeout(400);
  check('time filter under 1 hour keeps the 30-minute check-in', await vis(p.getByText('Check-in — Villa Margherita')));
  check('time filter under 1 hour hides the 2-hour dinner', !(await vis(p.getByText('Dinner at Da Vincenzo'), 600)));
  await p.getByRole('button', { name: /Any length/ }).click(); await p.waitForTimeout(300);
  await p.getByRole('tab', { name: /Vote/ }).click(); await p.waitForTimeout(600);
  check('vote tab lists limoncello', await vis(p.getByText('Limoncello tasting')));
  await p.getByRole('button', { name: /Lock in Limoncello/ }).click(); await p.waitForTimeout(500);
  check('lock in toast', await vis(p.getByText(/Locked in: Limoncello/)));
  check('vote count drops to 1', await vis(p.getByRole('tab', { name: /Vote \(1\)/ })));
  await p.getByLabel(/I’m in\. 0 of 4/).first().click(); await p.waitForTimeout(400);
  check('vote pill toggles to 1/4', await vis(p.getByLabel(/I’m in\. 1 of 4/)));
  await p.getByRole('tab', { name: 'Itinerary' }).click(); await p.waitForTimeout(500);
  check('locked item now in itinerary', await vis(p.getByText('Limoncello tasting')));
  // add activity validation
  await p.getByLabel('Add activity').first().click(); await p.waitForTimeout(800);
  await p.getByTestId('add-activity-submit').click(); await p.waitForTimeout(400);
  check('add-activity validation errors', await vis(p.getByText(/required|enter|name/i)));
  await p.getByTestId('activity-name').fill('Gelato run');
  await p.getByTestId('activity-location').fill('Positano centre');
  await p.getByTestId('add-activity-submit').click(); await p.waitForTimeout(1000);
  check('activity added', await vis(p.getByText('Gelato run')));
  await p.getByLabel('Delete Gelato run').click(); await p.waitForTimeout(500);
  await p.getByRole('button', { name: 'Delete', exact: true }).last().click(); await p.waitForTimeout(600);
  check('activity deleted', !(await vis(p.getByText('Gelato run'), 600)));
  // reminder bell
  await p.getByLabel(/Turn reminder (on|off)/).first().click().catch(() => {}); await p.waitForTimeout(500);
  check('reminder toast', await vis(p.getByText(/Reminder (on|turned off)/)));
  // ---- Budget
  await tab(p, 'Budget');
  check('budget per person', await vis(p.getByText('$522')));
  check('budget daily allowance', await vis(p.getByText('$319')));
  check('who owes who: treasurer gets back', await vis(p.getByText(/gets back \$1,129/)));
  await p.getByRole('tab', { name: /Local/ }).click(); await p.waitForTimeout(500);
  check('local view shows euros', await vis(p.getByText(/€/)));
  await p.getByRole('tab', { name: /^Home ·/ }).click();
  await p.getByTestId('add-expense-open').click(); await p.waitForTimeout(700);
  await p.getByTestId('add-expense').click(); await p.waitForTimeout(400);
  check('expense validation', await vis(p.getByText(/what it was for|Enter an amount|amount/i)));
  await p.getByTestId('expense-title').fill('Gelato');
  await p.getByTestId('expense-amount').fill('12.5');
  const [fc] = await Promise.all([p.waitForEvent('filechooser', { timeout: 4000 }).catch(() => null), p.getByRole('button', { name: /Att
ach receipt/ }).click()]);
  if (fc) { await fc.setFiles(fixture('receipt.jpg')); await p.waitForTimeout(900); }
  check('receipt attach thumbnail', !!fc && await vis(p.getByLabel(/receipt/i), 1500));
  await p.getByTestId('add-expense').click(); await p.waitForTimeout(800);
  check('expense added toast', await vis(p.getByText(/Added \$13|Added \$12/)));
  await p.getByLabel('Settle up').first().click().catch(async () => { await p.getByRole('button', { name: /Settle up/ }).first().click
(); }); await p.waitForTimeout(600);
  check('settle sheet lists payments', await vis(p.getByText(/pays/)));
  await p.getByRole('button', { name: 'Mark paid' }).first().click(); await p.waitForTimeout(500);
  check('settle recorded', await vis(p.getByText(/Payment recorded/)));
  await p.keyboard.press('Escape');
  // ---- Chat
  await tab(p, 'Chat');
  await p.getByTestId('chat-input').fill('Anyone up for gelato?'); await p.getByTestId('chat-send').click(); await p.waitForTimeout(50
0);
  check('crew message sent', await vis(p.getByText('Anyone up for gelato?')));
  await p.getByRole('tab', { name: 'Scout AI' }).click(); await p.waitForTimeout(500);
  await p.getByTestId('chat-input').fill('Who owes who?'); await p.getByTestId('chat-send').click(); await p.waitForTimeout(2200);
  check('scout ai answers offline', await vis(p.getByText(/owes|gets|square/i)));
  check('ai message not in crew tab', await (async () => { await p.getByRole('tab', { name: 'Crew' }).click(); await p.waitForTimeout(50
0); return !(await vis(p.getByText('Who owes who?'), 600)); })());
  // ---- More
  await tab(p, 'More');
  for (const [label, expect] of [['Crew & roles', /Invite code|CREW/i], ['Flights & arrivals', /Flights/i], ['Packing list', /Packing/
i], ['Trip recap', /Recap/i]]) {
    await p.getByRole('button', { name: new RegExp(label) }).first().click(); await p.waitForTimeout(1000);
    check(`more → ${label}`, await vis(p.getByText(expect)));
    await p.goBack(); await p.waitForTimeout(900);
```

```
    if (!(await vis(p.getByText('Maya Chen'), 800))) { await tab(p, 'More'); }
  }
  console.log(results.join('\n'));
  console.log('errors:', errors.length ? errors.join('\n') : 'none');
  await b.close();
})().catch((e) => { console.log(results.join('\n')); console.error('CRASH', e.message.slice(0, 500)); process.exit(1); });
```

e2e/verify2.js

```
const { BASE, start } = require('./lib');
const results = [];
const check = (name, ok, extra = '') => { results.push(`${ok ? 'PASS' : 'FAIL'}  ${name}${ok ? '' : '  ' + extra}`); };
const vis = async (loc, t = 2500) => loc.filter({ visible: true }).first().waitFor({ state: 'visible', timeout: t }).then(() => true).ca
tch(() => false);
const go = async (p, r) => { await p.goto(BASE + '/' + r, { waitUntil: 'networkidle' }); await p.waitForTimeout(1500); if (await p.getBy
Role('button', { name: 'Skip' }).count()) await p.getByRole('button', { name: 'Skip' }).first().click().catch(() => {}); };
(async () => {
  let { b, p, errors } = await start();
  // ---- Crew hub
  await go(p, 'crew');
  await p.getByLabel(/Diego Alvarez, Food Lead/).click(); await p.waitForTimeout(600);
  await p.getByRole('button', { name: 'Navigator' }).last().click().catch(() => {}); await p.waitForTimeout(500);
  check('role change toast', await vis(p.getByText(/Diego is now Navigator/)));
  await p.keyboard.press('Escape'); await p.waitForTimeout(300);
  await p.getByLabel(/Fornillo \(quiet, pebbles\)/).click().catch(() => {}); await p.waitForTimeout(500);
  check('poll vote registers', await vis(p.getByLabel(/Fornillo \(quiet, pebbles\), .*your vote/)));
  await p.getByLabel('Add task').click(); await p.waitForTimeout(400);
  check('empty task error', await vis(p.getByText(/enter|task|empty|required/i)));
  await p.getByLabel('New task').fill('Buy sunscreen'); await p.getByLabel('Add task').click(); await p.waitForTimeout(500);
  check('task added', await vis(p.getByText('Buy sunscreen')));
  await p.getByRole('button', { name: /New code/ }).click(); await p.waitForTimeout(500);
  await p.getByRole('button', { name: /^(Create|New code|Confirm|Yes)/ }).last().click().catch(() => {}); await p.waitForTimeout(600);
  // add member validation
  await p.getByText('+ Add').first().click(); await p.waitForTimeout(500);
  await p.getByRole('button', { name: 'Add member' }).click(); await p.waitForTimeout(400);
  check('add member validation', await vis(p.getByText(/name/i).filter({ hasText: /enter|required|letters/i })) || await vis(p.getByText
(/Enter .*name|required/i)));
  await p.keyboard.press('Escape');
  // ---- Flights
  await go(p, 'flights');
  check('flights Sam shows no flight', await vis(p.getByText('No flight added yet')));
  await p.getByTestId('add-flight').click(); await p.waitForTimeout(700);
  await p.getByRole('button', { name: 'Save flight' }).click(); await p.waitForTimeout(400);
  check('flight validation', await vis(p.getByText(/required|enter|airline/i)));
  await p.getByPlaceholder('ANA').fill('Delta'); await p.getByPlaceholder('NH 175').fill('DL 99'); await p.getByPlaceholder('JFK').fill
('XXX');
  await p.getByRole('button', { name: 'Save flight' }).click(); await p.waitForTimeout(400);
  check('unknown IATA rejected', await vis(p.getByText(/airport|IATA|recogni/i)));
  await p.getByPlaceholder('JFK').fill('BOS');
  await p.getByRole('button', { name: 'Save flight' }).click(); await p.waitForTimeout(900);
  check('flight saved', await vis(p.getByText(/Flight saved/)) || await vis(p.getByText('Delta DL 99')));
  // ---- Packing
  await go(p, 'packing');
  await p.getByRole('button', { name: /Split the rest fairly/ }).click(); await p.waitForTimeout(700);
  check('packing split toast', await vis(p.getByText(/Assigned \d+ item/)));
  await p.getByRole('tab', { name: 'Mine' }).click(); await p.waitForTimeout(500);
  check('mine filter shows my items', await vis(p.getByText(/Me$/)));
  // ---- Recap
  await go(p, 'recap');
  check('recap story card', await vis(p.getByLabel(/Story card/)));
  await p.getByRole('button', { name: /Share story card/ }).click(); await p.waitForTimeout(700);
  check('share sheet opens', await vis(p.getByText(/WhatsApp/)));
  await p.keyboard.press('Escape');
  // ---- How it works + help
  await go(p, 'how-it-works');
  check('how it works rubric map', await vis(p.getByText(/Rubric/i)));
  // ---- Language
  await go(p, 'settings');
  await p.getByRole('button', { name: /Español/ }).click(); await p.waitForTimeout(700);
  await go(p, ''); // settings has no tab bar: look at the tabs on Home
  await p.getByRole('tab', { name: 'Inicio' }).waitFor({ timeout: 3000 }).then(() => check('language switches tabs', true)).catch(() =>
check('language switches tabs', false));
  await go(p, 'settings');
  await p.getByRole('button', { name: /English/ }).click().catch(() => {}); await p.waitForTimeout(600);
  // ---- Demo timeline: in progress -> Today card
  await go(p, 'settings');
  await p.mouse.move(195, 500); await p.mouse.wheel(0, 1500); await p.waitForTimeout(500);
  await p.getByRole('button', { name: /in progress/i }).first().click().catch(() => {}); await p.waitForTimeout(500);
  await p.getByRole('button', { name: /Rebuild|Reset|Confirm|Apply/ }).last().click().catch(() => {}); await p.waitForTimeout(800);
  await go(p, '');
  check('active trip headline', await vis(p.getByText(/living it/i)));
  check('today card on home', await vis(p.getByText(/Today · Day \d of 6/)));
  // ---- Join
  await go(p, 'join');
  await p.getByRole('button', { name: /^Join/ }).first().click().catch(() => {});
  await p.getByPlaceholder(/ABC|TKY|code/i).first().fill('SEO-5521').catch(() => {});
```

```
  await p.getByRole('button', { name: /Join trip|Join$/ }).last().click().catch(() => {}); await p.waitForTimeout(1200);
  check('join SEO-5521', await vis(p.getByText(/Seoul/)));
  console.log(results.join('\n'));
  console.log('errors:', errors.length ? errors.join('\n') : 'none');
  await b.close();
})().catch((e) => { console.log(results.join('\n')); console.error('CRASH', e.message.slice(0, 600)); process.exit(1); });
```

e2e/verify3.js

```
const { BASE, launch, shot } = require('./lib');
const results = [];
const check = (name, ok, extra = '') => { results.push(`${ok ? 'PASS' : 'FAIL'}  ${name}${ok ? '' : '  ' + extra}`); };
const vis = async (loc, t = 2500) => loc.filter({ visible: true }).first().waitFor({ state: 'visible', timeout: t }).then(() => true).ca
tch(() => false);
(async () => {
  const b = await launch();
  const ctx = await b.newContext({ viewport: { width: 390, height: 844 }, isMobile: true, hasTouch: true });
  const p = await ctx.newPage(); const errs = []; p.on('pageerror', (e) => errs.push(e.message));
  await p.goto(BASE + '/', { waitUntil: 'networkidle' }); await p.waitForTimeout(2000);
  await p.getByRole('button', { name: 'Skip' }).click(); await p.waitForTimeout(700);
  // sign-up → survey prompt
  await p.getByRole('link', { name: 'Sign Up' }).click(); await p.waitForTimeout(900);
  await p.getByPlaceholder('e.g. Alex Rivera').fill('Jordan Smith'); await p.getByPlaceholder('you@email.com').fill('jordan@school.or
g');
  await p.getByPlaceholder('At least 8 characters').fill('Trip2027go'); await p.getByPlaceholder('Re-enter password').fill('Trip2027g
o');
  await p.getByRole('checkbox').click(); await p.getByRole('button', { name: 'Create account' }).click(); await p.waitForTimeout(2500);
  check('new account lands in survey', await vis(p.getByText(/make it\s+yours/i)));
  await p.getByRole('button', { name: 'Let’s go' }).click(); await p.waitForTimeout(600);
  await p.getByLabel('Sunset theme').click();
  await p.getByRole('button', { name: 'Continue' }).click(); await p.waitForTimeout(500);
  await p.getByRole('button', { name: 'Continue' }).click(); await p.waitForTimeout(500);
  check('survey needs an interest', await vis(p.getByText(/at least one interest/)));
  await p.getByRole('checkbox', { name: 'Beaches' }).click();
  await p.getByLabel('Skip to the end').click(); await p.waitForTimeout(600);
  // the last page lists every answer; tapping a row opens that step, and "Back to summary" (or the back arrow) returns
  check('last page lists the answers as tappable rows', await vis(p.getByLabel(/Change interests, now .*🏖/)));
  await p.getByLabel('Change interests').click(); await p.waitForTimeout(500);
  check('a summary row opens its step', await vis(p.getByRole('checkbox', { name: 'Nightlife' })));
  await p.getByRole('checkbox', { name: 'Nightlife' }).click();
  await p.getByRole('button', { name: 'Back to summary' }).click(); await p.waitForTimeout(600);
  check('"Back to summary" returns with the change applied', await vis(p.getByLabel(/Change interests, now .*🏖.*🌃/)));
  await p.getByLabel('Change card shape').click(); await p.waitForTimeout(500);
  await p.getByRole('radio', { name: 'Bubbly corners' }).click();
  await p.getByRole('button', { name: 'Back', exact: true }).click(); await p.waitForTimeout(600);
  check('the back arrow also returns to the summary', await vis(p.getByLabel(/Change card shape, now Bubbly corners/)));
  await p.getByRole('button', { name: 'Apply my style' }).click(); await p.waitForTimeout(5500);
  const saved = await p.evaluate(() => localStorage.getItem('togetherwego-appearance'));
  check('theme saved', /sunset/.test(saved || ''), saved);
  check('still signed in after reload (no trip card)', await vis(p.getByText(/Plan your first group trip/)));
  check('no-trip home has create/join', await vis(p.getByRole('button', { name: /Create trip/ })));
  await p.close();
  // tour for the demo user
  const p2 = await (await b.newContext({ viewport: { width: 390, height: 844 }, isMobile: true, hasTouch: true })).newPage();
  await p2.goto(BASE + '/', { waitUntil: 'networkidle' }); await p2.waitForTimeout(2000);
  await p2.getByRole('button', { name: 'Skip' }).click(); await p2.waitForTimeout(700);
  await p2.getByLabel('Use the demo account').click(); await p2.getByTestId('sign-in').click(); await p2.waitForTimeout(2500);
  check('guided tour shows on first run', await vis(p2.getByText(/at a glance/i)));
  for (let i = 0; i < 4; i++) { await p2.getByRole('button', { name: 'Next' }).last().click(); await p2.waitForTimeout(450); }
  check('tour last step', await vis(p2.getByRole('button', { name: 'Got it' })));
  await p2.getByRole('button', { name: 'Got it' }).click(); await p2.waitForTimeout(700);
  check('tour closes', !(await vis(p2.getByText(/at a glance/i), 500)));
  await p2.reload({ waitUntil: 'networkidle' }); await p2.waitForTimeout(2500);
  check('tour does not return', !(await vis(p2.getByRole('button', { name: 'Got it' }), 800)));
  // midnight theme screenshot set
  await p2.evaluate(() => localStorage.setItem('togetherwego-appearance', JSON.stringify({ theme: 'midnight', corners: 'bubbly' })));
  await p2.reload({ waitUntil: 'networkidle' }); await p2.waitForTimeout(2500);
  await p2.screenshot({ path: shot('v3-midnight-home.png') });
  for (const t of ['Plan', 'Budget', 'Chat', 'More']) { await p2.getByRole('tab', { name: t, exact: true }).click(); await p2.waitForTim
eout(1100); await p2.screenshot({ path: shot(`v3-midnight-${t}.png`) }); }
  console.log(results.join('\n')); console.log('errors:', errs.length ? errs.join('\n') : 'none');
  await b.close();
})().catch((e) => { console.log(results.join('\n')); console.error('CRASH', e.message.slice(0, 600)); process.exit(1); });
```

e2e/verify4.js

```
const { BASE, start } = require('./lib');
const results = [];
const check = (name, ok, extra = '') => results.push(`${ok ? 'PASS' : 'FAIL'}  ${name}${ok ? '' : '  ' + extra}`);
const vis = async (loc, t = 2500) => loc.filter({ visible: true }).first().waitFor({ state: 'visible', timeout: t }).then(() => true).ca
tch(() => false);
const heading = async (p) => Number((await p.getByText(/360° ·/).first().innerText()).match(/(\d+)°$/)[1]);
(async () => {
  const { b, p, errors } = await start();
  // Home chip opens the trip's city
  await p.getByLabel('Open the 360 degree city view').click(); await p.waitForTimeout(2000);
```

```
  check('home 360° chip opens Positano', await vis(p.getByRole('heading', { name: /Positano/i })));
  check('night/day toggle exists', await vis(p.getByLabel(/(Night|Day) view\./)));
  // day/night toggle flips
  const before = await p.getByLabel(/(Night|Day) view\./).getAttribute('aria-label');
  await p.getByLabel(/(Night|Day) view\./).click(); await p.waitForTimeout(500);
  const after = await p.getByLabel(/(Night|Day) view\./).getAttribute('aria-label');
  check('toggle flips day/night', before !== after, `${before} -> ${after}`);
  // city picker
  await p.getByLabel('Paris 360 degree view').click(); await p.waitForTimeout(1500);
  check('city picker → Paris', await vis(p.getByRole('heading', { name: /Paris/i })));
  check('landmark callout shows Eiffel fact', await vis(p.getByText(/330 m tall/)));
  // arrows change heading, auto pause
  await p.getByLabel(/Auto rotate/).click();
  const h0 = await heading(p);
  await p.getByLabel('Look right').click(); await p.waitForTimeout(1200);
  const h1 = await heading(p);
  check('arrow turns the view', h1 !== h0, `${h0} -> ${h1}`);
  // drag
  await p.mouse.move(150, 300); await p.mouse.down(); await p.mouse.move(300, 300, { steps: 6 }); await p.mouse.up(); await p.waitForTim
eout(600);
  check('drag turns the view', (await heading(p)) !== h1);
  // landmark jump
  await p.getByLabel('Look at Arc de Triomphe').click(); await p.waitForTimeout(1500);
  check('jump to landmark shows its fact', await vis(p.getByText(/Napoleon in 1806/)));
  // full wrap: 360° of arrows returns near start heading
  const start0 = await heading(p);
  for (let i = 0; i < 12; i++) { await p.getByLabel('Look right').click(); await p.waitForTimeout(900); }
  const end0 = await heading(p);
  const diff = Math.abs(((end0 - start0 + 540) % 360) - 180);
  check('360° wraps around (≈ full circle after 12 turns)', diff <= 25, `${start0} -> ${end0} (off by ${diff}°)`);
  console.log(`  (12 arrow turns: ${start0}° -> ${end0}°, off by ${diff}°)`);
  // destination page button
  await p.goto(BASE + '/destination/tokyo', { waitUntil: 'networkidle' }); await p.waitForTimeout(1800);
  await p.getByRole('button', { name: /360° view/ }).click(); await p.waitForTimeout(1500);
  check('destination page opens Tokyo 360', await vis(p.getByRole('heading', { name: /Tokyo/i })));
  // More menu entry
  await p.goto(BASE + '/profile', { waitUntil: 'networkidle' }); await p.waitForTimeout(1500);
  await p.getByRole('button', { name: /360° City Views/ }).first().click(); await p.waitForTimeout(1500);
  check('More → 360° City Views', await vis(p.getByLabel(/360 degree view/)));
  console.log(results.join('\n')); console.log('errors:', errors.length ? errors.join('\n') : 'none');
  await b.close();
})().catch((e) => { console.log(results.join('\n')); console.error('CRASH', e.message.slice(0, 500)); process.exit(1); });
```

scripts/source-pdf.js

```
// Prints all hand-written source code into docs/TogetherWeGo-source-code.pdf.
//   npm run source-pdf         (needs a browser: `npx playwright-core install chromium`, or set E2E_CHROME)
const fs = require('fs');
const path = require('path');
const { chromium } = require('playwright-core');

const root = path.join(__dirname, '..');
const SKIP_DIRS = new Set(['node_modules', '.git', '.expo', 'dist', '.dist', 'shots', 'data', 'assets', 'screenshots', 'fixtures']);
const KEEP = /\.(tsx?|jsx?|json|md)$/;
const ROOT_FILES = ['README.md', 'ATTRIBUTIONS.md', 'package.json', 'app.json', 'eas.json', 'tsconfig.json', 'eslint.config.js', 'jest.c
onfig.js', 'jest.setup.js'];
const FOLDERS = ['src', 'server', 'e2e', 'scripts', 'docs'];

function walk(dir, out = []) {
  for (const name of fs.readdirSync(dir).sort()) {
    const full = path.join(dir, name);
    if (fs.statSync(full).isDirectory()) {
      if (!SKIP_DIRS.has(name)) walk(full, out);
    } else if (KEEP.test(name)) {
      out.push(full);
    }
  }
  return out;
}

const files = [...ROOT_FILES.map((f) => path.join(root, f)), ...FOLDERS.flatMap((d) => walk(path.join(root, d)))].filter((f) => fs.exist
sSync(f));
const esc = (s) => s.replace(/&/g, '&amp;').replace(/</g, '&lt;').replace(/>/g, '&gt;');
const rel = (f) => path.relative(root, f).split(path.sep).join('/');
const lines = files.reduce((n, f) => n + fs.readFileSync(f, 'utf8').split('\n').length, 0);

const html = `<html><head><meta charset="utf-8"><style>
@page{size:Letter;margin:14mm 10mm}body{font-family:Helvetica,Arial,sans-serif;font-size:10pt}h1{font-size:22pt}
h2{font-size:10.5pt;background:#222;color:#fff;padding:3px 6px;margin:0 0 4px;page-break-after:avoid}
pre{font-family:DejaVu Sans Mono,Menlo,Consolas,monospace;font-size:6.6pt;line-height:1.25;white-space:pre-wrap;word-break:break-all;mar
gin:0 0 10px}
.f{margin-top:6px}.toc div{font-size:8.5pt;line-height:1.5}
</style></head><body>
<h1>TogetherWeGo: complete source code</h1>
<p>FBLA Mobile Application Development 2026-27, &ldquo;Together We Go: Group Trip Planner&rdquo;. Expo SDK 57 / React Native / TypeScrip
t, with a small Node sync server. ${files.length} files, ${lines.toLocaleString('en-US')} lines. Photos and fonts are listed in ATTRIBUT
```

```
IONS.md and are not printed.</p>
<div class="toc">${files.map((f, i) => `<div>${i + 1}. ${esc(rel(f))}</div>`).join('')}</div>
${files.map((f) => `<div class="f" style="page-break-before:auto"><h2>${esc(rel(f))}</h2><pre>${esc(fs.readFileSync(f, 'utf8'))}</pre></
div>`).join('\n')}
</body></html>`;

(async () => {
  const browser = await chromium.launch(process.env.E2E_CHROME ? { executablePath: process.env.E2E_CHROME } : {});
  const page = await browser.newPage();
  await page.setContent(html, { waitUntil: 'load' });
  const out = path.join(root, 'docs', 'TogetherWeGo-source-code.pdf');
  await page.pdf({
    path: out,
    format: 'Letter',
    printBackground: true,
    displayHeaderFooter: true,
    headerTemplate: '<span></span>',
    footerTemplate: '<div style="font-size:7px;width:100%;text-align:center">TogetherWeGo source code - page <span class="pageNumber"></
span> of <span class="totalPages"></span></div>',
    margin: { top: '14mm', bottom: '14mm', left: '10mm', right: '10mm' },
  });
  await browser.close();
  console.log(`Wrote ${rel(out)} (${files.length} files, ${lines} lines)`);
})();
```

docs/BUILDING.md

```
# Running TogetherWeGo without Expo Go

The competition rules ask for a solution that runs on its own. Expo Go is a development tool: it needs your laptop's dev server running
and reachable. This page lists the ways to get a standalone copy and what each one costs.

| Goal | Way | Cost |
|---|---|---|
| Android phone, no laptop needed | Build an **`.apk`** with EAS (below) | Free Expo account |
| iPhone, no laptop needed | EAS build for iOS or TestFlight | **Apple Developer Program** (paid, yearly) |
| iPhone, laptop running | Expo Go with the laptop on the same wifi or a phone hotspot (`npx expo start`, or `--tunnel` if the wifi bloc
ks devices from seeing each other). SDK 57 is not in Apple's App Store: build your own Expo Go with `npx eas-cli@latest go` (see the REA
DME) | Apple Developer Program for that TestFlight build, and the laptop must stay running |
| Any phone or laptop browser | Web export (below) | Free, fewer features |

## Android `.apk` with EAS (about 15 minutes, mostly waiting)

```bash
npm install
npx eas-cli@latest login            # free account at expo.dev
npx eas-cli@latest init             # links this folder to your account (writes a project id into app.json)
npx eas-cli@latest build -p android --profile preview
```

When the build finishes (about 10 to 20 minutes in the cloud queue) EAS prints a link to the `.apk`. Open it on the phone, allow install
ing from the browser, and install. The app then runs with **no laptop, no dev server and no internet**. The first build asks to create a
signing key: say yes.

You can also build on your own computer with Android Studio installed: `npx expo run:android`.

iPhone

An installable iPhone build always needs an Apple Developer account:

```bash
npx eas-cli@latest device:create     # registers a phone (internal distribution)
npx eas-cli@latest build -p ios --profile preview
```

Without that account, use the web build on the iPhone (below) or an Android phone with Expo Go. Tip for the competition: start Expo on t
he laptop **before** the set-up window, join a phone hotspot with both devices, and keep the screenshots in `docs/screenshots/` as a bac
kup.

Web build

```bash
npx expo export --platform web      # writes a static site to dist/
```

Host the `dist/` folder anywhere that serves static files. On an iPhone, *Share → Add to Home Screen* gives it an icon. The web build ru
ns the whole planner, but not the things that need a phone's hardware: notifications, the native map (it falls back to OpenStreetMap), t
he camera QR scanner, haptics and speech on some browsers.

Sync from an installed copy

The sync server speaks plain HTTP on your own network. The build settings allow that (`app.json`: Android `usesCleartextTraffic`, iOS `N
SAllowsLocalNetworking`), and the first time the app talks to the server iOS asks for local-network access. An installed copy cannot gue
ss the server's address the way Expo Go does, so either:

* open an invite link or scan an invite QR code that carries the address, or
```

* type the address once in *Settings → Sync across phones* (the server prints it when it starts).

Windows tips

* If PowerShell says *running scripts is disabled on this system*, run once: `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`, or u
se **Command Prompt** instead.
* The first time `npm run server` or `npx expo start` runs, Windows Firewall asks whether Node.js may use the network. Choose **Allow**
for private networks. Without it phones cannot reach ports 8081 (Expo) or 8787 (sync).
* The phone and the laptop must be on the same network, and some school or venue wifi blocks devices from talking to each other. A phone
hotspot always works.

Things to know

* The native map on a standalone Android build needs a Google Maps API key (`android.config.googleMaps.apiKey` in `app.json`). Expo Go a
lready includes one. The offline radar map needs nothing.
* Live weather and exchange rates need internet; offline the app shows the last saved data and says so.
* `npx expo prebuild` generates the native projects if you want to open them in Xcode or Android Studio. The generated folders are not c
ommitted.

docs/DEMO_SCRIPT.md

7-minute presentation script

Built around the rating sheet so every scored item gets shown. Times are cumulative. Split the speaking parts between teammates (all mem
bers must present).

Before the 3-minute set-up ends (do not talk to the judges while you do it):

* Laptop: slides on screen, and a second terminal running `npm run server`. Put the laptop and **both phones on a phone hotspot** (venue
wifi often stops devices seeing each other). That is exactly three devices.
* Phone A: signed in as the demo account, *Settings → Demo timeline* on **"Starts in 3 days"**, notifications allowed, volume up (phrase
book audio). *Settings → Sync across phones → Test connection* says **Connected**.
* Phone B: signed in with a second account (Sign Up with any name, skip the survey), on the *Join a trip* screen.
* Backup if the network fails: skip the two-phone minute and show `docs/screenshots/`.

Time	Show	Say (key points)	Rating-sheet item
0:00	Title / app icon	The problem: group trips scattered across chats, spreadsheets and screenshots. TogetherWeGo puts plan, mone
y, people and paperwork in one place that works offline.	Addresses the prompt
0:25	Planning doc (personas, journey, MoSCoW)	How we planned: personas → user journey → feature priorities → navigation map → data
model → test log.	Planning process
0:50	Sign-in → tap Demo Account → Sign In; then a validation error on sign-up	Expo SDK 57 / React Native for iOS **and** Android.
Passwords are salted and stretched (PBKDF2). Every form checks format *and* meaning.	Platform · Input validation · Data handling
1:10	Home: hero photo, ☀️/🌙 chip, countdown, code pill, weather + local time, budget, Up next	The hero follows the local time in
Positano. Everything you need before a trip is on one screen.	UX · Functionality
1:35	Home → 360° chip → spin Positano, switch to Paris, 🌙 (lights come on), tap a landmark chip	Any city in 360°; at night the wi
ndows, landmarks and cars light up. Drawn in code, so it works offline.	Innovation · UX
1:50	More → Crew & roles → **Share** → WhatsApp; change a role; vote in a poll	Invite codes, roles, polls and a shared checklist.
One tap shares into social apps; the invite is also a QR code (we do not ask you to scan it).	**Social media integration** · Collabora
tion
2:20	Crew & roles → **Share this trip across phones**; phone B: type the code → Join. Send a chat message from each phone, add an ex
pense on B, add a photo on A	Two real phones, one server we run. Changes arrive within a second; the badge says *Live · 2 phones*. Men
tion airplane mode: changes wait and catch up.	Collaboration · Communication · Innovation
3:00	Plan: filters (Food, Under 1 hour), expand a card → 🔔; Vote tab → **Lock in**	Time and category filters. Votes show who is i
n; the organizer locks in the winner and the reminder turns on.	Scheduling · Collaboration
3:30	➕ Add Activity → submit empty (errors) → fill → overlap warning	Syntactic and semantic validation (it even warns about sched
ule clashes).	Input validation
3:50	More → Flights & arrivals → tap the time-zone line	Flights are stored as exact instants, so arrivals convert between Rome tim
e and your time.	Scheduling / time zones
4:15	Budget: donut, Local · EUR, ➕ add a € expense + **receipt photo**, Who owes who → Settle up	Money is kept in whole cents; th
e settle-up finds the fewest payments; the receipt is attached (and reaches the other phone).	Budgeting
4:55	Chat → Crew, then **Scout AI** → "How's the budget?"	The group chat stays human; Scout answers from our real trip data, **off
line**. Optional Claude mode with a key in the secure keychain.	Innovation
5:20	Airplane mode → Offline Docs, Trip Map → Offline radar, Packing → Split the rest fairly	Built for bad wifi: documents, map, p
hrasebook and emergency info work offline. Everyone sees who is bringing what.	Data handling · Organization
5:45	Settings → Language → 日本語 → back; text size; mention Reduce Motion	6 languages, adjustable text, screen-reader labels, and
the phone's Reduce Motion setting is respected.	Accessibility
6:00	Home → Customize → Make it yours → Sunset/Midnight + interests → Apply	Every user makes the app theirs: themes incl. dark mod
e, interests drive "Picked for you".	Innovation · UX
6:15	Settings → Demo timeline "In progress" → Home "Today" card; then "Just finished" → Recap story card	The full lifecycle: plann
ing → travelling → recap with stats, balances and a shareable story card.	Addresses all parts
6:35	More → How it works; then the repo: README, `docs/`, tests	Layered MVVM, a sync engine with its own tests, strict TypeScript,
0 lint errors; unit tests and browser tests; every library, API and photo documented.	Code quality · Architecture · Documentation

Protocol reminders (full checklist in `docs/RATING_SHEET.md`): max 3 devices, no external speakers, no power, never ask judges to sc
an a QR code or click a link, leave nothing behind, stay silent during set-up.

Likely questions: short answers

- **How do friends' phones stay in sync?** Every phone keeps its own full copy and turns each edit into a tiny stamped change. A small s
erver we run passes the changes between phones, and each field takes its newest change, so all phones end up identical whatever the orde
r. Changes made offline wait in a queue and are sent when the connection is back.
- **How do you know the merge is correct?** We wrote a test that makes three simulated phones edit at random (add, change, delete, vote)
and swap changes in random orders, 150 times; they always end identical. There are also tests with the real server over real HTTP.
- **Why not Firebase?** It needs an account and keys that a school project cannot safely hold, and it does not give us offline merging f
or free. Our server is about 150 lines, has no dependencies and is part of the repo. A real release would put the same protocol behind H

TTPS and accounts.
- **Is the sync secure?** A trip is protected by a 24-character random id that only the invite code reveals, code look-ups are rate limited, and private things (your Scout chat, people's emails, flight tickets) never leave your phone. It is plain HTTP, so it is for a network you trust. That is a stated limit, not a surprise.
- **What if the venue wifi blocks it?** We use a phone hotspot. Without a network each phone still works fully on its own.
- **What did you test?** Unit tests for the money maths (exact settle-up), validation, time zones, password hashing against the official test vector, the store, the assistant and the whole sync stack; browser tests of every main flow, a sideways-overflow audit of every screen, and two phones syncing. We have **not** yet tapped the camera QR scanner, on-device file upload and "share as picture" on a physical phone.
- **Why Expo / React Native?** One TypeScript codebase for iOS and Android, with native modules for notifications, maps, camera, files and secure storage.
- **How is data protected?** Salted PBKDF2-HMAC-SHA256 password hashes (stretched, upgraded automatically), optional session persistence, API key in Keychain/Keystore, files in app-private storage, private data kept off the sync server.
- **What makes it accessible?** Labels and roles on every control, adjustable text size, colour contrast, haptics toggle, Reduce Motion respected, 6 languages, clear inline errors.
- **What happens with no internet?** Everything keeps working from local storage; weather and exchange rates show the last saved data (or an estimate) and say so.
- **How does the AI work offline?** It is our own intent-matching engine over the trip data (budget maths, settle-up algorithm, weather cache, place database). Claude is optional for open-ended questions.
- **Can I try to break it?** Yes: `docs/JUDGE_CHECKLIST.md` lists things to try and what should happen.

docs/JUDGE_CHECKLIST.md

```
# Try to break it: a checklist for judges and teammates

The sample trip is tidy on purpose. These checks use the app the way a real person (or a grumpy judge) would, so you can see that nothing is hard-wired. Each line says what *should* happen.

## Start from nothing
- [ ] **Create a new account** (Sign Up) with your own name. You land in a short survey, then an empty Home that says *Plan your first group trip*. There is no sample data.
- [ ] With the demo account: *Settings → Clear the sample trips*. The app is empty again. *Reset app data* brings the sample trips back.
- [ ] **Create a trip** from scratch (any city, dates, budget). Add an activity, a poll, an expense, a task, a flight.

## Type the wrong thing
- [ ] Sign up with `abc`, then `a@b`, then a 5-letter password: each gets a plain message under the field.
- [ ] Add an expense of `-5`, `abc`, `0`, `1000001`: refused with a clear message. `12.5` and `1,200` work.
- [ ] Join a trip with `ZZZ-9999`, `abc`, an empty code: refused. Join a trip you are already in: refused.
- [ ] Add a flight with an unknown airport (`XXX`), identical from/to airports, or arrival before departure: refused.
- [ ] Add an activity that overlaps another: you get a warning.
- [ ] Type a name with an accent or apostrophe (`Zoë`, `O'Brien`): accepted.

## Break the connection
- [ ] Turn on **airplane mode**. Plan, vote, add expenses, open documents and the offline map, ask Scout. Weather and rates show saved data and say so.
- [ ] With sync on (see below) make a change offline, then turn the connection back on: it reaches the other phone. The badge says *Offline · N waiting* and then *Live*.

## Two phones
- [ ] Start the server (`npm run server`). Phone A: *Crew & roles → Share this trip across phones*. Phone B: *Join a trip*, type the code. Both show **Live · 2 phones**.
- [ ] Send a chat message, cast a vote, add an expense, add a photo, a receipt, a PDF on one phone: each shows up on the other.
- [ ] Your **Scout AI** chat and other people's emails do not appear on the other phone.

## Accessibility and comfort
- [ ] *Settings → Text size → Extra large*: nothing is cut off or sideways-scrolling.
- [ ] Turn on the phone's **Reduce Motion**: the 360° view stops auto-rotating, entrances do not animate, Settings explains why.
- [ ] Switch language to 日本語, then back.
- [ ] Use a screen reader (VoiceOver / TalkBack): every button has a label.

## Look under the bonnet
- [ ] `npm test` and `npm run e2e` (see `e2e/README.md`): unit and browser tests pass.
- [ ] *More → How it works*: architecture, sync, security, validation, rubric map.
- [ ] `docs/RATING_SHEET.md`: every line of the rating sheet and where it is shown, including the weak spots.
```

docs/PLANNING.md

```
# TogetherWeGo — Planning documents

These are the planning artifacts behind the app (rating sheet: *Planning Process — tangible planning documents*).

## 1. Problem statement
Group trips fall apart in group chats: plans live in five apps, nobody knows who paid for what, flights land at different times in different time zones, and the one person with the booking PDF has no signal. **TogetherWeGo puts the whole trip — plan, money, people and paperwork — in one shared place that still works offline.**

## 2. Personas
| Persona | Needs | Features that serve them |
|---|---|---|
| **Maya, 17 — the organizer** | Get six friends to agree, keep everyone on schedule | Invite codes, roles, polls & activity voting, live reminders, printable planner |
| **Priya, 17 — the treasurer** | Track a shared budget fairly, settle up without awkwardness | Budget cap, multi-currency expenses, splits, fewest-payment settle-up |
| **Diego, 16 — the foodie** | Find great food nearby within budget | Nearby radius + price filter, budget finds, Trip Assistant “best food nearby?” |
| **Mr. Kim — a parent chaperone** | Know where everyone is, keep documents & emergency info handy | Flights & arrivals board, offline d
```

ocument vault, emergency numbers, location sharing |

3. User journey
```mermaid
flowchart LR
  A[Onboarding] --> B[Sign in / Sign up]
  B --> C{Has a trip?}
  C -- no --> D[Create trip<br/>or Join with code]
  C -- yes --> E[Home]
  D --> F[Invite group<br/>share code to socials]
  F --> G[Assign roles]
  G --> H[Build itinerary<br/>vote & confirm]
  H --> I[Add flights<br/>time-zone board]
  I --> J[Budget & expenses]
  J --> K[Travel: reminders,<br/>chat + assistant, map,<br/>photo dump]
  K --> L[Mark complete →<br/>Recap & settle up]
```

4. Navigation map
```mermaid
flowchart TB
  Root[Root Stack<br/>Stack.Protected] --> Auth[Onboarding · Sign in · Sign up · Reset password]
  Root --> Tabs[Bottom tabs]
  Tabs --> T1[Home]
  Tabs --> T2[Plan + Vote]
  Tabs --> T3[Budget]
  Tabs --> T4[Chat]
  Tabs --> T5[More menu]
  Root --> More[Stack screens]
  More --> M1[Destination · All destinations · Nearby · Map]
  More --> M2[Flights · Weather · Currency · Nearest airport]
  More --> M3[Photo dump · Documents · Packing · Bucket list]
  More --> M4[Quiz · Survey · Recap · Emergency · Phrasebook]
  More --> M5[My trips · New/Edit trip · Join · Notifications · Settings · Help · About]
  More --> M6[Make it yours — personalization survey]
  More --> M7[Crew & roles · How it works]
  More --> M8[360° City Views — day / night panoramas]
```

5. Architecture
```mermaid
flowchart LR
  subgraph View[Views — src/app + src/components]
    S[Screens] --> UI[UI kit: Button, Sheet, Chip, Input, Slider…]
  end
  subgraph VM[View-model — selector hooks]
    H[useActiveTrip · useCurrentUser · tripStatus · balances · settlePlan]
  end
  subgraph Model[Model — src/store]
    Z[(Zustand + Immer store)] --> P[(AsyncStorage<br/>offline persistence)]
  end
  subgraph Svc[Services — src/services]
    W[weather] & C[currency] & N[notifications] & AI[assistant] & F[files] & PR[print] & SH[share] & L[location] & SE[secrets]
  end
  subgraph Sync[Sync — src/sync + server]
    E[Sync engine<br/>diff · merge · outbox] --> SV[(Sync server<br/>Node, runs on your computer)]
  end
  S --> H --> Z
  S -- actions --> Z
  S --> Svc
  Z <--> E
  W --> OM[(Open-Meteo)]
  C --> ER[(ExchangeRate API)]
  AI -. optional .-> CL[(Claude API)]
  SE --> KC[(Keychain / Keystore)]
```

6. Data model
```mermaid
erDiagram
  ACCOUNT ||--o{ MEMBER : "is"
  TRIP ||--|{ MEMBER : has
  TRIP ||--o{ ACTIVITY : schedules
  TRIP ||--o{ EXPENSE : tracks
  TRIP ||--o{ FLIGHT : lists
  TRIP ||--o{ POLL : runs
  TRIP ||--o{ TASK : checklist
  TRIP ||--o{ DOC : vault
  TRIP ||--o{ MESSAGE : chat
  TRIP ||--o{ PHOTO : "photo dump"
  TRIP ||--o{ PACKING_ITEM : packs
  ACTIVITY }o--o{ MEMBER : "assigned / votes"
  EXPENSE }o--|| MEMBER : "paid by"
  EXPENSE }o--|{ MEMBER : "split between"
  FLIGHT }o--|| MEMBER : "belongs to"
```

Key decisions:
- **Dates vs. instants.** Itinerary times are wall-clock times in the destination’s zone (`2026-10-06` + `14:30`), because that is how people plan. Flights are stored as UTC instants so they convert correctly to any time zone.
- **Money in one base currency.** Expenses are stored in USD (with the original amount kept) so totals and splits are always comparable; views convert to home or local currency.
- **Majority voting.** A “voting” activity becomes “confirmed” when more than half of members vote yes.
- **Exact money.** Splits and balances are calculated in whole cents, and the fewest-payments settle-up is solved exactly for groups up to 12 people, so the books always balance to the cent.
- **Passwords.** Salted PBKDF2-HMAC-SHA256, the round count stored with each hash so it can be raised later; older hashes are upgraded the next time someone signs in.
- **Sync.** Every field of a shared trip is a small register stamped by a hybrid logical clock; the newest stamp wins, so phones that have exchanged their changes hold identical trips whatever the order. Private data (Scout AI chat, other people's emails, flight tickets) is never sent. Details in `docs/SYNC.md`.

7. Feature priorities (MoSCoW)
Must	Should	Could	Won’t (this version)
Trips + invite codes, roles, itinerary, voting, budget & splits, chat, flights & time zones, reminders, offline storage, validation	AI assistant, currency converter, weather, documents vault, map, nearby radius & filters, printable planner, social sharing	Quiz, survey recommendations, phrasebook TTS, recap, packing suggestions, emergency screen, 6 languages, personalization (themes, interests, home layout), animations, **live sync across phones** through a self-hosted server, invite QR codes	A hosted cloud service, accounts on a server, real Google OAuth, payments

8. Accessibility & UX rationale
- Reference-matched visual language: cream canvas, forest-green primary, soft-green highlights, pill navigation — calm and readable outdoors.
- Every icon-only control has an `accessibilityLabel`; toggles expose checked/selected state; sliders are adjustable with screen readers.
- Text size setting (Default / Large / Extra large) on top of the phone’s own font scaling.
- Personalization: 6 colour themes including a dark mode, adjustable card shapes and an *Animations* switch for people sensitive to motion.
- Errors appear inline under the field in red with a plain-language fix (“Codes look like ABC-1234”).
- Destructive actions always confirm; toasts confirm every successful action.

9. Test log

9a. Automated (`npm test`, runs in about 5 seconds)
Area	What the tests prove
Money (`src/utils/__tests__/money.test.ts`)	Cents never drift (even 0.1 + 0.2), splits add up to the total with the odd cents handed out fairly, settle-up leaves every balance at zero, never needs more than (people − 1) payments on random groups, and finds fewer payments than the simple largest-first method when a group splits into pairs
Validation, time zones, passwords	Email/name/amount/date/invite/flight/IATA rules, daylight-saving edges, PBKDF2 against the RFC 7914 test vector, old hashes upgrade
Store (`src/store/__tests__`)	Sign-up/sign-in/reset, trips, joining, itinerary and voting, budget, polls, packing, photos, docs, attaching files, clearing sample data
Assistant	Local answers (budget, who owes who, currency, emergency numbers, places) and the Claude request: what is sent, the last 12 messages only, wrong key, refusal, empty answer, the 45-second limit
Sync (`src/sync/__tests__`, `server/__tests__`)	Merge convergence on 150 random three-phone runs, sharing and joining, offline queue, restart, private data, files, the real server over real HTTP, time limits, address parsing
Invites and QR	Link round trip for every address form, QR codes that a decoder reads back
Home ranking, pretend friends, Reduce Motion	The "Picked for you" ranking, canned replies, the motion rule

9b. Automated browser tests (`npm run e2e`, see `e2e/README.md`)
Eight scripts drive the web build in a phone-sized window: every main flow, the personalisation survey (including tapping a summary row to change an answer), the 360° views, Reduce Motion, a **sideways-overflow audit of every screen at 390 px, 320 px and with larger text**, a **no-button-inside-a-button audit** (the web warning *"`<button>` cannot contain a nested `<button>`"*), and a **two-phone sync run** against the real server (share, join, chat, expenses, offline catch-up, restart, a photo, a receipt and a PDF).

9c. Manual + earlier browser runs
Area	Checks
Auth	Empty-field errors, wrong password, sign-up validation & strength meter, reset-code flow, old password rejected after reset
Plan	Add-activity validation, overlap warning, vote → auto-confirm, **Lock in** (organizer only), reminder toggle, edit, delete, category + time filters
Budget	Receipt attach, expense validation, who-owes-who, settle up, home/local currency view
360° views	Drag/arrows/auto-rotate wrap past 0°, day/night toggle, landmark jump + facts, city picker, entry points (Home, destination, More)
Extras	Day/night hero toggle, Scout AI tab separate from the crew chat, packing assignment + fair split, first-run tour, story card
Budget	Amount validation, ¥ expense converted to USD, settle-up records payment, budget-cap validation
Collaboration	Join with empty/unknown/valid code, add-member validation (name, email), polls (vote, close), checklist add + empty-task error
Flights	Missing-field errors, unknown IATA rejected, flight saved, time-zone toggle
Personalization	Survey requires ≥1 interest, nickname length check, theme/corner choice survives restart, sign-in kept across the theme reload, Explore sections hide/show, “Picked for you” ranking
Platform	`tsc --noEmit` (strict) ✔, ESLint 0 problems ✔, iOS and Android bundles compile ✔, `expo prebuild` produces the native projects with the local-network and camera settings ✔
Not run on a physical phone	On-device file upload/download, the camera QR scanner and "share as picture" (they are native-only; everything around them is covered above)

docs/RATING_SHEET.md

Rating sheet map: where each point is earned

The FBLA 2026-27 *Mobile Application Development* rating sheet has 16 scored lines (110 points). This page lists every line, what we sho

w the judges, and where to find it. It also lists the honest weak spots.

Design and code quality (25)

Line	Pts	Evidence	Where
Planning process	10	Personas, user journey, navigation map, MoSCoW priorities, data model, manual test log, all in one planning document. Terms used: persona, user journey, MoSCoW, MVP, MVVM, offline-first, one-way data flow.	`docs/PLANNING.md`
Classes, modules, components	5	UI kit of reusable components (`Screen`, `Card`, `Button`, `Press`, `Segmented`, `Donut`), trip widgets, one component per file, services for every outside system. The store is split into slices (`auth`, `trips`, `itinerary`, `budget`, `groupTools`, `social`, `files`, `demo`, `app`) and every large screen is a short composition of components (Home, Crew, Settings, Chat, Photos, Flights, Budget, Survey). No screen file is over 300 lines.	`src/components/ui/`, `src/components/home/`, `src/components/crew/`, `src/store/slices/`, `src/services/`
Architectural patterns	5	Layered MVVM: screens (views) read selector hooks and call store actions; services wrap APIs and device features; one-way data flow; Expo Router file-based navigation with `Stack.Protected` guards; platform-specific files (`TripMap.tsx` / `TripMap.web.tsx`). A separate **sync engine** that knows nothing about React or the network (store, storage and transport are passed in, so it is tested with two phones in memory), plus a small Node server.	`src/store/`, `src/app/`, `src/sync/`, `server/`, *More → How it works*
Innovation and creativity	5	**Live sync across phones** with a merge rule that gives identical trips whatever order changes arrive in (offline queue, photos and receipts included), 360° city views with day/night and night lights, skyline hero that follows the city's local time, offline Scout AI answering from trip data, "Lock in" voting, 7-step personalization survey with live themes, offline radar map, invite QR codes.	`src/sync/`, `src/app/panorama.tsx`, `src/services/assistant.ts`, `src/app/personalize.tsx`

User experience (20)

Line	Pts	Evidence	Where
UX design: user journey, rationale, accessibility	5	Journey map and rationale in the planning doc. Accessibility: labels and roles on every control, adjustable text size, high-contrast palette, haptics toggle, "Animations" off switch that also follows the phone's **Reduce Motion** setting, 6 languages. A browser audit checks that no screen scrolls sideways at 390 px, 320 px or with larger text.	`docs/PLANNING.md`, *Settings*, `e2e/overflow.js`
Intuitive UI or clear instructions	5	First-run guided tour, a 9-step *How to use* page with an FAQ, inline hints, empty states that say what to do next.	`src/components/GuidedTour.tsx`, `src/app/help.tsx`
Icons and graphics consistent	5	Custom app icon (a take-off plane on a forest-green tile) reused as the in-app logo on sign-in, onboarding, tour and recap; one icon set and one palette throughout.	`assets/images/icon.png`, `src/components/Logo.tsx`
Input validation (syntax **and** meaning)	5	Syntax: email, invite-code format (`AML-2026`), time `HH:MM`, flight number, IATA code, amount parsing. Meaning: trip end after start, amount between limits, activity time-clash warning, duplicate email is refused, joining a trip you are already in is refused.	`src/utils/validation.ts`

Application functionality (25)

Line	Pts	Evidence	Where
Addresses all parts of the prompt	10	The topic names collaboration, communication, scheduling, budgeting, organization and multi-user trip management. Each is a feature set, and the mapping is explained inside the app.	README section 2, *More → How it works → Rubric map*
Social media integration	5	Direct deep links into WhatsApp, X, Facebook, Messenger, Telegram, Instagram, SMS and email, plus the native share sheet. Photos and the **recap story card share as pictures**. Invites are a code, a link and a **QR code** (scan it inside the app or with a camera); the link opens the Join screen with the code filled in, and also carries the sync server's address. Invite link and QR round trips are unit tested. (Logging in with or posting to a social network from inside the app needs OAuth credentials and app review, which is out of reach for a student project.)	`src/services/share.ts`, `src/sync/invite.ts`, `src/components/crew/InviteQrSheet.tsx`, *Recap*
Data handling and storage	5	Zustand store persisted to local storage with versioned migrations (integrity); passwords salted and stretched with PBKDF2, session off when "Remember me" is off, API key in the Keychain/Keystore (protection); money kept in whole cents so totals never drift; export a trip, clear the sample data and reset all data (control); whole app works offline; shared trips are kept identical across phones and private data (Scout chat, emails, tickets) never leaves the phone.	`src/store/`, `src/sync/`, `src/services/secrets.ts`, `src/utils/password.ts`, *Settings*
Documentation and copyright	5	README, planning doc (with a test log), sync design, build guide, judge checklist and attributions: every library with its licence, every photo with its Unsplash source URL, fonts under OFL, no logos or trademarks.	`README.md`, `docs/`, `ATTRIBUTIONS.md`

Evidence that it works

* `npm test`: unit tests for money, validation, time zones, passwords, the store, the assistant, invites, QR codes and the whole sync stack (merge convergence on 150 random three-phone runs; the real server over HTTP).
* `npm run e2e`: browser tests of every main flow, a sideways-overflow audit of every screen, Reduce Motion, and two phones syncing through the real server.
* `docs/JUDGE_CHECKLIST.md`: things to try (wrong input, airplane mode, two phones, large text) and what should happen.

Presentation delivery (30)

Line	Pts	Prepared	Where
Organized, clearly stated	10	A timed 7-minute script built around this very sheet, with a speaker split.	`docs/DEMO_SCRIPT.md`
Confidence, body language, eye contact, voice	10	Rehearse with the table below. Judges mark this by watching you, not the app.	n/a
Answering questions	10	Short answers to the likely questions (sync, security, offline, why React Native, accessibility).	`docs/DEMO_SCRIPT.md`

Presentation protocols (10): the checklist

- [ ] At most **3 devices** (e.g. phone + laptop for slides + a tablet). Screens laptop-sized or smaller.
- [ ] **No external speakers**: audio (phrasebook voice) comes from the phone itself.
- [ ] **No power outlet**: charge everything beforehand.
- [ ] **Links and QR codes**: you may show them, but judges must never scan or click them. Do not make the demo depend on a QR code or link (for the two-phone demo type the invite code on the second phone).

- [ ] **Two-phone sync demo**: phone A, phone B and the laptop (which shows the slides *and* runs `npm run server`) are exactly 3 devices. Put all three on a **phone hotspot** (venue wifi often blocks devices from seeing each other), start the server before set-up ends and check *Settings → Sync across phones → Test connection* on both phones.
- [ ] Nothing left with the judges afterwards. Pack up printouts.
- [ ] No food or animals. Topic stays "Together We Go: Group Trip Planner".
- [ ] Do not talk to the judges during the 3-minute set-up.
- [ ] Every teammate presents a part.
- [ ] Venue wifi may fail: use *Settings → Demo timeline*, airplane-mode offline demo, and the screenshots in `docs/screenshots/` as a backup.

Honest weak spots (so you are not surprised in Q&A)

1. **Standalone run.** The rules say the solution must run standalone. Expo Go needs your laptop's dev server, so build an installed copy before the event (`docs/BUILDING.md`): `npx eas-cli@latest build -p android --profile preview` gives an `.apk` (free Expo account). iPhone standalone needs an Apple Developer account. **No installed build was made while writing this**: it needs an Expo account that only the team can create. The build configuration was checked with `expo prebuild`.
2. **Sync needs a server you run.** Phones share a trip through `npm run server` on a computer they can reach. There is no hosted service and no accounts, and the traffic is plain HTTP, so it belongs on a network you trust. Say so before a judge asks (`docs/SYNC.md`, "Limits").
3. **Not run on a physical phone.** The on-device file upload and download, the camera QR scanner and "share as picture" are native-only. They compile into the iPhone bundle and the logic around them is tested, but nobody has tapped them on a phone yet. Try them once before the event.
4. **Same-field edits.** If two people change the *same* field at the same instant, the later change wins and the other is lost. Chat, votes, expenses and photos are separate items, so they never clash.
5. **Passwords** use salted PBKDF2-HMAC-SHA256 with 10,000 rounds (a server would use 600,000 or Argon2id). The count is a judgement call for a phone's JavaScript engine: it takes about 20 ms in Node, and we expect well under a second on a phone, but we have **not timed it on a phone**. The work is split into 8 ms slices so the screen stays responsive, and the round count is stored in each hash so it can be raised later. Accounts live on the phone, not on a server.
6. **Live weather and exchange rates** need internet. Offline, the app shows the last saved data and labels it.
7. **"Continue with Google"** is a simulated account chooser and says so on screen.

docs/SYNC.md

How phones stay in sync

Friends can each plan on their own phone and see the same trip: chat, votes, the itinerary, expenses, the checklist, photos, documents and receipts. This page explains how it works, what is private, how it was tested and where it stops.

Try it (two minutes)

1. On the computer that runs the app, open a **second terminal** in the project folder and start the sync server:

   ```bash
   npm run server
   ```

   It prints addresses such as `http://192.168.1.20:8787`. On Windows, click **Allow access** (private networks) when the firewall asks.

2. Start the app as usual (`npx expo start`). While developing, phones **find the server by themselves** (they use the same computer address that Expo Go used). If a phone cannot, enter the address in **Settings → Sync across phones** and tap *Test connection*.

3. Phone A: **More → Crew & roles → Share this trip across phones**.
4. Phone B: **Join a trip**, type the invite code (or scan the QR code from *Crew & roles → QR code*; the QR code and invite link carry the server's address once the trip is shared, and the phone asks before using it). The trip appears with its chat, plan and photos, and the badge says **Live · 2 phones**.

Everything either phone changes shows up on the other within a second or two. Turn on airplane mode, make changes, turn it off: they catch up.

> Rules of the competition: judges must not scan QR codes or click links, so during the presentation type the code on the second phone instead.

What is shared and what is not

Shared with the group	Stays on your phone
Chat messages, polls and votes, the checklist, packing list	Your **Scout AI** conversation
Itinerary, activity votes, flights (without ticket files)	Other people's **email addresses**
Expenses and settle-up, members and roles	**Flight ticket files** (they hold personal details)
Photos, documents and receipt photos (as files)	Your password, settings, language, theme

The pieces

```
 phone A                                              phone B
┌──────────────────────┐                             ┌──────────────────────┐
│ app store (Zustand)  │                             │ app store (Zustand)  │
│        │ diff        │                             │        ▲ rebuild     │
│        ▼             │   ops    ┌──────────────┐   │        │             │
│  sync engine ──────────────────▶ │  sync server │ ops │        │             │
│  registers + outbox  │◀──────── │ (Node, no    │ ──────▶ sync engine       │
└──────────────────────┘   long   │  accounts)   │ ◀────── registers+outbox  │
                           poll   │ log per trip │ long  └──────────────────────┘
                                  │ files per trip│ poll
                                  └──────────────┘
```

* `src/sync/schema.ts` says which parts of a trip are shared and how each part merges.
* `src/sync/crdt.ts` turns "what changed" into small stamped changes (ops) and merges ops into **registers**.
* `src/sync/engine.ts` watches the store, queues ops, sends them, fetches friends' ops, and writes the merged trip back into the store. It knows nothing about React or the network library, which is why the tests can run two phones against one server in memory.
* `src/sync/transport.ts` is the HTTP client (every request has a time limit). `server/` is the server.
* `src/sync/blobs.ts` moves photos, documents and receipts (see below).

The merge rule

Every field of the trip is a **register** that remembers the stamp of the change that last wrote it. **The newest stamp wins**, field by field. Stamps are hybrid logical clocks (`<time>-<counter>-<phone>`), so they stay in order even when two phones' clocks disagree a little.

* Two people editing *different* things both win (one renames an activity, the other changes its time).
* Two people editing the *same* field: the later change wins, on every phone.
* Lists are lists of items with ids. Adding an item never conflicts. A delete leaves a small tombstone so a late change to a deleted item cannot bring it back.
* Votes and "who is assigned" are sets or maps, so two people voting at the same moment both count.

Because the rule is deterministic, phones that have seen the same changes hold the same trip, **no matter in what order the changes arrived**. The tests check this with random runs (see below).

Offline and restarts

Changes go into an **outbox** that is saved on the phone. If the server is out of reach the badge says *Offline · 3 waiting* and the app keeps working. When the connection returns the outbox is sent. If the app is closed before that, the unsent change is found again on the next start (the engine compares the saved trip with what it last sent).

Files

Photos, documents and receipts are too big to be changes. The phone that has the file uploads it (`PUT /v1/trips/:id/files/:fileId`) and then writes the server's file id on the item, which syncs like any other change. Friends' phones see the id, download the file into the app's private storage and show it; after that it opens offline too. Pictures that ship with the app are shared by name (`asset:amalfi`) instead of being uploaded. A file over 8 MB stays on the phone that has it, and that person is told once.

The server

`server/index.js` is about 150 lines of Node with no dependencies. It keeps, for every shared trip, an append-only log of ops and a folder of files, in `server/data/` (a JSON file and a `files/` folder). It never has to understand an op.

Request	What it does
`POST /v1/trips/:id/ops`	send changes (up to 500, 1 MB)
`GET /v1/trips/:id/ops?since=N&wait=ms`	get changes after N; if there are none, **wait** up to 25 s for one (long polling), so news arrives at once
`PUT` / `GET /v1/trips/:id/files/:fileId`	send / fetch a file (8 MB each, 150 MB per trip)
`GET /v1/codes/:code`	invite code to trip id (60 tries a minute per address)
`GET /health`	is it alive, how many trips

Security model (read this before you rely on it). There are no accounts. A trip is protected by its **id**: 24 random characters (about 124 bits) from the phone's secure random generator. A phone only learns the id by entering the invite code, and code look-ups are rate limited. Traffic is plain HTTP, so **only run the server on a network you trust** (home wifi, a phone hotspot), not on a public network. Anyone who can read the network could read the trip. A production version would use HTTPS and real accounts.

What was tested

* `src/sync/__tests__/crdt.test.ts`: the merge rule. **150 random runs** where three phones make random edits (add, change, delete, vote) and exchange changes in random orders always end with identical trips.
* `src/sync/__tests__/engine.test.ts`: sharing, joining, edits both ways, offline queueing, restart, a change made while the app was closed, private data staying private, paging a trip of 700 messages, an unknown code, a code that is already taken.
* `src/sync/__tests__/e2e.test.ts`: the same engines talking to the **real server over real HTTP**, including the long-poll wake-up.
* `src/sync/__tests__/blobs.test.ts`: uploading, downloading, retrying, too-big files, deleted files, bundled pictures.
* `src/sync/__tests__/transport.test.ts`: time limits, cancelling, error kinds, and server-address checks against **both** Node's `URL` and the phone's looser `URL` class.
* `server/__tests__/server.test.js`: the store and the HTTP API, including limits and restarts.
* Browser test (`e2e/twophones.js`): two separate browser profiles and one real server. Phone A shares, phone B joins from a decoded QR link, chat and expenses travel both ways, A goes offline and comes back, B restarts, and a photo, a receipt and a PDF reach the other phone.

Limits (the honest list)

* The server is a program you run on your own computer; there is no hosted service. Phones need to reach it (same wifi, hotspot, or the address typed in Settings).
* Files uploaded to the server are not deleted when the item is deleted (the 150 MB per trip limit keeps this bounded).
* The first version of "last writer wins" means two people editing the *same* field at the same moment lose one of the edits (the later one stays). Chat messages, votes and expenses are separate items, so they never clash.
* The on-device file upload and download code, the camera QR scanner and "share as picture" use the phone's native modules. They build into the iPhone bundle and are covered by the web tests of everything around them, but they were **not run on a physical phone** while this was written.